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QUESTION DISTRIBUTION

(CDS ARITHMETIC MATHS)

		2017		2018		2019		2020		2021		2022		2023		2024	
		1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2
1.	Ratio	3	3	4	3	3	1	1	3	1	1	0	2	2	1	5	
2.	Average	3	2	3	2	3	1	1	3	4	2	0	3	1	2	0	
3.	Time and work	3	2	4	5	2	3	2	1	2	3	1	3	4	2	4	
4.	Time and distance	4	5	4	4	4	1	3	5	3	2	4	4	2	2	5	
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6.	Profit loss	1	2	1	1	1	2	-	3	2	1	1	0	1	1	2	
7.	Interest	1	1	3	2	2	2	1	1	2	1	1	1	2	2	3	
8.	Number System	5	1	7	7	15	10	9	11	4	5	8	4	21	4	11	
9.	Simplification	1	3	3	-	1	2	2	2	3	1	3	1	2	0	3	
10.	LCM HCF	2	-	2	2	-	2	3	2	-	3	1	2	4	2	2	

QUESTION DISTRIBUTION

(CDS ADVANCE MATHS)

		2017		2018		2019		2020		2021		2022		2023		2024	
		1	2	1	2	1	2	1	2	1	2	1	2	1	2	1	2
11.	EQUATIONS	8	3	6	9	5	11	6	2	6	10	10	3	8	18	3	
12.	ALGEBRA	5	5	7	7	9	3	7	10	15	11	11	16	3	19	7	
13.	LCM HCF OF ALGEBRIC FACTORS	1	2	0	1	1	1	1	2	2	0	0	3	0	2	1	
14.	SETS	1	2	0	7	1	1	1	0	0	0	4	0	2	0	0	
15.	LOGARITHM	3	1	1	2	1	3	3	2	2	3	0	2	0	1	0	
16.	GEOMETRY	21	20	17	14	15	15	14	17	13	10	16	13	10	9	16	
17.	MENSURATION 2D	3	2	3	12	12	13	18	9	11	8	6	12	19	9	21	
18.	MENSURANTION 3D	3	8	13	7	5	10	5	7	11	13	9	9	6	7	3	
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PRACTICE SHEET 1

1. If $a : b = 7 : 9$ and $b : c = 15 : 7$, then what is $a : c$?
 (1) 5 : 3 (2) 3 : 5
 (3) 7 : 21 (4) 7 : 15
2. If $x = \frac{1}{3}y$ and $y = \frac{1}{2}z$, then $x : y : z$, is equal to:
 (1) 3 : 2 : 1 (2) 1 : 2 : 6
 (3) 1 : 3 : 6 (4) 2 : 4 : 6
3. If $x : y = 3 : 1$, then $x^3 - y^3 : x^3 + y^3 =$
 (1) 13 : 14 (2) 14 : 13
 (3) 10 : 11 (4) 11 : 10
4. The fourth proportional to 0.12, 0.21, 8 is :
 (1) 8.9 (2) 56
 (3) 14 (4) 17
5. The ratio $2^{1.5} : 2^{0.5}$ is the same as
 (1) 2 : 1 (2) 3 : 1
 (3) 6 : 1 (4) 3 : 2
6. If $A : B = 3 : 4$, $B : C = 5 : 7$ and $C : D = 8 : 9$ then $A : D$ is equal to
 (1) 3 : 7 (2) 7 : 3
 (3) 21 : 10 (4) 10 : 21
7. If b is the mean proportional of a and c , then $(a - b)^3 : (b - c)^3$ equals
 (1) $a^3 : c^3$ (2) $b^2 : c^2$
 (3) $a^2 : c^2$ (4) $a^3 : b^3$
8. If $a : b = 5 : 7$ and $c : d = 2a : 3b$, then $ac : bd$ is :
 (1) 20 : 38 (2) 50 : 147
 (3) 10 : 21 (4) 50 : 151
9. If $A : B : C = 2 : 3 : 4$, then ratio $\frac{A}{B} : \frac{B}{C} : \frac{C}{A}$ is equal to
 (1) 8 : 9 : 16 (2) 8 : 9 : 12
 (3) 8 : 9 : 24 (4) 4 : 9 : 16
10. If $a : b = c : d = e : f = 1 : 2$, then $(3a + 5c + 7e) : (3b + 5d + 7f)$ is equal to
 (1) 8 : 7 (2) 2 : 1
 (3) 1 : 4 (4) 1 : 2
11. The mean proportional between $(3 + \sqrt{2})$ and $(12 - \sqrt{32})$ is
 (1) 7 (2) $2\sqrt{7}$
 (3) 6 (4) $\frac{15 - 3\sqrt{2}}{2}$
12. If $a : b : c = 2 : 3 : 4$ and $2a - 3b + 4c = 33$, then the value of c is
 (1) 6 (2) 9
 (3) 12 (4) $66/7$
13. If two times of A is equal to three times of B and also equal to four times of C , then $A : B : C$ is
 (1) 2 : 3 : 4 (2) 3 : 4 : 2
 (3) 4 : 6 : 3 (4) 6 : 4 : 3
14. If A and B are in the ratio 3 : 4, and B and C in the ratio 12 : 13, then A and C will be in the ratio
 (1) 3 : 13 (2) 9 : 13
 (3) 36 : 13 (4) 13 : 9
15. If $3x = 5y = 4z$, then $x : y : z$ is equal to
 (1) 9 : 12 : 16 (2) 20 : 12 : 15
 (3) 15 : 10 : 9 (4) 8 : 5 : 3
16. If $A : B = 3 : 4$ and $B : C = 6 : 5$, then $A : (A + C)$ is equal to
 (1) 9 : 10 (2) 10 : 9
 (3) 9 : 19 (4) 19 : 9
17. If 78 is divided into three parts which are in the ratio $1 : \frac{1}{3} : \frac{1}{6}$, the middle part is
 (1) $9\frac{1}{3}$ (2) 13
 (3) $17\frac{1}{3}$ (4) $18\frac{1}{3}$
18. If two-third of A is four-fifth of B , then $A : B = ?$
 (1) 5 : 6 (2) 6 : 5
 (3) 10 : 9 (4) 9 : 10
19. If $A = \frac{4}{5}$ of B and $B = \frac{5}{2}$ of C , then the ratio of $A : C$ is
 (1) 1 : 2 (2) 2 : 1
 (3) 2 : 3 (4) 1 : 3
20. Marks of two candidates P and Q are in the ratio of 2 : 5. If the marks of P is 120, marks of Q will be

(1) 120
(3) 300

(2) 240
(4) 360

21. If $A : B = 4 : 9$ and $A : C = 2 : 3$, then $(A + B) : (A + C)$ is

(1) 15 : 13
(3) 13 : 10

(2) 10 : 13
(4) 13 : 15

22. If $(a + b) : (b + c) : (c + a) = 6 : 7 : 8$ and $(a + b + c) = 14$, then the value of c is

(1) 6
(3) 8

(2) 7
(4) 14

23. If 18, x and 50 are in continued proportion, then the value of x is

(1) 30
(3) 5

(2) 3
(4) 32

24. If $(a + b) : \sqrt{ab} = 4 : 1$, where $a > b > 0$, then $a : b$ is

(1) $(2 + \sqrt{3}) : (2 - \sqrt{3})$
(2) $(2 - \sqrt{3}) : (2 + \sqrt{3})$
(3) $(3 + \sqrt{2}) : (3 - \sqrt{2})$
(4) $(3 - \sqrt{2}) : (3 + \sqrt{2})$

25. Rs. 2420 were divided among A, B and C so that $A : B = 5 : 4$ and $B : C = 9 : 10$ then C gets

(1) Rs. 680
(3) Rs. 900

(2) Rs. 800
(4) Rs. 950

26. If $A = \frac{1}{4}B$ and $B = \frac{1}{2}C$, then $A : B : C$ is :

(1) 8 : 4 : 1
(3) 1 : 4 : 8

(2) 4 : 2 : 1
(4) 1 : 2 : 4

27. If $2A = 3B = 4C$, then $A : B : C$ is :

(1) 2 : 3 : 4
(3) 6 : 4 : 3

(2) 4 : 3 : 2
(4) 3 : 4 : 6

28. The ratio $4^{3.5} : 2^5$ is the same as

(1) 4 : 1
(3) 1 : 2

(2) 2 : 1
(4) 1 : 4

29. There are three numbers A, B, C such that twice A is equal to thrice B and four times B is equal to five times C. Then the ratio between A and C is

(1) 3 : 4
(3) 15 : 8

(2) 8 : 15
(4) 4 : 3

30. If $\frac{3x+5}{5x-2} = \frac{2}{3}$, then the value of x is

(1) 11
(3) 23

(2) 19
(4) 7

ANSWER KEY

1.	(1)	2.	(3)	3.	(1)	4.	(3)	5.	(1)	6.	(4)	7.	(4)	8.	(2)	9.	(3)	10.	(4)
11.	(2)	12.	(3)	13.	(4)	14.	(2)	15.	(2)	16.	(3)	17.	(3)	18.	(2)	19.	(2)	20.	(3)
21.	(3)	22.	(1)	23.	(1)	24.	(1)	25.	(2)	26.	(3)	27.	(3)	28.	(1)	29.	(3)	30.	(2)

PRACTICE SHEET 2

1. Ram, Shyam and Kamal together started a business in partnership. The ratio of their capitals is 3:4:7. If their annual profit be Rs. 21000, what will be Kamal's share in this profit?
 (1) Rs. 12500 (b) Rs. 10500
 (3) Rs. 15000 (4) Rs. 10000
2. Three partners A,B,C in a business invested money such that 6(A's capital)=8 (B's capital)=10 (C's capital) then the ratio of their capitals is:
 (1) 3:4:5 (2) 12:15:20
 (3) 20:15:12 (4) 6:8:10
3. If 4 (A's capital) = 6 (B's capital) = 10 (C's capital), then out of a profit of Rs. 4650, C will receive
 (1) 465 (2) 900
 (3) 1550 (4) 2250
4. Three partners A,B,C in a business invested money such that (A's capital)/3 = (B's capital)/1 = [C's capital]/5 then the ratio of their capitals is:
 (1) 3:1:5 (2) 12:15:20
 (3) 20:15:12 (4) 6:8:10
5. In a business A and b invested amounts in ratio 2:3, where as B and C invested amounts in the ration 3:4. If their annual profit be 45000 then B's share in profit is:
 (1) Rs. 15000 (2) Rs. 30000
 (3) Rs. 20000 (4) None
6. A,B and C invested Rs. 26000, Rs. 34000 and Rs. 10000 respectively in a business. At the end of the year, they earn a profit of Rs.3500. B's share of profit is:
 (1) Rs. 1200 (2) Rs. 1500
 (3) Rs. 1700 (4) Rs. 1900
7. Four milkmen rented a pasture A grazed 18 cows for 4 months B 25 cows for 2 months C 28 cows for 5 months and D 21 cows for 3 months. total rent of the field is 1625, A's share of rent is:
 (1) Rs. 250 (2) Rs. 300
 (3) Rs. 360 (4) Rs. 650
8. A,B,C rented a video cassette at á rent of 350. If they use if for 6 hrs, 10 hrs and 12 hrs. Rent to be paid C, is
 (1) Rs. 75 (2) Rs. 125
 (3) Rs. 135 (4) Rs. 150
9. Amit started a business investing Rs. 25000. After 3 months, Vinay joined him. With a capital of Rs. 30000. At the end of the year they make a profit of Rs. 19000. What will be Amit's share in the profit?
 (1) Rs. 9423 (2) Rs. 10000
 (3) Rs. 12500 (4) none
10. A,B and C start a business each investing Rs. 20000. After 5 months A withdrew Rs. 5000, B Rs. 4000 and C invested Rs. 6000 more. At the end of the year a total profit of Rs. 69900 was recorded. What is share of B?
 (1) Rs. 20500 (2) Rs. 21200
 (3) Rs. 27300 (4) Rs. 28200
11. A,B,C enter into partnership. A initially invests Rs. 25 lakhs and another Rs.10 Lakhs after one year. B initially invests Rs. 35 lakhs and withdraws Rs. 10 lakhs after 2 years and C invests Rs. 30 lakhs. In what ratio should the profits be divided at the end of 3 years?
 (1) 10:10:9 (2) 19:19:18
 (3) 20:20:19 (4) None
12. A,BC enter into partnership and their capitals are into the proportion 1/3:1/4:1/5, a withdraws half his capital after 4 months. Out of a total annual profit of Rs. 8470, A's share is:
 (1) Rs. 2520 (2) Rs. 2800
 (3) Rs. 3150 (4) Rs. 4120
13. In a partnership A invests 1/6 of the capital of 1/6 of the time B invests 1/3 of the capital for 1/3 of the time and C, the rest of the capital for the Whole time. Out of a profit of Rs. 23000, B's share is
 (1) Rs. 4000 (2) Rs. 5000
 (3) Rs. 3250 (4) Rs. 4800
14. A and B started a business with initial investments In ratio 12:11 and their annual profits were in the ratio 4:1. If A invested the

money for 11 months B invested the money for

- (1) 3 months (2) 4 months
(3) 5 months (4) 7 months

15. A began a business with Rs. 4500 and was joined afterwards by B with Rs. 5400. If the profits at the end of the year were divided in the ratio 2:1, B joined the business after:

- (1) 4 month (2) 5 month
(3) 6 month (4) 7 month

16. A and B partners in a business. A contributes $\frac{1}{4}$ of the capital for 15 months and B received $\frac{2}{3}$ of the profit. Find how long B's money was used.

- (1) 6 month (2) 9 month
(3) 10 month (4) 1 year

17. A and B enter into partnership. A Invests Rs. 16000 for 8 months and B remains in the business for 4 months. Out off a total profit, B claims $\frac{2}{7}$ of the profit. How much money was contributed by B?

- (1) 11900 (2) 10500
(3) 13600 (4) 12800

18. A is a working and B, a sleeping partner in a business. A puts in Rs. 12000 and B Rs. 20000. A receives 10% of the profits for managing, the rest being divided in proportion to their capitals. Out of a total profit of Rs. 18000 the money received by A is:

- (1) Rs. 6480 (2) Rs. 8400
(3) Rs. 7875 (4) Rs.8325

19. A is a working and B, a sleeping partner in a business. A puts in Rs. 20000 and B Rs. 30000. A receives 20% of the profits for managing, the rest being divided in proportion 'to their capitals. Out of total profit of Rs. 50000 the money received by A is:

- (1) Rs. 26000 (2) Rs. 24000
(3) Rs. 30000 (4) Rs.20000

20. A,B,C subscribe Rs. 50000 for a business. A subscribes Rs. 4000 more than B and B Rs. 5000 More than C. Out of a total profit of Rs. 35000, A receives:

- (1) 8400 (2) 11900
(3) 13600 (4) 14700

21. A,B,C subscribe Rs. 80000 for a business. A subscribes Rs. 10000 more than B and B

Rs. 20000 More than C. Out of a total profit of Rs. 40000, A receives:

- (1) 32000 (2) 31900
(3) 33600 (4) 20000

22. A,B and C enter into partnership. A invests some money at the begging B invests double the amount after 6 months and C invests thrice the amount after 8 months. If the annual profit be Rs. 27000, C's share is:

- (1) 8625 (2) 9000
(3) 10800 (4) 11250

23. A, B and C entered in to a partnership. A invested Rs. 2560 and B invested Rs. 2000. At the end of the year they gained Rs. 1105, out of which a got Rs. 320. C's capital was:

- (1) Rs. 2840 (2) Rs. 4028
(3) Rs. 4280 (4) Rs. 4820

24. A,B and Care three partners in a business. If A's capital is equal to twice B's capital and B 's capital is three times C's capital, ratio of the capitals of A,B,C is

- (1) 2:1:3 (2) 1:2:6
(3) 6:3:1 (4) 1:3:6

25. A,B and C start a business. A invests 3 times as much as B. invests and B invests two third of C invests. Then the ratio of capitals of A,B ,C is:

- (1) 3:9:2 (2) 6:10:15
(3) 5:3:2 (4) 6:2:3

26. A,B,C started a shop by investing Rs.27000, Rs. 81000 and Rs.72000 respectively. At end of the one year, B's share of total profit was Rs.36000. What total profit?

- (1) 108000 (2) 116000
(3) 80000 (4) 92000

27. A and B started a joint firm. A's investment was thrice the investment of B and the period of his investment was twice the period of investment of B. If B got Rs. 12000 as profit then the total profit was

- (1) 72000 (2) 48000
(3) 84000 (4) 60000

28. A and b entered into partnership with capitals in the ratio 4:5. After 3 months A withdraw $\frac{1}{4}$ of his capital and B. withdrew $\frac{1}{5}$ of this capital. The gain at the end of 10 months was Rs. 15200. A's share in this profit is:

(1) Rs. 7200
(3) Rs. 7800

(2) Rs. 6600
(4) Rs. 8200

29. A,B,C became partners in a business by investing money in ratio 5:7:6. Next year they increase their investments by 26%, 20% and 15%. profit earned during 2nd year should be distributed in ratio.

(1) 21:28:23
(3) 31:27:21

(2) 26:20:15
(4) None

30. A and B invest in business in tlhe ratio 3:2. If 5% of the total profit goes to charity and A's share is Rs. 8550, then total profit is:
(1) 15760
(2) 15735
(3) 15000
(4) 142500

ANSWER KEY

1.	(2)	2.	(3)	3.	(2)	4.	(1)	5.	(1)	6.	(3)	7.	(3)	8.	(4)	9.	(2)	10.	(2)
11.	(2)	12.	(2)	13.	(1)	14.	(1)	15.	(4)	16.	(3)	17.	(4)	18.	(3)	19.	(1)	20.	(4)
21.	(4)	22.	(2)	23.	(3)	24.	(3)	25.	(4)	26.	(3)	27.	(3)	28.	(2)	29.	(1)	30.	(3)

PRACTICE SHEET 3

- The ratio of the age of Ram and Rahim 10 years ago was 1 : 3. The ratio of their age five years hence will be 2 : 3. Then the ratio of their present age is
(1) 1 : 2
(2) 3 : 5
(3) 3 : 4
(4) 2 : 5
- The ratio of the age of a father to that of his son is 5: 2. If the product of their ages in years is 1000, then the father's age (in years) after 10 years will be :
(1) 50
(2) 60
(3) 80
(4) 100
- The ratio between Sumit's and Prakash's age at present is 2 : 3. Sumit is 6 years younger than Prakash. The ratio of Sumit's age to Prakash's age after 6 years will be
(1) 2 : 3
(2) 1 : 2
(3) 4 : 3
(4) 3 : 4
- Harsha is 40 years old and Ritu is 60 years old. How many years ago was the ratio of their ages 3 : 5 ?
(1) 10 years
(2) 20 years
(3) 37 years
(4) 5 years
- The ratio of present age of two brothers is 1 : 2 and 5 years back the ratio was 1 : 3. What will be the ratio of their age after 5 years ?
(1) 1 : 4
(2) 2 : 3
(3) 3 : 5
(4) 5 : 6
- Four years ago, the ratio of the age of A and B was 2 : 3 and after four years it will become 5 : 7. Find their present age.
(1) 36 years and 40 years
(2) 32 years and 48 years
(3) 40 years and 56 years
(4) 36 years and 52 years
- The average age of boys in the class is twice the number of girls in the class. The ratio of boys and girls in the class of 50 is 4 : 1. The total of the ages (in years) of the boys in the class is
(1) 2000
(2) 2500
(3) 800
(4) 400
- The ratio of age of two boys is 5 : 6. After two years the ratio will be 7 : 8. The ratio of their age after 12 years will be
(1) 22/24
(2) 15/16
(3) 17/18
(4) 11/12
- The ratio of the present age of Puneet and Appu is 2 : 3. After 3 years the ratio of their age will be 3 : 4. The present age of Puneet is:
(1) 3 years
(2) 6 years
(3) 9 years
(4) 4 years
- The ratio of the ages of a father and his son 10 years hence will be 5 : 3, while 10 years ago, it was 3:1. The ratio of the age of the son to that of the father today, is

- (1) 1 : 2 (2) 1 : 3
(3) 2 : 3 (4) 2 : 5
- 11.** Father is three times more than his son Ronit. After 8 years, he would be two and half times of Ronit's age, after further 16 years, how many times would he be of Ronit's age
(1) 2 times (b) 2 and half times
(3) 2.75 times (4) 3 times
- 12.** The sum of ages of 5 children born at the intervals of 3 years each is 50 years what is the age of the youngest child
(1) 4 years (2) 8 years
(3) 10 years (4) none of these
- 13.** A father said to his son "I was as old as you at the present at the time of your birth" if the father's age is 38 years now the son's age 5 years back was
(a) 14 years (b) 19 years
(c) 33 years (d) 38 years
- 14.** A is 2 years older than B who is twice as old as C. If the total of the ages of A, B and C to be 27, then how old is B
(a) 7 (b) 8
(c) 9 (d) 10
- 15.** Present age of Sameer and Anand are in the ratio 5 : 4 respectively. 3 years hence, the ratio of their ages will become 11:9 respectively. what is Anand's present age in years
(1) 24 (2) 27
(3) 40 (4) 50
- 16.** A man is 24 years older than his son. In two years, his age will be twice the age of his son. The present age of his son is
(1) 14 years (2) 18 years
(3) 20 years (4) 22 years
- 17.** Six years ago, the ratio of the ages of Kunal and Sagar was 6 : 5 . four years hence the ratio of their ages will be 11 :10 . what is present age of Sagar
(1) 16 years (2) 18 years
(3) 20 years (4) 22 years
- 18.** The sum of present age of father and his son is 60 years. Six years ago, father's age was five times the age of his son, after 6 years son's age will be
(1) 12 years (2) 14 years
(3) 18 years (4) 20 years
- 19.** At present the ratio between the ages of Arun and Deepak is 4:3, After 6 years, Arun's age will be 26 years. what is the age of Deepak at present
(1) 12 years (2) 15 years
(3) 19 and half (4) 21 years
- 20.** Sachin is younger than Rahul by 7 years. if their ages are in the respective ratio of 7:9, then how old is Sachin
(1) 16 years (2) 18 years
(3) 28 years (4) 24.5 years
- 21.** The present age of three persons in proportions 4 : 7 : 9 , Eight years ago the sum of their ages was 56. find their present age
(1) 8, 20, 28 (2) 16, 28, 36
(3) 20, 35, 45 (4) none of these
- 22.** Ayesha's father was 38 years of age when she was born while her mother was 36 years old when her brother was 4 years younger to her was born what is the difference between the ages of his parents
(1) 2 years (2) 4 years
(3) 6 years (4) 8 years
- 23.** Rs 180 contained in a box consists of one rupee, 50 paise and 25 paise coins in the ratio 2 : 3 : 4. What is the number of 50 paise coins?
(1) 60 (2) 120
(3) 150 (4) 180
- 24.** If 378 coins consist of rupees, 50 paise and 25 paise coins, whose values are in the ratio of 13 : 11 : 7, the number of 50 paise coins will be :
(1) 132 (2) 128
(3) 136 (4) 133
- 25.** A bag contains Rs 90 in coins of denominations of 50 paise, 25 paise and 10 paise. If coins of 50 paise, 25 paise and 10 paise are in the ratio 2 : 3 : 5, then the number of 25 paise coins in the bag is
(1) 80 (2) 120
(3) 100 (4) 135
- 26.** There are Rs 225 consisting of one rupee, 50 paise and 25 paise coins. The ratio of their numbers in that order is 8 : 5 : 3. The number of one-rupee coins is :
(1) 80 (2) 112
(3) 160 (4) 172

- 27.** A box contains 1-rupee, 50-paise and 25-paise coins in the ratio 8 : 5 : 3. If the total amount of money in the box is 112.50, the number of 50-paise coins is
 (1) 80 (2) 50
 (3) 30 (4) 42
- 28.** In a bag, there are three types of coins — 1-rupee, 50 paise and 25-paise in the ratio of 3 : 8 : 20. Their total value is Rs 372. The total number of coins is
 (1) 1200 (2) 961
 (3) 744 (4) 612
- 29.** A box has 210 coins of denominations one-rupee and fifty paise only. The ratio of their respective values is 13 : 11. The number of one-rupee coins is
 (1) 65 (2) 66
 (3) 77 (4) 78
- 30.** A boy has a few coins of denominations 50 paise, 25 paise and 10 paise in the ratio 1 : 2 : 3. If the total amount of the coins is 6.50, the number of 10 paise coins is
 (1) 5 (2) 10
 (3) 15 (4) 20

ANSWER KEY

1.	(2)	2.	(2)	3.	(4)	4.	(1)	5.	(3)	6.	(4)	7.	(3)	8.	(3)	9.	(2)	10.	(1)
11.	(1)	12.	(1)	13.	(1)	14.	(4)	15.	(1)	16.	(4)	17.	(1)	18.	(4)	19.	(2)	20.	(4)
21.	(2)	22.	(3)	23.	(2)	24.	(1)	25.	(2)	26.	(3)	27.	(2)	28.	(2)	29.	(4)	30.	(3)

PRACTICE SHEET 4

- 1.** The income of A, B and C are in the ratio 3 : 7 : 4 and their expenses in the ratio 4 : 3 : 5. If A saves 300 out of an income of 2,400, the savings of B and C are :
 (1) 4025 and 575 (2) 1575 and 2.625
 (3) 2750 and 1.525 (4) 3725 and 1.525
- 2.** Between two consecutive years my income are in the ratio of 2 : 3 and expenses in the ratio 5 : 9. If my income in the second year is 45000 and my expenses in the first year is 25000 my total savings for the two years is :
 (1) Nil (2) 15000
 (3) 10000 (4) 5000
- 3.** A and B have monthly incomes in the ratio 5 : 6 and monthly expenditures in the ratio 3 : 4. If they save 1800 and 1600 respectively, find the monthly income of B :
 (1) 3400 (2) 2700
 (3) 1720 (4) 7200
- 4.** The ratio of income of two persons is 5 : 3 and that of their expenditures is 9 : 5. Find the income of each person, if they save 1,300 and 900 respectively.
 (1) 4,000, 2,400 (2) 3,000, 1,800
 (3) 5,000, 3,000 (4) 4,500, 2,700
- 5.** The annual income of A and B are in the ratio 4 : 3 and the ratio of their expenditures is 3 : 2. If each of them saves 600 in the year, the annual income of A is
 (1) 4800 (2) 1800
 (3) 1200 (4) 2400
- 6.** The income of A, B and C are in the ratio 7 : 9 : 12 and their spending are in the ratio 8 : 9 : 15. If A saves $\frac{1}{4}$ th of his income, then the savings of A, B and C are in the ratio of :
 (1) 56 : 99 : 69 (2) 69 : 56 : 99
 (3) 99 : 56 : 69 (4) 99 : 69 : 56
- 7.** The ratio of income of P and Q is 3 : 4 and the ratio of their expenditures is 2 : 3. If both of them save 6000, the income of P is
 (1) 20000 (2) 12000
 (3) 18000 (4) 24000
- 8.** A man spends a part of his monthly income and saves a part of it. The ratio of his expenditure to his saving is 26 : 3. If his

monthly income is 7250, what is the amount of his monthly savings ?

- (1) 350 (2) 290
(3) 750 (4) 780

9. The monthly salaries of A, B and C are in the ratio 2 : 3 : 5. If C's monthly salary is 12,000 more than that of A, then B's annual salary is
(1) 1,20,000 (2) 1,44,000
(3) 1,80,000 (4) 2,40,000

10. The ratio of income of two persons is 5 : 3 and that of their expenditures is 9 : 5. If they save 2600 and 1800 respectively, their incomes are :
(1) 8000; 4800 (2) 6000; 3600
(3) 10000; 6000 (4) 9000; 5400

11. The monthly income of two persons are in the ratio 2 : 3 and their monthly expenses are in the ratio 5 : 9. If each of them saves 600 per month, then their monthly incomes are
(1) 1,500; 2,250 (2) 1,200; 1,800
(3) 1,600; 2,400 (4) 1,400; 2,100

12. A person bought some rice and wheat for 380. The ratio of weight of rice and wheat is 4 : 3 and the price of equal amount of rice and wheat is in the ratio 5 : 6. The rice was bought of worth
(1) 380 (2) 300
(3) 200 (4) 180

13. Divide 7,500 among A, B and C such that A's share to B's share is in ratio 5 : 2 and B's share to C's share is in the ratio 7 : 13. How much will B receive?
(1) 1,400 (2) 3,500
(3) 2,600 (4) 7,000

14. A sum of 1240 is distributed among A, B and C such that the ratio of amount received by A and B is 6 : 5 and that of B and C is 10 : 9 respectively. Find the share of C.
(1) 480 (2) 360
(3) 400 (4) 630

15. 3400 is divided among A, B, C, D in such a way that the share of A and B, B and C, C and D may be as 2 : 3, 4 : 3 and 2 : 3 respectively. The sum of shares of B and D is
(1) 2040 (2) 1680
(3) 2000 (4) 1720

16. 750 are divided among A, B and C in such a manner that A : B is 5 : 2 and B : C is 7 : 13. What is A's share ?

- (1) 350 (2) 260
(3) 140 (4) 250

17. In an army selection process, the ratio of selected to unselected candidates was 3 : 1. If 80 less had applied and 40 less selected, the ratio of selected to unselected candidates would have been 4 : 1. How many candidates had applied for the process?
(1) 480 (2) 960
(3) 240 (4) 1440

18. In an army selection process, the ratio of selected to unselected candidates was 4:1. If 90 less had applied and 20 less were selected, the ratio of selected to unselected candidates would have been 5:1. How many candidates had applied for the process ?
(1) 1650 (2) 3300
(3) 825 (4) 4950

19. In a cricket match there are three types of tickets say A, B and C each costing Rs. 1000, Rs. 500 and Rs. 200 respectively. The ratio of the tickets sold of categories A, B and C is 3 : 2 : 5. If the total collection from selling the tickets is Rs 2.5 crore, find the total number of tickets sold?
(1) 5000 (2) 4800
(3) 50000 (4) 52000

20. From each of the two given unequal numbers, half the smaller number is subtracted. Then, of the resulting numbers, the larger one is five times than the smaller one. Then the ratio of the larger to smaller one is
(1) 2 : 1 (2) 3 : 2
(3) 3 : 1 (4) 1 : 4

21. A person ordered 4 shirts of brand A and some shirts of brand B. The price of one shirt of brand A was twice that of brand B. When the order was executed, it was found that the numbers of the two brands has been interchanged. This increased the bill by 40%. The ratio of the number of brand A shirts to that of brand B shirts in the original order was
(1) 1 : 2 (2) 1 : 3
(3) 1 : 4 (4) 1 : 5

22. A and B together have 158. C has 101 less than what A and B together have, and B has 23 more than C. The amount of A is :
(1) 80 (2) 78
(3) 57 (4) 88

- 23.** A sum of 340.68 is distributed among L, M and N such that L gets 5.72 more than N and M gets Rs. 2.24 more than L. N gets
 (1) 109 (2) 110.90
 (3) 113.56 (4) 114.72
- 24.** The ratio of the first and second class fares between two railway stations is 4 : 1 and that of the number of passengers travelling by first and second classes is 1 : 40. If on a day 1,100 are collected as total fare, the amount collected from the first class passengers is
 (1) 315 (2) 275
 (3) 137.50 (4) 100
- 25.** Three persons walk from place A to place B. Their speeds are in the ratio 4 : 3 : 5. The ratio of the time taken by them to reach B will be :
 (1) 10 : 15 : 13 (2) 2 : 3 : 4
 (3) 15 : 20 : 12 (4) 16 : 18 : 15
- 26.** Two numbers are in the ratio of 3 : 5. If 9 be subtracted from each, then they are in the ratio of 12 : 23. Find the numbers.
 (1) 15, 28 (2) 36, 115
 (3) 33, 55 (4) 60, 69
- 27.** Three numbers are in the ratio 1 : 2 : 3. By adding 5 to each of them, the new numbers are in the ratio 2 : 3 : 4. The numbers are:
 (1) 10, 20, 30 (2) 15, 30, 45
 (3) 1, 2, 3 (4) 5, 10, 15
- 28.** Ram got twice as many marks in English as in Science. His total marks in English, Science and Maths are 180. If the ratio of his marks in English and Maths is 2 : 3, what is his marks in Science ?
 (1) 30 (2) 60
 (3) 72 (4) 90
- 29.** The ratio of number of boys to the number of girls in a school of 432 pupils is 5 : 4. When some new boys and girls are admitted, the number of boys increase by 12 and the ratio of the boys to girls changes to 7 : 6. The number of new girls admitted is
 (1) 12 (2) 14
 (3) 24 (4) 20
- 30.** If the ratio of two numbers is 1 : 5 and their product is 320, then the difference between the squares of these two numbers is :
 (1) 1024 (2) 1256
 (3) 1536 (4) 1435

ANSWER KEY

1.	(1)	2.	(4)	3.	(4)	4.	(1)	5.	(4)	6.	(1)	7.	(3)	8.	(3)	9.	(2)	10.	(1)
11.	(3)	12.	(3)	13.	(1)	14.	(2)	15.	(1)	16.	(1)	17.	(1)	18.	(1)	19.	(3)	20.	(3)
21.	(2)	22.	(2)	23.	(1)	24.	(4)	25.	(3)	26.	(3)	27.	(4)	28.	(1)	29.	(3)	30.	(3)

PRACTICE SHEET 5

- 1.** A mixture of 30 litres contain milk and water in the ratio of 7 : 3. How much water should be added to it so that the ratio of milk and water becomes 3 : 7 ?
 (1) 40 litres (2) 49 litres
 (3) 56 litres (4) 63 litres
- 2.** A barrel contains a mixture of wine and water in the ratio 3 : 1. How much fraction of the mixture must be drawn off and substituted by water so that the ratio of wine and water in the resultant mixture in the barrel becomes 1 : 1 ?
 (1) $\frac{1}{4}$ (2) $\frac{1}{30}$
 (3) $\frac{3}{4}$ (4) $\frac{2}{3}$
- 3.** There is 81 litres pure milk in a container. One-third of milk is replaced by water in the container. Again one-third of mixture is extracted and equal amount of water is added. What is the ratio of milk to water in the new mixture?
 (1) 1 : 2 (2) 1 : 1
 (3) 2 : 1 (4) 4 : 5
- 4.** In 80 litres mixture of milk and water the ratio of amount of milk to that of amount of water is 7 : 3. In order to make this ratio 2 : 1, how many litres of water should be added?
 (1) 5 (2) 6
 (3) 8 (4) 4

5. Vessels A and B contain mixtures of milk and water in the ratios 4 : 5 and 5 : 1 respectively. In what ratio should quantities of mixture be taken from A and B to form a mixture in which milk to water is in the ratio 5 : 4?
- (1) 2 : 5 (2) 4 : 3
(3) 5 : 2 (4) 2 : 3
6. The milk and water in a mixture are in the ratio 7 : 5. When 15 litres of water are added to it, the ratio of milk and water in the new mixture becomes 7 : 8. The total quantity of water in the new mixture is
- (1) 35 litres (2) 40 litres
(3) 60 litres (4) 96 litres
7. In a mixture of milk and water of volume 30 L, the ratio of milk and water is 7: 3. The quantity of water to be added to the mixture to make the ratio of milk and water 1: 2 is
- (a) 30 (b) 32
(c) 33 (d) 35
8. Two equal vessels are filled with the mixtures of water and milk in the ratio of 3:4 and 5:3 respectively. If the mixtures are poured into a third vessel, the ratio of water and milk in the third vessel will be
- (1) 15 : 12 (2) 53 : 59
(3) 20 : 9 (4) 59 : 53
9. A Can contains a mixture of two liquids A and B in the ratio 7:5. When 9 litres of mixture are drained off and the Can is filled with B, the ratio of A and B becomes 7 : 9. How many litres of liquid A was contained by the Can initially ?
- (1) 10 litres (2) 20 litres
(3) 21 litres (4) 25 litres
10. The ratio of milk and water in mixtures of four containers are 5 : 3, 2 : 1, 3 : 2 and 7 : 4 respectively. In which container is the quantity of milk, relative to water, minimum?
- (1) First (2) Second
(3) Third (4) Fourth
11. An alloy contains copper, zinc and nickel in the ratio of 5 : 3 : 2. The quantity of nickel (in kg) that must be added to 100 kg of this alloy to have the new ratio 5 : 3 : 3 is
- (1) 8 (2) 10
(3) 12 (4) 15
12. In an alloy, zinc and copper are in the ratio 1 : 2. In the second alloy, the same elements are in the ratio 2 : 3. If these two alloys be mixed to form a new alloy in which two elements are in the ratio 5 : 8, the ratio of these two alloys in the new alloy is
- (1) 3 : 10 (2) 3 : 7
(3) 10 : 3 (4) 7 : 3
13. A liquid 'P' is $1\frac{3}{7}$ times as heavy as water and water is $1\frac{2}{5}$ times as heavy as another liquid 'Q'. The amount of liquid 'P' that must be added to 7 litres of the liquid 'Q' so that the mixture may weigh as much as an equal volume of water, will be
- (1) 7 litres (2) $5\frac{1}{6}$ litres
(3) 5 litres (4) $4\frac{2}{3}$ litres
14. Two alloys contain tin and iron in the ratio of 1 : 2 and 2 : 3. If the two alloys are mixed in the proportion of 3 : 4 respectively (by weight), the ratio of tin and iron in the newly formed alloy is :
- (1) 10 : 21 (2) 13 : 22
(3) 14 : 25 (4) 12 : 23
15. 60 kg of an alloy A is mixed with 100 kg of alloy B. If alloy A has lead and tin in the ratio 3 : 2 and alloy B has tin and copper in the ratio 1 : 4, the amount of tin in the new alloy is
- (1) 53 kg (2) 44 kg
(3) 80 kg (4) 24 kg
16. Three glasses of equal volume contains acid mixed with water. The ratios of acid and water are 2 : 3, 3 : 4 and 4 : 5 respectively. Contents of these glasses are poured in a large vessel. The ratio of acid and water in the large vessel is
- (1) 411 : 540 (2) 401 : 544
(3) 417 : 564 (4) 407 : 560
17. Zinc and copper are in the ratio of 5 : 3 in 200 gm of an alloy. How much grams of copper be added to make the ratio as 3 : 5?
- (1) 400/3 (2) 1/200
(3) 72 (4) 66

- 18.** The ratio of copper and zinc in brass is 13 : 7. How much zinc will be there in 100 kg of brass ?
 (1) 20 kg. (2) 55 kg.
 (3) 35 kg. (4) 40 kg.
- 19.** In 30 litres mixture of acid, the ratio of acid and water is 2 : 3. What amount of water should be added to the mixture so that the ratio of acid and water becomes 2 : 5 ?
 (1) 10 litres (2) 15 litres
 (3) 18 litres (4) 12 litres
- 20.** There are three containers of equal capacity. The ratio of Sulphuric acid to water in the first container is 3 : 2, that in the second container is 7 : 3 and in the third container it is 11 : 4. If all the liquids are mixed together, then the ratio of Sulphuric acid to water in the mixture will be :
 (1) 61 : 29 (2) 61 : 28
 (3) 60 : 29 (4) 59 : 29
- 21.** 200 litres of a mixture contains milk and water in the ratio 17 : 3. After the addition of some more milk to it, the ratio of milk to water in the resulting mixture becomes 7 : 1. The quantity of milk added to it was
 (1) 20 litres (2) 40 litres
 (3) 60 litres (4) 80 litres
- 22.** The ratio in which a man must mix rice at 10.20 per kg and 14.40 per kg so as to make a mixture worth 12.60 per kg, is
 (1) 4 : 3 (2) 2 : 5
 (2) 18 : 24 (4) 3 : 4
- 23.** A mixture contains spirit and water in the ratio 3 : 2. If it contains 3 litres more spirit than water, the quantity of spirit in the mixture is
 (1) 10 litres (2) 12 litres
 (3) 8 litres (4) 9 litres
- 24.** A vessel is filled with liquid, 3 parts of which are water and 5 parts syrup. How much of the mixture must be drawn off and replaced with water so that the mixture may be half water and half syrup ?
 (1) 1/3 (2) 1/4 (3) 1/5 (4) 1/7
- 25.** Two vessels A and B contain milk and water mixed in the ratio 4 : 3 and 2 : 3. The ratio in which these mixtures be mixed to form a new mixture containing half milk and half water is
 (1) 7 : 5 (2) 6 : 5
 (3) 5 : 6 (4) 4 : 3
- 26.** A mixture contains alcohol and water in the ratio 4 : 3. If 5 litres of water is added to the mixture, the ratio becomes 4 : 5. The quantity of alcohol in the given mixture is
 (1) 3 litres (2) 4 litres
 (3) 15 litres (4) 10 litres
- 27.** In two alloys A and B, the ratio of zinc to tin is 5 : 2 and 3 : 4 respectively. Seven kg of the alloy A and 21 kg of the alloy B are mixed together to form a new alloy. What will be the ratio of zinc and tin in the new alloy ?
 (1) 2 : 1 (2) 1 : 2
 (3) 2 : 3 (4) 1 : 1
- 28.** A jar contained a mixture of two liquids A and B in the ratio 4 : 1. When 10 litres of the mixture was taken out and 10 litres of liquid B was poured into the jar, this ratio became 2 : 3. The quantity of liquid A contained in the jar initially was
 (1) 4 litres (2) 8 litres
 (3) 16 litres (4) 40 litres
- 29.** A vessel contains 20 litres of acid. 4 litres of acid is taken out of the vessel and replaced by the same quantity of water. Next 4 litres of the mixture are withdrawn, and again the vessel is filled with the same quantity of acid left in the vessel with the quantity of acid initially in the vessel is
 (1) 4 : 5 (2) 4 : 25
 (3) 16 : 25 (4) 1 : 5
- 30.** 729 ml of a mixture contains milk and water in the ratio 7 : 2. How much more water is to be added to get a new mixture containing milk and water in the ratio 7 : 3 ?
 (1) 81 ml (2) 60 ml
 (3) 71 ml (4) 52 ml

ANSWER KEY

1.	(1)	2.	(2)	3.	(4)	4.	(4)	5.	(3)	6.	(2)	7.	(4)	8.	(4)	9.	(3)	10.	(3)
11.	(2)	12.	(1)	13.	(4)	14.	(2)	15.	(2)	16.	(2)	17.	(1)	18.	(3)	19.	(4)	20.	(1)
21.	(2)	22.	(4)	23.	(4)	24.	(3)	25.	(1)	26.	(4)	27.	(4)	28.	(3)	29.	(3)	30.	(1)

CDS PYQ

1. In a certain school, the ratio of boys to girls is 7:5. If there are 2400 students in the school, then how many girls are there?
[CDS 2013 (I)]
(a) 500 (b) 700
(c) 800 (d) 1000
2. If a, b, c, d and e are in continued proportion then $a/e =$
[CDS 2013 (I)]
(a) a^3/b^3 (b) a^4/b^4
(c) b^3/a^3 (d) b^4/a^4
3. The present age of Ravi's father is 4 times of Ravi's present age. 5 years back Ravi father was 7 times as old as Ravi was at that time. What is the present age of Ravi's father?
[CDS 2014 (I)]
(a) 84 years (b) 70 years
(c) 40 years (d) 35 years
4. If $A:B = 2:3$, $B:C = 5:7$ and $C:D = 3:10$, then what is A: D equal to?
[CDS 2014 (I)]
(a) 1:7 (b) 2:7
(c) 1:5 (d) 5:1
5. $(x+y) : (x-y) = 3:5$ and $xy =$ positive imply that
[CDS 2014 (II)]
(a) x and y are both positive
(b) x and y are both negative
(c) one of them is positive and one of them is negative
(d) no real solution for x and y exists
6. The ratio of ages of A and B is 2:5 and the ratio of ages of B and C is 3:4. What is the ratio of ages of A, B and C?
[CDS 2014 (II)]
(a) 6 : 15 : 20 (b) 8 : 5 : 3
(c) 6 : 5 : 4 (d) 2 : 15 : 4
7. The height of a tree varies as the square root of its age (between 5 to 17 yr). When the age of the tree is 9 yr, its height is 4 ft. What will be the height of the tree at the age of 16 yr?
[CDS 2014 (II)]
(a) 5 ft 4 inch (b) 5 ft 5 inch
(c) 4 ft 4 inch (d) 4 ft 5 inch
8. 16 L of a mixture contains milk and water in the ratio 5:3. If 4 L of milk is added to this mixture, the ratio of milk to water in the new mixture would be
[CDS 2015 (I)]
(a) 2: 1 (b) 7:3
(c) 4:3 (d) 8:3
9. If $a:b=3:5$ and $b:c=7:8$, then $2a: 3b: 7c$ is equal to
[CDS 2015 (II)]
(a) 42: 105: 320 (b) 15: 21: 40
(c) 6:15: 40 (d) 30 21 350
10. In a mixture of milk and water of volume 30 L, the ratio of milk and water is 7: 3. The quantity of water to be added to the mixture to make the ratio of milk and water 1: 2 is
[CDS 2015 (II)]
(a) 30 (b) 32
(c) 33 (d) 35
11. The annual incomes of two persons are in the ratio 9:7 and their expenses are in the ratio 4:3. If each of them saves Rs 2000 per year, what is the difference in their annual incomes?
[CDS 2016 (I)]
(a) Rs 4000 (b) Rs 4500
(c) Rs 5000 (d) Rs 5500
12. If $\frac{a}{b} = \frac{b}{c} = \frac{c}{d}$, then which of the following is/are correct ?
I. $\frac{b^3+c^3+d^3}{a^3+b^3+c^3} = \frac{d}{a}$
II. $\frac{a^2+b^2+c^2}{b^2+c^2+d^2} = \frac{a}{d}$
Select the correct answer using the code given below
[CDS 2016 (I)]
(a) Only 1 (b) Only 2
(c) Both I and II (d) Neither I nor II
13. In an office, one third of the workers are women, half of the women are married and one-third of the married women have children, If three-fourth of the men are married and one third of married men have children, then what is the ratio of married women to married men ?
[CDS 2016 (II)]
(a) 1:2 (b) 2:1
(c) 3:1 (d) 1:3

14. There are twelve friends A, B, C, D, E, F, G, H, I, J, K and L who invested money in some business in the ratio of 1 : 2 : 3 : 4 : 5 : 6 : 7 : 8 : 9 : 10 : 11 : 12 and the duration for which they invested the money is in the ratio of 12 : 11 : 10 : 9 : 8 : 7 : 6 : 5 : 4 : 3 : 2 : 1 respectively. Who will get the maximum profit at the end of the year ?

[CDS 2016 (II)]

- (a) F only (b) G only
(c) Both F and G (d) Neither F nor G

15. Income of Mahesh and Kamal are in the ratio 1:2, and their expenses are in the ratio 1:3, Which one of the following statements are correct ?

[CDS 2016 (II)]

- (a) Mahesh saves more than what Kamal saves
(b) Saving of both of them are equal
(c) Kamal save more than what Mahesh saves
(d) It is not possible to determine who saves more

16. X and Y entered into partnership with Rs. 700 and Rs 600 respectively. After 3 months X withdraw $\frac{2}{7}$ of his stock but after 3 months, he puts back $\frac{3}{5}$ of what he had drawn. The profit at the end of the year is Rs. 726. How much of this should X receive ?

[CDS 2016 (II)]

- (a) Rs 336 (b) Rs 366
(c) Rs 633 (d) Rs 663

17. Leela got married 6 years ago. Today her age is $1\frac{1}{4}$ times her age at the time of her

marriage. Her son's age is $\frac{1}{10}$ times her age.

What is the present age of her son?

[CDS 2017 (I)]

- (a) 1 years (b) 2 years
(c) 3 years (d) 4 years

18. The ratio of two numbers is 1 : 5 and their product is 320. What is the difference between the squares of these two numbers?

[CDS 2017 (I)]

- (a) 1024 (b) 1256
(c) 1536 (d) 1640

19. 25 kg of alloy X is mixed with 125 of alloy Y. If the amount of lead and tin in the alloy X is in the ratio 1 : 2 and the amount of lead and

tin in the alloy Y is in the ratio 2 : 3, then what is the ratio of lead to tin in the mixture?

[CDS 2017 (I)]

- (a) 1 : 2 (b) 2 : 3
(c) 3 : 5 (d) 7 : 11

20. The cost of a diamond varies directly as the square of its weight. A diamond broke into four pieces with their weights in the ratio of 1:2:3:4. If the loss in total value of the diamond was Rs.70,000, what was the price of the original diamond?

[CDS 2017 (I)]

- (a) Rs 1,00,000 (b) Rs 1,40,000
(c) Rs 1,50,000 (d) Rs 1,75,000

21. In a class of 49 students, the ratio of girls to boys is 4:3. If 4 girls leave the class, the ratio of girls to boys would be:

[CDS 2017 (I)]

- (a) 11:7 (b) 8:7
(c) 6:5 (d) 9:8

22. Five year ago Ram was three times as old as Shyam. Four years from now, Ram will be only twice as old as Shyam what is the percentage of Ram?

[CDS 2017 (II)]

- (a) 30 years (b) 32 years
(c) 36 years (d) 40 years

23. If 78 is divided into 3 parts which are proportional to $1, \frac{1}{3}, \frac{1}{6}$, then what is the middle part

[CDS 2017 (II)]

- (a) $28/3$ (b) 13
(c) $52/3$ (d) $56/3$

24. There are 350 boys in the first three standards. The ratio of the number of boys in first and second standards is 2:3, while that of boys in second and third standards is 4:5. What is the total number of boys in first and third standards?

[CDS 2017 (II)]

- (a) 302 (b) 280
(c) 242 (d) 230

25. If A:B = 1:2, B:C = 3:4, C:D = 2:3 and D:E = 3:4, then what is B:E equal to?

[CDS 2018 (I)]

- (a) 3:2 (b) 1:8
(c) 3:8 (d) 4:1

26. Rs.120 are divided among A,B,C such that A's share is Rs. 20 more than B's and Rs.20 less than C's. What is B's share?

[CDS 2018 (I)]

- (a) ₹10 (b) ₹15
(c) ₹20 (d) ₹25

27. A, B, C, D and E start a partnership firm. Capital contributed by A is three times that contributed by D. E contributes half of A's contribution, B contributes one third of E's contribution and C contributes two third of A's contribution. If the difference between the combined shares of A, D and E and the combined shares of B and C in the total profit of the firm is Rs. 13500, what is the combined share of B, C and E? (The shares are supposed to be proportional to the contributions)

[CDS 2018 (I)]

- (a) Rs. 135000 (b) Rs. 18000
(c) Rs. 19750 (d) Rs. 20250

28. An alloy A contains two elements, copper and tin in the ratio of 2 : 3, whereas an alloy B contains the same elements in the ratio of 3:4. If 20 kg of alloy A, 28 kg of alloy B and some more pure copper are mixed to form a third alloy C which now contains copper and tin in the ratio of 6:7, then what is the quantity of pure copper mixed in the alloy C?

[CDS 2018 (I)]

- (a) 3kg (b) 4kg
(c) 5kg (d) 7kg

29. The sum of ages of a father a mother, a son Sonu and daughter Savita and Sonia is 96 years. Son is the youngest member of the family. The year Sonu was born, the sum of the ages of all the members of the family was 66 years. If the father's age now is six times that of Sonu's present age ,then 12 years hence the father's age will be

[CDS 2018 (II)]

- (a) 44 years (b) 45 years
(c) 46 years (d) 48 years

30. 10 years before the age of a mother and her daughter were in the ratio 3:1, in another 10 years from now the ratio of their ages will be 13:7. what are their present age

[CDS 2018 (II)]

- (a) 39 yr, 21 yr (b) 55 yr, 25 yr
(c) 75 yr, 25 yr (d) 49 yr, 31 yr

31. The ratio of the sum and difference of the ages of the father and the son is 11:3, Consider the following statements

1. The ratio of their age is 8 : 5
 2. The ratio of their ages after the son attains twice the present age will be 11:8
- Which of the statements given above are correct?

[CDS 2018 (II)]

- (a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2

32. If $a:b = c:d = 1:6$, then what is the value of $\frac{a^2 + c^2}{b^2 + d^2}$?

[CDS 2019 (I)]

- (a) 1/600 (b) 1/60
(c) 1/36 (d) 1/6

33. If $(4a+7b)(4c-7d)=(4a-7b)(4c+7d)$, then which one of the following is correct ?

[CDS 2019 (I)]

- (a) $\frac{a}{b} = \frac{c}{d}$ (b) $\frac{a}{d} = \frac{c}{b}$
(c) $\frac{a}{b} = \frac{d}{c}$ (d) $\frac{4a}{7b} = \frac{c}{d}$

34. Priya's age was cube of an integral number (different from 1) four years ago and square of an integral number after four years. How long should she wait so that her age becomes square of a number in the previous year and cube of a number in the next year?

[CDS 2019 (I)]

- (a) 7yr (b) 12 yr
(c) 14 yr (d) 21 yr

35. The monthly incomes of A and B are in the ratio 4:3. Each saves 600. If their expenditures are in the ratio 3 :2, then what is the monthly income of A

[CDS 2019 (II)]

- (a) 1800 (b) 2000
(c) 2400 (d) 3600

36. Radha and Rani are sisters, five years back the age of Radha was 3 times that of Rani, but one year back the age of Radha was 2 times that of Rani. What is age difference between them:

[CDS 2019 (II)]

- (a) 8 (b) 9
(c) 10 (d) 11

37. The population of two villages is 1525 and 2600 respectively. If the ratio of male to female population in the first village is 27:34 and the ratio of male to female population in the second village is 6:7, then what is the ratio of male to female population of these two villages taken together?

[CDS 2020 (I)]

- (a) $\frac{33}{41}$ (b) $\frac{85}{82}$
(c) $\frac{71}{90}$ (d) $\frac{5}{6}$

38. If $\frac{a}{b} = \frac{1}{3}$, $\frac{b}{c} = 2$, $\frac{c}{d} = \frac{1}{2}$, $\frac{d}{e} = 3$ and $\frac{e}{f} = \frac{1}{4}$,

then what is the value of $\frac{abc}{def}$?

[CDS 2020 (I)]

- (a) $\frac{1}{4}$ (b) $\frac{3}{4}$
(c) $\frac{3}{8}$ (d) $\frac{27}{4}$

39. Three persons start a business with capitals in the ratio $\frac{1}{3} : \frac{1}{4} : \frac{1}{5}$. The first person withdraws half his capital after 4 months. What is the share of profit of A if the business fetches an annual profit of Rs 96800?

[CDS 2020 (II)]

- (a) Rs 32000 (b) Rs 34500
(c) Rs 36000 (d) Rs 36800

40. Ena was born 4 years after her parents marriage. Her mother is 3 years younger than her father and 24 years older than Era, who is 13 years old. At what age did Rea's father get married?

[CDS 2020 (II)]

- (a) 25 years (b) 24 years
(c) 23 years (d) 22 years

41. Mahesh is 60 years old. Ram is 5 years younger to Mahesh and 4 years elder to Raju. Babu is younger brother of Raju and he is 6 years younger. What is age difference between Mahesh and Babu?

[CDS 2020 (II)]

- (a) 18 years (b) 15 years
(c) 13 years (d) 11 years

42. The income of A, B and C are in the ratio 7 : 9 : 12 and their expenditures are in the ratio 8 : 9 : 15. If A's saving is one-fourth of his income, then the ratio of savings of A, B and C is

[CDS 2021 (I)]

- (a) 56 : 99 : 69 (b) 99 : 56 : 69
(c) 69 : 56 : 99 (d) 99 : 69 : 56

43. If three positive numbers are in the ratio 2 : 3 : 5 and the sum of their squares is 1368, then what is the sum of the numbers?

[CDS 2021 (II)]

- (a) 30 (b) 45
(c) 60 (d) 75

44. If $a = 3b$, $4b = 5c$, $6c = 7d$, then what is

$\frac{d+a}{d-a}$ equal to?

[CDS 2022 (II)]

- (a) $\frac{27}{43}$ (b) $\frac{43}{27}$
(c) $-\frac{27}{43}$ (d) $-\frac{43}{27}$

45. Two glasses of equal volume are filled with a mixture of alcohol and water in the ratio 3:2 and 4:1, respectively. These glasses are emptied into a third glass. What is the ratio of alcohol and water in the third glass?

[CDS 2022 (II)]

- (a) 5:4 (b) 7:2
(c) 7:3 (d) 7:4

46. A, B, C, D and E enter into a business. They invest money in the ratio 2 : 3 : 4 : 5 : 6. However, the time invested by them is in the ratio 6 : 5 : 4 : 3 : 2. If the profit distributed is directly proportional to time and money invested, then who receives the highest amount of profit?

[CDS - 2023 (1)]

- (a) C
(b) both B and D
(c) both c and D
(d) All get equal profit

47. If a, b, c, d, e and f satisfy $2a = 3b = 6c = 9d = 12e = 18f$, then what is the value of $(a + b)/(c + d + e + f)$?

[CDS - 2023 (1)]

- (a) $\frac{4}{7}$ (b) 2
(c) $\frac{5}{2}$ (d) $\frac{9}{2}$

48. Three glasses P, Q and R have capacities in the ratio 1:2:3. All these glasses are completely filled with mixtures of milk and water. The ratio of milk to water in P is 1:2, in Q it is 2:3 and in R it is 3:1. If the content

of all three glasses are put into a bigger container, what will be the ratio of milk to water in the container?

[CDS - 2023 (2)]

- (a) 203:117 (b) 203:157
(c) 172:91 (d) 165:88

49. Rs 9400 is distributed among P, Q, R in such a way that if Rs 93, Rs 24, Rs 55 are deducted from their respective shares, then they have money in the ratio 3 : 4 : 5. What is the share of P?

[CDS - 2024 (1)]

- (a) Rs 2307 (b) Rs 2376
(c) Rs 2508 (d) Rs 12896

50. If $(a + b) : (b + c) : (c + a) = 5 : 7 : 6$, then what is the value of $(a - b + c) : (a + b - c)$?

[CDS - 2024 (1)]

- (a) 1 : 1 (b) 2 : 3
(c) 3 : 1 (d) 4 : 3

51. A bottle contains spirit and water in the ratio 1 : 4 and another identical bottle contains spirit and water in the ratio 4 : 1. In what ratio should the mixtures in the two bottles be mixed to get a new mixture in which the ratio of spirit to water is 1 : 3?

[CDS - 2024 (1)]

- (a) 5 : 1 (b) 6 : 1
(c) 10 : 1 (d) 11 : 1

52. If $a : b : c : d = \sqrt{4} : \sqrt{3} : \sqrt{2} : \sqrt{1}$, then what is the value of $\frac{(-a^2 + b^2 + c^2 + d^2)}{(a^2 - b^2 + c^2 - d^2)}$?

[CDS - 2024 (1)]

- (a) 1 (b) 2
(c) 3 (d) 6

53. If $m : n = 1 : 2$ and $p : q = 3 : 4$, then what is $(2m + 4p) : (n + 3q)$ equal to?

[CDS - 2024 (1)]

- (a) 1 : 1 (b) 1 : 3
(c) 2 : 1 (d) 2 : 3

ANSWER KEY

1.	d	2.	b	3.	c	4.	a	5.	d	6.	a	7.	a	8.	b	9.	c	10.	c
11.	a	12.	a	13.	d	14.	c	15.	d	16.	b	17.	c	18.	c	19.	d	20.	a
21.	b	22.	b	23.	c	24.	d	25.	c	26.	c	27.	b	28.	b	29.	d	30.	b
31.	b	32.	c	33.	a	34.	c	35.	c	36.	a	37.	d	38.	c	39.	a	40.	c
41.	b	42.	a	43.	c	44.	d	45.	c	46.	c	47.	b	48.	b	49.	a	50.	c
51.	d	52.	a	53.	a														

CAPF PYQ

1. 5 % of income of A is equal to 15% of income of B and 10% income of B is equal to 20% of income of C. If the income of C is Rs. 2000, then what is the total income of A,B and C is
[CAPF 2016]
(a) Rs. 12000 (b) Rs. 18000
(c) Rs. 14000 (d) Rs. 6000
2. One year ago, a father was four times as old as his son. After six years his age exceeds twice his son's age by 9 years. The ratio of their present age is
[CAPF 2018]
(a) 9 : 2 (b) 11 : 3
(c) 12 : 5 (d) 13 : 4
3. The ratio of ages of a man and his son is 3 : 1. After 15 years, the age ratio will be 2 : 1, What is the age of man ?
[CAPF 2018]
(a) 45 years (b) 40 years
(c) 35 years (d) 30 years
4. If 15% of A is double of 30% of B, then what is the ratio of A to B ?
[CAPF 2020]
- (a) 1 : 2 (b) 2: 1
(c) 1 : 4 (d) 4 : 1
5. Two years ago , the age of A was three times the age of B , If B is currently 9 years old , then after how many years The age of A will be double of the age of B ?
[CAPF 2020]
(a) 2 years (b) 3 years
(c) 4 years (d) 5 years
6. The ratio of monthly income of A and B is 7:10. The ratio of their expenditures is 2:3. If each of A and B saves Rs 1000 per month, then what will be the monthly income of B?
[CAPF 2021]
(a) Rs 9000 (b) Rs 10000
(c) Rs 15000 (d) Rs 12000
7. A, B and C start a business by investing Rs 7000, Rs 8000 and Rs 12000 respectively. After a year, B get Rs 3200 as his share of profit. What is the total profit?
[CAPF 2021]
(a) Rs 16600 (b) Rs 10000
(c) Rs 21600 (d) Rs 10800

ANSWER KEY

1.	(b)	2.	(b)	3.	(a)	4.	(d)	5.	(d)	6.	(b)	7.	(d)						
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PRACTICE SHEET 1

1. The average of the first 7 integers in a series of 13 consecutive odd integers is 37. What is the average of the entire series?
(1) 37 (2) 39
(3) 41 (4) 43
2. The average of 25 consecutive odd integers is 55. The highest of these integers is
(1) 79 (2) 105
(3) 155 (4) 109
3. The arithmetic mean of the following numbers 1, 2, 2, 3, 3, 3, 4, 4, 4, 4, 5, 5, 5, 5, 6, 6, 6, 6, 6 and 7, 7, 7, 7, 7, 7, 7 is
(1) 4 (2) 5
(3) 14 (4) 20
4. The average of all the numbers between 6 and 50 which are divisible by 5 is
(1) 27.5 (2) 30
(3) 28.5 (4) 22
5. The average of 1, 3, 5, 7, 9, 11, ----- to 25 terms is
(1) 125 (2) 25
(3) 625 (4) 50
6. The average of 7 consecutive numbers is 20. The largest of these numbers is
(1) 20 (2) 23
(3) 24 (4) 26
7. The average of 8 numbers is 27. If each of the numbers is multiplied by 8, find the average of new set of numbers.
(1) 1128 (2) 938
(3) 316 (4) 216
8. The average of eight successive numbers is 6.5. The average of the smallest and the greatest numbers among them will be :
(1) 4 (2) 6.5
(3) 7.5 (4) 9
9. The average of 7 consecutive numbers is 20. The largest of these numbers is :
(1) 24 (2) 23
(3) 22 (4) 20
10. The average of first nine prime numbers is
(1) 9 (2) 11
(3) $11\frac{2}{9}$ (4) $11\frac{1}{9}$
11. The average of 5 consecutive natural numbers is m . If the next three natural numbers are also included, how much more than m will the average of these 8 numbers be?
(1) 2 (2) 1
(3) 1.4 (4) 1.5
12. The average of the first 100 positive integers is
(1) 100 (2) 51
(3) 50.5 (4) 49.5
13. The average of odd numbers upto 100 is
(1) 50.5 (2) 50
(3) 49.5 (4) 49
14. The average of the squares of first ten natural numbers is
(1) 35.5 (2) 36
(3) 37.5 (4) 38.5
15. The average of seven consecutive positive integers is 26. The smallest of these integers is :
(1) 21 (2) 23
(3) 25 (4) 26
16. The average of 5 consecutive natural numbers is m . If the next three natural numbers are also included, how much more than m will the average of these 8 numbers be?
(1) 2 (2) 1
(3) 1.4 (4) 1.5
17. Average of first five odd multiples of 3 is
(1) 12 (2) 16
(3) 15 (4) 21
18. The average of nine consecutive numbers is n . If the next two numbers are also included the new average will
(1) increase by 2
(2) remain the same
(3) increase by 1.5
(4) increase by 1
19. The average of four consecutive even numbers is 9. Find the largest number.
(1) 12 (2) 6
(3) 8 (4) 10

- 20.** If a, b, c, d, e are five consecutive odd numbers, their average is
 (1) $5(a + 4)$
 (2) $abcde/5$
 (3) $5(a + b + c + d + e)$
 (4) $a + 4$
- 21.** Average of first five prime numbers is
 (1) 5.3 (2) 5.6
 (3) 5 (4) 3.6
- 22.** What is the average of the first six (positive) odd numbers each of which is divisible by 7?
 (1) 42 (2) 43
 (3) 47 (4) 49
- 23.** The average of first ten prime numbers is
 (1) 10.1 (2) 10
 (3) 12.9 (4) 13
- 24.** If the average of eight consecutive even numbers be 93, then the greatest number among them is
 (1) 100 (2) 86
 (3) 102 (4) 98
- 25.** The average of 6 consecutive natural numbers is K . If the next two natural numbers are also included, how much more than K will the average of these 8 numbers be?
 (1) 1.3 (2) 1
 (3) 2 (4) 1.8
- 26.** The average of five consecutive positive integers is n . If the next two integers are also included, the average of all these integers will
 (1) increase by 1.5
 (2) increase by 1
 (3) remain the same
 (4) increase by 2
- 27.** The average of all the odd integers between 2 and 22 is:
 (1) 14 (2) 12
 (3) 13 (4) 11
- 28.** The sum of three consecutive even numbers is 28 more than the average of these three numbers. Then the smallest of these three numbers is
 (1) 6 (2) 12
 (3) 14 (4) 16
- 29.** The average of 7 consecutive numbers is 20. The largest of these numbers is
 (1) 20 (2) 23
 (3) 24 (4) 26
- 30.** The average of 25 consecutive odd integers is 55. The highest of these integers is
 (1) 79 (2) 105
 (3) 155 (4) 10

ANSWER KEY

1.	(4)	2.	(1)	3.	(2)	4.	(1)	5.	(2)	6.	(2)	7.	(4)	8.	(2)	9.	(2)	10.	(4)
11.	(4)	12.	(3)	13.	(2)	14.	(4)	15.	(2)	16.	(4)	17.	(3)	18.	(4)	19.	(1)	20.	(4)
21.	(2)	22.	(1)	23.	(3)	24.	(1)	25.	(2)	26.	(2)	27.	(2)	28.	(2)	29.	(2)	30.	(1)

PRACTICE SHEET 2

1. 12 kg of rice costing 30 per kg is mixed with 8 kg of rice costing 40 per kg. The average per kg price of mixed rice is
 (1) 38 (2) 37
 (3) 35 (4) 34
2. A library has an average number of 510 visitors on Sunday and 240 on other days. The average number of visitors per day in a month of 30 days beginning with Sunday is:
 (a) 285 (b) 295
 (c) 300 (d) 290
3. The average weight of five persons sitting in a boat is 38 kg. The average weight of the boat and the persons sitting in the boat is 52kg. What is the weight of the boat?
 (1) 228 kg (2) 122 kg
 (3) 232 kg (4) 242 kg
4. The average marks of 32 boys of section A of class X is 60 whereas the average marks of 40 boys of section B of class X is 33. The average marks for both the sections combined together is
 (1) 44 (2) 45
 (3) $46\frac{1}{2}$ (4) $45\frac{1}{2}$
5. Six tables and twelve chairs were bought for 7,800. If the average price of a table is 750, then the average price of a chair would be
 (1) 250 (2) 275
 (3) 150 (4) 175
6. a, b, c, d, e, f, g are consecutive even numbers. j, k, l, m, n are consecutive odd numbers. The average of all the numbers is
 (1) $3\left(\frac{a+n}{2}\right)$ (2) $\left(\frac{5l+7d}{12}\right)$
 (3) $\left(\frac{a+b+m+n}{4}\right)$ (4) $\left(\frac{j+c+n+g}{4}\right)$
7. The average weight of 15 person in a boat is increased by 1.6 kg when one of the crew, who weighs 42 kg is replaced by a new man. Find the weight of the new man (in kg).
 (1) 67 (2) 65
 (3) 66 (4) 43
8. The average of 18 numbers is 37.5. If six numbers of average x are added to them, then the average of all the numbers increases by one. The value of x is:
 (a) 42 (b) 40
 (c) 38.5 (d) 41.5
9. The average of 9 numbers is 30. The average of first 5 numbers is 25 and that of the last 3 numbers is 35. What is the 6th number?
 (1) 20 (2) 30
 (3) 40 (4) 50
10. The average of 15 numbers is 7. If the average of the first 8 numbers be 6.5 and the average of last 8 numbers be 9.5, then the middle number is
 (1) 20 (2) 21
 (3) 23 (4) 18
11. Out of seven given numbers, the average of the first four numbers is 4 and that of the last four numbers is also 4. If the average of all the seven numbers is 3, fourth number is
 (1) 3 (2) 4
 (3) 7 (4) 11
12. The average temperature of the first 4 days of a week was 37°C and that of the last 4 days of the week was 41°C . If the average temperature of the whole week was 39°C , the temperature of the fourth day was
 (1) 38°C (2) 38.5°C
 (3) 39°C (4) 40°C
13. In a certain year, the average monthly income of a person was 3,400. For the first eight months of the year, his average monthly income was 3,160 and for the last five months, it was 4,120. His income in the eighth month of the year was
 (1) 3,160 (2) 5,080
 (3) 15,520 (4) 5,520
14. The average of 30 numbers is 12. The average of the first 20 of them is 11 and that of the next 9 is 10. The last number is
 (1) 60 (2) 45
 (3) 40 (4) 50
15. The average of 11 results is 50. If the average of the first six results is 49 and that of the last six is 52, the sixth no. is
 (1) 48 (2) 50
 (3) 52 (4) 56

- 16.** The average of five numbers is 27. If one number is excluded, the average becomes 25. The excluded number is :
 (1) 25 (2) 27
 (3) 30 (4) 35
- 17.** The average of marks of 28 students in Mathematics was 50; 8 students left the school, then this average increased by 5. What is the average of marks obtained by the students who left the school?
 (1) 50.5 (2) 37.5
 (3) 42.5 (4) 45
- 18.** The average weight of 12 parcels is 1.8 kg. Addition of another new parcel reduces the average weight by 50 g. What is the weight of the new parcel?
 (1) 1.50 kg (2) 1.10 kg
 (3) 1.15 kg (4) 1.01 kg.
- 19.** The average of 50 numbers is 38. If two numbers namely 45 and 55 are discarded, the average of the remaining numbers is :
 (1) 35 (2) 32.5
 (3) 37.5 (4) 36
- 20.** There are 50 students in a class. Their average weight is 45 kg. When one student leaves the class the average weight reduces by 100g. What is the weight of the student who left the class ?
 (1) 45 kg (2) 47.9 kg
 (3) 49.9 kg (4) 50.1 kg
- 21.** Of the three numbers whose average is 60, the first is one fourth of the sum of the others. The first number is :
 (1) 30 (2) 36
 (3) 42 (4) 45
- 22.** Of the three numbers, second is twice the first and also thrice the third. If the average of the three numbers is 44, the largest number is:
 (1) 24 (2) 72
 (3) 36 (4) 108
- 23.** The average of first three numbers is thrice the fourth number. If the average of all the four numbers is 5, then find the fourth number.
- (1) 4.5 (2) 5
 (3) 2 (4) 4
- 24.** Of the three numbers, first is twice the second and second is twice the third. The average of three numbers is 21. The smallest of the three numbers is
 (1) 9 (2) 6
 (3) 12 (4) 18
- 25.** Of the three numbers, the first is 3 times the second and the third is 5 times the first. If the average of the three numbers is 57, the difference between the largest and the smallest number is
 (1) 9 (2) 18
 (3) 126 (4) 135
- 26.** Of the three numbers, the first is twice the second and the second is 3 times the third. If their average is 100, the largest of the three numbers is :
 (1) 120 (2) 150
 (3) 180 (4) 300
- 27.** Of the three numbers, the first is twice the second and the second is thrice the third. If the average of the three numbers is 10, the largest number is:
 (1) 12 (2) 15
 (3) 18 (4) 30
- 28.** The average of first three numbers is double of the fourth number. If the average of all the four numbers is 12, find the 4th number.
 (1) 16 (2) 48/7
 (3) 20 (4) 18/7
- 29.** The average of three numbers is 77. The first number is twice the second and the second number is twice the third. The first number is:
 (1) 33 (2) 66
 (3) 77 (4) 132
- 30.** Out of three numbers, the first is twice the second and is half of the third. If the average of the three numbers is 56, then difference of first and third number is
 (1) 12 (2) 20
 (3) 24 (4) 48

ANSWER KEY

1.	(4)	2.	(1)	3.	(2)	4.	(2)	5.	(2)	6.	(2)	7.	(3)	8.	(4)	9.	(3)	10.	(3)
11.	(4)	12.	(3)	13.	(2)	14.	(4)	15.	(4)	16.	(4)	17.	(2)	18.	(3)	19.	(3)	20.	(3)
21.	(2)	22.	(2)	23.	(3)	24.	(1)	25.	(3)	26.	(3)	27.	(3)	28.	(2)	29.	(4)	30.	(4)

PRACTICE SHEET 3

- The average of marks of 14 student was calculated as 71. But it was later found that the marks of one student had been wrongly entered as 42 instead of 56 and of another as 74 instead of 32. The correct average is :
 (1) 67 (2) 68
 (3) 69 (4) 71
- The average weight of three men A, B and C is 84 kg. D joins them and the average weight of the four becomes 80 kg. If E whose weight is 3 kg more than that of D, replaces A, the average weight of B, C, D and E becomes 79 kg. The weight of A is
 (1) 65 kg. (2) 70 kg.
 (3) 75 kg. (4) 80 kg.
- The average of a collection of 20 measurements was calculated to be 56 cm. But later it was found that a mistake had occurred in one of the measurements which was recorded as 64 cm, but should have been 61 cm. The correct average must be
 (1) 53 cm (2) 54.5 cm
 (3) 55.85 cm (4) 56.15 cm
- The average of marks in Mathematics for 5 students was found to be 50. Later, it was discovered that in the case of one student the marks 48 were misread as 84. The correct average is:
 (1) 40.2 (2) 40.8
 (3) 42.8 (4) 48.2
- The average weight of 15 students in a class increases by 1.5kg when one of the students weighing 40 kg is replaced by a new student. What is the weight (in kg) of the new student?
 (1) 64.5 kg. (2) 56 kg.
 (3) 60 kg. (4) 62.5 kg.
- The average marks of 100 students were found to be 40. Later on it was discovered that a score of 53 was misread as 83. Find the correct average corresponding to the correct score.
 (1) 38.7 (2) 39
 (3) 39.7 (4) 41
- A cricketer whose bowling average is 24.85, runs per wicket, takes 5 wickets for 52 runs and thereby decreases his average by 0.85. The number of wickets taken by him till the last match was :
 (1) 64 (2) 72
 (3) 80 (4) 96
- The average of runs of a cricket player of 10 innings was 32. How many runs must he make in his next inning so as to increase his average of runs by 4?
 (1) 76 (2) 70
 (3) 4 (4) 2
- The bowling average of a cricketer was 12.4. He improves his bowling average by 0.2 points when he takes 5 wickets for 26 runs in his last match. The number of wickets taken by him before the last match was
 (1) 125 (2) 150
 (3) 175 (4) 200
- A cricketer had a certain average of runs for his 64 innings. In his 65th innings, he is bowled out for no score on his part. This brings down his average by 2 runs. His new average of runs is
 (1) 130 (2) 128
 (3) 70 (4) 68
- A cricketer has a certain average of runs for his 8 innings. In the ninth innings, he scores 100 runs, thereby increases his average by 9 runs. His new average of runs is
 (1) 20 (2) 24
 (3) 28 (4) 32

- 12.** The average of runs scored by a player in 10 innings is 50. How many runs should he score in the 11th innings so that his average is increased by 2 runs?
 (1) 80 runs (2) 72 runs
 (3) 60 runs (4) 54 runs
- 13.** A cricket batsman had a certain average of runs for his 11 innings. In the 12th innings, he made a score of 90 runs and thereby his average of runs was decreased by 5. His average of runs after 12th innings is :
 (1) 155 (2) 150
 (3) 145 (4) 140
- 14.** The batting average for 40 innings of a cricket player is 50 runs. His highest score exceeds his lowest score by 172 runs. If these two innings are excluded, the average of the remaining 38 innings is 48 runs. The highest score of the player is
 (1) 165 runs (2) 170 runs
 (3) 172 runs (4) 174 runs
- 15.** The batting average of a cricket player for 64 innings is 62 runs. His highest score exceeds his lowest score by 180 runs. Excluding these two innings, the average of remaining innings becomes 60 runs. His highest score was
 (1) 180 runs (2) 209 runs
 (3) 212 runs (4) 214 runs
- 16.** A cricket player after playing 10 tests scored 100 runs in the 11th test. As a result, the average of his runs is increased by 5. The present average of runs is
 (1) 45 (2) 40
 (3) 50 (4) 55
- 17.** A cricketer has a mean score of 60 runs in 10 innings. Find out how many runs are to be scored in the eleventh innings to raise the mean score to 62?
 (1) 83 (2) 82
 (3) 80 (4) 81
- 18.** In a 20 over match, the required run rate to win is 7.2. If the run rate is 6 at the end of the 15th over, the required run rate to win the match is
 (1) 1.2 (2) 13.2
 (3) 10.8 (4) 12
- 19.** Average age of 6 sons of a family is 8 years. Average age of sons together with their parents is 22 years. If the father is older than the mother by 8 years, the age of mother (in years) is :
 (1) 44 (2) 52
 (3) 60 (4) 68
- 20.** Out of 10 teachers of a school, one teacher retires and in his place, a new teacher of age 25 years joins. As a result, average age of teachers reduces by 3 years. The age of the retired teacher is
 (1) 50 years (2) 55 years
 (3) 58 years (4) 60 years
- 21.** 3 years ago, the average age of a family of 5 members was 17 years. A baby having been born, the average age of the family is the same today. The present age of the baby is :
 (1) 3 years (2) 2 years
 (3) 1.5 years (4) 1 year
- 22.** The average age of 45 persons is decreased by $\frac{1}{9}$ year when one of them of 60 years is replaced by a new comer. Then the age of the new comer is :
 (1) 45 years (2) 55 years
 (3) 59 years (4) 49 years
- 23.** When the average age of a husband and wife and their son was 42 years, the son got married and a child was born just one year after the marriage. When child turned to be five years then the average age of family became 36 years. What was the age of daughter-in-law at the time of marriage?
 (1) 26 years (2) 25 years
 (3) 24 years (4) 23 years
- 24.** The average age of 30 boys in a class is 15 years. One boy aged 20 years, left the class, but two new boys came in his place whose ages differ by 5 years. If the average age of all the boys now in the class still remains 15 years, the age of the younger newcomer is :
 (1) 20 years (2) 15 years
 (3) 10 years (4) 8 years
- 25.** The average age of 24 boys and their teacher is 15 years. When the teacher's age is excluded, the average age decreases by 1 year. The age of the teacher is
 (1) 38 years (2) 39 years
 (3) 40 years (4) 41 years

- 26.** There were 24 students in a class. One of them, who was 18 years old, left the class and his place was filled up by a newcomer. If the average of the class thereby, was lowered by one month, the age of the newcomer is
 (1) 14 years (2) 15 years
 (3) 16 years (4) 17 years
- 27.** The average age of 30 students is 9 years. If the age of their teacher is included, the average age becomes 10 years. The age of the teacher (in years) is
 (1) 27 (2) 31
 (3) 35 (4) 40
- 28.** From a class of 24 boys, a boy, aged 10 years, leaves the class and in his place a new boy is admitted. As a result, the average age of the class is increased by 2 months. What is the age of the new boy?
 (1) 12 years (2) 15 years
 (3) 14 years (4) 13 years
- 29.** 5 years ago, the average age of A, B, C and D was 45 years. With E joining them now, the average age of all the five is 49 years. How old is E?
 (1) 25 years (2) 40 years
 (3) 45 years (4) 64 years
- 30.** In a family, the average age of a father and a mother is 35 years. The average age of the father, mother and their only son is 27 years. What is the age of the son?
 (1) 12 years (2) 11 years
 (3) 10.5 years (4) 10 years

ANSWER KEY

1.	(3)	2.	(3)	3.	(3)	4.	(3)	5.	(4)	6.	(3)	7.	(3)	8.	(1)	9.	(3)	10.	(2)
11.	(3)	12.	(2)	13.	(3)	14.	(4)	15.	(4)	16.	(3)	17.	(2)	18.	(3)	19.	(3)	20.	(2)
21.	(2)	22.	(2)	23.	(2)	24.	(2)	25.	(2)	26.	(3)	27.	(4)	28.	(3)	29.	(3)	30.	(2)

CDS PYQ

1. The mean of 7 observations is 7. If each observation is increased by 2, then the new mean is

[CDS 2015 (II)]

[CDS 2013(I)]

(a) 12
(b) 10

(c) 9
(d) 8

(a) 10
(b) 15

(c) 20
(d) 25
2. There are 45 male and 15 female employees in an office. If the mean salary of 60 employees is Rs 4800 and the mean salary of the male employees is Rs 5000, then the mean salary of the females employees is

[CDS 2015 (II)]

[CDS 2013(I)]

(a) Rs 4200
(b) Rs 4500

(c) Rs 5600
(d) Rs 6000

(a) 32 years
(b) 28 years

(c) 36 years
(d) 40 years
3. The average age of male employees is 52 years and that of female employees is 42 years the mean age of all employees is 50 years. The percentage of male and female employees is:

[CDS 2016(II)]

[CDS 2014 (I)]

(a) 80% and 20%
(b) 20% and 80%

(c) 50% and 50%
(d) 52% and 48%

(a) 60
(b) 50

(c) 45
(d) 40
4. If the average of A and B is 30, the average of C and D is 20, then which of the following is/are correct?
 1.The average of B and C must be greater than 25.
 2.The average of A and D must be less than 25.
 Select the correct answer using the code given below:

[CDS 2016(II)]

[CDS 2014 (I)]

(a) 1 only
(b) 2 only

(c) either 1 or 2
(d) neither 1 nor 2

(a) 81.5
(b) 80.5

(c) 79.0
(d) 78.0
5. The average of m numbers is n^4 and the average of n numbers is m^4 . The average of (m + n) numbers is:

[CDS 2016(II)]

[CDS 2015 (II)]

(a) mn
(b) $m^2 + n^2$

(c) $mn(m^2 + n^2)$
(d) $mn(m^2 + n^2 - mn)$

(a) 42
(b) 47

(c) 48
(d) 60
6. The average weight of students in a class is 43kg. Four new students are admitted to the class whose weights are 42kg, 36.5kg, 39kg and 42.5kg respectively. Now the average weight-of the students of the class is 42.5kg. The number of students in the beginning was:

[CDS 2016(II)]

(a) 17.4
(b) 16.6

(c) 15.8
(d) 14.2

(a) 17.4
(b) 16.6

(c) 15.8
(d) 14.2
7. Four years ago, the average age of A and B was 18 years. Now the average age of A, B and C is 24 years. After 8 years, the age of C will be:

[CDS 2016(II)]

(a) 60
(b) 50

(c) 45
(d) 40

(a) 81.5
(b) 80.5

(c) 79.0
(d) 78.0
8. The mean age of combined group of boys and girls is 18 years and the mean of age of boys is 20 and that of girls is 16, then what is the percentage of boys in the group is

[CDS 2016(II)]

(a) 81.5
(b) 80.5

(c) 79.0
(d) 78.0

(a) 42
(b) 47

(c) 48
(d) 60
9. The average score of class X is 83. The average score of class Y is 76. The average score of class Z is 85. The average score of class X and Y is 79. The average score of class Y and Z is 81.What is the average score of class X, Y and Z ?

[CDS 2016(II)]

(a) 42
(b) 47

(c) 48
(d) 60

(a) 17.4
(b) 16.6

(c) 15.8
(d) 14.2
10. A cricketer has a certain average of 10 innings. In the eleventh inning he scored 108 runs, thereby increasing his average by 6 runs. What is his new average?

[CDS 2016(II)]

(a) 17.4
(b) 16.6

(c) 15.8
(d) 14.2

(a) 17.4
(b) 16.6

(c) 15.8
(d) 14.2
11. The mean of 20 observations is 17. On checking it was found that the two observations were wrongly copied as 3 and 6. If wrong observations are replaced by correct values 8 and 9, then what is the correct mean?

[CDS 2016(II)]

(a) 17.4
(b) 16.6

(c) 15.8
(d) 14.2

(a) 17.4
(b) 16.6

(c) 15.8
(d) 14.2
12. In a class of 100 students, there are 70 boys whose average marks in a subject are 75. If the average marks of the complete class is 72, then what is the average marks of girls?

[CDS 2016(II)]

- (a) 64 (b) 65
(c) 68 (d) 74

13. The mean of 5 numbers is 15, if one more number is included, the mean of 6 observations become 17. What is the included number?

[CDS 2017(I)]

- (a) 24 (b) 25
(c) 26 (d) 27

14. The mean marks obtained by 300 students in a subject are 60. The mean of top 100 students was found to be 80 and the mean of last 100 students was found to be 50. The mean marks of the remaining 100 students are

[CDS 2017(II)]

- (a) 70 (b) 65
(c) 60 (d) 50

15. Let a, b, c, d, e, f, g be consecutive even numbers and j, k, l, m, n be consecutive odd numbers. What is the average of all the numbers?

[CDS 2017(I)]

- (a) $\frac{3(a+n)}{2}$ (b) $\frac{5(l+7d)}{12}$
(c) $\frac{a+b+m+n}{4}$ (d) None

16. A small company pays each of its 5 category 'C' workers Rs 20000, each of its 3 category 'B' workers Rs 25000 and a category 'A' workers Rs 65000. The number of workers earning less than the mean salary is

[CDS 2017(II)]

- (a) 8 (b) 5
(c) 4 (d) 3

17. The average height of 22 students of a class is 140 cm, and the average height of 28 students of another class is 152cm. What is the average height of students of both the classes?

[CDS 2017(II)]

- (a) 144.32 cm (b) 145.52 cm
(c) 146.72 cm (d) 147.92 cm

18. At present the average of the ages of a father and a son is 25years After 7 year the son will be 17 year old. What will be the age of the father after 10 year?

[CDS 2018(I)]

- (a) 44yr (b) 45yr
(c) 50 yr (d) 52 yr

19. If the average of 9 consecutive positive integers is 55, then what is the largest integer?

[CDS 2018(I)]

- (a) 57 (b) 58
(c) 59 (d) 60

20. The average of the ages of 15 students in a class is 19 year. When 5 new students are admitted to the class, the average age of the class becomes 18.5 year. What is the average age of 5 newly admitted students?

[CDS 2018(I)]

- (a) 17 yr (b) 17.5 yr
(c) 18 yr (d) 18.5 yr

21. The arithmetic mean of 11 observations is 11. The arithmetic mean of the first 6 observations is 10.5 and the arithmetic mean of the last 6 observations is 11.5. What is the sixth observation?

[CDS 2018(I)]

- (a) 10 (b) 10.5
(c) 11 (d) 11.5

22. The average marks of section A are 65 and that of section B are 70. If the average of both the sections combined are 67, then the ratio of number of students of section A and to that of section B is

[CDS 2018(II)]

- (a) 3 : 2 (b) 1 : 3
(c) 3 : 1 (d) 2 : 3

23. Let $\bar{x}_1$ and $\bar{x}_2$ (where $\bar{x}_2 > \bar{x}_1$) be the means of two sets comparing n_1 and n_2 (where $n_2 < n_1$) observations respectively. If $\bar{x}$ is the mean when they are pooled. then which one of the following is correct ?

[CDS 2018(II)]

- (a) $\bar{x}_1 < \bar{x} < \bar{x}_2$
(b) $\bar{x} > \bar{x}_2$
(c) $\bar{x} < \bar{x}_1$
(d) $(\bar{x}_1 - \bar{x}) + (\bar{x}_2 - \bar{x}) = 0$

24. In a class of 100 students, the average weight is 30 kg. If the average weight of girls is 24 kg

and that of the boys is 32 kg, then what is the number of girls in class?

[CDS 2019(I)]

- (a) 25 (b) 26
(c) 27 (d) 28

25. The average of 50 consecutive natural numbers is x . what will be the new average when the next four numbers also included?

[CDS 2019(I)]

- (a) $x + 1$ (b) $x + 2$
(c) $x + 4$ (d) $x + (x/54)$

26. The mean of m observations out of n observations is n and the mean of remaining observations is m , then what is the mean of all n observations?

[CDS 2019(I)]

- (a) $2m - (m^2/n)$ (b) $2m + (m^2/n)$
(c) $m - (m^2/n)$ (d) $m + (m^2/n)$

27. A library has an average number of 510 visitors on Sunday and 240 on other days. What is the average number of visitors per day in a month of 30 days beginning with Saturday?

[CDS 2019(II)]

- (a) 276 (b) 282
(c) 285 (d) 375

28. In a class room of number of girls to that of boys is 3 : 4, The average height of students in the class is 4.6 feet. If average height of the boys in the class is 4.8feet, then what is the average height of the girls in the class?

[CDS 2020(I)]

- (a) Less than 4.2 kg
(b) More than 4.5 feet and less than 4.3 feet
(c) More than 4.3 feet but less than 4.4 feet
(d) More than 4.4 feet but less than 4.5 feet

29. The ages of 7 family members are 2, 5, 12, 18, 38, 40 and 60 years respectively. After 5 years a new member aged x years is added. If the mean age of the family now goes up by 1.5 years, then what is the value of x ?

[CDS 2020(II)]

- (a) 1 (b) 2
(c) 3 (d) 4

30. The mean weight of 100 students in a class is 46kg. The mean weight of boys is 50kg and that of girls is 40kg. The number of boys exceeds the number of girls by

[CDS 2020(II)]

- (a) 10 (b) 15

- (c) 20 (d) 25

31. The mean of five observations $x, x + 2, x + 4, x + 6, x + 8$ is m . What is the mean of first three observation.

[CDS 2020(II)]

- (a) m (b) $m - 1$
(c) $m - 2$ (d) $m - 3$

32. What is the arithmetic mean of the first ten composite numbers?

[CDS 2021(I)]

- (a) 8.5 (b) 9.5
(c) 10.2 (d) 11.2

33. The marks obtained by 5 students are 21, 27, 19, 26, 32. Later on 5 grace marks are added to each student. What are the average marks of the revised marks of the students?

[CDS 2021(I)]

- (a) 26 (b) 30
(c) 31 (d) 32

34. Let p be the mean of m observations and q be the mean of n observations, where $p \leq q$. If the combined mean of $(m + n)$ observations is c , then which one of the following is correct ?

[CDS 2021(I)]

- (a) $c \leq p$ (b) $c \geq q$
(c) $p \leq c \leq q$ (d) $q \leq c \leq p$

35. Let the average score of a class of boys and girls in an examination be p . The ratio of boys and girls in the class is 3 : 1. If the average score of the boys is $(p + 1)$, then what is the average score of girls?

[CDS 2021(I)]

- (a) $(p - 1)$ (b) $(p - 2)$
(c) $(p - 3)$ (d) p

Direction for next two:

Collect all the sequence of five consecutive integers such that their product is equal to one of these integers. Let X be the collection of all possible such sequences. Let P be the smallest integer and Q be the largest integer occurring in these sequences.

36. How many such sequences of five consecutive integers are possible?

[CDS 2021(II)]

- (a) One (b) Two
(c) Three (d) Five

37. What is the arithmetic mean of P and Q ?

[CDS 2021(II)]

- (a) 0

- (b) 1
(c) 2
(d) Cannot be determined due to insufficient data

38. Mean marks of 50 students were found to be 78.4. But later it was detected that 95 was misread as 59 and 25 was misread as 52. What is the difference between correct mean and incorrect mean?

[CDS 2022 (II)]

- (a) 0.04 (b) 0.08
(c) 0.12 (d) 0.18

39. If M is the mean of the first 100 even natural numbers, then what is the mean of the first 100 odd natural numbers?

[CDS 2022 (II)]

- (a) $M-1$ (b) M
(c) $M+1$ (d) $M+2$

40. Average marks in Mathematics of Section A comprising 30 students is 65 and that of Section B comprising 35 students is 70. What are the average marks (approximately) of both the sections if it was detected later that an entry of 47 marks was wrongly made as 74?

[CDS - 2023 (1)]

- (a) 67.28 (b) 67.58
(c) 68.11 (d) 68.63

41. The mean of p, q, r, s and t is 280. If the mean of p, r and t is 240, what is the mean of q and s ?

[CDS-2023 (2)]

- (a) 310 (b) 320
(c) 330 (d) 340

Consider the following for the next ten (10) items that follow:

Mark option (a) if the question can be answered by using one of the statements alone, but cannot be answered using the other statement alone.

Mark option (b) if the question can be answered by using either statements alone.

Mark option (c) if the question can be answered by using both the statements together, but cannot be answered using either statement alone.

Mark option (d) if the question cannot be answered even by using both the statements together.

[CDS-2023 (2)]

42. Question: Is average of the largest and the smallest of 4 given numbers greater than the average of the 4 numbers?

Statement I:

The difference between the largest and the second largest numbers is less than the difference between the second smallest and the smallest of the numbers.

Statement II:

The difference between the largest and the smallest numbers is greater than the difference between the second largest and the second smallest of the numbers.

ANSWER KEY

1.	c	2.	a	3.	a	4.	d	5.	d	6.	c	7.	c	8.	b	9.	a	10.	c
11.	a	12.	b	13.	d	14.	d	15.	b	16.	a	17.	c	18.	c	19.	c	20.	a
21.	c	22.	a	23.	a	24.	a	25.	b	26.	a	27.	c	28.	c	29.	b	30.	c
31.	c	32.	d	33.	b	34.	c	35.	c	36.	d	37.	d	38.	d	39.	a	40.	a
41.	d	42.	a																

CAPF PYQ

1. The average of 7 consecutive odd numbers is M. If the next 3 odd numbers are also included, the average
[CAPF 2017]
(a) Remains unchanged
(b) increase by 1.5
(c) increase by 2
(d) increase by 3
2. Suppose a, b, c d and e are five consecutive odd numbers in ascending order. Consider the following statements:
1. Their average is $(a + 4)$
2. Their average is $(b + 2)$
3. Their average is $(e - 4)$
Which of the statements given above is/are correct?
[CAPF 2018]
(a) 1 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3
3. The average age of the boys in a class is 12 years. The average age of the girls in the class is 11 years. There are 50% more girls than boys in the class. Which one of the following is the average age of the class (in years)?
[CAPF 2020]
(a) 11.2 years
(b) 11.4 years
(c) 11.6 years
(d) 11.8 years
4. The average age of raj and his father is 45 years. If the ages of the father and the grandfather of Raj are respectively two and three times that of Raj, then the age of Raj's grandfather is
[CAPF 2021]
(a) 75 years
(b) 90 years
(c) 81 years
(d) 84 years
5. The average age of father and elder son is 35 years, the average age of father and younger son is 32 years and the average age of the two sons is 17 years. What is the average age of the father and his sons?
[CAPF 2022]
(a) 30 years
(b) 27 years
(c) 28 years
(d) 29 years
6. Suppose A and B can complete a work together in 10 days. If B alone can complete the work in 15 days, then in how many days can A alone finish the work?
[CAPF 2022]
(a) 20 days
(b) 24 days
(c) 25 days
(d) 30 days
7. If the average of the first four of five numbers in decreasing order is 25 and the average of the last four numbers is 20, then what is the difference between the first and the last number?
[CAPF 2022]
(a) 5
(b) 10
(c) 15
(d) 20

ANSWER KEY

1.	(d)	2.	(d)	3.	(b)	4.	(b)	5.	(c)	6.	(d)	7.	(d)						
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PRACTICE SHEET 1

1. A and B together can finish a job in 24 days, while A, B and C together can finish the same job in 8 days. C alone will finish the job in
 (1) 12 days (2) 14 days
 (3) 16 days (4) 24 days
2. A and B together can do a piece of work in 10 days. A alone can do it in 30 days. The time in which B alone can do it is
 (1) 10 days (2) 12 days
 (3) 15 days (4) 20 days
3. A can do a piece of work in 4 hours; B and C can do it in 3 hours. A and C can do it in 2 hours. How long will B alone take to do it?
 (1) 10 hours (2) 12 hours
 (3) 8 hours (4) 24 hours
4. A and B can do a work in 12 days, B and C in 15 days and C and A in 20 days. If A, B and C work together, they will complete the work in :
 (1) 5 days (2) $7\frac{5}{6}$ days
 (3) 10 days (4) $15\frac{2}{3}$ days
5. A and B can do a piece of work in 72 days. B and C can do it in 120 days, A and C can do it in 90 days. In how many days all the three together can do the work?
 (1) 80 days (2) 100 days
 (3) 60 days (4) 150 days
6. A can do a work in 6 days and B in 9 days. How many days will both take together to complete the work?
 (1) 7.5 days (2) 5.4 days
 (3) 3.6 days (4) 3 days
7. A and B can do a piece of work in 10 days, B and C in 15 days and C and A in 20 days. C alone can do the work in :
 (1) 60 days (2) 120 days
 (3) 80 days (4) 30 days
8. P can complete $\frac{1}{4}$ of a work in 10 days, Q can complete 40% of the same work in 15 days, R, completes $\frac{1}{3}$ of the work in 13 days and S, $\frac{1}{6}$ of the work in 7 days. Who will be able to complete the work first?
 (1) P (2) Q
 (3) R (4) S
9. A, B and C can complete a piece of work in 24, 6 and 12 days respectively. Working together, they will complete the same work in
 (1) $\frac{1}{4}$ day (2) $\frac{7}{24}$ day
 (3) $3\frac{3}{7}$ days (4) 4 days
10. A and B together can complete a piece of work in 72 days, B and C together can complete it in 120 days, and A and C together in 90 days. In what time can A alone complete the work?
 (1) 80 days (2) 100 days
 (3) 120 days (4) 150 days
11. A and B together can complete a work in 8 days and B and C together in 12 days. All of the three together can complete the work in 6 days. In how much time will A and C together complete the work?
 (1) 8 days (2) 10 days
 (3) 12 days (4) 20 days
12. A alone can complete a work in 12 days. A and B together can complete it in 8 days. How long will B alone take to complete the work?
 (1) 24 days (2) 18 days
 (3) 16 days (4) 20 days
13. If A and B together can complete a work in 18 days, A and C together in 12 days and B and C together in 9 days, then B alone can do the work in
 (1) 18 days (2) 24 days
 (3) 30 days (4) 40 days
14. A can finish a work in 24 days, B in 9 days and C in 12 days. B and C start the work but

are forced to leave after 3 days. The remaining work was done by A in:

- (1) 5 days (2) 6 days
(3) 10 days (4) $10\frac{1}{2}$ days

15. A and B can together finish a work in 30 days. They worked together for 20 days and then B left. After another 20 days, A finished the remaining work. In how many days A alone can finish the job?

- (1) 50 days (2) 60 days
(3) 48 days (4) 54 days

16. A and B together can complete a work in 8 days. B alone can complete that work in 12 days. B alone worked for four days. After that how long will A alone take to complete the work?

- (1) 15 days (2) 18 days
(3) 16 days (4) 20 days

17. A can do a work in 20 days and B in 40 days. If they work on it together for 5 days, then the fraction of the work that is left is :

- (1) $\frac{5}{8}$ (2) $\frac{8}{15}$
(3) $\frac{7}{15}$ (4) $\frac{1}{10}$

18. A and B working separately can do a piece of work in 10 days and 15 days respectively. If they work on alternate days beginning with A, in how many days will the work be completed?

- (1) 18 days (2) 13 days
(3) 12 days (4) 6 days

19. A can finish a work in 18 days and B can do the same work in half the time taken by A. Then working together what part of the same work they can finish in a day?

- (1) $\frac{1}{6}$ (2) $\frac{2}{5}$
(3) $\frac{1}{9}$ (4) $\frac{2}{7}$

20. A work can be completed by P and Q in 12 days, Q and R in 15 days, R and P in 20 days. In how many days P alone can finish the work?

- (1) 10 days (2) 20 days
(3) 30 days (4) 60 days

21. A and B can do a work in 18 and 24 days respectively. They worked together for 8 days and then A left. The remaining work was finished by B in:

- (1) 5 days (2) $5\frac{1}{3}$ days
(3) 8 days (4) 10 days

22. A can do a piece of work in 12 days and B can do it in 18 days. They work together for 2 days and then A leaves. How long will B take to finish the remaining work?

- (1) 6 days (2) 8 days
(3) 10 days (4) 13 days

23. A and B can do a job in 6 and 12 days respectively. They began the work together but A leaves after 3 days. Then the total number of days needed for the completion of the work is:

- (1) 4 days (2) 5 days
(3) 6 days (4) 9 days

24. A and B can do a piece of work in 28 and 35 days respectively. They began to work together but A leaves after sometime and B completed remaining work in 17 days. After how many days did A leave?

- (1) $14\frac{2}{5}$ days (2) 9 days
(3) 8 days (4) $7\frac{5}{9}$ days

25. A and B can complete a work in 15 days and 10 days respectively. They started doing the work together but after 2 days, B had to leave and A alone completed the remaining work. The whole work was completed in:

- (1) 10 days (2) 8 days
(3) 12 days (4) 15 days

26. A and B can do a piece of work in 20 days and 12 days respectively. A started the work alone and then after 4 days B joined him till the completion of the work. How long did the work last?

- (1) 10 days (2) 20 days
(3) 15 days (4) 6 days

27. A does $\frac{7}{10}$ part of work in 15 days. After that he completes the remaining work in 4 days with the help of B. In how many days will A and B together do the same work?

- (1) $10\frac{1}{3}$ days (2) $12\frac{2}{3}$ days

(3) $13\frac{1}{3}$ days

(4) $8\frac{1}{4}$ days

28. A does $\frac{4}{5}$ of a piece of work in 20 days; He then calls in B and they finish the remaining work in 3 days. How long B alone will take to do whole work?

(1) $37\frac{1}{2}$ days

(2) 37 days

(3) 40 days

(4) 23 days

29. A and B can do a work in 45 days and 40 days respectively. They began the work together but A left after some time and B completed the remaining work in 23 days.

After how many days of the start of the work did A leave?

(1) 10 days

(2) 9 days

(3) 8 days

(4) 5 days

30. A can complete $\frac{1}{3}$ of a work in 5 days and B,

$\frac{2}{5}$ of the work in 10 days. In how many days both A and B together can complete the work?

(1) 10 days

(2) $9\frac{3}{8}$ days

(3) $8\frac{4}{5}$ days

(4) $7\frac{1}{2}$ days

ANSWER KEY

1.	(1)	2.	(3)	3.	(2)	4.	(3)	5.	(3)	6.	(3)	7.	(2)	8.	(2)	9.	(3)	10.	(3)
11.	(1)	12.	(1)	13.	(2)	14.	(3)	15.	(2)	16.	(3)	17.	(1)	18.	(3)	19.	(1)	20.	(3)
21.	(2)	22.	(4)	23.	(3)	24.	(3)	25.	(3)	26.	(1)	27.	(3)	28.	(1)	29.	(2)	30.	(2)

PRACTICE SHEET 2

1. A and B can do a work in 18 and 24 days respectively. They worked together for 8 days and then A left. The remaining work was finished by B in :

(1) 5 days

(2) $5\frac{1}{3}$ days

(3) 8 days

(4) 10 days

2. A can do a piece of work in 12Vdays and B can do it in 18 days. They work together for 2 days and then A leaves. How long will B take to finish the remaining work?

(1) 6 days

(2) 8 days

(3) 10 days

(4) 13 days

3. A and B can do a job in 6 and 12 days respectively. They began the work together but A leaves after 3 days. Then the total number of days needed for the completion of the work is:

(1) 4 days

(2) 5 days

(3) 6 days

(4) 9 days

4. A and B working separately can do a piece of work in 10 days and 15 days respectively. If they work on alternate days beginning with

A, in how many days will the work be completed?

(1) 18 days

(2) 13 days

(3) 12 days

(4) 6 days

5. A and B can do a piece of work in 28 and 35 days respectively. They began to work together but A leaves after sometime and B completed remaining work in 17 days. After how many days did A leave?

(1) $14\frac{2}{5}$ days

(2) 9 days

(3) 8 days

(4) $7\frac{5}{9}$ days

6. A and B can complete a work in 15 days and 10 days respectively. They started doing the work together but after 2 days, B had to leave and A alone completed the remaining work. The whole work was completed in:

(1) 10 days

(2) 8 days

(3) 12 days

(4) 15 days

7. A and B can complete a piece of work in 12 and 18 days respectively. A begins to do the

work and they work alternatively one at a time for one day each. The whole work will be completed in

- (1) $14\frac{1}{3}$ days (2) $15\frac{2}{3}$ days
(3) $16\frac{1}{3}$ days (4) $18\frac{2}{3}$ days

8. A, B and C can complete a work in 10, 12 and 15 days respectively. They started the work together. But A left the work before 5 days of its completion. B also left the work 2 days after A left. In how many days was the work completed?

- (1) 4 days (2) 5 days
(3) 7 days (4) 8 days

9. A can complete a piece of work in 10 days, B in 15 days and C in 20 days. A and C worked together for two days and then A was replaced by B. In how many days, altogether, was the work completed?

- (1) 12 days (2) 10 days
(3) 6 days (4) 8 days

10. A, B and C can do a piece of work in 20, 30 and 60 days respectively. In how many days can A do the work if he is assisted by B and C on every third day?

- (1) 10 days (2) 12 days
(3) 15 days (4) 20 days

11. A and B together can complete a work in 3 days. They start together. But, after 2 days, B left the work. If the work is completed after 2 more days, B alone could do the work in

- (1) 10 days (2) 4 days
(3) 6 days (4) 8 days

12. A can do a piece of work in 20 days and B in 30 days. They work together for 7 days and then both leave the work. Then C alone finishes the remaining work in 10 days. In how many days will C finish the full work?

- (1) 25 days (2) 30 days
(3) 24 days (4) 20 days

13. A and B can do a piece of work in 45 and 40 days respectively. They began the work together but A left after some days and B finished the remaining work in 23 days. A left after

- (1) 6 days (2) 9 days

- (3) 12 days (4) 5 days

14. A and B together can finish a work in 30 days. They worked on it for 20 days and then B left the work. The remaining work was done by A alone in 20 days more. In how many days can A alone finish the work?

- (1) 48 days (2) 50 days
(3) 54 days (4) 60 days

15. Ramesh and Rahman can do a work in 20 and 25 days respectively. After doing collectively for 10 days at the work, they leave the work due to illness and Suresh completes rest of the work in 3 days. How many days Suresh alone can take to complete the whole work?

- (1) 32 days (2) 28 days
(3) 29 days (4) 30 days

16. A can do a piece of work in 10 days and B can do it in 12 days. They work together for 3 days. Then B leaves and A alone continues. 2 days after that C joins and the work is completed in 2 days more. In how many days can C do it, if he works alone?

- (1) 30 days (2) 50 days
(3) 40 days (4) 60 days

17. Pratibha is thrice as efficient as Sonia and is therefore able to finish a piece of work in 60 days less than Sonia. Pratibha and Sonia can individually complete the work respectively in

- (1) 30 days, 60 days
(2) 60 days, 90 days
(3) 30 days, 90 days
(4) 40 days, 120 days

18. If A, B and C can complete a work in 6 days. If A can work twice faster than B and thrice faster than C, then the number of days C alone can complete the work is :

- (1) 33 days (2) 44 days
(3) 22 days (4) 11 days

19. A is twice as good as B and together they finish a piece of work in 16 days. The number of days taken by A alone to finish the work is

- (1) 20 days (2) 21 days
(3) 22 days (4) 24 days

20. A man does double the work done by a boy in the same time. The number of days that 3

men and 4 boys will take to finish a work which can be done by 10 men in 8 days is

(1) 4 (2) 16

(3) $7\frac{3}{11}$ (4) $8\frac{4}{5}$

21. A works twice as fast as B. If B can complete a piece of work independently in 12 days, then what will be the number of days taken by A and B together to finish the work?

(1) 4 (2) 6
(3) 8 (4) 18

22. A can complete a work in 6 days while B can complete the same work in 12 days. If they work together and complete it, the portion of the work done by A is

(1) $\frac{1}{3}$ (2) $\frac{2}{3}$
(3) $\frac{1}{4}$ (4) $\frac{1}{2}$

23. A can do $\frac{1}{2}$ of a piece of work in 5 days, B can do $\frac{3}{5}$ of the same work in 9 days and C can do $\frac{2}{3}$ of that work in 8 days. In how many days can three of them together o the work?

(1) 3 days (2) 5 days
(3) $4\frac{1}{2}$ days (4) 4 days

24. If 28 men complete $\frac{7}{8}$ of a piece of work in a week, then the number of men, who must be engaged to get the remaining work completed in another week, is

(1) 5 (2) 6
(3) 4 (4) 3

25. A can complete $\frac{1}{3}$ of a work in 5 days and B, $\frac{2}{5}$ of the work in 10 days. In how many days both A and B together can complete the work?

(1) 10 days (2) $9\frac{3}{8}$ days

(3) $8\frac{4}{5}$ days

(4) $7\frac{1}{2}$ days

26. A, B and C together can do a piece of work in 40 days. After working with B and C for 16 days, A leaves and then B and C complete the remaining work in 40 days more. A alone could do the work in

(1) 80 days (2) 90 days
(3) 100 days (4) 120 days

27. A and B together can complete a work in 8 days. B alone can complete that work in 12 days. B alone worked for four days. After that how long will A alone take to complete the work?

(1) 15 days (2) 18 days
(3) 16 days (4) 20 days

28. A and B can complete a piece of work in 12 and 18 days respectively. A begins to do the work and they work alternatively one at a time for one day each. The whole work will be completed in

(1) $14\frac{1}{3}$ days (2) $15\frac{2}{3}$ days
(3) $16\frac{1}{3}$ days (4) $18\frac{2}{3}$ days

29. A and B can separately complete a piece of work in 20 days and 30 days respectively. They worked together for some time, then B left the work. If A completed the rest of the work in 10 days, then B worked for

(1) 6 days (2) 8 days
(3) 12 days (4) 16 days

30. A and B together can complete a work in 12 days. A alone can complete in 20 days. If B does the work only half a day daily, then in how many days A and B together will complete the work?

(1) 10 days (2) 20 days
(3) 11 days (4) 15 days

ANSWER KEY

1.	(2)	2.	(4)	3.	(3)	4.	(3)	5.	(3)	6.	(3)	7.	(1)	8.	(3)	9.	(4)	10.	(3)
11.	(3)	12.	(3)	13.	(2)	14.	(4)	15.	(4)	16.	(3)	17.	(3)	18.	(1)	19.	(4)	20.	(2)
21.	(1)	22.	(2)	23.	(4)	24.	(3)	25.	(2)	26.	(3)	27.	(3)	28.	(1)	29.	(1)	30.	(4)

PRACTICE SHEET 3

1. 39 persons can repair a road in 12 days working 5 hours a day. In how many days will 30 persons working 6 hours a day complete the work?
 (1) 10 days (2) 13 days
 (3) 14 days (4) 15 days
2. If 72 men can build a wall of 280 m length in 21 days, how many men could take 18 days to build a similar type of wall of length 100 m?
 (1) 30 (2) 10
 (3) 18 (4) 28
3. A wall of 100 metres can be built by 7 men or 10 women in 10 days. How many days will 14 men and 20 women take to build a wall of 600 metres ?
 (1) 15 (2) 20
 (3) 25 (4) 30
4. 5 persons can prepare an admission list in 8 days working 7 hours a day. If 2 persons join them so as to complete the work in 4 days, they need to work per day for :
 (1) 10 hours (2) 9 hours
 (3) 12 hours (4) 8 hours
5. 4 mat-weavers can weave 4 mats in 4 days. At the same rate how many mats would be woven by 8 mat-weavers in 8 days ?
 (1) 4 (2) 8
 (3) 12 (4) 16
6. 10 men working 6 hours a day can complete a work in 18 days. How many hours a day must 15 men work to complete the same work in 12 days ?
 (1) 6 days (2) 10 days
 (3) 12 days (4) 15 days
7. 'x' number of men can finish a piece of work in 30 days. If there were 6 men more, the work could be finished in 10 days less. The original number of men is
 (1) 6 (2) 10
 (3) 12 (4) 15
8. Some carpenters promised to do a job in 9 days but 5 of them were absent and remaining men did the job in 12 days. The original number of carpenters was
 (1) 24 (2) 20
 (3) 16 (4) 18
9. Working 8 hours a day, Anu can copy a book in 18 days. How many hours a day should she work so as to finish the work in 12 days?
 (1) 12 hours (2) 10 hours
 (3) 11 hours (4) 13 hours
10. If the work done by $(x-1)$ men in $(x+1)$ days is to the work done by $(x+2)$ men in $(x-1)$ days are in the ratio 9 : 10, then the value of x is equal to :
 (1) 5 (2) 6
 (3) 7 (4) 8
11. 30 men can repair a road in 18 days. They are joined by 6 more workers. Now the road can be repaired in
 (1) 14 days (2) 15 days
 (3) 16 days (4) 17 days
12. If 6 men and 8 boys can do a piece of work in 10 days and 26 men and 48 boys can do the same in 2 days, then the time taken by 15 men and 20 boys to do the same type of work will be :
 (1) 5 days (2) 4 days
 (3) 6 days (4) 7 days
13. 5 men can do a piece of work in 6 days while 10 women can do it in 5 days. In how many days can 5 women and 3 men do it ?
 (1) 4 days (2) 5 days
 (3) 6 days (4) 8 days
14. If 3 men or 6 women can do piece of work in 16 days, in how many days can 12 men and 8 women do the same piece of work?
 (1) 4 days (2) 5 days
 (3) 3 days (4) 2 days
15. A man, a woman and a boy can complete a job in 3, 4 and 12 days respectively. How many boys must assist 1 man and 1 woman to complete the job in $1/4$ of a day?
 (1) 1 (2) 4
 (3) 19 (4) 41
16. If 16 men or 20 women can do a piece of work in 25 days. In what time will 28 men

- and 15 women do it?
 (1) 100/7 days (2) 100/3 days
 (3) 75/4 days (4) 10 days
- 17.** If 5 men or 8 women can do a piece of work in 12 days, how many days will be taken by 2 men and 4 women to do the same work?
 (1) 15 days (2) 27/2 days
 (3) 40/3 days (4) 10 days
- 18.** If 3 men or 4 women can plough a field in 43 days, how long will 7 men and 5 women take to plough it ?
 (1) 10 days (2) 11 days
 (3) 9 days (4) 12 days
- 19.** 6 men or 12 women can do a piece of work in 20 days. In how many days can 8 men and 16 women do twice as big as this work ?
 (1) 2 days (2) 5 days
 (3) 15 days (4) 10 days
- 20.** A man, a woman and a boy can together complete a piece of work in 3 days. If a man alone can do it in 6 days and a boy alone in 18 days, how long will a woman alone take to complete the work?
 (1) 9 days (2) 21 days
 (3) 24 days (4) 27 days
- 21.** 3 men or 5 women can do a work in 12 days. How long will 6 men and 5 women take to finish the work?
 (1) 20 days (2) 10 days
 (3) 4 days (4) 15 days
- 22.** Two workers A and B are engaged to do a piece of work. A working alone would take 8 hours more to complete the work than when work together. If B worked alone, would take 4 and half an hours more than when work together. The time required to finish the work together is
 (1) 5 hours (2) 4 hours
 (3) 8 hours (4) 6 hours
- 23.** A company employed 200 workers to complete a certain work in 150 days. If only one fourth of the work has been done in 50 days, then in order to complete the whole work in time, the number of additional workers to be employed was
 (1) 100 (2) 300
 (3) 600 (4) 200
- 24.** A contractor was engaged to construct a road in 16 days. After working for 12 days with 20 labours it was found that only 5/8th of the road had been constructed. To complete the work in stipulated time the number of extra labours required is :
 (1) 18 (2) 10
 (3) 12 (4) 16
- 25.** Janardan completes 2/3 of his work in 10 days. Time he will take to complete 3/5 of the same work, is
 (1) 8 days (2) 6 days
 (3) 9 days (4) 4 days
- 26.** Two workers A and B working together completed a job in 5 days. If A worked twice as efficiently as he actually did and B worked 1/3 as efficiently as he actually did, the work would have been completed in 3 days. To complete the job alone, A would require
 (1) 26/5 days (2) 25/4 days
 (3) 15/2 days (4) 35/4 days
- 27.** A man undertakes to do a certain work in 150 days. He employs 200 men. He finds that only a quarter of the work is done in 50 days. The number of additional men that should be appointed so that the whole work will be finished in time is :
 (1) 75 (2) 100
 (3) 125 (4) 50
- 28.** A contractor undertook to finish a certain work in 124 days and employed 120 men. After 64 days, he found that he had already done 2/3 of the work. How many men can be discharged now so that the work may finish in time ?
 (1) 48 (2) 56
 (3) 40 (4) 50
- 29.** If 7 men working 7 hrs a day for each of 7 days produce 7 units of work, then the units of work produced by 5 men working 5 hrs a day for each of 5 days is
 (1) 25/343 (2) 125/49
 (3) 49/125 (4) 343/25
- 30.** Seventy-five men are employed to lay down a railway line in 3 months. Due to certain emergency conditions, the work was to be finished in 18 days. How many more men should be employed to complete the work in the desired time ?

(1) 300

(2) 325

(3) 350

(4) 375

ANSWER KEY

1.	(2)	2.	(1)	3.	(1)	4.	(1)	5.	(4)	6.	(1)	7.	(3)	8.	(2)	9.	(1)	10.	(4)
11.	(2)	12.	(2)	13.	(2)	14.	(3)	15.	(4)	16.	(4)	17.	(3)	18.	(4)	19.	(3)	20.	(1)
21.	(3)	22.	(4)	23.	(1)	24.	(4)	25.	(3)	26.	(2)	27.	(2)	28.	(2)	29.	(2)	30.	(1)

PRACTICE SHEET 4

- A and B can complete a piece of work in 15 days and 10 days respectively. They contracted to complete the work for 30,000. The share of A in the contracted money will be
 (1) 18,000 (2) 16,500
 (3) 12,500 (4) 12,000
- A man and a boy received 800 as wages for 5 days for the work they did together. The man's efficiency in the work was three times that of the boy. What are the daily wages of the boy?
 (1) 76 (2) 56
 (3) 44 (4) 40
- A daily-wage labourer was engaged for a certain number of days for 5,750; but being absent on some of those days he was paid only 5,000. What was his maximum possible daily wage?
 (1) 125 (2) 250
 (3) 375 (4) 500
- A, B and C completed a work costing 1,800. A worked for 6 days, B for 4 days and C for 9 days. If their daily wages are in the ratio of 5 : 6 : 4, how much amount will be received by A?
 (1) 800 (2) 600
 (3) 900 (4) 750
- A labourer was appointed by a contractor on the condition that he would be paid 75 for each day of his work but would be fined at the rate of 15 per day for his absence, apart from losing his wages. After 20 days, the contractor paid the labourer 1140. The number of days the labourer abstained from work was
 (1) 3 (2) 5 (3) 4 (4) 2
- Two men undertook to do a job for 1400. One of them can do it alone in 7 days, and the other in 8 days. With the assistance of a boy they together completed the work in 3 days. How much money will the boy get?
 (1) 300 (2) 325
 (3) 275 (4) 250
- If 5 men or 7 women can earn 5,250 per day, how much would 7 men and 13 women earn per day?
 (1) 11,600 (2) 11,700
 (3) 16,100 (4) 17,100
- 2 men and 1 woman together can complete a piece of work in 14 days, while 4 women and 2 men together can do it in 8 days. If a man gets 600 per day, how much should a woman get per day?
 (1) 400 (2) 450
 (3) 480 (4) 360
- Two men undertake a job for 960. They can complete it in 16 days and 24 days respectively. They work along with a third man and take 8 days to complete it. Then the share of the third man should be
 (1) 155 (2) 165
 (3) 160 (4) 150
- If there is a reduction in the number of workers in a factory in the ratio 15 : 11 and an increment in their wages in the ratio 22 : 25, then the ratio by which the total wages of the workers should be decreased is
 (1) 6 : 5 (2) 5 : 6
 (3) 3 : 7 (4) 3 : 5

- 11.** Stanie and Paul take a piece of work for 28,800. One alone could do it in 36 days, the other in 48 days. With the assistance of an expert, they finish it in 12 days. How much remuneration the expert should get ?
 (1) 10000 (2) 18000
 (3) 16000 (4) 12000
- 12.** A and B were assigned to do a job for an amount of 1,200. A alone can do it in 15 days, while B can do it in 12 days. With the help of C, they can finish in 5 days. The share of amount that C earns is
 (1) 300 (2) 400
 (3) 500 (4) 600
- 13.** A sum of money is sufficient to pay A's wages for 21 days and B's wages for 28 days. The same money is sufficient to pay the wages of both for :
 (1) 12 days (2) 14 days 14
 (3) 24 days (4) 12 days 12
- 14.** A can do a piece of work in 12 days while B alone can do it in 15 days. With the help of C they can finish it in 5 days. If they are paid 960 for the whole work how much money A gets ?
 (1) 480 (2) 240
 (3) 320 (4) 400
- 15.** A, B and C together earn 150 per day while A and C together earn 94 and B and C together earn 76. The daily earning of 'C' is
 (1) 56 (2) 20
 (3) 34 (4) 75
- 16.** Two pipes A and B can fill a tank in 20 minutes and 30 minutes respectively. If both pipes are opened together, the time taken to fill the tank is :
 (1) 50 minutes (2) 12 minutes
 (3) 25 minutes (4) 15 minutes
- 17.** A tap can empty a tank in one hour. A second tap can empty it in 30 minutes. If both the taps operate simultaneously, how much time is needed to empty the tank?
 (1) 20 minutes (2) 30 minutes
 (3) 25 minutes (4) 15 minutes
- 18.** A cistern can be filled with water by a pipe in 5 hours and it can be emptied by a second pipe in 4 hours. If both the pipes are opened when the cistern is full, the time in which it will be emptied is :
 (1) 9 hours (2) 18 hours
 (3) 20 hours (4) 12 hours
- 19.** Two pipes A and B can separately fill a cistern in 60 minutes and 75 minutes respectively. There is a third pipe in the bottom of the cistern to empty it. If all the three pipes are simultaneously opened, then the cistern is full in 50 minutes. In how much time the third pipe alone can empty the cistern?
 (1) 110 minutes (2) 100 minutes
 (3) 120 minutes (4) 90 minutes
- 20.** Two pipes can fill a cistern in 3 hours and 4 hours respectively and a waste pipe can empty it in 2 hours. If all the three pipes are kept open, then the cistern will be filled in :
 (1) 5 hours (2) 8 hours
 (3) 10 hours (4) 12 hours
- 21.** 12 pumps working 6 hours a day can empty a completely filled reservoir in 15 days. How many such pumps working 9 hours a day will empty the same reservoir in 12 days ?
 (1) 15 (2) 9
 (3) 10 (4) 12
- 22.** Three pipes P, Q and R can separately fill a cistern in 4, 8 and 12 hours respectively. Another pipe S can empty the completely filled cistern in 10 hours. Which of the following arrangements will fill the empty cistern in less time than others?
 (1) Q alone is open.
 (2) P and S are open.
 (3) P, R and S are open.
 (4) P, Q, R, S are open all together
- 23.** Two pipes can fill a cistern separately in 10 hours and 15 hours. They can together fill the cistern in
 (1) 6 hours (2) 7 hours
 (3) 8 hours (4) 9 hours
- 24.** Three taps A, B and C together can fill an empty cistern in 10 minutes. The tap A alone can fill it in 30 minutes and the tap B alone in 40 minutes. How long will the tap C alone take to fill it ?
 (1) 16 minutes (2) 24 minutes
 (3) 32 minutes (4) 40 minutes
- 25.** One tap can fill a water tank in 40 minutes and another tap can make the filled tank empty in 60 minutes. If both the taps are

open, in how many hours will the empty tank be filled ?

- (1) 2 hours (2) 2.5 hours
(3) 3 hours (4) 3.5 hours

26. A tank can be filled by pipe A in 2 hours and pipe B in 6 hours. At 10 A.M. pipe A was opened. At what time will the tank be filled if pipe B is opened at 11 A.M.?

- (1) 12.45 A.M. (2) 5 P.M.
(3) 11.45 A.M. (4) 12 P.M.

27. Two pipes, P and Q, together can fill a cistern in 20 minutes and P alone can in 30 minutes. Then Q alone can fill the cistern in

- (1) 62 minutes (2) 60 minutes
(3) 61 minutes (4) 51 minutes

28. Two pipes A and B can fill a cistern in 3 hours and 5 hours respectively. Pipe C can empty in 2 hours. If all the three pipes are

open, in how many hours the cistern will be full?

- (1) can't be filled (2) 10 hours
(3) 15 hours (4) 30 hours

29. Two pipes can fill a tank with water in 15 and 12 hours respectively and a third pipe can empty it in 4 hours. If the pipes be opened in order at 8, 9 and 11 a.m. respectively, the tank will be emptied at

- (1) 11 : 40 a.m. (2) 12 : 40 p.m.
(3) 1 : 40 p.m. (4) 2 : 40 p.m.

30. A Boy and girl together fill a cistern with water. The boy pours 4 litres of water every 3 minutes and the girl pours 3 litres every 4 minutes. How much time will it take to fill 100 litres of water in the cistern ?

- (1) 36 minutes (2) 42 minutes
(3) 48 minutes (4) 60 minutes

ANSWER KEY

1.	(4)	2.	(4)	3.	(2)	4.	(2)	5.	(3)	6.	(3)	7.	(4)	8.	(1)	9.	(3)	10.	(1)
11.	(4)	12.	(1)	13.	(4)	14.	(4)	15.	(2)	16.	(2)	17.	(1)	18.	(3)	19.	(2)	20.	(4)
21.	(3)	22.	(4)	23.	(1)	24.	(2)	25.	(1)	26.	(3)	27.	(2)	28.	(4)	29.	(4)	30.	(3)

CDS PYQ

1. If 3 men and 4 boys can do a piece of work in 8 days then 6 men and 8 boys can do the same work in?

[CDS 2013 (I)]

- (a) 2 days (b) 4 days
(c) 6 days (d) 18 days

2. A mason can build a tank in 12 h. After working for 6 h, he took the help of a boy and finished the work in another 5 h. The time that the boy will take alone to complete the work is

[CDS 2013 (I)]

- (a) 30 h (b) 45 h
(c) 60 h (d) 64 h

3. X can complete a job in 12 days. If X and Y work together, they can complete the job in $6\frac{2}{3}$ days. Y alone can complete the job in

[CDS 2013 (I)]

- (a) 10 days (b) 12 days
(c) 15 days (d) 18 days

4. Pipe A can fill a tank in 10 min and pipe B can empty it in 15 min. If both the pipes are opened in an empty tank, the time taken to make it full is

[CDS 2013 (I)]

- (a) 20 min (b) 25 min
(c) 30 min (d) None of these

5. 4 goats or 6 sheep's can graze a field in 50 days. 2 goats and 9 sheep's can graze the field in

[CDS 2013 (II)]

- (a) 100 days (b) 75 days
(c) 50 days (d) 25 days

6. A can finish a work in 15 days, B in 20 days and C in 25 days. All these three worked together and earned Rs 4700. The share of C is

[CDS 2013 (II)]

- (a) 1200 (b) 1500
(c) 1800 (d) 2000

7. 20 workers working for 5 h per day complete work in 10 days. If 25 workers are employed to work 10 h per day, what is the time required to complete the work?

[CDS 2013 (II)]

- (a) 4 days (b) 5 days
(c) 6 days (d) 8 days

8. 18 men can earn Rs 360 in 5 days. How much money will 15 men earn in 9 days?

[CDS 2013 (II)]

- (a) 600 (b) 540
(c) 480 (d) 360

9. 2 men and 1 woman can complete a piece of work in 14 days, while 4 women and 2 men can do the same work in 8 days. If a man gets Rs 90 per day, what should be the wages per day of a woman?

[CDS 2013 (II)]

- (a) 48 (b) 60
(c) 72 (d) 135

10. A can do a piece of work in 4 days and B can complete the same work in 12 days. What is the number of days required to do the same work together?

[CDS 2013 (II)]

- (a) 2 days (b) 3 days
(c) 4 days (d) 5 days

11. X can do a work in 16 days in how many days will the work be completed by Y if efficiency of Y is 60% more than that of X?

[CDS 2013 (II)]

- (a) 10 days (b) 12 days
(c) 25 days (d) 30 days

12. 15 men complete a work in 16 days if 24 men are employed then the time required to complete that work will be:

[CDS 2014 (I)]

- (a) 7 days (b) 8 days
(c) 10 days (d) 12 days

13. A, B and C can do a piece of work individually in 8, 12 and 15 days, respectively. A and B start working but A quits after working for 2 days. After this, C joins B till the completion of work. In how many days will the work be completed?

[CDS 2014 (I)]

- (a) $5\frac{8}{9}$ days (b) $4\frac{6}{7}$ days
(c) $6\frac{7}{13}$ days (d) $3\frac{3}{4}$ days

14. A, B and C can do a piece of work individually in 8, 12 and 15 days, respectively. A and B start working but A quits after working for 2 days. After this, C joins B till the completion of work. In how many days will the work be completed?

[CDS 2014 (II)]

- (a) $3\frac{8}{9}$ days (b) $5\frac{8}{9}$ days
(c) $5\frac{2}{3}$ days (d) $6\frac{1}{18}$ days

15. Pipe A can fill a tank in 10 min. and pipe B can empty it in 15 min. If both the pipes are opened in an empty tank the time taken to make it full is:

[CDS 2014 (II)]

- (a) 20min (b) 25min
(c) 30min (d) None of these

16. A is thrice as efficient as B and hence completes a work in 40 days less than the number of days taken by B. What will be the number of days taken by both of them when working together?

[CDS 2014 (II)]

- (a) 22.5 days (b) 15 days
(c) 20 days (d) 18 days

17. If 10 persons can dig 8 ft trench in 12 days, then how many days will 8 persons take to dig 6 ft. trench?

[CDS 2014 (II)]

- (a) 10 days (b) 10.25 days
(c) 11 days (d) 11.25 days

18. The efficiency of P is twice that of Q, whereas the efficiency of P and Q together is three times that of R. If P, Q and R work together on a job, in what ratio should they share their earnings?

[CDS 2015 (I)]

- (a) 2:1:1 (b) 4:2:1
(c) 4:3:2 (d) 4:2:3

19. A and B are two taps which can fill a tank individually in 10 min and 20 min, respectively. However, there is a leakage at

the bottom, which can empty a filled tank in 40 min. If the tank is empty initially, then how much time will both the taps take to fill the tank with leakage?

[CDS 2015 (II)]

- (a) 2 min (b) 4 min
(c) 5 min (d) 8 min

20. If 4 men working 4h per day for 4 days complete 4 units of work, then how many units of work will be completed by 2 men working for 2 h per day in 2 days?

[CDS 2015 (II)]

- (a) 2 (b) 1
(c) $\frac{1}{2}$ (d) $\frac{1}{8}$

21. If m persons can paint a house in d days, then how many days will it take for (m + 2) persons to paint the same house?

[CDS 2015 (II)]

- (a) $md + 2$ (b) $md - 2$
(c) $\frac{m+2}{md}$ (d) $\frac{md}{m+2}$

22. Two pipes A and B can fill a tank in 60 min and 75 min, respectively. There is also an outlet C. If A, B and C are opened together, then the tank is full in 50 min. How much time will be taken by C to empty the full tank?

[CDS 2016 (I)]

- (a) 100 min (b) 110 min
(c) 120 min (d) 125 min

23. A can do 50% more work than B in the same time. B alone can do a piece of work in 30 hrs. B starts working and had already worked for 12 hours when A joins him. How many hours should B and A work together to complete the remaining work?

[CDS 2016 (II)]

- (a) 6 hours (b) 12 hours
(c) 4.8 hours (d) 7.2 hours

24. A tank can be filled by pipe X in 2 hours and pipe Y in 6 hours. At 10 a.m. Pipe X was opened. At what time will the tank be filled if pipe Y is opened at 11 a.m.?

[CDS 2016 (II)]

- (a) 12:45 hours (b) 5:00 p.m.
(c) 11:45 a.m. (d) 11:50 a.m.

25. If 15 men take 21 days of 8 hours each to do a piece of work, then what is the number of days of 6 hours each that 21 women would

take, if three women would do as much work as 2 men ?

[CDS 2017 (I)]

- (a) 18 (b) 20
(c) 25 (d) 30

26. A and B working together can finish a piece of work in 12 days while B alone can finish it in 30 days. In how many days A can alone finish the work?

[CDS 2017 (I)]

- (a) 18 days (b) 20 days
(c) 24 days (d) 25 days

27. If 5 men can do a piece of work in 10 days and 12 women can do the same work in 15 days, the number of days required to complete the work by 5 men and 6 women

[CDS 2017 (I)]

- (a) $7\frac{1}{2}$ days (b) 8 days
(c) $9\frac{1}{2}$ days (d) 12 days

28. 30 men can complete a job in 40 days. However, after 24 days some men left the job the remaining people took another 40 days to complete the job the number of men who left the job is

[CDS 2017 (II)]

- (a) 24 (b) 18
(c) 12 (d) 6

29. A tap can fill a tube in 10 hours. After opening the tap for 5 hours it was found that a small outlet at the bottom of the tub was open and water was leaking through it. It was then immediately closed. It took 7 hours to fill the tub after closing the outlet. What time will be taken by the outlet to empty the full tub of water?

[CDS 2017 (II)]

- (a) 35 hrs (b) 25 hrs
(c) 20 hr (d) 17 hrs

30. To maintain 8 cows for 60 days, a milkman has to spend Rs 6400. To maintain 5 cows for n days, he has to spend Rs 4800. What is the value of n ?

[CDS 2017 (II)]

- (a) 46 days (b) 50 days
(c) 58 days (d) 72 days

31. 4 goats or 6 sheep can graze a field in 50 days. 2 goats and 3 sheep will graze it in:

[CDS 2017 (II)]

- (a) 200 days (b) 150 days
(c) 100 days (d) 50 days

32. A work when done by 10 women is completed in 12 days. The same work can be completed in 8 days when done by 5 men. How many days will it take to complete when 6 women and three men are employed to perform the same job?

[CDS 2018 (I)]

- (a) 12 (b) 10
(c) 8 (d) 5

33. A man undertakes to do a certain work in 150 days he employs 200 man he finds that only a quarter of the work is done in 50 days. How many additional men should he employs so that the whole work is finished in the time?

[CDS 2018 (I)]

- (a) 75 (b) 85
(c) 100 (d) 120

34. If 5 tractors can plough 5 hectares of land in 5 days. then what is the number of tractors required to plough 100 hectares in 50 days

[CDS 2018 (I)]

- (a) 100 (b) 20
(c) 10 (d) 5

35. A water tank has been fitted with two taps P and Q and a drain pipe R. Taps P and Q fill at the rate of 12 litre per minute and 10 litre per minute respectively.

Consider the following statements S_1 , S_2 and S_3

S_1 : pipe R drains out at the rate of 6 litre per minutes

S_2 : if both the taps and the drain pipes are opened simultaneously then the water tank is filled in 5 hours 45 minutes

S_3 : Pipe R drains out (fully) the field tank in 15 hours 20 minutes.

To know what is the capacity of the tank which one of the following is correct ?

[CDS 2018 (I)]

- (a) S_2 is only sufficient
(b) S_1 , S_2 and S_3 are necessary
(c) any two out of S_1 , S_2 and S_3 are sufficient
(d) none of the above

36. Two workers A and B working together completed a job in 5 days had A work twice

as efficiently as he actually did and B worked one-third as efficiency as he actually did the work would have completed in 3 days. In how many days could a alone completed the job

[CDS 2018 (II)]

- (a) $3\frac{1}{2}$ days (b) $4\frac{1}{6}$ days
(c) $5\frac{1}{2}$ days (d) $6\frac{1}{4}$ days

37. A is thrice as good a workman as B and takes 10 days less to do a piece of work than B takes. the number of days taken by B alone to finish the work is

[CDS 2018 (II)]

- (a) 12 (b) 15
(c) 20 (d) 30

38. 12 men work 8 hr per day and require 10 days to build a wall, if 8 men are available, how many hours per day must they work to finish the work in 8 days

[CDS 2018 (II)]

- (a) 10 (b) 12
(c) 15 (d) 18

39. 12 person can paint 10 identical rooms in 16 days. In how many days can 8 persons paint 20 such rooms?

[CDS 2018 (II)]

- (a) 12 (b) 24
(c) 36 (d) 48

40. X and Y together can finish a job in 6 days, X can alone do the same job in 12 days. how long will we alone take to do the same job

[CDS 2018 (II)]

- (a) 16 days (b) 12 days
(c) 10 days (d) 8 days

41. A field can be reaped by 12 men or 18 women in 14 days. In how many days can 8 men and 16 women reap it?

[CDS 2019 (I)]

- (a) 26 days (b) 24 days
(c) 9 days (d) 8 days

42. Tushar takes 6 hours to complete a piece of work, while Amar completed the same work in 10 hours. if both of them work together then what is the time required to complete the work ?

[CDS 2019 (I)]

- (a) 3 h (b) 3 h 15 min
(c) 3 h 30 min (d) 3 hr 45 min.

43. A stock of food grains is enough for 240 men for 48 days. How long will the same stock last for 160 men?

[CDS 2019 (II)]

- (a) 72 days (b) 64 days
(c) 60 days (d) 54 days

44. Two taps X and Y are fixed to a water tank, if only X is opened, it drains out the full tank of water in 20 min. If both X and Y are opened, then they drain out the full tank of water in 15 min. If only Y is opened, how long does it take to drain out the full tank of water?

[CDS 2019 (II)]

- (a) 30min (b) 45min
(c) 60 min (d) 90 min

45. If the ratio of the work done by $(x + 2)$ workers in $(x - 3)$ days to the work done by $(x + 4)$ workers in $(x - 2)$ days is 3:4, then what is the value of x ?

[CDS 2019 (II)]

- (a) 8 (b) 10
(c) 12 (d) 15

46. If 6 men and 8 women can do a piece of work in 10 days and 13 men and 24 women can do the same work in 4 days, then what is the ratio of daily work done by a man to that of a woman?

[CDS 2020 (I)]

- (a) 2:1 (b) 1:2
(c) 4:3 (d) 3:4

47. In a water tank there are two outlets. It takes 20 min to empty the tank if both the outlets are opened. If the first outlet is opened, the tank is emptied in 30 min. What is the time taken to empty the tank by second outlet?

[CDS 2020 (I)]

- (a) 30 min (b) 40 min
(c) 50 min (d) 60 min

48. If x men working x hours per day can do x units of work in x days, then y men working y hours per day would be able to do k units of work in y days. What is the value of k ?

[CDS 2020 (II)]

- (a) x^2y^{-3} (b) x^3y^{-2}
(c) y^2x^{-3} (d) y^3x^{-2}

49. If 20 persons can clean 20 floors in 20 days, then how many days can 16 persons clean 16 floors?

[CDS 2021 (I)]

- (a) 25 days (b) 24 days
(c) 20 days (d) 16 days

50. Let the work done by $(x - 1)$ men in $(x + 1)$ days be y . Let the work done by $(x + 2)$ men in $(x - 1)$ days be z . If $y:z = 9:10$, then what is the value of x ?

[CDS 2021 (I)]

- (a) 8 (b) 9
(c) 10 (d) 12

51. 20 men are supposed to complete a work in 10 days. After working for 5 days, they realize that only one-fourth of the work is done. How many more men they need to employ to finish the work on time?

[CDS 2021 (II)]

- (a) 40 (b) 30
(c) 20 (d) 15

52. Let work done by $(3n - 1)$ men in $(2n + 1)$ days be x and work done by $(3n + 1)$ men in $(4n - 3)$ days be y . If $x : y = 6 : 11$, then what is the value of n ?

[CDS 2021 (II)]

- (a) 6 (b) 7
(c) 8 (d) 9

53. Sheela can stitch a suit in 2 days, while Meena can stitch a suit in $1\frac{1}{2}$ days. How many days will both take in stitching 30 suits?

[CDS 2021 (II)]

- (a) 32 days (b) 33 days
(c) 35 days (d) 40 days

54. 8 men or 12 women can do a piece of work in 24 days. In how many days can the work be done by 8 men and 12 women?

[CDS 2022 (I)]

- (a) 12 days
(b) 18 days
(c) 24 days
(d) cannot be determined due to insufficient data

55. X and Y can do a piece of work in 45 days and 40 days respectively. They begin to work together, but X leaves after n days and then

Y completes and remaining work in 23 days.
What is n equal to?

[CDS 2022 (II)]

- (a) 8 (b) 9
(c) 10 (d) 12

56. 480 persons working 10 hours per day complete one-fourth of work in 10 days. How many additional persons are to be employed in order to complete the remaining work in 20 days, working 8 hours per day?

[CDS 2022 (II)]

- (a) 400 (b) 420
(c) 480 (d) 500

57. 12 women and 16 men can do a piece of work in 5 days. 13 women and 24 men can do it in 4 days. How long will 25 women and 50 men take to do it?

[CDS 2022 (II)]

- (a) 1 day (b) 2 days
(c) 3 days (d) 4 days

58. If 17 women and 24 men can do a piece of work in 5 days and 12 women and 23 men can do it in 6 days, then which one of the following is correct?

[CDS - 2023 (1)]

- (a) efficiency of 13 women = efficiency of 18 men
(c) efficiency of 11 women = efficiency of 16 men
(c) efficiency of 13 women = efficiency of 17 men
(d) efficiency of 11 women = efficiency of 15 men

59. Three taps A, B and C together can fill a tank in 6 hours. Tap C alone can fill the tank in 12 hours. To fill the tank, when it is empty, all the three taps are started together. After working t hours, tap C is closed and the tank is filled in 8 more hours. What is t equal to?

[CDS - 2023 (1)]

- (a) 1 (b) 2
(c) 4 (d) 6

60. A, B and C can complete a work in x, $1.5x$ and $2x$ days respectively. If they complete the work together, in what ratio should they be paid?

[CDS - 2023 (1)]

- (a) 2:3:4 (b) 6:4:3
(c) 3:2:1 (d) 4:3:2

61. A can do a certain work at twice the speed of B. Further, B can do the same work at 1.5 times the speed of C. All of them together can finish the work in 12 days. In how many days can C alone finish the work?

[CDS - 2023 (1)]

- (a) 36 days (b) 45 days
(c) 60 days (d) 66 days

62. If A and B can finish a work in 10 days, B and C can finish the same work in 12 days, C and A can finish the same work in 15 days; then in how many days can A, B and C together finish half of the work?

[CDS-2023 (2)]

- (a) 8 days (b) 5 days
(c) 4 days (d) 3 days

63. Three persons A, B and C together can do a piece of work in 36 days. A and B together can do five times as much work as C alone, B and C together can do as much work as A alone. If A and C together can do n times as much work as B alone, then what is the value of n?

[CDS-2023 (2)]

- (a) 1.5 (b) 2
(c) 2.5 (d) 3

64. p number of men can finish a piece of work in q days. If there are 50% more men, then the work will be finished 12 days earlier. What is the value of q?

[CDS-2024 (1)]

- (a) 48
(b) 40
(c) 36
(d) Cannot be determined due to insufficient data

65. If the work done by x men in $(x + 1)$ days is equal to the work done by $(x + 5)$ men in $(x - 2)$ days, then what is the value of x?

[CDS-2024 (1)]

- (a) 5 (b) 6
(c) 7 (d) 8

66. 50 men can complete a work in 40 days. They being the work together but a batch of 5 men left after reach period of 10 days. What is the time to complete the work?

[CDS-2024 (1)]

- (a) 45 days (b) 50 days
(c) 55 days (d) 60 days

67. A, B, C, D can complete a work in 3, 6, 9, 12 hours respectively. Further, only one person can work a time in each hour and nobody can work for two consecutive hours. It is not necessary to engage all. What is the

minimum number of hours that they will take to finish the work?

[CDS-2024 (1)]

- (a) 36/25 (b) 12/5
(c) 4 (d) 2

ANSWER KEY

1.	(b)	2.	(c)	3.	(c)	4.	(c)	5.	(d)	6.	(a)	7.	(a)	8.	(b)	9.	(b)	10.	(b)
11.	(a)	12.	(c)	13.	(a)	14.	(b)	15.	(c)	16.	(b)	17.	(d)	18.	(a)	19.	(d)	20.	(c)
21.	(d)	22.	(a)	23.	(d)	24.	(c)	25.	(d)	26.	(b)	27.	(a)	28.	(c)	29.	(b)	30.	(d)
31.	(d)	32.	(a)	33.	(c)	34.	(c)	35.	(c)	36.	(d)	37.	(b)	38.	(c)	39.	(d)	40.	(b)
41.	(c)	42.	(d)	43.	(a)	44.	(c)	45.	(b)	46.	(a)	47.	(d)	48.	(d)	49.	(c)	50.	(a)
51.	(a)	52.	(b)	53.	(c)	54.	(a)	55.	(b)	56.	(b)	57.	(b)	58.	(a)	59.	(b)	60.	(b)
61.	(d)	62.	(c)	63.	(b)	64.	(c)	65.	(a)	66.	(b)	67.	(c)						

CAPF PYQ

1. If 5 persons can weave 160 mats in 8 days, how many mats will 8 persons weave in 6 days

[CAPF 2012]

- (a) 200 (b) 192
(c) 190 (d) 180

2. Two pipes A and B can fill a tank in 12 minutes and 16 minutes respectively, if both the pipes are opened together then after how much time B should be closed so that the tank is full in 9 minutes

[CAPF 2017]

- (a) 3 and half minutes
(b) 4 minutes

- (c) 4 and half minutes
(d) 4 minutes 45 sec

3. A and B together can finish a job in 20 days. B and C together can finish the same job in 30 days. If A and C together can finish it in 24 days, in how many days can A alone finish the job?

[CAPF 2021]

- (a) $35\frac{2}{7}$ (b) $37\frac{2}{7}$
(c) $34\frac{2}{7}$ (d) $33\frac{2}{7}$

ANSWER KEY

1.	(b)	2.	(b)	3.	(c)														
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PRACTICE SHEET 1

1. The speed of a bus is 72 km/hr. The distance covered by the bus in 5 seconds is
 (1) 100 m (2) 60 m
 (3) 50 m (4) 74.5 m
2. Two men start together to walk a certain distance, one at 4 km/h and another at 3 km/h. The former arrives half an hour before the latter. Find the distance.
 (1) 8 km (2) 7 km
 (3) 6 km (4) 9 km
3. A train starts from a place A at 6 a.m. and arrives at another place B at 4.30 p.m. on the same day. If the speed of the train is 40 km per hour, find the distance travelled by the train?
 (1) 420 km (2) 230 km
 (3) 320 km (4) 400 km
4. A boy runs 20 km in 2.5 hours. How long will he take to run 32 km at double the previous speed?
 (1) 2 hours (2) 2.5 hours
 (3) 4.5 hours (4) 5 hours
5. A train is moving with the speed of 180 km/hr. Its speed (in metres per second) is:
 (1) 5 (2) 40
 (3) 30 (4) 50
6. A man riding his bicycle covers 150 metres in 25 seconds. What is his speed in km per hour?
 (1) 25 (2) 21.6
 (3) 23 (4) 20
7. Walking at the rate of 4 km an hour, a man covers a certain distance in 3 hours 45 minutes. If he covers the same distance on cycle, cycling at the rate of 16.5 km/hour, the time taken by him is
 (1) 55.45 minutes (2) 54.55 minutes
 (3) 55.44 minutes (4) 45.55 minutes
8. A train covers a distance of 10 km in 12 minutes. If its speed is decreased by 5 km/hr, the time taken by it to cover the same distance will be:
 (1) 10 minutes
 (2) 13 minutes 20 sec
 (3) 13 minutes
 (4) 11 minutes 20 sec
9. A man rides at the rate of 18 km/hr, but stops for 6 mins. to change horses at the end of every 7th km. The time that he will take to cover a distance of 90 km is:
 (1) 6 hrs. (2) 6 hrs. 12 min.
 (3) 6 hrs. 18 min. (4) 6 hrs. 24 min.
10. A speed of 30.6 km/hr is the same as
 (1) 8.5 m/sec. (2) 10 m/sec.
 (3) 12 m/sec. (4) 15.5 m/sec.
11. A man covers $\frac{2}{15}$ of the total journey by train, $\frac{9}{20}$ by bus and the remaining 10 km on foot. His total journey (in km) is
 (1) 15.6 (2) 24
 (3) 16.4 (4) 12.8
12. A man travels for 5 hours 15 minutes. If he covers the first half of the journey at 60 km/h and rest at 45 km/h. Find the total distance travelled by him.
 (1) 1028.85 km. (2) 189 km.
 (3) 378 km. (4) 270 km.
13. A car can finish a certain journey in 10 hours at the speed of 42 kmph. In order to cover the same distance in 7 hours, the speed of the car (km/h) must be increased by:
 (1) 12 (2) 15
 (3) 18 (4) 24
14. A man cycles at the speed of 8 km/hr and reaches office at 11 am and when he cycles at the speed of 12 km/hr he reaches office at 9 am. At what speed should he cycle so that he reaches his office at 10 am?
 (1) 9.6 kmph
 (2) 10 kmph
 (3) 11.2 kmph
 (4) Cannot be determined
15. A bus travels at the speed of 36 km/hr, then the distance covered by it in one second is
 (1) 10 metre (2) 15 metre
 (3) 12.5 metre (4) 13.5 metre

- 16.** A boy rides his bicycle 10km at an average speed of 12 km/hr and again travels 12 km at an average speed of 10 km/hr. His average speed for the entire trip is approximately :
 (1) 10.4 km/hr (2) 10.8 km/hr
 (3) 11.0 km/hr (4) 12.2 km/hr
- 17.** A person travels 600 km by train at 80km/hr, 800 km by ship at 40 km/hr 500 km by aeroplane at 400 km/hr and 100 km by car at 50km/hr. What is the average speed for the entire distance ?
 (1) $65\frac{5}{123}$ vkm./hr. (2) 60 km./hr.
 (3) $60\frac{5}{123}$ km./hr. (4) 62 km./hr.
- 18.** A train moves with a speed of 30 kmph for 12 minutes and for next 8 minutes at a speed of 45 kmph. Find the average speed of the train:
 (1) 37.5 kmph (2) 36 kmph
 (3) 48 kmph (4) 30 kmph
- 19.** A man covers half of his journey at 6km/hr and the remaining half at 3km/hr. His average speed is
 (1) 9 km/hr (2) 4.5 km/hr
 (3) 4 km/hr (4) 3 km/hr
- 20.** A man goes from A to B at a uniform speed of 12 kmph and returns with a uniform speed of 4 kmph His average speed (in kmph) for the whole journey is :
 (1) 8 (2) 7.5
 (3) 6 (4) 4.5
- 21.** A train covers a distance of 3584 km in 2 days 8 hours. If it covers 1440 km on the first day and 1608 km on the second day, by how much does the average speed of the train for the remaining part of the journey differ from that for the entire journey ?
 (1) 3 km/hour more (2) 3 km/hour less
 (3) 4 km/hour more (4) 5 km/hour less
- 22.** A man travels a distance of 24 km at 6 kmph. Another distance of 24 km at 8 kmph and a third distance of 24 km at 12 kmph. His average speed for the whole journey (in kmph) is
 (1) $8\frac{2}{3}$ (2) 8
- (3) $2\frac{10}{13}$ (4) 9
- 23.** A constant distance from Chennai to Bangalore is covered by Express train at 100 km/hr. If it returns to the same distance at 80 km/hr, then the average speed during the whole journey is
 (1) 90.20 km/hr (2) 88.78 km/hr
 (3) 88.98 km/hr (4) 88.89 km/hr
- 24.** A person went from A to B at an average speed of x km/hr and returned from B to A at an average speed of y km/hr. What was his average speed during the total journey
 (1) $\frac{x+y}{xy}$ (2) $\frac{2xy}{x+y}$
 (3) $\frac{2}{x+y}$ (4) $\frac{1}{x} + \frac{1}{y}$
- 25.** A man goes from Mysore to Bangalore at a uniform speed of 40 km/hr and comes back to Mysore at a uniform speed of 60 km/hr. His average speed for the whole journey is
 (1) 48 km/hr (2) 50 km/hr
 (3) 54 km/hr (4) 55 km/hr
- 26.** A man goes from a place A to B at a speed of 12 km/hr and returns from B to A at a speed of 18 km/hr. The average speed for the whole journey is
 (1) $14\frac{2}{5}$ km/hr (2) 15 km/hr
 (3) $15\frac{1}{2}$ km/hr (4) 16 km/hr
- 27.** One third of a certain journey is covered at the rate of 25 km/ hour, one-fourth at the rate of 30 km/hour and the rest at 50 km/ hour. The average speed for the whole journey is
 (1) 35 km/hour (2) $33\frac{1}{3}$ km/hour
 (3) 30 km/hour (4) $37\frac{1}{12}$ km/hour
- 28.** A man completes 30 km of a journey at the speed of 6 km/hr and the remaining 40 km of the journey in 5 hours. His average speed for the whole journey is

- (1) 7 km/hr (2) $6\frac{4}{11}$ km/hr
(3) 8 km/hr (4) 7.5 km/hr

- 29.** A man covers the journey from a station A to station B at a uniform speed of 36 km/hr and returns to A with a uniform speed of 45 km/hr. His average speed for the whole journey is
(1) 40 km/hr (2) 40.5 km/hr

- (3) 41 km/hr (4) 42 km/hr

- 30.** The speed of a train going from Nagpur to Allahabad is 100 kmph while its speed is 150 kmph when coming back from Allahabad to Nagpur. Then the average speed during the whole journey is :
(1) 120 kmph (2) 125 kmph
(3) 140 kmph (4) 135 kmph

ANSWER KEY

1.	(1)	2.	(3)	3.	(1)	4.	(1)	5.	(4)	6.	(2)	7.	(2)	8.	(2)	9.	(2)	10.	(1)
11.	(2)	12.	(4)	13.	(3)	14.	(1)	15.	(1)	16.	(2)	17.	(1)	18.	(2)	19.	(3)	20.	(3)
21.	(1)	22.	(2)	23.	(4)	24.	(2)	25.	(1)	26.	(1)	27.	(2)	28.	(3)	29.	(1)	30.	(1)

PRACTICE SHEET 2

- The length of a train and that of a platform are equal. If with a speed of 90 km/hr the train crosses the platform in one minute, then the length of the train (in metres) is :
(1) 500 (2) 600
(3) 750 (4) 900
- A train passes two bridges of lengths 800 m and 400 m in 100 seconds and 60 seconds respectively. The length of the train is :
(1) 80 m (2) 90 m
(3) 200 m (4) 150 m
- A train 300 metres long is running at a speed of 25 metres per second. It will cross a bridge of 200 metres in
(1) 5 seconds (2) 10 seconds
(3) 20 seconds (4) 25 seconds
- A train 800 metres long is running at the speed of 78 km/hr. If it crosses a tunnel in 1 minute, then the length of the tunnel (in metres) is :
(1) 77200 (2) 500
(3) 1300 (4) 13
- A train is moving at a speed of 132 km/hour. If the length of the train is 110 metres, how long will it take to cross a railway platform 165 metres long?
(1) 5 seconds (2) 7.5 seconds
(3) 10 seconds (4) 15 seconds
- A train takes 18 seconds to pass through a platform 162 m long and 15 seconds to pass through another platform 120 m long. The length of the train (in m) is :
(1) 70 (2) 80
(3) 90 (4) 105
- A train, 150 m long, takes 30 seconds to cross a bridge 500 m long. How much time will the train take to cross a platform 370 m long ?
(1) 36 secs (2) 30 secs
(3) 24 secs (4) 18 secs
- A 120 metre long train is running at a speed of 90 km per hour. It will cross a railway platform 230 m long in
(1) 4.8 seconds (2) 9.2 seconds
(3) 7 seconds (4) 14 seconds
- A train travelling at a speed of 30 m/sec crosses a platform, 600 metres long, in 30 seconds. The length (in metres) of train is
(1) 120 (2) 150
(3) 200 (4) 300
- A train with a uniform speed passes a platform, 122 metres long, in 17 seconds

and a bridge, 210 metres long, in 25 seconds. The speed of the train is

- (1) 46.5 km/hour (2) 37.5 km/hour
(3) 37.6 km/hour (4) 39.6 km/hour

- 11.** A train running at $\frac{7}{11}$ of its own speed reached a place in 22 hours. How much time could be saved if the train would run at its own speed?

- (1) 14 hours (2) 7 hours
(3) 8 hours (4) 16 hours

- 12.** A man with $\frac{3}{5}$ of his usual speed reaches the destination 2 and half hours late. Find his usual time to reach the destination.

- (1) 4 hours (2) 3 hours
(3) 3.75 hours (4) 4.5 hours

- 13.** A car travelling with $\frac{5}{7}$ of its usual speed covers 42 km in 1 hour 40 min 48 sec. What is the usual speed of the car?

- (1) 17.8 km/hr (2) 35 km/hr
(3) 25 km/hr (4) 30 km/hr

- 14.** Walking at three-fourth of his usual speed, a man covers a certain distance in 2 hours more than the time he takes to cover the distance at his usual speed. The time taken by him to cover the distance with his usual speed is

- (1) 4.5 hours (2) 5.5 hours
(3) 6 hours (4) 5 hours

- 15.** By walking at $\frac{3}{4}$ of his usual speed, a man reaches his office 20 minutes later than his usual time. The usual time taken by him to reach his office is

- (1) 75 minutes (2) 60 minutes
(3) 40 minutes (4) 30 minutes

- 16.** If a man walks 20 km at 5 km/hr, he will be late by 40 minutes. If he walks at 8 km/hr, how early from the fixed time will he reach?

- (1) 15 minutes (2) 25 minutes
(3) 50 minutes (4) 90 minutes

- 17.** If a man reduces his speed to $\frac{2}{3}$, he takes 1 hour more in walking a certain distance. The time (in hours) to cover the distance with his normal speed is :

- (1) 2 (2) 1
(3) 3 (4) 1.5

- 18.** A student rides on bicycle at 8 km/hour and reaches his school 2.5 minutes late. The next day he increases his speed to 10 km/

hour and reaches school 5 minutes early. How far is the school from his house ?

- (1) $5\frac{1}{8}$ km (2) 8 km
(3) 5 km (4) 10 km

- 19.** A man covered a certain distance at some speed. Had he moved 3 km per hour faster, he would have taken 40 minutes less. If he had moved 2 km per hour slower, he would have taken 40 minutes more. The distance (in km) is :

- (1) 20 (2) 35
(3) 36.6 (4) 40

- 20.** If a train runs at 40 km/hour, it reaches its destination late by 11 minutes. But if it runs at 50 km/hour, it is late by 5 minutes only. The correct time (in minutes) for the train to complete the journey is

- (1) 13 (2) 15
(3) 19 (4) 21

- 21.** A student walks from his house at a speed of 2.5 km per hour and reaches his school 6 minutes late. The next day he increases his speed by 1 km per hour and reaches 6 minutes before school time. How far is the school from his house?

- (1) $5\frac{1}{4}$ km (2) $7\frac{1}{4}$ km
(3) $9\frac{1}{4}$ km (4) $11\frac{1}{4}$ km

- 22.** A boy is late by 9 minutes if he walks to school at a speed of 4 km/hour. If he walks at the rate of 5 km/hour, he arrives 9 minutes early. The distance to his school is

- (1) 9 km (2) 5 km
(3) 4 km (4) 6 km

- 23.** A car can cover a certain distance in 4.5 hours. If the speed is increased by 5 km/hour, it would take $\frac{1}{2}$ hour less to cover the same distance. Find the slower speed of the car.

- (1) 50 km/hour (2) 40 km/hour
(3) 45 km/hour (4) 60 km/hour

- 24.** Shri X goes to his office by scooter at a speed of 30 km/h and reaches 6 minutes earlier. If he goes at a speed of 24 km/h, he reaches 5 minutes late. The distance of his office is

- (1) 20 km (2) 21 km
(3) 22 km (4) 24 km

- 25.** Walking at 5 km/hr a student reaches his school from his house 15 minutes early and

walking at 3 km/hr he is late by 9 minutes. What is the distance between his school and his house ?

- (1) 5 km (2) 8 km
(3) 3 km (4) 2 km

26. A train 180 m long moving at the speed of 20 m/sec. over-takes a man moving at a speed of 10m/ sec in the same direction. The train passes the man in :

- (1) 6 sec (2) 9 sec
(3) 18 sec (4) 27 sec

27. A train 100m long is running at the speed of 30 km/hr. The time (in second) in which it will pass a man standing near the railway line is :

- (1) 10 (2) 11
(3) 12 (4) 15

28. How many seconds will a 500 metre long train take to cross a man walking with a

speed of 3 km/hr. in the direction of the moving train if the speed of the train is 63 km/hr ?

- (1) 25 sec (2) 30 sec
(3) 40 sec (4) 45 sec

29. A train is 125 m long. If the train takes 30 seconds to cross a tree by the railway line, then the speed of the train is :

- (1) 14 km/hr (2) 15 km/hr
(3) 16 km/hr (4) 12 km/hr

30. A 120 m long train takes 10 seconds to cross a man standing on a platform. What is the speed of the train?

- (1) 12 m/sec. (2) 10 m/sec.
(3) 15 m/sec. (4) 20 m/sec.

ANSWER KEY

1.	(3)	2.	(3)	3.	(3)	4.	(2)	5.	(2)	6.	(3)	7.	(3)	8.	(4)	9.	(4)	10.	(4)
11.	(3)	12.	(3)	13.	(2)	14.	(3)	15.	(2)	16.	(3)	17.	(1)	18.	(3)	19.	(4)	20.	(3)
21.	(2)	22.	(4)	23.	(2)	24.	(3)	25.	(3)	26.	(3)	27.	(3)	28.	(2)	29.	(2)	30.	(1)

PRACTICE SHEET 3

1. On increasing the speed of a train at the rate of 10 km per hr, 30 minutes is saved in a journey of 100 kms. Find the initial speed of train.

- (1) 40 kmph (2) 45 kmph
(3) 42 kmph (4) 44 kmph

2. A thief is noticed by a policeman from a distance of 200m. The thief starts running and the policeman chases him. The thief and the policeman run at the rate of 10 km./hr and 11 km./hr respectively. What is the distance between them after 6 minutes?

- (1) 100 m (2) 190 m
(3) 200 m (4) 150 m

3. A moving train, 66 metres long, overtakes another train of 88 metres long, moving in the same direction in 0.168 minutes. If the second train is moving at 30 km/hr, at what speed is the first train moving?

- (1) 85 km/hr. (2) 50 km/hr.
(3) 55 km/hr. (4) 25 km/hr.

4. A constable is 114 metres behind a thief. The constable runs 21 metres and the thief runs 15 metres in a minute. In what time will the constable catch the thief ?

- (1) 19 minutes (2) 18 minutes
(3) 17 minutes (4) 16 minutes

5. How much time does a train, 50 m long, moving at 68 km/ hour take to pass another train, 75 m long, moving at 50 km/ hour in the same direction.

- (1) 5 seconds (2) 10 seconds
(3) 20 seconds (4) 25 seconds

6. A constable follows a thief who is 200 m ahead of the constable. If the constable and the thief run at speed of 8 km/hour and 7

- km/hour respectively, the constable would catch the thief in
 (1) 10 minutes (2) 12 minutes
 (3) 15 minutes (4) 20 minutes
- 7.** Two trains are running with speed 30 km/hr and 58 km/hr in the same direction. A man in the slower train passes the faster train in 18 seconds. The length (in metres) of the faster train is :
 (1) 70 (2) 100
 (3) 128 (4) 140
- 8.** Two trains travel in the same direction at the speed of 56 km/h and 29 km/h respectively. The faster train passes a man in the slower train in 10 seconds. The length of the faster train (in metres) is
 (1) 100 (2) 80
 (3) 75 (4) 120
- 9.** A bus moving at a speed of 45 km/hr overtakes a truck 150 metres ahead going in the same direction in 30 seconds. The speed of the truck is
 (1) 27 km/hr (2) 24 km/hr
 (3) 25 km/hr (4) 28 km/hr
- 10.** A man covered a certain distance at some speed. Had he moved 3 km per hour faster, he would have taken 40 minutes less. If he had moved 2 km per hour slower, he would have taken 40 minutes more. The distance (in km) is :
 (1) 20 (2) 35
 (3) 36.6 (4) 40
- 11.** The distance between two cities A and B is 330 km. A train starts from A at 8 a.m. and travels towards B at 60 km/hr. Another train starts from B at 9 a.m. and travels towards A at 75 km/hr. At what time do they meet?
 (1) 10 a.m. (2) 10 : 30 a.m.
 (3) 11 a.m. (4) 11 : 30 a.m.
- 12.** Two men are standing on opposite ends of a bridge 1200 metres long. If they walk towards each other at the rate of 5m/minute and 10m/minute respectively, in how much time will they meet each other ?
 (1) 60 minutes (2) 80 minutes
 (3) 85 minutes (4) 90 minutes
- 13.** Two trains, one 160 m and the other 140 m long are running in opposite directions on parallel rails, the first at 77 km an hour and the other at 67 km an hour. How long will they take to cross each other?
 (1) 7 seconds (2) 7.5seconds
 (3) 6 seconds (4) 10 seconds
- 14.** Two trains are running in opposite direction with the same speed. If the length of each train is 120 metres and they cross each other in 12 seconds, the speed of each train (in km/hour) is
 (1) 72 (2) 10
 (3) 36 (4) 18
- 15.** Two trains 140 m and 160 m long run at the speed of 60 km/ hour and 40 km/hour respectively in opposite directions on parallel tracks. The time (in seconds) which they take to cross each other, is :
 (1) 10 sec. (2) 10.8 sec.
 (3) 9 sec. (4) 9.6 sec.
- 16.** A train 'B' speeding with 100 kmph crosses another train C, running in the same direction, in 2 minutes. If the length of the train B and C be 150 metre and 250 metre respectively, what is the speed of the train C (in kmph)?
 (1) 75 (2) 88
 (3) 95 (4) 110
- 17.** A passenger train running at the speed of 80 kms./hr leaves the railway station 6 hours after a goods train leaves and overtakes it in 4 hours. What is the speed of the goods train?
 (1) 32 kmph (2) 50 kmph
 (3) 45 kmph (4) 64 kmph
- 18.** Two trains start from a certain place on two parallel tracks in the same direction. The speed of the trains are 45 km/hr. and 40 km/ hr respectively. The distance between the two trains after 45 minutes will be
 (1) 2.5 km. (2) 2.75 km.
 (3) 3.7 km. (4) 3.75 km.
- 19.** A thief is stopped by a policeman from a distance of 400 metres. When the policeman starts the chase, the thief also starts running. Assuming the speed of the thief as 5 km/h and that of policeman as 9 km/h, how far the thief would have run, before he is over taken by the policeman?
 (1) 400 metre (2) 600 metre
 (3) 500 metre (4) 300 metre

- 20.** Two trains of equal length are running on parallel lines in the same direction at 46 km/hour and 36 km/hour. The faster train passes the slower train in 36 seconds. The length of each train is
 (1) 72 m (2) 80 m
 (3) 82 m (4) 50 m
- 21.** Two trains leave a railway station at the same time. The first train travels due west and the second train due north. The first train travels 5 km per hr. faster than the second train. If after two hours they are 50 km apart, find the average speed of faster train.
 (1) 18 kmph (2) 15 kmph
 (3) 20 kmph (4) 25 kmph
- 22.** A carriage driving in a fog passed a man who was walking at the rate of 6 km per hr. in the same direction. He could see the carriage for 4 minutes and it was visible to him up to a distance of 200 metres. Find the speed of the carriage.
 (1) 8.75 kmph (2) 8.5 kmph
 (3) 8 kmph (4) 9 kmph
- 23.** Two bullets were fired at a place at an interval of 12 minutes. A person approaching the firing point in his car hears the two sounds at an interval of 11 minutes 40 seconds. The speed of sound is 330 metres per second. What is the approximate speed of the car?
 (1) 34 kmph (2) 32 kmph
 (3) 36 kmph (4) 38 kmph
- 24.** A and B start simultaneously at 5 km per hr. and 4 km per hr. from P and Q, 180 kms apart, towards Q and P respectively. They cross each other at M and after reaching Q and P turn back immediately and meet again at N. Find the distance MN.
 (1) 45 km (2) 40 km
 (3) 35km (4) 42 km
- 25.** A car driving in the morning fog passes a man walking at 4 km per hr. in the same direction. The man can see the car for 3 minutes and visibility is upto a distance of 130 metres. Find the speed of the car.
 (1) 7.5 kmph (2) 6.6 kmph
 (3) 6 kmph (4) 7 kmph
- 26.** Ram starts his journey from Bombay to Pune and simultaneously Mohan starts from Pune to Bombay. After crossing each other they finish their remaining journey in $6\frac{1}{4}$ and 4 hours respectively. What is Mohan's speed if Ram's speed is 20 km per hr.
 (1) 28 kmph (2) 24 kmph
 (3) 25 kmph (4) 30 kmph
- 27.** Two guns were fired from same place at an interval of 5 minutes but a man sitting in the train going away from the place near the second fire 6 minute after the first if the speed of train is 40 km per hour find the speed of sound
 (a) 160 kmph (b) 200kmph
 (c) 240 kmph (d) 350 kmph
- 28.** The buses are departed after every 25 minutes but man going towards from the bus deport after every 20 minute get the buses find the speed of message if the speed of man is 30 km per hour?
 (a) 120 kmph (b) 200kmph
 (c) 150 kmph (d) 180kmph
- 29.** Two men A and B start from Delhi e and Agra at same time towards each other after meeting on the way they cover the remaining journey in in $7\frac{1}{9}$ and $6\frac{1}{4}$ hours respectively find the slower speed if faster speed is 40 kilometre more than the slower
 (a) 500 kmph (b) 600 kmph
 (c) 650 kmph (d) 720 kmph
- 30.** Two men A and B start from X and Y at same time towards each other after meeting on the way they cover the remaining journey in 9 and 4 hours respectively find the faster speed if slower speed is 40 kilometre
 (a) 50 kmph (b) 60 kmph
 (c) 65 kmph (d) 72 kmph

ANSWER KEY

1.	(1)	2.	(1)	3.	(1)	4.	(1)	5.	(4)	6.	(2)	7.	(4)	8.	(3)	9.	(1)	10.	(4)
11.	(3)	12.	(2)	13.	(2)	14.	(3)	15.	(2)	16.	(2)	17.	(1)	18.	(4)	19.	(3)	20.	(4)
21.	(3)	22.	(4)	23.	(1)	24.	(2)	25.	(2)	26.	(3)	27.	(3)	28.	(1)	29.	(2)	30.	(2)

PRACTICE SHEET 4

1. In a one-kilometre race A, B and C are the three participants. A can give B a start of 50 m. and C a start of 69 m. The start, which B can allow C is
 (1) 17 m. (2) 20 m.
 (3) 19 m. (4) 18 m.
2. A runs twice as fast as B and B runs thrice as fast as C. The distance covered by C in 72 minutes, will be covered by A in :
 (1) 18 minutes (2) 24 minutes
 (3) 16 minutes (4) 12 minutes
3. A is twice as fast runner as B, and B is thrice as fast runner as C. If C travelled a distance in 1 hour 54 minutes, the time taken by B to cover the same distance is
 (1) 19 minutes (2) 38 minutes
 (3) 51 minutes (4) 57 minutes
4. In a 100m race, Kamal defeats Bimal by 5 seconds. If the speed of Kamal is 18 Kmph, then the speed of Bimal is
 (1) 15.4 kmph (2) 14.5 kmph
 (3) 14.4 kmph (4) 14 kmph
5. In a race of 1000 m, A can beat B by 100m. In a race of 400 m, B beats C by 40m. In a race of 500m. A will beat C by
 (1) 95 m (2) 50 m
 (3) 45 m (4) 60 m
6. In a race of 800 metres, A can beat B by 40 metres. In a race of 500 metres, B can beat C by 5 metres. In a race of 200 metres, A will beat C by
 (1) 11.9 metre (2) 1.19 metre
 (3) 12.7 metre (4) 1.27 metre
7. In a race of 200 metres, B can give a start of 10 metres to A, and C can give a start of 20 metres to B. The start that C can give to A, in the same race, is
 (1) 30 metres (2) 25 metres
 (3) 29 metres (4) 27 metres
8. A can give 40 metres start to B and 70 metres to C in a race of one kilometre How many metres start can B give to C in a race of one kilometre ?
 (1) 30 metre (2) 31.25metre
 (3) 31.75 metre (4) 32 metre
9. A jeep is chasing a car which is 5km ahead. Their respective speed are 90 km/hr and 75 km/ hr. After how many minutes will the jeep catch the car?
 (1) 18 min. (2) 20 min.
 (3) 24 min. (4) 25 min.
10. A is twice as fast as B, and B is thrice as fast as C is. The journey covered by C in 1 and half hours will be covered by A in
 (1) 15 minutes (2) 30 minutes
 (3) 1 hour (4) 10 minutes
11. It takes 8 hours for a 600 km journey, if 120 km is done by train and the rest by car. It takes 20 minutes more if 200 km is done by train and the rest by car. The ratio of the speed of the train to that of the car is
 (1) 2 : 3 (2) 3 : 2
 (3) 3 : 4 (4) 4 : 3
12. It takes eight hours for a 600 km journey, if 120 km is done by train and the rest by car. It takes 20 minutes more, if 200 km is done by train and the rest by car. The ratio of the speed of the train to that of the car is :
 (1) 3 : 5 (2) 3 : 4
 (3) 4 : 3 (4) 4 : 5
13. A truck covers a distance of 550 metre in one minute where as a bus covers a distance of 33 km in 3/4 hour. Then the ratio of their speeds is
 (1) 1 : 3 (2) 2 : 3
 (3) 3 : 4 (4) 1 : 4
14. A car travels 80 km. in 2 hours and a train travels 180 km. in 3 hours. The ratio of the speed of the car to that of the train is :
 (1) 2 : 3 (2) 3 : 2
 (3) 3 : 4 (4) 4 : 3
15. The speeds of three cars are in the ratio of 1 : 3 : 5. The ratio among the time taken by these cars to travel the same distance is
 (1) 3 : 5 : 15 (2) 15 : 3 : 5
 (3) 15 : 5 : 3 (4) 5 : 3 : 1
16. A man rows a boat 18 kilometres in 4 hours downstream and returns upstream in 12 hours. The speed of the stream (in km per hour) is :

- (1) 1 (2) 1.5
(3) 2 (4) 1.75
- 17.** A motorboat in still water travels at a speed of 36 kmph. It goes 56 km upstream in 1 hour 45 minutes. The time taken by it to cover the same distance down the stream will be :
(1) 2 hours 25 minutes
(2) 3 hours
(3) 1 hour 24 minutes
(4) 2 hours 21 minutes
- 18.** A boat running downstream covers a distance of 20km in 2 hrs while it covers the same distance upstream in 5 hrs. Then speed of the boat in still water is
(1) 7 km/hr (2) 8 km/hr
(3) 9 km/hr (4) 10 km/hr
- 19.** A boatman rows 1 km in 5 minutes, along the stream and 6 km in 1 hour against the stream. The speed of the stream is
(1) 3 kmph (2) 6 kmph
(3) 10 kmph (4) 12 kmph
- 20.** A boat covers 24 km upstream and 36 km downstream in 6 hours, while it covers 36 km upstream and 24 km downstream in 6 and half hours. The speed of the current is
(1) 1 km/hr (2) 2 km/hr
(3) 1.5 km/hr (4) 2.5 km/hr
- 21.** A boat goes 6 km an hour in still water, but takes thrice as much time in going the same distance against the current. The speed of the current (in km/hour) is :
(1) 4 (2) 5
(3) 3 (4) 2
- 22.** In a fixed time, a boy swims double the distance along the current that he swims against the current. If the speed of the current is 3 km/hr, the speed of the boy in still water is
(1) 6 km/hr (2) 9 km/hr
(3) 10 km/hr (4) 12 km/hr
- 23.** A man can row 30 km downstream and return in a total of 8 hours. If the speed of the boat in still water is four times the speed of the current, then the speed of the current is
(1) 1 km/hour (2) 2 km/hour
(3) 4 km/hour (4) 3 km/hour
- 24.** A person can row 7.5 km an hour in still water and he finds that it takes him twice as long to row up as to row down the river. The speed of the stream is :
(1) 2 km/hr (2) 3 km/hr
(3) 2.5 km/hr (4) 3.5 km/hr
- 25.** A man can row at a speed of 4.5 km/hr in still water. If he takes 2 times as long to row a distance upstream as to row the same distance downstream, then, the speed of stream (in km/hr) is
(1) 1 (2) 1.5
(3) 2 (4) 2.5
- 26.** A man can row at 5 kmph. in still water. If the velocity of current is 1 kmph. and it takes him 1 hour to row to a place and come back, how far is the place ?
(1) 2.5 km (2) 3 km
(3) 2.4 km (4) 3.6 km
- 27.** The speed of a motor-boat is that of the current of water as 36 : 5. The boat goes along with the current in 5 hours 10 minutes. It will come back in
(1) 5 hours 50 minutes
(2) 6 hours
(3) 6 hours 50 minutes
(4) 12 hours 10 minutes
- 28.** A man goes downstream with a boat to some destination and returns upstream to his original place in 5 hours. If the speed of the boat in still water and the stream are 10 km/hr and 4 km/hr respectively, the distance of the destination from the starting place is
(1) 16 km (2) 18 km
(3) 21 km (4) 25 km
- 29.** Two boats A and B start towards each other from two places, 108 km apart. Speed of the boat A and B in still water are 12km/hr and 15km/hr respectively. If A proceeds down and B up the stream, they will meet after.
(1) 4.5 hours (2) 4 hours
(3) 5.4 hours (4) 6 hours
- 30.** A man can row 6 km/h in still water. If the speed of the current is 2 km/h, it takes 3 hours more in upstream than in the downstream for the same distance. The distance is
(1) 30 km (2) 24 km
(3) 20 km (4) 32 km

ANSWER KEY

1.	(2)	2.	(4)	3.	(2)	4.	(3)	5.	(1)	6.	(1)	7.	(3)	8.	(2)	9.	(2)	10.	(1)
11.	(3)	12.	(2)	13.	(3)	14.	(1)	15.	(1)	16.	(2)	17.	(3)	18.	(1)	19.	(1)	20.	(2)
21.	(1)	22.	(2)	23.	(2)	24.	(3)	25.	(2)	26.	(3)	27.	(3)	28.	(3)	29.	(2)	30.	(2)

CDS PYQ

- A person travels a certain distance at 3 km/h and reaches 15 min late. If he travels at 4 km/h, he reaches 15 min earlier. The distance he has to travel is
[CDS 2013(I)]
(a) 4.5 km (b) 6 km
(c) 7.2 km (d) 12 km
- If a body covers a distance at the rate of x km/h and another equal distance at the rate of y km/h, then the average speed (in km/h) is
[CDS 2013(I)]
(a) $\frac{(x+y)}{2}$ (d) $\sqrt{xy}$
(c) $\frac{2xy}{x+y}$ (d) $\frac{(x+y)}{xy}$
- A sailor sails a distance of 48 km along the flow of a river in 8 h. If it takes 12 h to return the same distance, then the speed of the flow of the river is
[CDS 2013(I)]
(a) 0.5 km/h (b) 1 km/h
(c) 1.5 km/h (d) 2 km/h
- A train running at the speed of 72 km/h goes past a pole in 15 s. What is the length of the train.
[CDS 2013(II)]
(a) 150 m (b) 200 m
(c) 300 m (d) 350 m
- Two cars A and B start simultaneously from a certain place at the speed of 30 km/h and 45 km/h, respectively. The car B reaches the destination 2 h earlier than A. What is the distance between the starting point and destination?
[CDS 2013(II)]
(a) 90 km (b) 180 km
(c) 270 km (d) 360 km
- A man cycles with a speed of 10 km/h and reaches his office at 1: 00 pm. However, when he cycles with a speed of 15 km/h, he reaches his Office at 11 00 am. At what speed should he cycle, so that he reaches his office at 12 noon?
[CDS 2013(II)]
(a) 12.5 km/h (b) 12 km/h
(c) 13 km/h (d) 13.5 km/h
- A train takes 9 s to cross a pole. If the speed of the train is 48 km/h, the length of the train is
[CDS 2014(I)]
(a) 150 m (b) 120m
(c) 90 m (d) 80 m
- A train takes 10 s to cross a pole and 20 s to cross a platform of length 200 m. What is the length of the train?
[CDS 2014(II)]
(a) 50 m (b) 100m
(c) 150 m (d) 200 m
- A train travels at a speed of 40 km/h and another train at a speed of 20 m/s. What is the ratio of speed of the first train to that of the second train?
[CDS 2014(II)]
(a) 2:1 (b) 5:9
(c) 5:3 (d) 9:5
- The distance between two points (A and B) is 110 km. X starts running from point A at a speed of 60 km/h and Y starts running from point B at a speed of 40 km/h at the same time. They meet at a point C, somewhere on the line OB. What is the ratio of AC to BC?
[CDS 2014(II)]

(a) 3: 2
(c) 3:4

(b) 2:3
(d) 4:3

11. A man rides one-third of the distance from A to B at the rate of x km/h and the remainder at the rate of $2y$ km/h. If he had travelled at a uniform rate of $6z$ km/h, then he could have ridden from A to B and back again in the same time. Which one of the following is correct?

[CDS 2014(II)]

(a) $z = x + y$

(b) $3Z = x + y$

(c) $\frac{1}{z} = \frac{1}{x} + \frac{1}{y}$

(d) $\frac{1}{2z} = \frac{1}{x} + \frac{1}{y}$

12. In a flight of 600 km, an aircraft was slowed down due to bad weather. Its average speed for the trip was reduced by 200 km/h and the time of flight increased by 30 min. The duration of the flight is

[CDS 2015(I)]

(a) 1h
(c) 3h

(b) 2h
(d) 4h

13. With a uniform speed, a car covers a distance in 8 h. Had the speed been increased by 4 km/h, the same distance could have been covered in 7 h and 30 min. What is the distance covered?

[CDS 2015(I)]

(a) 420 km
(c) 520 km

(b) 480 km
(d) 640 km

14. A car travels the first one-third of a certain distance with a speed of 10 km/h, the next one-third distance with a speed of 20 km/h and the last one-third distance with a speed of 60 km/h. The average speed of the car for the whole journey is

[CDS 2015(I)]

(a) 18 km/h
(c) 30 km/h

(b) 24 km/h
(d) 36 km/h

15. A man rows 32 km downstream and 14 km upstream, and he takes 6h to cover each distance. What is the speed of the current?

[CDS 2015(I)]

(a) 0.5 km/h
(c) 1.5 km/h

(b) 1 km/h
(d) 2 km/h

16. A runs $1\frac{2}{3}$ times as fast as B. If A gives B a start of 80 m, how far must the winning

post from the starting point be so that A and B might reach it at the same time?

[CDS 2015(I)]

(a) 200 m
(c) 270 m

(b) 300 m
(d) 160 m

17. A thief is noticed by a policeman from a distance of 200 m. The thief starts running and the thief and the policeman chase him. Policeman runs at the speed of 10 km/h and 11 km/h, respectively. What is the distance between them after 6 min?

[CDS 2015(I)]

(a) 100 m
(c) 150 m

(b) 120 m
(d) 160 m

18. Two persons A and B start simultaneously from two places c km apart and walk in the same direction. If A travels at the rate of p km/h and B travels at the rate of q km/h, then A has travelled before he overtakes B a distance of

[CDS 2015(I)]

(a) $\frac{qc}{p+q}$

(b) $\frac{pc}{p-q}$

(c) $\frac{qc}{p-q}$

(d) $\frac{pc}{p+q}$

19. By increasing the speed of his car by 15 km/h, a person covers 300 km distance by taking an hour less than before. The original speed of the car was

[CDS 2015(II)]

(a) 45 km/h
(c) 60 km/h

(b) 50 km/h
(d) 75 km/h

20. Two trains, one is of 121 m in length at the speed of 40 km/h and the other is of 99 m in length at the speed of 32 km/h are running in opposite directions. In how much time will they be completely clear from each other from the moment they meet?

[CDS 2015(II)]

(a) 10 s
(c) 16 s

(b) 11 s
(d) 21 s

21. The speeds of three buses are in the ratio 2:3:4. The time taken by these buses to travel the same distance will be in the ratio.

[CDS 2015(II)]

(a) 2:3:4
(c) 4:3:6

(b) 4:3:2
(d) 6:4:3

- 22.** Two trains are moving in the same direction at 1.5 km/min and 60 km/h, respectively. A man in the faster train observes that it takes 27 s to cross the slower train. The length of the slower train is

[CDS 2015(II)]

- (a) 225 m (b) 230 m
(c) 240 m (d) 250 m

- 23.** In a race of 100 m, A beats B by 4 m and A beats C by 2 m. By how many metres (approximately) would C beat B in another 100 m race assuming C and B run with their respective speeds as in the earlier race?

[CDS 2015(II)]

- (a) 2 (b) 2.04
(c) 2.08 (d) 3.2

- 24.** Three athletes run a 4 km race. Their speeds are in the ratio 16:15:11. When the winner wins the race, then the distance between the athlete in the second position to the athlete in the third position is

[CDS 2015(II)]

- (a) 1000 m (b) 800 m
(c) 750 m (d) 600 m

- 25.** A motorboat, whose speed is 15 km/h in still water goes 30 km downstream and comes back a total of 4h and 30 min. The speed of the stream is

[CDS 2015(II)]

- (a) 4 km/h (b) 5 km/h
(c) 6 km/h (d) 10 km/h

- 26.** In a race A, B, C take part. A beats B by 30m, B beats C by 20m and A beats C by 48m. Which of the following is/are correct?

1. The length of race is 300m.

2. The speeds of A, B, C are in the ratio 50:45:42.

Select the correct answer using codes given below

[CDS 2015(II)]

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

- 27.** A bike consumes 20 mL of petrol per kilometre if it is driven at a speed in the range of 25-50 km/h and consumes 40 mL of petrol per kilometre at any other speed. How much is consumed by the bike in travelling distance of 50 km, if the bike is driven at a speed of 40 km/h for the first 10 km, at a speed of 60 km/h for the next 30

km and at a speed of 30 km/h for the last 10 km?

[CDS 2016(I)]

- (a) 1 L (b) 1.2 L
(c) 1.4 L (d) 1.6 L

- 28.** A passenger train takes 1 h less for a journey of 120 km, if its speed is increased by 10 km/h from its usual speed. What is its usual speed?

[CDS 2016(I)]

- (a) 50 km/h (b) 40 km/h
(c) 35 km/h (d) 30 km/h

- 29.** A man walking at 5 km/h noticed that a 225 m long train coming in the opposite direction crossed him in 9s. The speed of the train is

[CDS 2016(I)]

- (a) 75 km/h (b) 80 km/h
(c) 85 km/h (d) 90 km/h

- 30.** A cyclist moves non-stop from A to B, a distance of 14 km, at a certain average speed. If his average speed reduces by 1 km/h, then he takes 20 min more to cover the same distance. The original average speed of the cyclist is

[CDS 2016(I)]

- (a) 5 km/h (b) 6 km/h
(c) 7 km/h (d) None of these

- 31.** In a race of 1000 m, A beats B by 100 m or 10 s. If they start a race of 1000 m simultaneously from the same point and if B gets injured after running 50 m less than half the race length and due to which his speed gets halved, then by how much time will A beat B?

[CDS 2016(I)]

- (a) 65 s (b) 60 s
(c) 50 s (d) 45 s

- 32.** Two men, A and B run a race on a course 0.25 km around. If their speeds are in the ratio 5:4, how often does the winner pass the others.

[CDS 2016(II)]

- (a) once (b) Twice
(c) Thrice (d) Four times

- 33.** When the speed of a train is increased by 20% , it takes 20 minutes less to cover the same distance. What is the time taken to cover the same distance with the original speed?

[CDS 2016(II)]

- (a) 140 min (b) 120 min
(c) 100 min (d) 80 min

34. A person can row downstream 20 km in 2 hours and upstream 4 km in 2 hours. What is the speed of the current?

[CDS 2016(II)]

- (a) 2km/hr (b) 2.5 km/hr
(c) 3 km/hr (d) 4km/hr

35. A train is travelling at 48 km/hr completely crosses another train having half its length and travelling in opposite direction at 42 km/hr in 12 s. It also passes a railway platform in 45 s. What is the length of platform?

[CDS 2016(II)]

- (a) 600 m (b) 400 m
(c) 300 m (d) 200 m

36. The speeds of three cars are in the ratio 2 : 3 : 4 . What is the ratio between the times taken by these cars to travel the same distance?

[CDS 2016(II)]

- (a) 4 : 3 : 2 (b) 2 : 3 : 4
(c) 4 : 3 : 4 (d) 6 : 4 : 3

37. A motorist travels to a place 150km away at an average speed of 50 km/hour and returns at 30km/hr. What is the average speed of the whole journey?

[CDS 2016(II)]

- (a) 35 km/hour (b) 37 km/hour
(c) 37.5 km/hour (d) 40 km/hour

38. In a 100m race. A runs at a speed of 5/3 m/s. If A gives a start of 4 m to B and still beats him by 12 seconds, What is the speed of B?

[CDS 2017(I)]

- (a) 5/4 m/s (b) 7/5 m/s
(c) 4/3 m/s (d) 6/5 m/s

39. A passenger train departs from delhi at 6 pm from Mumbai. At 9 pm, an express train, whose average speed exceeds that of the passenger train by 15km/hr leaves Mumbai for Delhi. Two trains meet each other mid-route. At what time do they meet, given that the distance between the cities is 1080 km?

[CDS 2017(I)]

- (a) 4 pm (b) 2 am
(c) 12 mid night (d) 6 am

40. A 225m long train is running at a speed of 30 km/hr. How much time does it take to cross a man running at 3 km/hr in the same direction?

[CDS 2017(II)]

- (a) 40 sec (b) 30 sec
(c) 25 sec (d) 15 sec

41. A thief is spotted by a policeman from a distance of 100m. When the policeman starts the chase, the thief also starts running. If the speed of the thief is 8km/hr and that of the policeman is 10 km/hr, then how far will the thief have to run before the overtaken?

[CDS 2017(II)]

- (a) 200m (b) 300m
(c) 400m (d) 500m

42. If a train crosses a km-stone in 12 seconds, how long will it take to cross 91 km-stones completely if its speed is 60 km/hr?

[CDS 2017(II)]

- (a) 1hr 30 min
(b) 1 hr 30 min 12 sec
(c) 1 hr 51 min
(d) 1 hr 1 min 3 sec

43. In a 100m race, A runs at 6km/hr. If A gives B a start of 8m and still beats him by 9 seconds. What is the speed of B?

[CDS 2017(II)]

- (a) 4.6 km/hr (b) 4.8 km/hr
(c) 5.2 km/hr (d) 5.4 km/hr

44. A boy went to school at a speed of 12km/hr and returned to his house at a speed of 8 km/hr. If he has taken 50 minutes for the whole journey, what was the total distance walked?

[CDS 2017(II)]

- (a) 4km (b) 8km
(c) 16km (d) 20 km

45. A man rows down a river at 18 km in 4 hours with the stream and returns in 10 hours.

Consider the following statements

1. The speed of man against the stream is 1.8 km/hr
 2. The speed of the man in still water is 3.15 km/hr
 3. The speed of the stream is 1.35 km/hr.
- Which of the above statements are correct?

[CDS 2017(II)]

- (a) 1 and 2 (b) 2 and 3

(c) 1 and 3

(d) 1,2 and 3

- 46.** A man travelled 12km at a speed of 4 km/hr and further 10 km at a speed of 5 km/hr. What was his average speed?

[CDS 2017(II)]

(a) 4.4 km/hr

(b) 4.5 km/hr

(c) 5.0 km/hr

(d) 2.5 km/hr

- 47.** A train moving with a speed of 60 km per hour crosses an electric pole in 30 second what is the length of the train in metres?

[CDS 2018(I)]

(a) 300

(b) 400

(c) 500

(d) 600

- 48.** A passenger train and a goods train are running in the same direction on parallel railway tracks, if the passenger train now takes three times as long to pass the goods train as when they are running in opposite directions, then what is the ratio of the speed of the passenger train to that of the goods train (assume that the trains run at uniform speed)

[CDS 2018(I)]

(a) 2:1

(b) 3:2

(c) 4:3

(d) 1:1

- 49.** A man can row at a speed of X kilometre per hour in still water if in a stream which is flowing at a speed of Y kilometre per hour it takes him z hours to row to a place and back then what is the distance between the two places ?

[CDS 2018(I)]

(a) $\frac{z(x^2 - y^2)}{2y}$

(b) $\frac{z(x^2 - y^2)}{2x}$

(c) $\frac{(x^2 - y^2)}{2zx}$

(d) $\frac{z(x^2 - y^2)}{x}$

- 50.** A car has an average speed of 60 km/hr while going from Delhi to Agra and has an average speed of y km/hr while returning to Delhi from Agra by travelling the same distance if the average speed of the car for the whole journey is 40 km per hour then what is the value of Y

[CDS 2018(I)]

(a) 30 km/hr

(b) 35km/hr

(c) 40 km/hr

(d) 45 km/hr

- 51.** A cyclist covers his first 20 km at an average speed of 40 kmph, another 10 km at an

average speed of 10 kmp/h and the last 30 km at an average speed of 40 kmp/h. Then, the average speed of the entire journey is

[CDS 2018(II)]

(a) 20 km/h

(b) 26.67 km/h

(c) 28.24 km/h

(d) 30 km/h

- 52.** A train 100 m long passes a platform 100 m long in 10 s. The speed of the train is

[CDS 2018(II)]

(a) 36 km/h

(b) 45 km/h

(c) 54 km/h

(d) 72 km/h

- 53.** In a race of 1000 m, A beats B by 150 m, while in another race of 3000 m, C beats D by 400 m. Speed of B is equal to that of D. (Assume that A, B, C and D run with uniform speed in all the events). If A and C participate in a race of 6000 m, then which one of the following is correct

[CDS 2018(II)]

(a) A beats C by 250 m

(b) C beats A by 250 m

(c) A beats C by 115.38 m

(d) C beats A by 115.38 m

- 54.** A thief steals a car parked in a house and goes away with a speed of 40 km/h. The theft was discovered after half an hour and immediately the owner sets off in another car with a speed of 60km/h. When will the owner meet the thief?

[CDS 2018(II)]

(a) 55 km from the owner's house and one hour after the theft

(b) 60 km from the owner's house and 1.5 h after the theft

(c) 60 km from the owner's house and 1.5 h after the discovery of the theft

(d) 55 km from the owner's house and 1.5 h after the theft

- 55.** Three cars A, B and C started from a point at 5 pm, 6 pm and 7 pm, respectively uniform speeds of 60 km/h, 80 km/h and x km/h, respectively in the same direction. If all the three travelled at and meet at another point at the same instant during their journey, then what is the value of x?

[CDS 2019(I)]

(a) 120

(b) 110

(c) 105

(d) 100

- 56.** Radha and Hema are neighbours and study in the same school. Both of them use bicycles to go to the school. Radha's speed is

8 km/h whereas Hema's speed is 10 km/h. Hema takes 9 min less than Radha to reach the school. How far is the school from the locality of Radha and Hema?

[CDS 2019(I)]

- (a) 5 km (b) 5.5 km
(c) 6 km (d) 6.5 km

57. A plane is going in circles around an airport. The plane takes 3 min to complete one round. The angle of elevation of the plane from a point P on the ground at time t is equal to that at time $(t + 30)$ s. At time $(t + x)$ s, the plane flies vertically above the point P. What is x equal to?

[CDS 2019(I)]

- (a) 75s (b) 90 s
(c) 105 s (d) 135 s

58. A race has three parts. The speed and time required to complete the individual parts for a runner is displayed on the following chart

	Part 1	Part 2	Part 3
Speed(km/hr)	9	8	7.5
Time (Min)	50	80	100

What is the average speed of this runner

[CDS 2019(I)]

- (a) 8.17 km/h (b) 8 km/h
(c) 7.80 km/h (d) 7.77 km/h

59. It takes 11 h for a 600 km journey if 120 km is done by train and the rest by car. It takes 40 min more if 200 km are covered by train and the rest by car. What is the ratio of speed of the car to that of the train?

[CDS 2019(II)]

- (a) 3:2 (b) 2:3
(c) 3:4 (d) 4:3

60. The ratio of speeds of X and Y is 5:6. If Y allows X a start of 70 m in a 1.2 km race, then who will win the race and by what distance?

[CDS 2020(I)]

- (a) X wins the race by 30 m
(b) Y wins the race by 90 m
(c) Y wins the race by 130 m
(d) The race finishes in a dead heat

61. A train takes two hours less for a journey of 300 km if its speed, is increased by 5 km/hr from its usual speed. What is its usual speed?

[CDS 2020(I)]

- (a) 50 km/hr (b) 40 km/hr
(c) 35 km/hr (d) 25 km/hr

62. In covering certain distance, the average speeds of X and Y are in the ratio 4:5. If X takes 45 minutes more than Y to reach the destination, then what is the time taken by Y to reach the destination?

[CDS 2020(I)]

- (a) 135 minutes (b) 150 minutes
(c) 180 minutes (d) 225 minutes

63. A train of length 110 m is moving at a uniform speed of 132 km/hr. the time required to cross a bridge of length 165m is

[CDS 2020(II)]

- (a) 6.5 sec (b) 7 sec
(c) 7.5 sec (d) 8.5 sec

64. By increasing the speed of his car by 15 km/hr, a person cover a distance of 300 km by taking an hour less than before. What was the original speed of car?

[CDS 2020(II)]

- (a) 45 km/hr (b) 50 km/hr
(c) 60 km/hr (d) 75 km/hr

65. X, Y and Z travel from the same place with uniform speeds 4km/hr, 5km/hr and 6 km/hr respectively. Y starts 2 hours after X. How long after Y must Z start in order that they overtake X at the same instant?

[CDS 2020(II)]

- (a) $3/2$ hours (b) $4/3$ hours
(c) $9/8$ hours (d) $11/8$ hours

66. A car did a journey in t hours, Had the average speed been x kmph greater, the journey would have taken y hours less. How long was the journey?

[CDS 2020(II)]

- (a) $x(t-y)ty$ (b) $x(t-y)ty^{-1}$
(c) $x(t-y)ty^{-2}$ (d) $x(t+y)ty$

67. When a ball is allowed to fall, the time it takes to fall any distance varies as the square root of the distance and it takes 4 seconds to fall 78.40 m. How long would it take to fall 122.50?

[CDS 2020(II)]

- (a) 5 sec (b) 5.5 sec
(c) 6 sec (d) 6.5 sec

68. A train 200 m long passes a platform 100 m long in 10 seconds. What is the speed of train?

- (a) 40 m/s (b) 30 m/s
(c) 25 m/s (d) 20 m/s
- 69.** A train travels 600 km in 5 hours and the next 900 km in 10 hours. What is the average speed of the train?
[CDS 2021(I)]
(a) 80 km/hr (b) 90 km/hr
(c) 100 km/hr (d) 120 km/hr
- 70.** Walking at $\frac{4}{5}$ of his usual speed, a man is 12 minute late for his office. What is the usual time taken by him to cover that distance?
[CDS 2021(I)]
(a) 48 min (b) 50 min
(c) 54 min (d) 60 min
- 71.** A boat has speed 30km/hr in still water. It goes 60km downstream and comes back in $9\frac{1}{2}$ hours. What is the speed of the stream?
[CDS 2021(II)]
(a) 5km/hr (b) 8km/hr
(c) 10km/hr (d) 12km/hr
- 72.** A person rode one-third of a journey at 60km/hr, one third at 50km/hr and the rest at 40km/hr. Had the person ridden half of the journey at 60km/hr and the rest at 40km/hr, he would have taken 4 minute longer to complete the journey. What is the person ride?
[CDS 2021(II)]
(a) 180 km (b) 210 km
(c) 240 km (d) 300 km
- 73.** A train X takes 2 hours less than a train Y to cover a distance of 192 km between two cities. Their average speed differ by 16km/hr. How long does the faster train, take to cover the journey?
[CDS 2022(I)]
(a) 3 hours (b) 4 hours
(c) 5 hours (d) 6 hours
- 74.** A car travels from A to B at a speed of 40 km /hr. travels back from B to A at a speed of 30km/hr and again goes from A to B at a speed of 60 km/hr. What is the average speed of the car?
[CDS 2022(I)]
(a) $\frac{130}{3}$ km/hr (b) 42 km/hr
(c) 40km/hr (d) $\frac{125}{3}$ km/hr
- 75.** A man walks at an average speed of 3km/her from his residence and reaches office 40 minutes early. If he walks at an average speed of 2 km/hr, he reaches 40 minutes late. What is the distance between his residence and office?
[CDS 2022(I)]
(a) 6km (b) 8km
(c) 10km (d) 12km
- 76.** A car takes p minutes to travel a distance of 350km with an average speed of u km/hr. Another can takes q minutes to travel the same distance with an average speed of v km/hr. If $u - v = 5$ and $q - p = 140$, then what is the value of u?
[CDS 2022(I)]
(a) 35 (b) 30
(c) 25 (d) 20
- 77.** In a flight of 2800 km, an aircraft was slowed down due to bad weather. Its average speed for the trip was reduced by 100 km/hr and time of flight increased by 30 minutes. What was the original average speed of the aircraft?
[CDS 2022 (II)]
(a) 700km/hr (b) 750km/hr
(c) 800km/hr (d) 900km/hr
- 78.** A person X starts from a place A and another person Y start simultaneously from another place B which is d km away from A. They walk in the same direction. X walks at an average speed of u km/hr and Y walks at an average speed of v km/hr. How far will X have walked before he overtakes Y?
[CDS 2022 (II)]
(a) $\frac{ud}{(u - v)}$ (b) $\frac{vd}{(u - v)}$
(c) $\frac{(ud - vd)}{(u - v)}$ (d) $\frac{(ud + vd)}{(u + v)}$
- 79.** X takes 3 hours longer than Y to walk 30 km. If X doubles his speed, he takes 2 hours less than Y. What is the speed of Y?
[CDS 2022 (II)]
(a) 3km/hr (b) 4km/hr
(c) $4\frac{2}{7}$ km/hr (d) $4\frac{3}{7}$ km/hr

80. There are two station X and Y, 1320 km apart. A train starts from station X at 6 a.m. and moves at an average speed of 60 km/hr. At 2 p.m. another train starts from Y towards X and moves at an average speed of 80km/hr. When do they meet?

[CDS 2022 (II)]

- (a) 6p.m. (b) 7p.m.
(c) 8p.m (d) 9p.m.

81. The speed of a boat in still water is 15 km/her. If it can travel 42 km downstream and 28 km upstream in the same time, then what is the speed of the stream?

[CDS – 2023 (1)]

- (a) 2.5 km/hr (b) 3 km/hr
(c) 4.5 km/hr (d) 6 km/hr

82. The time taken by a train to cross a man travelling in another train is 10 seconds, when the other train is travelling in the opposite direction. However, it takes 20 seconds, if both the trains are travelling in the same direction. The length of the first train is 200 m and that of the second train is 150m. What is the speed of the first train?

[CDS – 2023 (1)]

- (a) 60km/hr (b) 56km/hr
(c) 54km/hr (d) 52km/hr

83. A train completely overtakes two persons, walking in the same direction with speeds 3 km/hr and 4 km/hr in 9 seconds and 75/9 seconds respectively. What is the length of the train?

[CDS-2023 (2)]

- (a) 60m (b) 62.5m
(c) 55m (d) 67.5m

84. A man walks at an average speed of 3km/hr from his home and reaches office 40 minutes early. If he walks at an average speed of 2km/hr, he would reach office 40 minutes late. What is the distance between his home and office?

[CDS-2023 (2)]

- (a) 6km (b) 8km
(c) 10km (d) 12km

Consider the following for the next three (03) items that follow:

Two trains A and B started from stations P and Q respectively towards each other. Train A started at 7 p.m. at a speed of 60 km/hr and train B started at 4 am (next day) at a speed of 90 km/hr. The distance between the two station P and Q is 800 km

85. How far from station Q will the two trains meet?

[CDS – 2024 (1)]

- (a) 140 km (b) 144 km
(c) 156 km (d) 504 km

86. At what time will the two trains meet?

[CDS – 2024 (1)]

- (a) 5.28 a.m. (b) 5.44 a.m.
(c) 4.56 a.m. (d) 6.25 a.m.

87. If the lengths of the two trains A and B are 400 m and 500 m respectively, then what is the time taken by them to cross each other?

[CDS – 2024 (1)]

- (a) 21.6 seconds (b) 18.2 seconds
(c) 17.4 seconds (d) 15.4 seconds

88. Two trains A and B leave Delhi for Hyderabad at 7:00 a.m. and 7:50 a.m. on the same day and travel at 80 kmph and 100 kmph respectively. After how many kilometers from Delhi will the two trains be together?

[CDS – 2024 (1)]

- (a) $\frac{200}{3}$ km (b) 100 km
(c) $\frac{400}{3}$ km (d) $\frac{1000}{3}$ km

89. The speeds of four cars are $2u$, $3u$, $4u$ and xu and the time taken by them to cover the same distance is xt , $4t$, $3t$ and $2t$ respectively, where x , u , t are real numbers. What is the value of x ?

[CDS – 2024 (1)]

- (a) 8 (b) 6
(c) 5 (d) 2

ANSWER KEY

1.	(b)	2.	(c)	3.	(b)	4.	(c)	5.	(b)	6.	(b)	7.	(b)	8.	(d)	9.	(b)	10.	(a)
11.	(c)	12.	(a)	13.	(b)	14.	(a)	15.	(c)	16.	(a)	17.	(a)	18.	(c)	19.	(c)	20.	(b)
21.	(d)	22.	(a)	23.	(b)	24.	(a)	25.	(b)	26.	(c)	27.	(d)	28.	(d)	29.	(c)	30.	(c)
31.	(a)	32.	(b)	33.	(c)	34.	(d)	35.	(a)	36.	(c)	37.	(c)	38.	(b)	39.	(d)	40.	(b)
41.	(c)	42.	(b)	43.	(b)	44.	(b)	45.	(d)	46.	(a)	47.	(c)	48.	(a)	49.	(b)	50.	(c)
51.	(b)	52.	(d)	53.	(c)	54.	(b)	55.	(a)	56.	(c)	57.	(c)	58.	(b)	59.	(a)	60.	(c)
61.	(d)	62.	(c)	63.	(c)	64.	(c)	65.	(b)	66.	(b)	67.	(b)	68.	(b)	69.	(c)	70.	(a)
71.	(c)	72.	(c)	73.	(b)	74.	(c)	75.	(b)	76.	(b)	77.	(c)	78.	(a)	79.	(c)	80.	(c)
81.	(b)	82.	(c)	83.	(b)	84.	(b)	85.	(c)	86.	(b)	87.	(a)	88.	(d)	89.	(b)		

CAPF PYQ

- A vehicle with mileage 15 km per litre contains 2 L of fuel. The vehicle gets some defect as a result of which 5 L of fuel gets wasted per hour when the engine is on. With what minimum speed the vehicle has to move to travel 20 km with the existing amount of fuel, if it travels with a uniform speed?
[CAPF 2016]
(a) 100 km per hour
(b) 120 km per hour
(c) 150 km per hour
(d) 200 km per hour
- Two men set out at the time to walk towards each other from points A and B, 72 km apart. The first man walks at the speed of 4 kmph while the second walks 2 km in the first hour, 2.5 km in second hour, 3 km in the third hour and so on. The two men will meet
[CAPF 2017]
(a) in 8 hours
(b) nearer to A than B
(c) nearer to B than A
(d) midway between A and B
- A runner's average speed reduces by 25% every hour. If he runs 16km in the first hour and he runs for 3 hours, then what is his overall average speed?
[CAPF 2022]
(a) 12km/hr
(b) 12.33km/hr
(c) 10.33km/hr
(d) 13km/hr
- A car travels $\frac{3}{4}$ th of the distance at a speed of 60 km/hr and the remaining $\frac{1}{4}$ th of the distance at a speed of v km/hr. If the average speed for the full journey is 50km/hr, then the value of v is:
[CAPF 2022]
(a) 40
(b) 30
(c) 100/3
(d) 35
- Two friends 10km apart start running towards each other at speeds of 10km/hr and 14km/hr respectively. After how much time will they meet each other?
[CAPF 2022]
(a) 20 minutes
(b) 25 minutes
(c) 28 minutes
(d) 30 minutes

ANSWER KEY

1.	(c)	2.	(d)	3.	(b)	4.	(c)	5.	(b)										
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PRACTICE SHEET 1

1. If 80% of A = 50% of B and B = x% of A, then the value of x is :
 (1) 400 (2) 300
 (3) 160 (4) 150
2. If x is 80% of y, what percent of x is y ?
 (1) 75% (2) 80%
 (3) 100% (4) 125%
3. If 8% of x is the same as 4% of y, then 20% of x is the same as:
 (1) 10% of y (2) 16% of y
 (3) 80% of y (4) 50% of y
4. A student multiplied a number by $\frac{3}{5}$ instead of $\frac{5}{3}$. What is the percentage error in the calculation?
 (1) 44% (2) 34%
 (3) 54% (4) 64%
5. If p% of p is 36, then p is equal to :
 (1) 3600 (2) 600
 (3) 60 (4) 15
6. 2 is what percent of 50?
 (1) 2 (2) 2.5
 (3) 4 (4) 5
7. $\frac{2}{3}$ is what percent of $\frac{1}{3}$?
 (1) 50% (2) $33\frac{1}{3}\%$
 (3) 150% (4) 200%
8. 0.15% of $33\frac{1}{3}\%$ of 10000 is :
 (1) 5 (2) 150
 (3) 0.05 (4) 105
9. 30% of x is 72. The value of x is:
 (1) 216 (2) 240
 (3) 480 (4) 640
10. If 15% of (A + B) = 25% of (A - B), then what per cent of B is equal to A?
 (1) 10% (2) 60%
 (3) 200% (4) 400%
11. A supply of juice lasts for 35 days. If its use is increased by 40% the number of days would the same amount of juice lasts, is
 (1) 25 days (2) 30 days
 (3) 24 days (4) 27 days
12. If 60% of A = 30% of B, B = 40% of C, C = x% of A, then value of x is
 (1) 200 (2) 500
 (3) 800 (4) 300
13. In an office, 40% of the staff is female. 70% of the female staff and 50% of the male staff are married. The percentage of the unmarried staff in the office is
 (1) 64 (2) 60
 (3) 54 (4) 42
14. 50% of a number when added to 50 is equal to the number. The number is
 (1) 50 (2) 75
 (3) 100 (4) 150
15. $83\frac{1}{3}\%$ of Rs. 90 is equal to 60% of ?
 (1) Rs. 123 (2) Rs. 124
 (3) Rs. 122 (4) Rs. 125
16. 51% of a whole number is 714. 25% of that number is
 (1) 350 (2) 450
 (3) 550 (4) 250
17. What is to be added to 15% of 160 so that the sum may be equal to 25% of 240
 (1) 24 (2) 84
 (3) 60 (4) 36
18. What per cent of a day is 30 minutes?
 (1) 2.83 (2) 2.083
 (3) 2.09 (4) 2.075
19. A basket contains 300 mangoes. 75 mangoes were distributed among some students. Find the percentage of mangoes left in the basket.
 (1) 70% (2) 72%
 (3) 76% (4) 75%
20. The weights of two iron balls are 3.5 kg and 7.5 kg. What is the percentage weight of the first ball with respect to second ball?
 (1) $46\frac{2}{3}\%$ (2) 35%

(3) $46\frac{1}{3}\%$

(4) 45%

21. If x is less than y by 25% then y exceed x by:

(1) $33\frac{1}{3}\%$

(2) 25%

(3) 75%

(4) $66\frac{2}{3}\%$

22. If x is 10% more than y , then by what per cent is y less than x ?

(1) $9\frac{1}{11}\%$

(2) $7\frac{1}{11}\%$

(3) $8\frac{1}{11}\%$

(4) $10\frac{1}{11}\%$

23. If A's height is 10% more than B's height, by how much per cent less is B's height than that of A?

(1) 10%

(2) $10\frac{1}{9}\%$

(3) $10\frac{1}{11}\%$

(4) $9\frac{1}{11}\%$

24. B got 20% marks less than A. What per cent marks did A got more than B?

(1) 20%

(2) 25%

(3) 12%

(4) 80%

25. If x earns 25% more than y . What percent less does y earn than x ?

(1) 16%

(2) 10%

(3) 20%

(4) 25%

26. The difference of two numbers is 20% of the larger number. If the smaller number is 20, the larger number is :

(1) 25

(2) 45

(3) 50

(4) 80

27. If a number x is 10% less than another number y and y is 10% more than 125, then x is equal to :

(1) 150

(2) 143

(3) 140.55

(4) 123.75

28. Two numbers are respectively $12\frac{1}{2}\%$ and 25% more than a third number. The first number as percentage of second number is

(1) 50

(2) 60

(3) 75

(4) 90

29. Which number is 40% less than 90?

(1) 36

(2) 54

(3) 50

(4) 60

30. Two numbers are less than a third number by 30% and 37% respectively. The per cent by which the second number is less than the first is

(1) 10%

(2) 7%

(3) 4%

(4) 3%

ANSWER KEY

1.	(3)	2.	(4)	3.	(1)	4.	(4)	5.	(3)	6.	(3)	7.	(4)	8.	(1)	9.	(2)	10.	(4)
11.	(1)	12.	(2)	13.	(4)	14.	(3)	15.	(4)	16.	(1)	17.	(4)	18.	(2)	19.	(4)	20.	(1)
21.	(1)	22.	(1)	23.	(4)	24.	(2)	25.	(3)	26.	(1)	27.	(4)	28.	(4)	29.	(2)	30.	(1)

PRACTICE SHEET 2

1. If the total monthly income of 16 persons is 80,800 and the income of one of them is 120% of the average income, then his income is

(1) 5,050

(2) 6,060

(3) 6,160

(4) 6,600

2. Mita's income is 25% more than that of Sita. What percent is Sita's income less than that of Mita?

(1) 25%

(2) 24%

(3) $22\frac{1}{2}\%$

(4) 20%

3. A man spends $12\frac{1}{2}\%$ of his salary on items of daily use and 30% of the remainder on house rent. After that he is left with 2940. How much is his salary?

- (1) 4800 (2) 5200
(3) 4500 (4) 4000
4. The monthly income of a person was 13,500 and his monthly expenditure was 9,000. Next year his income increased by 14% and his expenditure increased by 7%. The per cent increase in his savings was
(1) 7% (2) 21%
(3) 28% (4) 35%
5. A worker suffers a 20% cut in his wages. He may regain his original wages by obtaining a rise of
(1) 27.5% (2) 25.0%
(3) 22.5% (4) 20.0%
6. Given that 10% of A's income = 15% of B's income = 20% of C's income. If sum of their income is 7800, then B's income is:
(1) 3600 (2) 3000
(3) 2400 (4) 1800
7. If A's income is 25% less than B's income, by how much percent is B's income more than that of A ?
(1) 25% (2) 30%
(3) $33\frac{1}{3}\%$ (4) $66\frac{2}{3}\%$
8. A's salary is 50% more than that of B. How much per cent is B's salary less than that of A?
(1) 50% (2) $33\frac{1}{3}\%$
(3) 45% (4) $66\frac{2}{3}\%$
9. Tulsiram's salary is 20% more than that of Kashyap. If Tulsiram saves 720 which is 4% of his salary, then Kashyap's salary is
(1) 15,000 (2) 12,000
(3) 10,000 (4) 22,000
10. A's salary is 40% of B's salary and B's salary is 25% of C's salary. What percentage of C's salary is A's salary?
(1) 5% (2) 10%
(3) 15% (4) 20%
11. The price of sugar is increased by 20%. If the expenditure on sugar has to be kept the same as earlier, the ratio between the reduction in consumption and the original consumption is
(1) 1 : 3 (2) 1 : 4
(3) 1 : 6 (4) 1 : 5
12. Rama's expenditures and savings are in the ratio 5 : 3. If her income increases by 12% and expenditure by 15%, then by how much per cent do her savings increase?
(1) 12% (2) 7%
(3) 8% (4) 13%
13. The ratio of two numbers is 4:5 when the first is increased by 20% and the second is decreased by 20%, the ratio of the resulting numbers is
(1) 4 : 5 (2) 5 : 4
(3) 5 : 6 (4) 6 : 5
14. If 60% of A = $\frac{3}{4}$ of B, then A : B is
(1) 9 : 20 (2) 20 : 9
(3) 4 : 5 (4) 5 : 4
15. If A exceeds B by 60% and B is less than C by 20%, then A : C is
(1) 32 : 25 (2) 25 : 32
(3) 8 : 5 (4) 4 : 5
16. If 30% of (B - A) = 18% of (B + A), then the ratio A : B is equal to
(1) 4 : 1 (2) 1 : 4
(3) 5 : 4 (4) 5 : 9
17. If 60% of A = 30% of B, B = 40% of C, C = x% of A, then value of x is
(1) 200 (2) 500
(3) 800 (4) 300
18. In an office, 40% of the staff is female. 70% of the female staff and 50% of the male staff are married. The percentage of the unmarried staff in the office is
(1) 64 (2) 60
(3) 54 (4) 42
19. 50% of a number when added to 50 is equal to the number. The number is
(1) 50 (2) 75
(3) 100 (4) 150
20. $83\frac{1}{3}\%$ of Rs. 90 is equal to 60% of ?
(1) Rs. 123 (2) Rs. 124
(3) Rs. 122 (4) Rs. 125

- 21.** 51% of a whole number is 714. 25% of that number is
(1) 350 (2) 450
(3) 550 (4) 250
- 22.** In a marriage party 32% are women, 54% are men and there are 196 children. How many men are there in the marriage party?
(1) 756 (2) 448
(3) 332 (4) 324
- 23.** What per cent of a day is 30 minutes?
(1) 2.83 (2) 2.083
(3) 2.09 (4) 2.075
- 24.** A basket contains 300 mangoes. 75 mangoes were distributed among some students. Find the percentage of mangoes left in the basket.
(1) 70% (2) 72%
(3) 76% (4) 75%
- 25.** The weights of two iron balls are 3.5 kg and 7.5 kg. What is the percentage weight of the first ball with respect to second ball?
(1) $46\frac{2}{3}\%$ (2) 35%
(3) $46\frac{1}{3}\%$ (4) 45%
- 26.** If A earns 25% more than B then how much percent does B earn less than A :
(1) 15% (2) 20%
(3) 25% (4) 30%
- 27.** What per cent of 1 day is 36 minutes?
(1) 25% (2) 2.5%
(3) 3.6% (4) 0.25%
- 28.** One number is 25% of another number. The larger number is 12 more than the smaller. The larger number is
(1) 48 (2) 16
(3) 4 (4) 12
- 29.** The number of students in a class is increased by 20% and the number now becomes 66. Initially the number was
(1) 45 (2) 50
(3) 55 (4) 60
- 30.** In a school there are 1500 students, 44% of them are girls. Monthly fee of each boy is Rs. 540 and the fee of each girl is 25% less than that of a boy. The sum of fees of boys and girls both is
(1) Rs. 720600 (2) Rs. 720800
(3) Rs. 720900 (4) Rs. 721000

ANSWER KEY

1.	(2)	2.	(4)	3.	(1)	4.	(3)	5.	(2)	6.	(3)	7.	(3)	8.	(2)	9.	(1)	10.	(2)
11.	(3)	12.	(2)	13.	(4)	14.	(4)	15.	(1)	16.	(2)	17.	(2)	18.	(4)	19.	(3)	20.	(4)
21.	(1)	22.	(1)	23.	(2)	24.	(4)	25.	(1)	26.	(2)	27.	(2)	28.	(2)	29.	(3)	30.	(3)

PRACTICE SHEET 3

- 1.** Salary of a person is first increased by 20%, then it is decreased by 20%. Percentage change in his salary is:
(1) 4% decreased (2) 4% increased
(3) 8% decreased (4) 20% increased
- 2.** A number is increased by 20% and then it is decreased by 10%. Find the net increase or decrease per cent.
(1) 10% increase (2) 10% decrease
(3) 8% increase (4) 8% decrease
- 3.** The tax imposed on an article is decreased by 10% and its consumption increases by 10%. Find the percentage change in revenue from it.
(1) 10% increase (2) 2% decrease
(3) 1% decrease (4) 11% increase
- 4.** The price of an article was decreased by 10% and again reduced by 10%. By what per cent should the price have been reduced once, in order to produce the same effect as these two successive reductions?
(1) 15% (2) 19%
(3) 20% (4) 25%

5. If price of a book is first decreased by 25% and then increased by 20%, the net change in the price of the book will be
 (1) 10% decrease (2) 5% decrease
 (3) no change (4) 5% increase
6. The price of an article is reduced by 25% but the daily sale of the article is increased by 30%. The net effect on the daily sale receipts is
 (1) 2.5% increase (2) 2.5% decrease
 (3) 2% increase (4) 2% decrease
7. Two successive price increases of 10% and 10% of an article are equivalent to a single price increase of
 (1) 19% (2) 20%
 (3) 21% (4) 22%
8. The price of an article was first increased by 10% and then again by 20%. If the last increased price be 33, the original price was
 (1) 30 (2) 27.50
 (3) 26.50 (4) 25
9. If a number is increased by 20% and the resulting number is again increased by 20%, what per cent is the total increase ?
 (1) 48% (2) 44%
 (3) 41% (4) 40%
10. A number is increased by 10% and then it is decreased by 10%. The net change in the number is
 (1) no increase or decrease
 (2) 2% decrease
 (3) 1% increase
 (4) 1% decrease
11. If radius of a circle is increased by 5%, then the increase in its area is
 (1) 10.25 % (2) 10 %
 (3) 5.75 % (4) 5 %
12. The percentage increase in the area of a rectangle, if each of its sides is increased by 20% is equal to
 (1) 32% (2) 34%
 (3) 42% (4) 44%
13. If the radius of a circle is decreased by 10%, then the area of the circle is decreased by
 (1) 89% (2) 18%
 (3) 19% (4) 25%
14. If the area of a square is increased by 44%, retaining its shape as a square, each of its sides increases by :
 (1) 19% (2) 21%
 (3) 22% (4) 20%
15. What will be the percentage increase in the area of a square when each of the its sides is increased by 10%?
 (1) 20 (2) 11
 (3) 121 (4) 21
16. If the radius of a cylinder is decreased by 50% and the height is increased by 50% to form a new cylinder, the volume will be decreased by
 (1) 0% (2) 25%
 (3) 62.5% (4) 75%
17. Each of the height and base-radius of a cone is increased by 100%. The percentage increase in the volume of the cone is
 (1) 700% (2) 400%
 (3) 300% (4) 100%
18. If both the radius and height of a right circular cone are increased by 20%, its volume will be increased by
 (1) 20% (2) 40%
 (3) 60% (4) 72.8%
19. If the height of a right circular cone is increased by 200% and the radius of the base is reduced by 50%, the volume of the cone
 (1) increases by 25% (2) increases by 50%
 (3) remains unaltered (4) decreases by 25%
20. If the height and the radius of the base of a cone are each increased by 100%, then the volume of the cone becomes
 (1) double that of the original
 (2) three times that of the original
 (3) six times that of the original
 (4) eight times that of the original
21. A sample of 50 litres of glycerin is found to be adulterated to the extent of 20%. How much pure glycerin should be added to it so as to bring down the percentage of impurity to 5% ?
 (1) 155 litres (2) 150 litres
 (3) 150.4 litres (4) 149 litres
22. 1 litre of water is added to 5 litres of alcohol-water solution containing 40% alcohol

strength. The strength of alcohol in the new solution will be

- (1) 30% (2) 33%
(3) 33.66% (4) 33.33%

23. If 4 litres of water is evaporated on boiling from 12 litres of salt solution containing 7 percentage salt, the percentage of salt in the remaining solution is

- (1) 10.5% (2) 11.5%
(3) 12% (4) 13%

24. A vessel has 60 litres of solution of acid and water having 80% acid. How much water be added to make it a solution in which acid forms 60%?

- (1) 48 litres (2) 20 litres
(3) 36 litres (4) None of these

25. 75 gm of sugar solution has 30% sugar in it. Then the quantity of sugar that should be added to the solution to make the quantity of the sugar 70% in the solution, is

- (1) 125 gm (2) 100 gm
(3) 120 gm (4) 130 gm

26. A litre of pure alcohol is added to 6 litres of 30% alcohol solution. The percentage of water in the solution is

- (1) 50% (2) 65%
(3) 60% (4) 40%

27. An ore contains 25% of an alloy that has 90% iron. Other than this, in the remaining 75% of the ore, there is no iron. To obtain 60 kg of pure iron, the quantity of the ore needed (in kgs) is approximately

- (1) 250.57 (2) 266.67
(3) 275.23 (4) 300

28. How much water must be added to 100 ml of 80 per cent solution of boric acid to reduce it to a 50 per cent solution?

- (1) 30 ml (2) 40 ml
(3) 50 ml (4) 60 ml

29. In one litre of a mixture of alcohol and water, water is 30%. The amount of alcohol that must be added to the mixture so that the part of water in the mixture becomes 15% is :

- (1) 1000 ml (2) 700 ml
(3) 300 ml (4) 900 ml

30. One type of liquid contains 20% water and the second type of liquid contains 35% of water. A glass is filled with 10 parts of first liquid and 4 parts of second liquid. The water in the new mixture in the glass is

- (1) 37% (2) 46%
(3) 12.14% (4) 24.28%

ANSWER KEY

1.	(1)	2.	(3)	3.	(3)	4.	(2)	5.	(1)	6.	(2)	7.	(3)	8.	(4)	9.	(2)	10.	(4)
11.	(1)	12.	(4)	13.	(3)	14.	(4)	15.	(4)	16.	(3)	17.	(1)	18.	(4)	19.	(4)	20.	(4)
21.	(2)	22.	(4)	23.	(1)	24.	(2)	25.	(2)	26.	(3)	27.	(2)	28.	(4)	29.	(1)	30.	(4)

PRACTICE SHEET 4

1. The present population of a city is 180000. If it increases at the rate of 10% per annum, its population after 2 years will be:

- (1) 207800 (2) 227800
(3) 217800 (4) 237800

2. The value of an equipment depreciates by 20% each year. How much less will the value of the equipment be after 3 years ?

- (1) 48.8% (2) 51.2%
(3) 54% (4) 60%

3. A district has 64000 inhabitants. If the population increases at the rate of 2.5% per annum, the number of inhabitants at the end of 3 years will be

- (1) 70000 (2) 69200
(3) 68921 (4) 68911

4. The value of a property depreciates every year by 10% of its value at the beginning of

the year. The present value of the property is 8100. What was its value 2 years ago?

- (1) 10,000 (2) 9000
(3) 9500 (4) 9801

5. The population of a town 2 years ago was 62,500. Due to migration to big cities, it decreases every year at the rate of 4%. The present population of the town is:

- (1) 57,600 (2) 56,700
(3) 76,000 (4) 75,000

6. The population of a town increases every year by 4%. If its present population is 50,000, then after 2 years it will be

- (1) 53,900 (2) 54,000
(3) 54,080 (4) 54,900

7. A man received 8,80,000 as his annual salary of the year 2007 which was 10% more than his annual salary in 2006. His annual salary in the year 2006 was

- (1) 4,80,000 (2) 8,00,000
(3) 4,00,000 (4) 8,40,000

8. Present population of a village is 67600. It has been increasing annually at the rate of 4%. What was the population of the village two years ago?

- (1) 62500 (2) 63000
(3) 64756 (4) 65200

9. The value of a machine depreciates by 5% every year. If its present value is 2,00,000, its value after 2 years will be

- (1) 1,80,500 (2) 1,99,000
(3) 1,80,000 (4) 2,10,000

10. The value of a property decreases every year at the rate of 5%. If its present value is 4,11,540, what was its value 3 years ago ?

- (1) 4,50,000 (2) 4,60,000
(3) 4,75,000 (4) 4,80,000

11. 8% of the voters in an election did not cast their votes. In this election, there were only two candidates. The winner by obtaining 48% of the total votes defeated his contestant by 1100 votes. The total number of voters in the election was:

- (1) 21000 (2) 23500
(3) 22000 (4) 27500

12. In an election between two candidates, 75% of the voters cast their votes, out of which 2% votes were declared invalid. A candidate got 9261 votes which were 75% of the valid

votes. The total number of voters enrolled in that election was

- (1) 16000 (2) 16400
(3) 16800 (4) 18000

13. In an election between two candidates, the candidate getting 60% of the votes polled, is elected by a majority of 14,000 votes. The number of votes polled by the winning candidate is

- (1) 28,000 (2) 32,000
(3) 42,000 (4) 46,000

14. In an office 40% of the staff is female, 40% of the females and 60% of the males voted for me. The percentage of votes I got was

- (1) 24% (2) 42%
(3) 50% (4) 52%

15. In an election there were only two candidates. One of the candidates secured 40% of votes and is defeated by the other candidate by 298 votes. The total number of votes polled is

- (1) 745 (2) 1460
(3) 1490 (4) 1500

16. In an assembly election, a candidate got 55% of the total valid votes. 2% of the total votes were declared invalid. If the total number of voters is 104000, then the number of valid votes polled in favour of the candidate is:

- (1) 56506 (2) 56650
(3) 56560 (4) 56056

17. Two candidates contested in an election. One got 60% of the votes and won by 1600 votes. What is the number of votes polled?

- (1) 9000 (2) 8000
(3) 10000 (4) 7500

18. In an election, three candidates contested. The first candidate got 40% votes and the second got 36% votes. If total number of votes polled were 36000, find the number of votes got by the 3rd candidate.

- (1) 8040 (2) 8640
(3) 9360 (4) 9640

19. Two persons contested an election of Parliament. The winning candidate secured 57% of the total votes polled and won by a majority of 42,000 votes. The number of total votes polled is

- (1) 5,00,000 (2) 6,00,000
(3) 3,00,000 (4) 4,00,000

- 20.** In an election, a candidate who gets 84 % of the votes is elected by a majority of 476 votes. What is the total number of votes polled?
 (1) 900 (2) 810
 (3) 600 (4) 700
- 21.** In an examination 34% failed in Mathematics and 42% failed in English. If 20% failed in both the subjects, the percentage of students who passed in both subjects was
 (1) 54% (2) 50%
 (3) 44% (4) 56%
- 22.** A candidate secured 30% marks in an examination and failed by 6 marks. Another secured 40% marks and got 6 marks more than the bare minimum to pass. The maximum marks are
 (1) 150 (2) 120
 (3) 100 (4) 180
- 23.** In an examination, 52% students failed in Hindi and 42% in English. If 17% failed in both the subjects, what percentage of students passed in both the subjects?
 (1) 38% (2) 33%
 (3) 23% (4) 18%
- 24.** In a group of students, 70% can speak English and 65% can speak Hindi. If 27% of the students can speak none of the two languages, then what per cent of the group can speak both the languages?
 (1) 38% (2) 62%
 (3) 28% (4) 23%
- 25.** 25% of the candidates who appeared in an examination failed to qualify and only 450 candidates qualified. The number of candidates, who appeared in the examination, was
 (1) 700 (2) 600
 (3) 550 (4) 500
- 26.** In two successive years 100 and 75 students of a school appeared at the final examination. Respectively, 75% and 60% of them passed. The average rate of pass is
 (1) 68.57 % (2) 78%
 (3) 80.5 % (4) 80%
- 27.** A student has to secure minimum 35% marks to pass in an examination. If he gets 200 marks and fails by 10 marks, then the maximum marks are
 (1) 300 (2) 400
 (3) 500 (4) 600
- 28.** A candidate who scores 30 per cent fails by 5 marks, while another candidate who scores 40 per cent marks gets 10 more than minimum pass marks. The minimum marks required to pass are
 (1) 50 (2) 70
 (3) 100 (4) 150
- 29.** In an examination, 60% of the candidates passed in English and 70% of the candidates passed in Mathematics, but 20% failed in both of these subjects. If 2500 candidates passed in both the subjects, the number of candidates who appeared at the examination was
 (1) 3000 (2) 3500
 (3) 4000 (4) 5000
- 30.** In a test a student got 30% marks and failed by 25 marks. In the same test another student got 40% marks and secured 25 marks more than the essential minimum pass marks. The maximum marks for the test were
 (1) 400 (2) 480
 (3) 500 (4) 580

ANSWER KEY

1.	(3)	2.	(1)	3.	(3)	4.	(1)	5.	(1)	6.	(3)	7.	(2)	8.	(1)	9.	(1)	10.	(4)
11.	(4)	12.	(3)	13.	(3)	14.	(4)	15.	(3)	16.	(4)	17.	(2)	18.	(2)	19.	(3)	20.	(4)
21.	(3)	22.	(2)	23.	(3)	24.	(2)	25.	(2)	26.	(1)	27.	(4)	28.	(1)	29.	(4)	30.	(3)

PRACTICE SHEET 5

1. Price of sugar rises by 20%. By how much percent should the consumption of sugar be reduced so that the expenditure does not change?
 (1) 20% (2) 10%
 (3) $16\frac{2}{3}\%$ (4) 15%
2. If food prices go up by 10%, by how much should a man reduce his consumption so as not to increase his expenditure?
 (1) $9\frac{1}{11}\%$
 (2) 10%
 (3) $11\frac{1}{9}\%$
 (4) The data is not sufficient
3. In the new budget, the price of kerosene oil rose by 25%. By how much per cent must a person reduce his consumption of kerosene oil so that his expenditure on it does not increase?
 (1) 20% (2) 25%
 (3) 50% (4) 40%
4. If the price of tea is increased by 20%, by how much per cent the consumption of tea be reduced so that there is no increase in the expenditure on it?
 (1) $83\frac{1}{3}\%$ (2) 20%
 (3) $16\frac{2}{3}\%$ (4) $8\frac{1}{3}\%$
5. If the price of a commodity is decreased by 20% and its consumption is increased by 20%, what will be the increase or decrease in the expenditure on the commodity?
 (1) 4% increase (2) 4% decrease
 (3) 8% decrease (4) 8% increase
6. The price of a certain item is increased by 15%. If a consumer wants to keep his expenditure on the item the same as before, how much per cent must he reduce his consumption of that item?
 (1) 15% (2) $13\frac{1}{23}\%$
- (3) $16\frac{2}{3}\%$ (4) $10\frac{20}{23}\%$
7. If the price of a commodity is increased by 50%, by what fraction must its consumption be reduced so as to keep the same expenditure on its consumption?
 (1) $\frac{1}{4}$ (2) $\frac{1}{3}$
 (3) $\frac{1}{2}$ (4) $\frac{2}{3}$
8. If the price of rice be raised by 25%, the percent by which a house-holder must reduce his consumption of rice so as not to increase his expenditure on rice is
 (1) 225% (2) 25.75%
 (3) 25% (4) 20%
9. If the duty on an article is reduced by 40% of its present rate, by how much per cent must its consumption increase in order that the revenue remains unaltered
 (1) 60% (2) $62\frac{1}{3}\%$
 (3) 72% (4) $66\frac{2}{3}\%$
10. Price of milk has increased by 20%. To keep the expenditure unchanged, the present consumption s to be reduced by :
 (1) 20% (2) 18%
 (3) 10% (4) $16\frac{2}{3}\%$
11. The Government reduced the price of sugar by 10 per cent. By this a consumer can buy 6.2 kg more sugar for 837. The reduced price per kg of sugar is
 (1) 12.50 (2) 13.00
 (3) 13.50 (4) 14.00
12. The price of sugar is reduced by 20%. Now a person can buy 500g more sugar for 36. The original price of the sugar per kilogram was
 (1) 14.40 (2) 18
 (3) 15.60 (4) 16.50
13. A reduction of 10% in the price of sugar enables a housewife to buy 6.2 kg more for 1116. The reduced price per kg is
 (1) 12 (2) 14
 (3) 16 (4) 18

- 14.** When the price of sugar decreases by 10%, a man could buy 1 kg more for 270. Then the original price of sugar per kg is
 (1) 25 (2) 30
 (3) 27 (4) 32
- 15.** A reduction of 20% in the price of an apple enables a man to buy 10 apples more for 54. The reduced price of apples per dozen is
 (1) 4.32 (2) 12.96
 (3) 10.80 (4) 14.40
- 16.** Due to an increase of 50% in the price of eggs, 4 eggs less are available for 24. The present rate of eggs per dozen is :
 (1) 24 (2) 27
 (3) 36 (4) 42
- 17.** A reduction of 20% in the price of wheat enables Lalita to buy 5 kg more wheat for ₹ 320. The original rate (in rupees per kg) of wheat was
 (1) 16 (2) 18
 (3) 20 (4) 21
- 18.** Due to an increase of 20% in the price of eggs, 2 eggs less are available for 24. The present rate of eggs per dozen is :
 (1) 25.00 (2) 26.20
 (3) 27.80 (4) 28.80
- 19.** A reduction of 25% in the price of rice enables a person to buy 10 kg more rice for Rs 600. The reduced per kg price of rice is
 (1) Rs 30 (2) Rs 25
 (3) Rs 20 (4) Rs 15
- 20.** A reduction in the price of apples enables a person to purchase 3 apples for ₹ 1 instead of ₹ 1.25. What is the % of reduction in price (approximately)?
 (1) 20% (2) 25%
 (3) 30% (4) $33\frac{1}{3}\%$
- 21.** A number, on subtracting 15 from it, reduces to its 80%. What is 40% of the number?
 (1) 75 (2) 60
 (3) 30 (4) 90
- 22.** A reduction of 21% in the price of an item enables a person to buy 3 kg more for ₹ 100. The reduced price of item per kg is
 (1) 5.50 (2) 7.50
 (3) 10.50 (4) 7.00
- 23.** A reduction of 20% in the price of sugar enables a purchaser to obtain 8 kg more for Rs. 160. Then the price per kg before reduction was
 (1) Rs. 5 (2) Rs. 6
 (3) Rs. 10 (4) Rs. 4
- 24.** The price of an item was increased by 10%. This reduced the monthly total sales by 20%. The overall effect on the value of monthly sales is a
 (1) 10% increase (2) 10% decrease
 (3) 12% increase (4) 12% decrease
- 25.** The price of rice has increased by 60%. In order to restore the original price, the new price must be reduced by
 (1) $33\frac{1}{3}\%$ (2) $37\frac{1}{2}\%$
 (3) 40% (4) 45%
- 26.** In a village, each of the 60% of families has a cow; each of the 30% of families has a buffalo and each of the 15% of families has both a cow and buffalo. In all there are 96 families in the village. How many families do not have a cow or a buffalo?
 (1) 20 (2) 24
 (3) 26 (4) 28
- 27.** A man invests a part of ₹ 10,000 at 5% and the remainder at 6%. The 5% investment yields annually ₹ 76.50 more than the 6% investment. The amount invested at 6% is
 (1) 3,600 (2) 3,550
 (3) 3,850 (4) 4,000
- 28.** A number is divided into two parts in such a way that 80% of 1st part is 3 more than 60% of 2nd part and 80% of 2nd part is 6 more than 90% of the 1st part. Then the number is
 (1) 125 (2) 130
 (3) 135 (4) 145
- 29.** Price of a one gram gold coin decreased by 10% on its initial price on Monday and increased by 20% on Tuesday and again increased by 8% on Wednesday, and 5% increase on Thursday. If the final price on Thursday is ₹ 5511.24, then the initial price (in Rs) of one gram gold coin on Monday was?
 (a) 4500 (b) 4250
 (c) 4000 (d) 5000

30. Radha saves $x\%$ of her income. If her income increases by 28% and the expenditure increases by 20%, then her savings increases by 40%. What is the value of x ?

- (a) 35 (b) 40
(c) 50 (d) 25

ANSWER KEY

1.	(3)	2.	(1)	3.	(1)	4.	(3)	5.	(2)	6.	(2)	7.	(2)	8.	(4)	9.	(4)	10.	(4)
11.	(3)	12.	(2)	13.	(4)	14.	(2)	15.	(2)	16.	(3)	17.	(1)	18.	(4)	19.	(4)	20.	(1)
21.	(3)	22.	(4)	23.	(1)	24.	(4)	25.	(2)	26.	(2)	27.	(3)	28.	(3)	29.	(1)	30.	(2)

CDS PYQ

1. A person's salary has increased from Rs 7200 to Rs 8100. What is percentage increase in his salary?

[CDS 2013 (II)]

- (a) 25% (b) 18%
(c) $16\frac{2}{3}\%$ (d) 12.5%

2. A water pipe is cut into two pieces. The longer piece is 70% of the length of the pipe. By how much percentage is the longer piece longer than the shorter piece.

[CDS 2014 (I)]

- (a) 140% (b) $400/3\%$
(c) 40% (d) None of these

3. If $m\%$ of $m + n\%$ of $n = 2\%$ of mn , then what percent of m is n

[CDS 2014 (II)]

- (a) 50%
(b) 75%
(c) 100%
(d) Cannot determined

4. If side of a cube is increased by 100% , then by what percentage is the surface area of the cube increased ?

[CDS 2014 (II)]

- (a) 150% (b) 200%
(c) 300% (d) 400%

5. The price of commodity increased by 5% from 2010 to 2011, 8% from 2011 to 2012 and 7% from 2012 to 2013. What is the average price increase(approx.) from 2010 to 2013?

[CDS 2014 (II)]

- (a) 26% (b) 32%
(c) 24% (d) 30%

6. The radius of a circle is increased by 6%, then the area will increase by

[CDS 2014 (II)]

- (a) 6% (b) 9%
(c) 12.36% (d) 16.64%

7. A person could save 10% of his income. But 2 years later, when his income increased by 20%, he could save the same amount only as before. By how much percentage has his expenditure increased?

[CDS 2015 (I)]

- (a) $22\frac{2}{9}\%$ (b) $23\frac{1}{3}\%$
(c) $24\frac{2}{9}\%$ (d) $25\frac{2}{9}\%$

8. 20% of a number when added to 20 becomes the number itself, then the number is

[CDS 2015 (II)]

- (a) 20 (b) 25
(c) 50 (d) 80

9. A's salary was increased by 40% and then decreased by 20%. On the whole A's salary is increased by

[CDS 2015 (II)]

- (a) 60% (b) 40%
(c) 20% (d) 12%

10. In an election 10% of the voters on the voter list did not cast their vote and 60 voters cast their ballot papers blank. There were only two candidates. The winner was supported by 47% of total voters in the voter list and he got 308 voters more than his rival. The number of voters on the voter list is

[CDS 2015 (II)]

- (a) 3600 (b) 6200
(c) 6028 (d) 6400

11. The salary of a person is increased by 10% of his original salary. But he received the same amount even after increment. What is the percentage of his salary he did not receive?

[CDS 2016 (I)]

- (a) 11% (b) 10%
(c) 100/11% (d) 90/11%

12. The expenditure of a household for a certain month is Rs 20,000 out of which Rs 8000 is spent on education, Rs 5900 on food, Rs 2800 on shopping and the rest on personal care. What percentage of expenditure is spent on personal care?

[CDS 2016 (I)]

- (a) 12% (b) 16.5%
(c) 18% (d) 21.8%

13. A candidate scoring $x\%$ marks in an examination fails by a marks, while another candidate who scores $y\%$ marks gets b marks more than the minimum required pass marks. What is the maximum marks for the examination?

[CDS 2016 (II)]

- (a) $\frac{100(a+b)}{x-y}$ (b) $\frac{100(a-b)}{x+y}$
(c) $\frac{100(a+b)}{y-x}$ (d) $\frac{100(a-b)}{x-y}$

14. If $a\%$ of $a + b\%$ of $b = 2\%$ of ab , then what percent of a is b ?

[CDS 2017 (I)]

- (a) 50%
(b) 75%
(c) 100%
(d) Cannot be determined

15. $\frac{5}{9}$ th part of the population in a village are males. If 30% of the males are married, the percentage of unmarried females in the total population is?

[CDS 2017 (I)]

- (a) $20\frac{2}{9}\%$ (b) $27\frac{2}{9}\%$
(c) $27\frac{7}{9}\%$ (d) $29\frac{2}{9}\%$

16. A fruit seller has a certain number of mangoes of which 5% are rotten. He sells 75% of the remainder and he is left with 95 mangoes. How many mangoes did he have originally?

[CDS 2017 (II)]

- (a) 500 (b) 450
(c) 400 (d) 350

17. A student has to secure 40% of marks to pass an examination. He gets only 45 marks and fails by 5 marks. The maximum marks are

[CDS 2017 (II)]

- (a) 120 (b) 125
(c) 130 (d) 150

18. When prices rise by 12%, if the expenditure is to be the same, what is the percentage of consumption to be reduced?

[CDS 2017 (II)]

- (a) $16\frac{2}{3}\%$ (b) $10\frac{2}{7}\%$
(c) $16\frac{3}{5}\%$ (d) $10\frac{5}{7}\%$

19. If the price of wheat rises by 25%, then by how much percent must a man reduce his consumption in order to keep his budget the same as before?

[CDS 2018 (I)]

- (a) 15% (b) 20%
(c) 25% (d) 30%

20. A person bought two articles X and Y from a departmental store. The sum of prices before sales tax was Rs 130. There was no sales tax on the article X and 9% sales tax on the article Y. The total amount the person paid, including the sales tax was Rs 136.75. What was the price of the article Y before sales tax?

[CDS 2018 (II)]

- (a) Rs 75 (b) Rs 85
(c) Rs 122 (d) Rs 125

21. In the hostel the rent per room is increased by 20%. If the number of rooms in the hostel is also increased by 20% and the hostel is always full, then what is the percentage change in the total collection at the cash counter?

[CDS 2019 (I)]

- (a) 30% (b) 40%
(c) 44% (d) 48%

- 22.** x, y and z are three numbers such that x is 30% of z and y is 40% of z. If x is p% of y, then what is the value of p?
[CDS 2019 (I)]
(a) 45 (b) 55
(c) 65 (d) 75
- 23.** The train fare and bus fare between two stations is in the ratio 3 : 4, If the train fare increases by 20% and bus fare increases by 30% , then what is the ratio between revised train fare and revised bus fare ?
[CDS 2019 (II)]
(a) 9/13 (b) 17/12
(c) 32/43 (d) 19/21
- 24.** If the annual income of X is 20% more than that of Y, then the income of Y is less than that of X by p%. What is the value of p?
[CDS 2020 (I)]
(a) 10 (b) $16\frac{2}{3}$
(c) $17\frac{1}{3}$ (d) 20
- 25.** The price of an article X is increases by 20% every year and price of article Y increases by 10% every year. In the year 2010, the price of article X was Rs 5000 and price of article Y was Rs 2000. In which year the difference in their prices exceeded Rs 5000 for the first time?
[CDS 2020 (I)]
(a) 2012 (b) 2013
(c) 2014 (d) 2015
- 26.** The number of items in a booklet is N. In the first year there is an increase of x% in this number and in the subsequent year there is a decrease of x%. At the end of the two years, what will be the number of items in the booklet?
[CDS 2020 (II)]
(a) Less than N (b) Equal to N
(c) More than N (d) It depends on N
- 27.** X sales his goods 25% cheaper than Y and 25% dearer than Z, How much percentage is Z's goods cheaper than Y?
[CDS 2021 (I)]
(a) 100/3% (b) 40%
(c) 50% (d) 200/3%
- 28.** In a mixture of 80 litres of a liquid and water, 25% of the mixture is the liquid. How much water should be added to the mixture so that the liquid becomes 20% of the mixture?
[CDS 2021 (I)]
(a) 15 litres (b) 20 litres
(c) 24 litres (d) 25 litres
- 29.** If p varies directly as q and inversely as square of r, what is the percentage increase in p due to an increase in q by 20% and a decrease in r by 20%?
[CDS 2021 (II)]
(a) 87.5% (b) 85%
(c) 82.5% (d) 80%

ANSWER KEY

1.	(d)	2.	(b)	3.	(c)	4.	(c)	5.	(a)	6.	(c)	7.	(a)	8.	(b)	9.	(d)	10.	(b)
11.	(c)	12.	(b)	13.	(c)	14.	(c)	15.	(c)	16.	(c)	17.	(b)	18.	(d)	19.	(b)	20.	(a)
21.	(c)	22.	(d)	23.	(a)	24.	(b)	25.	(b)	26.	(a)	27.	(b)	28.	(b)	29.	(a)		

CAPF PYQ

1. How much water to be added to 75ml of alcohol so that the mixture contains 25% of alcohol?

[CAPF 2021]

- (a) 100ml (b) 225 ml
(c) 250 ml (d) 125 ml

2. If 35% of a number is 416 more than 27% of the same number, then the number is

[CAPF 2021]

- (a) 5200 (b) 2600
(c) 3900 (d) 3328

3. A tree is at present 9 feet tall, every year it grows $\frac{1}{9}$ of its height, will be the height of the tree three years?

[CAPF 2021]

- (a) 12 feet (b) 12.34 feet
(c) 13 feet (d) 13.10 feet

ANSWER KEY

1.	(b)	2.	(a)	3.	(b)														
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PRACTICE SHEET 1

- A man buys a cycle for 1400 and sells it at a loss of 15%. What is the selling price of the cycle?
(1) 1202 (2) 1190
(3) 1160 (4) 1000
- On selling an article for 651, there is a loss of 7%. The cost price of that article is
(1) 744 (2) 751
(3) 793 (4) 700
- A milkman bought 70 litres of milk for 630 and added 5 litres of water. If he sells it at 9.00 per litre, his profit percentage is
(1) $8\frac{1}{5}\%$ (2) 7%
(3) $8\frac{2}{5}\%$ (4) $7\frac{1}{7}\%$
- In terms of percentage profit, which is the best transaction?

C.P. (in)	Profit (In)
(I) 36	17
(II) 50	24
(III) 40	19
(IV) 60	29
(1) I	(2) II
(3) III	(4) IV
- A man bought an old typewriter for 1200 and spent 200 on its repair. He sold it for 1680. His profit per cent is:
(1) 20% (2) 10%
(3) 8% (4) 16%
- If the cost price is 95% of the selling price, what is the profit percent?
(1) 4% (2) 4.75%
(3) 5% (4) 5.26%
- A merchant buys an article for 27 and sells it at a profit of 10% of the selling price. The selling price of the article is :
(1) 29.70 (2) 30
(3) 37 (4) 32
- If the cost price of an article is 80% of its selling price, the profit per cent is :
(1) 20 % (2) $22\frac{1}{2}\%$
- (3) 24% (4) 25%
- Krishnan bought a camera and paid 20% less than its original price. He sold it at 40% profit on the price he had paid. The percentage of profit earned by Krishnan on the original price was
(1) 22% (2) 32%
(3) 12% (4) 15%
- By selling a car for 64,000, Mr. Rao lost 20%. Then the cost price of the car is :
(1) 72,000 (2) 76,800
(3) 80,000 (4) 84,000
- By selling an article, a man makes a profit of 25% of its selling price. His profit per cent is
(1) 20% (2) 25%
(3) $16\frac{2}{3}\%$ (4) $33\frac{1}{3}\%$
- If there is a profit of 20% on the cost price of an article, the percentage of profit calculated on its selling price will be
(1) 24 (2) $16\frac{2}{3}$
(3) $8\frac{1}{3}$ (4) 20
- A man purchased a bed sheet for Rs 450 and sold it at a gain of 10% calculated on the selling price. The selling price of the bed sheet was
(1) Rs 460 (2) Rs 475
(3) Rs 480 (4) Rs 500
- By selling an article for 960 a man incurs a loss of 4%; what was the cost price?
(1) 1,000 (2) 784
(3) 498.4 (4) 300
- A salesman expects a gain of 13% on his cost price. If in a month his sale was 7,91,000, what was his profit ?
(1) 85,659 (2) 88,300
(3) 91,000 (4) 97,786
- A merchant purchases a wrist watch for 450 and fixes its list price in such a way that after allowing a discount of 10%, he earns a

- profit of 20%. Then the list price of the watch is
 (1) 650 (2) 700
 (3) 550 (4) 600
- 17.** The cost price of a radio is 600. The 5% of the cost price is charged towards transportation. After adding that, if the net profit to be made is 15%, then the selling price of the radio must be
 (1) 704.50 (2) 724.50
 (3) 664.50 (4) 684.50
- 18.** If there is a profit of 20% on the cost price, the percentage of profit on the sale price is
 (1) $16\frac{2}{3}\%$ (2) 12 %
 (3) $15\frac{1}{3}\%$ (4) 16 %
- 19.** Nisha bought a number of oranges at 2 for a rupee and an equal number at 3 for a rupee. To make a profit of 20% she should sell a dozen for
 (1) 6 (2) 8
 (3) 10 (4) 12
- 20.** If selling price of an article is $1\frac{1}{3}$ of cost price, find gain %.
 (1) 25% (2) $33\frac{1}{3}\%$
 (3) 1.33% (4) $66\frac{2}{3}\%$
- 21.** A merchant loses 10% by selling an article. If the cost price of the article is 15, then the selling price of the article is
 (1) 13.20 (2) 16.50
 (3) 12.30 (4) 13.50
- 22.** Pooja wants to sell a watch at a profit of 20%. She bought it at 10% less and sold it at 30 less, but still she gained 20%. The cost price of watch is
 (1) 240 (2) 220
 (3) 250 (4) 225
- 23.** A fruit merchant makes a profit of 25% by selling mangoes at a certain price. If he charges Rs.1 more on each mango, he would gain 50%. At first the price of one mango was
 (1) Rs. 5 (2) Rs. 7
 (3) Rs. 4 (4) Rs. 6
- 24.** There is 10% loss if an article is sold at Rs. 270. Then the cost price of the article is
 (1) Rs. 300 (2) Rs. 270
 (3) Rs. 320 (4) Rs. 250
- 25.** If bananas are bought at the rate of 4 for a rupee, how many must be sold for a rupee so as to gain $33\frac{1}{3}\%$?
 (1) 2.5 (2) 2
 (3) 3 (4) 4
- 26.** If the profit on selling an article for Rs. 425 is the same as the loss on selling it for Rs. 355, then the cost price of the article is
 (1) Rs. 410 (2) Rs. 380
 (3) Rs. 400 (4) Rs. 390
- 27.** The C.P of 10 articles is equal to the S.P. of 15 articles. What is the profit or loss percentage?
 (1) 25.5% (2) 35%
 (3) 10% (4) 33.3%
- 28.** The selling price of 6 bananas is equal to the cost price of 8 bananas. Then the percentage of profit is:
 (1) 20 (2) $33\frac{1}{3}$
 (3) 25 (4) 30
- 29.** By selling a bag at Rs. 230, profit of 15% is made. The selling price of the bag, when it is sold at 20% profit would be
 (1) Rs. 250 (2) Rs. 205
 (3) Rs. 240 (4) Rs. 200
- 30.** A man gains 20% by selling an article for a certain price. If he sells it at double the price, the percentage of profit will be
 (1) 40% (2) 100%
 (3) 120% (4) 140%

ANSWER KEY

1.	(2)	2.	(4)	3.	(4)	4.	(4)	5.	(1)	6.	(4)	7.	(2)	8.	(4)	9.	(3)	10.	(3)
11.	(4)	12.	(2)	13.	(4)	14.	(1)	15.	(3)	16.	(4)	17.	(2)	18.	(1)	19.	(1)	20.	(2)
21.	(4)	22.	(3)	23.	(1)	24.	(1)	25.	(3)	26.	(4)	27.	(4)	28.	(2)	29.	(3)	30.	(4)

PRACTICE SHEET 2

- The cost price of 36 books is equal to the selling price of 30 books. The gain per cent is :
 (1) 20% (2) $16\frac{2}{3}\%$
 (3) 18% (4) $82\frac{1}{3}\%$
- The cost price of 15 articles is same as the selling price of 10 articles. The profit percent is:
 (1) 30% (2) 40%
 (3) 50% (4) 45%
- The selling price of 5 articles is the same as the cost price of 3 articles. The gain or loss percent is:
 (1) 20% gain (2) 25% gain
 (3) 33.33% loss (4) 40% loss
- If the cost price of 15 tables be equal to the selling price of 20 tables, the loss per cent is:
 (1) 20% (2) 30%
 (3) 25% (4) 37.5%
- The cost price of 18 articles is equal to the selling price of 15 articles. The gain per cent is
 (1) 15% (2) 20%
 (3) 25% (4) 18%
- A person sells 400 mangoes at the cost price of 320 mangoes. His percentage of loss is
 (1) 10% (2) 15%
 (3) 20% (4) 25%
- If the cost price of 50 oranges is equal to the selling price of 40 oranges, then the profit per cent is
 (1) 5% (2) 10%
 (3) 20% (4) 25%
- If the cost price of 12 oranges is equal to selling price of 10 oranges, then the percentage of profit is
 (1) $16\frac{2}{3}\%$ (2) 20%
 (3) 18% (4) 25%
- If the cost price of 10 articles is equal to the selling price of 9 articles, the gain or loss per cent is
 (1) $11\frac{1}{9}\%$ profit (2) $7\frac{6}{17}\%$ profit
 (3) $11\frac{1}{9}\%$ loss (4) $1\frac{12}{13}\%$ loss
- A man sells 320 mangoes at the cost price of 400 mangoes. His gain percent is:
 (1) 15% (2) 20%
 (3) 25% (4) 10%
- If the cost price of 12 pens is equal to the selling price of 8 pens, the gain per cent is :
 (1) $33\frac{1}{3}\%$ (2) $66\frac{2}{3}\%$
 (3) 25% (4) 50%
- The cost price of 8 articles is equal to the selling price of 9 articles. The profit or loss per cent in the transaction is
 (1) $12\frac{1}{2}\%$ loss (2) $12\frac{1}{2}\%$ profit
 (3) $11\frac{1}{9}\%$ loss (4) $11\frac{1}{9}\%$ profit
- A sold an article to B at 20% profit and B sold to C at 15% loss. If A sold it to C at the selling price of B, then A would make

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| <p>(1) 5% profit (2) 2% profit
(3) 2% loss (4) 5% loss</p> <p>14. If the cost price of 10 articles is equal to the selling price of 7 articles, then the gain or loss per cent is:</p> <p>(1) 51% gain (2) $42\frac{6}{7}$ % gain
(3) 35 % loss (4) $42\frac{6}{7}$ % loss</p> <p>15. Mahesh purchased a radio at $\frac{9}{10}$ of its selling price and sold it at 8% more than its original selling price. His gain per cent is :</p> <p>(1) 20 % (2) 18%
(3) 10 % (4) 8 %</p> <p>16. A coconut merchant finds that the cost price of 2750 coconuts is the same as the selling price of 2500 coconuts. The loss or gain per cent is</p> <p>(1) 5% loss (2) 15% loss
(3) 20% gain (4) 10% gain</p> <p>17. If the cost price of 10 articles is equal to the selling price of 16 articles, then the loss per cent is</p> <p>(1) 30% (2) 37.5%
(3) 42.5% (4) 45%</p> <p>18. If the selling price of 4 articles is equal to the cost price of 5 articles, the profit percent is</p> <p>(1) 20% (2) $22\frac{1}{2}$ %
(3) 25% (4) 30%</p> <p>19. The selling price of 10 oranges is the cost price of 13 oranges. Then the profit percentage is</p> <p>(1) 30% (2) 10%
(3) 13% (4) 3%</p> <p>20. If the selling price of 10 articles is equal to the cost price of 11 articles, then the gain percent is</p> <p>(1) 10% (2) 11%
(3) 15% (4) 25%</p> <p>21. A cloth merchant on selling 33 metres of cloth obtains a profit equal to the selling price of 11 metres of cloth. The profit percent is</p> | <p>(1) 40% (2) 22%
(3) 50% (4) 11%</p> <p>22. A shopkeeper buys 144 items at 90 paise each. On the way 20 items are broken. He sells the remainder at 1.20 each. His gain per cent correct to one place of decimal is</p> <p>(1) 13.8% (2) 14.6%
(3) 14.8% (4) 15.8%</p> <p>23. If goods be purchased for 450 and one third sold at a loss of 10%. At what gain percent should the remainder be sold so as to gain 20% on the whole transaction?</p> <p>(1) 32% (2) 35%
(3) 28% (4) 30%</p> <p>24. A shoe company sold 50 pairs of shoes on a day costing 189.50 each for 10,000. Then the profit obtained in is</p> <p>(1) 522 (2) 525
(3) 573 (4) 612</p> <p>25. The cost price of 25 books is equal to the selling price of 20 books. The profit per cent is</p> <p>(1) 20% (2) 22%
(3) 24% (4) 25%</p> <p>26. If the selling price of 40 articles is equal to the cost price of 50 articles, the loss or gain per cent is</p> <p>(1) 25% gain (2) 20% gain
(3) 25% loss (4) 20% loss</p> <p>27. If the cost price of 20 books is the same as selling price of 25 books, then the loss percentage is</p> <p>(1) 20 (2) 25
(3) 22 (4) 24</p> <p>28. By what fraction selling price (S.P.) must be multiplied to get the cost price (C.P.) if the loss is 20%?</p> <p>(1) $\frac{4}{5}$ (2) $\frac{8}{5}$
(3) $\frac{5}{4}$ (4) $\frac{6}{5}$</p> <p>29. A store sells a watch for a profit of 25% of its cost price. Then the percentage of profit against selling price is:</p> <p>(1) 22% (2) 20%
(3) 18% (4) 15%</p> <p>30. The per cent profit made when an article is sold for Rs. 78 is twice as much as when it</p> |
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is sold for Rs. 69. The cost price of the article is

- (1) Rs. 60
(2) Rs. 51
(3) Rs. 55.50
(4) Rs. 70

ANSWER KEY

1.	(1)	2.	(3)	3.	(4)	4.	(3)	5.	(2)	6.	(3)	7.	(4)	8.	(2)	9.	(1)	10.	(3)
11.	(4)	12.	(3)	13.	(2)	14.	(2)	15.	(1)	16.	(4)	17.	(2)	18.	(3)	19.	(1)	20.	(1)
21.	(3)	22.	(3)	23.	(2)	24.	(2)	25.	(4)	26.	(1)	27.	(1)	28.	(3)	29.	(2)	30.	(1)

PRACTICE SHEET 3

- A table is sold at a profit of 13%. If it is sold for 25 more, profit is 18 %. Cost price of table is
(1) 100
(2) 500
(3) 200
(4) 1, 000
- A man sold his watch at a loss of 5%. Had he sold it for 56.25 more, he would have gained 10%. What is the cost price of the watch (in)?
(1) 370
(2) 365
(3) 375
(4) 390
- Kamala bought a bicycle for 1,650. She had to sell it at a loss of 8%. She sold it for
(1) 1, 581
(2) 1, 518
(3) 1, 510
(4) 1, 508
- A table is sold at Rs. 1,800 at a loss of 10%. At what price should it be sold to earn a profit of 15%?
(1) 2,070
(2) 1,890
(3) 2,000
(4) 2,300
- A manufacturer sells an item to a wholesale dealer at a profit of 18%. The wholesaler sells the same to a retailer at a profit of 20%. The retailer in turn sells it to a customer for 15045 thereby earning a profit of 25%. The cost price of the manufacturer is
(1) 8000
(2) 8500
(3) 9000
(4) 10000
- A man sold an article at a gain of 5%. Had he sold it for Rs. 240 more, he would have gained 8%. The cost price of the article is
(1) Rs. 6,000
(2) Rs. 10,000
(3) Rs. 12,000
(4) Rs. 8,000
- A radio is sold at a profit of 20%. Had it been sold for Rs. 60 more the profit would have been 30%. The cost price of the radio is
(1) Rs. 500
(2) Rs. 600
(3) Rs. 550
(4) Rs. 620
- A dealer sold a bicycle at a profit of 10%. Had he bought the bicycle at 10% less price and sold it at a price Rs. 60 more, he would have gained 25%. The cost price of the bicycle was
(1) Rs. 2400
(2) Rs. 2600
(3) Rs. 2000
(4) Rs. 2200
- If 3 articles are sold for the cost of 5 articles, then the profit percentage is :
(1) 50
(2) 60
(3) $66\frac{2}{3}$
(4) 65
- A sold a watch at a gain of 5% to B and B sold it to C at a gain of 4%. If C paid Rs. 91 for it, the price paid by A is:
(1) Rs. 83.33
(2) Rs. 84.33
(3) Rs. 83
(4) Rs. 82.81
- Arun buys one kilogram of apples for Rs. 120 and sells it to Swati gaining 25%. Swati sells it to Divya who again sells it for Rs. 198, making a profit of 10%. What is the profit percentage made by Swati?
(1) 25%
(2) 20%
(3) 16.67%
(4) 15%
- A dealer sold an article at 6% loss. Had he sold it for Rs. 64 more, he would have made a profit of 10%. Then the cost of the article is
(1) Rs. 400
(2) Rs. 200
(3) Rs. 164
(4) Rs. 464
- Profit after selling a commodity for 524 is the same as loss after selling it for 452. The cost price of the commodity is
(1) 480
(2) 500

- (3) 488 (4) 485
- 14.** A clock was sold for 144. If the percentage of profit was numerically equal to the cost price, the cost of the clock was
 (1) 72 (2) 80
 (3) 90 (4) 100
- 15.** By selling 144 hens Mahesh suffered a loss equal to the selling price of 6 hens. His loss percent is
 (1) 4% (2) 3%
 (3) 9% (4) $4\frac{1}{2}\%$
- 16.** If the profit per cent got on selling an article is numerically equal to its cost price in rupees and the selling price is 39, then cost price (in) will be
 (1) 20 (2) 22
 (3) 28 (4) 30
- 17.** By selling 1 dozen ball pens, a shopkeeper earned the profit equal to the selling price of 4 ball pens. His profit per cent is
 (1) 50% (2) 40%
 (3) $33\frac{1}{3}\%$ (4) $31\frac{1}{4}\%$
- 18.** A merchant sold an article for 75 at a profit percent equal to its cost price. The cost price of the article was:
 (1) 45 (2) 50
 (3) 54 (4) 60
- 19.** If the profit on selling an article for 425 is the same as the loss on selling it for 355, then the cost price of the article is
 (1) 370 (2) 380
 (3) 390 (4) 400
- 20.** The loss incurred on selling 21 articles equals the selling price of 3 articles. Then the loss per cent is
 (1) $9\frac{1}{11}\%$ (2) 10%
 (3) $12\frac{1}{2}\%$ (4) $11\frac{1}{9}\%$
- 21.** A man sold 250 chairs and had a gain equal to selling price of 50 chairs. His profit per cent is:
 (1) 20% (2) 25%
 (3) 50% (4) 15%
- 22.** On selling 17 balls at 720, there is a loss equal to the cost price of 5 balls. The cost price (in) of a ball is
 (1) 45 (2) 50
 (3) 55 (4) 60
- 23.** A dishonest dealer professes to sell his goods at cost price but uses a weight of 875 gms for the kilogram weight. His gain in percentage is a
 (1) 17% (2) $14\frac{5}{7}\%$
 (3) $14\frac{2}{7}\%$ (4) 14%
- 24.** A dishonest dealer professes to sell his goods at cost price but uses a weight of 800 gms for the kilogram weight. His gain in percentage is a
 (1) 20% (2) 25%
 (3) 30% (4) 35%
- 25.** A dishonest shopkeeper professes to sell goods at his cost price but uses a false weight of 50 gms, for each kilogram. His gain per cent is:
 (1) $6\frac{1}{4}\%$ (2) $5\frac{5}{19}\%$
 (3) $5\frac{3}{17}\%$ (4) $6\frac{2}{7}\%$
- 26.** A dishonest dealer defrauds to the extent of $x\%$ in buying as well as selling his goods by using faulty weight. What will be the gain per cent on his outlay?
 (1) $2x\%$ (2) $\left(\frac{10}{x} + x^2\right)\%$
 (3) $\left(\frac{x^2}{100} + x\right)\%$ (4) None of these
- 27.** A shopkeeper sells rice at 10% profit and uses weight 30% less than the actual measure. His gain per cent is
 (1) $57\frac{2}{3}\%$ (2) $57\frac{1}{7}\%$
 (3) $57\frac{2}{5}\%$ (4) $57\frac{3}{7}\%$
- 28.** A shopkeeper sells rice at 9% loss and uses weight 30% less than the actual measure. His gain per cent is

- (1) 30% (2) 25%
(3) 20% (4) 15%

29. A shopkeeper sells rice at 10% profit and uses weight 20% less than the actual measure. His gain per cent is

- (1) $57\frac{2}{3}\%$ (2) $37\frac{1}{2}\%$
(3) 25% (4) $57\frac{3}{7}\%$

30. A shopkeeper promise to sell his goods at x% profit but he uses 20% less weight thus gain

$37\frac{1}{2}\%$. Find x per cent is?

- (1) 10% (2) 25%
(3) 20% (4) 15%

ANSWER KEY

1.	(2)	2.	(3)	3.	(2)	4.	(4)	5.	(2)	6.	(4)	7.	(2)	8.	(1)	9.	(3)	10.	(1)
11.	(2)	12.	(1)	13.	(3)	14.	(2)	15.	(1)	16.	(4)	17.	(1)	18.	(2)	19.	(3)	20.	(3)
21.	(2)	22.	(4)	23.	(3)	24.	(2)	25.	(2)	26.	(4)	27.	(2)	28.	(1)	29.	(2)	30.	(1)

PRACTICE SHEET 4

- Applied to a bill for 1,00,000 the difference between a discount of 40% and two successive discounts of 36% and 4% is :
(1) Nil (2) 1,440
(3) 2,500 (4) 4,000
- Successive discounts of 10% and 30% are equivalent to a single discount of:
(1) 40% (2) 35%
(3) 38% (4) 37%
- The marked price of a watch was 720/-. A man bought the same for 550.80, after getting two successive discounts, the first at 10%. What was the second discount rate?
(1) 12% (2) 14%
(3) 15% (4) 18%
- An item is marked for 240 for sale. If two successive discounts of 10% and 5% are allowed on the sale price, the selling price of the article will be
(1) 205.20 (2) 204
(3) 34.80 (4) 36
- The price of an article is raised by 30% and then two successive discounts of 10% each are allowed. Ultimately the price of the article is
(1) increased by 10% (2) increased by 5.3%
(3) decreased by 3% (4) decreased by 5.3%
- A single discount equivalent to the successive discounts of 10%, 20% and 25% is
(1) 55% (2) 45%
(3) 46% (4) 60%
- A shopkeeper gives two successive discounts on an article marked 450. The first discount given is 10 per cent. If the customer pays 344.25 for the article, the second discount given is
(1) 14 per cent (2) 10 per cent
(3) 12 per cent (4) 15 per cent
- A company offers three types of successive discounts: (i) 25% and 15%, (ii) 30% and 10%, (iii) 35% and 5%. Which offer is the best for a customer?
(1) First offer
(2) Second offer
(3) Third offer
(4) Any one; all are equally good
- An article is listed at 900 and two successive discounts of 8% and 8% are given on it. How much would the seller gain or lose, if he

gives a single discount of 16%, instead of two discounts?

- (1) Gain of 4.76 (2) Loss of 5.76
(3) Gain of 5.76 (4) Loss of 4.76

- 10.** The marked price of a T.V. is 16,000. After two successive discounts it is sold for 11,400.

If the first discount is 5%, then the rate of second discount is

- (1) 15% (2) 20%
(3) 30% (4) 25%

- 11.** The difference between successive discounts of 40% followed by 30% and 45% followed by 20% on the marked price of an article is Rs.

12. The marked price of the article is:

- (1) 800 (2) 400
(3) 200 (4) 600

- 12.** Buy 4 Get 1 free is equivalent to a discount of

- (1) 15% (2) 20%
(3) 30% (4) 25%

- 13.** Buy 7 Get 3 free is equivalent to a discount of

- (1) 15% (2) 20%
(3) 30% (4) 25%

- 14.** Buy 3 Get 1 free + 20% more discount is equivalent to a discount of

- (1) 25% (2) 30%
(3) 35% (4) 40%

- 15.** Buy 4 Get 1 free + 10% more discount is equivalent to a discount of

- (1) 22% (2) 24%
(3) 26% (4) 28%

- 16.** A trader marked the selling price of an article at 10% above the cost price. At the time of selling, he allows certain discount and suffers a loss of 1%. He allowed the discount of:

- (1) 11% (2) 10%
(3) 9% (4) 10.5%

- 17.** A shopkeeper marks his goods at 30% above the cost price but allows a discount of 10% at the time of sale. His gain is

- (1) 21% (2) 20%
(3) 18% (4) 17%

- 18.** A shopkeeper marks the price of an item keeping 20% profit. If he offers a discount of

$12\frac{1}{2}\%$ on the marked price, his gain

percent will be

- (1) 4.5% (2) 5%
(3) 7.5% (4) 8%

- 19.** A seller marks his goods 30% above their cost price but allows 15% discount for cash payment. His percentage of profit when sold in cash is

- (1) 10.5% (2) 15%
(3) 9% (4) 8.5%

- 20.** A tradesman marks his goods at 25% above its cost price and allows purchasers a

discount of $12\frac{1}{2}\%$ for cash payment. The profit, he thus makes, is

- (1) $9\frac{3}{8}\%$ (2) $9\frac{1}{2}\%$
(3) $8\frac{1}{2}\%$ (4) $8\frac{3}{8}\%$

- 21.** What price should a shopkeeper mark on an article costing him 200 to gain 35% after allowing a discount of 25%?

- (1) 270 (2) 300
(3) 330 (4) 360

- 22.** A trader marks his goods 40% above cost price and allows a discount of 25%. The profit he makes, is:

- (1) 15% (2) 10%
(3) 5% (4) 2%

- 23.** A dealer marks his goods 20% above their cost price. He then allows some discount on marked price so that he makes a profit of 10%. The rate of discount is

- (1) $10\frac{1}{3}\%$ (2) $9\frac{1}{3}\%$
(3) $8\frac{2}{3}\%$ (4) $8\frac{1}{3}\%$

- 24.** In a shop, shirts are usually sold at 40% above the cost price. During a sale, the shopkeeper offers a discount of 10% off the usual selling price. If he manages to sell 72 shirts for 13,608, then his cost price per shirt, (in) is

- (1) 210 (2) 150
(3) 149 (4) 125

- 25.** If a shopkeeper marks the price of goods 50% more than their cost price and allows a discount of 40%, what is his gain or loss percent?
 (1) Gain of 10% (2) Loss of 10%
 (3) Gain of 20% (4) Loss of 20%
- 26.** The marked price of a shirt and trousers are in the ratio 1 : 2. The shopkeeper gives 40% discount on the shirt. If the total discount on the set of the shirt and trousers is 30%, the discount offered on the trousers is
 (1) 15% (2) 20%
 (3) 25% (4) 30%
- 27.** A retailer purchases a grinder at a discount of 15% and sells it for 1955 at a profit of 15%. The amount of discount received by the retailer from the wholesaler was
 (1) 270 (2) 290
 (3) 300 (4) 330
- 28.** A dozen pair of socks quoted at 80 are available at a discount of 10%. How many pairs of socks can be bought for 24?
 (1) 4 (2) 5
 (3) 3 (4) 6
- 29.** If an electricity bill is paid before due date, one gets a reduction of 4% on the amount of the bill. By paying the bill before due date a person got a reduction of 13. The amount of his electricity bill was
 (1) 125 (2) 225
 (3) 325 (4) 425
- 30.** A shopkeeper lists the price of an article as 500. But he gives a certain discount which allows the buyer to pay 500 for the article including 10% sales tax. The rate of discount is
 (1) 10% (2) $10\frac{1}{11}\%$
 (3) $9\frac{1}{11}\%$ (4) 11%.

ANSWER KEY

1.	(2)	2.	(4)	3.	(3)	4.	(1)	5.	(2)	6.	(3)	7.	(4)	8.	(3)	9.	(2)	10.	(4)
11.	(4)	12.	(2)	13.	(3)	14.	(4)	15.	(4)	16.	(2)	17.	(4)	18.	(2)	19.	(1)	20.	(1)
21.	(4)	22.	(3)	23.	(4)	24.	(2)	25.	(2)	26.	(3)	27.	(3)	28.	(1)	29.	(3)	30.	(3)

PRACTICE SHEET 5

- 1.** A person sells two machines at 396 each. On one he gains 10% and on the other he loses 10%. His profit or loss in the whole transaction is :
 (1) no gain no loss (2) 1% loss
 (3) 1% profit (4) 8% profit
- 2.** A dealer sold two types of goods for 10,000 each. On one of them, he lost 20% and on the other he gained 20%. His gain or loss per cent in the entire transaction was
 (1) 2% loss (2) 2% gain
 (3) 4% gain (4) 4% loss
- 3.** A television and a refrigerator were sold for 12,000 each. If the television was sold at a loss of 20% of the cost and the refrigerator at a gain of 20% of the cost, the entire transaction resulted in
 (1) No loss or gain (2) Loss of 1,000
 (3) Gain of 1,000 (4) Loss of 1,200
- 4.** A man had 100 kgs of sugar, part of which he sold at 7% profit and rest at 17% profit. He gained 10% on the whole. How much did he sell at 7% profit?
 (1) 65 kg (2) 35 kg
 (3) 30 kg (4) 70 kg
- 5.** A man bought two goats for 1008. He sold one at a loss of 20% and other at a profit of 44%. If each goat was sold for the same price, the cost price of the goat which was sold at loss, was:
 (1) 648 (2) 360
 (3) 568 (4) 440

6. Two bicycles were sold for 3990 each, gaining 5% on one and losing 5% on other. The gain or loss per cent on the whole transaction is :
 (1) neither gain nor loss
 (2) 2.5% gain
 (3) 2.5% loss
 (4) 0.25% loss
7. A man sold two watches for 240 each. On one he gains 20% and incurs a loss of 20% on another. What is his gain or loss per cent in this transaction?
 (1) 1% profit (2) 2% loss
 (3) 4% profit (4) 4% loss
8. When the price of cloth was reduced by 25%, the quantity of cloth sold increased by 20%. What was the effect on gross receipt of the shop?
 (1) 5% increase (2) 5% decrease
 (3) 10% increase (4) 10% decrease
9. A cloth merchant sold half of his cloth at 20% profit, half of the remaining cloth at 20% loss and the rest was sold at his cost price. In the total transaction, his gain or loss will be
 (1) 5% profit
 (2) Neither loss nor gain
 (3) 5% loss
 (4) 10% profit
10. The total cost price of two watches is 840. One is sold at a profit of 16 per cent and the other at a loss of 12 per cent. There is no loss or gain in the whole transaction. The cost price of the watch on which the shopkeeper gains, is
 (1) 360 (2) 370
 (3) 380 (4) 390
11. A shopkeeper purchased 510 eggs at the rate of Rs. 20 per dozen. 30 eggs were broken on the way. In order to make a gain of 20%, he must sell the remaining eggs at the rate of
 (1) Rs. 22.50 per dozen
 (2) Rs. 25.50 per dozen
 (3) Rs. 26 per dozen
 (4) Rs. 26.50 per dozen
12. A sells a watch to B and makes a loss of 12%. B makes a profit of $12\frac{1}{2}\%$ by selling the watch to C. If A sells the watch to B at the cost of which C purchased it, then the percentage of loss or profit of A will be
 (1) 1% loss (2) 1% profit
 (3) 2% loss (4) 2% profit
13. A man buys 3 type-I cakes and 6 type-II cakes for Rs. 900. He sells type-I cakes at a profit of 15% and type-II cakes at a loss of 10%. If his overall profit is Rs.30, the cost price (in Rs.) of a type-I and of a type-II cakes is
 (1) 100, 100 (2) 160, 70
 (3) 180, 60 (4) 120, 90
14. A merchant buys 25 litres of milk daily at the rate of Rs. 12 per litre. He mixes 5 litres of water in it and sells at the rate Rs. 10.40 per litre. His gain is :
 (1) 8% profit (2) 2% profit
 (3) 4% profit (4) 6% profit
15. A trader had 22 quintals of wheat. He sold a part of it at 23% profit and the rest at 33% profit, so that he made a total profit of 27%. How much wheat did he sell at 33% profit?
 (1) 1320 kg. (2) 440 kg.
 (3) 880 kg. (4) 1760 kg.
16. A milkman mixes water with milk and sells the mixture at the cost price of pure milk. The volume of water in litres to be mixed with each litre of milk to get a 25% profit is
 (1) $\frac{1}{4}$
 (2) $\frac{1}{5}$
 (3) $1\frac{1}{4}$
 (4) cannot be calculated without knowing the cost price of milk
17. The difference between the selling prices of an article at a profit of 15% and at a profit of 10% is 10. The cost price of the article is
 (1) 100 (2) 120
 (3) 150 (4) 200
18. The difference between the selling price and cost price of an article is 210. If the profit percent is 25, then the selling price of the article is
 (1) 950 (2) 1,050
 (3) 1,150 (4) 1,250

- 19.** A sells a cycle to B at a profit of 10%, B sells to C at a profit of 20%. If C pays 264 for it, how much did A pay for it?
 (1) 200 (2) 220
 (3) 225 (4) 234
- 20.** A man purchased an article and sold it to B at a profit of 25% and B sold it to C at a loss of 10% and C paid 675 for it. For how much did A purchase it (in)?
 (1) 625 (2) 575
 (3) 600 (4) 550
- 21.** A milkman makes 20% profit by selling milk mixed with water at 9 per litre. If the cost price of 1 litre pure milk is 10, then the ratio of milk and water in the said mixture is
 (1) 3 : 1 (2) 4 : 1
 (3) 3 : 2 (4) 4 : 3
- 22.** Nita blends two varieties of tea-one costing 180 per kg and another costing 200 per kg in the ratio 5 : 3. If she sells the blended variety at 210 per kg, then her gain percent is
 (1) 10% (2) 11%
 (3) 12% (4) 13%
- 23.** In what ratio Darjeeling Tea costing 320 per kg be mixed with Assam Tea costing 250 per kg so that there is a gain of 20% by selling the mixture at 324 per kg ?
 (1) 1 : 2 (2) 2 : 3
 (3) 3 : 2 (4) 2 : 5
- 24.** The ratio, in which tea costing 192 per kg is to be mixed with tea costing 150 per kg so that the mixed tea, when sold for 194.40 per kg, gives a profit of 20%, is
 (1) 2 : 5 (2) 3 : 5
 (3) 5 : 3 (4) 5 : 2
- 25.** The ratio of the quantities of sugar, in which sugar costing 20 per kg. and 15 per kg. should be mixed so that there will be neither loss nor gain on selling the mixed sugar at the rate of 16 per kg, is
 (1) 2 : 1 (2) 1 : 2
 (3) 4 : 1 (4) 1 : 4
- 26.** The ratio in which the Darjeeling tea at 32 per kg is mixed with the Assam tea at 25 per kg so as to gain 20% by selling the mixture at 32.40 per kg is
 (1) 4 : 3 (2) 3 : 4
 (3) 5 : 2 (4) 2 : 5
- 27.** 7 kg of tea costing 280 per kg is mixed with 9 kg of tea costing 240 per kg. The average price per kg of the mixed tea is
 (1) 255.80 (2) 257.50
 (3) 267.20 (4) 267.50
- 28.** A shopkeeper bought 15kg of rice at the rate of 29 per kg and 25kg of rice at the rate of 20 per kg. He sold the mixture of both types of rice at the rate of 27 per kg. His profit in this transaction is
 (1) 125 (2) 150
 (3) 140 (4) 145
- 29.** Costs of two watches were in the ratio of 16 : 23. The cost of first watch increases by 10% and that of second by Rs. 477. Now the costs of two watches are in a ratio of 11 : 20. The price of the second watch (in Rs.) in the beginning was
 (1) 932 (2) 1219
 (3) 1696 (4) 848
- 30.** The liquids, X and Y are mixed in the ratio of 3 : 2 and the mixture is sold at Rs. 11 per litre at a profit of 10%. If the liquid X costs Rs. 2 more per litre than Y, the cost of X per litre is (in Rs.) :
 (1) 10.80 (2) 11.75
 (3) 9.50 (4) 11

ANSWER KEY

1.	(2)	2.	(4)	3.	(2)	4.	(4)	5.	(1)	6.	(4)	7.	(4)	8.	(4)	9.	(1)	10.	(1)
11.	(2)	12.	(1)	13.	(2)	14.	(3)	15.	(3)	16.	(2)	17.	(4)	18.	(2)	19.	(1)	20.	(3)
21.	(1)	22.	(3)	23.	(4)	24.	(1)	25.	(4)	26.	(4)	27.	(2)	28.	(4)	29.	(2)	30.	(1)

CDS PYQ

1. Two lots of onions with equal quantity, one costing 10 per kg and the other costing Rs 15 per kg, are mixed together and whole lot is sold at Rs 15 per kg. What is the profit or loss?
[CDS 2013- II]
(a) 10% loss (b) 10% profit
(c) 20% profit (d) 20% loss
2. A person sold an article from Rs 3600 and got a profit of 20%. Had he sold the article for Rs 3150, how much profit would he have got?
[CDS 2013- II]
(a) 4% (b) 5%
(c) 6% (d) 10%
3. On a 20% discount sale, an article costs 596. What was the original price of the article?
[CDS 2014- I]
(a) Rs 720 (b) Rs 735
(c) Rs 745 (d) Rs 775
4. A person selling an article for 96 finds that his loss per cent is one-fourth of the amount of rupees that he paid for the article, what can be the cost price?
[CDS 2014- II]
(a) Only Rs 160
(b) Only Rs 240
(c) Either Rs 160 or Rs 240
(d) Neither Rs 160 nor Rs 240
5. A shopkeeper sells his articles at their cost price but uses a faulty balance which reads 1000g for 800g. What is his actual profit percentage?
[CDS 2014- II]
(a) 25% (b) 20%
(c) 40% (d) 30%
6. A man buys 200 oranges for R 1000. How many oranges for 100 can be sold, so that his profit percentage 25% is:
[CDS 2014- II]
(a) 10 (b) 14
(c) 16 (d) 20
7. When an article is sold at 20% discount, the selling price is Rs 24. What will be the selling price when the discount is 30%?
[CDS 2014- II]
(a) Rs. 25 (b) Rs 23
(c) Rs 21 (d) Rs 20
8. A milkman claims to sell milk at its cost price only, but he is making a profit of 20% since he has mixed some amount of water in the milk. What is the percentage of milk in the mixture?
[CDS 2015- I]
(a) 80% (b) $\frac{250}{3}\%$
(c) 75% (d) $\frac{200}{3}\%$
9. The value of a single discount on some amount which is equivalent to a series of discounts of 10%, 20% and 40% on the same amount, is equal to
[CDS 2015- II]
(a) 43.2% (b) 50%
(c) 56.8% (d) 70%
10. A milkman claims to sell milk at its cost price only. Still he is making a profit of 20% since he has mixed some amount of water in the milk. What is percentage of milk in the mixture?
[CDS 2015- II]
(a) $200/3\%$ (b) 75%
(c) 80% (d) $250/3\%$
11. A cloth merchant buys cloth from a weaver and cheats him by using a scale which is 10 cm longer than a normal metre scale. He claims to sell cloth at the cost price to his customers, but while selling uses a scale which is 10 cm shorter than a normal metre scale. What are his gains?
[CDS 2016- I]
(a) 20% (b) 21%
(c) $22\frac{2}{9}\%$ (d) 23%

- 12.** The cost of 2.5 kg rice is Rs 125. The cost of 9 kg rice is equal to that of 4 kg pulses. The cost of 14 kg pulses is equal to that of 1.5 kg tea. The cost of 2 kg tea is equal to that of 5 kg nuts. What is the cost of 11 kg nuts?

[CDS 2016- II]

- (a) Rs 2310 (b) Rs 3190
(c) Rs 4070 (d) Rs 4620

- 13.** A shopkeeper increases the cost price of an item by 20% and offers a discount of 10% on this marked price. What is his percentage gain?

[CDS 2016- II]

- (a) 15% (b) 12%
(c) 10% (d) 8%

- 14.** Rajendra bought a mobile with 25% discount on the selling price. If the mobile cost him 4,875, what is the original selling price of the mobile?

[CDS 2017- I]

- (a) Rs 6,300 (b) Rs 6,400
(c) Rs 6,500 (d) Rs 6,600

- 15.** Sudhir purchased a chair with three consecutive discounts of 20%, 12.5% and 5%. The actual deduction will be

[CDS 2017- II]

- (a) 33.5% (b) 30%
(c) 32% (d) 35%

- 16.** Ram buys 4 chairs and 9 stools for Rs 1,340. If he sells chairs at 10% profit and stools at 20% profit, he earns a total profit of Rs 188. How much money did he have to pay for the chairs?

[CDS 2017- II]

- (a) Rs 200 (b) Rs 400
(c) Rs 800 (d) Rs 1,600

- 17.** An article is sold at a profit at 32%. If the cost price is increased by 20% and the sale price remains the same, then the profit percentage becomes

[CDS 2018- I]

- (a) 10% (b) 12%
(c) 15% (d) 20%

- 18.** A milk vendor bought 28 litres of milk at the rate of Rs 8.50 per litre. After adding some water he sold the mixture at the same price.

If his gain is 12.5%, how much water did he add?

[CDS 2018- II]

- (a) 4.5 litres (b) 4 litres
(c) 3.5 litres (d) 3 litres

- 19.** If an article is at a gain of 6% instead of a loss of 6%, the seller gets Rs 6 more. What is the cost price of the article?

[CDS 2019- I]

- (a) Rs 18 (b) Rs 36
(c) Rs 42 (d) Rs 50

- 20.** A trader sells two computers at the same price, making a profit of 30% on one and a loss of 30% on the other. What is the net loss or profit percentage on the transaction?

[CDS 2019- II]

- (a) 6% loss (b) 6 % gain
(c) 9 % loss (d) 9 % gain

- 21.** A person sells two items each at Rs 990, one at a profit of 10% and another at a loss of 10%. What is the combined percentage of profit or loss for two items?

[CDS 2019- II]

- (a) 1% loss (b) 1% profit
(c) No profit no loss (d) 0.5% profit

- 22.** A shopkeeper sells his articles at their cost price but uses a faulty balance which reads 1000gm for 800gm. What is the actual profit percentage?

[CDS 2020- II]

- (a) 20% (b) 25%
(c) 30% (d) 40%

- 23.** If a television set is sold at Rs x, a loss of 28% would be incurred. If it is sold at Rs y, a profit of 12% would be incurred. What is the ratio of y to x?

[CDS 2020- II]

- (a) 41 : 9 (b) 31 : 9
(c) 23 : 9 (d) 14 : 9

- 24.** A person sold an article for Rs 75 which cost him Rs x. He finds that he realized x % profit on his outlay. What is x equal to?

[CDS 2020- II]

- (a) 20 (b) 25
(c) 50 (d) 100

25. A trader gives successive discounts of 20%, 10% and 5% respectively. What is the overall discount?

[CDS 2021- I]

- (a) 30% (b) 31.6%
(c) 32.5% (d) 35%

26. The cost price of 100 mangoes is equal to the selling price of 80 mangoes. What is the profit percentage?

[CDS 2021- I]

- (a) 16% (b) 20%
(c) 24% (d) 25%

27. A person bought chair and a table for Rs 750. He sold the chair at a gain of 5% and the table at a gain of 20%. He gained 16% of the whole. What is the original cost of the table?

[CDS 2021- II]

- (a) Rs 400 (b) Rs 450
(c) Rs 550 (d) Rs 600

28. A shopkeeper marks the price of an article at Rs 200. After allowing a discount of 10%, he still gains 20% on the cost price. What is the cost price of the article?

[CDS 2022 (I)]

- (a) Rs 170 (b) Rs 160
(c) Rs 150 (d) Rs 120

29. Twelve percent of bananas bought by a fruit vendor got lost during transportation. On

selling the remaining bananas, the vendor's overall profit turned out to be 4%. If the vendor had not lost any bananas and had sold them at the price of the remaining bananas, what would have been his profit percentage?

[CDS - 2023 (1)]

- (a) $8\frac{1}{9}\%$ (b) $9\frac{1}{11}\%$
(c) $18\frac{2}{11}\%$ (d) None of the above

30. A person bought an article and sold it at a profit of 20%. Had he bought it at 20% less what would have been the profit percentage if the selling price had been the same?

[CDS-2023 (2)]

- (a) 25% (b) 40%
(c) 50% (d) 60%

31. The cost price of y articles is equal to selling price of z articles. If $y : z = 5 : 4$, what is the profit percentage?

[CDS-2024 (1)]

- (a) 20% (b) 25%
(c) 30% (d) 40%

32. A person bought a book at $\frac{3}{4}$ th of its listed price and sold it at 50% more than its listed price. What is the percentage of gain in the transaction?

[CDS-2024 (1)]

- (a) 20% (b) 40%
(c) 75% (d) 100%

ANSWER KEY

1.	(c)	2.	(b)	3.	(c)	4.	(c)	5.	(a)	6.	(c)	7.	(c)	8.	(b)	9.	(c)	10.	(d)
11.	(c)	12.	(d)	13.	(d)	14.	(c)	15.	(a)	16.	(c)	17.	(a)	18.	(c)	19.	(d)	20.	(c)
21.	(a)	22.	(b)	23.	(d)	24.	(c)	25.	(b)	26.	(d)	27.	(c)	28.	(c)	29.	(c)	30.	(c)
31.	(b)	32.	(d)																

CAPF PYQ

1. If 9 mangoes cost as much as 5 oranges, 5 oranges as much as 3 apples, 4 apples as much as 9 pineapples and if 3 pineapples cost Rs 48, what will a mango cost?
[CAPF 2013]
(a) Rs 9 (b) Rs 12
(c) Rs 18 (d) Rs 27
2. By selling an article of Rs 270 a man loses 10%. If he would sell it at Rs 360 his gain percent is
[CAPF 2013]
(a) 10 (b) 15
(c) 20 (d) 25
3. A, B' and 'C' started their independence businesses with equal amounts of capital. During the first year 'A' made 10% profit, 'B' incurred 10% loss and 'C' made a profit of 5%. In the second year 'A' incurred of 20% loss, B' made profit of 20% and 'C' made profit of 5%. Which of the following is FALSE at the end of second year?
[CAPF 2015]
(a) 'C' is the richest (b) A' is the poorest
(c) 'B' is the richest (d) 'C' is richer than 'B'
4. By selling an article for Rs 2700, a man loses 10%. If he sells it for Rs 3600, his gain per cent is
[CAPF 2019]
(a) 15 (b) 18
(c) 20 (d) 25
5. The cost of gold varies directly as the cube of its weight. A gold piece weighing 20 decigram costs 1,000. If it is broken into two pieces whose weights are in the ratio 2:3, then what is the profit or loss incurred?
[CAPF 2020]
(a) 280 profit (b) 280 loss
(c) 720 profit (d) 720 loss
6. In a stock clearance sell, a shopkeeper gives 40% off on all his items. He incurs a loss of 25% by selling by selling a item of marked price Rs 2000, Which one of the following is the cost price of item?
[CAPF 2021]
(a) Rs 1800 (b) Rs 1200
(c) Rs 1600 (d) Rs 1400
7. A shopkeeper gives two consecutive discounts of 10% and 5% respectively on his items, He then add 20% GST on his items. If an item has marked price Rs 2000, how much more or less of the actual price of the item a customer has to pay?
[CAPF 2021]
(a) 2.6% less (b) 2.6% more
(c) Same price (d) 5.2% more
8. A person buys an item from a shop for which the shopkeeper offers a discount of 10% on the market price. The person pays using an e-wallet which gives 10% cash back. Which one of the following is the value of effective discount?
[CAPF 2022]
(a) 20% (b) 18%
(c) 19% (d) 21%

ANSWER KEY

1.	(b)	2.	(c)	3.	(c)	4.	(c)	5.	(d)	6.	(c)	7.	(b)	8.	(c)				
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PRACTICE SHEET-1

1. What sum of money must be given as simple interest for six months at 4% per annum in order to earn 150 interest?
 (1) 5000 (2) 7500
 (3) 10000 (4) 15000
2. A sum of 1600 gives a simple interest of 252 in 2 years and 3 months. The rate of interest per annum is:
 (1) 5.5 % (2) 8%
 (3) 7% (4) 6%
3. A sum of money lent at simple interest amounts to 880 in 2 years and to 920 in 3 years. The sum of money (in rupees) is
 (1) 700 (2) 760
 (3) 784 (4) 800
4. At some rate of simple interest, A lent 6,000 to B for 2 years and 1,500 to C for 4 years and received 9,00 as interest from both of them together. The rate of interest per annum was
 (1) 5% (2) 6%
 (3) 8% (4) 10%
5. A lent 5000 to B for 2 years and 3000 to C for 4 years on simple interest at the same rate of interest and received 2200 in all from both as interest. The rate of interest per annum is
 (1) 7% (2) 5%
 (3) 7.125% (4) 10%
6. What sum of money will amount to 520 in 5 years and to 568 in 7 years at simple interest?
 (1) 400 (2) 120
 (3) 510 (4) 220
7. 500 was invested at 12% per annum simple interest and a certain sum of money invested at 10% per annum simple interest. If the sum of the interest on both the sum after 4 years is 480, the latter sum of money is:
 (1) 450 (2) 750
 (3) 600 (4) 550
8. A money lender finds that due to fall in the annual rate of interest 8% to 7.75 %, his yearly income diminishes by 61.50. His capital is
 (1) 22400 (2) 23800
 (3) 24600 (4) 26000
9. A lends 2500 to B and a certain sum to C at the same time at 7% annual simple interest. If after 4 years, A altogether receives 1120 as interest from B and C, the sum lent to C is
 (1) 700 (2) 6500
 (3) 4000 (4) 1500
10. A certain sum of money amounts to 756 in 2 years and to 873 in 3 and half years at a certain rate of simple interest. The rate of interest per annum is
 (1) 10% (2) 11%
 (3) 12% (4) 13%
11. What sum will amount to 7000 in 5 years at $3\frac{1}{3}\%$ simple interest?
 (1) 6300 (2) 6500
 (3) 6000 (4) 5000
12. A man took a loan from a bank at the rate of 12% per annum at simple interest. After 3 years he had to pay 5,400 as interest only for the period. The principal amount borrowed by him was:
 (1) 2,000 (2) 10,000
 (3) 20,000 (4) 15,000
13. A sum of money at simple interest amounts to 1,012 in $2\frac{1}{2}$ years and to 1,067.20 in 4 years. The rate of interest per annum is :
 (1) 2.5% (2) 3%
 (3) 4% (4) 5%
14. A sum of money lent out at simple interest amounts to 720 after 2 years and to 1020 after a further period of 5 years. The sum is :
 (1) 500 (2) 600
 (3) 700 (4) 710
15. The sum of money, that will give 1 as interest per day at the rate of 5% per annum simple interest is
 (1) 3650 (2) 36500
 (3) 730 (4) 7300

- 16.** A sum of money becomes $\frac{7}{6}$ of itself in 3 years at a certain rate of simple interest. The rate per annum is:
 (1) $5\frac{5}{9}\%$ (2) $6\frac{5}{9}\%$
 (3) 18% (4) 25%
- 17.** A sum of money becomes $\frac{41}{40}$ of itself in $\frac{1}{4}$ years at a certain rate of simple interest. The rate of interest per annum is
 (1) 10% (2) 1%
 (3) 2.5% (4) 5%
- 18.** A certain sum of money becomes three times of itself in 20 years at simple interest. In how many years does it become double of itself at the same rate of simple interest?
 (1) 8 years (2) 10 years
 (3) 12 years (4) 14 years
- 19.** At what rate per cent per annum will the simple interest on a sum of money be $\frac{2}{5}$ of the amount in 10 years?
 (1) 4% (2) 6%
 (3) $5\frac{2}{3}\%$ (4) $6\frac{2}{3}\%$
- 20.** 6,000 become 7,200 in 4 years at a certain rate of simple interest. If the rate becomes 1.5 times of itself, the amount of the same principal in 5 years will be
 (1) 8,000 (2) 8,250
 (3) 9,250 (4) 9,000
- 21.** A sum of money at simple interest trebles itself in 15 years. It will become 5 times of itself in
 (1) 40 years (2) 36 years
 (3) 30 years (4) 25 years
- 22.** If a sum of money at simple interest doubles in 12 years, the rate of interest per annum is
 (1) $16\frac{2}{3}\%$ (2) 7.5%
 (3) $8\frac{1}{3}\%$ (4) 10%
- 23.** At what rate of simple interest per annum will a sum become $\frac{7}{4}$ of itself in 4 years?
 (1) 18% (2) $18\frac{1}{4}\%$
 (3) $18\frac{3}{4}\%$ (4) $18\frac{1}{2}\%$
- 24.** A sum of money at a certain rate per annum of simple interest doubles in the 5 years and at a different rate becomes three times in 12 years. The lower rate of interest per annum is
 (1) 15% (2) 20%
 (3) $15\frac{3}{4}\%$ (4) $16\frac{2}{3}\%$
- 25.** In how many years will a sum of money double itself at $6\frac{1}{4}\%$ simple interest per annum?
 (1) 24 years (2) 20 years
 (3) 16 years (4) 12 years
- 26.** At a certain rate of simple interest, a certain sum of money becomes double of itself in 10 years. It will become treble of itself in
 (1) 15 years (2) 18 years
 (3) 20 years (4) 30 years
- 27.** In how much time, will a sum of money become double of itself at 15% per annum simple interest?
 (1) $6\frac{1}{4}$ years (2) $6\frac{1}{2}$ years
 (3) $6\frac{1}{3}$ (4) $6\frac{2}{3}$ years
- 28.** In how many years will a sum of money double itself at 12% per annum?
 (1) 8 yrs. 6 months (2) 6 yrs. 9 months
 (3) 8 yrs. 4 months (4) 7 yrs. 6 months
- 29.** A sum amounts to double in 8 years by simple interest. Then the rate of simple interest per annum is
 (1) 10% (2) 12.5%
 (3) 15% (4) 20%

30. A sum doubles itself in 16 years, then in how many years will it triple itself; rate of interest being simple
- (1) 25 years (2) 24 years
(3) 48 years (4) 32 years

ANSWER KEY

1.	(2)	2.	(3)	3.	(4)	4.	(1)	5.	(4)	6.	(1)	7.	(3)	8.	(4)	9.	(4)	10.	(4)
11.	(3)	12.	(4)	13.	(3)	14.	(2)	15.	(4)	16.	(1)	17.	(1)	18.	(2)	19.	(4)	20.	(2)
21.	(3)	22.	(3)	23.	(3)	24.	(4)	25.	(3)	26.	(3)	27.	(4)	28.	(3)	29.	(2)	30.	(4)

PRACTICE SHEET-2

- In what time will the simple interest be $\frac{2}{5}$ of the principal at 8 per cent per annum?
(1) 8 years (2) 7 years
(3) 5 years (4) 6 years
- The simple interest on a sum after 4 years is $\frac{1}{5}$ of the sum. The rate of interest per annum is:
(1) 4% (2) 5%
(3) 6% (4) 8%
- Simple interest on a certain sum for 6 years is $\frac{9}{25}$ of the sum. The rate of interest is:
(1) 6% (2) $6\frac{1}{2}\%$
(3) 8% (4) $8\frac{1}{2}\%$
- The simple interest on a sum for 5 years is one fourth of the sum. The rate of interest per annum is
(1) 5% (2) 6%
(3) 4% (4) 8%
- On a certain sum, the simple interest at the end of $6\frac{1}{4}$ years becomes $\frac{3}{8}$ of the sum. The rate of interest is
(1) 5% (2) 6%
(3) 7% (4) 8%
- The simple interest on a certain sum at 5% per annum for 3 years and 4 years differ by 42. The sum is :
(1) 210 (2) 280
(3) 750 (4) 840
- The difference between the simple interest received from two different sources on 1500 for 3 years is 13.50. The difference between their rates of interest is:
(1) 0.1% (2) 0.2%
(3) 0.3% (4) 0.4%
- The simple interest on a sum of money is $\frac{4}{9}$ of the principal and the number of years is equal to the rate percent per annum. The rate per annum is:
(1) 5% (2) $6\frac{2}{3}\%$
(3) 6% (4) $7\frac{1}{5}\%$
- The simple interest on a certain sum for 8 months at 4% per annum is 129 less than the simple interest on the same sum for 15 months at 5% per annum. The sum is :
(1) 2,580 (2) 2400
(3) 2529 (4) 3600
- Mohan lent some amount of money at 9% simple interest and an equal amount of money at 10% simple interest each for two years. If his total interest was Rs. 760, what amount was lent in each case?
(1) 1700 (2) 1800
(3) 1900 (4) 2000

- 11.** A sum of 1550 was lent partly at 5% and partly at 8% simple interest. The total interest received after 3 years is 300. The ratio of money lent at 5% to that at 8% is :
 (1) 5 : 8 (2) 8 : 5
 (3) 31 : 6 (4) 16 : 15
- 12.** A person lent 5,000 partly at the rate of 4 per cent and partly at the rate of 5 per cent per annum simple interest. The total interest after 2 years is 440. To find the sum of money lent at each of the above rates, 5,000 is to be divided in the ratio :
 (1) 4 : 5 (2) 3 : 2
 (3) 5 : 4 (4) 2 : 3
- 13.** A person borrows some money for 5 years and loan amount: total interest amount is 5 : 2. The ratio of loan amount: interest rate is equal to :
 (1) 2 : 25 (2) 2 : 1
 (3) 5 : 2 (4) 25 : 2
- 14.** A person invests money in three different schemes for 6 years, 10 years and 12 years at 10 per cent, 12 per cent and 15 per cent simple interest respectively. At the completion of each scheme, he gets the same interest. The ratio of his investment is
 (1) 6 : 3 : 2 (2) 2 : 3 : 4
 (3) 3 : 4 : 6 (4) 3 : 4 : 2
- 15.** With a given rate of simple interest, the ratio of principal and amount for a certain period of time is 4 : 5. After 3 years, with the same rate of interest, the ratio of the principal and amount becomes 5 : 7. The rate of interest is
 (1) 4% (2) 6%
 (3) 5% (4) 7%
- 16.** A sum of 400 amounts to 480 in 4 years. What will it amount to if the rate of interest is increased by 2%?
 (1) 484 (2) 560
 (3) 512 (4) None of these
- 17.** A man loses 55.50 yearly when the annual rate of interest falls from 11.5% to 10%. His capital (in rupees) is
 (1) 3700 (2) 7400
 (3) 8325 (4) 11100
- 18.** If the annual rate of simple interest increases from 10% to $12\frac{1}{2}\%$, a man's yearly income increases by 1250. His principal (in rupees) is
 (1) 50,000 (2) 45,000
 (3) 60,000 (4) 65,000
- 19.** A sum was invested on simple interest at a certain rate for 2 years. Had it been put at 3% higher rate, it would have fetched 72 more. The sum is
 (1) 1,200 (2) 1,500
 (3) 1,600 (4) 1,800
- 20.** A sum of money was lent at simple interest at a certain rate for 3 years. Had it been lent at 2.5% per annum higher rate, it would have fetched 540 more. The money lent was:
 (1) 6400 (2) 6472
 (3) 6840 (4) 7200
- 21.** A sum of money was invested at a certain rate of simple interest for 2 years. Had it been invested at 1% higher rate, it would have fetched 24 more interest. The sum of money is :
 (1) 1200 (2) 1050
 (3) 1000 (4) 9600
- 22.** A person who pays income tax at the rate of 4 paise per rupee, find that a fall of interest rate from 4% to 3.75% diminishes his net yearly income by 48. What is his capital ?
 (1) 24,000 (2) 25,000
 (3) 20,000 (4) 18,000
- 23.** A sum was lent at simple interest at a certain rate for 2 years. Had it been lent at 3% higher rate, it would have fetched 300 more. The original sum of money was
 (1) 5000 (2) 6000
 (3) 7000 (4) 4000
- 24.** A sum of 2,400 amounts to 3,264 in 4 years at a certain rate of simple interest. If the rate of interest is increased by 1%, the same sum in the same time would amount to
 (1) 3,288 (2) 3,312
 (3) 3,340 (4) 3,360
- 25.** 800 amounts to 920 in 3 years at simple interest. If the interest rate is increased by 3%, it would amount to
 (1) 1,056 (2) 1,112
 (3) 1,182 (4) 992

- 26.** A sum of 10,000 is lent partly at 8% and remaining at 10% per annum. If the yearly interest on the average is 9.2%, the two parts are :
 (1) 4000, 6000 (2) 4500, 5500
 (3) 5000, 5000 (4) 5500, 4500
- 27.** A sum of 1000 is lent out partly at 8% and the remaining at 10% per annum. If the yearly income on the average is 9.2%, the two parts respectively are
 (1) 400, 600 (2) 450, 550
 (3) 500, 500 (4) 550, 450
- 28.** An old article is available for 12,000 at cash payment or is available for 7,000 cash payment and a monthly installment of 630 for 8 months. The rate per cent per annum is
 (1) 2.1 per cent (2) 3 per cent
 (3) 3.25 per cent (4) 3.3 per cent
- 29.** The effective annual rate of interest, corresponding to a nominal rate of 6% per annum payable half yearly, is :
 (1) 6.06% (2) 6.07%
 (3) 6.08% (4) 6.09%
- 30.** A person lends 40% of his sum of money at 15% per annum, 50% of rest at 10% per annum and the rest at 18% per annum rate of interest. What would be the annual rate of interest, if the interest is calculated on the whole sum?
 (1) 13.4% (2) 14.33%
 (3) 14.4% (4) 13.33%

ANSWER KEY

1.	(3)	2.	(2)	3.	(1)	4.	(1)	5.	(2)	6.	(4)	7.	(3)	8.	(4)	9.	(4)	10.	(4)
11.	(4)	12.	(2)	13.	(4)	14.	(1)	15.	(3)	16.	(3)	17.	(1)	18.	(1)	19.	(1)	20.	(4)
21.	(1)	22.	(3)	23.	(1)	24.	(4)	25.	(4)	26.	(1)	27.	(1)	28.	(1)	29.	(4)	30.	(3)

PRACTICE SHEET- 3

- 1.** At what percent per annum will 3000/- amounts to 3993/- in 3 years if the interest is compounded annually?
 (1) 9% (2) 10%
 (3) 11% (4) 13%
- 2.** The compound interest on 10,000 in 2 years at 4% per annum, the interest being compounded half-yearly, is :
 (1) 636.80 (2) 824.32
 (3) 912.86 (4) 828.82
- 3.** In how many years will 2,000 amounts to 2,420 at 10% per annum compound interest?
 (1) 3 years (2) $2\frac{1}{2}$ years
 (3) 2 years (4) $1\frac{1}{2}$ years
- 4.** In what time will 1000 becomes 1331 at 10% per annum compounded annually?
 (1) 3 years (2) $2\frac{1}{2}$ years
- (3) 2 years (4) $3\frac{1}{2}$ years
- 5.** The principal, who will amount to 270.40 in 2 years at the rate of 4% per annum compound interest, is
 (1) 200 (2) 225
 (3) 250 (4) 220
- 6.** A sum of money on compound interest amounts to 10648 in 3 years and 9680 in 2 years. The rate of interest per annum is
 (1) 5% (2) 10%
 (3) 15% (4) 20%
- 7.** At what rate per cent per annum will 2304 amount to 2500 in 2 years at compound interest ?
 (1) $4\frac{1}{2}\%$ (2) $4\frac{1}{5}\%$
 (3) $4\frac{1}{6}\%$ (4) $4\frac{1}{3}\%$

- 8.** A sum becomes 1,352 in 2 years at 4% per annum compound interest. The sum is
 (1) 1,225 (2) 1,270
 (3) 1,245 (4) 1,250
- 9.** The compound interest on 16,000 for 9 months at 20% per annum, interest being compounded quarterly, is
 (1) 2,520 (2) 2,524
 (3) 2,522 (4) 2,518
- 10.** If the rate of interest be 4% per annum for first year, 5% per annum for second year and 6% per annum for third year, then the compound interest of 10,000 for 3 years will be
 (1) 1,600 (2) 1,625.80
 (3) 1,575.20 (4) 2,000
- 11.** The compound interest on 2000 in 2 years if the rate of interest is 4% per annum for the first year and 3% per annum for the second year, will be
 (1) 142.40 (2) 140.40
 (3) 141.40 (4) 143.40
- 12.** At what rate per annum will 32000 yield a compound interest of 5044 in 9 months interest being compounded quarterly?
 (1) 20% (2) 32%
 (3) 50% (4) 80%
- 13.** The compound interest on 8,000 at 15% per annum for 2 years 4 months, compounded annually is:
 (1) 2980 (2) 3091
 (3) 3109 (4) 3100
- 14.** In what time will 10,000 amount to 13310 at 20% per annum compounded half yearly?
 (1) $1\frac{1}{2}$ years (2) 2 years
 (3) $2\frac{1}{2}$ years (4) 3 years
- 15.** A certain sum of money yields 1261 as compound interest for 3 years at 5% per annum. The sum is
 (1) 9000 (2) 8400
 (3) 7500 (4) 8000
- 16.** A certain sum, invested at 4% per annum compound interest, compounded half yearly, amounts to 7,803 at the end of one year. The sum is
 (1) 7,000 (2) 7,200
 (3) 7,500 (4) 7,700
- 17.** A certain sum amounts to 5,832 in 2 years at 8% per annum compound interest, the sum is
 (1) 5,000 (2) 5,200
 (3) 5,280 (4) 5,400
- 18.** The compound interest on 6,000 at 10% per annum for $1\frac{1}{2}$ years, when the interest being compounded annually, is
 (1) 910 (2) 870
 (3) 930 (4) 900
- 19.** In what time 8,000 will amount to 9,261 at 10% per annum compound interest, when the interest is compounded half yearly?
 (1) $3\frac{1}{2}$ years (2) $1\frac{1}{2}$ years
 (3) $2\frac{1}{2}$ years (4) 2 years
- 20.** At what rate per cent per annum will a sum of 1,000 amounts to 1,102.50 in 2 years at compound interest?
 (1) 5% (2) 5.5%
 (3) 6% (4) 6.5%
- 21.** In how many years will a sum of 800 at 10% per annum compound interest, compounded semi-annually becomes 926.10 ?
 (1) $1\frac{1}{2}$ years (2) $1\frac{2}{3}$ years
 (3) $2\frac{1}{3}$ years (4) $2\frac{1}{2}$ years
- 22.** An amount of 6,000 lent at 5% per annum compound interest for 2 years will become
 (1) 600 (2) 6,600
 (3) 6,610 (4) 6,615
- 23.** In what time will 1000 amounts to 1331 at 20% per annum, compounded half yearly?
 (1) $1\frac{1}{2}$ years (2) 2 years
 (3) 1 year (4) $2\frac{1}{2}$ years

- 24.** The compound interest on 30,000 at 7% per annum for a certain time is 4,347. The time is
 (1) 3 years (2) 4 years
 (3) 2 years (4) 2.5 years
- 25.** A sum of 8000 will amount to 8820 in 2 years if the interest is calculated every year. The rate of compound interest is
 (1) 6% (2) 7%
 (3) 3% (4) 5%
- 26.** A principal of 10,000, after 2 years compounded annually, the rate of interest being 10% per annum during the first year and 12% per annum during the second year (in rupees) will amount to :
 (1) 12,000 (2) 12,320
 (3) 12,500 (4) 11,320
- 27.** The sum of money that yields a compound interest of 420 during the second year at 5% p.a is
 (1) 4,000 (2) 42,000
- (3) 8,000 (4) 21,000
- 28.** A man saves 2000 at the end of each year and invests the money at 5% compound interest. At the end of 3 years he will have :
 (1) 4305 (2) 6305
 (3) 4205 (4) 2205
- 29.** The time in which 80,000 amounts to 92,610 at 10% p.a. compound interest, interest being compounded semi annually is:
 (1) $1\frac{1}{2}$ years (2) 2 years
 (3) $2\frac{1}{2}$ years (4) 3 years
- 30.** A man borrows 21000 at 10% compound interest. How much he has to pay annually at the end of each year, to settle his loan in two years?
 (1) 12000 (2) 12100
 (3) 12200 (4) 12300

ANSWER KEY

1.	(2)	2.	(2)	3.	(3)	4.	(1)	5.	(3)	6.	(2)	7.	(3)	8.	(4)	9.	(3)	10.	(3)
11.	(1)	12.	(1)	13.	(3)	14.	(1)	15.	(4)	16.	(3)	17.	(1)	18.	(3)	19.	(2)	20.	(1)
21.	(1)	22.	(4)	23.	(1)	24.	(3)	25.	(4)	26.	(2)	27.	(3)	28.	(2)	29.	(1)	30.	(2)

PRACTICE SHEET-4

- 1.** If the compound interest on a certain sum for 2 years at 3% per annum is 101.50, then the simple interest on the same sum at the same rate and for the same time will be
 (1) 90.00 (2) 95.50
 (3) 100.00 (4) 98.25
- 2.** If the compound interest on a sum of money for 3 years at the rate of 5% per annum is 252.20, the simple interest on the same sum at the same rate and for the same time is
 (1) 220 (2) 240
 (3) 245 (4) 250
- 3.** On a certain sum of money the compound interest for 2 years is 282.15 and the simple interest for the same period of time is 270. The rate of interest per annum is
 (1) 6.07% (2) 10%
 (3) 9% (4) 12.15%
- 4.** If the compound interest on a sum for 2 years at $12\frac{1}{2}$ % per annum is 510, the simple interest on the same sum at the same rate for the same period of time is:
 (1) 400 (2) 480
 (3) 450 (4) 460
- 5.** The compound interest on a certain sum of money at a certain rate for 2 years is 40.80 and the simple interest on the same sum is 40 at the same rate and for the same time. The rate of interest is
 (1) 2% per annum (2) 3% per annum
 (3) 4% per annum (4) 5% per annum
- 6.** The compound interest on a certain sum of money invested for 2 years at 5% per annum is 328. The simple interest on the sum, at

- the same rate and for the same period will be
 (1) 320 (2) 308
 (3) 300 (4) 287
- 7.** Compound interest on a sum of money for 2 years at 4 per cent per annum is 2, 448. Simple interest of the same sum of money at the same rate of interest for 2 years will be
 (1) 2,500 (2) 2,400
 (3) 2,360 (4) 2,250
- 8.** At a certain rate per annum, the simple interest on a sum of money for one year is 260 and the compound interest on the same sum for two years is 540.80. The rate of interest per annum is
 (1) 4% (2) 6%
 (3) 8% (4) 10%
- 9.** The simple interest on a sum of money at 4% per annum for 2 years is 80. The compound interest in the same sum for the same period is
 (1) 82.60 (2) 82.20
 (3) 81.80 (4) 81.60
- 10.** The compound interest on a certain sum of money at 5% per annum for 2 years is 246. The simple interest on the same sum for 3 years at 6% per annum is
 (1) 435 (2) 450
 (3) 430 (4) 432
- 11.** The simple interest and compound interest (compounded annually) on a certain sum of money with a given rate for a period of 2 years are 900 and 954 respectively. The sum of money is
 (1) 3700 (2) 3650
 (3) 3850 (4) 3750
- 12.** The compound interest on a certain sum of money for 2 years at 10% per annum is 420. The simple interest on the same sum at the same rate and for the same time will be
 (1) 350 (2) 375
 (3) 380 (4) 400
- 13.** If the compound interest on a certain sum for 2 years at 4% p.a. is 102, the simple interest at the same rate of interest for two years would be
 (1) 200 (2) 50
 (3) 150 (4) 100
- 14.** There is 100% increase to an amount in 8 years, at simple interest. Find the compound interest of 8000 after 2 years at the same rate of interest.
 (1) 2500 (2) 2000
 (3) 2250 (4) 2125
- 15.** If the compound interest on a certain sum for two years at 12% per annum is 2,544, the simple interest on it at the same rate for 2 years will be
 (1) 2,400 (2) 2,500
 (3) 2,480 (4) 2,440
- 16.** The difference between compound and simple interest on a certain sum for 3 years at 5% per annum is Rs. 122. The sum is
 (1) 16,000 (2) 15,000
 (3) 12,000 (4) 10,000
- 17.** The difference between simple interest and compound interest of a certain sum of money at 20% per annum for 2 years is 48. Then the sum is
 (1) 1,000 (2) 1, 200
 (3) 1, 500 (4) 2, 000
- 18.** The difference between the compound interest and simple interest on 10,000 for 2 years is 25. The rate of interest per annum is
 (1) 5% (2) 7%
 (3) 10% (4) 12%
- 19.** If the difference between S.I. and C.I. for 2 years on a sum of money lent at 5% is 6, then the sum is
 (1) 2200 (2) 2400
 (3) 2600 (4) 2000
- 20.** On a certain sum of money lent out at 16% p.a. the difference between the compound interest for 1 year, payable half yearly, and the simple interest for 1 year is 56. The sum is
 (1) 1080 (2) 7805
 (3) 8750 (4) 5780
- 21.** If the amount is 2.25 times of the sum after 2 years at compound interest (compound annually), the rate of interest per annum is :
 (1) 25% (2) 30%
 (3) 45% (4) 50%

- 22.** A sum of money doubles itself in 4 years at compound interest. It will amount to 8 times itself at the same rate of interest in :
 (1) 18 years (2) 12 years
 (3) 16 years (4) 24 years
- 23.** A sum borrowed under compound interest doubles itself in 10 years. When will it become fourfold of itself at the same rate of interest?
 (1) 15 years (2) 20 years
 (3) 24 years (4) 40 years
- 24.** A sum of money becomes eight times of itself in 3 years at compound interest. The rate of interest per annum is
 (1) 100% (2) 80%
 (3) 20% (4) 10%
- 25.** A sum of money invested at compound interest doubles itself in 6 years. At the same rate of interest it will amount to eight times of itself in :
 (1) 15 years (2) 12 years
 (3) 18 years (4) 10 years
- 26.** A sum of money placed at compound interest doubles itself in 5 years. In how many years, it would amount to eight times of itself at the same rate of interest?
 (1) 10 years (2) 15 years
 (3) 7 years (4) 20 years
- 27.** A sum of money at compound interest doubles itself in 15 years. It will become eight times of itself in
 (1) 45 years (2) 48 years
 (3) 54 years (4) 60 years
- 28.** A sum of 12,000, deposited at compound interest becomes double after 5 years. How much will it be after 20 years?
 (1) 1,44,000 (2) 1,20,000
 (3) 1,50,000 (4) 1,92,000
- 29.** At what rate percent per annum of compound interest, will a sum of money become four times of itself in two years?
 (1) 100% (2) 75%
 (3) 50% (4) 20%
- 30.** A sum of money becomes double in 3 years at compound interest compounded annually. At the same rate, in how many years will it become four times of itself?
 (1) 4 years (2) 6 years
 (3) 6.4 years (4) 7.5 years

ANSWER KEY

1.	(3)	2.	(2)	3.	(3)	4.	(2)	5.	(3)	6.	(1)	7.	(2)	8.	(3)	9.	(4)	10.	(4)
11.	(4)	12.	(4)	13.	(4)	14.	(4)	15.	(1)	16.	(1)	17.	(2)	18.	(1)	19.	(2)	20.	(3)
21.	(4)	22.	(2)	23.	(2)	24.	(1)	25.	(3)	26.	(2)	27.	(1)	28.	(4)	29.	(1)	30.	(2)

PRACTICE SHEET-5

- 1.** A sum of money amounts to 4,840 in 2 years and to 5,324 in 3 years at compound interest compounded annually. The rate of interest per annum is :
 (1) 10% (2) 9%
 (3) 11% (4) 8%
- 2.** A certain sum of money amounts to 2,420 in 2 years and 2,662 in 3 years at some rate of compound interest, compounded annually. The rate of interest per annum is
 (1) 6% (2) 8%
 (3) 9% (4) 10%
- 3.** An amount of money at compound interest grows up to 3,840 in 4 years and up to 3,936 in 5 years. Find the rate of interest.
 (1) 2.5% (2) 2%
 (3) 3.5% (4) 2.05%
- 4.** A certain amount of money at $r\%$, compounded annually after two and three years becomes 1440 and 1728 respectively. r is
 (1) 5 (2) 10
 (3) 15 (4) 20

5. The compound interest on a certain sum for two successive years are 225 and 238.50. The rate of interest per annum is:
 (1) $7\frac{1}{2}\%$ (2) 5%
 (3) 10% (4) 6%
6. An amount of money appreciates to 7,000 after 4 years and to 10,000 after 8 years at a certain compound interest compounded annually. The initial amount of money was
 (1) 4,700 (2) 4,900
 (3) 4,100 (4) 4,300
7. A sum of money invested at compound interest amounts to 650 at the end of first year and 676 at the end of second year. The sum of money is:
 (1) 600 (2) 540
 (3) 625 (4) 560
8. A sum of money invested at compound interest amounts in 3 years to 2,400 and in 4 years to 2,520. The interest rate per annum is
 (1) 5% (2) 6%
 (3) 10% (4) 12%
9. A sum becomes 4500 after two years and 6750 after four years at compound interest. The sum is
 (1) 4000 (2) 2500
 (3) 3000 (4) 3050
10. A sum of money at compound interest will amount to 650 at the end of the first year and 676 at the end of the second year. The amount of money is
 (1) 1,300 (2) 650
 (3) 1,250 (4) 625
11. A builder borrows 2550 to be paid back with compound interest at the rate of 4% per annum by the end of 2 years in two equal yearly installments. How much will each installment be?
 (1) 1352 (2) 1377
 (3) 1275 (4) 1283
12. A man buys a scooter on making a cash down payment of 16224 and promises to pay two more yearly installments of equivalent amount in next two years. If the rate of interest is 4% per annum, compounded yearly, the cash value of the scooter, is
 (1) 40000 (2) 46824
 (3) 46000 (4) 50000
13. Kamal took 6800 as a loan which along with interest is to be repaid in two equal annual installments. If the rate of interest is $12\frac{1}{2}\%$, compounded annually, then the value of each installment is
 (1) 8100 (2) 4150
 (3) 4050 (4) 4000
14. A loan of 12,300 at 5% per annum compound interest, is to be repaid in two equal annual installments at the end of every year. Find the amount of each installment.
 (1) 6,651 (2) 6,615
 (3) 6,516 (4) 6,156
15. A sum of 210 was taken as a loan. This is to be paid back in two equal installments. If the rate of interest be 10% compounded annually, then the value of each installment is
 (1) 127 (2) 121
 (3) 210 (4) 225
16. Rs. 16,820 is divided between two brothers of age 27 years and 25 years. They invested their money at 5% per annum compound interest in such a way that both will receive equal money at the age of 40 years. The share (in Rs.) of elder brother is
 (1) 8,280 (2) 8,410
 (3) 8,820 (4) 8,000
17. A sum of money is paid back in two annual installments of Rs. 17, 640 each, allowing 5% compound interest compounded annually. The sum borrowed was
 (1) Rs. 32,800 (2) Rs. 32,200
 (3) Rs. 32,000 (4) Rs. 32,400
18. Mr. Dutta desired to deposit his retirement benefit of Rs. 3 lacs partly to a post office and partly to a bank at 10% and 6% interests respectively. If his monthly interest income was Rs. 2000, then the difference of his deposits in the post office and in the bank was
 (1) Rs. 50,000 (2) Rs. 40,000
 (3) Nil (4) Rs. 1,00,000

- 19.** The income of a company increases 20% per year. If the income is Rs. 26,64,000 in the year 2012, then its income in the year 2010 was :
 (1) Rs. 28,55,000 (2) Rs. 18,50,000
 (3) Rs. 28,20,000 (4) Rs. 21,20,000
- 20.** A person deposited a sum of 6,000 in a bank at 5% per annum simple interest. Another person deposited 5,000 at 8% per annum compound interest. After two years, the difference of their interests will be
 (1) 230 (2) 232
 (3) 832 (4) 600
- 21.** A money-lender borrows money at 4% per annum and pays the interest at the end of the year. He lends it at 6% per annum compound interest compounded half yearly and receives the interest at the end of the year. In this way, he gains 104.50 a year. The amount of money he borrows, is
 (1) 6,000 (2) 5,500
 (3) 5,000 (4) 4,500
- 22.** A sum of 13,360 was borrowed at $8\frac{3}{4}\%$ per annum compound interest and paid back in two years in two equal annual installments. What was the amount of each installment?
 (1) 5,769 (2) 7,569
 (3) 7,009 (4) 7,500
- 23.** Sita deposited 5,000 at 10% simple interest for 2 years. How much more money will Sita have in her account at the end of two years, if it is compounded semiannually.
 (1) 50 (2) 40
 (3) 77.50 (4) 85.50
- 24.** What does 250 amounts to in 2 years with compound interest at the rate of 4% in the 1st year and 8% in the second year?
 (1) 280 (2) 280.80
 (3) 468 (4) 290.80
- 25.** A man gave 50% of his savings of 84,100 to his wife and divided the remaining sum among his two sons A and B of 15 and 13 years of age respectively. He divided it in such a way that each of his sons, when they attain the age of 18 years, would receive the same amount at 5% compound interest per annum. The share of B was
 (1) 20,000 (2) 20,050
 (3) 22,000 (4) 22,050
- 26.** Find the rate percent per annum, if Rs. 2000 amounts to Rs. 2,315.25 in a year and a half, interest being compounded half yearly.
 (1) 11.5% (2) 10%
 (3) 5% (4) 20%
- 27.** A sum of money placed at compound interest doubles itself in 5 years. It will amount to eight times of itself at the same rate of interest in
 (1) 20 years (2) 10 years
 (3) 12 years (4) 15 years
- 28.** The sum of money which when given on compound interest at 18% per annum would fetch Rs. 960 more when the interest is payable half yearly than when it was payable annually for 2 years is :
 (1) Rs.60,000 (2) Rs. 30,000
 (3) Rs. 40,000 (4) Rs. 50,000
- 29.** The amount on Rs. 25,000 in 2 years at annual compound interest, if the rates for the successive years be 4% and 5% per annum respectively is :
 (1) Rs. 30,000 (2) Rs. 26,800
 (3) Rs. 27,300 (4) Rs. 28,500
- 30.** The amount of Rs. 10,000 after 2 years, compounded annually with the rate of interest being 10% per annum during the first year and 12% per annum during the second year, would be (in rupees)
 (1) 11,320 (2) 12,000
 (3) 12,320 (4) 12,500

ANSWER KEY

1.	(1)	2.	(4)	3.	(1)	4.	(4)	5.	(4)	6.	(2)	7.	(3)	8.	(1)	9.	(3)	10.	(4)
11.	(1)	12.	(2)	13.	(3)	14.	(2)	15.	(2)	16.	(3)	17.	(1)	18.	(3)	19.	(2)	20.	(2)
21.	(3)	22.	(2)	23.	(3)	24.	(2)	25.	(1)	26.	(2)	27.	(4)	28.	(4)	29.	(3)	30.	(3)

CDS PYQ

1. The principal on whom a simple interest of Rs 55 will be obtained after 9 months at the rate of $3\frac{2}{3}\%$ per annum is.
[CDS 2013 (I)]
(a) Rs 1000 (b) Rs 1500
(c) Rs 2000 (d) Rs 2500
2. The difference between compound interest and simple interest for 2 yr at the rate of 10% over principal amount of X is Rs 10. What is the value of X?
[CDS 2014 (II)]
(a) Rs 100 (b) Rs 1000
(c) Rs 500 (d) Rs 5000
3. A sum of money becomes 3 times in 5 yr at simple interest. In how many years, will the same sum become 6 times at the same rate of simple interest?
[CDS 2014 (II)]
(a) 10 yr (b) 12 yr
(c) 12.5 yr (d) 10.5 yr
4. The difference between compound interest and simple interest at the same rate of interest R per cent per annum on Rs 15,000 for 2 years is Rs 96. What is the value of R?
[CDS 2015 (I)]
(a) 8
(b) 10
(c) 12
(d) Cannot be determined due to insufficient data
5. There is 60% increase in an amount in 6 years at simple interest. What will be the compound interest on 12,000 after 3 years at the same rate of interest?
[CDS 2015 (I)]
(a) 2160 (b) 3120
(c) 3972 (d) 6240
6. In how much time would the simple interest on a principle be 0.125 time the principal amount at 10% per annum?
[CDS 2015 (I)]
- (a) $1\frac{1}{4}$ yrs (b) $1\frac{3}{4}$ yrs
(c) $2\frac{1}{4}$ yrs (d) $2\frac{3}{4}$ yrs
7. An automobile finance claims to be lending money at simple interest, but he includes the interest every six months for calculating the principal. If he is charging an interest at the rate of 10%, the effective rate of interest becomes
[CDS 2015 (I)]
(a) 10.25% (b) 10.5%
(c) 10.75% (d) 11%
8. A sum of Rs 10,000 is deposited for 1 year at the rate of interest 10% compounded half yearly. What will be the interest at the end of one year?
[CDS 2015 (II)]
(a) 1000 (b) 1025
(c) 1050 (d) 1100
9. If a sum of money at a certain rate of simple interest per year doubles in 5 years and at a different rate of simple interest per year becomes three times in 12 years, then the difference in the two rates of Simple interest per year is
[CDS 2016 (I)]
(a) 2% (b) 3%
(c) $3\frac{1}{3}\%$ (d) 4%
10. A sum of Rs 8,400 was taken as a loan. This is to be paid in two equal installments. If the rate of interest is 10% per annum, compounded annually, then the value of each installment is
[CDS 2017 (I)]
(a) 4,200
(b) 4,480
(c) 4,840
(d) None of the above
11. The difference between the simple and the compound interest on a certain sum of

money at 4% per annum in 2 years is Rs 10. What is the sum?

[CDS 2017 (I)]

- (a) 5,000 (b) 6,000
(c) 6,250 (d) 7,500

12. The difference between the compound interest (compounded annually) and simple interest on a sum of money deposited for 2 years at 5% per annum is Rs 15. What is the sum of money deposited?

[CDS 2017 (II)]

- (a) 6,000 (b) 4,800
(c) 3,600 (d) 2,400

13. A person borrowed Rs 5,000 at 5% rate of interest per annum and immediately lent it at 5.5%. After two years he collected the amount and settled his loan. What is the amount gained by him in this transaction?

[CDS 2018 (I)]

- (a) 25 (b) 50
(c) 100 (d) 200

14. A merchant commences with a certain capital and gains annually at the rate of 25%. At the end of 3 years he has 10,000. What is the original amount that the merchant invested?

[CDS 2018 (I)]

- (a) 5,120 (b) 5,210
(c) 5,350 (d) 5,500

15. The annual income of a person decreases by Rs 64 if the annual rate of interest decreases from 4% to 3.75%. What is his original annual income?

[CDS 2018 (I)]

- (a) 24,000 (b) 24,000
(c) 25,600 (d) 24,600

16. What the principle amount is with earns Rs 210 as compound interest for the second year at 5% per annum?

[CDS 2018 (II)]

- (a) Rs 2,000 (b) Rs 3,200
(c) Rs 4,000 (d) Rs 4,800

17. Two equal amounts were borrowed at 5% and 4% simple interest. The total interest after 4 years amounted to Rs 405. What was the total amount borrowed?

[CDS 2018 (II)]

- (a) Rs 1075 (b) Rs 1100
(c) Rs 1125 (d) Rs 1150

18. What is the least number of complete years in which a sum of money put at 40% annual compound interest will be more than treble?

[CDS 2019 (I)]

- (a) 3 (b) 4
(c) 5 (d) 6

19. A person divided a sum of Rs 17200 into three parts and invested at 5%, 6% and 9% per annum simple interest. At the end of two years, he got the same interest on each part of money. What is the money invested at 9%?

[CDS 2019 (I)]

- (a) Rs 3,200 (b) Rs 4,000
(c) Rs 4,800 (d) Rs 5,000

20. The rate of interest on two different schemes is the same and it is 20%. But in one of the schemes, the interest is compounded half yearly: and in the other the interest is, compounded annually. Equal amounts are invested in the schemes. If the difference of the returns after 2 years is Rs 482, then what is the. Principal amount in each scheme?

[CDS 2019 (II)]

- (a) 10,000 (b) 16,000
(c) 20,000 (d) 24,000

21. A lent 25000 to B and at the same time lent some amount to C at same 7% simple interest. After 4 years A received 11200 as interest from B and C. How much did A lend to C?

[CDS 2019 (II)]

- (a) Rs 20000 (b) Rs 25000
(c) Rs 15000 (d) Rs 10000

22. A sum was put at simple interest at certain rate of 2 years. Had it been put at 1 % higher rate of interest, it would have fetched Rs 24 more. What is the sum?

[CDS 2020 (I)]

- (a) Rs 500 (b) Rs 600
(c) Rs 800 (d) Rs 1200

23. The simple interest on a certain sum is one-fourth of the sum. If the number of years and the rate of annual interest are numerically equal, then the number of years is
[CDS 2020 (II)]
(a) 2.5 (b) 3
(c) 3.5 (d) 5
24. Sum of money was invested at simple interest at a certain rate for 5 years. Had it been invested at a 5% higher rate, it would have fetched ₹ 500 more. What was the principal amount?
[CDS 2021 (I)]
(a) 2,000 (b) 1,800
(c) 1,600 (d) 1,200
25. The difference between the compound interest (compounded annually) and the of simple interest on a certain sum of money at 12% per annum for 2 years is ₹ 72. What is the principal amount?
[CDS 2021 (I)]
(a) 6,500 (b) 6,000
(c) 5,500 (d) 5,000
26. A sum of money compounded annually doubles itself in 5 years. In how many years will it become four times of itself?
[CDS 2021 (II)]
(a) 10 years (b) 12 years
(c) 15 years (d) 20 years
27. A person borrowed ₹ 9,000 at 7%, ₹ 12,000 at 8% and ₹ 15,000 at 9% simple interest per annum. He had to pay ₹ 50,700 at the end of n years. What is the value of n ?
[CDS 2022 (I)]
(a) 3 (b) 4
(c) 5 (d) 6
28. If C is the compound interest on ₹ 10,000 for one year at 4% per annum when compounded quarterly, then which one of the following is correct?
[CDS 2022 (II)]
(a) $C < ₹ 100$
(b) $₹ 100 < C < ₹ 200$
(c) $₹ 200 < C < ₹ 400$
(d) $C > ₹ 400$
29. What is the difference between simple interest and compound interest on ₹ 10,000 for two years at 20% per annum compounded half-yearly.
[CDS – 2023 (1)]
(a) ₹ 842 (b) ₹ 756
(c) ₹ 641 (d) ₹ 542
30. In a business dealing, A owes B ₹ 20,000 payable after 5 years, whereas B owes A ₹ 12,000 payable after 4 years. They want to settle it now at the rate of 5% simple interest. Who gives how much money in this settlement?
[CDS – 2023 (1)]
(a) both are at par
(b) B gives ₹ 6,000 to A
(c) A gives ₹ 6,000 to B
(d) A gives ₹ 4,000 to B
31. A sum of money at 20% rate of compound interest per annum becomes more than 100 times in n years. What is the least value of n ? (use $\log_{10} 2 = 0.301$, $\log_{10} 3 = 0.477$)
[CDS-2023 (2)]
(a) 23 (b) 24
(c) 25 (d) 26
32. A person borrowed ₹ 10,000 at 12% rate of interest per annum compounded quarterly for a period of 9 months. What is the interest paid by him to settle his account after 9 months?
[CDS-2023 (2)]
(a) ₹ 927.27 (b) ₹ 947.47
(c) ₹ 967.67 (d) ₹ 987.87
33. A sum of money invested at simple interest triples itself in 8 years and becomes n times in 20 years. What is the value of n ?
[CDS-2024 (1)]
(a) 5 (b) 6
(c) 7.5 (d) 9
34. Let x be the compound interest at the end of 3 years on a sum of ₹ 1000 at the rate of 10% compounded annually and y be the simple interest at the end of 3 years on a sum of ₹ 1000 at the annual rate of 11%. What is the difference between x and y ?
[CDS-2024 (1)]
(a) ₹ 16 (b) ₹ 15
(c) ₹ 5 (d) ₹ 1
35. If the rate of interest is 5%, then what would be the difference between compound

interest and simple interest received on Rs 10,000 (each) after 3 years from now?

[CDS-2024 (1)]

(a) Rs 175.25
(c) Rs 76.25

(b) Rs 152.25
(d) Rs 24.25

ANSWER KEY

1.	(c)	2.	(b)	3.	(c)	4.	(a)	5.	(c)	6.	(a)	7.	(a)	8.	(b)	9.	(c)	10.	(c)
11.	(c)	12.	(a)	13.	(b)	14.	(a)	15.	(c)	16.	(c)	17.	(c)	18.	(b)	19.	(b)	20.	(c)
21.	(c)	22.	(d)	23.	(d)	24.	(a)	25.	(d)	26.	(a)	27.	(c)	28.	(d)	29.	(c)	30.	(c)
31.	d	32.	(a)	33.	(b)	34.	(d)	35.	(c)										

CAPF PYQ

1. The difference between the compound interest and the simple interest for 2 years on a sum of money is Rs. 60. If the simple interest for 2 years is Rs. 1440, what is the rate of interest?

[CAPF 2017]

(a) $4\frac{1}{6}\%$

(b) $6\frac{1}{4}\%$

(c) 8%

(d) $8\frac{1}{3}\%$

2. A sum triples in ten years under compound interest at a certain rate of interest; the interest is being compounded annually. In how many years, it would become nine times

[CAPF 2020]

(a) 20 yrs
(c) 40 yrs

(b) 30 yrs
(d) 50 yrs

3. The difference of compound interest and simple interest of a sum of money at the rate of 5% per year for 2 years is Rs 250. The sum is

[CAPF 2021]

(a) Rs 100000
(c) Rs 40000

(b) Rs 80000
(d) Rs 120000

4. Suppose a bank gives an interest of 10% per annum compounded annually for a fixed deposit for a period of two years. What should be the simple interest rate per annum if the maturity amount after two years is to remain the same?

[CAPF 2022]

(a) 10%
(c) 11%

(b) 10.5%
(d) 12%

ANSWER KEY

1.	(d)	2.	(a)	3.	(a)	4.	(b)												
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PRACTICE SHEET 1

1. If 'n' be any natural number, then by which largest number ($n^3 - n$) is always divisible?
 (1) 3 (2) 6
 (3) 12 (4) 18
2. If n is an integer, then ($n^2 - n$) is always divisible by :
 (1) 4 (2) 5
 (3) 2 (4) 7
3. If the sum of the digits of any integer lying between 100 and 1000 is subtracted from the number, the result always is
 (1) Divisible by 6 (2) divisible by 2
 (3) Divisible by 9 (4) divisible by 5
4. If a number is divisible by both 11 and 13, then it must be necessarily:
 (1) Divisible by (11 + 13)
 (2) Divisible by (13 - 11)
 (3) Divisible by (11 × 13)
 (4) 429
5. If * is a digit such that 5824* is divisible by 11, then * equals:
 (1) 2 (2) 3
 (3) 5 (4) 6
6. If 78*3945 is divisible by 11, where * is a digit, then * is equal to
 (1) 1 (2) 0
 (3) 3 (4) 5
7. If the number 4 8 3 2 7 * 8 is divisible by 11, then the missing digit (*) is
 (1) 5 (2) 3
 (3) 2 (4) 1
8. Both the end digits of a 99 digit number N are 2. N is divisible by 11, then all the middle digits are:
 (1) 1 (2) 2
 (3) 3 (4) 4
9. If n is a whole number greater than 1, then $n^2(n^2 - 1)$ is always divisible by :
 (1) 16 (2) 12
 (3) 10 (4) 8
10. A 4-digit number is formed by repeating a 2-digit number such as 2525, 3232, etc. Any number of this form is always exactly divisible by :
 (1) 7
 (2) 11
 (3) 13
 (4) Smallest 3-digit prime number
11. What least number, of 5 digits is divisible by 41?
 (1) 10045 (2) 10004
 (3) 10041 (4) 41000
12. The digit in unit's place of the product $81 \times 82 \times 83 \times \dots \times 89$ is
 (1) 0 (2) 2
 (3) 6 (4) 8
13. The digit in unit's place of the product $(2153)^{167}$ is:
 (1) 1 (2) 3
 (3) 7 (4) 9
14. The digit in the unit's place of the product $(2464)^{1793} \times (615)^{317} \times (131)^{491}$ is
 (1) 0 (2) 2
 (3) 3 (4) 5
15. Unit digit in $(264)^{102} + (264)^{103}$ is:
 (1) 0 (2) 4
 (3) 6 (4) 8
16. The digit in the unit's place of $[(251)^{98} + (21)^{29} - (106)^{100} + (705)^{35} - 16^4 + 259]$ is:
 (1) 1 (2) 4
 (3) 5 (4) 6
17. The last digit of 3^{40} is
 (1) 1 (2) 3
 (3) 7 (4) 9
18. What will be the unit digit in the product 7^{105} ?
 (1) 5 (2) 7
 (3) 9 (4) 1
19. The unit digit in the expansion of $(2137)^{754}$ is
 (1) 1 (2) 3
 (3) 7 (4) 9
20. One's digit of the number $(22)^{23}$ is
 (1) 4 (2) 6
 (3) 8 (4) 2

- 21.** The unit digit in the product $(122)^{173}$ is
 (1) 2 (2) 4
 (3) 6 (4) 8
- 22.** Find the unit digit of $(7^{95} - 3^{58})$
 (1) 7 (2) 3
 (3) 4 (4) 0
- 23.** Find the unit digit of $111!$
 (1) 0 (2) 1
 (3) 5 (4) 3
- 24.** Unit digit in $(264)^{102} + (264)^{103}$ is:
 (1) 0 (2) 4
 (3) 6 (4) 8
- 25.** Find the unit digit in the product: $(4387)^{245} \times (621)^{72}$.
 (1) 1 (2) 2
 (3) 5 (4) 7
- 26.** The unit's digit in the product $7^{71} \times 6^{63} \times 3^{65}$ is
 (1) 1 (2) 2
 (3) 3 (4) 4
- 27.** Find the unit digit of the product of all the prime number between 1 and 99999
 (1) 9 (2) 7
 (3) 0 (4) N.O.T.
- 28.** Find the unit digit of $1^1 + 2^2 + 3^3 + \dots + 10^{10}$.
 (1) 9 (2) 7
 (3) 0 (4) N.O.T.
- 29.** Find the unit digit of
 $13^{24} \times 68^{57} + 24^{13} \times 57^{68} + 1234 + 5678$.
 (1) 4 (2) 7
 (3) 0 (4) 8
- 30.** The last digit of the following expression is:
 $(1!)^1 + (2!)^2 + (3!)^3 + (4!)^4 + \dots + (10!)^{10}$
 (1) 4 (2) 5
 (3) 6 (4) 7

ANSWER KEY

1.	(2)	2.	(3)	3.	(3)	4.	(3)	5.	(3)	6.	(4)	7.	(4)	8.	(4)	9.	(2)	10.	(4)
11.	(2)	12.	(1)	13.	(3)	14.	(1)	15.	(1)	16.	(2)	17.	(1)	18.	(2)	19.	(4)	20.	(3)
21.	(1)	22.	(3)	23.	(1)	24.	(1)	25.	(4)	26.	(4)	27.	(3)	28.	(2)	29.	(1)	30.	(2)

PRACTICE SHEET 2

1. Find the number of zeroes at the end of the product 1000!
(1) 200 (2) 249
(3) 248 (4) 250
2. Find the number of zeroes at the end of the product 2700!
(1) 673 (2) 670
(3) 669 (4) 675
3. Find the number of zeroes at the end of the product $10 \times 15 \times 44 \times 28 \times 70$
(1) 2 (2) 3
(3) 4 (4) 5
4. Find the number of zeroes at the end of the product $2 \times 4 \times 6 \times \dots \times 48 \times 50$
(1) 6 (2) 12
(3) 7 (4) 5
5. Find the number of zeroes at the end of the product $1 \times 3 \times 5 \times 7 \times 9 \times 11 \times \dots \times 99 \times 101$
(1) 24 (2) 5
(3) 2 (4) 0
6. Find the number of zeroes at the end of the product $35 \times 36 \times 37 \times \dots \times 89 \times 90$
(1) 21 (2) 7
(3) 14 (4) 20
7. Find the number of zeroes at the end of the product $41 \times 42 \times \dots \times 109 \times 110$
(1) 26 (2) 9
(3) 17 (4) 25
8. Find the number of zeroes at the end of the product $25! \times 32! \times 45!$
(1) 10 (2) 23
(3) 22 (4) 7
9. Find the number of zeroes at the end of the product $1140! \times 358! \times 171!$
(1) 282 (2) 325
(3) 411 (4) 370
10. Find the number of zeroes at the end of the product $1^{1!} \times 2^{2!} \times 3^{3!} \times 4^{4!} \times \dots \times 10^{10!}$
(1) 51 (2) 10
(3) $5! + 10!$ (4) N.O.T
11. Find the number of Factors of 1728
(1) 28 (2) 29
(3) 30 (4) 31
12. Find the Number of Factor of 1420
(1) 12 (2) 13
(3) 14 (4) 15
13. Find the Number of Divisors of 10800
(1) 30 (2) 60
(3) 120 (4) 180
14. Find the No. of Prime Factor of 240.
(1) 4 (2) 5
(3) 6 (4) 8
15. Find the No. of prime factor $(30)^{26} \times (25)^{51} \times (12)^{23}$
(1) 249 (2) 250
(3) 255 (4) 260
16. Find the No. of Prime Factor $(30)^{15} \times (22)^{11} \times (15)^{24}$
(1) 110 (2) 115
(3) 120 (4) 125
17. Find the No. of prime factor of $(56)^{20} \times (36)^{31} \times (42)^{13} \times (13)^{21}$
(1) 240 (2) 242
(3) 264 (4) 248
18. Find the prime Factors 210
(1) 3 (2) 4
(3) 5 (4) 6
19. Find the sum of odd factors of 544
(1) 16 (2) 18
(3) 20 (4) 22
20. For the Number 2450 find
(i) Number of all factors
(ii) Number of even factors
(iii) Number of odd factors
(1) 18,9,9 (2) 18,10,8
(3) 18,8,10 (4) 18,12,6
21. When a number is divided by 893, the remainder is 193. What will be remainder when it is divided by 47?
(1) 3 (2) 5
(3) 25 (4) 33
22. $8.\overline{31} + 0.\overline{6} + 0.00\overline{2}$ is equal to
(1) $8.\overline{912}$ (2) $8.\overline{912}$
(3) $8.9\overline{79}$ (4) $8.9\overline{79}$

- 23.** The value of $0.\overline{63} + 0.\overline{37}$ is
 (1) 1 (2) $100/99$
 (3) $99/100$ (4) $100/33$
- 24.** $(0.\overline{11} + 0.\overline{22}) \times 3$ is equal to
 (1) 3 (2) $1.\overline{9}$
 (3) 1 (4) $0.\overline{3}$
- 25.** $(1.\overline{2} \times 0.\overline{03}) =$
 (1) $0.\overline{04}$ (2) $0.\overline{036}$
 (3) $1.\overline{13}$ (4) $0.\overline{037}$
- 26.** The difference between $5.\overline{76}$ and $2.\overline{3}$ is
 (1) $2.\overline{54}$ (2) $3.\overline{73}$
 (3) $3.\overline{46}$ (4) $3.\overline{43}$
- 27.** The value of $0.\overline{2} + 0.\overline{3} + 0.\overline{32}$ is
 (1) $0.\overline{87}$ (2) $0.\overline{77}$
 (3) $0.\overline{82}$ (4) $0.\overline{86}$
- 28.** A number when divided by 280 leaves 115 as remainder. When the same number is divided by 35, the remainder is
 (1) 15 (2) 10
 (3) 20 (4) 17
- 29.** A certain number when divided by 175 leaves a remainder 132. When the same number is divided by 25, the remainder is:
 (1) 6 (2) 7
 (3) 8 (4) 9
- 30.** A positive integer when divided by 425 gives a remainder 45. When the same number is divided by 17, the remainder will be
 (1) 11 (2) 8
 (3) 9 (4) 10

ANSWER KEY

1.	(2)	2.	(1)	3.	(2)	4.	(1)	5.	(4)	6.	(3)	7.	(3)	8.	(2)	9.	(3)	10.	(3)
11.	(1)	12.	(1)	13.	(2)	14.	(3)	15.	(1)	16.	(2)	17.	(3)	18.	(2)	19.	(2)	20.	(1)
21.	(2)	22.	(3)	23.	(2)	24.	(3)	25.	(4)	26.	(4)	27.	(1)	28.	(2)	29.	(2)	30.	(1)

PRACTICE SHEET 3

1. Find the Remainder when $77 \times 85 \times 73$ is divided by 9
 (1) 1 (2) 2
 (3) 4 (4) 7
2. Find the Remainder when $273 + 375 + 478 + 657 + 597$ is divided by 25
 (1) 5 (2) 10
 (3) 9 (4) 8
3. Find the Remainder when $1330 \times 1356 \times 1363 \times 1368 \times 1397$ is divided by 13
 (1) 7 (2) 9
 (3) 11 (4) 8
4. Find the Remainder when $2327 + 2372 + 2394 + 4624 + 4650$ is divided by 23
 (1) 12 (2) 14
 (3) 13 (4) 10
5. Find the Remainder when 67^{32} is divided by 68
 (1) 67 (2) 66
 (3) 1 (4) 0
6. Find the Remainder when 99^{99} is divided by 100
 (1) 99 (2) 98
 (3) 1 (4) 3
7. Find the Remainder 197^{130} is divided by 196
 (1) 1 (2) 195
 (3) 7 (4) 5
8. Find the Remainder 6^{36} is divided by 215
 (1) 214 (2) 6
 (3) 5 (4) 1
9. Find the Remainder 75^{75} is divided by 37
 (1) 1 (2) 36
 (3) 3 (4) 7
10. Find the Remainder 43^{197} is divided by 7
 (1) 42 (2) 41
 (3) 1 (d) 6
11. Find the Remainder when $(59)^{28}$ is divided by 7
 (1) 2 (2) 4
 (3) 6 (4) 1
12. Find the Remainder when 41^{77} is divided by 17
 (1) 2 (2) 1
 (3) 6 (4) 4
13. Find the Remainder when 2^{49} is divided by 7
 (1) 1 (2) 2
 (3) 3 (4) 4
14. Find the Remainder when $(51^{203} + 2^{49})$ is divided by 17
 (1) 4 (2) 5
 (3) 6 (4) None of these
15. Find the Remainder when 8^{77} is divided by 7
 (1) 8 (2) 1
 (3) 13 (4) 7
16. Find the Remainder when $1! + 2! + 3! + 4! + \dots + 100!$ is divided by 5 is
 (1) 0 (2) 1
 (3) 2 (4) 3
17. Find the Remainder when $1! + 2! + 3! + 4! + \dots + 100!$ is divided by 6 is
 (1) 3 (2) 4
 (3) 2 (4) 1
18. Find the Remainder when 9^{111} is divided by 11:
 (1) 2 (2) 9
 (3) 7 (4) 6
19. Find the Remainder when 5^{2450} is divided by 126
 (1) 5 (2) 25
 (3) 125 (4) 1
20. Find the Remainder when 40^{1012} is divided by 7
 (1) 5 (2) 4
 (3) 3 (4) 2
21. Find the Remainder when $10^1 + 10 + 10^3 + \dots + 10^{100}$ is divided by 6
 (1) 4 (2) 6
 (3) 2 (4) 3
22. Find the Remainder when $10^1 + 10^2 + 10^3 + \dots + 10^{1000} + 10^{1001}$ is divided by 6
 (1) 4 (2) 6
 (3) 2 (4) 3

- 23.** Find the Remainder when 666666
134 times is divided by 13
(1) 1 (2) 3
(3) 11 (4) 9
- 24.** Find the Remainder when 555555
244 times is divided by 37
(1) 18 (2) 5
(3) 36 (4) 0
- 25.** Find the Remainder when 777777
363 times is divided by 11
(1) 0 (2) 7
(3) 1 (4) 3
- 26.** Find the Remainder when 888888 184
times is divided by 37
(1) 1 (2) 8
(3) 36 (4) 7
- 27.** Find the Remainder when 999999999 is
divided by 13
(1) 8 (2) 11
(3) 5 (4) 12
- 28.** Find the Remainder when 7^{99} is divided by
2400
(1) 1 (2) 49
(3) 343 (4) 7
- 29.** Find the Remainder when 3^{1989} is divided by
7
(1) 2 (2) 6
(3) 4 (4) 5
- 30.** Find the Remainder when 54^{124} is divided by
17
(1) 4 (2) 5
(3) 3 (4) 15

ANSWER KEY

1.	(2)	2.	(1)	3.	(2)	4.	(2)	5.	(3)	6.	(1)	7.	(1)	8.	(4)	9.	(1)	10.	(3)
11.	(2)	12.	(3)	13.	(2)	14.	(4)	15.	(1)	16.	(4)	17.	(1)	18.	(2)	19.	(2)	20.	(4)
21.	(1)	22.	(3)	23.	(1)	24.	(2)	25.	(2)	26.	(2)	27.	(2)	28.	(3)	29.	(2)	30.	(1)

PRACTICE SHEET 4

1. $(7^{19} + 2)$ is divided by 6, the remainder is :
 (1) 5 (2) 3
 (3) 2 (4) 1
2. If two numbers are each divided by the same divisor, the remainders are respectively 3 and 4. If the sum of the two numbers be divided by the same divisor, the remainder is 2. The divisor is
 (1) 9 (2) 7
 (3) 5 (4) 3
3. The remainder when 3^{21} is divided by 5 is
 (1) 1 (2) 2
 (3) 3 (4) 4
4. $9^6 - 11$ when divided by 8 would leave a remainder of :
 (1) 0 (2) 1
 (3) 2 (4) 6
5. If 17^{200} is divided by 18, the remainder is—
 (1) 17 (2) 16
 (3) 1 (4) 2
6. When 2^{31} is divided by 5 the remainder is
 (1) 4 (2) 3
 (3) 2 (4) 1
7. $(4^{61} + 4^{62} + 4^{63})$ is divisible by
 (1) 3 (2) 11
 (3) 13 (4) 17
8. $(2^{71} + 2^{72} + 2^{73} + 2^{74})$ is divisible by
 (1) 9 (2) 10
 (3) 11 (4) 13
9. $(3^{25} + 3^{26} + 3^{27} + 3^{28})$ is divisible by
 (1) 11 (2) 16
 (3) 25 (4) 30
10. $(49)^{15} - 1$ is exactly divisible by :
 (1) 50 (2) 51
 (3) 29 (4) 8
11. The sum of three consecutive odd natural numbers is 147. Then, the middle number is:
 (1) 47 (2) 48
 (3) 49 (4) 51
12. The sum of first 20 odd natural numbers is equal to :
 (1) 210 (2) 300
 (3) 400 (4) 420
13. The sum of all natural numbers from 75 to 97 is :
 (1) 1598 (2) 1798
 (3) 1958 (4) 1978
14. The sum of all natural numbers between 100 and 200, which are multiples of 3 is :
 (1) 5000 (2) 4950
 (3) 4980 (4) 4900
15. The sum of the squares of three consecutive natural numbers is 2030. Then, what is the middle number?
 (1) 25 (2) 26
 (3) 27 (4) 28
16. The sum of three consecutive odd natural numbers is 87. The smallest of these numbers is :
 (1) 29 (2) 31
 (3) 23 (4) 27
17. Sum of three consecutive even integers is 54. Find the least among them.
 (1) 18 (2) 15
 (3) 14 (4) 16
18. The sum of three consecutive numbers is 87. The middle number is
 (1) 27 (2) 29
 (3) 30 (4) 28
19. What is the sum of two consecutive even numbers, the difference of whose square is 84?
 (1) 38 (2) 34
 (3) 42 (4) 46
20. The sum of all the natural numbers from 51 to 100 is
 (1) 5050 (2) 4275
 (3) 4025 (4) 3775
21. A number when divided successively by 4 and 5 leaves remainder 1 and 4 respectively.

When it is successively divided by 5 and 4 the respective remainders will be

- (1) 4, 1 (2) 3, 2
(3) 2, 3 (4) 1, 2

22. The expression $2^{6n} - 4^{2n}$, where n is a natural number is always divisible by

- (1) 15 (2) 18
(3) 36 (4) 48

23. When an integer K is divided by 3, the remainder is 1, and when $K + 1$ is divided by 5, the remainder is 0. Of the following, a possible value of K is

- (1) 62 (2) 63
(3) 64 (4) 65

24. Which one of the numbers is divisible by 25?

- (1) 303310 (2) 373355
(3) 303375 (4) 22040

25. For any integral value of n , $3^{2n} + 9n + 5$ when divided by 3 will leave the remainder

- (1) 1 (2) 2
(3) 0 (4) 5

26. The maximum value of F in the following equation $5E9 + 2F8 + 3G7 = 1114$ is where E, F, G each stands for any digit.

- (1) 8 (2) 9
(3) 7 (4) 5

27. The number $334 \times 545 \times 7p$ is divisible by 3340 if p is at least.

- (1) 2 (2) 4
(3) 3 (4) 1

28. The sum of three numbers is 252. If the first number is thrice the second and third number is two-third of the first, then the second number is

- (1) 41 (2) 21
(3) 42 (4) 84

29. The difference between the greatest and the least five-digit numbers formed by the digits 2, 5, 0, 6 and 8 is (repetition of digits is not allowed)

- (1) 69552 (2) 65925
(3) 65952 (4) 63952

30. A man has some hens and some cows. If the total number of heads of hens and cows together is 50 and the number of feet of hens and cows together is 142, then the number of cows is

- (1) 21 (2) 25
(3) 27 (4) 29

ANSWER KEY

1.	(2)	2.	(3)	3.	(3)	4.	(4)	5.	(3)	6.	(2)	7.	(1)	8.	(2)	9.	(4)	10.	(4)
11.	(3)	12.	(3)	13.	(4)	14.	(2)	15.	(2)	16.	(4)	17.	(4)	18.	(2)	19.	(3)	20.	(4)
21.	(3)	22.	(4)	23.	(3)	24.	(3)	25.	(2)	26.	(4)	27.	(1)	28.	(3)	29.	(3)	30.	(1)

PRACTICE SHEET 5

1. A number divided by 13 leaves a remainder 1 and if the quotient, thus obtained, is divided by 5, we get a remainder of 3. What will be the remainder if the number is divided by 65 ?
 (1) 28 (2) 16
 (3) 18 (4) 40
2. If two numbers are each divided by the same divisor, the remainders are respectively 3 and 4. If the sum of the two numbers be divided by the same divisor, the remainder is 2. The divisor is
 (1) 9 (2) 7
 (3) 5 (4) 3
3. A number consists of two digits. If the number formed by interchanging the digits is added to the original number, the resulting number (i.e. the sum) must be divisible by
 (1) 11 (2) 9
 (3) 5 (4) 3
4. What is the value of x for which $x, x + 1, x + 3$ are all prime numbers?
 (1) 0 (2) 1
 (3) 2 (4) 101
5. Which one of the following is correct regarding the number 222222 ?
 (1) It is divisible by 3 but not divisible by 7
 (2) It is divisible by 3 and 7 but not divisible by 11
 (3) It is divisible by 2 and 7 but not divisible by 11
 (4) It is divisible by 3, 7 and 11
6. What is the sum of positive integers less than 100 which leave remainder 1 when divided by 3 and leave a remainder 2 when divided by 4?
 (1) 416 (2) 620
 (3) 1250 (4) 1314
7. A three-digit number is divisible by 11 and has its digit in the unit's place equal to 1. The number is 297 more than the number obtained by reversing the digits. What is the number?
 (1) 121 (2) 231
 (3) 561 (4) 451
8. The remainder on dividing given integers a and b by 7 are, 5 and 4 respectively. What is the remainder when ab is divided by 7?
 (1) 3 (2) 4
 (3) 5 (4) 6
9. Which one of the following is the largest 4 digit number divisible by 88?
 (1) 9988 (2) 9966
 (3) 9944 (4) 8888
10. The number 2784936 is divisible by which of the following number?
 (1) 86 (2) 87
 (3) 88 (4) 89
11. If k is any even positive integer, then $(k^2 + 2k)$ is
 (1) divisible by 24
 (2) divisible by 8 but may not be divisible by 24
 (3) divisible by 4 but may not be divisible by 8
 (4) divisible by 2 but may not be divisible by 4
12. What is the number of total three digit numbers with unit digit 7 and divisible by 11 ?
 (1) 6 (2) 7
 (3) 8 (4) 9
13. In a two digit number if it is known that its units digit exceeds its tens digit by 2 and that the product of the given number and the sum of its digits is equal to 144, then the number is
 (1) 46 (2) 42
 (3) 26 (4) 24
14. The largest integer that divides product of any four consecutive integers is
 (1) 4 (2) 6
 (3) 12 (4) 24
15. A ten digit number is divisible by 4 as well as by 5. What could be the possible digit at ten's place in the given number?
 (1) 0, 1, 2, 4 and 6 (2) 1, 2, 4, and 8
 (3) 2, 3, 4, 6 and 8 (4) 0, 2, 4, 6 and 8

16. If $\frac{49}{15} = 3 + \frac{1}{x + \frac{1}{y + \frac{1}{z}}}$

Where x, y, z are natural numbers, then what is z equal to?

- (a) 1
- (b) 2
- (c) 3
- (d) Cannot be determined due to insufficient data

17. Consider the following statements :

If p is a prime number such that $p + 2$ is also a prime then

I. $p(p + 2) + 1$ is a perfect square.

II. 12 is a divisor of $p + (p + 2)$, if $p > 3$

Which of the statements given above is/are correct

- (a) Only I
- (b) Only II
- (c) Both I and II
- (d) Neither I nor II

18. A number consists of two digits such that the digit in the ten's place is less by 2 than the digit in the unit's place. Three times the number added to $\frac{6}{7}$ times the number obtained by reversing the digits equals 108. The sum of digits in the number is :

- (1) 8
- (2) 9
- (3) 6
- (4) 7

19. Of the three numbers, the second is twice the first and it is also thrice the third. If the average of three numbers is 44, the difference of the first number and the third number is :

- (1) 24
- (2) 18
- (3) 12
- (4) 6

20. A two digit number is five times the sum of its digits. If 9 is added to the number, the digits interchange their positions. The sum of digits of the number is :

- (1) 11
- (2) 9
- (3) 7
- (4) 6

21. How many numbers less than 1000 are multiples of both 10 and 13?

- (1) 9
- (2) 8
- (3) 6
- (4) 7

22. The number 1, 2, 3, 4, ..., 1000 are multiplied together. The number of zeros at the end (on the right) of the product must be:

- (1) 30
- (2) 200
- (3) 211
- (4) 249

23. Consider the following assumption and two statements:

Assumption: A number, 'ABCDE' is divisible by 11.

Statement I: $E - D + C - B + A$ is divisible by 11.

Statement II: $E - D + C - B + A = 0$

Which one of the following is correct ?

- (a) Only statement I can be drawn from the assumption
- (b) Only statement II can be drawn from the assumption
- (c) Both the statements can be drawn from the assumption
- (d) Neither of the statements can be drawn from the assumption

24. The product of two numbers is 0.008. One of the number is $\frac{1}{5}$ of the other. The smaller number is

- (1) 0.2
- (2) 0.4
- (3) 0.02
- (4) 0.04

25. I multiplied a natural number by 18 and another by 21 and added the products. Which one of the following could be the sum?

- (1) 2007
- (2) 2008
- (3) 2006
- (4) 2002

26. If the sum of two numbers be multiplied by each number separately, the products so obtained are 247 and 114. The sum of the numbers is

- (1) 19
- (2) 20
- (3) 21
- (4) 23

27. If a and b are odd numbers, then which of the following is even ?

- (1) $a + b + ab$
- (2) $a + b - 1$
- (3) $a + b + 1$
- (4) $a + b + 2ab$

28. If two numbers x and y separately divided by a number d, remainders obtained are 4375 and 2986 respectively. If the sum of the numbers i.e. $(x+y)$ is divided by the same number d remainder obtained is 2361. The value of number d is:

- (1) 7361
- (2) 5000
- (3) 4000
- (4) 2542

- 29.** A farmer divides his herd of n cows among his four sons so that the first son gets one – half the herd, the second son gets one – fourth, the third son gets one – fifth and the fourth son gets 7 cows. The value of n is
 (1) 80 (2) 100
 (3) 140 (4) 180

- 30.** In an examination, a student scores 4 marks for every correct answer and loses 1 mark

for every wrong answer. A student attempted all the 200 questions and scored in all 200 marks. The number of questions, he answered correctly was

- (1) 82 (2) 80
 (3) 68 (4) 60

ANSWER KEY

1.	(4)	2.	(3)	3.	(1)	4.	(3)	5.	(4)	6.	(1)	7.	(4)	8.	(4)	9.	(3)	10.	(3)
11.	(2)	12.	(3)	13.	(4)	14.	(4)	15.	(4)	16.	(3)	17.	(3)	18.	(3)	19.	(3)	20.	(2)
21.	(4)	22.	(4)	23.	(3)	24.	(4)	25.	(1)	26.	(1)	27.	(4)	28.	(2)	29.	(3)	30.	(2)

CDS PYQ

1. The two digit number, which when divided by sum of the digit and product of the digits, respectively. The remainder is same and the difference of quotients is one, is
[CDS 2013 (I)]
(a) 14 (b) 23
(c) 32 (d) 41
2. Which one of the following numbers is divisible by 11?
[CDS 2013 (1)]
(a) 45678940 (b) 54857266
(c) 87524398 (d) 93455120
3. If N, (N + 2) and (N + 4) are prime number, then the number of possible solution for N are:
[CDS 2013 (1)]
(a) 1 (b) 2
(c) 3 (d) None of these
4. The smallest positive prime (say p) such that $2^p - 1$ is not a prime is:
[CDS 2013 (1)]
(a) 5 (b) 11
(c) 17 (d) 29
5. Consider the following statements:
I. There are a finite number of rational numbers between any two rational numbers.
II. There is an infinite number of rational numbers between any two rational numbers
III. There are a finite number of irrational numbers between any two rational numbers
Which of the above statements is/are correct?
[CDS 2013 (1)]
(a) Only I (b) Only II
(c) Only III (d) Both I and II
6. Representation of $0.\overline{2341}$ in the form p/q , where p and q are integers, $q \neq 0$, is
[CDS]
(a) $781/3330$ (b) $1171/4995$
(c) $2341/9990$ (d) $2339/9990$
7. If m and n are natural number, then $\sqrt[n]{m}$ is:
[CDS 2013 (2)]
(a) Always irrational
(b) Irrational unless n is the math power of an integer
(c) Irrational unless m is the nth power of an integer
(d) Irrational unless m is the n are co prime
8. If x is positive even integer and y is negative odd integer, then x^y is:
[CDS 2013 (2)]
(a) Odd integer (b) Even integer
(c) Rational number (d) None of these
9. The paid of rational number that lies between $\frac{1}{4}$ and $\frac{3}{4}$ is:
[CDS 2013 (2)]
(a) $\frac{262}{1000}, \frac{752}{1000}$ (b) $\frac{24}{100}, \frac{78}{100}$
(c) $\frac{9}{40}, \frac{31}{40}$ (d) $\frac{252}{1000}, \frac{748}{1000}$
10. What is the last digit in $7^{402} + 3^{402}$?
[CDS 2013 (2)]
(a) 0
(b) 4
(c) 8
(d) None of the above
11. What is $26^2 + 97^2$ equal to?
[CDS 2013 (2)]
(a) $27^2 + 93^2$ (b) $34^2 + 93^2$
(c) $82^2 + 41^2$ (d) $79^2 + 62^2$
12. Every prime number of the form $3k + 1$ can be represented in the form $6m + 1$ (where, k and m are integer), when:
[CDS 2013 (2)]
(a) k is odd
(b) k is even
(c) k can be both odd and even
(d) No such form is possible
13. If k is a positive integer, then every square integer is of the form:
[CDS 2013 (2)]
(a) Only $4k$ (b) $4k$ or $4k + 3$
(c) $4k + 1$ or $4k + 3$ (d) $4k$ or $4k + 1$
14. If b is the largest square divisor of c and a^2 divides c, then which one of the following is correct (where a, b and c are integers)?
[CDS 2013 (2)]

- (a) b divides a
(b) a does not divide b
(c) a divided b
(d) Neither 10 nor 20
- 15.** $19^5 + 21^5$ is divisible by: **[CDS 2013 (2)]**
- (a) Only 10
(b) Only 20
(c) Both 10 and 20
(d) Neither 10 nor 20
- 16.** Consider the following statements:
I. 7710312401 is divisible by 11
II. 173 is a primer number
Which of the statements given above is/are correct? **[CDS 2013 (2)]**
- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
- 17.** Consider the following statements:
I. To obtain prime number less than 121, we have to reject all the multiples of 2,3,5 and 7
II. Every composite number less than 121 is divisible by a prime number less than 11
Which of the statements given above is/are correct? **[CDS 2013 (2)]**
- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
- 18.** Consider the following statements:
I. No integer of the form $4k + 3$, where k is an integer can be expressed as the sum of two squares.
II. Square of an odd integer can express in the form $8k + 1$, where k is an integer.
Which of the statements given above is/are correct? **[CDS 2014 (1)]**
- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
- 19.** If n is a whole number greater than 1, then $n^2 (n^2 - 1)$ is always divisible by: **[CDS 2014 (1)]**
- (a) 12 (b) 24
(c) 48 (d) 60
- 20.** What is the remainder when 4^{1000} is divided by 7? **[CDS 2014 (1)]**
- (a) 1 (b) 2
(c) 4 (d) None of these
- 21.** The difference of two consecutive cubes: **[CDS 2014 (1)]**
- (a) Is odd or even
(b) Is never divisible by 2
(c) Is always even
(d) None of these
- 22.** The product of four consecutive natural numbers plus one is: **[CDS 2014 (1)]**
- (a) A non square
(b) Always sum of two spare numbers
(c) A square
(d) None of the above
- 23.** $7^{10} - 5^{10}$ is divisible by: **[CDS 2014 (2)]**
- (a) 10 (b) 7
(c) 5 (d) 11
- 24.** The multiplication of a three digits number XY5, with digit Z yields X215. What is $X + Y + Z$ equal to? **[CDS 2014 (2)]**
- (a) 13 (b) 15
(c) 17 (d) 18
- 25.** If $N^2 - 33$, $N^2 - 31$ and $N^2 - 29$ are prime numbers, then what is the number of possible values of N, where N is integer? **[CDS 2014 (2)]**
- (a) 1 (b) 2
(c) 6 (d) None of these
- 26.** How many pairs of positive integers m and n satisfy the equation $\frac{1}{m} + \frac{4}{n} = \frac{1}{12}$, where n is an odd inters less than 60? **[CDS 2014 (2)]**
- (a) 7 (b) 5
(c) 4 (d) 3
- 27.** Consider all those two digits positive integers less than 50, which when divided by 4 yield unity as remainder. What is their sum? **[CDS 2014 (2)]**
- (a) 310 (b) 314
(c) 218 (d) 323
- 28.** How many pairs of X and Y are possible in the number 763X4Y2, if the number is divisible by 9? **[CDS 2014 (2)]**

(a) 8
(c) 10

(b) 9
(d) 11

29. What is the remainder when 4^{1012} is divided by 7?

[CDS 2014 (2)]

(a) 1
(c) 3

(b) 2
(d) 4

30. What is the remainder when $(17^{23} + 23^{23} + 29^{23})$ is divided by 23?

[CDS 2014 (2)]

(a) 0
(c) 2

(b) 1
(d) 3

31. p, q and r are prime numbers such that $p < q < r < 13$. In how many cases would $(p + q + r)$ also be a prime number?

[CDS 2014 (2)]

(a) 1
(c) 3

(b) 2
(d) None of these

32. What is the number of divisors of 360?

[CDS 2014 (2)]

(a) 12
(c) 24

(b) 18
(d) None of these

33. If $a_n = 3 - 4n$, then what is $a_1 + a_2 + a_3 + \dots + a_n$ equal to?

$$\left[1 + 2 + 3 + 4 + \dots + n = \frac{n(n+1)}{2} \right]$$

[CDS 2014 (2)]

(a) $-n(4n-3)$
(c) $-n^2$

(b) $-n(2n-1)$
(d) $-n(2n+1)$

34. What is the remainder when $(1235 \times 4523 \times 2451)$ is divided by 12?

[CDS 2014 (2)]

(a) 1
(c) 5

(b) 3
(d) 7

35. A certain number of two digits is three times the sum of its digits. If 45 is added to the number, the digits will be reversed. What is the sum of the squares of the two digits of the number?

[CDS 2014 (2)]

(a) 41
(c) 53

(b) 45
(d) 64

36. If $\frac{37}{13} = 2 + \frac{1}{x + \frac{1}{y + \frac{1}{z}}}$

Where x, y, z are natural numbers, then what is z equal to?

[CDS 2015 (1)]

(a) 1
(b) 2
(c) 3
(d) Cannot be determined due to insufficient data

37. The digit in the units place of the product $81 \times 82 \times 83 \times 84 \times \dots \times 99$ is:

[CDS 2015 (1)]

(a) 0
(c) 6

(b) 4
(d) 8

38. A student was asked to multiply a number by 25. He instead multiplied the number by 52 and got the answer 324 more than the correct answer. The number to be multiplied was:

[CDS 2015 (1)]

(a) 12
(c) 25

(b) 15
(d) 32

39. What is the number of possible pairs of (P, Q) if the number 357P25Q is divisible by both 3 and 5?

[CDS 2015 (1)]

(a) 7
(b) 6
(c) 5
(d) None of the above

40. The difference between the square of two consecutive odd integers is always divisible by:

[CDS 2015 (1)]

(a) 3
(c) 8

(b) 7
(d) 16

41. What is the maximum value of m if the number $N = 35 \times 45 \times 55 \times 60 \times 124 \times 75$ is divisible by 5^m ?

[CDS 2015 (1)]

(a) 4
(c) 6

(b) 5
(d) 7

42. When a ball bounces, it rises to $\frac{2}{3}$ of the height from which it fell. If the ball is dropped from a height of 36m, how high will it rise at the third bounce?

[CDS 2015 (1)]

(a) $10\frac{1}{3}m$

(b) $10\frac{2}{3}m$

(c) $12\frac{1}{3}m$

(d) $12\frac{2}{3}m$

43. A light was seen regularly at an interval of 13 seconds. It was seen for the first time at 1 hour 54 minutes 50 seconds (am) and the last time at 3 hours 17 minutes 49 seconds (am). How many times was the light seen?

[CDS 2015 (1)]

(a) 375
(c) 283

(b) 378
(d) 384

44. If n is natural number and $n = p_1^{x_1} p_2^{x_2} p_3^{x_3}$, where p_1, p_2, p_3 are distinct prime factors, then the number of prime factors for n is:

[CDS 2015 (1)]

(a) $x_1 + x_2 + x_3$

(b) $x_1 x_2 x_3$

(c) $(x_1 + 1)(x_2 + 1)(x_3 + 1)$

(d) None of the above

45. Consider the following statements for the sequence of numbers given below:

11, 111, 1111, 11111

1. Each number can be expressed in the form $(4m + 3)$, where m is a natural number

2. Some numbers are squares.

Which of the above statements is/are correct?

[CDS 2015 (1)]

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

46. Consider the following statements:

1. There exists only one prime number p such that $(17p + 1)$ is a square

2. If x is the product of 10 consecutive prime numbers starting from 2, then $(x+1)$ is also prime number.

Which of the above statements is/are correct?

[CDS 2015 (1)]

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

47. The last digit in the expansion of 17^{256} is:

[CDS 2015 (1)]

(a) 9

(b) 7

(c) 3

(d) 1

48. What is the remainder obtained when $1421 \times 1423 \times 1425$ is divided by 12?

[CDS 2015 (1)]

(a) 1

(b) 2

(c) 3

(d) 4

49. What is the remainder when 4^{96} is divided by 6?

[CDS 2015 (1)]

(a) 4

(b) 3

(c) 2

(d) 1

50. Let p be a prime number other than 2 or 5. One would like to express the vulgar fraction $1/p$ in the form of a recurring decimal. Then the decimal will be

[CDS 2015 (1)]

(a) a pure recurring decimal and its period will be necessarily $(p - 1)$

(b) a mixed recurring decimal and its period will be necessarily $(p - 1)$

(c) a pure recurring decimal and its period will be some factor of $(p - 1)$

(d) a mixed recurring decimal and its period will be some factor of $(p - 1)$

51. The number of values of x satisfying $x + \frac{100}{x} > 50$, where x is a natural number less than or equal to 100 is:

[CDS 2015 (2)]

(a) 51

(b) 53

(c) 55

(d) 57

52. The largest natural number which divides every natural number of the form $(n^3 - n)$ ($n - 2$), where n is a natural number greater than 2 is:

[CDS 2015 (2)]

(a) 6

(b) 12

(c) 24

(d) 48

53. The digit in the units place of the resulting number of the expression $(234)^{100} + (234)^{101}$ is:

[CDS 2015 (2)]

(a) 6

(b) 4

(c) 2

(d) 0

54. The seven digit number $876p37q$ is divisible by 225. The values of p and q can be respectively:

[CDS 2015 (2)]

(a) 9, 0

(b) 0, 0

(c) 0, 5

(d) 9, 5

55. Let x and y be positive integers such that $x > y$. The expressions $3x + 2y$ and $2x + 3y$ when

divided by 5 leave remainder 2 and 3 respectively. What is the remainder when $(x-y)$ is divided by 5?

[CDS 2015 (2)]

- (a) 4 (b) 2
(c) 1 (d) 0

56. A number consists of two digits, whose sum is 7. If the digits are reversed, the number is increased by 27. The product of digits of the number is:

[CDS 2015 (2)]

- (a) 6 (b) 8
(c) 10 (d) 12

57. Consider all positive two digit numbers each of which when divided by 7 leaves a remainder 3. What is their sum?

[CDS 2015 (2)]

- (a) 661 (b) 666
(c) 676 (d) 777

58. The value of $0.\overline{63} + 0.\overline{37}$ is

[CDS 2015 (2)]

- (a) 1 (b) $100/91$
(c) $100/99$ (d) $1000/99$

59. Consider the following statements in respect of the expression

$$S_n = \frac{n(n+1)}{2}, \text{ where } n \text{ is an integer.}$$

1. There are exactly two values of n for which $S_n = 861$

2. $S_n = S_{-(n+1)}$ and hence for any integer m , we have two values of n for which $S_n = m$.

Which of the above statement is/are correct?

[CDS 2016 (1)]

- (a) 1 Only (b) 2 Only
(c) Both 1 and 2 (d) neither 1 nor 2

60. Consider the following statements in respect of two different non-zero integers' p and q .
I. For $(p+q)$ to be less than $(p-q)$, q must be negative.

II. For $(p+q)$ to be greater than $(p-q)$, both p and q must be positive.

Which of the above statements is/are correct?

[CDS 2016 (1)]

- (a) 1 Only (b) 2 Only
(c) Both 1 and 2 (d) neither 1 nor 2

61. $7^{10} - 5^{10}$ is divisible by:

[CDS 2016 (1)]

- (a) 5 (b) 7

- (c) 10 (d) none

62. Let a two digit number be k times the sum of its digits. If the number formed by interchanging the digit is m times the sum of the digits, then the value of m is:

[CDS 2016 (1)]

- (a) $9-k$ (b) $10-k$
(c) $11-k$ (d) $k-1$

63. Let S be a set of first fourteen natural numbers. The possible number of pairs (a, b) , where $a, b, \in S$ and $a \neq b$ such that ab leaves remainder 1 when divided by 15, is:

[CDS 2016 (1)]

- (a) 3
(b) 5
(c) 6
(d) None of the above

64. A clock strikes once 1 O'clock, twice at 2 O'clock and thrice at 3 O'clock and so on. If it takes 8 seconds to strike at 5 O'clock, the time taken by it to strike at 10 O'clock is:

[CDS 2016 (1)]

- (a) 14 seconds
(b) 16 seconds
(c) 18 seconds
(d) None of the above

65. What is the maximum value of m , if the number $N = 90 \times 42 \times 324 \times 55$ is divisible by 3^m ?

[CDS 2016 (1)]

- (a) 8 (b) 7
(c) 6 (d) 5

66. Consider the following statements:

1. Every natural number is a real number
 2. Every real number is a rational number
 3. Every integer is a real number
 4. Every rational number is a real number
- Which of the above statements are correct?

[CDS 2016 (1)]

- (a) 1, 2 and 3 (b) 1, 3 and 4
(c) 2 and 3 only (d) 3 and 3 only

67. Consider the following statements:

1. There exists a positive real number m such that $\cos x = 2^{m+1}$.
2. $mn \geq m + n$ for all m, n belonging to set of natural numbers.

Which of the above statements is/are correct?

[CDS 2016 (1)]

- (a) 1 Only (b) 2 Only

- (c) Both 1 and 2 (d) neither 1 nor 2

68. If m and n are distinct natural numbers, then which of the following is/are integer/integers?

1. $\frac{m}{n} + \frac{n}{m}$
2. $mn \left(\frac{m}{n} + \frac{n}{m} \right) (m^2 + n^2)$
3. $\frac{mn}{m^2 + n^2}$

Select the correct answer using the code given below:

[CDS 2016 (1)]

- (a) 1 and 2 (b) 2 only
(c) 2 and 3 (d) 3 only

69. Outside a meeting room, Madhukar was told by a person that each meeting takes place after $13/4$ hours. The last meeting has been over just 45 minutes ago and the next meeting will take place at 2 pm. At what time did Madhukar receive this information?

[CDS 2016 (1)]

- (a) 10:20 am (b) 11:30 am
(c) 11:40 am (d) 11:50 am

70. If $\frac{61}{19} = 3 + \frac{1}{x + \frac{1}{y + \frac{1}{z}}}$

Where x, y, z are natural numbers, then what is z equal to?

[CDS 2016 (1)]

- (a) 1
(b) 2
(c) 3
(d) Cannot be determined due to insufficient data

71. What would be the maximum value of Q in the equation $5P9 + 3R7 + 2Q8 = 1114$?

[CDS 2016 (2)]

- (a) 9 (b) 8
(c) 5 (d) 4

72. In an examination, a student was asked to divide a certain number by 8. By mistake he multiplied it by 8 and got the answer 2016 more than the correct answer. What was the number?

[CDS 2016 (2)]

- (a) 252 (b) 256
(c) 258 (d) 260

73. A boy saves Rs 4.65 daily. What is the least number of days in which he will be able to save an exact number of rupees?

[CDS 2016 (2)]

- (a) 10 (b) 20
(c) 21 (d) 25

74. What is the unit digit of 7^{139} ?

[CDS 2016 (2)]

- (a) 9 (b) 7
(c) 6 (d) 3

75. Which of the following rational number has non-terminating and repeating decimal expansion?

[CDS 2016 (2)]

- (a) $15/1600$ (b) $23/8$
(c) $35/50$ (d) $17/6$

76. The sum of squares of two positive integers is 208. If the square of the larger number is 18 times the smaller number, then what is the difference of the larger and smaller number?

[CDS 2016 (2)]

- (a) 2 (b) 3
(c) 4 (d) 6

77. What is the remainder when 2^{100} is divided by 101?

[CDS 2016 (2)]

- (a) 1 (b) 11
(c) 99 (d) 100

78. Which one of the following is correct in respect of the number 1729?

[CDS 2016 (2)]

- (a) It cannot be written as the sum of the cube of two positive integers
(b) It can be written as the sum of the cubes of two positive integers in one way only
(c) It can be written as the sum of the cubes of two positive integers in two ways only
(d) It can be written as the sum of the cubes of two positive integers in three ways only

79. Consider the following statements in respect of positive odd integers x and y :

1. $x^2 + y^2$ is even integer.
2. $x^2 + y^2$ is divisible by 4

Which of the above statements is/are correct?

[CDS 2016 (2)]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

- 80.** What is the remainder when $13^5 + 14^5 + 15^5 + 16^5$ is divided by 29?
[CDS 2016 (2)]
(a) 8 (b) 5
(c) 3 (d) 0
- 81.** What is the difference between the sum of the cubes and that of squares of first ten natural numbers?
[CDS 2016 (2)]
(a) 2280 (b) 2640
(c) 3820 (d) 4130
- 82.** $2^{122} + 4^{62} + 8^{42} + 4^{64} + 2^{130}$ is divisible by which one of the following integers?
[CDS 2016 (2)]
(a) 3 (b) 5
(c) 7 (d) 11
- 83.** What is the remainder when the number $(4444)^{4444}$ is divided by 9?
[CDS 2017 (1)]
(a) 4 (b) 6
(c) 7 (d) 8
- 84.** The number of prime numbers which are less than 100 is:
[CDS 2017 (1)]
(a) 24 (b) 25
(c) 26 (d) 27
- 85.** Consider the following statements:
1. Of two consecutive integers, one is even
2. Square of an odd integer is of the form $8n+1$
Which of the above statements is/are correct?
[CDS 2017 (1)]
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
- 86.** $(N^{p-1} - 1)$ is a multiple of p, if N is prime to p and p is a:
[CDS 2017 (1)]
(a) Prime number
(b) Rational number
(c) Real number
(d) Composite number
- 87.** Which one of the following is largest
[CDS 2017 (1)]
(a) $7/9$ (b) $11/14$
(c) $3/4$ (d) $10/13$
- 88.** How many numbers between 500 and 1000 are divisible by 13?
- (a) 36 (b) 37
(c) 38 (d) 39
- 89.** If the points P and Q represent the real numbers $0.8\bar{3}$ and $0.6\bar{2}$ on the number line, then the distance between P and Q is
[CDS 2017 (2)]
(a) $21/90$ (b) $19/90$
(c) $21/100$ (d) $56/90$
- 90.** Three numbers which are co-prime to each other, are such that the product of the first two is 286 and that of the last two is 770. What is the sum of the three numbers?
[CDS 2018 (1)]
(a) 85 (b) 80
(c) 75 (d) 70
- 91.** All odd prime numbers upto 110 are multiplied together. What is the unit digit in this product?
[CDS 2018 (1)]
(a) 0
(b) 80
(c) 5
(d) None of the above
- 92.** Which one of the following is correct?
[CDS 2018 (1)]
(a) Decimal expansion of a rational number is terminating.
(b) Decimal expansion of a rational number is non-terminating.
(c) Decimal expansion of an irrational number is terminating.
(d) Decimal expansion of an irrational number is non-terminating and non-repeating.
- 93.** What is the difference between $0.\bar{9}$ and 0.9?
[CDS 2018 (1)]
(a) 0 (b) 0.099
(c) 0.1 (d) 0.09
- 94.** Which one of the following decimal number is a rational number with denominator 37?
[CDS 2018 (1)]
(a) 0.459459459... (b) 0.459459459
(c) 0.045949459... (d) 0.00459459...
- 95.** What is the largest power of 10 that divides the product $1 \times 2 \times 3 \times 4 \times \dots \times 23 \times 24 \times 25$?
[CDS 2018 (1)]

- (a) 2
(b) 4
(c) 5
(d) None of the above

96. $5^{17} + 5^{18} + 5^{19} + 5^{20}$ is divisible by:

[CDS 2018 (1)]

- (a) 7 (b) 9
(c) 11 (d) 13

97. How many five digit numbers of the form $XXYXX$ is/are divisible by 33?

[CDS 2018 (2)]

- (a) 1 (b) 3
(c) 5 (d) Infinite

98. A five digit number $XY235$ is divisible by 3 where X and Y are digits satisfying $X + Y \leq 5$. What is the number of possible pairs of values of (X, Y) ?

[CDS 2018 (2)]

- (a) 5 (b) 6
(c) 7 (d) 9

99. The number of divisors of the number 38808, exclusive of the divisors 1 and itself, is:

[CDS 2018 (2)]

- (a) 74 (b) 72
(c) 70 (d) 68

100. If α and β are two real numbers such that $\alpha + \beta = -\frac{q}{p}$ and $\alpha\beta = \frac{r}{p}$, where $1 < p < q < r$, then which one of the following is the greatest?

[CDS 2018 (2)]

- (a) $\frac{1}{\alpha + \beta}$ (b) $\frac{1}{\alpha} + \frac{1}{\beta}$
(c) $-\frac{1}{\alpha\beta}$ (d) $\frac{\alpha\beta}{\alpha + \beta}$

101. Consider the following statements in respect of three 3-digit numbers XYZ , YZX and ZXY :

1. The sum of the numbers is not divisible by $(X+Y+Z)$.

2. The sum of the numbers is divisible by 111.

Which of the above statements is/are correct?

[CDS 2018 (2)]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

102. The number of all pairs (m, n) , where m and n are positive integers, such that

$$\frac{1}{m} + \frac{1}{n} - \frac{1}{mn} = \frac{2}{5} \text{ is:}$$

[CDS 2018 (2)]

- (a) 6 (b) 5
(c) 4 (d) 2

103. The sum of all possible product taken two at a time out of the numbers $\pm 1, \pm 2, \pm 3, \pm 4$ is:

[CDS 2018 (2)]

- (a) 0 (b) -30
(c) 30 (d) 55

104. If a, b and c are positive integers such that

$$\frac{1}{a + \frac{1}{b + \frac{1}{c + \frac{1}{2}}}} = \frac{16}{23}, \text{ then what is the mean of } a, b \text{ and } c?$$

[CDS 2019 (1)]

- (a) 1 (b) 2
(c) 1.33 (d) 2.33

105. What is the remainder when $(17^{29} + 19^{29})$ is divisible by 18?

[CDS 2019 (1)]

- (a) 6 (b) 2
(c) 1 (d) 0

106. What is the largest value of n such that 10^n divides the product:

$$2^5 \times 3^3 \times 4^8 \times 5^3 \times 6^7 \times 7^6 \times 8^{12} \times 9^9 \times 10^6 \times 15^{12} \times 20^{14} \times 22^{11} \times 25^{15} ?$$

[CDS 2019 (1)]

- (a) 65 (b) 55
(c) 50 (d) 45

107. How many pairs (A, B) are possible in the number 479865AB if the number is divisible by 9 and it is given that the last digit of the number is odd?

[CDS 2019 (1)]

- (a) 5 (b) 6
(c) 9 (d) 11

108. Consider the multiplication $999 \times abc = def132$ in decimal notation, where a, b, c, d, e and f are digits. What are the values of a, b, c, d, e and respectively?

[CDS 2019 (1)]

- (a) 6,6,8,6,8,7 (b) 8,6,8,6,7,8
(c) 6,8,8,7,8,6 (d) 8,6,8,8,6,7

- 109.** Which of the following statements is not true?
[CDS 2019 (1)]
(a) The difference of two prime numbers, both greater than 2, is divisible by 2.
(b) For two different integers m , n and a prime number p , if p divides the product $m \times n$, then p divides either m or n .
(c) If a number is of the form $6n-1$ (n being a natural number), then it is a prime number.
(d) There is only one set of three prime numbers such that there is a gap of 2 between two adjacent prime numbers.
- 110.** The sum of three prime numbers is 100. If one of them exceeds another by 36, then one of the numbers is:
[CDS 2019 (1)]
(a) 17
(b) 29
(c) 43
(d) None of the above
- 111.** What is the unit place digit in the expansion of 7^{73} ?
[CDS 2019 (1)]
(a) 1
(b) 3
(c) 7
(d) 9
- 112.** Suppose n is a positive integer such that $(n^2 + 48)$ is a perfect square. What is the number of such n ?
[CDS 2019 (1)]
(a) One
(b) Two
(c) Three
(d) Four
- 113.** Consider the following statements in respect of two integers p and q (both > 1) which are relatively prime:
1. Both p and q may be prime numbers
2. Both p and q may be composite numbers
3. One of p and q may be prime and the other composite
Which of the above statements are correct?
[CDS 2019 (1)]
(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3
- 114.** Which is $0.\overline{53} + 0.5\overline{3}$ equal to?
[CDS 2019 (1)]
(a) $1.\overline{068}$
(b) $1.0\overline{68}$
(c) $1.\overline{068}$
(d) 1.068
- 115.** Which one of the following is an irrational number?
[CDS 2019 (1)]
(a) $\sqrt{59049}$
(b) $\frac{231}{593}$
(c) 0.45454545.....
(d) 0.12112211122211112222.....
- 116.** The number 3^{521} is divided by 8. What is the remainder?
[CDS 2019 (1)]
(a) 1
(b) 3
(c) 7
(d) 9
- 117.** A prime number contains the digit X at unit's place. How many such digits of X are possible?
[CDS 2019 (1)]
(a) 3
(b) 4
(c) 5
(d) 6
- 118.** The expression $5^{2n} - 2^{3n}$ has a factor:
[CDS 2019 (1)]
(a) 3
(b) 7
(c) 17
(d) None of the above
- 119.** Consider two digit numbers which remain the same when the digits interchange their positions. What is the average of such digit numbers?
[CDS 2019 (1)]
(a) 33
(b) 44
(c) 55
(d) 66
- 120.** If $\frac{36}{11} = 3 + \frac{1}{x + \frac{1}{y + \frac{1}{z}}}$, when x , y and z are natural numbers, then what is $(x + y + z)$ equal to?
[CDS 2019 (2)]
(a) 6
(b) 7
(c) 8
(d) 9
- 121.** Consider the following statements:
1. Unit digit in 17^{174} is 7:
2. Difference of the squares of any two odd numbers is always divisible by 8.
3. Adding 1 to the product of two consecutive odd numbers makes it a perfect square.
Which of the above statements are correct?

- (a) 1, 2 and 3 (b) 1 and 2 only
(c) 2 and 3 only (d) 1 and 3 only
- 122.** When N is divided by 17, the quotient is equal to 182. The difference between the quotient and the remainder is 175. What is the value of N?
[CDS 2019 (2)]
(a) 2975 (b) 3094
(c) 3101 (d) 3269
- 123.** Consider the following statements:
1.If p is relatively prime to each of q and r, then p is relatively prime to the product qr.
2.If p divides the product qr and if p divides q, then p must divide r.
Which of the above statements is/are correct?
[CDS 2019 (2)]
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
- 124.** Let x be the smallest positive integer such that when 14 divides x, the remainder is 7; and when 15 divides x, the remainder is 5. Which one of the following is correct?
[CDS 2019 (2)]
(a) $20 < x < 30$ (b) $30 < x < 40$
(c) $40 < x < 50$ (d) $x > 50$
- 125.** Consider the following statements:
1. $\sqrt{75}$ is a rational number
2. There exists at least a positive integer x such that $-\frac{4x}{5} < -\frac{7}{8}$.
3. $\frac{x-2}{x} < 1$ for all real values of x.
4. 4.232323..... can be expressed in the form $\frac{p}{q}$ where p and q are integer.
Which of the above statements are correct?
[CDS 2019 (2)]
(a) 1 and 2 (b) 2 and 3
(c) 3 and 4 (d) 2 and 4
- 126.** If 10^n divides $6^{23} \times 75^9 \times 105^2$, then what is the largest value of n?
[CDS 2019 (2)]
(a) 20 (b) 22
(c) 23 (d) 28
- 127.** What is the digit in the unit's place of the number represented by $3^{98} - 3^{89}$?
- (a) 3 (b) 6
(c) 7 (d) 9
- 128.** How many pairs of natural numbers are there such that the difference of their squares is 35?
[CDS 2019 (2)]
(a) 1 (b) 2
(c) 3 (d) 4
- 129.** Which one of the following is not correct?
[CDS 2019 (2)]
(a) 1 is neither prime nor composite
(b) 0 is neither positive nor negative
(c) If $p \times q$ is even, then p and q are always even
(d) $\sqrt{2}$ is an irrational number
- 130.** What is the sum of all integer values of n for which $n^2 + 19n + 92$ is a perfect square?
[CDS 2019 (2)]
(a) 21 (b) 19
(c) 0 (d) -19
- 131.** The number $2 \times 3 \times 5 \times 7 \times 11 + 1$ is:
[CDS 2020 (1)]
(a) A prime number
(b) Not a prime, but power of prime
(c) Not a power of a prime, but a composite even number
(d) Not a power of a prime, but a composite odd number
- 132.** If the sum of digit of a number $10^n - 1$, where n is natural number is equal to 3798, then what is the value of n?
[CDS 2020 (1)]
(a) 421 (b) 422
(c) 423 (d) 424
- 133.** If m is the number of prime numbers between 0 and 50; and n is the number of prime numbers between 50 and 100, then what is (m - n) equal to?
[CDS 2020 (1)]
(a) 4 (b) 5
(c) 6 (d) 7
- 134.** Which one of the following is the largest divisor of $3^x + 3^{x+1} + 3^{x+2}$, if x is any natural number?
[CDS 2020 (1)]
(a) 3 (b) 13
(c) 39 (d) 117

- 135.** A two digit number is 9 more than four times of the number obtained by interchanging its digits. If the product of digits in the two digit number is 8, then what is the number?

[CDS 2020 (1)]

- (a) 81 (b) 42
(c) 24 (d) 18

- 136.** If m and n are positive integers such that $m^n = 1331$, then what is the value of $(m-1)^{n-1}$?

[CDS 2020 (1)]

- (a) 1 (b) 100
(c) 121 (d) 125

- 137.** If the points P and Q represent real numbers $0.\overline{73}$ and $0.5\overline{6}$ on the number line, then what is the distance between P and Q ?

[CDS 2020 (1)]

- (a) $\frac{1}{6}$ (b) $\frac{1}{5}$
(c) $\frac{16}{45}$ (d) $\frac{11}{90}$

- 138.** The number of three digit numbers (all digits are different) which are divisible by 7 and also divisible by 7 on reversing the order of the digits, is:

[CDS 2020 (1)]

- (a) Six (b) Five
(c) Four (d) Three

- 139.** Let XYZ be a 3-digit number. Let $S = XYZ + YZX + ZXY$. What of the following statements is/are correct?

1. S is always divisible by 3 and $(X+Y+Z)$
2. S is always divisible by 9
3. S is always divisible by 37

Select the correct answer using the code given below:

[CDS 2020 (1)]

- (a) 1 only (b) 2 only
(c) 1 and 2 (d) 1 and 3

- 140.** $x^3 + x^2 + 16$ is exactly divisible by x , where x is a positive integer. The number of all such possible values of x is:

[CDS 2020 (2)]

- (a) 3 (b) 4
(c) 5 (d) 6

- 141.** The number of (a, b, c) where, a, b, c are positive integers such that $abc = 30$, is:

[CDS 2020 (2)]

- (a) 30 (b) 27
(c) 9 (d) 8

- 142.** If the number 23P62971335 is divisible by the smallest odd composite number, then what is the value of P ?

[CDS 2020 (2)]

- (a) 4 (b) 5
(c) 6 (d) 7

- 143.** What is the remainder when the sum $1^5 + 2^5 + 3^5 + 4^5 + 5^6$ is divided by 4?

[CDS 2020 (2)]

- (a) 0 (b) 1
(c) 2 (d) 3

- 144.** What is the digit in the unit place of 3^{99} ?

[CDS 2020 (2)]

- (a) 1 (b) 3
(c) 7 (d) 9

- 145.** If 17^{2020} is divided by 18, then what is the remainder?

[CDS 2020 (2)]

- (a) 1 (b) 2
(c) 16 (d) 17

- 146.** The sum of all possible products taken two at a time out of the numbers $\pm 1, \pm 2, \pm 3, \pm 4, \pm 5$ is:

[CDS 2020 (2)]

- (a) 0 (b) -30
(c) -55 (d) 55

- 147.** Let $d(n)$ denote the number of positive divisors of a positive integer n . Which of the following are correct?

1. $d(5) = d(11)$
2. $d(5) \cdot d(11) = d(55)$
3. $d(5) + d(11) = d(16)$

Select the correct answer using the code given below:

[CDS 2020 (2)]

- (a) 1 and 3 only (b) 1 and 2 only
(c) 2 and 3 only (d) 1, 2 and 3

- 148.** If $A_n = P_n + 1$, where P_n is the product of the first n prime numbers, then consider the following statements:

1. A_n is always a composite number
2. $A_n + 2$ is always an odd number
3. $A_n + 1$ is always an even number

Which of the above statements is/are correct?

[CDS 2020 (2)]

(a) 1 only
(c) 3 only

(b) 2 only
(d) 2 and 3 only

- 149.** The sum of the digits of a two digit number is 13 and the difference between the number and that formed by reversing the digits is 27. What is the product of the digits of the number?

[CDS 2020 (2)]

(a) 35
(c) 45

(b) 40
(d) 54

- 150.** If radius of a sphere is rational, then which of the following is/are correct?

1.Its surface area is rational
2.Its volume is rational

Select the correct answer using the code given below:

[CDS 2020 (2)]

(a) 1 only
(c) Both 1 and 2

(b) 2 only
(d) Neither 1 nor 2

- 151.** If the number 413283P759387 is divisible by 13, then what is the value of P?

[CDS 2021 (1)]

(a) 3
(c) 7

(b) 6
(d) 8

- 152.** What is the remainder when $2^{1000000}$ is divided by 7?

[CDS 2021 (1)]

(a) 1
(c) 4

(b) 2
(d) 6

- 153.** Consider the pairs of prime numbers (m,n) between 50 and 100 such that $m-n=6$. How many such pairs are there?

[CDS 2021 (1)]

(a) 2
(c) 4

(b) 3
(d) 5

- 154.** What is the unit digit in the expansion of 67^{32} ?

[CDS 2021 (1)]

(a) 1
(c) 7

(b) 3
(d) 9

- 155.** What are distinct prime factors of the number 26381?

[CDS 2021 (2)]

(a) 29, 17, 37
(c) 19, 37, 13

(b) 31, 17, 47
(d) 23, 31, 37

- 156.** What is the remainder after dividing the number 37^{1000} by 9?

[CDS 2021 (2)]

(a) 1
(c) 7

(b) 3
(d) 9

- 157.** If x is negative real number, then which of the following are not correct?

1. There is some natural number k such that $kx > 0$
2. $x^2 + x > 0$ always
3. $2x < x < -x$
4. x^2 is always a rational number

Select the correct answer using the code given below:

[CDS 2021 (2)]

(a) 1, 2 and 3
(c) 1, 3 and 4

(b) 1, 2 and 4
(d) 2, 3 and 4

- 158.** How many zeros are there in the product $150 \times 249 \times 348 \times \dots \times 501$?

[CDS 2021 (2)]

(a) 262
(c) 246

(b) 261
(d) 235

- 159.** A two digit number is such that the product of the digits is 8. If 63 is added to this number, the digits interchanged their places. What is the sum of the digit in the number?

[CDS 2021 (2)]

(a) 6
(c) 8

(b) 7
(d) 9

- 160.** Consider the following statements:

1. The sum of the cubes of three consecutive natural numbers is divisible by 9.
2. Every even power of every odd number (>1) when divided by 8 gives 1 as remainder.

Which of the above statements is/are correct?

[CDS 2022 (1)]

(a) 1 only
(c) Both 1 and 2

(b) 2 only
(d) Neither 1 nor 2

- 161.** What is the number of divisors of 1000 (excluding 1 and 1000)?

[CDS 2022 (1)]

(a) 12
(c) 14

(b) 13
(d) 16

- 162.** $27^5 + 3^{13}$ is divisible by:

[CDS 2022 (1)]

(a) 8
(c) 12

(b) 10
(d) 21

- 163.** Let $p = 2^{2n+2} + m$ and $q = 2^{4n} - m$ (where n is even natural number). What should be

the least value of m such that p as well as q is divisible by 5?

[CDS 2022 (1)]

- (a) -1 (b) 1
(c) 4 (d) 6

164. Let a, b, c, d be positive integers. If

$$\frac{1}{a + \frac{1}{b + \frac{1}{c + \frac{1}{d}}}} = \frac{17}{60}$$

then what is the product of a, b, c, d ?

[CDS 2022 (1)]

- (a) 24 (b) 51
(c) 68 (d) 102

165. What is the value of least value of n if $194480 + n = m^4$, where m and n are natural numbers?

[CDS 2022 (1)]

- (a) 1 (b) 2
(c) 3 (d) 4

166. $41^{43} + 43^{43}$ is divisible by

[CDS 2022 (1)]

- (a) 80 (b) 84
(c) 86 (d) 88

167. $46^1 + 46^2 + 46^3 + 46^4$ is divisible by:

[CDS 2022 (1)]

- (a) 7 (b) 9
(c) 11 (d) 17

168. There are two natural numbers x and y , where $x > y$. when x is divided by 6, it leaves the remainder 2 and; when y is divided by 6, it leaves the remainder 3. What is the remainder when $(x-y)$ is divided by 6?

[CDS 2022 (II)]

- (a) 1
(b) 3
(c) 5
(d) Remainder cannot be determined

169. If three times the greater of two numbers is divided by the smaller number, the quotient will be 6 and the remainder will be 6. If five times the smaller number is divided by the greater number, the quotient will be 2 and the remainder will be 3. What is the difference between the numbers?

[CDS 2022 (II)]

- (a) 8 (b) 9

(c) 10

(d) 12

170. Consider the number $N = 12^6 \times 3^8 \times 5^3$. Which of the following statements is/are correct?

1. the number of odd factors of N is 60
 2. the number of even factors of N is 720
- Select the correct answer using the code given below:

[CDS 2022 (II)]

- (a) only 1 (b) only 2
(c) both 1 and 2 (d) neither 1 nor 2

171. Consider the question and two statements given below:

A 2-digit number is added to the number formed by reversing the digits of the 2-digit number.

Question: What is the 2-digit number?

Statement-1: The sum is divisible by 9

Statement-2: The sum is divisible by 2.

Which one of the following is correct in respect of the question and the statements?

[CDS 2022 (II)]

- (a) Statement-1 alone is sufficient to answer the question
(b) Statement-2 alone is sufficient to answer the question
(c) Both statement-1 and statement-2 are sufficient to answer the question
(d) Both statement-1 and statement-2 are not sufficient to answer the question

172. Let a, b, c and d be four positive integers such that $a + b + c + d = 200$. If $S = (-1)^a + (-1)^b + (-1)^c + (-1)^d$, then what is the number of possible values of S ?

[CDS - 2023 (1)]

- (a) one (b) two
(c) three (d) four

173. The number $97^{30} - 14^{30}$ is divisible by:

[CDS - 2023 (1)]

- (a) 37 but not 83 (b) 83 but not 37
(c) both 37 and 83 (d) neither 37 nor 83

174. Consider the following statements:

1. $\log_{10} 50$ is a rational number

2. $\log_{100} 10$ is an irrational number

Which of the statements given above is/are correct?

[CDS - 2023 (1)]

- (a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2

175. Consider the following statements:

1. $n^3 - n$ is divisible by 6
2. $n^5 - n$ is divisible by 5
3. $n^5 - 5n^3 + 4n$ is divisible by 120.

Which of the statements given above are correct?

[CDS - 2023 (1)]

- | | |
|------------------|------------------|
| (a) 1 and 2 only | (b) 2 and 3 only |
| (c) 1 and 3 only | (d) 1, 2 and 3 |

176. What is the last digit of the sum $S = 9^{27} + 27^9$?

[CDS - 2023 (1)]

- | | |
|-------|-------|
| (a) 3 | (b) 6 |
| (c) 7 | (d) 9 |

177. Consider the following statements in respect of all factors of 360?

1. The number of factors is 24
2. The sum of all factors is 1170

Which of the above statements is/are correct?

[CDS - 2023 (1)]

- | | |
|------------------|---------------------|
| (a) 1 only | (b) 2 only |
| (c) both 1 and 2 | (d) neither 1 nor 2 |

178. Consider a 6-digit number of the form XYXYXY. The number is divisible by:

[CDS - 2023 (1)]

- | | |
|-----------------------|---------------------|
| (a) 3 and 7 only | (b) 7 and 13 only |
| (c) 3, 13 and 37 only | (d) 3, 7, 13 and 37 |

179. Let m and n be natural numbers. What is the minimum value of $(m + n)$ such that $33m + 22n$ is divisible by 121?

[CDS - 2023 (1)]

- | | |
|-------|--------|
| (a) 3 | (b) 4 |
| (c) 5 | (d) 10 |

180. If a , b and c are the sides of a triangle ABC, then $\sqrt{a} + \sqrt{b} - \sqrt{c}$ is always:

[CDS - 2023 (1)]

- | | |
|------------------|------------------|
| (a) Negative | (b) Positive |
| (c) non-negative | (d) non-positive |

181. When every even power of every odd integer (greater than 1) is divided by 8, what is the remainder?

[CDS - 2023 (1)]

- | |
|--|
| (a) 3 |
| (b) 2 |
| (c) 1 |
| (d) the remainder is not necessarily 1 |

182. Consider the following statements:

1. If n is a natural number, then the number $\frac{n(n^2 + 2)}{3}$ is also a natural number

2. If m is an odd integer, then the number $\frac{m^4 + 4m^2 + 11}{16}$ is an integer.

Which of the statements given above is/are correct?

[CDS - 2023 (1)]

- | | |
|------------------|---------------------|
| (a) 1 only | (b) 2 only |
| (c) both 1 and 2 | (d) neither 1 nor 2 |

183. It is given that 5 does not divide $n-1$, n and $n+1$, where n is a positive integer. Which one of the following is correct?

[CDS - 2023 (1)]

- | | |
|---------------------------|---------------------------|
| (a) 5 divide $(n^2 + 1)$ | (b) 5 divides $(n^2 - 1)$ |
| (c) 5 divides $(n^2 + n)$ | (d) 5 divides $(n^2 - n)$ |

184. What is the largest power of 10 that divides the product $29 \times 28 \times 27 \times \dots \times 2 \times 1$?

[CDS - 2023 (1)]

- | | |
|-------|-------|
| (a) 4 | (b) 5 |
| (c) 6 | (d) 7 |

185. What is the remainder when 65^{99} is divided by 11?

[CDS - 2023 (1)]

- | | |
|-------|--------|
| (a) 0 | (b) 5 |
| (c) 9 | (d) 10 |

186. Consider the following numbers:

1.437
2.797
3.1073

How many of the above numbers are prime?

[CDS - 2023 (1)]

- | | |
|---------------|--------------|
| (a) Only one | (b) Only two |
| (c) All three | (d) None |

For the next ten (4) items that follows:

Each item contains a question followed by two Statement. Answer each item using the following instructions:

[CDS - 2023 (1)]

(a) choose this option if the question can be answered by one of the statements alone but not by the other.

(b) choose this option if the question can be answered by either statement alone.

(c) Choose this option if the question can be answered by using both the statements

together, but cannot be answered by using either statement alone
(d) Choose this option if the question cannot be answered even by using both statements together.

- 187.** A number 277XY5 (where X, Y are digits) is divisible by 25.

Question: what is the value of X?

Statement I: The given number is divisible by 9

Statement II: $X > 5$.

Each item contains a question followed by two Statement. Answer each item using the following instructions:

- 188.** Question: Is $m > n$, where m, n are non-zero numbers?

Statement I: $\frac{m}{n} > 1$

Statement II: $m > 2n$

- 189.** x and y are consecutive odd integers.

Question: Can the value of $(x+y)$ be determined uniquely?

Statement I: $(x+y)^4 = 256$

Statement II: $(x+y)^3 < 16$

- 190.** Question: Is $p^2 + q^2 + q$ odd, where p, q are positive integers?

Statement I: $2p + q$ is odd

Statement II: $q - 2p$ is odd

Consider the following for the next three items that follow:

Mark option (a) if the question can be answered by using one of the statements alone, but cannot be answered using the other statement alone.

Mark option (b) if the question can be answered by using either statement alone.

Mark option (c) if the question can be answered by using both the statements together, but cannot be answered using either statement alone.

Mark option (d) if the question cannot be answered even by using both the statements together.

[CDS-2023 (2)]

- 191.** Question: Is $m > n$ if m, n are real numbers?

Statement I:

$$M = (1-p)(p^2 + p + 1) \text{ and}$$

$$n = (p + 1)(p^2 - p + 1)$$

Statement II:

$$m = pn$$

- 192.** Question: What is the 3 digit number which is divisible by 10?

Statement I:

If the digits in hundred's place and ten's place of the number are interchanged, the resulting number is diminished by 180.

Statement II:

If the digit in hundred's place is halved and digit in ten's place and unit place of the number are interchanged, the resulting number is diminished by 336.

- 193.** Question: Is $(a-b+c) > (a+b-c)$, where a, b and c are real numbers?

Statement I:

b is negative

Statement II:

c is negative

- 194.** A 2-digit number is such that the sum of the number and the number obtained by reversing the order of the digits of the number is 55. Further, the difference of the given number and the number obtained by reversing the order of the digits of the number is 45. What is the product of the digits?

[CDS-2023 (2)]

(a) 5

(b) 2

(c) 1

(d) 0

Consider the following for the next four items that follow:

Each item contains a Question followed by two Statements. Answer each item using the following instructions:

Choose option

(a) If the Question can be answered by one of the Statements alone, but not by the other.

(b) If the Question can be answered by either Statement alone

(c) If the Question can be answered by using both the Statements together, but cannot be answered by using either Statement alone.

(d) If the Question can be answered even by using both Statements together

- 195.** Let a, b, c and d be positive integers.

Question: Which one of a, b, c, d is closest to the product abcd?

Statement-I: $a > b > c$

Statement-II: c is not the smallest

[CDS-2024 (1)]

196. Let $mn = k$, where m and n are prime numbers and k is an even number.

Question: What is the value of $mn - n + 1$?

Statement-I: $m > n$

Statement-II: One of the numbers is 2

[CDS-2024 (1)]

197. Question: If p is a positive integer, then what is the remainder when p^n is divided by $p + 1$?

Statement-I: n is even

Statement-II: p is even

[CDS-2024 (1)]

198. Question: Is xy positive?

Statement-I: $x = \sqrt[3]{-0.19683}$

Statement-II: $y = \sqrt[3]{x}$

[CDS-2024 (1)]

199. What is the smallest natural number n such that $(n + 1) \times n \times (n - 1) \times (n - 2) \times \dots \times 3 \times 2 \times 1$ is divisible by 910?

[CDS-2024 (1)]

- | | |
|--------|--------|
| (a) 91 | (b) 90 |
| (c) 13 | (d) 12 |

200. The expression $555^{777} + 777^{555}$ is divisible by which of the following?

1. 2
2. 3
3. 37

Select the correct answer using the code given below:

[CDS-2024 (1)]

- | | |
|------------------|------------------|
| (a) 1 and 2 only | (b) 2 and 3 only |
| (c) 1 and 3 only | (d) 1, 2 and 3 |

201. Consider the following statements

1. If $(3m^3 + 2m^2 + 5m + n)/m$ is not an integer, where m and n are integers, then n is not divisible by m

2. $5(8^m) + 2^{3m}$ is divisible by 48 for all whole numbers m

Which of the statements given above is/are correct?

[CDS-2024 (1)]

- | | |
|------------------|---------------------|
| (a) 1 only | (b) 2 only |
| (c) Both 1 and 2 | (d) Neither 1 nor 2 |

202. What is the digit at the 100th place of number $(225)^{40}$?

[CDS-2024 (1)]

- | | |
|-------|-------|
| (a) 6 | (b) 5 |
| (c) 4 | (d) 2 |

203. If a, b, c, d are natural numbers, then how many possible remainders are there when $1^a + 2^b + 3^c + 4^d$ is divided by 10?

[CDS-2024 (1)]

- | | |
|-------|-------|
| (a) 3 | (b) 4 |
| (c) 5 | (d) 6 |

204. If n is a natural number, then what is the sum of all distinct remainders of $4^n + 6^n + 9^n + 11^n$ when divided by 10 for various values of n?

[CDS-2024 (1)]

- | | |
|-------|-------|
| (a) 3 | (b) 4 |
| (c) 6 | (d) 7 |

205. When the number (12345678910111213 ... 99100) is divided by 16, what will be the remainder?

[CDS-2024 (1)]

- | | |
|--------|---------|
| (a) 15 | (b) 12. |
| (c) 4 | (d) 3 |

ANSWER KEY

1.	c	2.	d	3.	a	4.	b	5.	b	6.	d	7.	b	8.	c	9.	d	10.	c
11.	d	12.	b	13.	d	14.	c	15.	c	16.	c	17.	c	18.	a	19.	a	20.	c
21.	b	22.	c	23.	d	24.	a	25.	c	26.	d	27.	a	28.	d	29.	d	30.	a
31.	b	32.	c	33.	b	34.	b	35.	c	36.	b	37.	a	38.	a	39.	a	40.	c
41.	c	42.	b	43.	d	44.	b	45.	a	46.	c	47.	d	48.	c	49.	a	50.	a
51.	c	52.	c	53.	d	54.	c,d	55.	a	56.	c	57.	c	58.	c	59.	c	60.	c
61.	d	62.	c	63.	d	64.	B	65.	b	66.	b	67.	d	68.	b	69.	b	70.	c
71.	a	72.	b	73.	b	74.	d	75.	d	76.	c	77.	a	78.	c	79.	a	80.	d
81.	b	82.	d	83.	c	84.	b	85.	c	86.	a	87.	b	88.	c	89.	b	90.	d
91.	c	92.	d	93.	c	94.	a	95.	d	96.	d	97.	b	98.	c	99.	c	100.	c
101.	b	102.	c	103.	b	104.	b	105.	d	106.	a	107.	a	108.	d	109.	c	110.	d
111.	c	112.	c	113.	d	114.	a	115.	d	116.	b	117.	c	118.	c	119.	c	120.	a
121.	c	122.	c	123.	a	124.	b	125.	d	126.	a	127.	b	128.	b	129.	c	130.	d
131.	a	132.	b	133.	b	134.	b	135.	a	136.	b	137.	a	138.	c	139.	d	140.	c
141.	b	142.	a	143.	b	144.	c	145.	a	146.	a	147.	b	148.	d	149.	b	150.	c
151.	d	152.	b	153.	d	154.	a	155.	d	156.	a	157.	b	158.	a	159.	d	160.	b
161.	c	162.	b	163.	b	164.	a	165.	a	166.	b	167.	d	168.	c	169.	b	170.	c
171.	d	172.	c	173.	c	174.	d	175.	d	176.	b	177.	c	178.	d	179.	b	180.	b
181.	c	182.	c	183.	a	184.	c	185.	d	186.	a	187.	c	188.	d	189.	c	190.	d
191.	d	192.	c	193.	c	194.	d	195.	c	196.	d	197.	a	198.	a	199.	d	200.	d
201.	a	202.	a	203.	c	204.	b	205.	b										

CAPF PYQ

1. If the product of n positive numbers is unity, then their sum is:
[CAPF 2017]
(a) A positive integer (b) Divisible by n
(c) Equal to $n + 1/n$ (d) Never less than n
2. Consider the following number $n = [(6374)^{1793} \times (625)^{317} \times (313)^{49}]$. Which one of the following is the digit at unit place of n .
[CAPF 2017]
(a) 0 (b) 1
(c) 2 (d) 5
3. What is the largest value four n (natural number) such that 6^n divides the product of the first 100 natural numbers.
[CAPF 2018]
(a) 18 (b) 33
(c) 44 (d) 48
4. Suppose $72 = m \times n$, where m and n are positive integers such that $1 < m < n$. How many possible values of m are there?
[CAPF 2018]
(a) 5 (b) 6
(c) 10 (d) 12
5. Suppose xyz are three positive integers such that $x \leq y \leq z$ and $xyz = 72$. Which one of the following values of S yields more than one solution to the equation $x+y+z=S$?
[CAPF 2018]
(a) 13 (b) 14
(c) 15 (d) 16
6. If the number $2^2 \times 5^4 \times 4^6 \times 10^8 \times 6^{10} \times 15^{12} \times 8^{14} \times 20^{16} \times 10^{18} \times 25^{20}$ is divisible by 10^n . Then which one of the following is maximum value of n ?
[CAPF 2018]
(a) 78 (b) 85
(c) 89 (d) 98
7. Which one of the following is the remainder when 10^{20} is divided by 7?
[CAPF 2018]
(a) 1 (b) 2
(c) 4 (d) 6
8. On simplifications the product $(x_1 + y_1)(x_2 + y_2) \dots (x_{10} + y_{10})$, how many such terms are there which will have only single x and rest y 's?
[CAPF 2018]
(a) 2^{10} (b) 10
(c) 20 (d) 1
9. What is the natural number n for which $3^9 + 3^{12} + 3^{15} + 3^n$ is a perfect cube of an integer?
[CAPF 2019]
(a) 10 (b) 11
(c) 13 (d) 14
10. When the digits of two-digit numbers are reversed, the number increases by 27, the sum of such two-digit numbers is:
[CAPF 2019]
(a) 235 (b) 249
(c) 213 (d) 180
11. Which one of the following is the greatest number by which the product of three consecutive even numbers would be exactly divisible?
[CAPF 2020]
(a) 12 (b) 24
(c) 48 (d) 64
12. A is the smallest positive integer which when divided by 9 and 12 leaves remainder 8. B is the smallest positive integer which when divided by 9 and 12 leaves remainder 5. Which one of the following is the value of $A-B$?
[CAPF 2020]
(a) 3 (b) 2
(c) 1 (d) 0
13. Suppose the n th term of a series is $1 + \frac{n}{2} + \frac{n^2}{2}$. If there are 20 terms in the series, then the sum of the series is equal to:
[CAPF 2021]
(a) 1360 (b) 1450
(c) 1500 (d) 1560

ANSWER KEY

1.	d	2.	a	3.	d	4.	c	5.	b	6.	d	7.	b	8.	b	9.	d	10.	b
11.	b	12.	a	13.	d														

(c) 2

(d) 3

21. What number must be added to the expression $16a^2 - 12a$ to make it a perfect square?

(1) $9/4$

(2) $11/2$

(3) $13/2$

(4) 16

22. What value should come in place of x in

$$\frac{\frac{9}{x^4}}{\frac{1}{x^4}} = \frac{1}{9}$$

(a) 27

(b) 36^2

(c) 6

(d) 36

23. One-third of the square root of which number is 0.001?

(1) 0.0009

(2) 0.000001

(3) 0.00009

(4) None of the above

24. $\sqrt{7 + \sqrt{48}} =$

(a) $4 + \sqrt{2}$

(b) $2 + \sqrt{3}$

(c) $3 - \sqrt{2}$

(d) $4 - \sqrt{2}$

25. If the sum of the squares of three consecutive natural numbers is 110, then the smallest of these natural numbers is:

(1) 8

(2) 6

(3) 7

(4) 5

26. Which of the following is true?

(1) $\sqrt{5} + \sqrt{3} > \sqrt{6} + \sqrt{2}$

(2) $\sqrt{5} + \sqrt{3} < \sqrt{6} + \sqrt{2}$

(3) $\sqrt{5} + \sqrt{3} = \sqrt{6} + \sqrt{2}$

(4) $(\sqrt{5} + \sqrt{3})(\sqrt{6} + \sqrt{2}) = 1$

27. If $5416 * 6$ is a perfect square, then the digit at * is :

(1) 9

(2) 4

(3) 6

(4) 5

28. $4(10 + 15 \div 5 \times 4 - 2 \times 2) =$

(a) 56

(b) 24

(c) 48

(d) 72

29. The sum of squares of three positive integers is 323. If the sum of squares of two numbers is twice the third, their product is

(1) 255

(2) 260

(3) 265

(4) 270

30. The difference between two numbers is 9 and the difference between their squares is 207. The numbers are:

(1) 17 and 8

(2) 16 and 7

(3) 15 and 6

(4) 23 and 14

ANSWER KEY

1.	(b)	2.	(b)	3.	(c)	4.	(a)	5.	(d)	6.	(b)	7.	(c)	8.	(c)	9.	(a)	10.	(d)
11.	(c)	12.	(d)	13.	(b)	14.	(c)	15.	(c)	16.	(c)	17.	(a)	18.	(c)	19.	(d)	20.	(c)
21.	(a)	22.	(c)	23.	(d)	24.	(b)	25.	(d)	26.	(a)	27.	(a)	28.	(d)	29.	(a)	30.	(b)

PRACTICE SHEET 2

1. If $x = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{2}}}}$ then the value of $2x + \frac{7}{4}$ is

(1) 3

(2) 4

(3) 5

(4) 6

2. The simplification of $\frac{5}{3 + \frac{3}{1 - \frac{2}{3}}}$ gives

(a) 5

(b) $5/3$

(c) $5/12$

(d) $3/5$

3. If $2 = x + \frac{1}{1 + \frac{1}{3 + \frac{1}{4}}}$, then the value of x is

(a) $18/17$

(b) $21/17$

(c) $13/17$

(d) $12/17$

4. Simplify $1 + \frac{4}{2 + \frac{3}{5 - \frac{1}{2}}} - \frac{1}{2}(10 \div 2)$

- (a) 1
(c) - 15/2
- (b) 0
(d) - 1/2

5. The value of $1 - \frac{a}{1 - \frac{1}{1 + \frac{a}{1 - a}}}$ is

- (a) a
(c) 1
- (b) 1 - a
(d) 0

6. Simplify:

$$8\frac{1}{2} - \left[3\frac{1}{4} \div \left\{ 1\frac{1}{4} - \frac{1}{2} \left(1\frac{1}{2} - \frac{1}{3} - \frac{1}{6} \right) \right\} \right]$$

- (a) $4\frac{1}{2}$
(c) $9\frac{1}{2}$
- (b) $4\frac{1}{6}$
(d) $\frac{2}{9}$

7. The value of $0.008 \times 0.01 \times 0.072 \div (0.12 \times 0.0004)$ is:

- (1) 1.2
(3) 0.012
- (2) 0.12
(4) 1.02

8. The value of $\frac{2}{3} \times \frac{3}{\frac{5}{6} \div \frac{2}{3} \text{ of } 1\frac{1}{4}}$ is

- (a) 2
(c) 1/2
- (b) 1
(d) 2/3

9. $9 - 1\frac{2}{9} \text{ of } 3\frac{3}{11} \div 5\frac{1}{7} \text{ of } \frac{7}{9}$ is equal to

- (a) 8
(c) 7
- (b) 9
(d) $\frac{3}{4}$

10. The value of $1 \div [1 + 1 \div \{1 + 1 \div (1 + 1 \div 2)\}]$ is

- (1) 1
(3) 2
- (2) 5/8
(4) 1/2

11. $\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \frac{1}{30} + \frac{1}{42} =$

- (a) 5/7
(c) 2/7
- (b) 6/7
(d) 1

12. $\frac{1}{12} + \frac{1}{20} + \frac{1}{30} + \frac{1}{42} + \frac{1}{56} + \frac{1}{72} =$

- (a) 2/9
(c) 3/4
- (b) 4/9
(d) 5/9

13. $\frac{1}{1.3} + \frac{1}{3.5} + \frac{1}{5.7} + \frac{1}{7.9} + \frac{1}{9.11} =$

- (a) 5/11
(c) 6/11
- (b) 10/11
(d) 3/11

14. $\frac{1}{15} + \frac{1}{35} + \frac{1}{63} + \frac{1}{99} + \frac{1}{143} =$

- (a) 6/39
(c) 5/39
- (b) 10/39
(d) 4/39

15. $\frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \frac{1}{30} + \frac{1}{42} + \frac{1}{56} =$

- (a) 3/8
(c) 7/8
- (b) 5/8
(d) 1/8

16. $(16)^{0.16} \times (16)^{0.04} \times (2)^{0.2}$ is equal to :

- (1) 1
(3) 4
- (2) 2
(4) 16

17. The square root of $14 + 6\sqrt{5}$ is

- (1) $2 + \sqrt{5}$
(3) $5 + \sqrt{3}$
- (2) $3 + \sqrt{5}$
(4) $3 + 2\sqrt{5}$

18. $(\sqrt{2} + \sqrt{7 - 2\sqrt{10}})$ is equal to

- (a) $\sqrt{2}$
(c) $\sqrt{5}$
- (b) $\sqrt{7}$
(d) $2\sqrt{5}$

19. Which of the following is largest $\sqrt[3]{4}$, $\sqrt[4]{6}$, $\sqrt[6]{15}$, $\sqrt[12]{245}$

- (a) $\sqrt[3]{4}$
(c) $\sqrt[6]{15}$
- (b) $\sqrt[4]{6}$
(d) $\sqrt[12]{245}$

20. The value of $\frac{1}{\sqrt{2}+1} + \frac{1}{\sqrt{3}+\sqrt{2}} + \frac{1}{\sqrt{4}+\sqrt{3}}$

$$+ \frac{1}{\sqrt{5}+\sqrt{4}} + \frac{1}{\sqrt{6}+\sqrt{5}} + \frac{1}{\sqrt{7}+\sqrt{6}} + \frac{1}{\sqrt{8}+\sqrt{7}} + \frac{1}{\sqrt{9}+\sqrt{8}}$$

(a) 0
(c) 2

(b) 1
(d) 3

21. What is the square root of $9 + 2\sqrt{14}$?

- (1) $1 + 2\sqrt{2}$
(3) $\sqrt{2} + \sqrt{7}$
- (2) $\sqrt{3} + \sqrt{6}$
(4) $\sqrt{2} + \sqrt{5}$

22. The sum of the square of a number and the square of the reciprocal of the number is thrice the difference of the square of the number and the square of the reciprocal of the number. What is the number?

- (1) 1
(3) $(3)^{1/3}$
- (2) $(2)^{1/4}$
(4) $(4)^{1/4}$

23. What is/are the real value(s) of

$$(256)^{0.16} \times (16)^{0.18} ?$$

(1) Only - 4
(3) 4, - 4

(2) Only 4
(4) 2, - 2

24. Assertion (A): $\sqrt{\frac{5041}{6889}}$ is rational

Reason (R): The square root of a rational number is always rational.

- (1) A and R are correct and R is correct explanation of A
(2) A and R are correct but R is not correct explanation of A
(3) A is correct but R is wrong
(4) A is wrong but R is correct

25. What is the square of $(2 + \sqrt{2})$?

- (1) A rational number
(2) An irrational number
(3) A natural number
(4) A whole number

26. If $(ab^{-1})^{2x-1} = (ba^{-1})^{x-2}$

- (1) 1
(2) 2
(3) 3
(4) 4

27. What is the value of $\sqrt{7.84} + \sqrt{0.0784} + \sqrt{0.000784} + \sqrt{0.00000784}$?

- (1) 3.08
(2) 3.108
(3) 3.1008
(4) 3.1108

28. What is the value of

$$\frac{1}{1+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \dots + \frac{1}{\sqrt{15}+\sqrt{16}}$$

- (1) 0
(2) 1
(3) 2
(4) 3

29. If $27 \times (81)^{2n+3} - 3^m = 0$, then what is m equal to?

- (1) $2n + 5$
(2) $5n + 6$
(3) $8n + 3$
(4) $8n + 15$

30. The number $\sqrt{0.0001}$ is

- (a) a rational number less than 0.01
(b) a rational number
(c) an irrational number
(d) neither a rational number nor an irrational number

ANSWER KEY

1.	(3)	2.	(3)	3.	(2)	4.	(2)	5.	(4)	6.	(2)	7.	(2)	8.	(1)	9.	(1)	10.	(2)
11.	(2)	12.	(1)	13.	(1)	14.	(2)	15.	(1)	16.	(2)	17.	(2)	18.	(3)	19.	(1)	20.	(3)
21.	(3)	22.	(2)	23.	(2)	24.	(3)	25.	(2)	26.	(1)	27.	(4)	28.	(4)	29.	(4)	30.	(2)

CDS PYQ

1. What is the value of $\sqrt{6 + \sqrt{6 + \sqrt{6 + \dots}}}$

[CDS 2013(I)]

- (a) 2
(b) 3
(c) 3.5
(d) 4

2. If $16 \times 8^{n+2} = 2^m$, then m is equal to

[CDS 2013(I)]

- (a) $n + 8$
(b) $2n + 10$
(c) $3n + 2$
(d) $3n + 10$

3. The expression $\left[(\sqrt{2})^{\sqrt{2}} \right]^{\sqrt{2}}$ gives

[CDS 2013(I)]

- (a) a natural number
(b) an integer and not a natural number
(c) a rational number but not an integer
(d) a real number but not a rational number

4. If $x^2 = 6 + \sqrt{6 + \sqrt{6 + \sqrt{6 + \dots}}}$, then what is one of the values of x equal to

[CDS 2013(II)]

- (a) 6
(b) 5
(c) 4
(d) 3

5. Which is the smallest number among the following?

[CDS 2013(II)]

- (a) $\left[(5^{-2})^{-2} \right]^{-2}$
(b) $\left[(5^{-2})^2 \right]^2$
(c) $\left[(2^{-5})^{-2} \right]^{-2}$
(d) $\left[(2^{-5})^2 \right]^{-2}$

6. Consider the following in respect of the numbers $\sqrt{2}$, $\sqrt[3]{3}$, $\sqrt[6]{6}$

1. $\sqrt[6]{6}$ is the greatest number

[CDS 2018(I)]

- (a) $-2\sqrt{15}$ (b) $2\sqrt{15}$
(c) $\sqrt{15}$ (d) $-\sqrt{15}$

20. What is the value of $2 + \sqrt{2 + \sqrt{2 + \sqrt{2 + \dots}}}$

[CDS 2019(I)]

- (a) 1 (b) 2
(c) 3 (d) 4

21. What is the square root of $16 + 6\sqrt{7}$?

[CDS 2019(II)]

- (a) $4 + \sqrt{7}$ (b) $4 - \sqrt{7}$
(c) $3 + \sqrt{7}$ (d) $3 - \sqrt{7}$

22. If $a = \sqrt{7 + 4\sqrt{3}}$, then what is the value of $a + \frac{1}{a}$

[CDS 2019(II)]

- (a) 2 (b) 3
(c) 4 (d) 7

23. What is $\frac{1}{a^{m-n}-1} + \frac{1}{a^{n-m}-1}$?

[CDS 2020(I)]

- (a) 1 (b) -1
(c) 0 (d) $2a^{m-n}$

24. If $x = \sqrt{2}$, $y = \sqrt[3]{3}$ and $z = \sqrt[5]{6}$ then which one the following is correct ?

[CDS 2020(I)]

- (a) $y < x < z$ (b) $z < x < y$
(c) $z < y < x$ (d) $x < y < z$

25. What is the value of

$$\frac{1}{1+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \dots + \frac{1}{\sqrt{99}+\sqrt{100}}$$

[CDS 2020(II)]

- (1) 1 (2) 5
(3) 9 (4) 10

26. If $x^m = \sqrt[14]{x\sqrt{x}\sqrt[3]{x}}$, then what is the value of m?

[CDS 2020(II)]

- (a) $1/8$ (b) $1/4$
(c) $3/4$ (d) $7/4$

27. $\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{n(n+1)} = \frac{99}{100}$ then what is the value of n ?

[CDS 2021(I)]

- (a) 98 (b) 99
(c) 100 (d) 101

28. For what integral value of x is

$$\frac{12}{7 - \frac{6}{7 - \frac{3}{5-x}}} = x?$$

[CDS 2021(I)]

- (a) 4 (b) 3
(c) 2 (d) 1

29. What is the square root of $15 - 4\sqrt{14}$?

[CDS 2021(II)]

- (a) $2\sqrt{2} - \sqrt{7}$ (b) $3\sqrt{2} - 2\sqrt{17}$
(c) $\sqrt{15} - \sqrt{7}$ (d) $\sqrt{5} - \sqrt{3}$

30. What is the square root of $23 - 4\sqrt{15}$?

[CDS 2021(II)]

- (a) $\sqrt{6} - 3\sqrt{2}$ (b) $7 - 3\sqrt{5}$
(c) $\sqrt{3} - 2\sqrt{5}$ (d) $\sqrt{5} - 4\sqrt{3}$

31. What is $\frac{1}{1+\sqrt{2}} + \frac{1}{\sqrt{2}+\sqrt{3}} + \frac{1}{\sqrt{3}+\sqrt{4}} + \dots +$

$$\frac{1}{\sqrt{2020} + \sqrt{2021}}$$
 equal to?

[CDS 2022(I)]

- (a) $\sqrt{2020} + 1$ (b) $\sqrt{2021} + 1$
(c) $\sqrt{2020} + \sqrt{2021} - 1$ (d) $\sqrt{2021} - 1$

32. What is the smallest natural number from the following which must be subtracted from 9410 to make the remaining number a perfect square?

[CDS 2022(I)]

- (a) 4 (b) 3
(c) 2 (d) 1

33. What is the value of the following?

$$\frac{1}{5\sqrt{4} + 4\sqrt{5}} + \frac{1}{6\sqrt{5} + 5\sqrt{6}} + \frac{1}{7\sqrt{6} + 6\sqrt{7}} + \frac{1}{8\sqrt{7} + 7\sqrt{8}} + \frac{1}{9\sqrt{8} + 8\sqrt{9}}$$

[CDS 2022(I)]

- (a) $\frac{1}{\sqrt{6}}$ (b) $\frac{1}{2}$
(c) 1 (d) $\frac{1}{6}$

34. If $x^2 - 20$

$$= \sqrt{20 + \sqrt{20 + \sqrt{20 + \sqrt{20 + \dots \text{infinite terms}}}}} \text{ then what is x equal to?}$$

[CDS 2022 (II)]

- (a) 4 (b) 5

PRACTICE SHEET 1

- | | |
|---|---|
| <p>1. The LCM of two numbers is 30 and their HCF is 5. One of the numbers is 10. The other is
 (1) 20 (2) 25
 (3) 15 (4) 5</p> <p>2. The product of two numbers is 1280 and their H.C.F. is 8. The L.C.M. of the number will be:
 (1) 160 (2) 150
 (3) 120 (4) 140</p> <p>3. The H.C.F. and L.C.M. of two numbers are 8 and 48 respectively. If one of the numbers is 24, then the other number is
 (1) 48 (2) 36
 (3) 24 (4) 16</p> <p>4. The H.C.F and L.C.M of two numbers are 12 and 336 respectively. If one of the numbers is 84, the other is
 (1) 36 (2) 48
 (3) 72 (4) 96</p> <p>5. The product of two numbers is 216. If the HCF is 6, then their LCM is
 (1) 72 (2) 60
 (3) 48 (4) 36</p> <p>6. The HCF and LCM of two numbers are 18 and 378 respectively. If one of the numbers is 54, then the other number is
 (1) 126 (2) 144
 (3) 198 (4) 238</p> <p>7. The HCF and product of two numbers are 15 and 6300 respectively. The number of possible pairs of the numbers is
 (1) 4 (2) 3
 (3) 2 (4) 1</p> <p>8. The HCF of two numbers is 15 and their LCM is 225. If one of the number is 75, then the other number is:
 (1) 105 (2) 90
 (3) 60 (4) 45</p> <p>9. The LCM of two numbers is 520 and their HCF is 4. If one of the number is 52, then the other number is</p> | <p>(1) 40 (2) 42
 (3) 50 (4) 52</p> <p>10. The H.C.F. of two numbers is 96 and their L.C.M. is 1296. If one of the numbers is 864, the other is
 (1) 132 (2) 135
 (3) 140 (4) 144</p> <p>11. The LCM of two numbers is 4 times their HCF. The sum of LCM and HCF is 125. If one of the number is 100, then the other number is
 (1) 5 (2) 25
 (3) 100 (4) 125</p> <p>12. Product of two co-prime numbers is 117. Then their L.C.M. is
 (1) 117 (2) 9
 (3) 13 (4) 39</p> <p>13. The product of two numbers is 2160 and their HCF is 12. Number of such possible pairs is
 (1) 1 (2) 2
 (3) 3 (4) 4</p> <p>14. LCM of two numbers is 2079 and their HCF is 27. If one of the number is 189, the other number is
 (1) 297 (2) 584
 (3) 189 (4) 216</p> <p>15. The product of two numbers is 2028 and their HCF is 13. The number of such pairs is
 (1) 1 (2) 2
 (3) 3 (4) 4</p> <p>16. The LCM and the HCF of the numbers 28 and 42 are in the ratio:
 (1) 6 : 1 (2) 2 : 3
 (3) 3 : 2 (4) 7 : 2</p> <p>17. If the ratio of two numbers is 2 : 3 and their L.C.M. is 54, then the sum of the two numbers is
 (1) 5 (2) 15
 (3) 45 (4) 270</p> <p>18. The ratio of two numbers is 4 : 5 and their L.C.M. is 120. The numbers are</p> |
|---|---|

- (1) 30, 40 (2) 40, 32
(3) 24, 30 (4) 36, 20

19. Three numbers are in the ratio 2 : 3 : 4 and their H.C.F. is 12. The L.C.M. of the numbers is

- (1) 144 (2) 192
(3) 96 (4) 72

20. Two numbers are in the ratio 3 : 4. If their LCM is 240, the smaller of the two number is

- (1) 100 (2) 80
(3) 60 (4) 50

21. Two numbers are in the ratio 3 : 4. Their L.C.M. is 84. The greater number is

- (1) 21 (2) 24
(3) 28 (4) 84

22. Two numbers are in the ratio 3 : 4. If their HCF is 4, then their LCM is

- (1) 48 (2) 42
(3) 36 (4) 24

23. The ratio of the sum to the LCM of two natural numbers is 7 : 12. If their HCF is 4, then the smaller number is :

- (1) 20 (2) 16
(3) 12 (4) 8

24. Two numbers are in the ratio 3 : 4. The product of their H.C.F. and L.C.M. is 2028. The sum of the numbers is

- (1) 68 (2) 72
(3) 86 (4) 91

25. The LCM of two numbers is 48. The numbers are in the ratio 2 : 3. The sum of the numbers is

- (1) 28 (2) 32
(3) 40 (4) 64

26. The ratio of two numbers is 4 : 5 and their H.C.F. is 8. Then their L.C.M. is

- (1) 130 (2) 140
(3) 150 (4) 160

27. The ratio of two numbers is 3 : 4 and their HCF is 5. Their LCM is :

- (1) 10 (2) 60
(3) 15 (4) 12

28. Three numbers are in the ratio 1 : 2 : 3 and their HCF is 12. The numbers are

- (1) 12, 24, 36 (2) 5, 10, 15
(3) 4, 8, 12 (4) 10, 20, 30

29. If $x : y$ be the ratio of two whole numbers and z be their HCF, then the LCM of those two numbers is

- (1) yz (2) xz/y
(3) xy/z (4) xyz

30. The H.C.F. and L.C.M. of two numbers are 21 and 84 respectively. If the ratio the two numbers is 1 : 4, then the larger of the two numbers is

- (1) 12 (2) 108
(3) 48 (4) 84

ANSWER KEY

1.	(3)	2.	(1)	3.	(4)	4.	(2)	5.	(4)	6.	(1)	7.	(3)	8.	(4)	9.	(1)	10.	(4)
11.	(2)	12.	(1)	13.	(2)	14.	(1)	15.	(2)	16.	(1)	17.	(3)	18.	(3)	19.	(1)	20.	(3)
21.	(3)	22.	(1)	23.	(3)	24.	(4)	25.	(3)	26.	(4)	27.	(2)	28.	(1)	29.	(4)	30.	(4)

PRACTICE SHEET 2

1. The least number which when divided by 4, 6, 8, 12 and 16 leaves a remainder of 2 in each case is:
 (1) 46 (2) 48
 (3) 50 (4) 56
2. The least number, which when divided by 12, 15, 20 or 54 leaves a remainder of 4 in each case, is:
 (1) 450 (2) 454
 (3) 540 (4) 544
3. Find the greatest number of five digits which when divided by 3, 5, 8, 12 have 2 as remainder:
 (1) 99999 (2) 99958
 (3) 99960 (4) 99962
4. The least multiple of 13, which on dividing by 4, 5, 6, 7 and 8 leaves remainder 2 in each case is:
 (1) 2520 (2) 842
 (3) 2522 (4) 840
5. A, B, C start running at the same time and at the same point in the same direction in a circular stadium. A completes a round in 252 seconds, B in 308 seconds and C in 198 seconds. After what time will they meet again at the starting point ?
 (1) 26 minutes 18 seconds
 (2) 42 minutes 36 seconds
 (3) 45 minutes
 (4) 46 minutes 12 seconds
6. Find the largest number of four digits such that on dividing by 15, 18, 21 and 24 the remainders are 11, 14, 17 and 20 respectively.
 (1) 6557 (2) 7556
 (3) 5675 (4) 7664
7. The least perfect square, which is divisible by each of 21, 36 and 66 is
 (1) 214344 (2) 214434
 (3) 213444 (4) 231444
8. The least number, which when divided by 4, 5 and 6 leaves remainder 1, 2 and 3 respectively, is
 (1) 57 (2) 59
 (3) 61 (4) 63
9. Let the least number of six digits which when divided by 4, 6, 10, 15 leaves in each case same remainder 2 be N. The sum of digits in N is :
 (1) 3 (2) 5
 (3) 4 (4) 6
10. Which is the least number which when doubled will be exactly divisible by 12, 18, 21 and 30?
 (1) 2520 (2) 1260
 (3) 630 (4) 196
11. The smallest square number divisible by 10, 16 and 24 is
 (1) 900 (2) 1600
 (3) 2500 (4) 3600
12. If the students of a class can be grouped exactly into 6 or 8 or 10, then the minimum number of students in the class must be
 (1) 60 (2) 120
 (3) 180 (4) 240
13. The least number which when divided by 4, 6, 8 and 9 leave zero remainder in each case and when divided by 13 leaves a remainder of 7 is :
 (1) 144 (2) 72
 (3) 36 (4) 85
14. The smallest number, which when divided by 12 and 16 leaves remainder 5 and 9 respectively, is :
 (1) 55 (2) 41
 (3) 39 (4) 29
15. The greatest number of four digits which when divided by 12, 16 and 24 leave remainders 2, 6 and 14 respectively is
 (1) 9974 (2) 9970
 (3) 9807 (4) 9998
16. When a number is divided by 15, 20 or 35, each time the remainder is 8. Then the smallest number is
 (1) 428 (2) 427
 (3) 328 (4) 338
17. The smallest number, which, when divided by 12 or 10 or 8, leaves remainder 6 in each case, is

- (1) 246 (2) 186
(3) 126 (4) 66
- 18.** The traffic lights at three different road crossings change after 24 seconds, 36 seconds and 54 seconds respectively. If they all change simultaneously at 10 : 15 : 00 AM, then at what time will they again change simultaneously?
(1) 10 : 16 : 54 AM
(2) 10 : 18 : 36 AM
(3) 10 : 17 : 02 AM
(4) 10 : 22 : 12 AM
- 19.** From a point on a circular track 5 km long A, B and C started running in the same direction at the same time with speed of $2\frac{1}{2}$ km per hour, 3 km per hour and 2 km per hour respectively. Then on the starting point all three will meet again after
(1) 30 hours (2) 6 hours
(3) 10 hours (4) 15 hours
- 20.** Four runners started running simultaneously from a point on a circular track. They took 200 seconds, 300 seconds, 360 seconds and 450 seconds to complete one round. After how much time do they meet at the starting point for the first time ?
(1) 1800 seconds (2) 3600 seconds
(3) 2400 seconds (4) 4800 seconds
- 21.** The maximum number of students among whom 1001 pens and 910 pencils can be distributed in such a way that each student gets same number of pens and same number of pencils, is :
(1) 91 (2) 910
(3) 1001 (4) 1911
- 22.** The greatest number, which when divide 989 and 1327 leave remainders 5 and 7 respectively, is :
(1) 8 (2) 16
(3) 24 (4) 32
- 23.** H.C.F of $\frac{2}{3}$, $\frac{4}{5}$, $\frac{6}{7}$ is
(1) $\frac{48}{105}$ (2) $\frac{2}{105}$
(3) $\frac{1}{105}$ (4) $\frac{24}{105}$
- 24.** Let N be the greatest number that will divide 1305, 4665 and 6905 leaving the same remainder in each case. Then, sum of the digits in N is :
(1) 4 (2) 5
(3) 6 (4) 8
- 25.** What is the greatest number that will divide 307 and 330 leaving remainders 3 and 7 respectively?
(1) 19 (2) 16
(3) 17 (4) 23
- 26.** Which greatest number will divide 3026 and 5053 leaving remainders 11 and 13 respectively?
(1) 18 (2) 30
(3) 45 (4) 60
- 27.** A farmer has 945 cows and 2475 sheep. He farms them into flocks, keeping cows and sheep separate and having the same number of animals in each flock. If these flocks are as large as possible, then the maximum number of animals in each flock and total number of flocks required for the purpose are respectively
(1) 15 and 228 (2) 9 and 380
(3) 45 and 76 (4) 46 and 75
- 28.** A milk vendor has 21 litres of cow milk, 42 litres of toned milk and 63 litres of double toned milk. If he wants to pack them in cans so that each can contains same litres of milk and does not want to mix any two kinds of milk in a can, then the least number of cans required is
(1) 3 (2) 6
(3) 9 (4) 12
- 29.** The greatest number that divides 411, 684, 821 and leaves 3, 4 and 5 as remainders, respectively, is
(1) 254 (2) 146
(3) 136 (4) 204
- 30.** Find the greatest number which will exactly divide 200 and 320.
(1) 10 (2) 20
(3) 16 (4) 40

ANSWER KEY

1.	(3)	2.	(4)	3.	(4)	4.	(3)	5.	(4)	6.	(2)	7.	(3)	8.	(1)	9.	(2)	10.	(3)
11.	(4)	12.	(2)	13.	(2)	14.	(2)	15.	(1)	16.	(1)	17.	(3)	18.	(2)	19.	(3)	20.	(1)
21.	(1)	22.	(3)	23.	(2)	24.	(1)	25.	(1)	26.	(3)	27.	(3)	28.	(2)	29.	(3)	30.	(4)

CDS PYQ

- The LCM of two numbers is 2376 while their HCF is 33. If one of the numbers is 297, then the other number is
[CDS 2013(I)]
(a) 216 (b) 264
(c) 642 (d) 792
- The HCF of two numbers is 98 and their LCM is 2352. The sum of the numbers may be
[CDS 2013(II)]
(a) 1372 (b) 1398
(c) 1426 (d) 1484
- If for integers a, b and c, if $HCF(a, b) = 1$ and $HCF(a, c) = 1$, then which one of the following is correct?
[CDS 2013(II)]
(a) $HCF(a, bc) = 1$ (b) $HCF(a, bc) = a$
(c) $HCF(a, bc) = b$ (d) None of these
- In a fire range, 4 shooters are firing at their respective targets. The first, the second, the third and the fourth shooters hit the target once is every 5 s, 6 s, 7s and 8s, respectively. If all the them hit their target at 9:00 am, when will they hit their target together again?
[CDS 2014(I)]
(a) 9:04 am (b) 9:08 am
(c) 9:14 am (d) None of these
- For any integer n, what is $HCF(22n + 7, 33n + 10)$ equal to?
[CDS 2014(I)]
(a) n (b) 1
(c) 11 (d) None of these
- For any integers 'a' and 'b' with $HCF(a, b) = 1$, what is $HCF(a + b, a - b)$ equal to?
[CDS 2014(I)]
(a) It is always 1 (b) It is always 2
(c) Either 1 or 2 (d) None of these
- If a and b be positive integers, then $HCF\left(\frac{a}{HCF(a, b)}, \frac{b}{HCF(a, b)}\right)$ equal to ?
[CDS 2014(II)]
(a) a (b) b
(c) 1 (d) $\frac{a}{HCF(a, b)}$
- What is the number of integral solutions of the equations $HCF(a, b) = 5$ and $a + b = 65$?
[CDS 2014(II)]
(a) None (b) Infinitely many
(c) Less than 65 (d) Exactly one
- The LCM of two integers is 1237. What is their HCF ?
[CDS 2014(II)]
(a) 37 (b) 19
(c) 1 (d) Cannot be determined
- There are 48 cricket balls, 72 hockey balls and 84 tennis balls and they have to be arranged in several rows in such a way that every row contains the same number balls of balls of one type. What is the minimum number of rows required for this to happen?
[CDS 2014(II)]
(a) 12 (b) 16
(c) 17 (d) 19
- The HCF of two natural numbers m and n is 24 and their product is 552. How many sets of values of m and n are possible?
[CDS 2014(II)]
(a) 1
(b) 2

(c) 4

(d) No set of m and n is possible satisfying the given conditions

- 12.** The LCM of two numbers is 90 times their HCF. The sum of LCM and HCF is 1456. If one of the numbers is 160, then what is the other number?

[CDS 2014(II)]

- (a) 120 (b) 136
(c) 144 (d) 184

- 13.** What is the sum of digits of the least multiple of 13, which when divided by 6, 8 and 12 leaves 5, 7 and 11 respectively as the remainders ?

[CDS 2015(II)]

- (a) 5 (b) 6
(c) 7 (d) 8

- 14.** The LCM of two numbers is 12 times their HCF. The sum of HCF and LCM is 403. If one of the numbers is 93, then the other number is

[CDS 2015(II)]

- (a) 124 (b) 128
(c) 134 (d) 138

- 15.** Consider the following in respect of natural numbers a , b and c :

1. $\text{LCM}(ab, ac) = a \text{ LCM}(b, c)$
2. $\text{HCF}(ab, ac) = a \text{ HCF}(b, c)$
3. $\text{HCF}(a, b) < \text{LCM}(a, b)$
4. $\text{HCF}(a, b)$ divides $\text{LCM}(a, b)$

Which of the above are correct?

[CDS 2016(I)]

- (a) 1 only (b) 3 and 4 only
(c) 1, 2 and 4 only (d) 1, 2, 3 and 4

- 16.** There are two numbers p and q such that their HCF is 1. Which of the following statements are correct?

1. Both p and q may be prime.
 2. One number may be prime and the other composite,
 3. Both the numbers may be composite.
- Select the correct answer using the code given below;

[CDS 2016(II)]

- (a) 1 and 2 only (b) 6) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

- 17.** Consider the following statements:

I. If $a = bc$ with $\text{HCF}(b, c) = 1$, then $\text{HCF}(c, bd) = \text{HCF}(c, d)$.

II. $a = bc$ with $\text{HCF}(b, c) = 1$, then $\text{LCM}(a, d) = \text{LCM}(c, bd)$.

Which of the above statements is/are correct?

[CDS 2017(I)]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

- 18.** The HCF of two expressions p and q is 1. What is the reciprocal of their LCM?

[CDS 2017(II)]

- (a) $p + q$ (b) $p - q$
(c) pq (d) $(pq)^{-1}$

- 19.** How many numbers from 1 to 1000 are divisible by 2, 3, 4 and 5?

[CDS 2017(II)]

- (a) 16 (b) 17
(c) 32 (d) None of these

- 20.** There are two numbers which are greater than 21 and their LCM and HCF are 3003 and 21 respectively. What is the sum of these numbers?

[CDS 2018(I)]

- (a) 504 (b) 508
(c) 514 (d) 528

- 21.** Walls (excluding their roofs and floors) of 5 identical rooms having length, breadth and height 6 m, 4 m and 2.5 respectively are to be painted. Paints are available only in cans of 1 L and one litre of paint can be used for painting 20 square metres. What is the number of cans required painting?

[CDS 2018(II)]

- (a) 10 (b) 12
(c) 13 (d) 14

- 22.** The highest four digit number which is divisible by each of the numbers 16, 36, 45, 48 is

[CDS 2018(II)]

- (a) 9180 (b) 9360
(c) 9630 (d) 9840

- 23.** The product of two integers p and q , where $p > 60$ and $q > 60$, is 7168 and their HCF is 16. Then the sum of two integers is

[CDS 2018(II)]

- (a) 256 (b) 184
(c) 176 (d) 164

24. HCF of two numbers is 12. Which one of the following can never be their LCM

[CDS 2019(II)]

- (a) 80 (b) 60
(c) 36 (d) 24

25. What is the LCM of $\frac{1}{3}, \frac{5}{6}, \frac{2}{9}, \frac{4}{27}$?

[CDS 2019(II)]

- (a) 5/18 (b) 1/27
(c) 10/27 (d) 20/3

26. X, Y and Z start at same point and same time in the same direction to run around a circular stadium. X completes a round in 252 seconds, Y in 308 seconds and Z in 198 seconds. After what time will they meet again at the starting point?

[CDS 2019(II)]

- (a) 26 minutes 18 seconds
(b) 42 minutes 36 seconds
(c) 45 minutes
(d) 46 minutes 12 seconds

27. A number divides 12288, 28200 and 44333, so as to leave the same remainder in each case. What is that number?

[CDS 2020(I)]

- (a) 272 (b) 232
(c) 221 (d) 120

28. Two unequal pairs of numbers satisfy the following conditions

(i) The product of the two numbers in each pair is 2160

(ii) The HCF of the two numbers in each pair is 12.

If x is the mean of the numbers in the first pair and y is the mean of the numbers in the second pair, then what is the mean of x and y?

[CDS 2020(I)]

- (a) 60 (b) 72
(c) 75 (d) 78

29. What is the least perfect square which is divisible by 3, 4, 5, 6 and 7?

[CDS 2020(I)]

- (a) 1764 (b) 17640
(c) 44100 (d) 176400

30. The LCM of two numbers is 28 times their HCF. The sum of LCM and HCF is 1740. If one of the number is 240, then the other number is

[CDS 2020(II)]

- (a) 420 (b) 640
(c) 820 (d) 1040

31. The Euclidean algorithm is used to calculate the

[CDS 2020(II)]

- (a) Square root of an integer
(b) Cube root of an integer
(c) Square of an integer
(d) HCF of two integers

32. The sum of LCM and HCF of two numbers is 536 and the difference between LCM and HCF is 296. If one of the numbers is 104, then what is the other number?

[CDS 2021(II)]

- (a) 420 (b) 480
(c) 484 (d) 506

33. The LCM of two prime numbers p and q is 2231, where $p > q$. What is the value of $p - q$?

[CDS 2021(II)]

- (a) 67 (b) 70
(c) 74 (d) 81

34. Three runners are running in a circular track, and they complete one round in 20, 30 and 35 minutes respectively. When will they next meet at the starting point?

[CDS 2021(II)]

- (a) After 3 hour 30 minutes
(b) After 4 hour 30 minutes
(c) After 3 hours
(d) After 7 hours

35. Suppose p and q are LCM and HCF respectively of two positive numbers. If $p : q = 14 : 1$ and $pq = 1134$, then what is the difference between the two numbers?

[CDS 2022(I)]

- (a) 27
(b) 35
(c) 45
(d) cannot be determined due to insufficient data

36. If x is the HCF and y is the LCM of $\frac{3}{5}, \frac{6}{25}, \frac{9}{20}, \frac{27}{50}$, then which one of the following is correct?

[CDS 2022 (II)]

- (a) $y=90x$ (b) $y=180x$
(c) $y=270x$ (d) $y=360x$

37. Consider the question and two statements given below:

LCM of two numbers x and y is 481 where $x - y$.

Question: What is the value of $(3x - 2y)$?

Statement-1: $y > 1$

Statement-2: HCF of x and y is 1.

Which one of the following is correct in respect of the question and the statements?

[CDS 2022 (II)]

- (a) Statement-1 alone is sufficient to answer the question
(b) Statement-2 alone is sufficient to answer the question
(c) Both statement-1 and statement-2 are sufficient to answer the question
(d) Both statement-1 and statement-2 are not sufficient to answer the question

38. What is the HCF of $3^{29} - 9$ and $3^{38} - 9$?

[CDS - 2023 (1)]

- (a) $3^9 - 1$ (b) $3^{11} - 1$
(c) $3^{11} - 3$ (d) $3^{11} - 9$

39. The product of two numbers is 2160 and their HCF is 12. If the sum of the square of the two numbers is 4896, then what is the mean of the two numbers?

[CDS - 2023 (1)]

- (a) 24 (b) 36
(c) 48 (d) 96

40. There are four bells which ring at an interval of 15 minutes, 35 minutes and 45 minutes respectively. If all of them ring at 9 a.m., how many more times will they ring together in the next 72 hours.

[CDS - 2023 (1)]

- (a) 0 (b) 1
(c) 2 (d) 3

41. What is the largest number which divides both $2^{35}-1$ and $2^{91}-1$?

[CDS - 2023 (1)]

- (a) 34 (b) 90
(c) 127 (d) 129

42. What is the largest 5-digit number, which leaves remainder 7, when divided by 18 as well as by 11?

[CDS - 2023 (1)]

- (a) 99981 (b) 99988
(c) 99997 (d) 99999

43. Consider the following for the next one items that follow:

Mark option (a) if the question can be answered by using one of the statements alone, but cannot be answered using the other statement alone

Mark option (b) if the question can be answered by using either statement alone.

Mark option (c) if the question can be answered by using both the statements together, but cannot be answered using either statement alone.

Mark option (d) if the question cannot be answered even by using both the statements together.

[CDS-2023 (2)]

Question: What is the sum of two natural numbers?

Statement I:

LCM of the two numbers is 144.

Statement II:

One of the numbers is 72

44. Let L be the LCM and H be the HCF of two given numbers. L and H are in the ratio 3:2. If the sum of the two numbers is 45, then what is the product of the numbers?

[CDS-2023 (2)]

- (a) 243
(b) 486
(c) 504
(d) cannot be determined due to insufficient data

45. A plank of wood 4.25 m long and 3.4 m wide is to be cut into square pieces of equal size. How many square pieces of largest size can be cut from the plank, if no wastage is allowed?

[CDS-2024 (1)]

- (a) 45 (b) 90
(c) 400 (d) 500

46. If HCF of 768 and x^6y^2 is $32xy$ for natural numbers $x \geq 2$, $y \geq 2$, then what is the value of $(x + y)$?

[CDS-2024 (1)]

- (a) 5 (b) 7
(c) 9 (d) 11

ANSWER KEY

1.	(b)	2.	(a)	3.	(a)	4.	(c)	5.	(b)	6.	(c)	7.	(c)	8.	(c)	9.	(c)	10.	(c)
11.	(d)	12.	(c)	13.	(d)	14.	(a)	15.	(c)	16.	(d)	17.	(c)	18.	(d)	19.	(a)	20.	(a)
21.	(c)	22.	(b)	23.	(c)	24.	(a)	25.	(d)	26.	(d)	27.	(c)	28.	(b)	29.	(c)	30.	(a)
31.	(d)	32.	(b)	33.	(c)	34.	(d)	35.	(d)	36.	(d)	37.	(c)	38.	(d)	39.	(c)	40.	(c)
41.	(c)	42.	(c)	43.	d	44.	d	45.	(c)	46.	(d)								

CAPF PYQ

1. The smallest number which when added to 10000 becomes divisible by 20, 24 and 30 is:

[CAPF 2021]

- (a) 120 (b) 800
(c) 80 (d) 83

2. Which one of the following is the difference of the sum of cubes of first ten natural numbers and the sum of squares of first ten natural numbers?

[CAPF 2022]

- (a) 2400 (b) 2640
(c) 2880 (d) 2000

ANSWER KEY

1.	c	2.	b																
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PRACTICE SHEET-1

1. A train started from a station with a certain number of passengers. At the first halt, $\frac{1}{3}$ rd of its passengers got down and 120 passengers got in. At the second halt, half of the passengers got down and 100 persons got in. Then, the train left for its destination with 240 passengers. How many passengers were there in the train when it started?
(a) 540 (b) 480
(c) 360 (d) 240
2. A person bought a certain number of books for Rs 80. If he had bought 4 more books for the same sum, each book would have cost Rs 1 less. What is the price of each book?
(a) Rs 10 (b) Rs 8
(c) Rs 5 (d) Rs 4
3. A person bought 5 tickets from a station P to a station Q and 10 tickets from the station P to a station R . He paid Rs 350. If the sum of a ticket from P to Q and a ticket from P to R is Rs 42, then what is the fare from P to Q ?
(a) Rs 12 (b) Rs 14
(c) Rs 16 (d) Rs 18
4. Pooja started her job with certain monthly salary and gets a fixed increment every year. If her salary was Rs 4200 after 3 years and Rs 6800 after 8 years of service, then what are her initial salary and the annual increment, respectively?
(a) Rs 2640, Rs 320 (b) Rs 2460, Rs 320
(c) Rs 2460, Rs 520 (d) Rs 2640, Rs 520
5. A number consists of two digits, whose sum is 10. If 18 is subtracted from the number, digits of the number are reversed. What is the product?
(a) 15 (b) 18
(c) 24 (d) 32
6. A railway ticket for a child costs half the full fare but the reservation charge is the same on half tickets as much as on full ticket. One reserved first class ticket for a journey between two stations is Rs 362, one full and one half reserved first class tickets cost Rs 554. What is the reservation charge?
(a) Rs 18 (b) Rs 22
(c) Rs 38 (d) Rs 46
7. Let there be three simultaneous linear equations in two unknowns, which are non-parallel and non-collinear them. What can be the number of solutions (if they do exist)?
(a) One or infinite (b) Only one
(c) Exactly two (d) exactly three
8. If 1 is added to the denominator of a fraction, it becomes $\frac{1}{2}$ and if 1 is added to the numerator, the fraction becomes 1. What is the fraction?
(a) $\frac{5}{9}$ (b) $\frac{2}{3}$
(c) $\frac{4}{7}$ (d) $\frac{10}{11}$
9. If $\frac{2}{x} + \frac{3}{y} = \frac{9}{xy}$ and $\frac{4}{x} + \frac{9}{y} = \frac{21}{xy}$, where $x \neq 0$ and $y \neq 0$, then what is the value of $x + y$?
(a) 2 (b) 3
(c) 4 (d) 8
10. The sum of two numbers is 80. If the larger number exceeds four times the smaller by 5, what is the smaller number?
(a) 5 (b) 15
(c) 20 (d) 25
11. If $2^a + 3^b = 17$ and $2^{a+2} - 3^{b+1} = 5$, then,
(a) $a = 2, b = 3$ (b) $a = -2, b = 3$
(c) $a = 2, b = -3$ (d) $a = 3, b = 2$
12. If $\frac{2x-3y+1}{2} = \frac{x+4y+8}{3} = \frac{4x-7y+2}{5}$ then what is $(x + y)$ equal to?
(a) 3 (b) 2
(c) 0 (d) -2
13. Sum of two numbers is 21 and their difference is 11, then the greatest number is
(a) 5 (b) 16
(c) 9 (d) 10
14. The area of a rectangle remains the same if the length is increased by 7 m and the breadth is decreased by 3 m. The area remains unaffected if the length is decreased by 7 m and the breadth is increased by 5 m, then area of rectangle is
(a) 280 m^2 (b) 320 m^2
(c) 420 m^2 (d) 400 m^2

- 15.** A and B each have a certain number of mangoes. A says to B: "If you give 30 of your mangoes, I will have twice as many as left with you" B replies "If you give me 10, I will have thrice as many as left with you". How many mangoes did A has?
(a) 41 (b) 62
(c) 34 (d) 32
- 16.** There are two examination rooms A and B. If 10 candidates are sent from room A to room B the number of students in each room is the same. If 20 candidates are sent from B to A, the number of students in A is double the number in of students in B. Then, number of students room B is
(a) 40 (b) 100
(c) 80 (d) 60
- 17.** The numerator of a fraction is 2 less than the denominator. If one is added to its denominator, it becomes $\frac{1}{2}$, the fraction is
(a) $\frac{2}{3}$ (b) $\frac{3}{5}$
(c) $\frac{3}{4}$ (d) $\frac{5}{7}$
- 18.** The sum of digits of a two-digit number is 8 and the difference between the number and that formed by reversing the digits is 18. What is three difference between the digits of the number?
(a) 1 (b) 2
(c) 3 (d) 4
- 19.** A length of rectangle is 5 cm less than twice its width. if the length is decreased by 5 cm, and width is decreased by 2 cm. the perimeter of resulting rectangle will be 18 cm, Find the length of original rectangle ?
(a) 9 (b) 7
(c) 6 (d) 5
- 20.** The present of Sita's father is three times the present age of Sita. After six years sum of their ages will be 72 years. Find the present age of Sita ?
(a) 12 yrs (b) 15 yrs
(c) 16 yrs (d) 10 yrs
- 21.** The Community Relief fund receives a large donation of \$ 2800. The foundation agrees to spend the money on \$ 20 school bags, \$ 25 sweaters, \$ 5 b0oks. They want to buy 200 items and send them to schools in earthquake-hit areas. They must order as many books as school bages and sweaters combined. How many of each item should they order?
(a) (40, 60, 100) (b) (20, 30, 80)
(c) (50, 100, 60) (d) (40, 80, 25)
- 22.** A student was asked to find $\frac{5}{16}$ of a number. By mistake he found $\frac{5}{6}$ of that number. His answer was 250 more than the correct answer. Find the given number.
(1) 300 (2) 480
(3) 450 (4) 500
- 23.** A tin of oil was $\frac{4}{5}$ full. When 6 bottles of oil was taken out and 4 bottles of oil was poured into it, it was $\frac{3}{4}$ full. How many bottles of oil can the tin contain?
(1) 10 (2) 20
(3) 30 (4) 40
- 24.** A candidate in an examination was asked to find $\frac{5}{14}$ of a certain number. By mistake he found $\frac{5}{4}$ of it. Thus, his answer was 25 more than the correct answer. The number was:
(1) 28 (2) 56
(3) 84 (4) 140
- 25.** A man read $\frac{2}{5}$ th of a book on the first day. He read $\frac{1}{3}$ rd more on second day than he read on the first day. 15 pages were left for the third day. The number of pages in the book is
(1) 100 (2) 105
(3) 225 (4) 250
- 26.** A girl was asked to multiply a number by $\frac{7}{8}$, instead she divided the number by $\frac{7}{8}$ and got the result 15 more than the correct result. The sum of the digits of the number was:
(1) 4 (2) 8
(3) 6 (4) 11

- 27.** A, B, C and D purchase a gift worth Rs 60. A pays $\frac{1}{2}$ of what others are paying, B pays $\frac{1}{3}$ of what others are paying and C pays $\frac{1}{4}$ of what others are paying. What is the amount paid by D?
 (1) 16 (2) 13
 (3) 14 (4) 15
- 28.** A number of boys raised 400 for a famine relief fund, each boy giving as many 25 paise coins as there were boys. The number of boys was:
 (1) 40 (2) 16
 (3) 20 (4) 100
- 29.** Thrice the square of a natural number decreased by four times the number is equal to 50 more than the number. The number is:
 (1) 4 (2) 5
 (3) 10 (4) 6
- 30.** A farmer divides his herd of n cows among his four sons so that the first son gets one – half the herd, the second son gets one – fourth, the third son gets one – fifth and the fourth son gets 7 cows. The value of n is
 (1) 80 (2) 100
 (3) 140 (4) 180

ANSWER KEY

1.	(d)	2.	(c)	3.	(b)	4.	(d)	5.	(c)	6.	(b)	7.	(b)	8.	(b)	9.	(c)	10.	(b)
11.	(d)	12.	(d)	13.	(b)	14.	(c)	15.	(c)	16.	(c)	17.	(b)	18.	(b)	19.	(a)	20.	(b)
21.	(a)	22.	(b)	23.	(d)	24.	(a)	25.	(c)	26.	(d)	27.	(b)	28.	(a)	29.	(b)	30.	(c)

PRACTICE SHEET-2

1. A man has some hens and cows. If the number of heads be 48 and the number of feet equals 140, then the number of hens will be
 (1) 23 (2) 22
 (3) 24 (4) 26
2. Slope of the line $2x - y + 4 = 0$ is
 (1) 2 (2) 1
 (3) 0 (4) $1/2$
3. A number consists of two digits. If the number formed by interchanging the digits is added to the original number, the resulting number (i.e. the sum) must be divisible by
 (1) 11 (2) 9
 (3) 5 (4) 3
4. If 1 is added to the denominator of a fraction it becomes $\frac{1}{2}$. If 1 is added to the numerator it becomes 1. The product of numerator and denominator of the fraction is
 (1) 6 (2) 10
 (3) 12 (4) 14
5. A fraction becomes $\frac{1}{3}$ when 1 is subtracted from both the numerator and the denominator. The same fraction becomes $\frac{1}{2}$ when 1 is added to both the numerator and the denominator. The sum of numerator and denominator of the fraction is
 (1) 10 (2) 18
 (3) 7 (4) 16
6. If 1 is added to both the numerator and the denominator of a fraction, it becomes $\frac{1}{4}$. If 2 is added to both the numerator and the denominator of that fraction, it becomes $\frac{1}{3}$. The sum of numerator and denominator of the fraction is:
 (1) 8 (2) 13
 (3) 22 (4) 27
7. In a two digit number if it is known that its units digit exceeds its tens digit by 2 and that the product of the given number and the sum of its digits is equal to 144, then the number is
 (1) 46 (2) 42
 (3) 26 (4) 24
8. In a test, 1 mark is awarded for each correct answer and one mark is deducted for each wrong answer. If a boy answers all 20 items of the test and gets 8 marks, the number of questions answered correct by him was
 (1) 16 (2) 14
 (3) 12 (4) 8
9. The difference between two positive numbers is 3. If the sum of their squares is 369, then the sum of the numbers is:
 (1) 81 (2) 33
 (3) 27 (4) 25
10. A number consists of two digits such that the digit in the ten's place is less by 2 than the digit in the unit's place. Three times the number added to $\frac{6}{7}$ times the number obtained by reversing the digits equals 108. The sum of digits in the number is:
 (1) 8 (2) 9
 (3) 6 (4) 7
11. Of the three numbers, the second is twice the first and it is also thrice the third. If the average of three numbers is 44, the difference of the first number and the third number is:
 (1) 24 (2) 18
 (3) 12 (4) 6
12. A two digit number is five times the sum of its digits. If 9 is added to the number, the digits interchange their positions. The sum of digits of the number is:
 (1) 11 (2) 9
 (3) 7 (4) 6
13. Solution of equation $x + y = 5$ and $2x + y - 6 = 0$ is
 (1) $x = 1, y = 2$ (2) $x = 1, y = 4$
 (3) $x = 2, y = 3$ (4) $x = 4, y = 2$

- 14.** Straight line $2x - 3y - 6 = 0$ will cut the line $4x + ky = 15$ if
 (1) $k = 1$ (2) $k = 2$
 (3) $k = 3$ (4) All of above
- 15.** The product of two numbers is 24 times the difference of these two numbers. If the sum of these numbers is 14, the larger number is
 (1) 9 (2) 8
 (3) 7 (4) 10
- 16.** If a and b are negative real numbers and c is a positive real number, then which of the following is/are correct?
 1. $a - b < a - c$
 2. If $a < b$ then $\frac{a}{c} < \frac{b}{c}$
 3. $\frac{1}{b} < \frac{1}{c}$
 Select the correct answer using the code given below:
 (1) 1 (2) 2 only
 (3) 3 only (4) 2 and 3
- 17.** Equation $x + 3y - 6 = 0$ will cut the Y axis
 (1) 6 unit above the origin
 (2) 2 unit above the origin
 (3) 6 unit below the origin
 (4) 2 unit below the origin
- 18.** The graph of equation $y = 5$ is parallel to
 (1) Y axis (2) X axis
 (3) Both axis (4) None of these
- 19.** Which of the following is not a method to solving linear equation of two variables?
 (1) Elimination method
 (2) Substitution method
 (3) Cross multiplication method
 (4) Division method
- 20.** Which of the following is not a linear equation?
 (1) $ax + by + c = 0$
 (2) $ax + c = 0$
 (3) $ax + by + cz + d = 0$
 (4) $(ax + b)x + c = 0$
- 21.** In an examination, a student scores 4 marks for every correct answer and loses 1 mark for every wrong answer. A student attempted all the 200 questions and scored in all 200 marks. The number of questions, he answered correctly was
 (1) 82 (2) 80
 (3) 68 (4) 60
- 22.** What is/are the solutions of the set of homogeneous equations $(4x + 2y = 0)$ and $(6x + 3y = 0)$?
 (a) Only $x = 0, y = 0$
 (b) Only $x = 0, y = 0$ and $x = 1, y = 2$
 (c) An infinite number of solutions
 (d) No solution
- 23.** The cost of 4 books and 3 pencils is same as that of 8 books and 1 pencil. This cost will be same as that of which one of the following?
 (a) 2 books and 6 pencils
 (b) 5 books and 5 pencils
 (c) 6 books and 2 pencils
 (d) 12 books and 4 pencils
- 24.** If one-third of a two-digit number exceeds its one-fourth by 8, then what is the sum of the digits of the number?
 (a) 6 (b) 13
 (c) 15 (d) 17
- 25.** What is the sum of two numbers whose difference is 45 and the quotient of the greater number by the lesser number is 4?
 (a) 100 (b) 90
 (c) 80 (d) 75
- 26.** The solution of the equations $\frac{3x - y + 1}{3} = \frac{2x + y + 2}{5} = \frac{3x + 2y + 1}{6}$ is given by which one of the following?
 (a) $x = 2, y = 1$
 (b) $x = 1, y = 1$
 (c) $x = -1, y = -1$
 (d) $x = 1, y = 2$
- 27.** What is the solution of the equations $x - y = 0.9$ and $11(x + y)^{-1} = 2$?
 (a) $x = 3.2$ and $y = 2.3$
 (b) $x = 1$ and $y = 0.1$
 (c) $x = 2$ and $y = 1.1$
 (d) $x = 1.2$ and $y = 0.3$
- 28.** What is the value of k for which the system of equations $x + 2y - 3 = 0$ and $5x + ky + 7 = 0$ has no solution?
 (a) $-3/14$ (b) $-14/3$
 (c) $1/10$ (d) 10

29. Under what condition do the equations $kx - y = 2$ and $6x - 2y = 3$ have a unique solution?
 (a) $k = 3$ (b) $k \neq 3$
 (c) $k = 0$ (d) $k \neq 0$
30. For what value of k , the following equation will be inconsistent? $2x - ky = 4$ and $3x + 2y = 6$?
 (a) $4/3$ (b) $-4/3$
 (c) $2/3$ (d) $3/2$

ANSWER KEY

1.	(4)	2.	(1)	3.	(1)	4.	(1)	5.	(1)	6.	(1)	7.	(4)	8.	(2)	9.	(3)	10.	(3)
11.	(3)	12.	(2)	13.	(2)	14.	(4)	15.	(2)	16.	(4)	17.	(2)	18.	(2)	19.	(4)	20.	(4)
21.	(2)	22.	(3)	23.	(3)	24.	(3)	25.	(4)	26.	(2)	27.	(1)	28.	(4)	29.	(2)	30.	(2)

PRACTICE SHEET-3

- Every quadratic equation $ax^2 + bx + c = 0$, where $a, b, c \in \mathbb{R}$, $a \neq 0$ has:
 (a) Exactly one real root
 (b) At least one real root
 (c) At least two real roots
 (d) At most two real roots
- If α, β are roots of $ax^2 + bx + c = 0$ and $\alpha + h, \beta + h$ are the roots of $px^2 + qx + r = 0$, then what is h equal to?
 (a) $\frac{1}{2} \left(\frac{b}{a} - \frac{q}{p} \right)$ (b) $\frac{1}{2} \left(-\frac{b}{a} + \frac{q}{p} \right)$
 (c) $\frac{1}{2} \left(\frac{b}{p} + \frac{q}{a} \right)$ (d) $\frac{1}{2} \left(-\frac{b}{a} + \frac{q}{a} \right)$
- If α and β are the roots the equation $ax^2 + bx + c = 0$, where $a \neq 0$, then $(a\alpha + b)(a\beta + b)$ is equal to:
 (a) ab (b) bc
 (c) ca (d) abc
- The roots of the equation $2a^2 x^2 - 2abx + b^2 = 0$, when $a < 0$ and $b > 0$ are:
 (a) Sometimes complex (b) Always irrational
 (c) Always complex (d) Always real
- The quadratic equation $x^2 + bx + 4 = 0$ will have real roots, if:
 (a) Only $b \leq -4$ (b) Only $b \geq 4$
 (c) $-4 < b < 4$ (d) $b \leq -4, b \geq 4$
- If α and β are the roots of the equation $x^2 + x + 2 = 0$, then what is $\frac{\alpha^{10} + \beta^{10}}{\alpha^{-10} + \beta^{-10}}$ equal to?
 (a) 4096 (b) 2048
 (c) 1024 (d) 512
- What is the difference in the roots of the equation $x^2 - 10x + 9 = 0$.
 (a) 2 (b) 3
 (c) 5 (d) 8
- If α and β are the roots of the equation $ax^2 + bx + c = 0$, then what is the value of $\sqrt{\frac{\alpha}{\beta}} + \sqrt{\frac{\beta}{\alpha}} + \sqrt{\frac{b}{a}} = ?$
 (a) -10 (b) 0
 (c) 1 (d) 2
- The roots of the equation $x^2 - 8x + 16 = 0$:
 (a) Are imaginary
 (b) Are distinct and real
 (c) Are equal and real
 (d) Cannot be determined
- If a and b are rational and b is not perfect square, then the quadratic equation with rational coefficients whose one root is $3a + \sqrt{b}$ is:
 (a) $x^2 - 6ax + 9a^2 - b = 0$
 (b) $3ax^2 + x - \sqrt{b} = 0$
 (c) $x^2 + 3ax + \sqrt{b} = 0$
 (d) $\sqrt{b} x^2 + x - 3a = 0$
- What is the sum of the squares of the roots of the equation $x^2 + 2x - 143 = 0$?
 (a) 170 (b) 180
 (c) 190 (d) 290
- If one of the roots of the equation $x^2 + ax - b = 0$ is, 1, then what is the value of $(a-b)$?
 (a) -1 (b) 1
 (c) 2 (d) -2

13. If α and β are the roots of the equation. $x^2 - q(1+x) - r = 0$ then what is the value of $(1+\alpha)(1+\beta)$?
 (a) $1 - r$ (b) $q - r$
 (c) $1 + r$ (d) $q + r$
14. The solution of the simultaneous linear equations $2x + y = 6$ and $3y = 8 + 4x$ will also be satisfied by which one of the following linear equation?
 (a) $x + y = 5$ (b) $2x + y = 5$
 (c) $2x - 3y = 10$ (d) $2x + 3y = 6$
15. If the difference between the roots of $ax^2 + bx + c = 0$ is 1, then which one of the following is correct?
 (a) $b^2 = a(a+4c)$ (b) $a^2 = b(b+4c)$
 (c) $a^2 = c(a+4c)$ (d) $b^2 = a(b+4c)$
- Direction (for next two):** The equation formed by multiplying each root of $ax^2 + bx + c = 0$ by 2 is $x^2 + 36x + 24 = 0$.
16. What is the value of $b:c$?
 (a) 3:1 (b) 1:2
 (c) 1:3 (d) 3:2
17. Which one of the following correct?
 (a) $bc = a^2$ (b) $bc = 36a^2$
 (c) $bc = 72a^2$ (d) $bc = 108a^2$
18. The sum of the reciprocals of two alternate natural numbers is $\frac{7}{24}$. What is the sum of the numbers?
 (a) 12 (b) 13
 (c) 14 (d) 16
19. What is the value of $y = \sqrt{8+2\sqrt{8+2\sqrt{8+2\sqrt{8+\dots}}}}$
 (a) 10 (b) 8
 (c) 6 (d) 4
20. If α and β be the roots of the equation $(x-a)(x-b) = c$, $c \neq 0$. Then, the roots of the equation $(x-\alpha)(x-\beta) + c = 0$ are:
 (a) a, c (b) b, c
 (c) a, b (d) $a+b, a+c$
21. What is the sum of the roots of the equation $(2-\sqrt{3})x^2 - (7-4\sqrt{3})x + (2+\sqrt{3}) = 0$?
 (a) $2-\sqrt{3}$ (b) $2+\sqrt{3}$
 (c) $7-4\sqrt{3}$ (d) 4
22. If one root of the equation $ax^2 + bx + c = 0$, $a \neq 0$ is reciprocal of the other root, then which one of the following is correct?
 (a) $a = c$ (b) $b = c$
 (c) $a = -c$ (d) $b = 0$
23. If sum of squares of the roots of the equation $x^2 + kx - b = 0$ is $2b$, then what is k equal to?
 (a) 1 (b) b
 (c) $-b$ (d) 0
24. If 3 is the root of the equation $x^2 - 8x + k = 0$, then what is the value of k ?
 (a) -15 (b) 9
 (c) 15 (d) 24
25. What is the condition that one root of the equation $ax^2 + bx + c = 0$, $a \neq 0$ should be double the other?
 (a) $2a^2 = 9bc$ (b) $2b^2 = 9ac$
 (c) $2c^2 = 9ab$ (d) None of these
26. What is the solution set for the equation $x^4 - 26x^2 + 25 = 0$?
 (a) $\{-5, -1, 1, 5\}$ (b) $\{-5, -1\}$
 (c) $\{1, 5\}$ (d) $\{-5, 0, 1, 5\}$
27. If p, q, r are rational numbers, then the roots of the equation $x^2 - 2px + p^2 - q^2 + 2qr - r^2 = 0$ are:
 (a) Complex (b) Pure imaginary
 (c) Irrational (d) Rational
28. If $x + y \leq 4$, then the how many non-zero positive integer ordered pair (x, y) ?
 (a) 4 (b) 5
 (c) 6 (d) 8
29. What is the set of points (x, y) satisfying the equations $x^2 + y^2 = 4$ and $x + y = 2$?
 (a) $\{(2, 0), (-2, 0), (0, 2)\}$
 (b) $\{(0, -2), (0, 2)\}$
 (c) $\{(2, 0), (0, 2)\}$
 (d) $\{(2, 0), (-2, 0), (0, 2), (0, -2)\}$
30. If α and β are the roots of equation $4x^2 + 3x + 7 = 0$, then what is the value of $\alpha^{-2} + \beta^{-2}$?
 (a) $47/49$ (b) $49/47$
 (c) $-47/49$ (d) $-49/47$

ANSWER KEY

1.	(c)	2.	(a)	3.	(c)	4.	(c)	5.	(d)	6.	(c)	7.	(d)	8.	(b)	9.	(c)	10.	(a)
11.	(d)	12.	(a)	13.	(a)	14.	(a)	15.	(a)	16.	(a)	17.	(d)	18.	(c)	19.	(d)	20.	(c)
21.	(a)	22.	(a)	23.	(d)	24.	(c)	25.	(b)	26.	(a)	27.	(d)	28.	(c)	29.	(c)	30.	(c)

PRACTICE SHEET-4

- If the product of the roots of $x^2 - 3kx + 2k^2 - 1 = 0$ is 7 for a fixed k , then what is the nature of roots?
(a) Integral and positive
(b) Integral and negative
(c) Irrational
(d) Rational but not integral
- Which one of the following is the quadratic equation whose roots are reciprocal to the roots of the quadratic equation $2x^2 - 3x - 4 = 0$?
(a) $3x^2 - 2x - 4 = 0$ (b) $4x^2 + 3x - 2 = 0$
(c) $3x^2 - 4x - 2 = 0$ (d) $4x^2 - 2x - 3 = 0$
- The value of y which will satisfy the equations $2x^2 + 6x + 5y + 1$ and $2x + y + 3 = 0$ may be found by solving which one of the following equations?
(a) $y^2 + 14y - 7 = 0$ (b) $y^2 + 8y + 1 = 0$
(c) $y^2 + 10y - 7 = 0$ (d) $y^2 - 8y + 7 = 0$
- If a polynomial equation has rational coefficient and has exactly three real roots, then what is the degree of the polynomial?
(a) Equal to 3
(b) Greater than or equal to 3
(c) Strictly greater than 3
(d) Less than 3
- If α and β are the roots of $ax^2 + bx + c = 0$, then what is the value of $\left(\frac{1}{\alpha^2} - \frac{1}{\beta^2}\right)^2$?
(a) $\frac{b^2(b^2 - 4ac)}{c^4}$ (b) $\frac{b(b^2 - 4ac)}{c^2}$
(c) $\frac{(b^2 - 4ac)}{c^2}$ (d) $\frac{(b^2 - 4ac)}{c^4}$
- Which one of the following is one of the two consecutive positive integers, the sum of whose squares is 761?
(a) 15 (b) 20 (c) 24 (d) 25
- If $3^x + 27(3^{-x}) = 12$, then what is the value of x ?
(a) Only 1 (b) Only 2
(c) 1 or 2 (d) 0 or 1
- What is the magnitude of difference of the roots of $x^2 - ax + b = 0$?
(a) $\sqrt{a^2 - 4b}$ (b) $\sqrt{b^2 - 4a}$
(c) $2\sqrt{a^2 - 4b}$ (d) $\sqrt{b^2 - 4ab}$
- If $a + b = 2m^2$, $b + c = 6m$, $a + c = 2$, where m is a real number and $a \leq b \leq c$, then which one of the following is correct?
(a) $0 \leq m \leq 1/2$ (b) $-1 \leq m \leq 0$
(c) $1/3 \leq m \leq 1$ (d) $1 < m \leq 2$
- Students of a class are made to stand in rows. If one student is extra in a row, there would be two rows less. If one student is less in a row, there would be three rows more. Then, what is the number of students in the class?
(a) 65 (b) 55
(c) 60 (d) 50
- What is the ratio of sum of squares of roots to the product of the roots of the equation $7x^2 + 12x + 18 = 0$?
(a) 6 : 1 (b) 1 : 6
(c) -6 : 1 (d) -6 : 7
- If the roots of the equation $\frac{x(x-1)-(m+1)}{(x-1)(m-1)} = \frac{x}{m}$ are equal, then what is the value of m ?
(a) 1 (b) $1/2$
(c) 0 (d) $-1/2$

- 13.** What is the least integral value of k for which equation $x^2 - 2(k - 1)x + (2k + 1) = 0$ has both the roots positive?
 (a) 1 (b) $-1/2$
 (c) 4 (d) 0
- 14.** If one of the roots of the equation $ax^2 + x - 3 = 0$ is -1.5 , then what is the value of a ?
 (a) 4 (b) 3
 (c) 2 (d) -2
- 15.** r is a non-zero real number such that $r^{75} > r^{90}$. This is possible only when
 (a) $-1 < r < 0$ (b) $0 < r < 1$
 (c) $1 < r$ (d) $-1 < r < 1$
- 16.** When the roots of the quadratic equation $ax^2 + bx + c = 0$ are negative of reciprocals of each other, then which one of the following is correct?
 (a) $b = 0$ (b) $c = 0$
 (c) $a = c$ (d) $a = -c$
- 17.** What are the roots of the equation $\log_{10}(x^2 - 6x + 45) = 2$?
 (a) 9, -5 (b) -9 , 5
 (c) 11, -5 (d) -11 , 5
- 18.** The sum of the roots of the equation $\frac{1}{x+a} + \frac{1}{x+b} = \frac{1}{c}$ is zero. What is the product of the roots of the equation?
 (a) $-\frac{(a+b)}{2}$ (b) $\frac{(a+b)}{2}$
 (c) $-\frac{(a^2+b^2)}{2}$ (d) $\frac{(a^2+b^2)}{2}$
- 19.** For what value of k , will the roots of the equation $kx^2 - 5x + 6 = 0$ be in the ratio of 2 : 3?
 (a) 0 (b) 1
 (c) -1 (d) 2
- 20.** If α and β are the roots of the equation $x^2 + px + q = 0$, then $-\alpha^{-1}$, $-\beta^{-1}$ are the roots of which one of the following equations?
 (a) $qx^2 - px + 1 = 0$ (b) $q^2 + px + 1 = 0$
 (c) $x^2 + px - q = 0$ (d) $x^2 - px + q = 0$
- 21.** If one root of the equation $ax^2 + x - 3 = 0$ is -1 , then what is the other root?
 (a) $1/4$ (b) $1/2$
 (c) $3/4$ (d) 1
- 22.** If the equation $(a^2 + b^2)x^2 - 2(ac + bd)x + (c^2 + d^2) = 0$ has equal roots, then which one of the following is correct?
 (a) $ab = cd$ (b) $ad = bc$
 (c) $a^2 + c^2 = b^2 + d^2$ (d) $ac = bd$
- 23.** What is the solution of the equation

$$\sqrt{\frac{x}{x+3}} - \sqrt{\frac{x+3}{x}} = -\frac{3}{2}$$

 (a) 1 (b) 2
 (c) 4 (d) None of these
- 24.** What are the roots of the equation $4x - 3.2x^2 + 2 + 32 = 0$?
 (a) 1, 2 (b) 3, 4
 (c) 2, 3 (d) 1, 3
- 25.** If α and β are the roots of the equation $x^2 - x - 1 = 0$, then what is the value of $(\alpha^4 + \beta^4)$?
 (a) 7 (b) 0
 (c) 2 (d) None of these
- 26.** If sum as well as product of roots of a quadratic equation is 9, then what is the equation?
 (a) $x^2 + 9x - 18 = 0$ (b) $x^2 - 18x + 9 = 0$
 (c) $x^2 + 9x + 9 = 0$ (d) $x^2 - 9x + 9 = 0$
- 27.** If one root of the equation $2x^2 + 3x + c = 0$ is 0.5, then what is the value of c ?
 (a) -1 (b) -2
 (c) -3 (d) -4
- 28.** What is the condition that the equation $ax^2 + bx + c = 0$, where $a \neq 0$ has both the roots positive?
 (a) a , b and c are of same sign.
 (b) a and b are of same sign.
 (c) b and c have the same sign opposite to that of a .
 (d) a and c have the same sign opposite to that of b .
- 29.** The equation $(1 + n^2)x^2 + 2ncx + (c^2 - a^2) = 0$ will have equal roots, if
 (a) $c^2 = 1 + a^2$ (b) $c^2 = 1 - a^2$
 (c) $c^2 = 1 + n^2 + a^2$ (d) $c^2 = (1 + n^2)a^2$
- 30.** The equation whose roots are twice the roots of the equation $x^2 - 2x + 4 = 0$ is
 (a) $x^2 - 2x + 4 = 0$ (b) $x^2 - 2x + 16 = 0$
 (c) $x^2 - 4x + 8 = 0$ (d) $x^2 - 4x + 16 = 0$

ANSWER KEY

1.	(c)	2.	(b)	3.	(c)	4.	(a)	5.	(a)	6.	(b)	7.	(c)	8.	(a)	9.	(c)	10.	(c)
11.	(d)	12.	(d)	13.	(c)	14.	(c)	15.	(b)	16.	(d)	17.	(c)	18.	(c)	19.	(b)	20.	(a)
21.	(c)	22.	(b)	23.	(a)	24.	(c)	25.	(a)	26.	(d)	27.	(b)	28.	(d)	29.	(d)	30.	(d)

CDS PYQ

- X, Y and Z had taken a dinner together. The cost of the meal of Z was 20% more than that of Y and the cost of the meal of X was $\frac{5}{6}$ as much as the cost of the meal of Z. If Y paid Rs 100, then what was the total amount that all the three of them had paid?
[CDS 2013 (1)]
(a) Rs 285
(b) Rs 300
(c) Rs 355
(d) None of the above
- The system of equations $3x + y - 4 = 0$ and $6x + 2y - 8 = 0$ has:
[CDS 2013 (1)]
(a) A unique solution $x = 1, y = 1$
(b) A unique solution $x = 0, y = 4$
(c) No solution
(d) Infinite solution
- For what value of k, $(x + 5)$ is a factor of $6x^2 + kx + 10$?
[CDS 2013 (1)]
(a) 5
(b) 32
(c) 36
(d) 40
- A number consists of two digit whose sum is 10. If the digit of the number are reversed, then the number decreased by 36. Which of the following is/are correct?
I. The number is divisible by a composite number
II. The number is a multiple of a prime number
Select the correct answer using the codes given below:
[CDS 2013 (1)]
(a) Only I
(b) Only II
(c) Both I and II
(d) Neither I nor II
- Ten chairs and six tables together cost Rs 6200, three chairs and two tables together cost Rs 1900. The cost of 4 chairs and 5 tables is:
[CDS 2013 (1)]
(a) Rs 3000
(b) Rs 3300
(c) Rs 3500
(d) Rs 3800
- The difference in the root of the equation $2x^2 - 11x + 5 = 0$ is
[CDS 2013 (1)]
(a) 4.5
(b) 4
(c) 3.5
(d) 3
- If one of the roots of the equation $x^2 - bx + c = 0$ is the square of the other, then which of the following option is correct?
[CDS 2013 (1)]
(a) $b^3 = 3bc + c^2 + c$
(b) $c^3 = 3bc + b^2 + b$
(c) $3bc = c^3 + b^2 + b$
(d) $3bc = c^3 + b^3 + b^2$
- If $x + y - 7 = 0$ and $3x + y - 13 = 0$, then what is $4x^2 + y^2 + 4xy$ equal to?
[CDS 2013 (2)]
(a) 75
(b) 85
(c) 91
(d) 100
- A bus starts with some passengers. At the first stop, one-fifth of the passengers gets down and 40 passengers get in. At the second stop, half of the passengers gets down and 30 get in. The number of passengers now in 70. The number of passengers with which the bus started was:
[CDS 2013 (2)]

(a) 40
(c) 60

(b) 50
(d) 70

10. Consider the following statements in respect of the quadratic equation $ax^2 + bx + c = 0$, where $a \neq 0$.

I. The product of roots is equal to the sum of the roots

II. The roots of the equation are always unequal and real

Which of the statements given above is/are correct?

[CDS 2013 (2)]

(a) Only I
(c) Both I and II

(b) Only II
(d) Neither I nor II

11. If α and β are the roots of the equation $x^2 - x - 1 = 0$, then what is $\frac{\alpha^2 + \beta^2}{(\alpha^2 - \beta^2)(\alpha - \beta)}$ equal to?

[CDS 2013 (2)]

(a) $\frac{2}{5}$

(b) $\frac{3}{5}$

(c) $\frac{4}{5}$

(d) 4

12. Which one is one of the factors of:

$$x^2 + \frac{1}{x^2} + 8\left(x + \frac{1}{x}\right) + 14$$

[CDS 2013 (2)]

(a) $x + \frac{1}{x} + 1$

(b) $x + \frac{1}{x} + 3$

(c) $x + \frac{1}{x} + 6$

(d) $x + \frac{1}{x} + 7$

13. A positive number, when increased by 10 equals 200 times its reciprocal. What is number?

[CDS 2014 (1)]

(a) 100
(c) 20

(b) 10
(d) 200

14. The sum of two positive numbers x and y is 2.5 times their difference. If the product of numbers is 84, then what is the sum of those two numbers?

[CDS 2014 (1)]

(a) 26
(c) 22

(b) 24
(d) 20

15. Ravi's brother is 3 years elder to him. His father was 28 years of age when his sister was born while his mother was 26 years of age when he was born. If his sister was 4 years of age when his brother was born, the ages of Ravi's father and mother, respectively, when his brother was born were:

[CDS 2014 (1)]

(a) 32 years and 23 years
(b) 32 years and 29 years
(c) 35 years and 29 years
(d) 35 years and 33 years

16. Two chairs and one table cost Rs 700 and 1 chair and 2 tables costs Rs 800. If the cost m tables and m chairs is Rs 30,000, then what is m equal to?

[CDS 2014 (1)]

(a) 60
(c) 50

(b) 55
(d) 45

17. In solving a problem, one student makes a mistake in the coefficient of the first degree term and obtain -9 and -1 for the roots. Another student makes a mistake in the constant term of the equation and obtains 8 and 2 for the roots. The correct equation was:

[CDS 2014 (1)]

(a) $x^2 + 10x + 9 = 0$
(c) $x^2 - 10x + 9 = 0$

(b) $x^2 - 10x + 16 = 0$
(d) None of these

18. If m and n are the roots of the equation $ax^2 + bx + c = 0$, then the equation whose roots are $\frac{(m^2 + 1)}{m}$ and $\frac{(n^2 + 1)}{n}$ is:

[CDS 2014 (1)]

(a) $acx^2 + (ab + bc)x + b^2 + (a - c)^2 = 0$
(b) $acx^2 + (ab - bc)x + b^2 + (a - c)^2 = 0$
(c) $acx^2 - (ab - bc)x + b^2 - (a - c)^2 = 0$
(d) $acx^2 - (ab + bc)x + b^2 - (a - c)^2 = 0$

19. The value of $x^2 - 4x + 11$ can never be less than:

[CDS 2014 (1)]

(a) 7
(c) 11

(b) 8
(d) 22

- 20.** If the roots of the equation $(a^2 - bc)x^2 + 2(b^2 - ac)x + (c^2 - ab) = 0$ are equal, where $a, b, c \neq 0$, then which one of the following is correct?

[CDS 2014 (1)]

(a) $a+b+c=abc$ (b) $a^2+b^2+c^2=0$
(c) $a^3+b^3+c^3=0$ (d) $a^3+b^3+c^3=abc$

- 21.** If the roots of the equation $Ax^2 + Bx + C = 0$ are -1 and 1 , then which one of the following is correct?

[CDS 2014 (1)]

(a) A and C are both zero
(b) A and B are both positive
(c) A and C are both negative
(d) A and C are of opposite sign

- 22.** If m and n ($m > n$) are the roots of the equation $7(x + 2a)^2 + 3a^2 = 5a(7x + 23a)$, where $a > 0$, then what is $3m-n$ equal to?

[CDS 2014 (2)]

(a) $12a$ (b) $14a$
(c) $15a$ (d) $18a$

- 23.** If one the roots of the equation $px^2 + qx + r = 0$ is three times the other, then which one of the following relations is correct?

[CDS 2014 (2)]

(a) $3q^2 = 16pr$ (b) $a^2 = 24pr$
(c) $p = q + r$ (d) $p + q + r = 1$

- 24.** If m and n are the roots of the equation $x^2 + ax + b = 0$ and m^2, n^2 are the roots of the equation $x^2 - cx + d = 0$, then which of the following is/are correct?

1. $2b - a^2 = c$
2. $b^2 = d$

Select the correct answer using the codes given below:

[CDS 2014 (2)]

(a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

- 25.** If $(x+k)$ is the common factor of $x^2 + ax + b$ and $x^2 + cx + d$ of and then what is k equal to?

[CDS 2014 (2)]

(a) $(d-b)/(c-a)$
(c) $(d+b)/(c+a)$

(b) $(d-b)/(a-c)$
(d) $(d-b)/(c+a)$

- 26.** If the equation $x^2 + 2(1+k)x + k^2 = 0$ has equal roots, then what is the value of k ?

[CDS 2014 (2)]

(a) $1/2$ (b) $-1/2$
(c) 1 (d) -1

- 27.** Consider the following statements:

1.The equation $1990x - 173y = 11$ has no solution in integers for x and y .
2.The equation $3x - 12y = 7$ has no solution in integers for x and y .

Which of the above statements is/are correct?

[CDS 2015 (1)]

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

- 28.** For which value of k does the pair of equations $x^2 - y^2 = 0$ and $(x - k)^2 + y^2 = 1$ yield a unique positive solution of x ?

[CDS 2015 (1)]

(a) 2 (b) 0
(c) $\sqrt{2}$ (d) $-\sqrt{2}$

- 29.** The sign of the quadratic polynomial $ax^2 + bx + c$ is always positive if:

[CDS 2015 (1)]

(a) a is positive and $b^2 - 4ac \leq 0$
(b) a is positive and $b^2 - 4ac \geq 0$
(c) a can be any real number and $b^2 - 4ac \leq 0$
(d) a can be any real number and $b^2 - 4ac \geq 0$

- 30.** If $k = x - y + 2z$ where $-2 \leq x \leq 1$, $-1 \leq y \leq 2$ and $3 \leq z \leq 6$, then which one of the following is correct?

[CDS 2015 (2)]

(a) $0 \leq k \leq 9$ (b) $5 \leq k \leq 11$
(c) $2 \leq k \leq 14$ (d) $2 \leq k \leq 11$

- 31.** The number of pairs (x, y) where x, y are integers satisfying the equation $21x + 48y = 5$ is:

[CDS 2015 (2)]

(a) Zero (b) One
(c) Two (d) Infinity

- 32.** Let x and y be positive integers such that x is prime and y is composite. Which of the following statements are correct?

1. $(y-x)$ can be an even integer
 2. xy can be an even integer
 3. $0.5(x+y)$ can be an even integer
- Select the correct answer using the code given below:

[CDS 2015 (2)]

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

33. If the roots of the quadratic equations $x^2 - 4 - \log_{10} N = 0$ are all real, then the minimum value of N is:

[CDS 2015 (2)]

- (a) $\frac{1}{100}$ (b) $\frac{1}{1000}$
(c) $\frac{1}{10000}$ (d) 10000

34. The difference of maximum values of the expressions $(6 + 5x - x^2)$ and $(y - 6 - y^2)$ for any real values of x and y is:

[CDS 2015 (2)]

- (a) 16 (b) 17
(c) 18 (d) 19

35. What is $\sqrt{4 + \sqrt{4 - \sqrt{4 + \sqrt{4 - \dots}}}}$ equal to?

[CDS 2015 (2)]

- (a) 3 (b) $\frac{\sqrt{13}-1}{2}$
(c) $\frac{\sqrt{13}+1}{2}$ (d) 0

36. A tin of oil was $\frac{4}{5}$ full. When 6 bottles of oil were taken out from this tin and 4 bottles of oil were poured into it, it was $\frac{3}{4}$ full. Oil of how many bottles can the tin contain? (All bottles are of equal volume)

[CDS 2015 (2)]

- (a) 35 (b) 40
(c) 45 (d) 50

37. If the equations $x^2 - px + q = 0$ and $x^2 + qx - p = 0$ have a common root, then which one of the following is correct?

[CDS 2016 (1)]

- (a) $p-q=0$ (b) $p+q-2=0$
(c) $p+q-1=0$ (d) $p-q-1=0$

38. The value of k , for which the system of equation $3x - ky - 20 = 0$ and $6x - 10y + 40 = 0$ has no solution, is:

[CDS 2016 (1)]

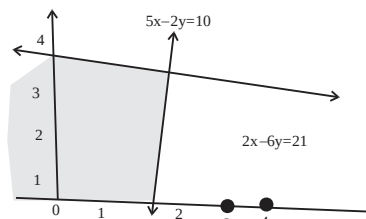
- (a) 10 (b) 6
(c) 5 (d) 3

39. There are three brothers. The sums of ages of two of them at a time are 4 years, 6 years and 8 years. The age difference between the eldest and the youngest is:

[CDS 2016 (1)]

- (a) 3years (b) 4years
(c) 5 years (d) 6 years

40.



The linear in equations, for which the shaded area in the figure given above is the solution set, are:

[CDS 2016 (1)]

- (a) $2x+6y \leq 21, 5x-2y \leq 10$
(b) $2x+6y \leq 21, 5x-2y \geq 10$
(c) $2x+6y \geq 21, 5x-2y \leq 10$
(d) $2x+6y \geq 21, 5x-2y \geq 10$

41. If a and b are negative real numbers and c is a positive real number, then which of the following is/are correct?

1. $a - b < a - c$
2. If $a < b$ then $\frac{a}{c} < \frac{b}{c}$
3. $\frac{1}{b} < \frac{1}{c}$

Select the correct answer using the code given below:

[CDS 2016 (1)]

- (a) 1 (b) 2 only
(c) 3 only (d) 2 and 3

42. If the roots of the equation $lx^2 + mx + m = 0$ are in the ratio $p : q$, then:

$\sqrt{\frac{p}{q}} + \sqrt{\frac{q}{p}} + \sqrt{\frac{m}{l}}$ is equal to:

[CDS 2016 (1)]

- (a) 0 (b) 1
(c) 2 (d) 3

43. If $\sqrt{3x^2 - 7x - 30} - \sqrt{2x^2 - 7x - 5} = x - 5$ has α and β as its roots, then the value of $\alpha\beta$ is:

[CDS 2016 (1)]

- (a) -15 (b) -5
(c) 0 (d) 5

44. Let p and q be non-zero integers. Consider the polynomial $A(x) = x^2 + px + q$. It is given that $(x - m)$ and $(x - km)$ are linear factors of $A(x)$, where m is a non-zero integer and k is a positive integer, $k \geq 2$. Which one of the following is correct?

[CDS 2016 (1)]

- (a) $(k+1)^2 p^2 = kg$ (b) $(k+1)^2 q = kp^2$
(c) $k^2 q = (k+1) p^2$ (d) $k^2 p^2 = (k+1)^2 q$

45. If the linear factors of $ax^2 - (a^2 + 1)x + a$ are p and q then $p + q$ is equal to:

[CDS 2016 (1)]

- (a) $(x - 1)(a + 1)$ (b) $(x + 1)(a + 1)$
(c) $(x - 1)(a - 1)$ (d) $(x + 1)(a - 1)$

46. If the sum of the roots of $ax^2 + bx + c = 0$ is equal to the sum of the squares of their reciprocals, then which one of the following relations is correct?

[CDS 2016 (1)]

- (a) $ab^2 + bc^2 = 2a^2 c$ (b) $ac^2 + bc^2 = 2b^2 a$
(c) $ab^2 + bc^2 = a^2 c$ (d) $a^2 + b^2 + c^2 = 1$

47. Under what condition on p and q , one of the roots of the equation $x^2 + px + q = 0$ is the square of the other?

[CDS 2016 (1)]

- (a) $1 + q + q^2 = 3pq$ (b) $1 + q + p^2 = 3pq$
(c) $p^3 + q + q^2 = 3pq$ (d) $q^3 + p + p^2 = 3pq$

48. The solution of the in-equation:

$$1 + \frac{1}{x} - \frac{1}{x^2} \geq 0 \text{ is (given that } x \neq 0)$$

[CDS 2016 (1)]

- (a) $x > 0$
(b) $x < 0$
(c) $\frac{-1-\sqrt{5}}{2} \leq x \leq \frac{-1+\sqrt{5}}{2}$
(d) $x \leq \frac{-1-\sqrt{5}}{2}$ or $x \geq \frac{-1+\sqrt{5}}{2}$

49. If $4 \times 2^y = 128$ and $3^3 \times 3^{2y} - 9^{xy} = 0$, then the value of $x + y$ can be equal to:

[CDS 2016 (1)]

- (a) 7 (b) 5
(c) 3 (d) 1

50. Which of the points $P(5, -1)$, $Q(3, -2)$ and $R(1, 1)$ lie in the solution of the system of in-equations $x + y \leq 4$ and $x - y \geq 2$?

[CDS 2016 (1)]

- (a) Q and R only (b) P and R only
(c) P and Q only (d) P , Q and R

51. If λ is an integer and α, β are the roots of $4x^2 - 16x + \frac{\lambda}{4} = 0$ such that $1 < \alpha < 2$ and $2 < \beta < 3$, then how many values can λ take?

[CDS 2016 (2)]

- (a) 3 (b) 9
(c) 14 (d) 15

52. If α and β are the two zeros of the polynomial $25x^2 - 15x + 2$, then what is a quadratic polynomial whose zeros are $(2\alpha)^{-1}$ and $(2\beta)^{-1}$?

[CDS 2016 (2)]

- (a) $x^2 + 30x + 2$ (b) $8x^2 - 30x + 25$
(c) $8x^2 - 30x$ (d) $x^2 + 30x$

53. If p and q are the roots of $x^2 + px + q = 0$, then which of the following is correct?

[CDS 2016 (2)]

- (a) $p=0$ or 1 (b) $p = 1$ only
(c) $p = -2$ or 0 (d) $p = -2$ only

54. The pair of linear equations $kx + 3y + 1 = 0$ and $2x + y + 3 = 0$ intersect each other, if:

[CDS 2017 (1)]

- (a) $k=6$ (b) $k \neq 6$
(c) $k=0$ (d) $k \neq 0$

55. The system of equations $2x + 4y = 6$ and $4x + 8y = 8$ is:

[CDS 2017 (1)]

- (a) Consistent with a unique solution
(b) Consistent with infinitely many solution
(c) Inconsistent
(d) None of the above

56. If α and β are the roots of the quadratic equation $2x^2 + 6x + k = 0$, where $k < 0$, then what is the maximum value of $\left(\frac{\alpha}{\beta} + \frac{\beta}{\alpha}\right)$?

[CDS 2017 (1)]

- (a) 2 (b) -2
(c) 9 (d) -9

57. If one root of $(a^2 - 5a + 3)x^2 + (3a - 1)x + 2 = 0$ is twice the other, then what is the value of 'a'?

[CDS 2017 (1)]

(a) $\frac{2}{3}$

(b) $-\frac{2}{3}$

(c) $\frac{1}{3}$

(d) $-\frac{1}{3}$

58. $(x+4)$ is a factor of which one of the following expressions?

[CDS 2017 (1)]

(a) $x^2 - 7x + 44$

(b) $x^2 - 7x - 44$

(c) $x^2 - 7x - 44$

(d) $x^2 + 7x + 44$

59. If α and β are the roots of the equation $x^2 + px + q = 0$, then what is $\alpha^2 + \beta^2$ equal to?

[CDS 2017 (1)]

(a) $p^2 - 2q$

(b) $a^2 - 2p$

(c) $p^2 + 2q$

(d) $q^2 - q$

60. Aman and Alok attempted to solve a quadratic equation. Aman made a mistake in writing down the constant term and ended up in roots (4,3). Alok made a mistake in writing down the coefficient of x to get roots (3,2). The correct roots of the equation are:

[CDS 2017 (1)]

(a) -4, -3

(b) 6, 1

(c) 4, 3

(d) -1, -1

61. What number must be subtracted from both the numerator and the denominator of the fraction $\frac{27}{35}$ so that it becomes $\frac{2}{3}$?

[CDS 2017 (1)]

(a) 6

(b) 8

(c) 9

(d) 11

62. The value of x which satisfy the equation $5^{1+x} + 5^{1-x} = 26$ are:

[CDS 2017 (1)]

(a) -1, 1

(b) 0, 1

(c) 1, 2

(d) -1, 0

63. Sunil wants to spend Rs200 on two types of sweets, costing Rs 7 and Rs 10 respectively. What is the maximum number of sweet he can get so that no money is left over?

[CDS 2017 (1)]

(a) 25

(b) 26

(c) 27

(d) 28

64. What is the value of u in the system of equations $3(2u + v) = 7uv$, $3(u + 3v) = 11uv$?

[CDS 2017 (2)]

(a) 0

(b) $\frac{1}{4}$

(c) $\frac{1}{2}$

(d) 1

65. What is the positive value of m for which the roots of the equation $12x^2 + mx + 5 = 0$ are in the ratio 3:2?

[CDS 2017 (2)]

(a) $5\sqrt{10}$

(b) $\frac{5\sqrt{10}}{12}$

(c) $\frac{5}{12}$

(d) $\frac{12}{5}$

66. If the roots of the equation $a(b-c)x^2 + b(c-a)x + c(a-b) = 0$ are equal, then which one of the following is correct?

[CDS 2017 (2)]

(a) $2b = a + c$

(b) $b^2 = ac$

(c) $\frac{2}{b} = \frac{1}{a} + \frac{1}{c}$

(d) $\frac{1}{b} = \frac{1}{a} + \frac{1}{c}$

67. If k is an integer, then $x^2 + 7x - 14 \left(k^2 - \frac{7}{8} \right) = 0$ has:

[CDS 2017 (2)]

(a) Both integral roots

(b) At least one integral root

(c) No integral root

(d) Both positive integral roots

68. What is the value of α ($\alpha \neq 0$) for which $x^2 - 5x + \alpha$, and $x^2 - 7x + 2\alpha$ have a common factor?

[CDS 2017 (2)]

(a) 6

(b) 4

(c) 3

(d) 2

69. The value of $\sqrt{1 + \sqrt{1 + \sqrt{1 + \dots}}}$:

[CDS 2017 (2)]

(a) Equals to 1

(b) Lies between 0 and 1

(c) Lies between 1 and 2

(d) Is greater than 2

70. If $65x - 33y = 97$ and $33x - 65y = 1$, then what is xy equal to?

[CDS 2018 (1)]

(a) 2

(b) 3

(c) -2

(d) -3

71. If the roots of the equation $px^2 + x + r = 0$ are reciprocal to each other, then which one of the following is correct?

[CDS 2018 (1)]

(a) $p=2r$

(b) $p=r$

(c) $2p=r$

(d) $p=4r$

72. If α and β are the roots of the equation $ax^2 + bx + c = 0$, then what is the value of expression $(\alpha+1)(\beta+1)$?

[CDS 2018 (1)]

- (a) $\frac{a+b+c}{a}$ (b) $\frac{b+c-a}{a}$
(c) $\frac{a-b+c}{a}$ (d) $\frac{a+b-c}{a}$

73. A quadratic polynomial $ax^2 + bx + c$ is such that when it is divided by x , $(x-1)$ and $(x+1)$, the remainders are 3, 6 and 4 respectively. What is the value of $(a+b)$?

[CDS 2018 (1)]

- (a) 3 (b) 2
(c) 1 (d) -1

74. $\frac{1}{25}$ of the students who registered did not appear for the examination, $\frac{11}{20}$ of those who appeared passed. If the number of registered students is 2000, the number who passed is:

[CDS 2018 (1)]

- (a) 1920 (b) 1056
(c) 1020 (d) 864

75. The age of a woman is a two-digit integer. On reversing this integer, the new integer is the age of her husband who is elder to her. The difference between their ages is one-eleventh of their sum. What is the difference between their ages?

[CDS 2018 (1)]

- (a) 8 yrs (b) 9 yrs
(c) 10 yrs (d) 11 yrs

76. If $x^2 - 6x - 27 > 0$, then which one of the following is correct?

[CDS 2018 (2)]

- (a) $-3 < x < 9$ (b) $x < 9$ or $x > -3$
(c) $x > 9$ or $x < -3$ (d) $x < -3$

77. If the sum of the square of three consecutive natural numbers is 110, then the sum of their cubes is:

[CDS 2018 (2)]

- (a) 625 (b) 654
(c) 684 (d) 725

78. If α and β are the roots of the equation $ax^2 + bx + c = 0$, then the value of $\frac{1}{a\alpha+b} + \frac{1}{a\beta+b}$ is:

[CDS 2018 (2)]

- (a) $\frac{a}{bc}$ (b) $\frac{b}{ac}$
(c) $\frac{c}{ab}$ (d) $\frac{1}{abc}$

79. The minimum value of the expression $2x^2 + 5x + 5$ is:

[CDS 2018 (2)]

- (a) 5 (b) $15/8$
(c) $-15/8$ (d) 0

80. According to Mr. Sharma's will, half of his property goes to his wife and the rest is equally divided between his two sons, Ravi and Raj. Some years later, Ravi dies and leaves half of his property to his widow and rest to his brother Raj. When Raj dies he leaves half of his property to his widow and remaining to his mother, who is still alive. The mother now owns Rs 88,000 worth of the property. The total worth of the property of Mr. Sharma was :

[CDS 2018 (2)]

- (a) Rs 1,00,000 (b) Rs 1,24,000
(c) Rs 1,28,000 (d) Rs 1,32,000

81. X bought 4 bottles of lemon juice and Y bought one bottle of orange juice. Orange juice per bottle costs twice the cost of lemon juice per bottle. Z bought nothing but contributed Rs 50 for his share of the drink which they mixed together and shared the cost equally. If Z's Rs 50 is covered from his share, then what is the cost of one bottle of orange juice?

[CDS 2018 (2)]

- (a) Rs 75 (b) Rs 50
(c) Rs 46 (d) Rs 30

82. The solution of linear inequalities $x + y \geq 5$ and $x - y \leq 3$ lies:

[CDS 2018 (2)]

- (a) Only in the first quadrant
(b) In the first and second quadrants
(c) In the second and third quadrants
(d) In the third and fourth quadrants

83. It is given that the equations $x^2 - y^2 = 0$ and $(x-a)^2 + y^2 = 1$ have single positive solution. For this, the value of 'a' is:

[CDS 2018 (2)]

- (a) $\sqrt{2}$ (b) 2
(c) $-\sqrt{2}$ (d) 1

- 84.** Which of the following pair of numbers is the solution of the equation $3^{x+2} + 3^{-x} = 10$?
[CDS 2019 (1)]
(a) 0, 2 (b) 0, -2
(c) 1, -1 (d) 1, 2
- 85.** The inequality $3^N > N^3$ holds when:
[CDS 2019 (1)]
(a) N is any natural number
(b) N is a natural number greater than 2
(c) N is a natural number greater than 3
(d) N is a natural number except 3
- 86.** A man who recently died left a sum of Rs 3,90,000 to be divided among his wife, five sons and four daughters. He directed that each son should receive 3 times as much as each daughter receives and that each daughter should receive twice as much as their mother receives. What was the wife's share?
[CDS 2019 (1)]
(a) Rs 14,000 (b) Rs 12,000
(c) Rs 10,000 (d) Rs 9,000
- 87.** If p and q are the roots of the equation $x^2 - 15x + r = 0$ and $p - q = 1$, then what is the value of r?
[CDS 2019 (1)]
(a) 55 (b) 56
(c) 60 (d) 64
- 88.** For the in-equality $x^2 - 7x + 12 > 0$, which one of the following is correct?
[CDS 2019 (1)]
(a) $3 < x < 4$
(b) $-\infty < x < 3$ only
(c) $4 < x < \infty$ only
(d) $-\infty < x < 3$ or $4 < x < \infty$
- 89.** If the sum of a real number and its reciprocal is $\frac{26}{5}$, then how many such numbers are possible?
[CDS 2019 (2)]
(a) None (b) one
(c) Two (d) Four
- 90.** If the equations $x^2 + 5x + 6 = 0$ and $x^2 + kx + 1 = 0$ have a common root, then what is the value of k?
[CDS 2019 (2)]
(a) $-\frac{5}{2}$ or $-\frac{10}{3}$ (b) $\frac{5}{2}$ or $\frac{10}{3}$
(c) $\frac{5}{2}$ or $-\frac{10}{3}$ (d) $-\frac{5}{2}$ or $\frac{10}{3}$
- 91.** What is $(x - a)(x - b)(x - c)$ equal to?
[CDS 2019 (2)]
(a) $x^3 - (a + b + c)x^2 + (bc + ca + ab)x - abc$
(b) $x^3 - (a + b + c)x^2 + (bc + ca + ab)x + abc$
(c) $x^3 - (bc + ca + ab)x^2 + (a + b + c)x - abc$
(d) $x^3 + (bc + ca + ab)x^2 - (a + b + c)x - abc$
- 92.** A person carries Rs500 and wants to buy apples and oranges out of it. If the cost of one apple is Rs5 and the cost of one orange is Rs7, then what is the number of ways in which a person can buy both apples and oranges using total amount?
[CDS 2019 (2)]
(a) 10 (b) 14
(c) 15 (d) 17
- 93.** A real number x is such that $(x - x^2)$ is maximum. What is x equal to?
[CDS 2019 (2)]
(a) -1.5 (b) -0.5
(c) 0.5 (d) 1.5
- 94.** The sum of the squares of four consecutive natural numbers is 294. What is the sum of the numbers?
[CDS 2019 (2)]
(a) 38 (b) 34
(c) 30 (d) 26
- 95.** The equation $x^2 + px + q = 0$ has roots equal to p and q where $q \neq 0$. What are the values of p and q respectively?
[CDS 2019 (2)]
(a) 1, -2 (b) 1, 2
(c) -1, 2 (d) -1, -2
- 96.** If (b-6) is one root of the quadratic equation $x^2 - 6x + b = 0$, where b is an integer, then what is the maximum value of b^2 ?
[CDS 2019 (2)]
(a) 36 (b) 49
(c) 64 (d) 81
- 97.** What is the maximum value of the expression $\frac{1}{x^2 + 5x + 10}$?
[CDS 2019 (2)]
(a) $\frac{15}{4}$ (b) $\frac{15}{2}$
(c) 1 (d) $\frac{4}{15}$
- 98.** Two numbers p and q are such that the quadratic equation $px^2 + 3x + 2q = 0$ has -6

as the sum and the product of the roots.
What is the value of $(p-q)$?

[CDS 2019 (2)]

- (a) -1 (b) 1
(c) 2 (d) 3

99. If α and β are the roots of the quadratic equation $x^2 + kx - 15 = 0$ such that $\alpha - \beta = 8$, then what is the positive value of k ?

[CDS 2020 (1)]

- (a) 2 (b) 3
(c) 4 (d) 5

100. Students of a class are made to sit in rows of equal number of chairs. If number of students is increased by 2 in each row, then the number of rows decreases by 3. If number of students is increased by 4 in each row, then the number of rows decreases by 5. What is the number of students in the class?

[CDS 2020 (1)]

- (a) 100 (b) 105
(c) 110 (d) 120

101. How many integral values of x and y satisfy the equation $5x + 9y = 7$, where $-500 < x < 500$ and $-500 < y < 500$?

[CDS 2020 (1)]

- (a) 110 (b) 111
(c) 112 (d) None

102. If $x^2 + 9y^2 = 6xy$, then what is $y : x$ equal to?

[CDS 2020 (1)]

- (a) 1 : 3 (b) 1 : 2
(c) 2 : 1 (d) 3 : 1

103. The number of different solutions of the equations $x + y + z = 12$, where each of x , y and z is a positive integer, is:

[CDS 2020 (2)]

- (a) 53 (b) 54
(c) 55 (d) 56

104. A 60 page book has n lines per page. If the number of lines were reduced by 3 in each page, the number of pages would have to be increased by 10 to give the same writing space. What is the value of n ?

[CDS 2020 (2)]

- (a) 18 (b) 21
(c) 24 (d) 30

105. If the equation $4x^2 - 2kx + 3k = 0$ has equal roots, then what are the values of k ?

[CDS 2021 (1)]

- (a) 4, 12 (b) 4, 8
(c) 0, 12 (d) 0, 8

106. If the sum as well as the product of the roots of the equation $px^2 - 6x + q = 0$ is 6, then what is $(p+q)$ equal to?

[CDS 2021 (1)]

- (a) 8 (b) 7
(c) 6 (d) 5

107. Which one of the following fractions will have minimum change in its value if 3 is added to both the numerator and the denominator of all the fractions?

[CDS 2021 (1)]

- (a) $\frac{2}{3}$ (b) $\frac{3}{4}$
(c) $\frac{4}{5}$ (d) $\frac{5}{6}$

108. The sum of the reciprocals of two alternate natural numbers is $\frac{7}{24}$. What is the sum of the numbers?

[CDS 2021 (1)]

- (a) 12 (b) 13
(c) 14 (d) 16

109. Which one of the following equations does not have real roots?

[CDS 2021 (2)]

- (a) $2x^2 + 16x + 3 = 0$ (b) $2x^2 + 10x - 1 = 0$
(c) $x^2 - 8x + 1 = 0$ (d) $4x^2 + 9x + 6 = 0$

110. The sum and the product of the roots of a quadratic equation are 7 and 12 respectively. If the bigger root is halved and the smaller root is doubled, then what is the resulting quadratic equation?

[CDS 2021 (2)]

- (a) $x^2 - 6x + 12 = 0$ (b) $x^2 - 8x + 12 = 0$
(c) $x^2 + 8x + 12 = 0$ (d) $x^2 - 10x + 12 = 0$

111. For which values of k , does the equation $x^2 - kx + 2 = 0$ have real and distinct solutions?

[CDS 2021 (2)]

- (a) $-2\sqrt{2} < k < 2\sqrt{2}$ (b) $k < -2\sqrt{2}$ only
(c) $k > 2\sqrt{2}$ only (d) $k < -2\sqrt{2}$ or $k > 2\sqrt{2}$

112. 5 pencils, 6 notebooks and 7 erasers Cost Rs250; whereas 6 pencils, 4 notebooks and 2 erasers cost Rs180. What is the cost of 2 notebooks and 4 erasers?

[CDS 2021 (2)]

- (a) Rs 90 (b) Rs 75
(c) Rs 60 (d) Rs 40

113. If p and q ($p > q$) are the roots of the equation $x^2 - 60x + 899 = 0$, then which one of the following is correct?

[CDS 2021 (2)]

- (a) $p - q - 1 = 0$ (b) $p - 2q + 27 = 0$
(c) $2p - q - 30 = 0$ (d) $3p - 2q - 43 = 0$

114. If one of the roots of the equation $ax^2 - 4ax + 15 = 0$ is $\frac{3}{2}$, then what is the sum of the squares of the roots?

[CDS 2021 (2)]

- (a) $\frac{15}{2}$ (b) $\frac{17}{2}$
(c) $\frac{19}{2}$ (d) $\frac{21}{2}$

115. The present age of a father is equal to sum of the ages of his 4 children. After ten years the sum of the ages of the children will be 1.6 times the age of their father. What is the present age of father?

[CDS 2021 (2)]

- (a) 36 years (b) 40 years
(c) 42 years (d) 45 years

116. The sum of numerator and denominator of a fraction is 10. If the numerator is increased by 3 and denominator is decreased by 1, the fraction becomes 1. What is the difference between numerator and denominator of the fraction?

[CDS 2021 (2)]

- (a) 2 (b) 3
(c) 4 (d) 5

117. If the system of equation $7x + ky = 27$ and $kx + 7y = 19$ have unique solution, then which one of the following is correct?

[CDS 2021 (2)]

- (a) $k \neq 7$ (b) $k \neq 13$
(c) $k = 7$ (d) $k = 13$

118. A person agrees to work for 30 days, on a condition for every day's work he should receive Rs500, and that for every day's absence from work he should forfeit Rs100. At the end of the time he received Rs 11,400. How many days did he work?

[CDS 2021 (2)]

- (a) 20 (b) 21
(c) 24 (d) 25

119. Consider a question and two statements:
Question:

Is $3x + 2y$ positive

Statement-I: $x^3 = -29.8$

Statement-II: $y^3 = 3x$

Which one of the following is correct in respect of the question and the statements?

[CDS-2022-1]

- (a) Statement-I alone is sufficient to answer the question
(b) Statement-II alone sufficient to answer the question
(c) Both Statements-I & II are together sufficient to answer the question.
(d) Both Statement-I & II are not sufficient to answer the question.

120. Consider a question and two statements:

Question:

Does the equation $ax^2 + bx + c^2 = 0$

Have real roots of opposite sign

Statement-I: The discriminates $D > 0$

Statement-II: $c/a > 0$

Which one of the following is correct in respect of the question and the statements?

[CDS-2022-1]

- (a) Statement-I alone is sufficient to answer the question
(b) Statement-II alone sufficient to answer the question
(c) Both Statements-I & II are together sufficient to answer the question.
(d) Both Statement-I & II are not sufficient to answer the question.

121. How many quadratic equations have the sum of their roots equal to the product of their roots?

[CDS-2022-1]

- (a) Zero (b) One
(c) Two (d) Infinitely many

122. If α and β are the roots of the quadratic equation $x^2 + \alpha x + \beta = 0$, where $\beta \neq 0$, then what is the value of $\alpha - \beta$?

[CDS-2022-1]

- (a) 4 (b) 3
(c) -1 (d) -3

123. If the sum of the roots of the equation $x^2 - k^2x + 30kx - 161x - 64 = 0$ is zero, then what is the difference of the roots?

[CDS-2022-1]

- (a) 15 (b) 16
(c) 17 (d) 18

124. A piece of cloth costs ₹10,000. If a 2 m longer piece of the same cloth is purchased for the same amount, it would cost ₹250

less per meter. What is the original length of the piece of cloth?

[CDS-2022-1]

- (a) 8m (b) 10m
(c) 12m (d) 16m

125. What is the condition that the roots of the equation $ax^2+bx+c=0$ are in the ratio $c : 1$?

[CDS-2022-1]

- (a) $b^2 = a(c + 1)^2$ (b) $a^2 = bc(c + 1)^2$
(c) $b^2 = a(c - 1)^2$ (d) $ab^2 = (c + 1)^2$

126. Two sides of a triangle forming a right angle are $6x^2$ and $(2x^2 - 1)$. If the area of the triangle is 84 square units, then what is the perimeter of the triangle?

[CDS-2022-1]

- (a) 51 units (b) 53 units
(c) 56 units (d) 59 units

127. If squaring a positive real number x is same as adding 12, then what is x equal to?

[CDS-2022-1]

- (a) 2 (b) 3
(c) 4 (d) 5

128. How many minutes are there in x weeks and x days?

[CDS-2022-1]

- (a) $11520x$ (b) $5760x$
(c) $480x$ (d) $192x$

129. What are the values of k for which the polynomial $(k - 3)x^2 - kx - 1$ has no real linear factors?

[CDS 2022 (II)]

- (a) $k < -6$ (b) $-6 < k < 2$
(c) $2 < k < 6$ (d) $k > 6$

130. The value of a 2-digit number is 5 times the sum of the digits. What is the product of the digits?

[CDS 2022 (II)]

- (a) 15 (b) 18
(c) 20 (d) 27

131. If the sum of the squares of the roots of the equation $x^2 - 14x + k = 0$ is 100, then what is the value of k ?

[CDS 2022 (II)]

- (a) 42 (b) 48
(c) 52 (d) 56

132. Consider the following statements in respect of the polynomial $a(b - c)(x - b)(x - c) + b(c - a)(x - c)(x - a) + c(a - b)(x - a)(x - b)$:

1. the coefficient of x^2 is 0

2. The coefficient of x is $(a - b)(b - c)(c - a)$
Which of the statements given above is/are correct?

[CDS - 2023 (1)]

- (a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2

133. For what values of m , is $mx^2 + mx + 8x + 9$ a perfect square?

[CDS - 2023 (1)]

- (a) 1, 4 (b) 4, 9
(c) 9, 16 (d) 4, 16

134. If α and β are the roots of the equation $x^2 - 7x + 1 = 0$, then what is the value of $\alpha^4 + \beta^4$?

[CDS - 2023 (1)]

- (a) 2207 (b) 2247
(c) 2317 (d) 2337

135. The age of Q exceeds the age of P by 3 years. The age of R is twice the age of S. Further, the age difference of R and S is 30 years. What is the sum of the ages of P and Q?

[CDS - 2023 (1)]

- (a) 35 years (b) 38 years
(c) 39 years (d) 45 years

136. If a, b, c are non-zero real numbers such that $a + b + c = 0$, then what are the roots of the equation $ax^2 + bx + c = 0$?

[CDS - 2023 (1)]

- (a) 2, $1 + (c/a)$ (b) 1, a/c
(c) 1, c/a (d) 2, $(c/a) - 1$

137. If the roots of the equation $x^2 - bx + c = 5$ differ by 5, then which one of the following is correct?

[CDS - 2023 (1)]

- (a) $b^2 = 4c + 5$ (b) $c^2 = 4b - 5$
(c) $b^2 + c^2 = 5$ (d) $b^2 - c^2 = 5$

138. The sum of digits of a 2-digit number is 12. When the digits are reversed, the number becomes greater by eighteen. What is the difference between the digits in the number?

[CDS - 2023 (1)]

- (a) 1 (b) 2
(c) 3 (d) 4

139. Question: What are the unique values of a, b and c if 2 is a root of the equation $ax^2 + bx + c = 0$?

Statement I: Ratio of c to a is 1

Statement II: Ratio of b to a is $(-5/2)$

[CDS - 2023 (1)]

(a) choose this option if the question can be answered by one of the statements alone but not by the other.

(b) choose this option if the question can be answered by either statement alone.

(c) Choose this option if the question can be answered by using both the statements together, but cannot be answered by using either statement alone

(d) Choose this option if the question cannot be answered even by using both statements together

- 140.** Let α and β be the roots of the equation $\frac{1}{x+a+b} = \frac{1}{x} + \frac{1}{a} + \frac{1}{b}$; $a \neq 0$, $b \neq 0$, $x \neq 0$. Which one of the following is a quadratic equation whose roots are α^2 and β^2 ?

[CDS-2023 (2)]

(a) $x^2 + (a^2 + b^2)x + a^2b^2 = 0$

(b) $x^2 - (a^2 + b^2)x + a^2b^2 = 0$

(c) $x^2 - (a^2 + b^2)x - a^2b^2 = 0$

(d) $x^2 + (a^2 + b^2)x - a^2b^2 = 0$

- 141.** If $x = \frac{6}{7 - \frac{6}{7 - \frac{6}{7 - \frac{6}{7 - x}}}}$ the value of $x^2 - 3x + 2$

equal to?

[CDS-2023 (2)]

(a) 0

(b) 1

(c) 18

(d) 20

Consider the following the next (05) items that follow:

A, B, C, D, E, F and G are cousins. D is thrice as old as A. Further, C is as many years younger to B, as G to E and E to D. The average age of D and G is 16 years; the average age of A and E is 11 years; the average of B and C is also 11 years. B and C have equal weight. A's weight is 10 kg less than that of B; D is 4 kg heavier than E; E is 4 kg heavier than F; F is 4 kg heavier than G. Further, D has age-weight ratio of 9:20, where age is in years and weight in kg; A has age-weight ratio of 2:5. Moreover, none of them is more than 40kg.

- 142.** What is D's age (in years)?

[CDS-2023 (2)]

(a) 15

(b) 16

(c) 17

(d) 18

- 143.** What is the average age (in years) of B, C, D, E and G?

[CDS-2023 (2)]

(a) 12

(b) 13

(c) 14

(d) 15

- 144.** What is the difference between the weights (in kg) of G and C?

[CDS-2023 (2)]

(a) 4

(b) 3

(c) 2

(d) 1

- 145.** What is the average weight (in kg) of A, B, C, D, E, F and G?

[CDS-2023 (2)]

(a) 201/7

(b) 197/7

(c) 30

(d) 32

- 146.** Consider the following statements:

1. The age of F cannot be determined due to insufficient data

2. The average weight of D and F is equal to weight of E

3. The weight difference is maximum for D and A

Which of the statements given above are correct?

[CDS-2023 (2)]

(a) 1 and 2 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

Consider the following for the next two items that follow:

Mark option (a) if the question can be answered by using one of the statements alone, but cannot be answered using the other statement alone.

Mark option (b) if the question can be answered by using either statements alone.

Mark option (c) if the question can be answered by using both the statements together, but cannot be answered using either statement alone.

Mark option (d) if the question cannot be answered even by using both the statements together.

- 147.** Question: What is the other root of the quadratic equation with real coefficients if one of the roots is $\frac{-4 - \sqrt{10}}{2}$?

Statement I:

The product of the roots is $-3/2 (3 + \sqrt{10})$

Statement II:

The sum of roots of quadratic equation is -1.

148. Question: What is the cost of 15 pens, 21 pencils and 18 note books?

Statement I:

The cost of 7 pens, 6 pencils and 5 note books is Rs.200

Statement II:

The of 3 pens, 8 pencils and 7 note books is Rs.210

149. The combined age of a man and his wife is 6 times the combined age of their children. Two years ago their combined age of their children; and six years later their combined age will be 3 times the combined age of their children. How many children do they have if each child is at least 2 years old?

[CDS-2023 (2)]

- (a) 2 (b) 3
(c) 4 (d) 5

150. How many real roots does the equation $\sqrt{x+9} = x-3$ have?

[CDS-2023 (2)]

- (a) only one (b) only two
(c) only three (d) none

151. If $3^{x-1} + 3^{3-x} = 6$, then what is $2^{x-1} + 2^{3-x}$ equal to?

[CDS-2023 (2)]

- (a) 4 (b) 3
(c) 2 (d) 1

Consider the following for the next two (02) items that follow:

Let $\frac{(x-a)(x-b)}{(x-m)(x-nb)} = \frac{(x+a)(x+b)}{(x+ma)(x+nb)}$; $m, a, b > 0$.

152. What is $\frac{x^2+ab}{x^2+m^2ab}$ equal to?

[CDS-2023 (2)]

- (a) $-\frac{1}{m^2}$ (b) $\frac{1}{m^2}$
(c) $\frac{2}{m}$ (d) $\frac{1}{m}$

153. What is x equal to?

[CDS-2023 (2)]

- (a) $\pm\sqrt{mab}$ (b) $\pm\sqrt{ab}$
(c) $\pm\sqrt{2mab}$ (d) $\pm\sqrt{2ab}$

Consider the following for the next two (02) items that follow:

The total monthly electricity bill for a house consists of the sum of two parts, one part is proportional to number of rooms and the other part is proportional to number of

units consumed. Rs.400 is the monthly electricity bill for the a house with 8 rooms and consuming 240 units and Rs.320 is the monthly electricity bill for a house with 6 rooms and consuming 200 units.

154. What is the monthly electricity bill for a house with m rooms and consuming n units?

[CDS-2023 (2)]

- (a) Rs.(40m + n) (b) Rs.(20m + n)
(c) Rs. $\frac{(40m+n)}{2}$ (d) Rs. $\frac{(30m+n)}{2}$

155. What is the monthly electricity bill for a house with 7 rooms consuming 300 units?

[CDS-2023 (2)]

- (a) Rs.500 (b) Rs.440
(c) Rs.340 (d) Rs.300

Consider the following for the next two (02) items that follow:

A quadratic equation is given by $(a+b+c)x^2 - (2a+2b)x + (a+b-c)=0$; where a, b and c are real and distinct.

156. What are the roots of the equation?

[CDS-2023 (2)]

- (a) $1, \frac{(a+b-c)}{(a+b+c)}$ (b) $1, \frac{(a-b+c)}{(a+b+c)}$
(c) $-1, \frac{(-a-c+c)}{(a+b+c)}$ (d) $-1, \frac{(a+b-c)}{(a+b+c)}$

157. Consider the following statements:

- 1.One of the roots of the equation is always less than 1 if a, b and c are all positive
 - 2.One of two roots of the equation is always negative if a, b and c are all negative.
- Which of the statements given above is/are correct?

[CDS-2023 (2)]

- (a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2

158. If the equation $x \cos \theta = x^2 + p$ has a real solution for every θ where $0 \leq \theta \leq \frac{\pi}{4}$, then which one of the following is correct?

[CDS-2024 (1)]

- (a) $p = 1/8$ (b) $p \leq 1/8$
(c) $p \geq 1/8$ (d) $p \leq 1/4$

159. What is the greatest value of k for which $2x^2 - 4x + k = 0$ has real roots?

[CDS-2024 (1)]

- (a) 1 (b) 2

(c) 3

(d) 4

160. If the sum and product of the roots of a quadratic equation are 2 and -100 respectively, then which one of the following is correct?

[CDS-2024 (1)]

(a) There are infinitely many such equations having differing roots

(b) There is only one such equation which is $x^2 + 2x - 100 = 0$

(c) There is only one such equation which is $x^2 - 2x - 100 = 0$

(d) There is no such equation

ANSWER KEY

1.	a	2.	d	3.	b	4.	d	5.	a	6.	a	7.	a	8.	d	9.	b	10.	d
11.	b	12.	c	13.	b	14.	d	15.	a	16.	a	17.	c	18.	a	19.	a	20.	d
21.	d	22.	c	23.	a	24.	b	25.	a	26.	b	27.	b	28.	c	29.	a	30.	c
31.	a	32.	d	33.	c	34.	c	35.	c	36.	b	37.	d	38.	c	39.	a	40.	c
41.	b	42.	a	43.	a	44.	b	45.	a	46.	a	47.	c	48.	d	49.	b	50.	c
51.	d	52.	b	53.	b	54.	b	55.	c	56.	a	57.	a	58.	c	59.	a	60.	b
61.	d	62.	a	63.	b	64.	d	65.	a	66.	c	67.	c	68.	c	69.	c	70.	a
71.	b	72.	c	73.	a	74.	b	75.	b	76.	c	77.	c	78.	b	79.	b	80.	c
81.	b	82.	b	83.	a	84.	b	85.	d	86.	c	87.	b	88.	d	89.	c	90.	b
91.	a	92.	b	93.	c	94.	b	95.	a	96.	d	97.	d	98.	c	99.	a	100.	d
101.	b	102.	a	103.	c	104.	b	105.	c	106.	b	107.	d	108.	c	109.	d	110.	b
111.	d	112.	b	113.	b	114.	b	115.	b	116.	c	117.	a	118.	c	119.	c	120.	c
121.	d	122.	b	123.	b	124.	a	125.	b	126.	c	127.	c	128.	a	129.	b	130.	c
131.	b	132.	a	133.	d	134.	a	135.	a	136.	c	137.	a	138.	b	139.	b	140.	a
141.	d	142.	d	143.	c	144.	b	145.	a	146.	a	147.	b	148.	c	149.	a	150.	b
151.	a	152.	d	153.	a	154.	b	155.	b	156.	a	157.	a	158.	b	159.	b	160.	c

CAPF PYQ

1. A number is 124 more than its one-third. What is that number?

[CAPF 2022]

(a) 194
(c) 189

(b) 180
(d) 186

2. A test consists of 25 MCQs. Each correct answer gives +4 marks and incorrect answer gives -1 mark. If a candidate scores 74 marks, then how many questions were left un-attempted?

(a) 4
(c) 5

(b) 3
(d) 9

3. A person has a total of 100 coins consisting of ₹2 and ₹5 coins. If the total value of the coins is ₹320, then the number of ₹2 coins is:

[CAPF 2022]

(a) 40
(c) 60

(b) 50
(d) 70

ANSWER KEY

1.	d	2.	a	3.	c														
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PRACTICE SHEET-1

1. If $\frac{2p}{p^2-2p+1} = \frac{1}{4}$, then the value of $p + \frac{1}{p}$ is
 (a) 4 (b) 5
 (c) 10 (d) 12
2. If $x^2 - 3x + 1 = 0$, then the value of $x + \frac{1}{x}$ is
 (a) 0 (b) 1
 (c) 2 (d) 3
3. If $\frac{x^2-x+1}{x^2+x+1} = \frac{2}{3}$ then the value of $x + \frac{1}{x}$ is
 (a) 4 (b) 5
 (c) 6 (d) 8
4. If $2x + \frac{1}{3x} = 5$, find the value of $\frac{5x}{6x^2+20x+1}$
 (a) $\frac{1}{4}$ (b) $\frac{1}{6}$
 (c) $\frac{1}{5}$ (d) $\frac{1}{7}$
5. If $\frac{x}{2x^2+5x+2} = \frac{1}{6}$, then value of $x + \frac{1}{x}$ is
 (a) 2 (b) $\frac{1}{2}$
 (c) $-\frac{1}{2}$ (d) -2
6. If for non-zero x , $x^2 - 4x - 1 = 0$, then the value of $x^2 + \frac{1}{x^2}$ is
 (a) 4 (b) 10
 (c) 12 (d) 18
7. If $x = 3 + \sqrt{8}$, then $x^2 + \frac{1}{x^2}$ is equal to
 (a) 38 (b) 36
 (c) 34 (d) 30
8. If $x^2 - 4x - 1 = 0$, then the value of $x^2 + \frac{1}{x^2} + 3x - \frac{3}{x}$ is
 (a) 0 (b) 10
 (c) 20 (d) 30
9. If $x - \frac{1}{x} = 4$, then $x + \frac{1}{x}$ is equal to
 (a) $5\sqrt{2}$ (b) $2\sqrt{5}$
 (c) $4\sqrt{2}$ (d) $4\sqrt{5}$
10. If $x = 5 + 2\sqrt{6}$, then the value of $\sqrt{x} + \frac{1}{\sqrt{x}}$ is
 (a) $2\sqrt{2}$ (b) $3\sqrt{2}$
 (c) $2\sqrt{3}$ (d) $3\sqrt{3}$
11. For $a > b$, If $a + b = 5$ and $ab = 6$, then the value of $a^2 - b^2$ is
 (a) 1 (b) 3
 (c) 5 (d) 7
12. If $x - \frac{1}{x} = 5$, then $x^2 + \frac{1}{x^2}$ is equal to
 (a) 5 (b) 25
 (c) 27 (d) 23
13. If $x = 3 + 2\sqrt{2}$, then $\sqrt{x} - \frac{1}{\sqrt{x}}$ is equal to
 (a) 1 (b) 2
 (c) $2\sqrt{2}$ (d) $3\sqrt{3}$
14. If $x = \sqrt{3} + \sqrt{2}$, then $x^2 + \frac{1}{x^2}$ is equal to
 (a) 4 (b) 6
 (c) 9 (d) 10
15. If $x + \frac{9}{x} = 6$, then $x^2 + \frac{9}{x^2}$ is equal to
 (a) 8 (b) 9
 (c) 10 (d) 12
16. If $m + \frac{1}{m-2} = 4$, then $(m-2)^2 + \frac{1}{(m-2)^2}$ is equal to
 (a) -2 (b) 0
 (c) 2 (d) 4
17. If $x^2 + x + 1 = 0$, then the value of $x^3 + 1$ is
 (a) 0 (b) 1
 (c) 2 (d) 3
18. If $x + \frac{1}{x} = 1$, then $x^{17} + \frac{1}{x^{17}}$ is equal to
 (a) 0 (b) 1
 (c) 2 (d) 3
19. If $x + \frac{1}{x} = 1$, then $x^{16} + x^{13}$ is equal to

(a) 0
(c) 2

(b) 1
(d) 3

20. If $x + \frac{1}{x} = 1$, then $x^{91} + x^{90} + x^{89} + x^{88} + x^{87} + x^{86}$ is equal to

(a) 0
(c) 2

(b) 1
(d) 3

21. If $x + \frac{1}{x} = \sqrt{3}$, then

$x^{93} + x^{91} + x^{87} + x^{85} + x^{83} + x^{89}$ is equal to

(a) 0
(c) 2

(b) 1
(d) 3

22. If $x^2 + x + 1 = 0$, then

$x^{93} + x^5 + x^4 + 1 = ?$ is equal to

(a) 0
(c) 2

(b) 1
(d) 3

23. If $x + \frac{1}{x} = \sqrt{3}$, then

$x^{30} + x^{24} + x^{18} + x^{12} + x^6$ is equal to

(a) -3
(c) -1

(b) 1
(d) 2

24. What is the value of $\left[\frac{(2.3)^3 - 0.027}{(2.3)^2 + 0.69 + 0.09} \right]$

(a) 2.6
(c) 1.3

(b) 2
(d) 1

25. The value of

$\frac{0.1 \times 0.1 \times 0.1 + 0.2 \times 0.2 \times 0.2 + 0.3 \times 0.3 \times 0.3 - 3 \times 0.1 \times 0.2 \times 0.3}{0.1 \times 0.1 + 0.2 \times 0.2 + 0.3 \times 0.3 - 0.1 \times 0.2 - 0.2 \times 0.3 - 0.3 \times 0.1}$ is

(a) 0.006
(c) 0

(b) 0.6
(d) 0.2

26. The value of $\frac{0.9 \times 0.9 \times 0.9 + 0.2 \times 0.2 \times 0.2 + 0.3 \times 0.3 \times 0.3 - 3 \times 0.9 \times 0.2 \times 0.3}{0.9 \times 0.9 + 0.2 \times 0.2 + 0.3 \times 0.3 - 0.9 \times 0.2 - 0.2 \times 0.3 - 0.3 \times 0.9}$ is

(a) 1.4
(c) 0.8

(b) 0.054
(d) 1.0

27. The value of

$\frac{(2.697 - 0.498)^2 + (2.697 + 0.498)^2}{2.697 \times 2.697 + 0.498 \times 0.498}$ is

(a) 4
(c) 2.199

(b) 2
(d) 3.195

28. The value of

$\frac{0.0347 \times 0.0347 \times 0.0347 + (0.9653)^3}{(0.0347)^2 - 0.0347 \times 0.9653 + (0.9653)^2}$ is

(a) 0.9306
(c) 1.0050

(b) 1.0009
(d) 1

29. The value of

$\frac{(3.2)^3 - 0.008}{(3.2)^2 + 0.64 + 0.04}$ is

(a) 0
(c) 3.208

(b) 2.994
(d) 3

30. $(113^2 + 115^2 + 117^2 - 113 \times 115 - 115 \times 117 - 117 \times 113)$ is equal to

(a) 0
(c) 8

(b) 4
(d) 12

ANSWER KEY

1.	(c)	2.	(d)	3.	(b)	4.	(d)	5.	(b)	6.	(d)	7.	(c)	8.	(d)	9.	(b)	10.	(c)
11.	(c)	12.	(c)	13.	(b)	14.	(d)	15.	(c)	16.	(c)	17.	(c)	18.	(b)	19.	(a)	20.	(a)
21.	(a)	22.	(a)	23.	(c)	24.	(b)	25.	(b)	26.	(a)	27.	(b)	28.	(d)	29.	(d)	30.	(d)

PRACTICE SHEET-2

1. If $\frac{a}{1-a} + \frac{b}{1-b} + \frac{c}{1-c} = 1$ then the value of $\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c}$ is
 (a) 1 (b) 2
 (c) 3 (d) 4
2. If $x + y = 2z$ then the value of $\frac{x}{x-z} + \frac{z}{y-z}$ is
 (a) 1 (b) 2
 (c) $1/2$ (d) 2
3. If $x^2 + y^2 + 2x + 1 = 0$, then the value of $x^{31} + y^{35}$ is
 (a) -1 (b) 0
 (c) 1 (d) 2
4. If a, b, c are real and $a^2 + b^2 + c^2 = 2(a-b-c) - 3$, then the value of $2a - 3b + 4c$ is
 (a) -1 (b) 0
 (c) 1 (d) 2
5. If $(3a + 1)^2 + (b-1)^2 + (2c-3)^2 = 0$, then the value of $(3a + b + 2c)$ is equal to
 (a) 3 (b) -1
 (c) 2 (d) 5
6. The value of expression $\frac{(a-b)^2}{(b-c)(c-a)} + \frac{(b-c)^2}{(a-b)(c-a)} + \frac{(c-a)^2}{(a-b)(b-c)}$
 (a) 0 (b) 3
 (c) $1/3$ (d) 2
7. If $a = \frac{\sqrt{5}+1}{\sqrt{5}-1}$ and $b = \frac{\sqrt{5}-1}{\sqrt{5}+1}$, then the value of $\frac{a^2+ab+b^2}{a^2-ab+b^2}$ is
 (a) $3/4$ (b) $4/3$
 (c) $3/5$ (d) $5/3$
8. If $a + b + c + d = 1$, then what is the maximum value of $(1+a)(1+b)(1+c)(1+d)$ is
 (a) 1 (b) $1/8$
 (c) $27/64$ (d) $625/256$
9. If $x = \frac{\sqrt{3}}{2}$, then $\frac{\sqrt{1+x}}{1+\sqrt{1+x}} + \frac{\sqrt{1-x}}{1-\sqrt{1-x}}$ is equal to
 (a) 1 (b) $2/\sqrt{3}$
 (c) $2 - \sqrt{3}$ (d) 2
10. If $a^2 + b^2 + c^2 + 3 = 2(a-b-c)$, then the value of $2a-b+c$ is
 (a) 3 (b) 4
 (c) 0 (d) 2
11. If $\frac{4x-3}{x} + \frac{4y-3}{y} + \frac{4z-3}{z} = 0$, then the value of $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$ is
 (a) 9 (b) 3
 (c) 4 (d) 6
12. If $\frac{xy}{x+y} = a$, $\frac{xz}{x+z} = b$ and $\frac{yz}{y+z} = c$, where a, b, c are all non-zero numbers, then x is equal to
 (a) $\frac{2abc}{ab+bc-ac}$ (b) $\frac{2abc}{ab+ac-bc}$
 (c) $\frac{2abc}{ac+bc-ab}$ (d) $\frac{2abc}{ab+bc+ac}$
13. If $a + \frac{1}{b} = 1$ and $b + \frac{1}{c} = 1$ then $c + \frac{1}{a}$ is equal to
 (a) 0 (b) $1/2$
 (c) 1 (d) 2
14. If $x^2 + y^2 + \frac{1}{x^2} + \frac{1}{y^2} = 4$, then the value of $x^2 + y^2$ is
 (a) 2 (b) 4
 (c) 8 (d) 16
15. If $x^2 = y + z$, $y^2 = z + x$, $z^2 = x + y$ then the value of $\frac{1}{1+x} + \frac{1}{1+y} + \frac{1}{1+z}$ is
 (a) -1 (b) 1
 (c) 2 (d) 4
16. If $a^2 + b^2 = 2$ and $c^2 + d^2 = 1$, then the value of $(ad-bc)^2 + (ac+bd)^2$ is
 (a) $4/9$ (b) $1/2$

- (c) 1 (b) 2
17. If a, b, c are real and $a^2 + b^2 + c^2 + 3 = 2(a + b + c)$, then the value of $a + b + c$ is
(a) 2 (b) 3
(c) 4 (d) 5
18. If $x = \frac{4ab}{a+b}$ ($a \neq b$), then the value of $\frac{x+2a}{x-2a} + \frac{x+2b}{x-2b}$ is
(a) a (b) b
(c) $2ab$ (d) 2
19. If $a^2 + b^2 + 2b + 4a + 5 = 0$, then the value of $\frac{a-b}{a+b}$ is
(a) 3 (b) -3
(c) $1/3$ (d) $-1/3$
20. If $x - y = \frac{x+y}{7} = \frac{xy}{4}$, the numerical value of xy is
(a) $4/3$ (b) $3/4$
(c) $1/4$ (d) $1/3$
21. If $x + y + z = 0$, then $\frac{x^2}{yz} + \frac{y^2}{zx} + \frac{z^2}{xy} = ?$
(a) $(xyz)^2$ (b) $x^2 + y^2 + z^2$
(c) 9 (d) 3
22. If $a + b + c = 0$, then $\frac{1}{(a+b)(b+c)} + \frac{1}{(a+c)(b+a)} + \frac{1}{(c+a)(c+b)} = ?$
(a) 1 (b) 0
(c) -1 (d) -2
23. If $a + b + c = 0$, then $\frac{a^2 + b^2 + c^2}{a^2 - bc}$ is
(a) 0 (b) 1
(c) 2 (d) 3
24. If $p + q = 10$ and $pq = 5$, then the numerical value of $\frac{p}{q} + \frac{q}{p}$ will be
(a) 16 (b) 20
- (c) 22 (d) 18
25. If $x = 3 + 2\sqrt{2}$ and $xy = 1$, then the value of $\frac{x^2 + 3xy + y^2}{x^2 - 3xy + y^2}$ is
(a) $30/31$ (b) $70/31$
(c) $35/31$ (d) $37/31$
26. If $\frac{x}{b+c} = \frac{y}{c+a} = \frac{z}{a+b}$, then
(a) $\frac{x-y}{b-a} = \frac{y-z}{c-b} = \frac{z-x}{a-c}$
(b) $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$
(c) $\frac{x-y}{a} = \frac{y-z}{b} = \frac{z-x}{c}$
(d) None of these
27. If $a + b + c = 0$, then the value of $\left(\frac{a+b}{c} + \frac{b+c}{a} + \frac{c+a}{b}\right)\left(\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b}\right)$ is
(a) 8 (b) -3
(c) 9 (d) 0
28. If a, b, c are non-zero, $a + \frac{1}{b} = 1$ and $b + \frac{1}{c} = 1$ then abc is equal to
(a) -1 (b) 3
(c) -3 (d) 1
29. If $a + b + c = 2s$, then $\frac{(s-a)^2 + (s-b)^2 + (s-c)^2 + s^2}{a^2 + b^2 + c^2}$ is equal to
(a) $a^2 + b^2 + c^2$ (b) 0
(c) 1 (d) 2
30. If $xy + yz + zx = 0$, then $\frac{1}{x^2 - yz} + \frac{1}{y^2 - zx} + \frac{1}{z^2 - xy} = ?$
(a) 3 (b) 1
(c) $x + y + z$ (d) 0

ANSWER KEY

1.	(d)	2.	(a)	3.	(a)	4.	(c)	5.	(a)	6.	(b)	7.	(b)	8.	(d)	9.	(b)	10.	(d)
11.	(c)	12.	(c)	13.	(c)	14.	(a)	15.	(b)	16.	(d)	17.	(b)	18.	(d)	19.	(c)	20.	(a)
21.	(d)	22.	(b)	23.	(c)	24.	(d)	25.	(d)	26.	(a)	27.	(c)	28.	(a)	29.	(c)	30.	(d)

PRACTICE SHEET-3

- The value of the expression $x^4 - 17x^3 + 17x^2 - 17x + 17$ at $x = 16$ is
(a) 0 (b) 1
(c) 2 (d) 3
- For what value of k , will the expression $3x^3 - kx^2 + 4x + 16$ be divisible by $\left(x - \frac{k}{2}\right)$?
(a) 4 (b) -4
(c) 2 (d) 0
- Which one of the following is the factor of $x^4 + xy^3 + xz^3 + x^3y + y^4 + yz^3$?
(a) $x + y + z$ (b) $x^2 + y^2 + z^2$
(c) $x^3 + y^3 + z^3$ (d) $x^2 + y^2$
- If one of the two factors of an expression which is the difference of two cubes, is $(x^4 + x^2y + y^2)$, then what is the other factor?
(a) $x + y$ (b) $x - y$
(c) $x^2 + y$ (d) $x^2 - y$
- Which one of the following is one of the factors of $x^2(y - z) + y^2(z - x) - z(xy - yz - zx)$?
(a) $x - y$ (b) $x + y - z$
(c) $x - y - z$ (d) $x + y + z$
- If $(3x^3 - 2x^2y - 13xy^2 + 10y^3)$ is divided by $(x - 2y)$, then what is the remainder?
(a) 0 (b) $y + 5$
(c) $y + 1$ (d) $y^2 + 3$
- If $(x + y + z = 0)$, then what is $(x + y)(y + z)(z + x)$ equal to?
(a) $-xyz$ (b) $x^2 + y^2 + z^2$
(c) $x^3 + y^3 + z^3 + 3xyz$ (d) xyz
- If $(x^3 + 5x^2 + 10k)$ leaves remainder $-2x$ when divided by $(x^2 + 2)$, then what is the value of k ?
(a) -2 (b) -1
(c) 1 (d) 2
- If $(5x^2 + 14x + 2)^2 - (4x^2 - 5x + 7)^2$ is divided by $(x^2 + x + 1)$, what is the remainder?
(a) -1 (b) 0
(c) 1 (d) 2
- What is/are the factor(s) of $(x^{29} - x^{24} + x^{13} - 1)$?
(a) Only $(x - 1)$
(b) Only $(x + 1)$
(c) $(x - 1)$ and $(x + 1)$
(d) Neither $(x - 1)$ nor $(x + 1)$
- What is the value of the polynomial $r(x)$, so that $f(x) = g(x)q(x) + r(x)$ and $\deg r(x) < \deg g(x)$, where $f(x) = x^2 + 1$ and $g(x) = x + 1$?
(a) 1 (b) -1
(c) 2 (d) -2
- If a is a rational number such that $(x - a)$ is a factor of the polynomial $x^3 - 3x^2 - 3x + 9$, then
(a) a can be any integer
(b) a is an integer dividing 9
(c) a cannot be an integer
(d) a can take three values
- If $x^2 - 11x + a$ and $x^2 - 14x + 2a$ have a common factor, then what are the values of a ?
(a) 0, 7 (b) 5, 20
(c) 0, 24 (d) 1, 3
- Which one of the following is a factor of $2x^3 - 3x^2 - 11x + 6$?
(a) $x + 1$ (b) $x - 1$
(c) $x + 2$ (d) $x - 2$
- What should be subtracted from $27x^3 - 9x^2 - 6x - 5$ to make it exactly divisible by $(3x - 1)$?
(a) -5 (b) -7
(c) 5 (d) 7

16. What is $x(y-z)(y+z) + y(z-x)(z+x) + z(x-y)(x+y)$ equal to?
 (a) $(x+y)(y+z)(z+x)$
 (b) $(x-y)(x-z)(z-y)$
 (c) $(x+y)(z-y)(x-z)$
 (d) $(y-x)(z-y)(x-z)$
17. If the remainder of the polynomial $a_0 + a_1x + a_2x^2 + \dots + a_nx^n$ when divided by $(x-1)$ is 1, then which one of the following is correct?
 (a) $a_0 + a_2 + \dots = a_1 + a_3 + \dots$
 (b) $a_0 + a_2 + \dots = 1 + a_1 + a_3 + \dots$
 (c) $1 + a_0 + a_2 + \dots = -(a_1 + a_3 + \dots)$
 (d) $1 - a_0 - a_2 - \dots = a_1 + a_3 + \dots$
18. When $(x^3 - 2x^2 + px - q)$ is divided by $(x^2 - 2x - 3)$, the remainder is $(x - 6)$. What are the values of p, q respectively?
 (a) -2, -6 (b) 2, -6
 (c) -2, 6 (d) 2, 6
19. If $x = 1 + \sqrt{2}$, then what is the value of $x^4 - 4x^3 + 4x^2$?
 (a) -1 (b) 0
 (c) 1 (d) 2
20. When $x^{40} + 2$ is divided by $x^4 + 1$, what is the remainder?
 (a) 1 (b) 2
 (c) 3 (d) 4
21. If the expression $px^3 + 3x^2 - 3$ and $2x^3 - 5x + p$ when divided by $x - 4$ leave the same remainder, then what is the value of p ?
 (a) -1 (b) 1
 (c) -2 (d) 2
22. If u, v and w are real numbers such that $u^3 - 8v^3 - 27w^3 = 18uvw$, then which one of the following is correct?
 (a) $u - v + w = 0$ (b) $u = -v = -w$
 (c) $u - 2v = 3w$ (d) $u + 2v = -3w$
23. $x^4 + 4y^4$ is divisible by which one of the following?
 (a) $(x^2 + 2xy + 2y^2)$ (b) $(x^2 + 2y^2)$
 (c) $(x^2 - 2y^2)$ (d) None of these
24. What are the factors of $x^2 + 4y^2 + 4y - 4xy - 2x - 8$?
 (a) $(x - 2y - 4)$ and $(x - 2y + 2)$
 (b) $(x - y + 2)$ and $(x - 4y + 4)$
 (c) $(x - y + 2)$ and $(x - 4y - 4)$
 (d) $(x + 2y - 4)$ and $(x + 2y + 2)$
25. If the expression $(px^3 + x^2 - 2x - q)$ is divisible by $(x - 1)$ and $(x + 1)$, what are the values of p and q , respectively?
 (a) 2, -1 (b) -2, 1
 (c) -2, -1 (d) 2, 1
26. If $x^5 - 9x^2 + 12x - 14$ is divisible by $(x - 3)$, then what is the remainder?
 (a) 0 (b) 1
 (c) 56 (d) 184
27. If $3x^3 - 2x^2y - 13xy^2 + 10y^3$ is divided by $x - 2y$, then what is the remainder?
 (a) 0 (b) x
 (c) $y + 5$ (d) $x - 3$
28. What is the remainder when $(x^{11} + 1)$ is divided by $(x + 1)$?
 (a) 0 (b) 2
 (c) 11 (d) 12
29. If $(x - 3)$ is a factor of $(x^2 + 4px - 11p)$, then what is the value of p ?
 (a) -9 (b) -3
 (c) -1 (d) 1
30. What is the value of k that $(2x - 1)$ may be a factor of $4x^4 - (k - 1)x^3 + kx^2 - 6x + 1$?
 (a) 8 (b) 9
 (c) 12 (d) 13

ANSWER KEY

1.	(b)	2.	(b)	3.	(c)	4.	(d)	5.	(c)	6.	(a)	7.	(a)	8.	(c)	9.	(b)	10.	(a)
11.	(c)	12.	(d)	13.	(c)	14.	(c)	15.	(b)	16.	(b)	17.	(d)	18.	(c)	19.	(c)	20.	(c)
21.	(b)	22.	(c)	23.	(a)	24.	(a)	25.	(d)	26.	(d)	27.	(a)	28.	(a)	29.	(a)	30.	(d)

CDS PYQ

1. The sum of two numbers is 7 and the sum of their squares is 25. The product of the two numbers is:
[CDS 2013 (1)]
(a) 6 (b) 10
(c) 12 (d) 15
2. One of the factors of the polynomial $x^4 - 7x^3 + 5x^2 - 6x + 81$ is:
[CDS 2013 (1)]
(a) $x+2$ (b) $x-2$
(c) $x+3$ (d) $x-3$
3. $(a+1)^4 - a^4$ is divisible by:
[CDS 2013 (1)]
(a) $-2a^2 + 2a - 1$ (b) $2a^3 - 2a - 1$
(c) $2a^3 - 2a + 1$ (d) $2a^2 + 2a + 1$
4. The factor(s) of $5px - 10qy + 2rpx = 4qy$ is/are:
[CDS 2013 (1)]
(a) Only $(5 + 2r)$
(b) Only $(px - 2qy)$
(c) Both $(5 + 2r)$ and $(px - 2qy)$
(d) Neither $(5 + 2r)$ nor $(px - 2qy)$
5. If the expression $x^3 + 3x^2 + 4x + p$ contains $(x + 6)$ as a factor, then the value of p is:
[CDS 2013 (1)]
(a) 132 (b) 141
(c) 144 (d) 151
6. What is the value of :
$$\frac{725 \times 725 \times 725 + 371 \times 371 \times 371}{725 \times 725 - 725 \times 371 + 371 \times 371} ?$$

[CDS 2013 (1)]
(a) 9610 (b) 1960
(c) 1096 (d) 1016
7. If $x + \frac{1}{x} = 2$, then what is value of $x - \frac{1}{x}$?
[CDS 2013 (1)]
(a) 0 (b) 1
(c) 2 (d) -2
8. Consider the following statements:
I. $x+3$ is the factor of $x^3 + 2x^2 + 3x + 8$
II. $x-2$ is the factor of $x^3 + 2x^2 + 3x + 8$
Which of the statements given above is/are correct?
[CDS 2013 (2)]
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
9. What is $\frac{(x^2 + y^2)(x - y) - (x - y)^3}{x^2y - xy^2}$ equal to?
[CDS 2013 (2)]
(a) 1 (b) 2
(c) 4 (d) -2
10. If the expression $x^3 + 3x^2 + 4x + k$ has a factor $x + 5$, then what is the value of k ?
[CDS 2013 (2)]
(a) -70 (b) 70
(c) 48 (d) -48
11. The quantity which must be added to $(1-x)(1+x^2)$ to obtain x^3 is:
[CDS 2013 (2)]
(a) $2x^3 + 3x^2 + x + 1$ (b) $2x^3 + x^2 + x - 1$
(c) $2x^3 - x^2 + x - 1$ (d) $-x^2 + x - 1$
12. If $(49)^2 - (25)^2 = 37x$, then what is x equal to?
[CDS 2013 (2)]
(a) 64 (b) 74
(c) 48 (d) 42
13. For what values of k will:
 $4x^5 + 9x^4 - 7x^3 - 5x^2 - 4kx + 3k^2$ contain $x-1$ as a factor?
[CDS 2013 (2)]
(a) 3, $-\frac{1}{2}$ (b) 3, -1
(c) 0, $\frac{1}{3}$ (d) 1, $\frac{1}{3}$
14. $x(y^2 - z^2) + y(z^2 - x^2) + z(x^2 - y^2)$ is divisible by:
[CDS 2013 (2)]

- (a) Only $(y-z)$
 (b) Only $(z-x)$
 (c) Both $(y-z)$ and $(z-x)$
 (d) Neither $(y-z)$ nor $(z-x)$

15. If $\frac{x}{2} + \frac{y}{3} = 4$ and $\frac{2}{x} + \frac{3}{y} = 1$, then what is $x + y$ equal to?

[CDS 2014 (1)]

- (a) 11 (b) 10
 (c) 9 (d) 8

16. $x^3 + 6x^2 + 11x + 6$ is divisible by:

[CDS 2014 (1)]

- (a) Only $(x+1)$ (b) Only $(x+2)$
 (c) Only $(x+3)$ (d) All of these

17. What should be added to be $x(x+a)$ $(x+2a)$ $(x+3a)$, so that, the sum be a perfect square?

[CDS 2014 (1)]

- (a) $9a^2$ (b) $4a^2$
 (c) a^4 (d) None of these

18. If $3x^4 - 2x^3 + 3x^2 - 2x + 3$ is divided by $(3x + 2)$, then the remainder is:

[CDS 2014 (1)]

- (a) 0 (b) $\frac{185}{27}$
 (c) $\frac{181}{25}$ (d) $\frac{3}{4}$

19. The expression $2x^3 + x^2 - 2x - 1$ is divisible by:

[CDS 2014 (1)]

- (a) $x+2$ (b) $2x+1$
 (c) $x-2$ (d) $2x-1$

20. If $\left(x^2 + \frac{1}{x^2}\right) = \frac{17}{4}$, then what is $\left(x^3 - \frac{1}{x^3}\right)$ equal to?

[CDS 2014 (1)]

- (a) $\frac{75}{16}$ (b) $\frac{63}{8}$
 (c) $\frac{95}{8}$ (d) None of these

21. If $x + y = 5$, $y + z = 10$ and $z + x = 15$, then which one of the following is correct?

[CDS 2014 (1)]

- (a) $x > y > z$ (b) $z < y > x$
 (c) $z > y > x$ (d) $x > z > y$

22. What is the remainder when $x^5 - 5x^2 + 125$ is divided by $x+5$?

[CDS 2014 (2)]

- (a) 0 (b) 125
 (c) -3125 (d) 3125

23. Consider the following statements:

1. $(a-b-c)$ is one of the factor of $3abc + b^3 + c^3 - a^3$
 2. $(b+c-1)$ is one of the factor of $3bc + b^3 + c^3 - 1$

Which of the above statement is/are correct?

[CDS 2014 (2)]

- (a) Only 1 (b) Only 2
 (c) Both 1 and 2 (d) Neither 1 nor 2

24. If $ax + by - 2 = 0$ and $abxy = 1$, where $a \neq 0$, $b \neq 0$, then what is $(a^2x + b^2y)$ equal to?

[CDS 2014 (2)]

- (a) $a + b$ (b) $2ab$
 (c) $a^3 + b^3$ (d) $a^4 + b^4$

25. For what value of k is $(x-5)$ a factor of $x^3 - 3x^2 + kx - 10$?

[CDS 2015 (1)]

- (a) -8 (b) 4
 (c) 2 (d) 1

26. The expression $x^3q^2 - x^3pt + 4x^2pt - 4x^2q^2 + 3xq^2 - 3xpt$ is divisible by:

[CDS 2015 (1)]

- (a) $(x-1)$ only
 (b) $(x-3)$ only
 (c) Both $(x-1)$ and $(x-3)$
 (d) Neither $(x-1)$ nor $(x-3)$

27. If $x + y + z = 0$ then $x^3 + y^3 + z^3 + 3xyz$ is equal to:

[CDS 2015 (1)]

- (a) 0 (b) $6xyz$
 (c) $12xyz$ (d) xyz

28. What are the possible solution for x of the equation $x^{\sqrt{x}} = \sqrt[n]{x^x}$, where n and x are positive.

[CDS 2015 (1)]

- (a) 0, n (b) 1, n^2
(c) n, n (d) 1, n

29. The square root of $\frac{(0.75)^3}{1-0.75} + [0.75 + (0.75)^2 + 1]$
[CDS 2015 (1)]

- (a) 1 (b) 2
(c) 3 (d) 4

30. Out of 532 saving accounts held in a post office, 218 accounts have deposits over Rs 10,000 each. Further, in 302 accounts, the first or sole depositors are men, of which the deposits exceed Rs 10,000 in 102 accounts. In how many accounts the first or sole depositors are women and the deposits are up to Rs 10,000 only?

[CDS 2015 (1)]

- (a) 116
(b) 114
(c) 100
(d) Cannot be determined from the given data

31. Consider the following in respect of the equation $y = \frac{\sqrt{(x-1)^2}}{x-1}$.

1. $y = 1$ if $x > 1$
2. $y = -1$ if $x < 1$
3. y exists for all values of x

Which of the above statements is/are correct?

[CDS 2015 (2)]

- (a) 1 only (b) 2 only
(c) 1 and 2 only (d) 1, 2 and 3

32. If a , b and c satisfy the equation $x^3 - 3x^2 + 2x + 1 = 0$ then what is the value of $\frac{1}{a} + \frac{1}{b} + \frac{1}{c}$?

[CDS 2015 (2)]

- (a) $-\frac{1}{2}$ (b) 2
(c) -2 (d) $\frac{1}{2}$

33. Which one of the following is correct?

[CDS 2015 (2)]

- (a) $(x+2)$ is a factor of $x^4 - 6x^3 + 12x^2 - 24x + 32$
(b) $(x+2)$ is a factor of $x^4 - 6x^3 - 12x^2 + 24x - 32$

- (c) $(x-2)$ is a factor of $x^4 - 6x^3 + 12x^2 - 24x + 32$
(d) $(x-2)$ is a factor of $x^4 + 6x^3 - 12x^2 + 24x - 32$

34. If $x = \frac{91}{216}$, then the value of $3 - \frac{1}{(1-x)^{1/3}}$ is:

[CDS 2015 (2)]

- (a) $\frac{9}{5}$ (b) $\frac{a^2-1}{a} = 5$
(c) $\frac{4}{9}$ (d) $\frac{4}{5}$

35. If $\frac{p}{x} + \frac{q}{y} = m$ and $\frac{q}{x} + \frac{p}{y} = n$, then what is $\frac{x}{y}$ equal to?

[CDS 2016 (1)]

- (a) $\frac{np+mq}{mp+nq}$ (b) $\frac{np+mq}{mp-nq}$
(c) $\frac{np-mq}{mp-nq}$ (d) $\frac{np-mq}{mp+nq}$

36. If $a^2 - by - cz = 0$, $ax - b^2 + cz = 0$ and $ax + by - c^2 = 0$, then the value of $\frac{x}{a+x} + \frac{y}{b+y} + \frac{z}{c+z}$ will be:

[CDS 2016 (1)]

- (a) $a + b + c$ (b) 3
(c) 1 (d) 0

37. If $(s-a) + (s-b) + (s-c) = s$, then the value of $\frac{(s-a)^2 + (s-b)^2 + (s-c)^2 + s^2}{a^2 + b^2 + c^2}$, will be:

[CDS 2016 (1)]

- (a) 3 (b) 1
(c) 0 (d) -1

38. If the polynomial $x^6 + px^5 + qx^4 - x^2 - x - 3$ is divisible by $(x^4 - 1)$, then the value of $p^2 + q^2$ is:

[CDS 2016 (1)]

- (a) 1 (b) 9
(c) 10 (d) 13

39. Let m be a non-zero integer and n be a positive integer. Let R be the remainder obtained on dividing the polynomial $x^n + m^n$ by $(x - m)$. Then:

[CDS 2016 (1)]

- (a) R is a non zero even integer
(b) R is odd, if m is odd
(c) $R = s^2$ for some integer s, if n is even
(d) $R = t^3$ for some integer t, if 3 divides n

40. If $x = \frac{\sqrt{a+2b} + \sqrt{a-2b}}{\sqrt{a+2b} - \sqrt{a-2b}}$ then $bx^2 - ax + b$ is equal to (given that $b \neq 0$):

[CDS 2016 (1)]

- (a) 0 (b) 1
(c) ab (d) 2ab

41. If $a^3 = 117 + b^3$ and $a = 3 + b$, then the value of $a + b$ is (given that $a > 0$ and $b > 0$)

[CDS 2016 (1)]

- (a) 7 (b) 9
(c) 11 (d) 13

42. For what value of k is $(x+1)$ a factor of $x^3 + kx^2 - x + 2$?

[CDS 2016 (1)]

- (a) 4 (b) 3
(c) 1 (d) -2

43. If $\sqrt{\frac{x}{y}} = \frac{10}{3} - \sqrt{\frac{y}{x}}$ and $x - y = 8$, then value of xy is equal to:

[CDS 2016 (1)]

- (a) 36 (b) 24
(c) 16 (d) 9

44. If $x = 2^{\frac{1}{3}} + 2^{-\frac{1}{3}}$, then the value of $2x^3 - 6x - 5$ is equal to:

[CDS 2016 (1)]

- (a) 0 (b) 1
(c) 2 (d) 3

45. If $x^2 = y + z$, $y^2 = z + x$ and $z^2 = x + y$, then what is the value of $\frac{1}{x+1} + \frac{1}{y+1} + \frac{1}{z+1}$?

[CDS 2016 (2)]

- (a) -1 (b) 1
(c) 2 (d) 4

46. If $4x + 3a = 0$, then what is the value of

$$\frac{x^2 + ax + a^2}{x^3 - a^3} - \frac{x^2 - ax + a^2}{x^3 + a^3}?$$

[CDS 2016 (2)]

- (a) $-\frac{4}{7a}$ (b) $\frac{7}{a}$
(c) $-\frac{32}{7a}$ (d) $\frac{24}{7a}$

47. If $2p + 3q = 12$ and $4p^2 + 4pq - 3q^2 = 126$, then what is the value of $p + 2q$?

[CDS 2016 (2)]

- (a) 5 (b) $\frac{21}{4}$
(c) $\frac{25}{4}$ (d) $\frac{99}{16}$

48. What is the square root of $\frac{(0.35)^2 + 0.70 + 1}{2.25} + 0.19$?

[CDS 2017 (1)]

- (a) 1 (b) 2
(c) 3 (d) 4

49. If $x = \frac{\sqrt{a+b} - \sqrt{a-b}}{\sqrt{a+b} + \sqrt{a-b}}$, then what is $bx^2 - 2ax + b$ equal to ($b \neq 0$)?

[CDS 2017 (1)]

- (a) 0 (b) 1
(c) ab (d) 2ab

50. If $x = t^{\frac{1}{t-1}}$ and $y = t^{\frac{t}{t-1}}$, $t > 0$, $t \neq 0$, then what is the relation between x and y?

[CDS 2017 (1)]

- (a) $y^x = x^{1/y}$ (b) $x^{1/y} = y^{1/x}$
(c) $x^y = y^x$ (d) $x^y = y^{1/x}$

51. If $A : B = 3 : 4$, then what is the value of the

expression $\left(\frac{3A^2 + 4B}{3A - 4B^2} \right) ?$

[CDS 2017 (1)]

- (a) $\frac{43}{55}$
(b) $-\frac{43}{55}$
(c) $\frac{47}{55}$
(d) Cannot be determined

52. If $x = 2 + 2^{2/3} + 2^{1/3}$, then what is the value of $x^3 - 6x^2 + 6x$?

[CDS 2017 (1)]

- (a) 3 (b) 2
(c) 1 (d) 0

53. If $\sqrt{\frac{x}{y}} = \frac{24}{5} + \sqrt{\frac{y}{x}}$ and $x + y = 26$, then what is the value of xy ?

[CDS 2017 (1)]

- (a) 5 (b) 15
(c) 25 (d) 30

54. If $a + b = 5$ and $ab = 6$, then what is the value of $a^3 + b^3$?

[CDS 2017 (1)]

- (a) 35 (b) 40
(c) 90 (d) 125

55. What is the value of $\frac{(443+547)^2 + (443-547)^2}{443 \times 443 + 547 \times 547}$?

[CDS 2017 (1)]

- (a) 0 (b) 1
(c) 2 (d) 3

56. If $9^x 3^y = 2187$ and $2^{3x} 2^{2y} - 4^{xy} = 0$, then what can be the value of $(x + y)$?

[CDS 2017 (1)]

- (a) 1 (b) 3
(c) 5 (d) 7

57. What are the factors of $x^3 + 4x^2 - 11x - 30$?

[CDS 2017 (2)]

- (a) $(x-2)$, $(x+3)$ and $(x+5)$
(b) $(x+2)$, $(x+3)$ and $(x-5)$
(c) $(x+2)$, $(x-3)$ and $(x+5)$
(d) $(x+2)$, $(x-3)$ and $(x-5)$

58. If $x = 111\dots1$ (20 digits), $y = 333\dots3$ (10 digits) and $z = 222\dots2$ (10 digits), then what is $\frac{x-y^2}{z}$ equal to?

[CDS 2017 (2)]

- (a) $1/2$ (b) 1
(c) 2 (d) 3

59. If $5x^3 + 5x^2 - 6x + 9$ is divided by $(x + 3)$, then the remainder is:

[CDS 2017 (2)]

- (a) 135 (b) -135
(c) 63 (d) -63

60. The quotient of $8x^3 - y^3$ when divided by $2xy + 4x^2 + y^2$ is:

[CDS 2017 (2)]

- (a) $2x + y$ (b) $x + 2y$
(c) $2x - y$ (d) $4x - y$

61. If $(x + 2)$ is a common factor of $x^2 + ax + b$ and $x^2 + bx + a$, then the ratio $a : b$ is equal to:

[CDS 2017 (2)]

- (a) 1 (b) 2
(c) 3 (d) 4

62. Let $f(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_{n-1} x + a_n$, where $a_0, a_1, a_2, \dots, a_n$ are real numbers.

If $f(x)$ is divided by $(ax-b)$, then the remainder is:

[CDS 2017 (2)]

- (a) $f\left(\frac{b}{a}\right)$ (b) $f\left(-\frac{b}{a}\right)$
(c) $f\left(\frac{a}{b}\right)$ (d) $f\left(-\frac{a}{b}\right)$

63. The product of the polynomials $(x + 2)$, $(x - 2)$, $(x^3 - 2x^2 + 4x - 8)$ and $(x^3 + 2x^2 + 4x + 8)$ is:

[CDS 2017 (2)]

- (a) $x^8 - 256$ (b) $(x^4 - 16)^2$
(c) $(x^4 + 16)^2$ (d) $(x^2 - 4)^4$

64. The factor of $x(x+2)(x+3)(x+5) - 72$ are

[CDS 2017 (2)]

- (a) $x, (x+3)$, $(x+4)$ and $(x-6)$
(b) $(x-1)$, $(x+6)$ and $(x^2 - 2x - 12)$
(c) $(x-1)$, $(x+6)$ and $(x^2 + 5x + 12)$
(d) $(x-1)$, $(x-6)$ and $(x^2 - 5x - 12)$

65. a, b, c, d are non-zero integers such that (ab) divides (cd) . If a and c co-prime, then which one of the following is correct?

[CDS 2017 (2)]

- (a) a is a factor of c
(b) a is a factor of b
(c) a is a factor of d
(d) d is the factor of a

66. If $ab + bc + ca = 0$, then what is the value of

$$\frac{a^2}{a^2 - bc} + \frac{b^2}{b^2 - ca} + \frac{c^2}{c^2 - ab} ?$$

[CDS 2017 (2)]

- (a) 3 (b) 0
(c) 1 (d) -1

67. What is $\frac{(x-y)(x-z)(z-x)}{(x-y)^3 + (y-z)^3 + (z-x)^3}$ equal to?

[CDS 2017 (2)]

- (a) $-\frac{1}{3}$ (b) $\frac{1}{3}$
(c) 3 (d) -3

68. Let $f(x)$ and $g(x)$ be two polynomials (with real coefficients) having degrees 3 and 4 respectively. What is the degree of $f(x)g(x)$?

[CDS 2017 (2)]

- (a) 12 (b) 7
(c) 4 (d) 3

69. The non-zero solution of the equation $\frac{a-x^2}{bx} - \frac{b-x}{c} = \frac{c-x}{b} - \frac{b-x^2}{cx}$, where $b \neq 0$, $c \neq 0$ is:

[CDS 2017 (2)]

- (a) $\frac{b^2 + ac}{b^2 + c^2}$ (b) $\frac{b^2 - ac}{b^2 - c^2}$
(c) $\frac{b^2 - ac}{b^2 + c^2}$ (d) $\frac{b^2 + ac}{b^2 - c^2}$

70. What is the value of $\frac{1}{1+x^{b-a}+x^{c-a}} + \frac{1}{1+x^{a-b}+x^{c-b}} + \frac{1}{1+x^{a-c}+x^{b-c}}$?

[CDS 2018 (1)]

- (a) -1 (b) 0
(c) 1 (d) 3

71. If $a + b = 2c$, then what is the value of

$$\frac{a}{a-c} + \frac{c}{b-c} ?$$

[CDS 2018 (1)]

- (a) -1 (b) 0
(c) 1 (d) 2

72. If $x + y + z = 0$, then what is $(y + z - x)^3 + (z + x - y)^3 + (x + y - z)^3$ equal to?

[CDS 2018 (1)]

- (a) $(x + y + z)^3$
(b) $3(x + y)(y + z)(z + x)$

- (c) $24xyz$
(d) $-24xyz$

73. If $\frac{b}{y} + \frac{z}{c} = 1$ and $\frac{c}{z} + \frac{x}{a} = 1$, then what is $\frac{ab + xy}{bx}$ equal to:

[CDS 2018 (1)]

- (a) 1 (b) 2
(c) 0 (d) -1

74. If $\frac{a^2 - 1}{a} = 5$, then what is the value of $\frac{a^6 - 1}{a^3}$?

[CDS 2018 (1)]

- (a) 125 (b) -125
(c) 140 (d) -140

75. If $(x + 3)$ is a factor of $x^3 + 3x^2 + 4x + k$, then what is the value of k ?

[CDS 2018 (1)]

- (a) 12 (b) 24
(c) 36 (d) 72

76. Which one of the following is a zero of the polynomial $3x^3 + 4x^2 - 7$?

[CDS 2018 (1)]

- (a) 0 (b) 1
(c) 2 (d) -1

77. The remainder when $3x^3 + kx^2 + 5x - 6$ is divided by $(x + 1)$ is -7 . What is the value of k ?

[CDS 2018 (1)]

- (a) -14 (b) 14
(c) -7 (d) 7

78. If $f(x)$ and $g(x)$ are polynomials of degree p and q respectively, then the degree of $\{f(x) \pm g(x)\}$ (if it is non-zero) is:

[CDS 2018 (1)]

- (a) Greater than $\min(p, q)$
(b) Greater than $\max(p, q)$
(c) Less than or equal to $\max(p, q)$
(d) Equal to $\min(p, q)$

79. If $x = y^{1/a}$, $y = z^{1/b}$ and $z = x^{1/c}$ where $x \neq 1$, $y \neq 1$, $z \neq 1$, then what is the value of abc ?

[CDS 2018 (1)]

- (a) -1 (b) 1
(c) 0 (d) 3

80. If $2b = a + c$ and $y^2 = xz$, then what is $x^{b-c} y^{c-a} z^{a-b}$ equal to?

[CDS 2018 (1)]

- (a) 3 (b) 2
(c) 1 (d) -1

81. The sum of a number and its square is 20. Then the number is

[CDS 2018 (1)]

- (a) -5 or 4 (b) 2 or 3
(c) -5 only (d) 5 or -4

82. If $x = y^a$, $y = z^b$ and $z = x^c$, then the value of abc is:

[CDS 2018 (2)]

- (a) 1 (b) 2
(c) -1 (d) 0

83. If $x = 2 + 2^{2/3} + 2^{1/3}$, then the value of the expression $x^3 - 6x^2 + 6x$ will be:

[CDS 2018 (2)]

- (a) 2 (b) 1
(c) 0 (d) -2

84. If $x^6 + \frac{1}{x^6} = k \left(x^2 + \frac{1}{x^2} \right)$, then k is equal to:

[CDS 2018 (2)]

- (a) $\left(x^2 - 1 + \frac{1}{x^2} \right)$ (b) $\left(x^4 - 1 + \frac{1}{x^4} \right)$
(c) $\left(x^4 + 1 + \frac{1}{x^4} \right)$ (d) $\left(x^4 - 1 - \frac{1}{x^4} \right)$

85. If $a^x = b^y = c^z$ and $abc = 1$, then the value of $\frac{1}{x} + \frac{1}{y} + \frac{1}{z}$ will be equal to:

[CDS 2018 (2)]

- (a) -1 (b) 0
(c) 1 (d) 3

86. If $a = xy^{p-1}$, $b = yz^{q-1}$, $c = zx^{r-1}$, then $a^{q-r} b^{r-p} c^{p-q}$ is equal to:

[CDS 2018 (2)]

- (a) abc
(b) xyz
(c) 0
(d) none of the above

87. The remainder when $3x^3 - 2x^2y - 13xy^2 + 10y^3$ is divided by $(x-2y)$ is equal to:

[CDS 2018 (2)]

- (a) zero (b) y
(c) $y-5$ (d) $y+3$

88. If $ab + bc + ca = 0$, then the value of

$$\frac{(b^2 - ca)(c^2 - ab) + (a^2 - bc)(c^2 - ab) + (a^2 - bc)(b^2 - ca)}{(a^2 - bc)(b^2 - ca)(c^2 - ab)}$$

is:

[CDS 2018 (2)]

- (a) -1 (b) 0
(c) 1 (d) 2

89. If α , β and γ are the zeros of the polynomial $f(x) = ax^3 + bx^2 + cx + d$, then $\alpha^2 + \beta^2 + \gamma^2$ is equal to:

[CDS 2018 (2)]

- (a) $\frac{b^2 - ac}{a^2}$ (b) $\frac{b^2 - 2ac}{a}$
(c) $\frac{b^2 - 2ac}{b^2}$ (d) $\frac{b^2 - 2ac}{a^2}$

90. For $x > 0$, what is minimum value of $x + \frac{x+2}{2x}$?

[CDS 2019 (1)]

- (a) 1
(b) 2
(c) $2\frac{1}{2}$
(d) cannot be determined

91. If $\frac{1+px}{1-px} \sqrt{\frac{1-qx}{1+qx}} = 1$, then what are the non-solutions of x ?

[CDS 2019 (1)]

- (a) $\pm \frac{1}{p} \sqrt{\frac{2p-q}{q}}$, $2p \neq q$ (b) $\pm \frac{1}{pq} \sqrt{p-q}$, $p \neq q$
(c) $\pm \frac{p}{q} \sqrt{p-q}$, $p \neq q$ (d) $\pm \frac{q}{p} \sqrt{2p-q}$, $2p \neq q$

92. For $x = \frac{4\sqrt{6}}{\sqrt{2} + \sqrt{3}}$, what is the value of

$$\frac{x + 2\sqrt{2}}{x - 2\sqrt{2}} + \frac{x + 2\sqrt{3}}{x - 2\sqrt{3}} ?$$

[CDS 2019 (1)]

- (a) 1 (b) $\sqrt{2}$
(c) $\sqrt{3}$ (d) 2

93. For any two numbers a and b , $\sqrt{(a-b)^2} + \sqrt{(b-a)^2}$ is:

[CDS 2019 (1)]

- (a) Always zero
- (b) Never zero
- (c) Positive only if $a \neq b$
- (d) Positive if and only if $a > b$

94. If $\frac{a}{b+c} + \frac{b}{c+a} = \frac{c}{a+b}$, then which one of the following statements is correct?

[CDS 2019 (1)]

- (a) Each fraction is equal to 1 or -1
- (b) Each fraction is equal to $1/2$ or 1
- (c) Each fraction is equal to $1/2$ or -1
- (d) Each fraction is equal to $1/2$ only

95. If $3^x = 4^y = 12^z$, then z is equal to:

[CDS 2019 (1)]

- (a) xy
- (b) $x + y$
- (c) $\frac{xy}{x+y}$
- (d) $4x + 3y$

96. Given that the polynomial $(x^2 + ax + b)$ leaves the same remainder when divided by $(x-1)$ or $(x+1)$. What are the values of a and b respectively?

[CDS 2019 (1)]

- (a) 4 and 0
- (b) 0 and 3
- (c) 3 and 0
- (d) 0 and any integer

97. If $a^x = b^y = c^z$ and $b^2 = ac$, then what is $\frac{1}{x} + \frac{1}{z}$ equal to?

[CDS 2019 (1)]

- (a) $\frac{1}{y}$
- (b) $-\frac{1}{y}$
- (c) $\frac{2}{y}$
- (d) $-\frac{2}{y}$

98. What is $\frac{(x-y)^3 + (y-z)^3 + (z-x)^3}{3(x-y)(y-z)(z-x)}$ equal to?

[CDS 2019 (1 & 2)]

- (a) 1
- (b) 0
- (c) $1/3$
- (d) 3

99. For what value of k can the expression $x^3 + kx^2 - 7x + 6$ be resolved into three linear factors?

[CDS 2019 (2)]

- (a) 0
- (b) 1
- (c) 2
- (d) 3

100. The quotient when $x^4 - x^2 + 7x + 5$ is divided by $(x+2)$ is $ax^3 + bx^2 + cx + d$. What are the values of a , b , c and d respectively?

[CDS 2019 (2)]

- (a) 1, -2, 3, 1
- (b) -1, 2, 3, 1
- (c) 1, -2, -3, -1
- (d) -1, 2, -3, -1

101. Let a and b two positive real numbers such that $a\sqrt{a} + b\sqrt{b} = 32$ and $a\sqrt{a} + b\sqrt{b} = 31$. What is the value of $\frac{5(a+b)}{7}$?

[CDS 2019 (2)]

- (a) 5
- (b) 7
- (c) 9
- (d) Cannot be determined

102. If $x = \frac{1+\sqrt{3}}{2}$ and $y = x^3$, the y satisfies which one of the following equations?

[CDS 2019 (2)]

- (a) $8y^2 - 20y - 1 = 0$
- (b) $8y^2 + 20y - 1 = 0$
- (c) $8y^2 - 20y + 1 = 0$
- (d) $8y^2 - 20y + 1 = 0$

103. Give y is inversely proportional to $\sqrt{x}$, and $x = 36$, when $y = 36$. What is the value of x when $y = 54$?

[CDS 2019 (2)]

- (a) 54
- (b) 27
- (c) 16
- (d) 8

104. What are the values of p and q respectively, if $(x-1)$ and $(x+2)$ divide the polynomial $x^3 + 4x^2 + px + q$?

[CDS 2020 (1)]

- (a) 1, -6
- (b) 2, -6
- (c) 1, 6
- (d) 2, 6

105. What is the point on the xy -plane satisfying $5x + 2y = 7xy$ and $10x + 3y = 8xy$?

[CDS 2020 (1)]

- (a) $\left(-1, \frac{1}{6}\right)$
- (b) $\left(\frac{1}{6}, -1\right)$

(c) $\left(1, \frac{1}{6}\right)$

(d) $\left(-\frac{1}{6}, -1\right)$

- 106.** Which one of the following is a set of solutions of the equation $x^{\sqrt{x}} = \sqrt[n]{x^x}$ if n is a positive integer?

[CDS 2020 (1)]

(a) $\{1, n^2\}$

(b) $\{1, \sqrt{n}\}$

(c) $\{1, n\}$

(d) $\{n, n^2\}$

- 107.** If $5^{x+1} - 5^{x-1} = 600$, then what is the value of 10^{2x} ?

[CDS 2020 (1)]

(a) 1

(b) 1000

(c) 100000

(d) 1000000

- 108.** If $f(x)$ is divided by $(x - \alpha)(x - \beta)$ where $\alpha \neq \beta$, then what is the remainder?

[CDS 2020 (1)]

(a) $\frac{(x - \alpha)f(\alpha) - (x - \beta)f(\beta)}{\alpha - \beta}$

(b) $\frac{(x - \alpha)f(\beta) - (x - \beta)f(\alpha)}{\alpha - \beta}$

(c) $\frac{(x - \beta)f(\alpha) - (x - \alpha)f(\beta)}{\alpha - \beta}$

(d) $\frac{(x - \beta)f(\beta) - (x - \alpha)f(\alpha)}{\alpha - \beta}$

- 109.** If the area of a square is $2401x^4 + 196x^2 + 4$, then what is its side length?

[CDS 2020 (1)]

(a) $49x^2 + 3x + 2$

(b) $49x^2 - 3x + 2$

(c) $49x^2 + 2$

(d) $59x^2 + 2$

- 110.** If x varies as yz , then y varies inversely as:

[CDS 2020 (1)]

(a) xz

(b) $\frac{x}{z}$

(c) $\frac{z}{x}$

(d) $\frac{1}{(xz)}$

- 111.** If $\left(x^8 + \frac{1}{x^8}\right) = 47$, what is the value of $\left(x^6 + \frac{1}{x^6}\right)$?

[CDS 2020 (1)]

(a) 36

(b) 27

(c) 18

(d) 9

- 112.** If $(x^2 - 1)$ is a factor of $ax^4 + bx^3 + cx^2 + dx + e$, then which one of the following is correct?

[CDS 2020 (1)]

(a) $a + b + c = d + e$

(b) $a + b + e = c + d$

(c) $b + c + d = a + e$

(d) $a + c + e = b + d$

- 113.** If $I = a^2 + b^2 + c^2$, where a and b are consecutive integers and $c = ab$, then I is:

[CDS 2020 (2)]

(a) An even number and it is not a square of an integer

(b) An odd number and it is not a square of an integer

(c) Square of an even integer

(d) Square of an odd integer

- 114.** $(x^n - a^n)$ is divisible by $(x - a)$, where $x \neq a$, for every:

[CDS 2020 (2)]

(a) Natural number n

(b) Even natural number n only

(c) Odd natural number n only

(d) Prime number n only

- 115.** If x varies as y , then which of the following is/are correct?

1. $x^2 + y^2$ varies as $x^2 - y^2$

2. $\frac{x}{y^2}$ varies inversely as y

3. $\sqrt[n]{x^2y}$ varies as $\sqrt[n]{x^4y^2}$

Select the correct answer using the code given below:

[CDS 2020 (2)]

(a) 1 and 2 only

(b) 2 and 3 only

(c) 3 only

(d) 1, 2 and 3

- 116.** If $ab + xy - xb = 0$ and $bc + yz - cy = 0$, then what is $\frac{x}{a} + \frac{c}{z}$ equal to?

[CDS 2020 (2)]

(a) $\frac{y}{b}$

(b) $\frac{b}{y}$

(c) 1

(d) 0

- 117.** If $(p + 2)(2q - 1) = 2pq - 10$ and $(p - 2)(2q - 1) = 2pq - 10$, then what is pq equal to?

[CDS 2020 (2)]

- (a) -10 (b) -5
(c) 5 (d) 10

- 118.** What is the value of :

$$\frac{a^2 + ac}{a^2c - c^3} - \frac{a^2 - c^2}{a^2c + 2ac^2 + c^3} - \frac{2c}{a^2 - c^2} + \frac{3}{a + c} ?$$

[CDS 2020 (2)]

- (a) 0 (b) 1
(c) $\frac{ac}{a^2 + c^2}$ (d) $\frac{6}{a + c}$

- 119.** What is the square root of $4x^4 + 8x^3 - 4x + 1$?

[CDS 2020 (2)]

- (a) $2x^2 - 2x - 1$ (b) $2x^2 - x - 1$
(c) $2x^2 + 2x + 1$ (d) $2x^2 + 2x - 1$

- 120.** If $\frac{x}{b+c} = \frac{y}{c+a} = \frac{z}{b-a}$, then which one of the following is correct?

[CDS 2020 (2)]

- (a) $x + y + z = 0$ (b) $x - y - z = 0$
(c) $x + y - z = 0$ (d) $x + 2y + 3z = 0$

- 121.** $1 - x - x^n + x^{n+1}$, where n is a natural number, is divisible by:

[CDS 2020 (2)]

- (a) $(1+x)^2$ (b) $(1-x)^2$
(c) $1 - 2x - x^2$ (d) $1 + 2x - x^2$

- 122.** $4x^3 + 12x^2 - x - 3$ is divisible by:

[CDS 2021 (1)]

- (a) $(2x + 1)$ only
(b) $(2x - 1)$ only
(c) Both $(2x + 1)$ and $(2x - 1)$
(d) Neither $(2x + 1)$ nor $(2x - 1)$

- 123.** If $\frac{x}{a} + \frac{y}{b} = a + b$ and $\frac{x}{a^2} + \frac{y}{b^2} = 2$, then what is

$$\frac{x}{a^2} - \frac{y}{b^2} \text{ equal to?}$$

[CDS 2021 (1)]

- (a) -2 (b) -1
(c) 0 (d) 1

- 124.** How many pairs of (x, y) can be chosen from the set $\{2, 3, 6, 8, 9\}$ such that

$$\frac{x}{y} + \frac{y}{x} = 2, \text{ where } x \neq y ?$$

[CDS 2021 (1)]

- (a) Zero (b) One
(c) Two (d) Three

- 125.** How many terms are there in following product?

$$(a_1 + a_2 + a_3)(b_1 + b_2 + b_3 + b_4)(c_1 + c_2 + c_3 + c_4 + c_5)$$

[CDS 2021 (1)]

- (a) 15 (b) 30
(c) 45 (d) 60

- 126.** What is the remainder when $27^{27} - 15^{27}$ is divided by 6?

[CDS 2021 (1)]

- (a) 0 (b) 1
(c) 3 (d) 4

- 127.** If $a + b + c = 0$ then which of the following are correct?

1. $a^3 + b^3 + c^3 = 3abc$
2. $a^2 + b^2 + c^2 = -2(ab + bc + ca)$
3. $a^3 + b^3 + c^3 = -3ab(a + b)$

Select the correct answer using the code given below:

[CDS 2021 (1)]

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

- 128.** If $p = \frac{\sqrt{3q+2} + \sqrt{3q-2}}{\sqrt{3q+2} - \sqrt{3q-2}}$, then what is the value of $p^2 - 3pq + 2$?

[CDS 2021 (1)]

- (a) 0 (b) 1
(c) 2 (d) 3

- 129.** What is the value of x , if: $\frac{b + \sqrt{b^2 - 2bx}}{b - \sqrt{b^2 - 2bx}} = a$?

[CDS 2021 (1)]

- (a) $\frac{ab}{(a+b)}$ (b) $\frac{2ab}{(a+1)}$
(c) $\frac{2ab}{(a+1)^2}$ (d) $\frac{ab}{(a+b)^2}$

130. For how many real values of k is $6kx^2 + 12kx - 24x + 16$ a perfect square for every integer x ?

[CDS 2021 (1)]

- (a) Zero (b) One
(c) Two (d) Four

131. If $x + \frac{1}{x} = \frac{5}{2}$, then what is $x^4 - \frac{1}{x^4}$ equal to?

[CDS 2021 (1)]

- (a) $\frac{195}{16}$ (b) $\frac{255}{16}$
(c) $\frac{625}{16}$ (d) 0

132. The expression $\frac{(x^3 - 1)(x^2 - 9x + 14)}{(x^2 + x + 1)(x^2 - 8x + 7)}$

simplifies to:

[CDS 2021 (1)]

- (a) $(x-1)$ (b) $(x-2)$
(c) $(x-7)$ (d) $(x+2)$

133. What should be added to

$$\frac{1}{(x-2)(x-4)} \text{ to get } \frac{2x-5}{(x^2-5x+6)(x-4)}?$$

[CDS 2021 (1)]

- (a) $\frac{1}{(x^2-7x+12)}$ (b) $\frac{1}{(x^2+7x+12)}$
(c) $\frac{1}{(x^2-7x-12)}$ (d) $\frac{1}{(x^2+7x-12)}$

134. Consider the following statements:

1.If x is directly proportional to z and y is directly proportional to z , then $(x^2 - y^2)$ is directly proportional to z^2 ?

2.If x is inversely proportional to z and y is inversely proportional to z , then (xy) is inversely proportional to z^2 ?

Which of the above statements is/are correct?

[CDS 2021 (1)]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

135. What is $\frac{8x}{1-x^4} - \frac{4x}{x^2+1} + \frac{x+1}{x-1} - \frac{x-1}{x+1}$ equal to?

[CDS 2021 (1)]

- (a) 0 (b) 1
(c) 2 (d) 4

136. If $x(x-1)(x-2)(x-3) + 1 = k^2$, then which one of the following is a possible expression for k ?

[CDS 2021 (1)]

- (a) $x^2 - 3x + 1$ (b) $x^2 - 3x - 1$
(c) $x^2 + 3x - 1$ (d) $x^2 - 2x - 1$

137. What is

$$\frac{1}{bc(a-b)(a-c)} + \frac{1}{ca(b-c)(b-a)} + \frac{1}{ab(c-a)(c-b)}$$

equal to:

[CDS 2021 (1)]

- (a) $a + b + c$ (b) 3
(c) $ab + bc + ca$ (d) 0

138. If n is any natural number, then $5^{2n} - 1$ is always divisible by how many natural numbers?

[CDS 2021 (1)]

- (a) One (b) Four
(c) Six (d) Eight

139. Which one of the following is a factor of the polynomial $(x-1)(x-2)(x-4) - 90$?

[CDS 2021 (2)]

- (a) $x + 14$ (b) $x - 14$
(c) $x - 6$ (d) $x - 7$

140. What is the sum of linear factors (in x and y) of the expression $2x^2 + xy - 3y^2$?

[CDS 2021 (2)]

- (a) $2x-3y$ (b) $3x-2y$
(c) $3x+2y$ (d) $2x+3y$

141. If $\alpha + \beta + \gamma = \alpha\beta + \beta\gamma + \gamma\alpha$, then what is $(1-\alpha)(1-\beta)(1-\gamma)$ equal to?

[CDS 2021 (2)]

- (a) $1-\alpha\beta\gamma$ (b) $1+\alpha\beta\gamma$
(c) $\alpha^2 + \beta^2 + \gamma^2$ (d) $(\alpha-\beta)(\beta-\gamma)(\gamma-\alpha)$

142. If $96 - 64a^3 + \frac{8}{a^6} - \frac{48}{a^3} - t^3 = 0$, then what is $a^2t + 4a^3$ equal to?

[CDS 2021 (2)]

- (a) 0 (b) 1
(c) 2 (d) 3

143. If $A + B = \frac{x^2 - 8}{x + 2}$ and $A - B = \frac{-x^2 + 2x + 4}{x + 2}$, then what is B equal to?

[CDS 2021 (2)]

- (a) $\frac{x^2 - 4}{x^2 + 4x + 4}$ (b) $\frac{x^2 - 4}{x^2 - 4x + 4}$
(c) $\frac{2x^2 - 7x + 3}{2x - 1}$ (d) $\frac{2x^2 + 7x - 3}{2x - 1}$

144. What is $\frac{x^4}{(x^2 - y^2)(x^2 - z^2)} + \frac{y^4}{(y^2 - x^2)(y^2 - z^2)} + \frac{z^4}{(z^2 - x^2)(z^2 - y^2)}$ equal to?

[CDS 2021 (2)]

- (a) -1 (b) 0
(c) 1 (d) $x^2 + y^2 + z^2$

145. If $(2ab - b^2) : (6a^2 - ab) = 1 : 6$, then what is $(a + b) : (a - b)$ equal to?

[CDS 2021 (2)]

- (a) 3 only (b) 5 only
(c) -3 or 3 (d) -5 or 5

146. If $\frac{x}{b + c - a} = \frac{y}{b - c - a} = \frac{z}{a + b - c} = k$, then what is:

$x^2 + y^2 + z^2 - 2xy - 2yz + 2zx$ equal to?

[CDS 2021 (2)]

- (a) $k^2 (a^2 + b^2 + c^2)$ (b) $k^2 (a^2 - b^2 + c^2)$
(c) $k^2 (a + b + c)^2$ (d) $k^2 (a - b + c)^2$

147. Consider the following inequalities:

1. $\frac{a^2 - b^2}{a^2 + b^2} > \frac{a - b}{a + b}$ where $a > b > 0$
2. $\frac{a^3 + b^3}{a^2 + b^2} > \frac{a^2 + b^2}{a + b}$ only when $a > b > 0$

Which of the above is/are correct?

[CDS 2021 (2)]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

148. If $\frac{ay - bx}{c} = \frac{cx - az}{b} = \frac{bz - cy}{a}$, then which of the following is/are correct?

1. $\frac{x}{a} = \frac{y}{b}$
2. $\frac{x + y + z}{a + b + c} = \frac{z}{c}$

Select the correct answer using the code given below:

[CDS 2021 (2)]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

149. If $2x - 3y - 7 = 0$, then what is the value of $8x^3 - 36x^2y + 54xy^2 - 27y^3 - 340$?

[CDS 2021 (2)]

- (a) -1 (b) 0
(c) 1 (d) 3

150. Consider a question and the two statements:

Question:

Is $\frac{x^6 + y^6}{x^4 + y^4}$ always greater than $\frac{x^4 + y^4}{x^2 + y^2}$ ($x \neq y \neq 0$)?

Statement-I: $x > y$

Statement-II: $x^2 + y^2 > 2xy$

Which one of the following is correct in respect of the question and the statements?

[CDS-2022-1]

- (a) Statement-I alone is required to answer the question
(b) Statement-II alone is required to answer the question
(c) Both Statements-I & II are required to answer the question.
(d) Both Statement-I nor statement-II is required to answer the question.

151. If $x + \frac{1}{x} = \frac{5}{2}$ then what is the value of the following? $\frac{5x}{7x^2 - 3x + 7}$

[CDS-2022-1]

- (a) $\frac{3}{7}$ (b) $\frac{5}{12}$
(c) $\frac{3}{14}$ (d) $\frac{10}{29}$

152. If $a + b = 2$, $\frac{1}{a} + \frac{1}{b} = 2$ then what is the value of $a^3 + b^3$?

[CDS-2022-1]

- (a) 2 (b) 4
(c) 6 (d) 8

153. If $43^x + 47^y = (2021)^2$, $x \neq 0$, $y \neq 0$, then what is the value of the following?

$$\frac{4xy + x + y}{2xy - x - y}$$

[CDS-2022-1]

- (a) 5 (b) 15
(c) 25 (d) 45

154. If $\frac{x-y}{x\sqrt{y}+y\sqrt{x}} = \frac{1}{\sqrt{x}}$; ($x > 0$, $y > 0$), then what is the value of $\frac{x}{y}$?

[CDS-2022-1]

- (a) 1 (b) 2
(c) 4 (d) 8

155. If $x = 7 + 4\sqrt{3}$, then what is the value of $\sqrt{x} + \frac{1}{\sqrt{x}}$?

[CDS-2022-1]

- (a) 1 (b) 2
(c) 3 (d) 4

156. What is the value of the following?

$$\frac{(5.4)^3 - 0.064}{(5.4)^2 + 2.16 + 0.16}$$

[CDS-2022-1]

- (a) 4 (b) 4.4
(c) 5 (d) 5.4

157. If $\left(x + \sqrt{1+x^2}\right)\left(y + \sqrt{1+y^2}\right) = 1$, where x and y are real numbers, then what is the value of $(x+y)^2$?

[CDS-2022-1]

- (a) 0 (b) 1
(c) 4 (d) 9

158. If $x = 9999$, then what is the value of the following?

$$\frac{4x^3 - x}{(2x+1)(6x-3)}$$

[CDS-2022-1]

- (a) 1111 (b) 2222
(c) 3333 (d) 6666

159. Consider a question and two statements:
Question: $a^2 + b^2 + c^2 - ab - bc - ca$ (a, b, c are distinct real numbers), always positive?
Statement-I: $a > b > c$
Statement-II: $a + b + c = 0$

Which one of the following is correct in respect of the question and the statements?

[CDS-2022-1]

- (a) Statement-I alone is required to answer the question
(b) Statement-II alone is required to answer the question
(c) Both Statements-I & II are required to answer the question.
(d) Both Statement-I nor statement-II is required to answer the question.

160. If $x^2 = 17x + y$ and $y^2 = x + 17y$, $x \neq y$, then what is the value of $\sqrt{x^2 + y^2 + 1}$?

[CDS-2022-1]

- (a) 17 (b) 19
(c) 23 (d) 27

Consider the following for the next two (02) items that follow:

$$\text{Consider } 6x^2 - 25x + \frac{6}{x^2} + \frac{25}{x} + 12 = 0$$

161. What is one of the possible values of $x - \frac{1}{x}$?

[CDS 2022 (II)]

- (a) $\frac{1}{2}$ (b) $\frac{3}{2}$
(c) 2 (d) $\frac{5}{2}$

162. What is one of the possible values of $x^2 + \frac{1}{x^2}$?

[CDS 2022 (II)]

- (a) 6 (b) $\frac{62}{9}$
(c) 8 (d) $\frac{82}{9}$

163. If $bc + cd = 2bd$ and $a + c = 2b$, then which one of the following is correct?

[CDS 2022 (II)]

- (a) $ab - cd = 0$ (b) $ac - bd = 0$
(c) $ad - bc = 0$ (d) $ad + bc = 0$

164. If $x = b + c$, $y = c + a$, $z = a + b$, then what is $(x + y + z)^3 - 24xyz$ equal to?

[CDS 2022 (II)]

- (a) $a^3 + b^3 + c^3$ (b) $2(a^3 + b^3 + c^3)$
(c) $8(a^3 + b^3 + c^3)$ (d) none of the above

165. If p is the difference between a number and its reciprocal and q is the difference between the square of the same number

and the square of its reciprocal, then what is $p^4 + 4p^2$ equal to?

[CDS 2022 (II)]

- (a) $4q$ (b) $8q$
(c) $4q^2$ (d) q^2

166. If $x = 2 + 2^{1/2}$, then what is the value of $x^4 + 16x^{-4}$?

[CDS 2022 (II)]

- (a) 152 (b) 144
(c) 136 (d) 132

167. Consider the following statements:

- $x^4(y-z) + y^4(z-x) + z^4(x-y)$ is positive if $x > y > z$.
- $x^4(y-z) + y^4(z-x) + z^4(x-y)$ is negative if $x < y < z$.

Which of the above statements is/are correct?

[CDS 2022 (II)]

- (a) only 1 (b) only 2
(c) both 1 and 2 (d) neither 1 nor 2

168. What is $\frac{x^6 - 24x^4 + 144x^2}{(x^2 + 4\sqrt{3}x + 12)(x - 2\sqrt{3})^2}$ equal to?

[CDS 2022 (II)]

- (a) x^2 (b) $x^2 - 2$
(c) $x^2 + 2\sqrt{3}$ (d) $x^2 - 2\sqrt{3}$

169. If $\frac{a+b}{b+c} = \frac{c+d}{d+a}$ where $a \neq c$, then which one of the following is correct?

[CDS 2022 (II)]

- (a) $a + b = c + d$ (b) $a + c = b + d$
(c) $a - b - c + d = 0$ (d) $a + b + c + d = 0$

170. If $(a^3 + b^3)$ is proportional to $(a^2 - b^2)$, then $(a^2 - ab + b^2)$ is proportional to:

[CDS 2022 (II)]

- (a) $(a - b)$ (b) $(a + b)$
(c) $(a + ab + b)$ (d) $(a^3 - b^2)$

171. If $\frac{\sqrt{x+20} + \sqrt{x-1}}{\sqrt{x+20} - \sqrt{x-1}} = \frac{7}{3}$, then what is the value of $\sqrt{(x+20)(x-1)}$?

[CDS 2022 (II)]

- (a) 8 (b) 9
(c) 10 (d) 12

172. Consider the following statements:

- $(ab + bc + ca)$ is a factor of $a^2(b - c)^3 + b^2(c - a)^3 + c^2(a - b)^3$
- $(a + b + c)$ is a factor of $a^2(b - c)^3 + b^2$

$$(c - a)^3 + c^2(a - b)^3$$

Which of the statements given above is/are correct?

[CDS 2022 (II)]

- (a) only 1 (b) only 2
(c) both 1 and 2 (d) neither 1 nor 2

173. What is

$$\frac{1}{x(x-y)(x-z)} + \frac{1}{y(y-z)(y-x)} + \frac{1}{z(z-x)(z-y)}$$

equal to?

[CDS 2022 (II)]

- (a) 0 (b) 1
(c) $\frac{1}{xyz}$ (d) $-\frac{1}{xyz}$

174. How many of the following values of x would satisfy the equation $qx^2 - 2px + q = 0$?

1. $x = \frac{\sqrt{p+q} + \sqrt{p-q}}{\sqrt{p+q} - \sqrt{p-q}}$, where $p > q$

2. $x = \frac{p + \sqrt{p^2 - q^2}}{q}$, where $p > q$

3. $x = \frac{\sqrt{p+q}}{\sqrt{p-q}}$, where $p > q$

Select the correct answer using the code given below:

[CDS 2022 (II)]

- (a) only one value (b) only two values
(c) all three values (d) none

175. Consider the question and two statements given below:

Question: Is $(x^n + y^n)$ divisible by $(x + y)$?

Statement-1: n is a natural number.

Statement-2: n is an even natural number. Which one of the following is correct in respect of the question and the statements?

[CDS 2022 (II)]

- (a) Statement-1 alone is sufficient to answer the question
(b) Statement-2 alone is sufficient to answer the question
(c) Both statement-1 and statement-2 are sufficient to answer the question
(d) Both statement-1 and statement-2 are not sufficient to answer the question

176. Consider the question and two statements given below:

Let x and y be two real numbers.

Question: Is $xy > 0$?

Statement-1: $x^8 y^9 < 0$.

Statement-2: $x^9 y^{10} < 0$

Which one of the following is correct in respect of the question and the statements?

[CDS 2022 (II)]

- (a) Statement-1 alone is sufficient to answer the question
 (b) Statement-2 alone is sufficient to answer the question
 (c) Both statement-1 and statement-2 are sufficient to answer the question
 (d) Both statement-1 and statement-2 are not sufficient to answer the question

177. Consider the following statements in respect of the polynomial $1 - x - x^n + x^{n+1}$, where n is a natural number:

1. It is divisible by $1 - 2x + x^2$

2. It is divisible by $1 - x^n$

Which of the statements given above is/are correct?

[CDS - 2023 (1)]

- (a) 1 only (b) 2 only
 (c) both 1 and 2 (d) neither 1 nor 2

178. If $x = a + b + \frac{(a-b)^2}{4a+4b}$ and $y = \frac{a+b}{4} + \frac{ab}{a+b}$, then what is the value of $(x-a)^2 - (y-b)^2$?

[CDS - 2023 (1)]

- (a) a^2 (b) b^2
 (c) ab (d) a^2b^2

179. If $x = \frac{\sqrt{3}+1}{\sqrt{3}-1}$ and $y = \frac{\sqrt{3}-1}{\sqrt{3}+1}$, then what is the value $x^3 - y^3$?

[CDS - 2023 (1)]

- (a) 60 (b) $45\sqrt{3}$
 (c) $30\sqrt{3}$ (d) 90

180. If $p = \frac{a^2}{(b-c)(c-a)}$, $q = \frac{b^2}{(c-b)(a-b)}$, $r = \frac{c^2}{(a-c)(b-c)}$, then what is $(p+q+r)^2$ equal to?

[CDS-2023 (2)]

- (a) 9 (b) 4
 (c) 1 (d) 0

181. Which one of the following is a factor of $a^2 - b^2 - c^2 + 2bc + a + b - c$?

[CDS-2023 (2)]

- (a) $a + b + c + 1$ (b) $a - b - c + 1$
 (c) $a + b + c - 1$ (d) $a - b + c + 1$

182. If $2s = a + b + c$, then what is $s^2 + (s-a)(s-b) + (s-b)(s-c) + (s-c)(s-a)$ equal to?

[CDS-2023 (2)]

- (a) $(a+b+c)^2$ (b) $ab + bc + ca$

- (c) $2(ab+bc+ca)$ (d) $3(ab+bc+ca)$

183. The cube root of x varies inversely as the square root of y . $x = 8$ when $y = 3$. What is the value of x when $y = \sqrt[3]{3}$?

[CDS-2023 (2)]

- (a) 18 (b) 21
 (c) 24 (d) 27

184. What is $\frac{x^2 - y^2 - z^2 - 2yz}{x^2 + y^2 - z^2 + 2xy} + \frac{x^2 - y^2 - z^2 - 2yz}{x^2 - y^2 + z^2 - 2xz}$ equal to?

[CDS-2023 (2)]

- (a) $\frac{x}{x+y-z}$ (b) $\frac{y+z}{x+y-z}$
 (c) $\frac{2x}{x+y-z}$ (d) $\frac{2y+2z}{x+y-z}$

185. If a, b, c, x, y, z are real numbers such that $(a+b+c)^2 - 3(ab+bc+ca) + 3(x^2+y^2+z^2) = 0$, then which one of the following is correct?

[CDS-2023 (2)]

- (a) $a = b = c, x = y = z \neq 0$
 (b) $a = b = c = 0, x = y = z$
 (c) $a = b = c, x = y = z = 0$
 (d) $a \neq b \neq c, x = y = z = 0$

Consider the following for the next two items that follow:

Mark option (a) if the question can be answered by using one of the statements alone, but cannot be answered using the other statement alone.

Mark option (b) if the question can be answered by using either statements alone.

Mark option (c) if the question can be answered by using both the statements together, but cannot be answered using either statement alone.

Mark option (d) if the question cannot be answered even by using both the statements together.

186. Question: Are x, y, z equal, where x, y, z are real numbers?

Statement I:

$$x^2 + y^2 + z^2 - xy - yz - zx = 0$$

Statement II:

$$x^3 + y^3 + z^3 - 3xyz = 0$$

187. Question: What is the ratio $x : y : z$ equal to if $x, y, z \neq 0$?

Statement I:

$$\frac{x+z}{y} = \frac{z}{x}$$

Statement II:

$$\frac{z-y}{x} = \frac{x}{z}$$

- 188.** If $a^2 - bc = \alpha$, $b^2 - ac = \beta$, $c^2 - ab = \gamma$, then what is $\frac{a\alpha + b\beta + c\gamma}{(a+b+c)(\alpha+\beta+\gamma)}$ equal to?

[CDS-2023 (2)]

- (a) $a + b - c$ (b) $a - b + c$
(c) $-a + b + c$ (d) 1

- 189.** If $(x - 1)^3$ is a factor of $x^4 + \alpha x^3 + \beta x^2 + \gamma x - 1$, then the other factor will be:

[CDS-2023 (2)]

- (a) $x + 1$ (b) $x - 3$
(c) $x + 2$ (d) x

- 190.** If $\frac{2a}{3} = \frac{4b}{5} = \frac{3c}{4}$, then what is the value of $\frac{18}{a}\sqrt{a^2 + c^2 - b^2}$?

[CDS-2023 (2)]

- (a) $3\sqrt{5}$ (b) $\sqrt{355}$
(c) $\sqrt{375}$ (d) $3\sqrt{15}$

- 191.** Consider the following statements:

1.If $(a+b)$ is directly proportional to $(a-b)$, then (a^2+b^2) is directly proportional to ab
2.If a is directly proportional to b , then (a^2-b^2) is directly proportional to ab .
Which of the statements given above is/are correct?

[CDS-2023 (2)]

- (a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2

- 192.** If $(3a + 6b + c + 2d) \times (3a - 6b - c + 2d) = (3a - 6b + c - 2d) \times (3a - c - 2d)$, then which one of the following is correct?

[CDS-2023 (2)]

- (a) $ab = cd$ (b) $ac = bd$
(c) $ad = bc$ (d) $ad + bc = 0$

- 193.** If $x = 97 + 56\sqrt{3}$, then what is the value of $\sqrt[4]{x} + \frac{1}{\sqrt[4]{x}}$?

[CDS-2023 (2)]

- (a) 7 (b) 6
(c) 5 (d) 4

- 194.** If $x\left(a - b + \frac{ab}{a-b}\right) = y\left(a + b - \frac{ab}{a+b}\right)$ and $x+y=2a^3$, then what is $x-y$ equal to?

[CDS-2023 (2)]

- (a) $-2b^3$ (b) $-2ab^3$
(c) $2b^3$ (d) $2ab^3$

- 195.** Which one of the following is a factor of $3\sqrt{3}x^3 + 2\sqrt{2}y^3 - 18xy + 6\sqrt{6}$?

[CDS-2023 (2)]

- (a) $\sqrt{3}x + \sqrt{2}y - \sqrt{3}$
(b) $\sqrt{3}x + \sqrt{2}y - \sqrt{6}$
(c) $3x^2 + 2y^2 - \sqrt{18}x - \sqrt{12}y - \sqrt{6}xy + 6$
(d) $3x^2 + 2y^2 + \sqrt{18}x + \sqrt{12}y - \sqrt{6}xy + 6$

- 196.** If $x^n - py^n + qz^n$ is divisible by $x^2 + aby - bzx - axy$, then what is $\frac{p}{a^n} - \frac{q}{b^n}$ equal to?

[CDS-2023 (2)]

- (a) -1 (b) 0
(c) 1 (d) 2

Consider the following for the next two (02) items that follow:

Let $p = x^4 - y^2z^2$, $q = y^4 - z^2x^2$, $r = z^4 - x^2y^2$

- 197.** What is $px^2 + qy^2 + rz^2$ equal to?

[CDS-2023 (2)]

- (a) $(x^2 + y^2 + z^2)(p+q+r)$ (c) $-(x^2 + y^2 + z^2)(p+q+r)$
(c) $(y^2 + z^2 + x^2)(p-q-r)$ (d) $(x^2 + y^2 - z^2)(p-q-r)$

- 198.** What is $x^2(px^2 + qy^2 + rz^2) + qr - p^2$ equal to?

[CDS-2023 (2)]

- (a) 0 (b) 1
(c) $p+q+r$ (d) $x^2 + y^2 + z^2$

- 199.** Let a , b and c be the sides of a triangle ABC
Question: Is the triangle equilateral?

Statement-I: $a^2 + b^2 + c^2 = (ab + bc + ca)$

Statement-II: $3a^2 + 3b^2 + 4c^2 = 2ab + 4bc + 4ca$

[CDS-2024 (1)]

(a) If the Question can be answered by one of the Statements alone, but not by the other.

(b) If the Question can be answered by either Statement alone

(c) If the Question can be answered by using both the Statements together, but cannot be answered by using either Statement alone.

(d) If the Question can be answered even by using both Statements together

- 200.** If P^2 varies as R and Q^2 varies as R , ($P \neq Q$), then which of the following are correct?

1. $P^2 + Q^2$ varies as R
2. PQ varies as R
3. $P^2 - Q^2$ varies as R

Select the correct answer using the code given below:

[CDS-2024 (1)]

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

201. What is the minimum value of

$$\left(\frac{a^2 + 3a + 1}{a}\right)\left(\frac{b^2 + 3b + 1}{b}\right) \text{ for } a, b > 0?$$

[CDS-2024 (1)]

- (a) 1 (b) 9
(c) 16 (d) 25

202. Consider the following in respect of the polynomial $x^{4k} + x^{4k+2} + x^{4k+4} + x^{4k+6}$;

1. The remainder is zero when the polynomial is divided by $x^2 + 1$

2. The remainder is zero when the polynomial is divided by $x^4 + 1$

Which of the statements given above is/are correct?

[CDS-2024 (1)]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

203. If $p = \sqrt[3]{a + \sqrt{a^2 + b^3}} + \sqrt[3]{a - \sqrt{a^2 + b^3}}$, then what is $p^3 + 3bp$ equal to?

[CDS-2024 (1)]

- (a) $-2a$ (b) a
(c) $2a$ (d) $3a$

204. If 2 is a zero of the polynomial

$p(x) = x^3 + 3x^2 - 6x - a$, then what is the sum of the squares of the other zeroes of the polynomial?

[CDS-2024 (1)]

- (a) 10 (b) 17
(c) 21 (d) 37

205. Suppose $p(x) = x^4 + a_3x^3 + a_2x^2 + a_1x + a_0$ and $q(x) = x^4 + b_3x^3 + b_2x^2 + b_1x + b_0$ are the polynomials. If $\alpha, \beta, \gamma, \delta$ are zeros of $p(x)$ and $\alpha, \beta, \gamma, \lambda$ are zeros of $q(x)$, then what is

$$\frac{p(x) - q(x)}{(x - \alpha)(x - \beta)(x - \gamma)}$$
 equal to?

[CDS-2024 (1)]

- (a) $-\lambda + \delta$ (b) $\lambda - \delta$
(c) $\lambda + \delta$ (d) $-\lambda - \delta$

ANSWER KEY

1.	c	2.	d	3.	d	4.	c	5.	a	6.	c	7.	a	8.	d	9.	b	10.	b
11.	c	12.	c	13.	d	14.	c	15.	b	16.	d	17.	c	18.	b	19.	b	20.	b
21.	a	22.	c	23.	c	24.	a	25.	a	26.	c	27.	b	28.	d	29.	d	30.	b
31.	c	32.	c	33.	c	34.	a	35.	c	36.	c	37.	b	38.	c	39.	c	40.	a
41.	a	42.	d	43.	d	44.	a	45.	b	46.	c	47.	a	48.	a	49.	a	50.	c
51.	d	52.	b	53.	c	54.	a	55.	c	56.	c	57.	c	58.	b	59.	d	60.	c
61.	a	62.	a	63.	b	64.	c	65.	c	66.	c	67.	d	68.	a	69.	b	70.	c
71.	c	72.	d	73.	a	74.	c	75.	a	76.	b	77.	d	78.	a	79.	b	80.	c
81.	a	82.	a	83.	a	84.	b	85.	b	86.	d	87.	a	88.	b	89.	d	90.	c
91.	a	92.	d	93.	c	94.	c	95.	c	96.	d	97.	c	98.	a	99.	a	100.	a
101.	c	102.	a	103.	c	104.	a	105.	b	106.	b	107.	d	108.	c	109.	c	110.	c
111.	c	112.	d	113.	d	114.	a	115.	b	116.	c	117.	c	118.	d	119.	d	120.	b
121.	b	122.	c	123.	c	124.	a	125.	d	126.	a	127.	c	128.	b	129.	c	130.	c
131.	b	132.	b	133.	a	134.	c	135.	a	136.	a	137.	d	138.	d	139.	d	140.	c
141.	a	142.	c	143.	c	144.	c	145.	d	146.	c	147.	c	148.	c	149.	d	150.	d
151.	d	152.	a	153.	a	154.	c	155.	d	156.	c	157.	a	158.	c	159.	a	160.	a
161.	b	162.	d	163.	c	164.	c	165.	d	166.	c	167.	c	168.	a	169.	d	170.	a
171.	c	172.	a	173.	c	174.	b	175.	b	176.	d	177.	c	178.	b	179.	c	180.	c
181.	d	182.	b	183.	c	184.	c	185.	b	186.	a	187.	c	188.	d	189.	a	190.	b
191.	c	192.	c	193.	d	194.	c	195.	c	196.	b	197.	a	198.	a	199.	b	200.	d
201.	d	202.	c	203.	c	204.	b	205.	b										

PRACTICE SHEET

1. What is the HCF of polynomials $x^3+3x^2y+2xy^2$ and $x^4+6x^3y+8x^2y^2$?
 (1) $x(x+2y)$ (2) $x(x+3y)$
 (3) $(x+2y)$ (4) None of these
2. If the HCF of x^3+mx^2-x+2m and x^2+mx-2 is a linear polynomial, then what is the value of m ?
 (1) 1 (2) 2
 (3) 3 (4) 4
3. What is the HCF of $(x^2+bx-x-b)$ and $x^2+x(a-1)-a$?
 (1) $x+b(b)$ (2) $x+a$
 (3) $x+1(d)$ (4) $x-1$
4. If the HCF of x^2+x-12 and $2x^2-kx-9$ is $x-k$, then what is the value of k ?
 (1) -3 (2) 3
 (3) -4 (4) 4
5. What is the LCM of $(6x^3 + 60x^2+150x)$ and $(3x^4+12x^3-15x^2)$?
 (1) $6x^2(x+5)^2(x-1)$ (2) $3x^2(x+5)^2(x-1)$
 (3) $6x^2(x+5)^2(x-1)^2$ (4) $3x^2(x+5)(x-1)^2$
6. What is the LCM of $(x+2)^2(x-2)$ and $x^2-4x-12$?
 (1) $(x+2)(x-2)$
 (2) $(x+2)^3(x-2)(x-6)$
 (3) $(x+2)(x-2)^2$
 (4) $(x+2)2(x-2)(x-6)$
7. What is the value of k for which the HCF of $2x^2+kx-12$ and $x^2+x-2k-2$ is $(x+4)$?
 (1) 5 (2) 7
 (3) 10 (4) -4
8. What is the LCM of $(x^2-y^2-z^2-2yz)$, $(x^2-y^2+z^2+2xz)$ and $(x^2+y^2-z^2-2xy)$?
 (a) $(x+y+z)(x+y-z)(x-y+z)$
 (b) $(x+y+z)(x-y-z)(x-y+z)$
 (c) $(x+y+z)(x+y-z)(x-y-z)$
 (d) $(x+y-z)(x-y-z)(x-y+z)$
9. If $(x+2)$ is the HCF of x^2+ax+b and x^2+cx+d then which one of the following is correct?
 (1) $a+c=b+d$ (2) $2a+b=2c+d$
 (3) $b+2c=2a+d$ (4) $b-2c=2a-d$
10. What are the values of c when the HCF of x^3+cx^2-x+2c and x^2+cx-2 over the rational is a linear polynomial?
 (1) ± 1 (2) ± 2
 (3) ± 3 (4) ± 4
11. If the HCF of x^3-27 and $x^3+4x^2+12x+k$ is a quadratic polynomial, then what is the value of k ?
 (1) 27 (2) 9
 (3) 3 (d) -3
12. What is the LCM of $3(a^3-b^3)$ and $11(a^4-b^4)$?
 (1) $33(a^3-b^3)(a^2+b^2)(a+b)$
 (2) $(a^3-b^3)(a^2+b^2)(a-b)$
 (3) $33(a-b)(a^2+b^2-ab)(a^3-b^3)$
 (4) $33(a^3-b^3)(a^4-b^4)$
13. I. The HCF of $x+y$ and $x^{10}-y^{10}$ is $x+y$.
 II. The HCF of $x+y$ and $x^{10}+y^{10}$ is $x+y$.
 III. The HCF of $x-y$ and $x^{10}+y^{10}$ is $x-y$.
 IV. The HCF of $x-y$ and $x^{10}-y^{10}$ is $x-y$.
 Which of the statement given above are correct?
 (1) I and II (2) II and III
 (3) I and IV (4) II and IV
14. If $f(x)$ and $g(x)$ are two polynomials with integral co-efficient which vanish at $x=1/2$, then what is the factor of HCF of $f(x)$ and $g(x)$?
 (1) $x-1$ (2) $x-2$
 (3) $2x-1$ (4) $2x+1$
15. What is the HCF of $4x^3+3x^2y-9xy^2+2y^3$ and $x^2+xy-2y^2$?
 (1) $x-2y$ (2) $x-y$
 (3) $(x+2y)(x-y)$ (4) $(x-2y)(x-y)$
16. The HCF of two polynomials $p(x)$ and $q(x)$ is $2x(x+2)$ and LCM is $24x(x+2)^2(x-2)$. If $p(x)=8x^3+32x^2+32x$, then what is $q(x)$ equal to?
 (1) $4x^3-16x$ (2) $6x^3-24x$
 (3) $12x^3+24x$ (4) $12x^3-24x$

17. What is the HCF of $(x^4 - x^2 - 6)$ and $(x^4 - 4x^2 + 3)$?
 (1) $x^2 - 3$ (2) $x + 2$
 (3) $x + 3$ (4) $x^2 + 3$
18. What is the LCM of $a^3b - ab^3$, $a^3b^2 + a^2b^3$ and $ab(a + b)$?
 (1) $a^2b^2(a^2 - b^2)$ (2) $ab(a^2 - b^2)$
 (3) $a^2b^2 + ab^3$ (4) $a^3b^3(a^2 - b^2)$
19. What is the HCF of the polynomials $x^4 - 3x + 2$, $x^3 - 3x^2 + 3x - 1$ and $x^4 - 1$?
 (1) $x - 1$ (2) $x + 1$
 (3) $x^2 - 1$ (4) None of these
20. What is the HCF of $(x^2 - x - 6)$ and $(x^2 - 4x + 3)$?
 (1) $x^2 - 3$ (2) $x + 2$
 (3) $x - 3$ (4) $x^2 + 3$

ANSWER KEY

1.	(1)	2.	(1)	3.	(4)	4.	(2)	5.	(1)	6.	(4)	7.	(1)	8.	(2)	9.	(3)	10.	(1)
11.	(2)	12.	(1)	13.	(3)	14.	(3)	15.	(3)	16.	(2)	17.	(1)	18.	(1)	19.	(1)	20.	(3)

CDS PYQ

1. What is the HCF of the polynomials x^3+8 , x^2+5x+6 and x^3+2x^2+4x+8 ?
[CDS 2013(I)]
 (a) $x+2$ (b) $x+3$
 (c) $(x+2)^2$ (d) None of these
2. The LCM of (x^3-x^2-2x) and (x^3+x^2) is
[CDS 2013(I)]
 (a) x^3-x^2-2x (b) x^2+x
 (c) $x^4-x^3-2x^2$ (d) $x-2$
3. The HCF of (x^4-y^4) and (x^6-y^6) is
[CDS 2013(I)]
 (a) x^2-y^2 (b) $x-y$
 (c) x^3-y^3 (d) x^4-y^4
4. What is the LCM of x^2+2x-8 , x^3-4x^2+4x and x^2+4x ?
[CDS 2013(II)]
 (a) $x(x+4)(x-2)^2$ (b) $x(x+4)(x-2)$
 (c) $x(x+4)(x+2)^2$ (d) $x(x+4)^2(x-2)$
5. What is the HCF of $a^2b^4 + 2a^2b^2$ and $(ab)^7-4a^2b^9$?
[CDS 2013(II)]
 (a) ab (b) a^2b^3
 (c) a^2b^2 (d) a^3b^3
6. What is the HCF of $8(x^5-x^3+x)$ and $28(x^6+1)$?
[CDS 2014(I)]
 (a) $4(x^4-x^2+1)$ (b) $2(x^4-x^2+1)$
 (c) (x^4-x^2+1) (d) None of these
7. What is the lowest common multiple of $ab(x^2 + 1) + x(a^2 + b^2)$ and $ab(x^2 - 1) + x(a^2 - b^2)$?
[CDS 2014(I)]
 (a) $(a^2x^2 - b^2)(a + bx)$
 (b) $(a^2x^2 - b^2)(a + bx)^2$
 (c) $(a^2x^2 - b^2)(a - bx)$
 (d) $(a^2x^2 - b^2)(a - bx)^2$
8. What is the highest common factor of $2x^3 + x^2 - x - 2$ and $3x^3 - 2x^2 + x - 2$?
[CDS 2014 (2)]
 (a) $x-1$ (b) $x + 1$
 (c) $2x+1$ (d) $2x-1$
9. The HCF and LCM of two polynomials are $(x+y)$ and $(3x^5+5x^4y+2x^3y^2-3x^2y^3-5xy^4-2y^5)$ respectively. If one root of the polynomial is (x^2-y^2) , then the other polynomial is
[CDS 2015(I)]
 (a) $3x^4-8x^3y+10x^2y^2+7xy^3-2y^4$
 (b) $3x^4-8x^3y-10x^2y^2+7xy^3+2y^4$
 (c) $3x^4+8x^3y+10x^2y^2+7xy^3+2y^4$
 (d) $3x^4+8x^3y-10x^2y^2+7xy^3+2y^4$
10. If $(x+1)$ is the HCF of Ax^2+Bx+C and Bx^2+Ax+C where $A \neq B$, then the value of C is
[CDS 2015(II)]
 (a) A (b) B
 (c) $A-B$ (d) 0
11. The sum and difference of two expressions are $5x^2-x-4$ and $x^2+9x-10$ respectively. The HCF of the two expression will be
[CDS 2016(I)]
 (a) $(x+1)$ (b) $(x-1)$
 (c) $(3x+7)$ (d) $(2x-3)$

12. What is the LCM of x^3+8 , x^2+5x+6 and x^3+4x^2+4x ?

[CDS 2017(I)]

- (a) $x(x+2)^2(x+3)(x^2-2x+4)$
(b) $x(x-2)^2(x-3)(x^2+2x+4)$
(c) $(x+2)^2(x+3)(x^2-2x+4)$
(d) $(x-2)^2(x-3)(x^2-2x+4)$

13. The product of two non zero expressions is $(x+y+z)p^3$. If their LCM is

[CDS 2017(II)]

- (a) $(x+y+z)$ (b) $(x+y+z)p^2$
(c) $(x+y+z)p^5$ (d) $(x+y+z)p$

14. If the HCF of polynomials $f(x) = (x-1)(x^2+3x+a)$ and $g(x) = (x+2)(x^2+2x+b)$ is (x^2+x-2) , then what are the values of a and b respectively?

[CDS 2017(II)]

- (a) 2, 2 (b) 2, -3
(c) -1, -3 (d) -2, -1

15. HCF and LCM of two polynomials are $(x+3)$ and $(x^3-9x^2-x+105)$ respectively. If one of the two polynomials is $x^2-4x-21$, then the other is

[CDS 2018(II)]

- (a) $x^2-2x-21$ (b) $x^2+2x+15$
(c) $x^2-2x-15$ (d) x^2-x-15

16. What is the LCM of the polynomials x^3+3x^2+3x+1 , x^3+5x^2+5x+4 and x^2+5x+4 ?

[CDS 2019(II)]

- (a) $(x+1)^3(x+4)(x^2+x+1)$
(b) $(x+4)(x^2+x+1)$
(c) $(x+1)(x^2+x+1)$
(d) $(x+1)^2(x+4)(x^2+x+1)$

17. If $(x+k)$ is the HCF of x^2+5x+6 and $x^2+8x+15$, then what is the value of k

[CDS 2020(I)]

- (a) 5 (b) 3
(c) 2 (d) 1

18. What is the HCF of the polynomials $x^6-3x^4+3x^2-1$ and x^3+3x^2+3x+1

[CDS 2020(II)]

- (a) $(x+1)$ (b) $(x+1)^2$
(c) x^2+1 (d) $(x+1)^3$

19. HCF and LCM of two polynomials are $(3x+1)$ and $(30x^3+7x^2-10x-3)$ respectively. If one of the two polynomials is $6x^2+5x+1$, then the other is

[CDS 2020(II)]

- (a) $15x^2+4x+3$ (b) $15x^2+4x-3$

- (c) $15x^2-4x+3$ (d) $15x^2-4x-3$

20. If $(x-k)$ is the HCF of x^2+ax+b and x^2+cx+d , then what is the value of k?

[CDS 2021(I)]

- (a) $\frac{d-b}{c-a}$ (b) $\frac{d-b}{a-c}$
(c) $\frac{d+b}{c+a}$ (d) $\frac{d-b}{c+a}$

21. What is the HCF of $x^3-19x+30$ and x^2-5x+6 ?

[CDS 2021(II)]

- (a) $(x+2)(x-3)$ (b) $(x-2)(x+3)$
(c) $(x+2)(x-1)$ (d) $(x-2)(x-3)$

22. What is the HCF of $(x^8 - y^8)$ and $(x^7 - y^7 + x^5y^2 - x^2y^5)$?

[CDS 2022 (II)]

- (a) $(x^2 + y^2)$
(b) $(x^2 - y^2)$
(c) $(x^3 - y^3 - x^2y + xy^2)$
(d) $(x^3 - y^3 + x^2y - xy^2)$

23. If $(x + k)$ is the HCF of $x^2 + px + q$ and $x^2 + qx + p$, where $p \neq q$, then what is the value of k?

[CDS 2022 (II)]

- (a) -1 (b) 0
(c) $\frac{1}{2}$ (d) 1

24. Let $p(x) = x^4 + x^2 + 1$, $q(x) = x^4 - 2x^3 + 3x^2 - 2x + 1$. If GCD of $p(x)$ and $q(x)$ is $x^2 - x + 1$, then what is their LCM?

[CDS 2022 (II)]

- (a) $(x^2 + x + 1)(x^2 - x + 1)^2$
(b) $(x^4 + x^2 + 1)^2(x^2 - x + 1)$
(c) $(x^4 + x^2 + 1)(x^2 + x + 1)^2$
(d) $(x^4 + x^2 + 1)(x^2 - x + 1)^2$

25. What is the HFC of $acx^3 + bcx^2 + adx^2 + acdx + bdx + bcd$ and $adx^3 + acx^3 + bdx^2 + bcx + acdx + bcd$ if HCF $(c, d)=1$, $c \neq d$?

[CDS-2023 (2)]

- (a) $bx + c$ (b) $cx + d$
(d) $ax + d$ (d) $ax + b$

26. What is the LCM of $x^4 + x^2y^2 + y^4$, $x^3 + y^3$, $x^3 - y^3$?

[CDS-2023 (2)]

- (a) $(x^2 - y^2)(x^4 + x^2y^2 + y^4)^2$
(b) $(x^2 - y^2)(x^4 + 2x^2y^2 + y^4)$
(c) $(x^6 - y^6)$
(d) $(x^6 - y^6)$

27. What is the HCF of $x^4 - 13x^2y^2 - 300y^4$, $x^3 - 4x^2y - 4xy^2 - 5y^3$ and $x^3 - 125y^3$?

(a) $x - 5y$ (b) $x + 5y$ [CDS-2024 (1)] (c) $x^2 + 5xy + 25y^2$ (d) 1

ANSWER KEY

1.	(a)	2.	(c)	3.	(a)	4.	(a)	5.	(c)	6.	(a)	7.	(a)	8.	(a)	9.	(c)	10.	(d)
11.	(b)	12.	(a)	13.	(b)	14.	(b)	15.	(c)	16.	(a)	17.	(b)	18.	(d)	19.	(d)	20.	(b)
21.	(d)	22.	(a)	23.	(a)	24.	(a)	25.	d	26.	(c)	27.	(a)						

PRACTICE SHEET-1

1. If $n(A) = 115$, $n(B) = 326$, $n(A - B) = 47$, then what is $n(A \cup B)$ equal to?
 (1) 373 (2) 165
 (3) 370 (4) 394
2. If $P(A)$ denotes the power set of A and A is the void set, then what is number of elements in $P\{P\{P(A)\}\}$?
 (1) 0 (2) 1
 (3) 4 (4) 16
3. If $N_a = \{ax \mid x \in N\}$, then what is $N_{12} \cap N_8$ equal to?
 (1) N_{12} (2) N_{20}
 (3) N_{24} (4) N_{48}
4. If $X = \{4^n - 3n - 1 \mid n \in N\}$ and $Y = \{9(n - 1) \mid n \in N\}$, then $X \cup Y$ equals to
 (1) X (2) Y
 (3) N (4) A null set
5. What is the number of proper subsets of a given finite set with n elements?
 (1) $2n - 1$ (2) $2n - 2$
 (3) $2^n - 1$ (4) $2^n - 2$
6. If A , B and C are three finite sets, then $[(A \cup B) \cap C]'$ equals to
 (1) $A' \cup B' \cap C'$ (2) $A' \cap B' \cap C'$
 (3) $A' \cap B' \cup C'$ (4) $A \cap B \cap C$
7. If A and B are subsets of a set X , then what is the value of $\{A \cap (X - B)\} \cup B$?
 (1) $A \cup B$ (2) $A \cap B$
 (3) A (4) B
8. Which one of the following is correct?
 (1) $A \cup P(A) = P(A)$
 (2) $A \cap P(A) = A$
 (3) $A - P(A) = A$
 (4) $P(A) - \{A\} = P(A)$
 Here, $P(A)$ denotes the power set of a set A .
9. If A , B and C are three sets, then $A - (B - C)$ equals to
 (a) $A - (B \cap C)$ (b) $(A - B) \cup C$
 (c) $(A - B) \cup (A \cap C)$ (d) $(A - B) \cup (A - C)$
10. Let $S = \{2, 4, 6, \dots, 20\}$, what is the number of subsets does S have?
 (a) 10 (b) 20
 (c) 512 (d) 1024
11. If A , B and C are subsets of a given set, then which one of the following is relations is
 (a) $A \cup (A \cap B) = A \cup B$
 (b) $A \cap (A \cup B) = A$
 (c) $(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$
 (d) $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$
12. If a set A contains 3 elements and another set B contains 6 elements, then what is the minimum number of elements that $(A \cup B)$ can have?
 (a) 3 (b) 6
 (c) 8 (d) 9
13. If $A = \{x: 0 \leq x \leq 2\}$ and $B = \{y: y \text{ is a prime number}\}$, then what is $A \cap B$ equal to?
 (a) $\emptyset$ (b) $\{1\}$
 (c) $\{2\}$ (d) $\{1, 2\}$
14. If $A = \{x : x \text{ is a multiple of } 2\}$, $B = \{x : x \text{ is a multiple of } 5\}$ and $C = \{x : x \text{ is a multiple of } 10\}$, then $A \cap \{B \cap C\}$ is equal to:
 (a) A
 (b) B
 (c) C
 (d) $\{x : x \text{ is a multiple of } 100\}$
15. If $S = \{x : x^2 + 1 = 0, x \text{ is real}\}$, then S is:
 (a) $\{-1\}$ (b) $\{0\}$
 (c) $\{1\}$ (d) An empty set
16. Consider the following in respect of sets A and B :
 1. $(A - B) \cup B = A$
 2. $(A - B) \cup A = A$
 3. $(A - B) \cap B = \phi$
 4. $A \subseteq B \Rightarrow A \cup B = B$
 Which of the above are correct?
 (1) 1, 2 and 3 (2) 2, 3 and 4
 (3) 1, 3 and 4 (4) 1, 2 and 4
17. Let S be a set of all distinct numbers of the form $\frac{p}{q}$ where $p, q \in (1, 2, 3, 4, 5, 6)$. What is the cardinality of the set S ?
 (1) 21 (2) 23
 (3) 32 (d) 36
18. If $A = \{x : x \text{ is a multiple of } 3\}$ and $B = \{x : x \text{ is a multiple of } 4\}$ and $C = \{x : x \text{ is a multiple of } 12\}$

- of 12}, then which one of the following is null set?
 (1) $(A \setminus B) \cup C$ (2) $(A \setminus B) \setminus C$
 (3) $(A \cap B) \cap C$ (4) $(A \cap B) \setminus C$
19. Which one of the following is a null set?
 (a) $\{0\}$
 (b) $\{\{\}\}$
 (c) $\{\{\}\}$
 (d) $\{x \mid x^2 + 1 = 0, x \in \mathbb{R}\}$
20. If $A = \{1, 3, 5, 7\}$, then what is the cardinality of the power set $P(A)$?
 (1) 8 (2) 15
 (3) 16 (4) 17
21. If A and B are any two sets, then what is the value of $A \cap (A \cup B)$?
 (a) Complement of A
 (b) Complement of B
 (c) B
 (d) A
22. Let $A = \{x : x \text{ is a square of a natural number and } x \text{ is less than } 100\}$ and B is a set of even natural numbers. What is the cardinality of $A \cap B$?
 (a) 4 (b) 5
 (c) 9 (d) None of these
23. Let $U = \{x \in \mathbb{N} : 1 \leq x \leq 10\}$ be the universal set, N being the set of natural numbers. If $A = \{1, 2, 3, 4\}$ and $B = \{2, 3, 6, 10\}$, then what is the complement of $(A - B)$?
 (a) $\{6, 10\}$
 (b) $\{1, 4\}$
 (c) $\{2, 3, 5, 6, 7, 8, 9, 10\}$
 (d) $\{5, 6, 7, 8, 9, 10\}$
24. Let $A = \{x : x \text{ is a square of a natural number and } x \text{ is less than } 100\}$ and B is set of even natural numbers. What is the cardinality of $A \cap B$?
 (a) 4 (b) 5
 (c) 9 (d) None of these
25. Let $U = \{x \in \mathbb{N} : 1 \leq x \leq 10\}$ be the universal set, N being the set of natural numbers. If $A = \{1, 2, 3, 4\}$ and $B = \{2, 3, 6, 10\}$, then what is the complement of $(A - B)$?
 (a) $\{6, 10\}$
 (b) $\{1, 4\}$
 (c) $\{2, 3, 5, 6, 7, 8, 9, 10\}$
 (d) $\{5, 6, 7, 8, 9, 10\}$
26. For a set A , consider the following statements
 I. $A \cup P(A) = P(A)$
 II. $\{A\} \cap P(A) = A$
 III. $P(A) - \{A\} = P(A)$
 Where P denotes power set.
 Which of the statements given above is/are not correct?
 (1) I only (2) II only
 (3) III only (4) I, II and III
27. If $A = \{a, b, c\}$, then what is the number of proper subsets of A ?
 (a) 5 (b) 6
 (c) 7 (d) 8
28. If $A = \{4n + 2 \mid n \text{ is natural number}\}$ and $B = \{3n \mid n \text{ is a natural number}\}$, then what is $(A \cap B)$ equal to?
 (1) $\{12n^2 + 6n \mid n \text{ is a natural number}\}$
 (2) $\{24n - 12 \mid n \text{ is a natural number}\}$
 (3) $\{12n^2 - 6n \mid n \text{ is a natural number}\}$
 (4) $\{12n - 6 \mid n \text{ is a natural number}\}$
29. If the cardinality of a set A is 4 and that of a set B is 3, then what is the cardinality of the set $A \Delta B$?
 (a) 1
 (b) 5
 (c) 7
 (d) Cannot be determined as the sets A and B are not given.
30. If $A = P\{1, 2\}$, where P denotes the power set, then which one of the following is correct?
 (a) $\{1, 2\} \subset A$ (b) $1 \in A$
 (c) $\phi \notin A$ (d) $\{1, 2\} \in A$

ANSWER KEY

1.	(a)	2.	(d)	3.	(c)	4.	(b)	5.	(c)	6.	(c)	7.	(a)	8.	(c)	9.	(c)	10.	(d)
11.	(b)	12.	(b)	13.	(c)	14.	(c)	15.	(d)	16.	(b)	17.	(b)	18.	(d)	19.	(d)	20.	(c)
21.	(d)	22.	(a)	23.	(c)	24.	(a)	25.	(c)	26.	(d)	27.	(c)	28.	(d)	29.	(d)	30.	(d)

PRACTICE SHEET-2

1. In an examination 45% of the students pass in English, 35% pass in Hindi and 20% pass in both. What percent of students fail in both English and Hindi?
(a) 10% (b) 20%
(c) 25% (d) 40%
 2. In an examination 60% of the students pass in English, 70% pass in Hindi and 40% pass in both. What percent of students fail in both English and Hindi?
(a) 10% (b) 20%
(c) 25% (d) 30%
 3. In an examination 70% of the candidates passed in English 80% passed in Mathematics 10% failed in both the subjects. If 144 candidates passed in both, the total number of candidates were:
(a) 125 (b) 200
(c) 240 (d) 375
 4. In an examination, 52% students failed in Hindi and 42% in English. If 17% failed in both the subjects, what percentage of students passed in both the subjects?
(a) 38% (b) 33%
(c) 23% (d) 18%
 5. In a group of students, 70% can speak English and 65% can speak Hindi. If 27% of the students can speak none of the two languages, then what percent of the group can speak both the languages.
(a) 38% (b) 62%
(c) 28% (d) 23%
 6. In an examination, 35% of the candidates failed in Mathematics and 25% in English. If 10% failed in both Mathematics and English, then how much percent of candidates passed in both the subjects?
(a) 50% (b) 55%
(c) 57% (d) 60%
 7. In an examination, 93% of students passed and 259 failed. The total number of students appearing at the examination was
(a) 3700 (b) 3850
(c) 3950 (d) 4200
 8. 90% of the students in a school passed in English, 85% passed in Mathematics and 150 students passed in both the subjects. If no student failed in both the subjects, find the total number of students.
(a) 120 (b) 220
(c) 200 (d) 300
 9. In an examination 75% candidates passed in English and 60% passed in Mathematics. 25% failed in both and 240 passed the examination. Find the total number of candidates.
(a) 492 (b) 300
(c) 500 (d) 400
 10. In an examination, 35% of total students failed in Hindi, 45% failed in English and 20% failed in both. Find the percentage of those students who passed in both the subjects?
(a) 45% (b) 35%
(c) 20% (d) 40%
- Direction (for next five):** The Students of a class are offered three languages (Hindi, English and French). 15 students learn all the three language whereas 28 students do not learn any language. The number of students learning Hindi and English but not French is twice the number of students learning Hindi and French but not English. The number of students learning English and French but not Hindi is thrice the number of students learning Hindi and French but not English. 23 students learn only Hindi and 17 students learn only English. The total number of students learning French is 46 and the total number of students learning only French is 11.
11. How many students learn precisely two languages?
(a) 55 (b) 40
(c) 30 (d) 13
 12. How many students learn at least two language?
(a) 15 (b) 30
(c) 45 (d) 55

13. What is the total strength of the class?

- (a) 124 (b) 100
(c) 96 (d) 66

14. How many students learn English and French?

- (a) 30 (b) 43
(c) 45 (d) 73

15. How many students learn at least one language?

- (a) 45 (b) 51
(c) 96 (d) None of these

Direction (for next three): Read the following information carefully to answer the questions that follow.

In a survey of 25 students, it was found that 15 have taken Mathematics, 12 have taken Physics and 11 taken Chemistry, 5 have taken Mathematics and Chemistry, 9 have taken Mathematics and Physics, 4 have taken Physics and Chemistry and 3 have taken all the three subjects.

16. The number of students who have taken only Physics, is

- (a) 2 (b) 3 (c) 5 (d) 6

17. The number of students who have taken only two subjects, is

- (a) 7 (b) 8
(c) 9 (d) 10

18. Consider the following statements

I. The number of students who have taken only one subject is equal to the number of students who have taken only two subjects.
II. The number of students who have taken at least two subjects is four times the number of students who have taken all the three subjects.

Which of the above statement(s) is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

Direction for (next two): In a school, all the students play at least one of the games - chess, carom and table tennis. 60 play chess, 50 play table tennis, 48 play carom, 12 play chess and carom, 15 play carom and table tennis, 20 play table tennis and chess.

19. What can be the minimum number of students in the school?

- (a) 123 (b) 111
(c) 95 (d) 63

20. What can be the maximum number of students in the school?

- (a) 111 (b) 123
(c) 125 (d) 135

ANSWER KEY

1.	(d)	2.	(a)	3.	(c)	4.	(c)	5.	(b)	6.	(a)	7.	(a)	8.	(c)	9.	(d)	10.	(d)
11.	(c)	12.	(c)	13.	(a)	14.	(a)	15.	(c)	16.	(a)	17.	(c)	18.	(b)	19.	(b)	20.	(b)

CDS PYQ

1. In a school there are 30 teachers who teach Mathematics or Physics. Of these teachers, 20 teach Mathematics and 15 teach Physics, 5 teach both Mathematics and Physics. The number of teachers teaching only Mathematics is
[CDS 2013(I)]
(a) 5 (b) 10
(c) 15 (d) 20
2. If $A = \{x : x \text{ is an odd integer}\}$ and $B = \{x : x^2 - 8x + 15 = 0\}$. Then, which one of the following is correct?
[CDS 2013(I)]
(a) $A = B$ (b) $A \subseteq B$
(c) $B \subseteq A$ (d) $A \subseteq B^c$
3. Consider the following in respect of the sets A and B .
I. $(A \cap B) \subseteq A$
II. $(A \cap B) \subseteq B$
III. $A \subseteq (A \cup B)$
Which of the above are correct?
[CDS 2013(I)]
(a) I and II (b) II and III
(c) I and III (d) I, II and III
4. The set of natural numbers is closed under
I. addition
II. subtraction
III. multiplication
IV. division
Which of the above is/are correct?
[CDS 2013(I)]
(a) Only I (b) Both I and III
(c) I, II and III (d) Both III and IV
5. In a class of 110 students, x students take both Mathematics and Statistics, $2x + 20$ students take Mathematics and $2x + 30$ students take Statistics. There are no students who take neither Mathematics nor Statistics. What is x equal to?
[CDS 2013(II)]
(a) 15 (b) 20
(c) 25 (d) 30
6. Out of 105 students taking an examination English and Mathematics, 80 students pass in English, 75 students pass in Mathematics 10 students fail in both the subjects. How many students fail in only one subject?
[CDS 2014(I)]
(a) 26 (b) 30
(c) 35 (d) 45
7. If A and B are any two non-empty subsets of a set E , then what is $A \cup (A \cap B)$ equal to?
[CDS 2014(I)]
(a) $A \cap B$ (b) $A \cup B$
(c) A (d) B
8. If A is a non-empty subset of a set E , then what is $E \cup (A \cap \phi) - (A - \phi)$ equal to?
[CDS 2014(I)]
(a) A
(b) Complement of A
(c) ϕ
(d) E
9. Let A and B be finite non-empty sets with the number of elements in $A = m$ and number of elements in $B = n$. Let $m > n$. If for some integer $k \geq 1$, the number of non-empty subsets of $A = 2^k +$ the number of non-empty subsets of B , then which one of the following is correct?
[CDS 2015(I)]
(a) $m = n + 2$
(b) $m = n + 1$
(c) $m = n + p$ for some odd prime number p
(d) $m = n + t$ for some composite number t
10. Let A denote the set of quadrilaterals having two diagonals equal and bisecting each other. Let B denote the set of quadrilaterals having diagonals bisecting each other at 90° . Then $A \cap B$ denotes
[CDS 2015(II)]
(a) the set of parallelograms
(b) the set of rhombuses
(c) the set of squares
(d) the set of rectangles
11. In a gathering of 100 people, 70 of them can speak Hindi, 60 can speak English and 30 can speak French. Further, 30 of them can speak both Hindi and English. 20 can speak both Hindi and French. If x is the number of people who can speak both English and French, then which one of the following is correct? (Assume that everyone can speak at least one of the three languages)
[CDS 2016(I)]

- (a) $9 < x \leq 30$ (b) $0 \leq x < 8$
(c) $x = 9$ (d) $x = 8$

12. A is a set of positive integers such that when divided by 2, 3, 4, 5 and 6 leaves the remainder 1, 2, 3, 4 and 5 respectively. How many integers between 0 and 100 belong to the set A?

[CDS 2016(II)]

- (a) No integer (b) One
(c) Two (d) Three

13. In the quadratic equation $x^2 + ax + b = 0$, a and b can take any value from the set {1, 2, 3, 4}. How many pairs of values of a and b are possible in order that the quadratic equation has real roots?

[CDS 2016(II)]

- (a) 6 (b) 7
(c) 8 (d) 16

14. Let $A = \{7, 8, 9, 10, 11, 12\}$ and $B = \{7, 10, 14, 15\}$. What is the number of elements in $(A-B)$ and $(B-A)$ respectively?

[CDS 2016(II)]

- (a) 2 and 4 (b) 4 and 2
(c) 2 and 2 (d) 4 and 4

15. Let S be a set of first ten natural numbers. What is the possible number of pairs (a, b) where $a, b \in S$ and $a \neq b$ such that the product ab (>12) leaves remainder 4 when divided by 12?

[CDS 2016(II)]

- (a) 4 (b) 6
(c) 8 (d) 10

16. If $A = \{x : x \text{ is a multiple of } 7\}$, $B = \{x : x \text{ is a multiple of } 5\}$ and $C = \{x : x \text{ is a multiple of } 35\}$, then which one of the following is a null set?

[CDS 2017(I)]

- (a) $(A - B) \cup C$ (b) $(A - B) - C$
(c) $(A \cap B) \cap C$ (d) $(A \cap B) - C$

17. Which one of the following is a correct statement?

[CDS 2017(II)]

- (a) $\{x : x + 5 = 5\} = \phi$
(b) $\{x : x + 5 = 5\} = \{0\}$
(c) $\{x : x + 5 = 5\} = 0$
(d) $\{x : x + 5 = 5\} = \{\phi\}$

18. In an examination, 35% students failed in Hindi, 45% students failed in English and 20% students failed in both the subjects.

What is the percentage of students passing in both the subjects?

[CDS 2017(II)]

- (a) 0 (b) 20
(c) 30 (d) 40

19. In an examination 50% of candidates failed in English, 40% failed in Hindi and 15% failed in both the subjects. The percentage of candidates who passed in both English and Hindi is:

[CDS 2018(II)]

- (a) 20% (b) 25%
(c) 60% (d) 75%

20. Out of 85 children playing badminton or table tennis or both, the total number of girls in the group is 70% of the total number of boys in the group. The number of boys playing only badminton is 50% of the number of boys and the total number of boys playing badminton is 60% of the total number of boys. The number of children playing only table tennis is 40% of the total number of children and a total of 12 children play badminton and table tennis both. The number of girls playing only badminton is

[CDS 2018(II)]

- (a) 14 (b) 16
(c) 17 (d) 35

21. In a class of 60 boys, there are 45 boys who play chess and 30 boys who play carom. If every boy of the class plays at least one of the two games, then how many boys play carom only?

[CDS 2018(II)]

- (a) 30 (b) 20
(c) 15 (d) 10

Consider the following for the next 04 (four) items that follow:

In an examination of Class XII, 55% students passed in Biology, 62% passed in Physics, 60% passed in Chemistry, 25% passed in Physics and Biology, 30% passed in Physics and Chemistry, 28% passed in Biology and Chemistry. Only 2% failed in all the subjects.

[CDS 2018(II)]

22. What percentage of students passed in all the three subjects?

- (a) 6 (b) 5
(c) 4 (d) 3

- 23.** What percentage of students passed in exactly one subject?
(a) 21 (b) 23
(c) 25 (d) 27
- 24.** If the number of students is 360, then how many passed in at least two subjects?
(a) 270 (b) 263
(c) 265 (d) 260
- 25.** What is the ratio of number of students who passed in both Physics and Chemistry to number of students who passed in both Biology and Physics but not Chemistry?
(a) 7:10 (b) 10:7
(c) 9:7 (d) 7:9
- 26.** In an examination, 52% candidates failed in English and 42% failed in Mathematics. If 17% failed in both the subjects, then what percent passed in both subjects?
[CDS 2019(I)]
(a) 77 (b) 58
(c) 48 (d) 23
- 27.** If $X = \{a, \{b\}, c\}$, $Y = \{\{a\}, b, c\}$ and $Z = \{a, b, \{c\}\}$, then $(X \cap Y) \cap Z$ equals to
[CDS 2019(II)]
(a) $\{a, b, c\}$ (b) $\{\{a\}, \{b\}, \{c\}\}$
(c) $\{\phi\}$ (d) ϕ
- 28.** In a competitive examination, 250 students have registered. Out of these, 50 students have registered for Physics, 75 students for Mathematics and 35 students for both Mathematics and Physics. What is the number of students who have registered neither for Physics nor for Mathematics?
[CDS 2020(I)]
(a) 90 (b) 100
(c) 150 (d) 160
- Direction: Consider the following for the four (04) items that follow:**
500 candidates appeared in an examination comprising tests in English, Hindi and Mathematics. 30 candidates failed in English only; 75 failed in Hindi only; 50 failed in Mathematics only; 15 failed in both English and Hindi; 17 failed in both Hindi and Mathematics; 17 failed in both Mathematics, English; 5 failed in all three tests.
- 29.** What is the percentage of candidates who failed in at least two subjects?
[CDS 2022(I)]
(a) 5.4% (b) 6.4%
(c) 6.8% (d) 7.8%
- 30.** What is the percentage of candidates who failed in only one subject?
[CDS 2022(I)]
(a) 28% (b) 31%
(c) 35.8% (d) 38.8%
- 31.** What is the percentage of candidates who failed in at least one subject?
[CDS 2022(I)]
(a) 31% (b) 35.4%
(c) 38.8% (d) 41.5%
- 32.** How many candidates passed in two or more subjects?
[CDS 2022(I)]
(a) 461 (b) 405
(c) 345 (d) 306
- 33.** Let $X = \{x | x = 2 + 4k, \text{ where } k = 0, 1, 2, 3, \dots, 24\}$. Let S be a subset of X such that the sum of no two elements of S is 100. What is the maximum possible number of elements in S ?
[CDS - 2023 (1)]
(a) 10 (b) 11
(c) 12 (d) 13
- 34.** In a party of 150 persons, 75 persons take tea, 60 persons take coffee and 50 persons take milk. 15 of them take both tea and coffee, but no one taking milk takes tea. If each person in the party takes at least one drink, then what is the number of persons taking milk only.
[CDS - 2023 (1)]
(a) 50 (b) 40
(c) 30 (d) 20

ANSWER KEY

1.	(c)	2.	(c)	3.	(d)	4.	(b)	5.	(b)	6.	(c)	7.	(c)	8.	(b)	9.	(b)	10.	(c)
11.	(a)	12.	(b)	13.	(b)	14.	(b)	15.	(c)	16.	(d)	17.	(b)	18.	(d)	19.	(b)	20.	(a)
21.	(c)	22.	(c)	23.	(b)	24.	(a)	25.	(b)	26.	(d)	27.	(d)	28.	(d)	29.	(d)	30.	(b)
31.	(c)	32.	(a)	33.	d	34.	c												

PRACTICE SHEET

1. The value of $\log 40$ is
 (1) 1.6020 (2) 1.2020
 (3) 1.94 (4) 1.3256
2. The value of $\log 8$ is equivalent to
 (1) $\log 12 + \log 4 - \log 8$
 (2) $\log 16 + \log 6 - \log 8$
 (3) $\log 6 + \log 4 - \log 2$
 (4) $\log 12 + \log 4 - \log 6$
3. Which one of the following is not correct
 (1) $\log 1 = 0$ (2) $\log_e 10 = 1$
 (3) $\log_{10} 10 = 1$ (4) $\log_{0.1} 0.1 = 1$
4. In $y = \log_e x$, value of x can never be
 (1) 2 (2) $1/2$
 (3) -1 (4) 1.6
5. The value of $\log 64$ is
 (1) 1.7658 (2) 1.8060
 (3) 1.2365 (4) 1.2546
6. If $\log 5 = k$ and $\log 125 = t$, then which of the following relation is correct?
 (1) $k = t^3$ (2) $k = 3t$
 (3) $t = 3k$ (4) $t = k^3$
7. The value of $7 \log \frac{16}{15} + 5 \log \frac{25}{24} + 3 \log \frac{81}{80}$ is :
 (1) 0.3010
 (2) 0.3512
 (3) 0.412
 (4) None of the above
8. The value of $\log \frac{70}{33} + \log \frac{22}{135} - \log \frac{7}{18}$ is:
 (given $\log 2 = 0.3010$, $\log 3 = 0.4771$)
 (a) -0.0512 (b) 0.4213
 (c) 0.3010 (d) None of these
9. $\log_{25} 625 - \log_{31} 961 + \log_{29} 841 = ?$
 (a) 2 (b) 6
 (c) 0 (d) 4
10. If $\log x + \log y = \log (x + y)$ then :
 (a) $x = y$ (b) $xy = 1$
 (c) $y = \frac{x-1}{x}$ (d) $y = \frac{x}{x-1}$
11. The value of $2^{2-\log_2 5}$ is equal to:
 (1) $4/5$ (2) $5/4$
 (3) $2/5$ (4) $5/2$
12. What is the value of $2 \log_8 2 - \frac{1}{3} \log_3 9$?
 (a) 0 (b) 1
 (c) 2 (d) $1/3$
13. What is the value of $\log_2 (\log_3 81)$?
 (a) 2 (b) 3
 (c) 4 (d) 9
14. What is $\log(a + \sqrt{a^2 + 1}) + \log\left(\frac{1}{a + \sqrt{a^2 + 1}}\right)$ equal to?
 (a) 1 (b) 0
 (c) 2 (d) $\frac{1}{2}$
15. What is the value of:
 $\log_{10} \left(\frac{9}{8}\right) - \log_{10} \left(\frac{27}{32}\right) + \log_{10} \left(\frac{3}{4}\right)$?
 (a) 3 (b) 2
 (c) 1 (d) 0
16. If $\log_3 \log_3 [\log_3 x] = \log_3 3$, then what is the value of x ?
 (a) $3a$ (b) 27
 (c) 3^9 (d) 3^{27}
17. What is the least integral value of $2 \log_{10} x - \log_x (0.01)$?
 (1) 0 (2) 2
 (3) 4 (4) 8
18. What is the value of $\log_{\sqrt{b}} a \log_{(c)^{1/3}} b \log_{(a)^{1/4}} c$?
 (1) 12 (2) 24
 (3) $1/12$ (4) $1/2$
19. The number of solutions of $\log_2 (x-1) = 2 \log_2 (x-3)$:
 (a) 2 (b) 1
 (c) 6 (d) 7
20. The value of $e^{(\log_{10} \tan 1^\circ + \log_{10} \tan 2^\circ + \dots + \log_{10} \tan 89^\circ)}$ is equal to:

- (a) 0 (b) 1
(c) e (d) $\frac{1}{e}$
21. What is the value of $\log_{100} 0.1$?
(1) $1/2$ (b) $-1/2$
(3) -2 (4) 2
22. The value of $\frac{1}{1+\log_c ab} + \frac{1}{1+\log_b ac} + \frac{1}{1+\log_a bc}$
(1) 0 (2) 1
(3) abc (4) $a + b + c$
23. The value of $\frac{1}{\log_{xy} xyz} + \frac{1}{\log_{yz} xyz} + \frac{1}{\log_{xz} xyz}$
(1) xyz (2) 2
(3) 0 (4) 1
24. If $\log_4 x + \log_2 x = 6$, then the value of x is ?
(1) 16 (2) 4
(3) 2 (4) 1
25. What is the value of $[\log_{10}(5\log_{10}100)]^2$?
(1) 4 (2) 3
(3) 2 (4) 1
26. If $\log_r p = 2$, $\log_r q = 3$ then the value of $\log_{pq}$ is
(1) $1/3$ (2) $2/3$
(3) $3/2$ (4) 6
27. If $\log_8 x + \log_4 x + \log_2 x = 11$, then the value of x is
(1) 128 (2) 16
(3) 32 (4) 64
28. $5^{2\log_5 4}$ is equal to
(1) 8 (2) 64
(3) 16 (4) 5
29. x triples every second. How will $\log_2 x$ change every second?
(1) It will double every second
(2) It will triple every second
(3) It increases by constant amount every second
(4) None of these
30. What is $\log_{81} 243$ equal to?
(1) 0.75 (2) 1.25
(3) 1.5 (4) 3

ANSWER KEY

1.	(1)	2.	(4)	3.	(2)	4.	(3)	5.	(2)	6.	(3)	7.	(1)	8.	(1)	9.	(1)	10.	(4)
11.	(1)	12.	(1)	13.	(1)	14.	(2)	15.	(4)	16.	(4)	17.	(3)	18.	(2)	19.	(2)	20.	(2)
21.	(2)	22.	(2)	23.	(2)	24.	(1)	25.	(4)	26.	(3)	27.	(4)	28.	(3)	29.	(3)	30.	(2)

CDS PYQ

1. The value of $\frac{1}{5}\log_{10} 3125 - 4\log_{10} 2 + \log_{10} 32$ is
[CDS 2016(I)]
(a) 0 (b) 1
(c) 2 (d) 4
2. What is the solution of $\log_{10} \left[1 - \left\{ 1 - (1 - x^2)^{-1} \right\}^{-1} \right]^{\frac{1}{2}} = 1$
[CDS 2016(II)]
(a) $x = 100$ (b) $x = 10$
(c) $x = 1$ (d) $x = 0$
3. What is the value of the equation $2^{x+2} \cdot 27^{\frac{x}{x-1}} = 9$.
[CDS 2016(II)]
(a) $2, 1 - \log\left(\frac{3}{2}\right)$ (b) $2, 1 - \log\left(\frac{2}{3}\right)$
(c) $-2, 1 - \frac{\log 3}{\log 2}$ (d) $-2, 1 - \left(\frac{\log 2}{\log 3}\right)$
4. The number of digits in 3^{30} is n and it is given that $\log_{10} 3 = 0.4771$. What is the value of n ?
[CDS 2016(II)]
(a) 13 (b) 14
(c) 15 (d) 16
5. What is the number of digits in 2^{40} ? (Given that $\log_{10} 2 = 0.3010$)
[CDS 2017(I)]
(a) 14 (b) 13
(c) 12 (d) 11
6. What is the solution of the equation $x \log_{10} \left(\frac{10}{3}\right) + \log_{10} 3 = \log_{10} (2 + 3^x) + x$?
[CDS 2017(II)]
(a) 10 (b) 3
(c) 1 (d) 0
7. If $\log_{10} 6 = 0.7782$ and $\log_{10} 8 = 0.9031$, then what is the value of $\log_{10} 8000 + \log_{10} 600$?
[CDS 2017(II)]
(a) 4.6813 (b) 5.5813
(c) 1.5813 (d) 6.6813
8. For $0 < m < 1$, which one of the following is correct?
[CDS 2018(I)]
(a) $\log_{10} m < m^2 < m < m^{-1}$
(b) $m < m^{-1} < m^2 < \log_{10} m$
(c) $\log_{10} m < m < m^{-1} < m^2$
(d) $\log_{10} m < m^{-1} < m < m^2$
9. If $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$, then what is the value of $\log_{100} (0.72)$?
[CDS 2018(II)]
(a) 0.9286 (b) $\bar{1}.9286$
(c) 1.8572 (d) $\bar{1}.8572$
10. There are n zeros appearing immediately after the decimal point in the value of $(0.2)^{25}$. It is given that value of $\log_{10} 2 = 0.30103$. The value of n is
[CDS 2018(II)]
(a) 25 (b) 19
(c) 18 (d) 17
11. It is given that $\log_{10} 2 = 0.301$ and $\log_{10} 3 = 0.477$. How many digits are there in $(108)^{10}$?
[CDS 2019(I)]
(a) 19 (b) 20
(c) 21 (d) 22
12. The sides of a triangle are 30cm, 28cm and 16cm respectively. In order to determine its area, the logarithm of which of the quantities are required?
[CDS 2019(II)]
(a) 37, 11, 28, 16 (b) 21, 30, 28, 7
(c) 37, 21, 11, 9 (d) 37, 21, 9, 7
13. If $\log_{10} 1995 = 3.3000$ then what is the value of $(0.001995)^{\frac{1}{8}}$?
[CDS 2019(II)]
(a) $\frac{1}{10^{0.3475}}$ (b) $\frac{1}{10^{0.3375}}$
(c) $\frac{1}{10^{0.3275}}$ (d) $\frac{1}{10^{0.3735}}$
14. Let XYZ be an equilateral triangle in which $XY = 7$ cm. If A denotes the area of the triangle, then what is the value of $\log_{10} A$? (Given that $\log_{10} 1050 = 3.0212$ and $\log_{10} 35 = 1.5441$)
[CDS 2019(II)]
(a) 4.75 cm (b) 5.50 cm
(c) 6.25 cm (d) 6.75 cm

15. What is the number of digits in 7^{25} , 8^{23} and 9^{20} respectively?
[Given $\log_{10} 2 = 0.301$, $\log_{10} 3 = 0.477$, $\log_{10} 7 = 0.845$]

[CDS 2019(II)]

- (a) 21, 20, 19 (b) 20, 19, 18
(c) 22, 21, 20 (d) 22, 20, 21

16. How many digits are there in $(54)^{10}$? ($\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$)

[CDS 2020(I)]

- (a) 16 (b) 18
(c) 19 (d) 27

17. If $\log x = 1.2500$ and $y = x^{\log x}$, then what is the value of $\log y$ equal to?

[CDS 2020(I)]

- (a) 40.2500 (b) 2.5625
(c) 1.5625 (d) 1.2500

18. Which one of the following is the largest number among 2222^2 , 222^{22} , 22^{222} , 2^{2222} ?

[CDS 2020(I)]

- (a) 2^{2222} (b) 22^{222}
(c) 222^{22} (d) 2222^2

19. If the roots of quadratic equation $x^2 - 4x - \log_{10} N = 0$ are real, then what is the minimum value of N ?

[CDS 2020(II)]

- (a) 1 (b) $1/10$
(c) $1/100$ (d) $1/10000$

20. If $6^{3-4x} 4^{x+5} = 8$ ($\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$), then which one the following is correct?

[CDS 2020(II)]

- (a) $0 < x < 1$ (b) $1 < x < 2$
(c) $2 < x < 3$ (d) $3 < x < 4$

21. What is the value of $\log_{10} 31.25$ equal to?

[CDS 2021(I)]

- (a) $3 - 5\log_{10} 2$ (b) $3 - 2\log_{10} 2$
(c) $5 - 5\log_{10} 2$ (d) $5 - 3\log_{10} 2$

22. If $5^{x-3} = 8$, then what is x equal to?

[CDS 2021(II)]

(a) $\frac{3}{1 - \log_{10} 2}$

(b) $\frac{3}{1 + \log_{10} 2}$

(c) $\frac{2}{1 - \log_{10} 2}$

(d) $\frac{5}{1 - \log_{10} 2}$

23. If $\log_{10} x + \log_{10} x^2 = 2 \log_{10} x + 1$, then what is the value of x ?

[CDS 2021(II)]

- (a) 1 (b) 2
(c) 5 (d) 10

24. If the roots of the equation $x^2 - 4x - \log_{10} N = 0$ are real, then what is the minimum value of N ?

[CDS 2021(II)]

- (a) 0.1 (b) 0.01
(c) 0.001 (d) 0.0001

25. If $5^{x-1} = (2.5)^{\log_{10} 5}$, then what is the value of x ?

[CDS 2021(II)]

- (a) 1 (b) $\log_{10} 2$
(c) $\log_{10} 5$ (d) $2\log_{10} 2$

26. If $\frac{\log_{10} a}{b-c} = \frac{\log_{10} b}{c-a} = \frac{\log_{10} c}{a-b}$, ($a \neq b \neq c$), then what is the value of abc ?

[CDS 2022 (II)]

- (a) -1 (b) 0
(c) 1 (d) 3

27. If $\log_{10} \left[995 + \sqrt{x^2 - 12x + 60} \right] = 3$, then what is the sum of the roots of the equation?

[CDS 2022 (II)]

- (a) 12 (b) 11
(c) 10 (d) 9

28. What is the number of digits in the expansion of 125^{100} ? (Given $\log_{10} 2 = 0.301$)

[CDS-2023 (2)]

- (a) 69 (b) 70
(c) 209 (d) 210

ANSWER KEY

1.	(b)	2.	(b)	3.	(c)	4.	(c)	5.	(b)	6.	(d)	7.	(d)	8.	(a)	9.	(b)	10.	(d)
11.	(c)	12.	(d)	13.	(b)	14.	(a)	15.	(c)	16.	(b)	17.	(c)	18.	(a)	19.	(d)	20.	(b)
21.	(a)	22.	(a)	23.	(d)	24.	(d)	25.	(d)	26.	(c)	27.	(a)	28.	(d)				

PRACTICE SHEET-1

1. The sides of a triangle are in the ratio 3 : 4 : 6. The triangle is:
 (1) acute-angled
 (2) right-angled
 (3) obtuse-angled
 (4) either acute-angled or right angled
2. Taking any three of the line segments out of segments of length 2 cm, 3 cm, 5 cm and 6 cm, the number of triangles that can be formed is:
 (1) 3
 (2) 2
 (3) 1
 (4) 4
3. If two angles of a triangle are 21° and 38° , then the triangle is
 (1) Right-angled triangle
 (2) Acute-angled triangle
 (3) Obtuse-angled triangle
 (4) Isosceles triangle
4. A man goes 24 m due west and then 10 m due north. Then the distance of him from the starting point is
 (1) 17 m
 (2) 26 m
 (3) 28 m
 (4) 34 m
5. If in $\triangle ABC$, $\angle ABC = 5 \angle ACB$ and $\angle BAC = 3 \angle ACB$, then $\angle ABC = ?$
 (1) 130°
 (2) 80°
 (3) 100°
 (4) 120°
6. AD is the median of a triangle ABC and O is the centroid such that $AO = 10$ cm. The length of OD (in cm) is
 (1) 4
 (2) 5
 (3) 6
 (4) 8
7. The measure of the angle between the internal and external bisectors of an angle is
 (1) 60°
 (2) 70°
 (3) 80°
 (4) 90°
8. If the sides of a triangle are extended in both the sides then the sum of the exterior angles so formed in both sides is
 (1) 360°
 (2) 540°
 (3) 720°
 (4) 180°
9. The measures of two angles of a triangle are in the ratio 4 : 5. If the sum of these two measures is equal to the measure of the third angle, find the smallest angle.
 (1) 10°
 (2) 50°
 (3) 90°
 (4) 40°
10. The sides of a triangle are in the ratio of 7 : 9 : 12. The difference between the lengths of largest and smallest sides is 15 cm. The length of the largest side would be :
 (1) 36 cm
 (2) 12 cm
 (3) 60 cm
 (4) 24 cm
11. The side BC of a triangle ABC is produced to D. If $\angle ACD = 112^\circ$ and $\angle B = (3/4) \angle A$, then the measure of $\angle B$ is
 (1) 30°
 (2) 48°
 (3) 45°
 (4) 64°
12. In a triangle ABC, if $\angle A + \angle C = 140^\circ$ and $\angle A + 3\angle B = 180^\circ$, then $\angle A$ is equal to
 (1) 80°
 (2) 40°
 (3) 60°
 (4) 20°
13. Which of the set of three sides can't form a triangle?
 (1) 5 cm, 6 cm, 7 cm
 (2) 5 cm, 8 cm, 15 cm
 (3) 8 cm, 15 cm, 18 cm
 (4) 6 cm, 7 cm, 11 cm
14. Possible measures of three angles of a triangle are
 (1) $33^\circ, 42^\circ, 115^\circ$
 (2) $40^\circ, 70^\circ, 80^\circ$
 (3) $30^\circ, 60^\circ, 100^\circ$
 (4) $50^\circ, 60^\circ, 70^\circ$
15. Suppose that the medians BD, CE and AF of a triangle ABC meet at G. Then $AG : GF$ is
 (1) 1 : 2
 (2) 2 : 1
 (3) 1 : 3
 (4) 2 : 3
16. In case of an acute angled triangle, its orthocenter lies
 (1) inside the triangle
 (2) outside the triangle
 (3) on the triangle
 (4) on one of the vertices of the triangle

17. The centroid of a triangle is the point where
(1) the medians meet
(2) the altitudes meet
(3) the right bisectors of the sides of the triangle meet
(4) the bisectors of the angles of the triangle meet
18. An exterior angle of a triangle is 115° and one of the interior opposite angles is 45° . Then the other two angles are
(1) $65^\circ, 70^\circ$ (2) $60^\circ, 75^\circ$
(3) $45^\circ, 90^\circ$ (4) $50^\circ, 85^\circ$
19. If the difference between the measures of the two smaller angles of a right angled triangle is 8° , then the smallest angle is :
(1) 37° (2) 41°
(3) 42° (4) 49°
20. Three sides of a triangle are 5 cm, 9 cm and x cm. The minimum integral value of x is:
(1) 2 (2) 3
(3) 4 (4) 5
21. If the measures of the angles of a triangle are in the ratio. 1 : 2 : 3 and if the length of the smallest side of the triangle is 10 cm, then the length of the longest side is
(1) 20 cm. (2) 25 cm.
(3) 30 cm. (4) 35 cm.
22. The sum of two angles of a triangle is 116° and their difference is 24° . The measure of the smallest angle of the triangle is :
(1) 38° (2) 28°
(3) 46° (4) 64°
23. In a triangle, the distance of the centroid from the three vertices is 4 cm, 6 cm and 8 cm respectively. Then the length of the smallest median is:
- (1) 8 (2) 7
(3) 6 (4) 5
24. If the complement of an angle is one-fourth of its supplementary angle, then the angle is
(1) 60° (2) 30°
(3) 90° (4) 120°
25. If two supplementary angles differ by 44° , then one of the angles is
(1) 68° (2) 65°
(3) 102° (4) 72°
26. The measure of an angle whose supplement is three times as large as its complement, is
(1) 75° (2) 30°
(3) 45° (4) 60°
27. In a triangle ABC, BC is produced to D so that $CD = AC$. If $\angle BAD = 111^\circ$ and $\angle ACB = 80^\circ$, then the measure of $\angle ABC$ is:
(1) 31° (2) 33°
(3) 35° (4) 29°
28. In $\triangle ABC$, $\angle A + \angle B = 145^\circ$ and $\angle C + 2\angle B = 180^\circ$. State which one of the following relations is true?
(1) $CA = AB$ (2) $CA < AB$
(3) $BC < AB$ (4) $CA > AB$
29. $\angle A, \angle B, \angle C$ are three angles of a triangle. If $\angle A - \angle B = 15^\circ$, $\angle B - \angle C = 30^\circ$, then $\angle A, \angle B$ and $\angle C$ are
(1) $80^\circ, 60^\circ, 40^\circ$ (2) $70^\circ, 50^\circ, 60^\circ$
(3) $80^\circ, 65^\circ, 35^\circ$ (4) $80^\circ, 55^\circ, 45^\circ$
30. In a $\triangle ABC$ $\angle A : \angle B : \angle C = 2 : 3 : 4$. A line CD drawn $\parallel$ to AB, then the $\angle ACD$ is
(1) 40° (2) 60°
(3) 80° (4) 20°

ANSWER KEY

1.	(3)	2.	(2)	3.	(3)	4.	(2)	5.	(3)	6.	(2)	7.	(4)	8.	(3)	9.	(4)	10.	(1)
11.	(2)	12.	(3)	13.	(2)	14.	(4)	15.	(2)	16.	(1)	17.	(1)	18.	(1)	19.	(2)	20.	(4)
21.	(1)	22.	(3)	23.	(3)	24.	(1)	25.	(1)	26.	(3)	27.	(4)	28.	(4)	29.	(3)	30.	(1)

PRACTICE SHEET-2

1. O and C are respectively the ortho-centre and circum-centre of an acute-angled triangle PQR. The points P and O are joined and produced to meet the side QR at S. If $\angle PQS = 60^\circ$ and $\angle QCR = 130^\circ$, then $\angle RPS =$
 (1) 30° (2) 35°
 (3) 100° (4) 60°
2. In $\triangle ABC$, AD is the internal bisector of $\angle A$, meeting the side BC at D. If $BD = 5$ cm, $BC = 7.5$ cm, then $AB : AC$ is
 (1) $2 : 1$ (2) $1 : 2$
 (3) $4 : 5$ (4) $3 : 5$
3. If the circum-centre of a triangle lies outside it, then the triangle is
 (1) Equilateral (2) Acute angled
 (3) Right angled (4) Obtuse angled
4. If the length of the sides of a triangle are in the ratio $4 : 5 : 6$ and the in radius of the triangle is 3 cm, then the altitude of the triangle corresponding to the largest side as base is :
 (1) 7.5 cm (2) 6 cm
 (3) 10 cm (4) 8 cm
5. ABC is a triangle. The bisectors of the internal angle $\angle B$ and external angle $\angle C$ intersect at D. If $\angle BDC = 50^\circ$, then $\angle A$ is
 (1) 100° (2) 90°
 (3) 120° (4) 60°
6. In a triangle ABC, the side BC is extended up to D. Such that $CD = AC$, if $\angle BAD = 109^\circ$ and $\angle ACB = 72^\circ$ then the value of $\angle ABC$ is
 (1) 35° (2) 60°
 (3) 40° (4) 45°
7. The sum of three altitudes of a triangle is
 (1) equal to the sum of three sides
 (2) less than the sum of sides
 (3) greater than the sum of sides
 (4) twice the sum of sides
8. I is the incentre of $\triangle ABC$, $\angle ABC = 60^\circ$ and $\angle ACB = 50^\circ$. Then $\angle BIC$ is:
 (1) 55° (2) 125°
 (3) 70° (4) 65°
9. In $\triangle ABC$, $\angle C$ is an obtuse angle. The bisectors of the exterior angles at A and B meet BC and AC produced at D and E respectively. If $AB = AD = BE$, then $\angle ACB$
 (1) 105° (2) 108°
 (3) 110° (4) 135°
10. The exterior angles obtained on producing the base BC of a triangle ABC in both ways are 120° and 105° , then the vertical $\angle A$ of the triangle is of measure
 (1) 36° (2) 40°
 (3) 45° (4) 55°
11. O is the in centre of $\triangle ABC$ and $\angle A = 30^\circ$, then $\angle BOC$ is
 (1) 100° (2) 105°
 (3) 110° (4) 90°
12. Let O be the in-centre of a triangle ABC and D be a point on the side BC of $\triangle ABC$, such that $OD \perp BC$. If $\angle BOD = 15^\circ$, then $\angle ABC =$
 (1) 75° (2) 45°
 (3) 150° (4) 90°
13. The external bisector of $\angle B$ and $\angle C$ of $\triangle ABC$ (where AB and AC extended to E and F respectively) meet at point P. If $\angle BAC = 100^\circ$, then the measure of $\angle BPC$ is
 (1) 50° (2) 80°
 (3) 40° (4) 100°
14. The points D and E are taken on the sides AB and AC of $\triangle ABC$ such that $AD = \frac{1}{3} AB$, $AE = \frac{1}{3} AC$. If the length of BC is 15 cm, then the length of DE is :
 (1) 10 cm (2) 8 cm
 (3) 6 cm (4) 5 cm
15. In a triangle ABC, $AB + BC = 12$ cm, $BC + CA = 14$ cm and $CA + AB = 18$ cm. Find the radius of the circle (in cm) which has the same perimeter as the triangle.
 (1) $\frac{5}{2}$ (2) $\frac{7}{2}$
 (3) $\frac{9}{2}$ (4) $\frac{11}{2}$
16. In $\triangle ABC$, D is the mid-point of BC. Length AD is 27 cm. N is a point in AD such that the length of DN is 12 cm. The distance of N from the centroid of $\triangle ABC$ is equal to
 (1) 3 cm (2) 6 cm
 (3) 9 cm (4) 15 cm
17. In $\triangle ABC$, $B = 60^\circ$, $\angle C = 40^\circ$, AD is the bisector of $\angle A$ and AE is drawn

- perpendicular on BC from A. Then the measure of $\angle EAD$ is
 (1) 40° (2) 30°
 (3) 10° (4) 80°
18. O is the orthocentre of $\triangle ABC$. Then $\angle BOC + \angle BAC$ is equal to
 (1) 120° (2) 135°
 (3) 180° (4) 90°
19. ABC is a triangle and the sides AB, BC and CA are produced to E, F and G respectively. If $\angle CBE = \angle ACF = 130^\circ$ then the value of $\angle GAB$ is
 (1) 100° (2) 130°
 (3) 80° (4) 90°
20. G is the centroid of $\triangle ABC$. The medians AD and BE intersect at right angles. If the lengths of AD and BE are 9 cm and 12 cm respectively; then the length of AB (in cm) is
 (1) 9.5 (2) 10
 (3) 11 (4) 10.5
21. The internal bisectors of the $\angle B$ and $\angle C$ of the $\triangle ABC$, intersect at O. If $\angle A = 100^\circ$, then the measure of $\angle BOC$ is:
 (1) 140° (2) 120°
 (3) 110° (4) 130°
22. AD is perpendicular to the internal bisector of $\angle ABC$ of $\triangle ABC$. DE is drawn through D and parallel to BC to meet AC at E. If the length of AC is 12 cm, then the length of AE (in cm.) is
 (1) 3 (2) 8
 (3) 4 (4) 6
23. What is the position of the circum-centre of an obtuse-angled triangle?
 (1) It lies inside the triangle.
 (2) It lies outside the triangle.
 (3) It is the mid-point of the largest side.
 (4) It is the vertex opposite to the largest side.
24. E is the mid-point of the median AD of $\triangle ABC$. BE is joined and produced to meet AC at F. F divides AC in the ratio:
 (1) 2 : 3 (2) 2 : 1
 (3) 1 : 3 (4) 3 : 2
25. G is the centroid of $\triangle ABC$. If $AG = BC$, then measure of $\angle BGC$ is
 (1) 45° (2) 60°
 (3) 90° (4) 120°
26. B_1 is a point on the side AC of $\triangle ABC$ and B_1B is joined. A line is drawn through A parallel to B_1B meeting BC at A_1 and another line is drawn through C parallel to B_1B meeting AB produced at C_1 . Then
 (1) $\frac{1}{CC_1} - \frac{1}{AA_1} = \frac{1}{BB_1}$
 (2) $\frac{1}{CC_1} + \frac{1}{AA_1} = \frac{1}{BB_1}$
 (3) $\frac{1}{BB_1} - \frac{1}{AA_1} = \frac{2}{CC_1}$
 (4) $\frac{1}{AA_1} - \frac{1}{CC_1} = \frac{2}{BB_1}$
27. In $\triangle PQR$, straight line parallel to the base QR cuts PQ at X and PR at Y. If $PX : XQ = 5 : 6$, then $XY : QR$ will be
 (1) 5 : 11 (2) 6 : 5
 (3) 11 : 6 (4) 11 : 5
28. In a triangle the length of the side opposite the angle which measures 45° is 8 cm, what is the length of the side opposite to the angle which measures 90° ?
 (1) $8\sqrt{2}$ cm. (2) $4\sqrt{2}$ cm.
 (3) $8\sqrt{3}$ cm. (4) $4\sqrt{3}$ cm.
29. In a triangle ABC, $\angle A = 70^\circ$, $\angle B = 80^\circ$ and D is the in centre of $\triangle ABC$. $\angle ACB = 2x^\circ$ and $\angle BDC = y^\circ$. The values of x and y , respectively are
 (1) 15, 130 (2) 15, 125
 (3) 35, 40 (4) 30, 150
30. If O is the ortho-centre of a triangle ABC and $\angle BOC = 100^\circ$, the measure of $\angle BAC$ is
 (1) 100° (2) 180°
 (3) 80° (4) 200°

ANSWER KEY

1.	(2)	2.	(1)	3.	(4)	4.	(1)	5.	(1)	6.	(1)	7.	(2)	8.	(2)	9.	(2)	10.	(3)
11.	(2)	12.	(3)	13.	(3)	14.	(4)	15.	(2)	16.	(1)	17.	(3)	18.	(3)	19.	(1)	20.	(2)
21.	(1)	22.	(4)	23.	(2)	24.	(2)	25.	(3)	26.	(2)	27.	(1)	28.	(1)	29.	(2)	30.	(3)

PRACTICE SHEET-3

1. If the in-centre of an equilateral triangle lies inside the triangle and its radius is 3 cm, then the side of the equilateral triangle is
 (1) $9\sqrt{3}$ cm (2) $6\sqrt{3}$ cm
 (3) $3\sqrt{3}$ cm (4) 6 cm
2. ABC is an equilateral triangle and CD is the internal bisector of $\angle C$. If DC is produced to E such that $AC = CE$, then $\angle CAE$ is equal to
 (1) 45° (2) 75°
 (3) 30° (4) 15°
3. G is the centroid of the equilateral $\triangle ABC$. If $AB = 10$ cm then length of AG is
 (1) $5\sqrt{3}/3$ cm (2) $10\sqrt{3}/3$ cm
 (3) $5\sqrt{3}$ cm (4) $10\sqrt{3}$ cm
4. The side BC of a triangle ABC is extended to D. If $\angle ACD = 120^\circ$ and $\angle ABC = (1/2) \angle CAB$, then the value of $\angle ABC$ is
 (1) 80° (2) 40°
 (3) 60° (4) 20°
5. If the three angles of a triangle are : $(x + 15^\circ)$, $(6x/5 + 6^\circ)$, and $(2x/3 + 30^\circ)$, then the triangle is :
 (1) Isosceles (2) Right angled
 (3) Equilateral (d) None of these
6. Let ABC be an equilateral triangle and AD perpendicular to BC. Then $AB^2 + BC^2 + CA^2 = ?$
 (1) $2AD^2$ (2) $3AD^2$
 (3) $4AD^2$ (4) $5AD^2$
7. O is the ortho-centre of $\triangle ABC$, and if $\angle BOC = 110^\circ$, then $\angle BAC$ will be
 (1) 110° (2) 70°
 (3) 100° (4) 90°
8. ABC is an equilateral triangle. Points D, E, F are taken in sides AB, BC, CA respectively, so that $AD = BE = CF$. Then AE, BF, CD enclosed a triangle which is:
 (1) equilateral triangle
 (2) isosceles triangle
 (3) right angle triangle
 (4) None of these
9. The lengths of the sides of a triangle are a , b and c respectively. If $a^2 + b^2 + c^2 = ab + bc + ca$, then the triangle is:
 (1) Isosceles (2) Equilateral
 (3) Scalene (4) Right-angled
10. PQR is an equilateral triangle. MN is drawn parallel to QR such that M is on PQ and N is on PR. If $PN = 6$ cm, then the length of MN is
 (1) 3 cm (2) 6 cm
 (3) 12 cm (4) 4.5 cm
11. Two medians AD and BE of $\triangle ABC$ intersect at G at right angles. If $AD = 9$ cm and $BE = 6$ cm, then the length of BD (in cm) is
 (1) 10 (2) 6
 (3) 5 (4) 3
12. In $\triangle ABC$, $\angle BAC = 90^\circ$ and $AB = (1/2)BC$. Then the measure of $\angle ACB$ is:
 (1) 60° (2) 30°
 (3) 45° (4) 15°
13. If the length of the three sides of a triangle are 6 cm, 8 cm and 10 cm, then the length of the median to its greatest side is
 (1) 8 cm (2) 6 cm
 (3) 5 cm (4) 4.8 cm
14. The length of the three sides of a right angled triangle are $(x - 2)$ cm, x cm and $(x + 2)$ cm respectively. Then the value of x is
 (1) 10 (2) 8
 (3) 4 (4) 0
15. In a triangle ABC, $\angle BAC = 90^\circ$ and AD is perpendicular to BC. If $AD = 6$ cm and $BD = 4$ cm, then the length of BC is
 (1) 8 cm (2) 10 cm
 (3) 9 cm (4) 13 cm
16. In a right angled $\triangle ABC$, $\angle ABC = 90^\circ$; BN is perpendicular to AC, $AB = 6$ cm, $AC = 10$ cm. Then $AN : NC$ is
 (1) 3 : 4 (2) 9 : 16
 (3) 3 : 16 (4) 1 : 4
17. In $\triangle ABC$, $\angle A = 90^\circ$ and $AD \perp BC$ where D lies on BC. If $BC = 8$ cm, $AC = 6$ cm, then $\triangle ABC : \triangle ACD = ?$
 (1) 4 : 3 (2) 25 : 16
 (3) 16 : 9 (4) 25 : 9

18. In a right-angled triangle ABC, $\angle ABC = 90^\circ$, $AB = 5$ cm and $BC = 12$ cm. The radius of the circum-circle of the triangle ABC is
(1) 7.5 cm (2) 6 cm
(3) 6.5 cm (4) 7 cm
19. In a right-angled triangle, the product of two sides is equal to half of the square of the third side i.e., hypotenuse. One of the acute angle must be
(1) 60° (2) 30°
(3) 45° (4) 15°
20. D and E are two points on the sides AC and BC respectively of $\triangle ABC$ such that $DE = 18$ cm, $CE = 5$ cm and $\angle DEC = 90^\circ$. If $\tan \angle ABC = 3.6$, then $AC : CD =$
(1) $BC : 2 CE$ (2) $2 CE : BC$
(3) $2 BC : CE$ (4) $CE : 2 BC$
21. If $\triangle ABC$ is an isosceles triangle with $\angle C = 90^\circ$ and $AC = 5$ cm, then AB is:
(1) 5 cm (2) 10 cm
(3) $5\sqrt{2}$ cm (4) 2.5 cm
22. ABC is an isosceles triangle such that $AB = AC$ and $\angle B = 35^\circ$. AD is the median to the base BC. Then $\angle BAD$ is:
(1) 70° (2) 35°
(3) 110° (4) 55°
23. ABC is an isosceles triangle with $AB = AC$. A circle through B touching AC at the middle point intersects AB at P. Then $AP : AB$ is :
(1) 4 : 1 (2) 2 : 3
(3) 3 : 5 (4) 1 : 4
24. In an isosceles triangle, if the unequal angle is twice the sum of the equal angles, then each equal angle is
(1) 120° (2) 60° (3) 30° (4) 90°
25. $\triangle ABC$ is an isosceles triangle and $AB = AC = 2a$ unit, $BC = a$ unit. Draw $AD \perp BC$, and find the length of AD .
(1) $15 a$ unit (2) $\sqrt{15}/2 a$ unit
(3) $17 a$ unit (4) $17/2 a$ unit
26. ABC is an isosceles triangle inscribed in a circle. If $AB = AC = 12\sqrt{5}$ cm and $BC = 24$ cm then the radius of circle is
(1) 10 cm. (2) 15 cm.
(3) 12 cm. (4) 14 cm.
27. The vertical angle A of an isosceles triangle $\triangle ABC$ is three times the angle B of it. The measure of the angle A is
(1) 90° (2) 108°
(3) 100° (4) 36°
28. $\triangle ABC$ is isosceles having $AB = AC$ and $\angle A = 40^\circ$. Bisectors PO and OQ of the exterior angles ABD and $\angle ACE$ formed by producing BC on both sides, meet at O. Then the value of $\angle BOC$ is
(1) 70° (2) 110°
(3) 80° (4) 55°
29. $\triangle ABC$ is an isosceles triangle with $AB = AC = 15$ cm and altitude from A to BC is 12 cm. The length of side BC is:
(1) 9 cm. (2) 12 cm.
(3) 18 cm. (4) 20 cm.
30. In an isosceles $\triangle ABC$, AD is the median to the unequal side meeting BC at D. DP is the angle bisector of $\angle ADB$ and PQ is drawn parallel to BC meeting AC at Q. Then the measure of $\angle PDQ$ is:
(1) 130° (2) 90°
(3) 180° (4) 45°

ANSWER KEY

1.	(2)	2.	(4)	3.	(2)	4.	(2)	5.	(3)	6.	(3)	7.	(2)	8.	(1)	9.	(2)	10.	(2)
11.	(3)	12.	(2)	13.	(3)	14.	(2)	15.	(4)	16.	(2)	17.	(3)	18.	(3)	19.	(3)	20.	(3)
21.	(3)	22.	(4)	23.	(4)	24.	(3)	25.	(2)	26.	(2)	27.	(2)	28.	(1)	29.	(3)	30.	(2)

PRACTICE SHEET-4

1. In $\triangle ABC$ and $\triangle DEF$, $AB = DE$ and $BC = EF$. Then one can infer that $\triangle ABC \cong \triangle DEF$, when
 (1) $\angle BAC = \angle EDF$ (2) $\angle ACB = \angle EDF$
 (3) $\angle ACB = \angle DFE$ (4) $\angle ABC = \angle DEF$
2. In $\triangle PQR$, S and T are points on sides PR and PQ respectively such that $\angle PQR = \angle PST$. If $PT = 5$ cm, $PS = 3$ cm and $TQ = 3$ cm, then length of SR is
 (1) 5 cm (2) 6 cm
 (3) $31/3$ cm (4) $41/3$ cm
3. The perimeters of two similar triangles $\triangle ABC$ and $\triangle PQR$ are 36 cm and 24 cm respectively. If $PQ = 10$ cm, then AB is
 (1) 15 cm (2) 12 cm
 (3) 14 cm (4) 26 cm
4. In $\triangle ABC$, two points D and E are taken on the lines AB and BC respectively in such a way that AC is parallel to DE . Then $\triangle ABC$ and $\triangle DBE$ are
 (1) similar only if D lies outside the line segment AB
 (2) congruent only if D lies outside the line segment AB
 (3) always similar
 (4) always congruent
5. Inside a triangle ABC , a straight line parallel to BC intersects AB and AC at the point P and Q respectively. If $AB = 3$ PB, then $PQ : BC$ is
 (1) 1 : 3 (2) 3 : 4
 (3) 1 : 2 (4) 2 : 3
6. In $\triangle ABC$, D and E are points on AB and AC respectively such that $DE \parallel BC$ and DE divides the $\triangle ABC$ into two parts of equal areas. Then ratio of AD and BD is
 (1) 1 : 1 (2) $1 : \sqrt{2} - 1$
 (3) $1 : \sqrt{2}$ (4) $1 : \sqrt{2} + 1$
7. In $\triangle ABC$, $DE \parallel AC$. D and E are two points on AB and CB respectively. If $AB = 10$ cm and $AD = 4$ cm, then $BE : CE$ is
 (1) 2 : 3 (2) 2 : 5
 (3) 5 : 2 (4) 3 : 2
8. For a triangle $ABCD$ and E are two points on AB and AC such that $AD = (1/4) AB$, $AE = (1/4) AC$. If $BC = 12$ cm, then DE is
 (1) 5 cm (2) 4 cm
 (3) 3 cm (4) 6 cm
9. In triangle ABC a straight line parallel to BC intersects AB and AC at D and E respectively. If $AB = 2AD$ then $DE : BC$ is
 (1) 2 : 3 (2) 2 : 1
 (3) 1 : 2 (4) 1 : 3
10. In a $\triangle ABC$, D and E are two points on AB and AC respectively such that $DE \parallel BC$, DE bisects the $\triangle ABC$ in two equal areas. Then the ratio $DB : AB$ is
 (1) $1 : \sqrt{2}$ (2) 1 : 2
 (3) $(\sqrt{2} - 1) : \sqrt{2}$ (4) $\sqrt{2} : 1$
11. In $\triangle ABC$, E and D are points on sides AB and AC respectively such that $\angle ABC = \angle ADE$. If $AE = 3$ cm, $AD = 2$ cm and $EB = 2$ cm, then length of DC is
 (1) 4 cm (2) 4.5 cm
 (3) 5.0 cm (4) 5.5 cm
12. $\triangle ABC$ and $\triangle DEF$ are similar. Also $\angle A = \angle D$ and $\angle B = \angle E$. If $4AB = DE$ and $BC = 12$ cm, then EF is equal to
 (1) 3 cm (2) 24 cm
 (3) 16 cm (4) 48 cm
13. In $\triangle ABC$ the straight line parallel to the side BC meets AB and AC at the points P and Q respectively. If $AP = QC$, the length of AB is 12 units and the length of AQ is 2 units, then the length (in units) of CQ is
 (1) 4 (2) 6
 (3) 8 (4) 10
14. ABC is a triangle in which $DE \parallel BC$ and $AD : DB = 5 : 4$. Then $DE : BC$ is
 (1) 4 : 5 (2) 4 : 9
 (3) 9 : 5 (4) 5 : 9
15. If in a triangle ABC , BE and CF are two medians perpendicular to each other and if $AB = 19$ cm and $AC = 22$ cm then the length of BC is
 (1) 19.5 cm (2) 26 cm
 (3) 20.5 cm (4) 13 cm

16. The medians CD and BE of a triangle ABC intersect each other at O. The ratio $\Delta ODE : \Delta ABC$ is equal to
 (1) 12 : 1 (2) 4 : 3
 (3) 3 : 4 (4) 1 : 12
17. ΔABC and ΔDEF are two similar triangles and the perimeters of ΔABC and ΔDEF are 30 cm and 18 cm respectively. If the length of DE = 36 cm, then length of AB is
 (1) 60 cm. (2) 40 cm.
 (3) 45 cm. (4) 50 cm.
18. If ΔPQR and ΔLMN are similar and $3PQ = LM$ and $MN = 9$ cm, then QR is equal to:
 (1) 12 cm (2) 6 cm
 (3) 9 cm (4) 3 cm
19. The perimeter of two similar triangles ABC and PQR are 36 cms and 24 cms respectively. If PQ = 10 cm then the length of AB is
 (1) 18 cm (2) 12 cm
 (3) 15 cm (4) 30 cm
20. Which of the following is a true statement?
 (1) Two similar triangles are always congruent.
 (2) Two similar triangles have equal areas
 (3) Two triangles are similar if their corresponding sides are proportional.
 (4) Two polygons are similar if their corresponding sides are proportional.
21. The perimeter of two similar triangles ΔABC and ΔPQR are 60 cm and 36 cm respectively. If PQ = 18 cm, then AB is :
 (1) 20 cm (2) 24 cm
 (3) 36 cm (4) 30 cm
22. D and E are the points on the sides AB and AC respectively of a ΔABC and AD = 8 cm, DB = 12 cm, AE = 6cm and EC = 9cm, then BC is equal to
 (1) $\frac{2}{5} DE$ (2) $\frac{5}{2} DE$
 (3) $\frac{3}{2} DE$ (4) $\frac{2}{3} DE$
23. A vertical stick 15cm long casts a shadow 12m long on the ground. At the same time, a tower casts a shadow 50m long on the ground. The height of the tower is
 (1) 60m (2) 62m
 (3) 62.5m (4) 63m
24. The areas of two similar triangles are 81 cm^2 and 49 cm^2 respectively,. The ratio of their corresponding heights is
 (1) 9:7 (2) 7 :9
 (3) 6:5 (4) 81:49
25. If D and E are points on the sides AB and AC respectively of a ΔABC such that $DE \parallel BC$. If AD = x, DB = x - 2, AE = x + 2 and EC = x - 1. The value of x is
 (1) 2.5 (2) 2
 (3) 3 (4) 4
26. In a ΔABC , AE is bisector of exterior $\angle CAD$ meeting BC produced in E. If AB = 10cm, AC = 6cm, BC = 12cm, then CE is equal to
 (1) 6,8 and 20cm (2) 10/3,9,12cm
 (3) 4,6 and 8cm (4) 5,6 and 8cm
27. In ΔPQR , $\angle Q = 3a$, $\angle P = a$, $\angle R = b$ and $3b - 5a = 30$, then triangle is
 (1) Scalene (2) Isosceles
 (3) Equilateral (4) Right angled
28. The points D and E are taken on the sides AB and AC of ΔABC such that AD = $(1/3)$ AB, AE = $(1/3)$ AC. If the length of BC is 15 cm, then the length of DE is:
 (1) 10 cm (2) 8 cm
 (3) 6 cm (4) 5 cm
29. D is any point on side AC of ΔABC . If P, Q, X, Y are the midpoints of AB, BC, AD and DC respectively, then the ratio of PX and QY is
 (1) 1 : 2 (2) 1 : 1
 (3) 2 : 1 (4) 2 : 3
30. In ΔABC , PQ is parallel to BC. If AP : PB = 1 : 2 and AQ = 3 cm; AC is equal to
 (1) 6 cm (2) 9 cm
 (3) 12 cm (4) 8 cm

ANSWER KEY

1.	(4)	2.	(3)	3.	(1)	4.	(3)	5.	(4)	6.	(2)	7.	(4)	8.	(3)	9.	(3)	10.	(3)
11.	(4)	12.	(4)	13.	(1)	14.	(4)	15.	(4)	16.	(4)	17.	(1)	18.	(4)	19.	(3)	20.	(3)
21.	(4)	22.	(2)	23.	(3)	24.	(1)	25.	(4)	26.	(4)	27.	(4)	28.	(4)	29.	(2)	30.	(2)

PRACTICE SHEET-5

1. Q is a point in the interior of a rectangle ABCD. If QA = 3 cm, QB = 4 cm and QC = 5 cm, then the length of QD (in cm) is
 (1) $3\sqrt{2}$ (2) $5\sqrt{2}$
 (3) $\sqrt{34}$ (4) $\sqrt{41}$
2. ABCD is a rectangle where the ratio of the length of AB and BC is 3 : 2. If P is the mid-point of AB, then the value of $\sin \angle CPB$ is
 (1) $\frac{3}{5}$ (2) $\frac{2}{5}$
 (3) $\frac{1}{5}$ (4) $\frac{4}{5}$
3. If the opposite sides of a quadrilateral and also its diagonals are equal, then each of the angles of the quadrilateral is
 (1) 90° (2) 120°
 (3) 100° (4) 60°
4. The length of the two adjacent sides of a rectangle inscribed in a circle are 5 cm and 12 cm respectively. Then the radius of the circle will be
 (1) 6 cm (2) 6.5 cm
 (3) 8 cm (4) 8.5 cm
5. PQRA is a rectangle, AP = 22 cm, PQ = 8 cm. $\triangle ABC$ is a triangle whose vertices lie on the sides of PQRA such that BQ = 2 cm and QC = 16 cm. Then the length of the line joining the mid points of the sides AB and BC is
 (1) $4\sqrt{2}$ cm. (2) 5 cm.
 (3) 6 cm. (4) 10 cm.
6. Inside a square ABCD, $\triangle BEC$ is an equilateral triangle. If CE and BD intersect at O, then $\angle BOC$ is equal to
 (1) 60° (2) 75°
 (3) 90° (4) 120°
7. A square is inscribed in a quarter-circle in such a manner that two of its adjacent vertices lie on the two radii at an equal distance from the centre, while the other two vertices lie on the circular arc. If the square has sides of length x , then the radius of the circle is
 (1) $\sqrt{2}x$ (2) $\frac{16}{(\pi+4)}$
 (3) $2x/\sqrt{\pi}$ (4) $\sqrt{5}x/\sqrt{2}$
8. Each interior angle of a regular polygon is three times its exterior angle, then the number of sides of the regular polygon is :
 (1) 9 (2) 8
 (3) 10 (4) 7
9. In a regular polygon, the exterior and interior angles are in the ratio 1 : 4. The number of sides of the polygon is
 (1) 10 (2) 12
 (3) 15 (4) 16
10. The difference between the exterior and interior angles at a vertex of a regular polygon is 150° . The number of sides of the polygon is
 (1) 10 (2) 15
 (3) 24 (4) 30
11. Each interior angle of a regular polygon is 144° . The number of sides of the polygon is
 (1) 8 (2) 9
 (3) 10 (4) 11
12. If the sum of the interior angles of a regular polygon be 1080° , the number of sides of the polygon is
 (1) 6 (2) 8
 (3) 10 (4) 12
13. The number of sides in two regular polygons are in the ratio 5 : 4 and the difference between each interior angle of the polygons is 6° . Then the numbers of sides are
 (1) 15, 12 (2) 5, 4
 (3) 10, 8 (4) 20, 16
14. Each internal angle of regular polygon is two times its external angle. Then the number of sides of the polygon is:
 (1) 8 (2) 6
 (3) 5 (4) 7
15. Ratio of the number of sides of two regular polygons is 5 : 6 and the ratio of their each interior angle is 24 : 25. Then the number of sides of these two polygons are:
 (1) 20, 24 (2) 15, 18
 (3) 10, 12 (4) 5, 6

16. Measure of each interior angle of a regular polygon can never be:
 (1) 150° (2) 105°
 (3) 108° (4) 144°
17. The sum of all interior angles of a regular polygon is twice the sum of all its exterior angles. The number of sides of the polygon is
 (1) 10 (2) 8
 (3) 12 (4) 6
18. The ratio between the number of sides of two regular polygons is 1 : 2 and the ratio between their interior angles is 2 : 3. The number of sides of these polygons is respectively
 (1) 6, 12 (2) 5, 10
 (3) 4, 8 (4) 7, 14
19. There are two regular polygons with number of sides equal to $(n - 1)$ and $(n + 2)$. Their exterior angles differ by 6° . The value of n is
 (1) 14 (2) 12
 (3) 13 (4) 11
20. If each interior angle of a regular polygon is 150° , the number of sides of the polygon is
 (1) 8 (2) 10
 (3) 15 (4) None of these
21. The sum of interior angles of a regular polygon is 1440° . The number of sides of the polygon is
 (1) 10 (2) 12
 (3) 6 (4) 8
22. Among the angles 30° , 36° , 45° , 50° one angle cannot be an exterior angle of a regular polygon. The angle is
 (1) 30° (2) 36°
 (3) 45° (4) 50°
23. If the sum of interior angles of a regular polygon is equal to two times the sum of exterior angles of that polygon, then the number of sides of that polygon is
 (1) 5 (2) 6
 (3) 7 (4) 8
24. An interior angle of a regular polygon is 5 times its exterior angle. Then the number of sides of the polygon is
 (1) 14 (2) 16
 (3) 12 (4) 18
25. The interior angle of a regular polygon is 140° . The number of sides of that polygon is
 (1) 9 (2) 8
 (3) 7 (4) 6
26. In a regular polygon if one of its internal angle is greater than the external angle by 132° , then the number of sides of the polygon is
 (1) 14 (2) 12
 (3) 15 (4) 16
27. If the ratio of an external angle and an internal angle of a regular polygon is 1 : 17, then the number of sides of the regular polygon is
 (1) 20 (2) 18
 (3) 36 (4) 12
28. The length of the diagonal BD of the parallelogram ABCD is 18 cm. If P and Q are the centroid of the $\triangle ABC$ and $\triangle ADC$ respectively then the length of the line segment PQ is
 (1) 4 cm (2) 6 cm
 (3) 9 cm (4) 12 cm
29. The side AB of a parallelogram ABCD is produced to E in such way that $BE = AB$. DE intersects BC at Q. The point Q divides BC in the ratio
 (1) 1 : 2 (2) 1 : 1
 (3) 2 : 3 (4) 2 : 1
30. In a parallelogram PQRS, angle P is four times of angle Q, then the measure of $\angle R$ is
 (1) 144° (2) 36°
 (3) 72° (4) 130°

ANSWER KEY

1.	(1)	2.	(4)	3.	(1)	4.	(2)	5.	(2)	6.	(2)	7.	(4)	8.	(2)	9.	(1)	10.	(3)
11.	(3)	12.	(2)	13.	(1)	14.	(2)	15.	(3)	16.	(2)	17.	(4)	18.	(3)	19.	(3)	20.	(4)
21.	(1)	22.	(4)	23.	(2)	24.	(3)	25.	(1)	26.	(3)	27.	(3)	28.	(2)	29.	(2)	30.	(1)

PRACTICE SHEET-6

1. Two equal circles of radius 4 cm intersect each other such that each passes through the centre of the other. The length of the common chord is :
 (1) $2\sqrt{3}$ cm (2) $4\sqrt{3}$ cm
 (3) $2\sqrt{2}$ cm (4) 8 cm
2. One chord of a circle is known to be 10.1 cm. The radius of this circle must be:
 (1) 5 cm
 (2) greater than 5 cm
 (3) greater than or equal to 5 cm
 (4) less than 5 cm
3. The length of the chord of a circle is 8 cm and perpendicular distance between centre and the chord is 3 cm. Then the radius of the circle is equal to:
 (1) 4 cm (2) 5 cm
 (3) 6 cm (4) 8 cm
4. The length of a chord of a circle is equal to the radius of the circle. The angle which this chord subtends in the major segment of the circle is equal to
 (1) 30° (2) 45°
 (3) 60° (4) 90°
5. $AB = 8$ cm and $CD = 6$ cm are two parallel chords on the same side of the centre of a circle. The distance between them is 1 cm. The radius of the circle is
 (1) 5 cm (2) 4 cm
 (3) 3 cm (4) 2 cm
6. The length of two chords AB and AC of a circle are 8 cm and 6 cm and $\angle BAC = 90^\circ$, then the radius of circle is
 (1) 25 cm (2) 20 cm
 (3) 4 cm (4) 5 cm
7. The distance between two parallel chords of length 8 cm each in a circle of diameter 10 cm is
 (1) 6 cm (2) 7 cm
 (3) 8 cm (4) 5.5 cm
8. The length of the common chord of two intersecting circles is 24 cm. If the diameter of the circles are 30 cm and 26 cm, then the distance between the centre (in cm) is
 (1) 13 (2) 14 (c) 15 (d) 16
9. In a circle of radius 21 cm, an arc subtends an angle of 72° at the centre. The length of the arc is
 (1) 21.6 cm (2) 26.4 cm
 (3) 13.2 cm (4) 19.8 cm
10. A unique circle can always be drawn through x number of given non-collinear points, then x must be:
 (1) 2 (2) 3
 (3) 4 (4) 5
11. A 8 cm long perpendicular is drawn from the centre of a circle to a 12 cm long chord. The diameter of the circle is:
 (1) 10 cm. (2) 12 cm.
 (3) 16 cm. (4) 20 cm.
12. The length of the radius of a circle with centre O is 5 cm and the length of the chord AB is 8 cm. The distance of the chord AB from the point O is
 (1) 2 cm. (2) 3 cm.
 (3) 4 cm. (4) 15 cm.
13. Two circles touch each other externally. The distance between their centres is 7 cm. If the radius of one circle is 4 cm, then the radius of the other circle will be
 (1) 3 cm. (2) 4 cm.
 (3) 5.5 cm. (4) 3.5 cm.
14. Points P, Q and R are on a circle such that $\angle PQR = 40^\circ$ and $\angle QRP = 60^\circ$. Then the subtended angle by arc QR at the centre is :
 (1) 80° (2) 120°
 (3) 140° (4) 160°
15. The length of a chord which is at a distance of 12 cm from the centre of a circle of radius 13 cm is
 (1) 10 cm. (2) 5 cm.
 (3) 6 cm. (4) 12 cm.
16. Number of circles that can be drawn through three non-collinear points is:
 (1) Exactly one (2) Two
 (3) Three (4) more than three

17. Two circles touch each other internally. The radius of the smaller circle is 6 cm and the distance between the centre of two circles is 3 cm. The radius of the larger circle is :
 (1) 7.5 cm (2) 9 cm
 (3) 8 cm (4) 10 cm
18. Length of a chord PQ of a circle with centre O is 4 cm. If the distance of PQ from the point O is 2 cm, then the length of the diameter is:
 (1) $2\sqrt{2}$ cm. (2) $3\sqrt{2}$ cm.
 (3) $5\sqrt{2}$ cm. (4) $4\sqrt{2}$ cm.
19. A chord of length 39 cm is at a distance of 10.4 cm from the centre of a circle. Find the radius of the circle.
 (1) 19.5 cm. (appr.) (2) 22.1 cm.(appr.)
 (3) 28.6 cm. (appr.) (4) 2. 21 cm. (appr.)
20. A chord of length 10 cm subtends an angle 120° at the centre of a circle. Distance of the chord from the centre is
 (1) $5\sqrt{3}$ cm. (2) $5\sqrt{3}/2$ cm
 (3) $5/\sqrt{3}$ cm. (4) 5 cm.
21. The radius of two concentric circles are 9 cm and 15 cm. If the chord of the greater circle be a tangent to the smaller circle, then the length of that chord is
 (1) 24 cm (2) 12 cm
 (3) 30 cm (4) 18 cm
22. If a chord of a circle of radius 5 cm is a tangent to another circle of radius 3 cm, both the circles being concentric, then the length of the chord is
 (1) 10 cm (2) 12.5 cm
 (3) 8 cm (4) 7 cm
23. The tangents are drawn at the extremities of diameter AB of a circle with centre P. If a tangent to the circle at the point C intersects the other two tangents at Q and R, then the measure of the $\angle QPR$ is
 (1) 45° (2) 60°
 (3) 90° (4) 180°
24. AB is a chord to a circle and PAT is the tangent to the circle at A. If $\angle BAT = 75^\circ$ and $\angle BAC = 45^\circ$, C being a point on the circle, then $\angle ABC$ is equal to
 (1) 40° (2) 45°
 (3) 60° (4) 70°
25. The tangents at two points A and B on the circle with centre O intersect at P ; If in quadrilateral PAOB, $\angle AOB : \angle APB = 5 : 1$, then measure of $\angle APB$ is :
 (1) 30° (2) 60°
 (3) 45° (4) 15°
26. Two circles touch each other externally at point A and PQ is a direct common tangent which touches the circles at P and Q respectively. Then $\angle PAQ =$
 (1) 45° (2) 90°
 (3) 80° (4) 100°
27. PR is tangent to a circle, with centre O and radius 4 cm, at point Q. If $\angle POR = 90^\circ$, OR = 5 cm and OP = $20/3$ cm, then (in cm) the length of PR is :
 (1) 3 (2) $16/3$
 (3) $23/3$ (4) $25/3$
28. Two circles touch each other externally at P. AB is a direct common tangent to the two circles, A and B are point of contact and $\angle PAB = 35^\circ$. Then $\angle ABP$ is
 (1) 35° (2) 55°
 (3) 65° (4) 75°
29. If the radii of two circles be 6 cm and 3 cm and the length of the transverse common tangent be 8 cm, then the distance between the two centres is
 (1) $\sqrt{145}$ cm (2) $\sqrt{140}$ cm
 (3) $\sqrt{150}$ cm (4) $\sqrt{135}$ cm
30. The distance between the centre of two equal circles, each of radius 3 cm, is 10 cm. The length of a transverse common tangent is
 (1) 8 cm (2) 10 cm
 (3) 4 cm (4) 6 cm

ANSWER KEY

1.	(2)	2.	(2)	3.	(2)	4.	(1)	5.	(1)	6.	(4)	7.	(1)	8.	(2)	9.	(2)	10.	(2)
11.	(4)	12.	(2)	13.	(1)	14.	(4)	15.	(1)	16.	(1)	17.	(2)	18.	(4)	19.	(2)	20.	(3)
21.	(1)	22.	(3)	23.	(3)	24.	(3)	25.	(1)	26.	(2)	27.	(4)	28.	(2)	29.	(1)	30.	(1)

CDS PYQ

1. If the arms of one angle are respectively parallel to the arms of another angle, then the two angles are

[CDS 2013(I)]

- (a) Neither equal nor supplementary
(b) Not equal but supplementary
(c) Equal but not supplementary
(d) Either equal or supplementary

2. The side BC of a $\triangle ABC$ is produced to D, bisectors of the $\angle ABC$ and $\angle ACD$ meet at P. if $\angle BPC = x^\circ$ and $\angle BAC = y^\circ$, then which one of the following option is correct?

[CDS 2013(I)]

- (a) $x^\circ = y^\circ$ (b) $x^\circ + y^\circ = 90^\circ$
(c) $x^\circ + y^\circ = 180^\circ$ (d) $2x^\circ = y^\circ$

3. ABC is a right angled triangle such that $AB = a - b$, $BC = a$ and $CA = a + b$. D is a point on BC such that $BD = AB$. The ratio of BD : DC for any value of a and b is given by:

[CDS 2013(I)]

- (a) 3 : 2 (b) 4 : 3
(c) 5 : 4 (d) 3 : 1

4. Let ABC be an equivalent triangle. If the side BC is proportional to the point D so that $BC = 2 CD$, then AD^2 is equal to:

[CDS 2013(I)]

- (a) $3 CD^2$ (b) $4 CD^2$
(c) $5 CD^2$ (d) $7 CD^2$

5. ABC is a triangle, where $BC = 2AB$, $\angle C = 30^\circ$ and $\angle A = 90^\circ$. The magnitude of the side AC is:

[CDS 2013(I)]

- (a) $\frac{2BC}{3}$ (b) $\frac{3BC}{4}$
(c) $\frac{BC}{\sqrt{3}}$ (d) $\frac{\sqrt{3}BC}{2}$

6. The bisectors BI and CI of the $\angle B$ and $\angle C$ of a $\triangle ABC$ meet in I. What is $\angle BIC$ equal to?

[CDS 2013(I)]

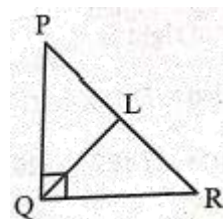
- (a) $90^\circ - \frac{A}{4}$ (b) $90^\circ + \frac{A}{4}$
(c) $90^\circ - \frac{A}{2}$ (d) $90^\circ + \frac{A}{2}$

7. In a $\triangle ABC$, $\angle BCA = 90^\circ$ and CD is perpendicular to AB. If $AD = 4$ cm and $BD = 9$ cm, then the value of DC will be:

[CDS 2013(I)]

- (a) $\sqrt{18}$ cm (b) $\sqrt{20}$ cm
(c) $\sqrt{65}$ cm (d) 6 cm

8.



In the figure given above, $\angle PQR = 90^\circ$ and QL is a median, $PQ = 5$ cm and $QR = 12$ cm. Then, QL is equal to:

[CDS 2013(I)]

- (a) 5 cm (b) 5.5 cm
(c) 6 cm (d) 6.5 cm

9. In a right angled $\triangle ABC$, $\angle C = 90^\circ$ and CD is perpendicular to AB. If $AB \times CD = CA \times CB$, then $\frac{1}{CD^2}$ is equal to:

[CDS 2013(I)]

- (a) $\frac{1}{AB^2} - \frac{1}{CA^2}$
(b) $\frac{1}{AB^2} - \frac{1}{CB^2}$
(c) $\frac{1}{BC^2} + \frac{1}{CA^2}$
(d) $\frac{1}{BC^2} - \frac{1}{CA^2}$, if $CA > CB$

10. Each side of the equilateral triangle is 6 cm. Its altitude is:

[CDS 2013(I)]

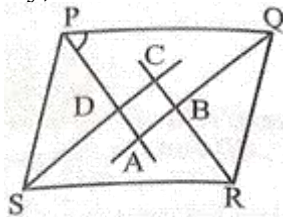
- (a) $6\sqrt{3}$ cm (b) $3\sqrt{3}$ cm
(c) $2\sqrt{3}$ cm (d) $\sqrt{3}$ cm

11. ABCD is a trapezium with parallel sides $AB = 2$ cm and $DC = 3$ cm. E and F are the mid points of the non parallel sides. The ratio of area of ABFE to area of EFCD is:

[CDS 2013(I)]

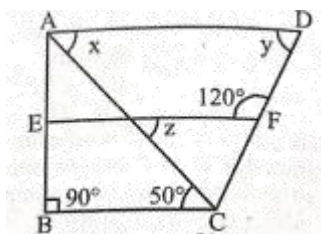
- (a) 9 : 10 (b) 8 : 9
(c) 9 : 11 (d) 11 : 9

12. In the figure given below, PQRS is a parallelogram. If AP, AQ, CR and CS are bisectors of $\angle P$, $\angle Q$, $\angle R$ and $\angle S$ respectively, then ABCD is a:



[CDS 2013(I)]

- (a) Square (b) Rhombus
(c) Rectangle (d) None of these
13. In the figure given above, ABCD is a trapezium. EF is parallel to AD and BG. $\angle y$ is equal to:



[CDS 2013(I)]

- (a) 30° (b) 45°
(c) 60° (d) 65°
14. Consider a circle with centre at O and radius r. Points A and B lies on its circumference and a point M lies outside of it such that M, A and O lies on the same straight line. Then, the ratio of MA to MB is: Then, the ratio of MA to MB is:
- [CDS 2013(I)]
- (a) Equal to 1 (b) Equal to r
(c) Greater than 1 (d) Less than 1
15. Consider the following statements:
I. The tangent of a circle is a line that meets the circle in one and only one point.
II. The tangent of a circle at the end point of the diameter is perpendicular to the diameter.
Which of the above statements is/are correct?
- [CDS 2013(I)]
- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

16. Consider the following statements:
I. If G is the centroid of $\triangle ABC$, then $GA = CB = GC$

II. If H is the ortho-centre of $\triangle ABC$, then $HA = BH = HC$
Which of the statements given above is/are correct?

[CDS 2013(II)]

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

17. If the bisector BI and CI of the angles B and C of a $\triangle ABC$ meet at the point I, then what is $\angle BIC$ equal to?

[CDS 2013(II)]

- (a) $2A$ (b) $90^\circ + \frac{A}{2}$
(c) $90^\circ - \frac{A}{2}$ (d) $90^\circ + A$

18. E is the mid-point of the median AD of a $\triangle ABC$, if BE produced meets the side AC at F, then CF is equal to:

[CDS 2013(II)]

- (a) $\frac{AC}{3}$ (b) $\frac{2AC}{3}$
(c) $\frac{AC}{3}$ (d) None of these

19. A $\triangle DEF$ is formed by joining the mid points of the sides of $\triangle ABC$. Similarly, a $\triangle PQR$ is formed by joining the mid points of the sides of the $\triangle DEF$. If the sides of the $\triangle PQR$ are of length 1, 2 and 3 units, what is the perimeter of the $\triangle ABC$?

[CDS 2013(II)]

- (a) 18 units
(b) 24 units
(c) 48 units
(d) Cannot be determined

20. PQR is an equilateral triangle is the point of intersection of altitudes PL, QM and RN. If $OP = 8\text{cm}$, then what is the perimeter of the $\triangle PQR$?

[CDS 2013(II)]

- (a) $8\sqrt{3}\text{cm}$ (b) $12\sqrt{3}\text{cm}$
(c) $16\sqrt{3}\text{cm}$ (d) $24\sqrt{3}\text{cm}$

21. In a $\triangle ABC$, $\angle B = 90^\circ$ and $\angle C = 2\angle A$, then what is AB^2 equal to?

[CDS 2013(II)]

- (a) $2BC^2$ (b) $3BC^2$
(c) $4BC^2$ (d) $5BC^2$

- 22.** The side AC of a $\triangle ABC$ is produced to D such that $BC = CD$. If $\angle ACB$ is 70° , then what is $\angle ADB$ equal to?
[CDS 2013(II)]
(a) 35° (b) 45°
(c) 70° (d) 110°
- 23.** Consider the following statements:
I. If the diagonals of a parallelogram ABCD are perpendicular, then ABCD may be a rhombus
II. If the diagonals of a quadrilateral ABCD are equal and perpendicular, then ABCD is a square
Which of the statements given below is/are correct?
[CDS 2013(II)]
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
- 24.** A quadrilateral ABCD is inscribed in a circle. If AB is parallel to CD and $AC = BD$, then the quadrilateral must be a:
[CDS 2013(II)]
(a) Parallelogram (b) Rhombus
(c) Trapezium (d) None of these
- 25.** ABCD is a quadrilateral such that $BC = BA$ and $CD > AD$. Which one of the following is correct?
[CDS 2013(II)]
(a) $\angle BAD = \angle BCD$ (b) $\angle BAD < \angle BCD$
(c) $\angle BAD > \angle BCD$ (d) None of these
- 26.** Consider the following statements in respect of two chords XY and ZT of a circle intersecting at P.
I. $PX \cdot PY = PZ \cdot PT$
II. PXZ and PTY are similar triangles
Which of the above statements is/are correct?
[CDS 2013(II)]
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
- 27.** The diameter of a circle with centre at C is 50 cm. CP is a radial segment of the circle. AB is a chord perpendicular to CP and passes through P. CP produced intersects the circle at D. If $DP = 18$ cm, then what is the length of AB?
[CDS 2013(II)]
(a) 24cm (b) 32cm
(c) 40cm (d) 48cm
- 28.** Consider the following statements:
I. The perpendicular bisector of a chord of a circle does not pass through the centre of the circle
II. The angle in a semi-circle is a right angle
Which of the above statements is/are correct?
[CDS 2013(II)]
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
- 29.** ABC is an equilateral triangle inscribed in a circle. D is any point on the arc BC. What is $\angle ADB$ equal to?
[CDS 2013(II)]
(a) 90° (b) 60°
(c) 45° (d) None of the above
- 30.** ABC and XYZ are two similar triangles with $\angle C = \angle Z$, whose areas are respectively 32cm^2 and 60.5cm^2 . If $XY = 7.7$ cm, then what is AB equal to?
[CDS 2013(II)]
(a) 5.6cm (b) 5.8cm
(c) 6.0cm (d) 6.2cm
- 31.** ABC is a triangle right angled at A and a perpendicular AD is drawn on the hypotenuse BC. What is $BC \cdot AD$ equal to?
[CDS 2013(II)]
(a) $AB \cdot AC$ (b) $AB \cdot AD$
(c) $CA \cdot CD$ (d) $AD \cdot BD$
- 32.** The height of two trees are x and y, where $x > y$. the tops of the trees are at a distance z apart. If s is the shortest distance between the trees, then what is s^2 equal to?
[CDS 2013(II)]
(a) $x^2 + y^2 - z^2 - 2xy$
(b) $x^2 + y^2 - z^2$
(c) $x^2 + y^2 + z^2 - 2xy$
(d) $z^2 - x^2 - y^2 + 2xy$
- 33.** The three sides of a triangle are 15, 25, x units. Which one of the following is correct?
[CDS 2014(II)]
(a) $10 < x < 40$ (b) $10 \leq x \leq 40$
(c) $10 \leq x < 40$ (d) $10 < x \leq 40$
- 34.** Which one of the following is Pythagorean triple in which one side differs from the hypotenuse by two units?
[CDS 2014(I)]
(a) $(2n+1, 4n, 2n^2 + 2n)$
(b) $(2n, 4n, n^2 + 1)$
(c) $(2n^2, 2n, 2n + 1)$
(d) $(2n, n^2 - 1, n^2 + 1)$

Where, n is a positive real number?

35. The sides of a right angled triangle are equal to three consecutive numbers expressed in centimeter. What can be the area of such a triangle?

[CDS 2014(I)]

- (a) 6cm^2 (b) 8cm^2
(c) 10cm^2 (d) 12cm^2

36. If triangles ABC and DEF are similar such that $2AB = DE$ and $BC = 8\text{cm}$, then what is EF equal to?

[CDS 2014(I)]

- (a) 16cm (b) 12cm
(c) 10cm (d) 8cm

37. The sides of a triangle are in geometric progression with common ratio $r < 1$. If the triangle is a right angled triangle, the square of common ratio is given by:

[CDS 2014(I)]

- (a) $\frac{\sqrt{5}+1}{2}$ (b) $\frac{\sqrt{5}-1}{2}$
(c) $\frac{\sqrt{3}+1}{2}$ (d) $\frac{\sqrt{3}-1}{2}$

38. In a $\triangle ABC$, AD is a perpendicular of BC and BE is perpendicular to AC. Which of the following is correct?

[CDS 2014(I)]

- (a) $CE \times CB = CA \times CD$
(b) $CE \times CA = CD \times CB$
(c) $AD \times BD = AE \times BE$
(d) $AB \times AC = AD \times BE$

39. Let ABC is triangle right angled at B. If $AB = 6\text{cm}$ and $BC = 8\text{cm}$, then what is the length of the circum-radius of the $\triangle ABC$?

[CDS 2014(I)]

- (a) 10cm (b) 7cm
(c) 6cm (d) 5cm

40. If AD is the internal angular bisector of $\triangle ABC$ with $AB = 3\text{cm}$ and $AC = 1\text{cm}$ then what is $BD : BC$ equal to?

[CDS 2014(I)]

- (a) $1:3$ (b) $1:4$
(c) $2:3$ (d) $3:4$

41. AB is a straight line, C and D are points the same side of AB such that AC is perpendicular to AB and BD is

perpendicular to AB. Let AD and BC meet at E. What is $\frac{AE}{AD} + \frac{BE}{BC}$ equal to?

[CDS 2014(I)]

- (a) 2 (b) 1.5
(c) 1 (d) None of these

42. Two light rods $AB = a + b$, $CD = a - b$ symmetrically lying on a horizontal AB. There are kept intact by two strings AC and BD. The perpendicular distance between rods is a . The length of AC is given by:

[CDS 2014(I)]

- (a) a (b) b
(c) $\sqrt{a^2 - b^2}$ (d) $\sqrt{a^2 + b^2}$

43. If PQRS be a rectangle such $PQ = \sqrt{3} QR$. Then, what is $\angle PRS$ equal to?

[CDS 2014(I)]

- (a) 60° (b) 45°
(c) 30° (d) 15°

44. In a trapezium, the two non-parallel sides are equal in length, each being of 5cm . The parallel sides are at a distance of 3cm apart. If the smaller side of the parallel sides is of length 2cm , then the sum of the diagonals of the trapezium is:

[CDS 2014(I)]

- (a) $10\sqrt{5}\text{cm}$ (b) $6\sqrt{5}\text{cm}$
(c) $5\sqrt{5}\text{cm}$ (d) $3\sqrt{5}\text{cm}$

45. Let ABCD be a parallelogram. Let P, Q, R and S be the mid points of sides AB, BC, CD and DA respectively. Consider the following statements.

I. Area of triangle APS < Area of triangle DSR, if $BD < AC$.

II. Area of triangle ABC = 4 (Area of triangle BPQ)

Select the correct answer using the codes given below:

[CDS 2014(I)]

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

46. Consider the following statement:

I. Let ABCD be a parallelogram which is not a rectangle, then $2(AB^2 + BC^2) \neq AC^2 + BD^2$

II. If ABCD is a rhombus with $AB = 4\text{cm}$, then $AC^2 + BD^2 = n^3$ for some positive integer n .

Which of the above statements is/are correct?

- [CDS 2014(I)]**
- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
- 47.** ABCD is a parallelogram, E is a point on BC such that $BE : EC = m : n$. If AE and DB intersect in F, then what is the ratio of the area of $\triangle FEB$ to the area of $\triangle AFD$?
- [CDS 2014(I)]**
- (a) m/n (b) $(m/n)^2$
(c) $(n/m)^2$ (d) $[m/(m+n)]^2$
- 48.** A circle of radius 10 cm has an equilateral triangle inscribed in it. The length of the perpendicular drawn from the centre to any side of the triangle is
- [CDS 2014(I)]**
- (a) $2.5\sqrt{3}$ cm (b) $5\sqrt{3}$ cm
(c) $10\sqrt{3}$ cm (d) None of these
- 49.** AB and CD are two chords of a circle meeting externally at P. Then, which of the following is/are correct?
I. $PA \times PD = PC \times PB$
II. $\triangle PAC$ and $\triangle PDB$ are similar.
Select the correct answer using the codes given below.
- [CDS 2014(I)]**
- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II
- 50.** In a $\triangle ABC$, $AB = BC = CA$. The ratio of the radius of the circum-circle to that of the incircle is
- [CDS 2014(I)]**
- (a) 2 : 1 (c) 3 : 1
(c) 3 : 2 (d) None of these
- 51.** Three straight lines are drawn through the three vertices of a $\triangle ABC$, the line through each vertex being parallel to the opposite side. The $\triangle DEF$ is bounded by these parallel lines:
Consider the following statements in respect of the $\triangle DEF$.
1. Each side of $\triangle DEF$ is double the side of the $\triangle ABC$ to which it is parallel
2. Area of $\triangle DEF$ is four times the area of $\triangle ABC$
Which of the above statements is/are correct?
- [CDS 2014(II)]**
- (a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2
- 52.** In a $\triangle ABC$, if $\angle B = 2\angle C = 2\angle A$. Then what is the ratio of AC to BC?
- [CDS 2014(II)]**
- (a) $\sqrt{2} : 1$ (b) $\sqrt{3} : 1$
(c) 1 : 1 (d) $1 : \sqrt{2}$
- 53.** For a triangle, the radius of the circum-circle is double the radius of the inscribed circle, then which one of the following is correct?
- [CDS 2014(II)]**
- (a) The triangle is a right angled
(b) The triangle is an isosceles
(c) The triangle is an equilateral
(d) None of the above
- 54.** Consider the following statements in respect of an equilateral triangle:
- [CDS 2014(II)]**
- The altitudes are congruent
 - The three medians are congruent
 - The centroid bisects the altitude
- Which of the above statements are correct?
- (a) 1 and 2 (b) 2 and 3
(c) 1 and 3 (d) 1, 2 and 3
- 55.** Consider the following:
1. ABC and DEF are triangles in a plane such that AB is parallel to DE, BC is parallel to EF and CA is parallel to FD.
Statement I if $\angle ABC$ is a right angle, then $\angle DEF$ is also a right angle
Statement II Triangles of the type ABC and DEF are always congruent
Which one of the following is correct in respect of the above statements?
- [CDS 2014(II)]**
- (a) Statement I and II are correct and Statement II is the correct explanation of Statement I
(b) Statement I and II are correct and Statement II is not the correct explanation of Statement I
(c) Statement I is correct and Statement II is incorrect
(d) Statement I is incorrect and Statement II is correct
- 56.** Let the incircle to a $\triangle ABC$ touch BC, AC and AB respectively at the point X, Y and Z.
Statement I if $AB > BC$ then $AB + AZ < BC + XC$
Statement II $AZ = AY$
- [CDS 2014(II)]**

Which one of the following is correct in respect of the above statements?

- (a) Statement I and II are correct and Statement II is the correct explanation of Statement I
- (b) Statement I and II are correct and Statement II is not the correct explanation of Statement I
- (c) Statement I is correct and Statement II is incorrect
- (d) Statement I is incorrect and Statement II is correct

- 57.** Let ABC be a triangle in which $\angle ACB = 60^\circ$ and $AC = x < BC$. Let the circle with centre at C and radius x meet BC at D. Let CF be the perpendicular drawn from C meeting AD at F.

Statement I $\triangle ACD$ is isosceles but not equilateral

Statement II $DF = x/2$

Which one of the following is correct in respect of the above statements?

[CDS 2014(II)]

- (a) Statement I and II are correct and Statement II is the correct explanation of Statement I
- (b) Statement I and II are correct and Statement II is not the correct explanation of Statement I
- (c) Statement I is correct and Statement II is incorrect
- (d) Statement I is incorrect and Statement II is correct

- 58.** Let ABCD be a parallelogram. Let X and Y be the mid points of the sides BC and AD, respectively. Let M and N be the mid points of the sides AB and CD, respectively.

Consider the following statements:

1. The straight line MX cannot be parallel to YN
2. The straight lines AC, BD, XY and MN meet at a point.

Which of the above statements is/are correct?

[CDS 2014(II)]

- (a) Only 1
- (b) Only 2
- (c) Both 1 and 2
- (d) Neither 1 nor 2

Direction (for next two): Read the following information carefully and answer the given questions that follow.

A piece of land is in the form of a parallelogram and the perimeter of the land is 86m. The length of one side exceeds the

other by 13 m and one of the diagonals is 41m.

[CDS 2014(II)]

- 59.** What is the area of the parallelogram?

- (a) $63m^2$
- (b) $96m^2$
- (c) $126m^2$
- (d) $252m^2$

- 60.** What is the shorter height of the parallelogram?

- (a) 9.0m
- (b) 7.5m
- (c) 5.5m
- (d) 4.5m

- 61.** Consider the following statements:

1. The difference between the diagonals of the parallelogram is more than 20 m
2. The difference between the height of the parallelogram is more than 10 m.

Which of the above statements is/are correct?

[CDS 2014(II)]

- (a) Only 1
- (b) Only 2
- (c) Both 1 and 2
- (d) Neither 1 nor 2

Direction (for next two): Read the following information carefully and answer the given questions that follow.

ABCD is a trapezium, in which AB is parallel to CD. Let M be the mid-point of BC.

- 62.** Consider the following statements:

I. Area of $\triangle ADM$ + Area of $\triangle DCM$ is equal to three fourth of the area of trapezium ABCD, if $AB = CD$

II. Area of $\triangle DCM$ + area of $\triangle ABM$ is always greater than half of the area of trapezium ABCD.

Which of the above statements is/are correct?

[CDS 2014(II)]

- (a) Only 1
- (b) Only 2
- (c) Both 1 and 2
- (d) Neither 1 nor 2

- 63.** Consider the following statements:

I. Area of $\triangle ADM$ – Area of $\triangle ABM$ is always equal to area of $\triangle DCM$, if $AB = CD$.

II. Area of $\triangle ABM$ is equal to one eighth of area of trapezium ABCD, if $AB = CD$.

Which of the above statements is/are correct?

[CDS 2014(II)]

- (a) Only 1
- (b) Only 2
- (c) Both 1 and 2
- (d) Neither 1 nor 2

- 64.** ABCD is a parallelogram. P and R are the mid points of DC and BC, respectively. The

line PR intersects the diagonal AC at Q. The distance CQ will be :

[CDS 2014(II)]

- (a) AC/4 (b) BD/3
(c) BD/4 (d) AC/3

65. Bisectors of two adjacent angles A and B of a quadrilateral ABCD intersect each other at a point P. Which one of the following is correct?

[CDS 2014(II)]

- (a) $2\angle APB = \angle C + \angle D$
(b) $\angle APB = \angle C + \angle D$
(c) $\angle APB = 180^\circ - (\angle C + \angle B)$
(d) $\angle APB = 180^\circ - (\angle C + \angle D)$

66. If the chord of an arc of a circle is of length x, the height of the arc is y and the radius of the circle is z. Then, which one of the following is correct?

[CDS 2014(II)]

- (a) $y(2z-y) = x^2$ (b) $y(2z-y) = 4x^2$
(c) $2y(2z-y) = x^2$ (d) $4y(2z-y) = x^2$

67. The sides of a triangle are in the ratio $\frac{1}{2} : \frac{1}{3} : \frac{1}{4}$. If its perimeter is 52cm, then what is the length of the smallest side?

[CDS 2014(II)]

- (a) 9cm (b) 10cm
(c) 11cm (d) 12cm

68. In a triangle ABC, AD is the median through A and E is the midpoint of AD, and BE produced meets AC at F. Then AF is equal to:

[CDS 2014(II)]

- (a) AC/5 (b) AC/4
(c) AC/3 (d) AC/2

69. Let OA, OB, OC and OD are rays in the anticlockwise direction such that $\angle AOB = \angle COD = 100^\circ$, $\angle BOC = 82^\circ$ and $\angle AOD = 78^\circ$. Consider the following statements :

I. AOC and BOD are lines

II. $\angle BOC$ and $\angle AOD$ are supplementary

Which of the above statements is/are correct?

[CDS 2015(I)]

- (a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

70. The angles of a triangle are in the ratio 4 : 1 : 1. Then the ratio of the largest side to the perimeter is:

[CDS 2015(I)]

- (a) $\frac{2}{3}$ (b) $\frac{1}{2+\sqrt{3}}$
(c) $\frac{\sqrt{3}}{2+\sqrt{3}}$ (d) $\frac{2}{1+\sqrt{3}}$

71. Let a, b, c be the sides of a right triangle, where c is the hypotenuse. The radius of the circle which touches the sides of the triangle is:

[CDS 2015(I)]

- (a) $(a + b - c)/2$ (b) $(a + b + c)/2$
(c) $(a + 2b + 2c)/2$ (d) $(2a + 2b - c)/2$

72. The area of the largest triangle that can be inscribed in a semicircle of radius r is:

[CDS 2015(I)]

- (a) r^2 (b) $2r^2$
(c) $3r^2$ (d) $4r^2$

73. Consider the following statements:

1. Let D be a point on the side BC of a triangle ABC. If area of triangle ABD = area of triangle ACD, then for all points O on AD, area of triangle ABO = area of triangle ACO.

2. If G is the point of concurrence of the medians of a triangle ABC, then area of triangle ABG = area of triangle BCG = area of triangle ACG.

Which of the above statements is/area correct?

[CDS 2015(I)]

- (a) 1 Only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

74. If each interior angle of a regular polygon is 135° , then the number of diagonals of the polygon is equal to:

[CDS 2015(I)]

- (a) 54 (b) 48
(c) 20 (d) 18

75. AD is the diameter of a circle and AB is a chord. If AD = 34 cm, AB = 30cm, the distance of AB from the center of the circle is:

[CDS 2015(I)]

- (a) 17cm (b) 15cm
(c) 13cm (d) 8cm

76. If a star figure is formed by elongating the sides of a regular pentagon, then the measure of each angle at the angular points of the star figure is:

[CDS 2015(I)]

- (a) 36° (b) 35°
(c) 32° (d) 30°
- 77.** The area of a rhombus with side 13cm and one diagonal 10cm will be:
[CDS 2015(I)]
(a) 140 square cm (b) 13 square cm
(c) 120 square cm (d) 110 square cm
- 78.** The diagonals of a trapezium are at right angles, and the slant sides, if produced, form an equilateral triangle with the greater of the two parallel sides. If the area of the trapezium is 16 square cm, then the distance between the parallel sides is:
[CDS 2015(I)]
(a) 2cm
(b) 4cm
(c) 8cm
(d) cannot be determined due to insufficient data
- 79.** Two circles, each of radius r , with centres P and O , are such that each circle passes through the centre of the other circle. Then the area common to the circles is less than one-third of the sum of the areas of the two circles.
[CDS 2015(I)]
(a) $\frac{\sqrt{3}r^2}{4}$ (b) $\frac{\sqrt{3}r^2}{3}$
(c) $\frac{\sqrt{3}r^2}{2}$ (d) $\sqrt{3}r^2$
- 80.** Three equal circles each of diameter d are drawn on a plane in such a way that each circle touches the other two circles. A big circle is drawn in such a manner that it touches each of the small circles internally. The area of the big circle is
[CDS 2015(I)]
(a) πd^2 (b) $\pi d^2(2 - \sqrt{3})^2$
(c) $\frac{\pi d^2(\sqrt{3} + 1)^2}{2}$ (d) $\frac{\pi d^2(\sqrt{3} + 2)^2}{12}$
- 81.** If the angle between the radii of a circle is 130° , then the angle between the tangents at the ends of the radii is
[CDS 2015(I)]
(a) 90° (b) 70°
(c) 50° (d) 40°
- 82.** Out of two concentric circles, the diameter of the outer circle is 26 cm and the chord MN of length 24cm is tangent to the inner circle. The radius of the inner circle
[CDS 2015(I)]
(a) 5 cm (b) 6 cm
(c) 8 cm (d) 10 cm
- 83.** $ABCD$ is a parallelogram, where $AB : AD = 2:1$. One of the angles of the parallelogram is 60° . The two diagonals are in the ratio.
[CDS 2015(I)]
(a) 7 : 3
(b) $\sqrt{7} : \sqrt{3}$
(c) 7 : 5
(d) None of the above
- 84.** The sides of a triangle are 25cm, 39cm and 56cm. The perpendicular from the opposite vertex on the side of 56cm is:
[CDS 2015(I)]
(a) 10cm (b) 12cm
(c) 15cm (d) 16cm
- 85.** From a circular piece of cardboard of radius 3cm, two sectors of 40° each have been cut off. The area of the remaining portion is:
[CDS 2015(I)]
(a) 11 square cm (b) 22 square cm
(c) 33 square cm (d) 44 square cm
- 86.** Consider the following statements:
1. If non-parallel sides of a trapezium are equal, then it is cyclic.
2. If the chord of a circle is equal to its radius, then the angle subtended by this chord at a point in major segment is 30° .
Which of the above statements is/are correct?
[CDS 2015(I)]
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2
- 87.** The point O is equidistant from the three sides of a triangle ABC . Consider the following statements:
1. $\angle OAC + \angle OCB + \angle OBA = 90^\circ$
2. $\angle BOC = 2 \angle BAC$
3. The perpendiculars drawn from any point on OA to AB and AC are always equal
Which of the above statements are correct?
[CDS 2015(II)]
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

88. ABCD is a parallelogram with AB and AD as adjacent sides. If $\angle A = 60^\circ$ and $AB = 2AD$, then the diagonal BD will be equal to:

[CDS 2015(II)]

- (a) $\sqrt{2}$ AD (b) $\sqrt{3}$ AD
(c) 2AD (d) 3AD

89. If X is any point within a square ABCD and on AX a square AXYZ is described, which of the following is/are correct?

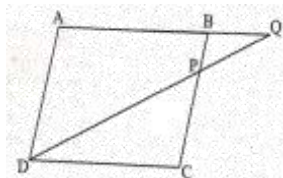
1. $BX = DZ$ or $BZ = DX$
2. $\angle ABX = \angle ADZ$ or $\angle ADX = \angle ABZ$

Select the correct answer using the code given below:

[CDS 2015(II)]

- (a) 1 Only (b) 2 Only
(c) Both 1 and 2 (d) Neither 1 nor 2

90.



In the above figure, ABCD is a parallelogram. P is a point on BC such that $PB : PC = 1 : 2$. DP and AB when both produced meet at Q. If area of triangle BPQ is 20 square unit, the area of triangle DCP is:

[CDS 2015(II)]

- (a) 20 square unit
(b) 30 square unit
(c) 40 square unit
(d) None of the above

91. A circle of radius r is inscribed in a regular polygon with n sides (the circle touches all sides of the polygon). If the perimeter of the polygon is p , then the area of the polygon is:

[CDS 2015(II)]

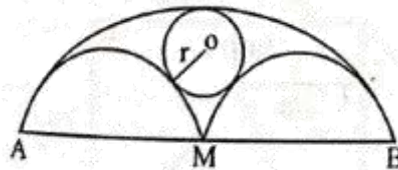
- (a) $(p + n)r$
(b) $(2p - n)r$
(c) pr
(d) None of the above

92. The two adjacent sides of a cyclic quadrilateral are 2 cm and 5 cm and the angle between them is 60° . If the third side is 3 cm, then the fourth side is of length.

[CDS 2015(II)]

- (a) 2 cm (b) 3 cm
(c) 4 cm (d) 5 cm

93.

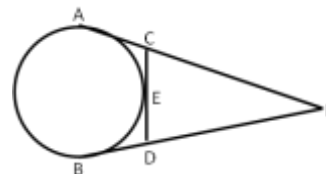


AB is a line segment of length $2a$, with M as mid-point. Semicircles are drawn on one side with AM, MB and AB as diameter as shown in the above figure. A circle with centre O and radius r is drawn such that this circle touches all the three semicircles. The value of r is:

[CDS 2015(II)]

- (a) $\frac{2a}{3}$ (b) $\frac{a}{2}$
(c) $\frac{a}{3}$ (d) $\frac{a}{4}$

94.



From an external point P tangents PA and PB are drawn to the circle as shown in the above figure. CD is the tangent to the circle at E. If $AP = 16$ cm, then the perimeter of the triangle PCD is equal to:

[CDS 2015(II)]

- (a) 24 cm (b) 28 cm
(c) 30 cm (d) 32 cm

95. Chord CD intersects the diameter AB of a circle at right angle at a point P in the ratio $1 : 2$. If diameter of circle is D then CD is equal to:

[CDS 2015(II)]

- (a) $\frac{\sqrt{2d}}{3}$ (b) $\frac{2d}{3}$
(c) $\frac{2\sqrt{2d}}{3}$ (d) $\frac{2\sqrt{3d}}{3}$

96. An equilateral triangle BOC is drawn inside a square ABCD. If angle AOD = 2θ , what is $\tan\theta$ equal to?

[CDS 2015(II)]

- (a) $2 - \sqrt{3}$ (b) $1 + \sqrt{2}$
(c) $4 - \sqrt{3}$ (d) $2 + \sqrt{3}$

97. Two poles are placed at P and Q on either side of a road such that the line joining P and Q is perpendicular to the length of the road. A person moves x meter away from P parallel to the road and places another pole at R . Then the person moves further x meter in the same direction and turns and moves a distance y meter away from the road perpendicularly, where he finds himself, Q and R on the same line. The distance between P and Q (i.e. the width of the road in meter is)

[CDS 2016(I)]

- (a) x (b) $\frac{x}{2}$
(c) y (d) $2y$

98. There are five lines in a plane, no two of which are parallel. The maximum number of the points in which they can intersect is.

[CDS 2016(I)]

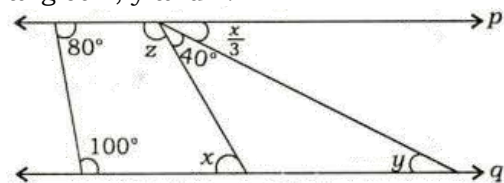
- (a) 4
(b) 6
(c) 10
(d) None of the above

99. If a transversal intersects four parallel straight lines, then the number of distinct values of the angles formed will be

[CDS 2016(I)]

- (a) 2 (b) 4
(c) 8 (d) 16

100. In the figure given below, P and q are parallel lines. What are the values of the angles x , y and z ?



[CDS 2016(I)]

- (a) $x = 80^\circ, y = 40^\circ, z = 100^\circ$
(b) $x = 80^\circ, y = 50^\circ, z = 105^\circ$
(c) $x = 70^\circ, y = 40^\circ, z = 110^\circ$
(d) $x = 60^\circ, y = 20^\circ, z = 120^\circ$

101. Let ABC and $A'B'C'$ be two triangles in which $AB > A'B'$, $BC > B'C'$ and $CA > C'A'$. Let D , E and F be the mid points of the sides BC , CA and AB respectively. Let $D'E'$ and F be the midpoints of the sides $B'C'$, $C'A'$ and $A'B'$ respectively.

Consider the following statements:

Statement I. $AD > A'D'$, $BE > B'E'$ and $CF > C'F'$ are always true

Statement II.

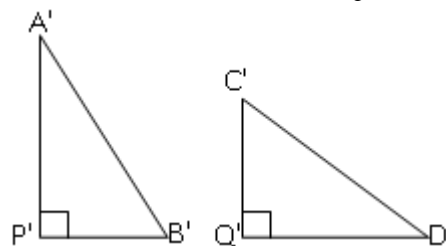
$$\frac{AB^2 + BC^2 + CA^2}{AD^2 + BE^2 + CF^2} = \frac{A'B'^2 + B'C'^2 + C'A'^2}{A'D'^2 + B'E'^2 + C'F'^2}$$

Which one of the following is correct in respect of the above statements?

[CDS 2016(I)]

- (a) Both Statement I and statement II are true and statement II are true and statement II is the correct explanation of statement I.
(b) Both Statement I and statement II are true but statement II is not the correct explanation of statement I.
(c) Statement I is true but statement II is false
(d) Statement I is false but statement II is true

102. Suppose chords AB and CD of a circle intersect at a point P inside the circle. Two right angled triangles $A'P'B'$ and $C'Q'D'$ are formed as shown in the figures below such that $A'P' = AP$, $B'P' = BP$, $C'Q' = CP$, $D'Q' = DP$ and $\angle A'P'B' = 90^\circ = \angle C'Q'D'$.



Which of the following statements are not correct?

1. $A'P'B'$ and $C'Q'D'$ are similar triangles but need not be congruent
2. $A'P'B'$ and $C'Q'D'$ are congruent triangles
3. $A'P'B'$ and $C'Q'D'$ are triangles of same area
4. $A'P'B'$ and $C'Q'D'$ are triangles of same perimeter

Select the correct answer using the code given below:

[CDS 2016(I)]

- (a) 2 and 3 only (b) 1 and 3 only
(c) 1, 2 and 4 only (d) 1, 2, 3 and 4

103. Suppose ABC is a triangle with AB of unit length D and E are the points lying on AB and AC respectively such that BC and DE are parallel. If the area of triangle ABC is twice the area of triangle ADE , then the length of AD is:

[CDS 2016(I)]

- (a) $\frac{1}{2}$ unit (b) $\frac{1}{3}$ unit
(c) $\frac{1}{\sqrt{2}}$ unit (d) $\frac{1}{\sqrt{3}}$ unit

104. Let the triangles ABC and DEF be such that $\angle ABC = \angle DEF$, $\angle ACB = \angle DFB$ and $\angle BAC = \angle EDF$. Let L be the midpoint of BC and M be the midpoint of EF. Consider the following statements:

Statements I. Triangle ABL and DEM are similar

Statement II. Triangle ALC is congruent to triangle DMF even in $AC \neq DF$
Which one of the following is correct in respect of the above statement?

[CDS 2016(I)]

- (a) Statement I and II are correct and Statement II is the correct explanation of Statement I
(b) Statement I and II are correct and Statement II is not the correct explanation of Statement I
(c) Statement I is correct and Statement II is incorrect
(d) Statement I is incorrect and Statement II is correct

105. ABC and DEF are similar triangles. If the ratio of side AB to side DE is $(\sqrt{2}+1:\sqrt{3})$, then the ratio of area of triangle ABC to that of triangle DEF is:

[CDS 2016(I)]

- (a) $(3-2\sqrt{2}):3$ (b) $(9-6\sqrt{2}):2$
(c) $1(9-6\sqrt{2})$ (d) $(3+2\sqrt{2}):3$

106. In a triangle ABC if $A-B = \frac{\pi}{2}$, then $C + 2B$ is equal to:

[CDS 2016(I)]

- (a) $\frac{2\pi}{3}$ (b) $\frac{3\pi}{4}$
(c) π (d) $\frac{\pi}{2}$

107. Let ABC be a triangle in which $AB = AC$. Let L be the locus of point X inside or on the triangle such that $BX = CX$. Which of the following statements are correct?

1. L is a straight line passing through A and in center of triangle ABC is on L

2. L is a straight line passing through A and orthocenter of triangle ABC is on a point L
3. L is a straight line passing through A and in center of triangle ABC is point on L
Select the correct answer using the code given below:

[CDS 2016(I)]

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

108. In a triangle PQR, point X is on PQ and point Y is on PR such that $XP = 1.5$ units, $XQ = 6$ unit, $PY = 2$ units and $YR = 8$ units. Which of the following are correct?

1. $QR = 5XY$
2. QR is parallel to XY
3. Triangle PYX is similar to triangle PRQ
Select the correct answer using the code given below:

[CDS 2016(I)]

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

109. A person travels 7km north and then turns right and travels 3 km and further turns right and travels 13km. What is the shortest distance of the present position of the person from his starting point?

[CDS 2016(I)]

- (a) 6km (b) $3\sqrt{5}$ km
(c) 7km (d) $4\sqrt{5}$ km

110. ABC is a triangle in which D is the midpoint of BC and E is the midpoint of AD. Which of the following statements is/are correct?

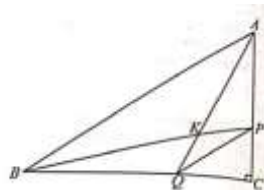
1. The area of triangle ABC is equal to four times the area of triangle BED
2. The area of triangle ADC is twice the area of triangle BED

Select the correct answer using the code given below:

[CDS 2016(I)]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

111.

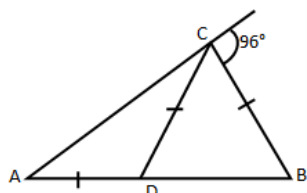


ABC is a triangle right angled at C as shown in the figure above. Which one of the following is correct?

[CDS 2016(I)]

- (a) $AQ^2 + AB^2 = BP^2 + PQ^2$
- (b) $AQ^2 + PQ^2 = AB^2 + BP^2$
- (c) $AQ^2 + BP^2 = AB^2 + PQ^2$
- (d) $AQ^2 + AP^2 = BK^2 + KQ^2$

112.



In the figure given above, $AD = CD = BC$. What is the value of $\angle CBD$.

[CDS 2016(I)]

- (a) 32°
- (b) 64°
- (c) 78°
- (d) Cannot be determined due to insufficient data

113. ABC is an equilateral triangle and X, Y and Z are the points of BC, CA and AB respectively such that $BX = CY = AZ$. Which of the following is/are correct?

- 1. XYZ is an equilateral triangle
 - 2. Triangle XYZ is similar to triangle ABC
- Select the correct answer using the code given below:

[CDS 2016(I)]

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

114. A rhombus is formed by joining midpoints of the sides of a rectangle in the suitable order. If the area of the rhombus is 2 square units, then the area of the rectangle is:

[CDS 2016(I)]

- (a) $2\sqrt{2}$ square units
- (b) 4 square units
- (c) $4\sqrt{2}$ square units
- (d) 8 square units

115. If each interior angle of a regular polygon is 140° , then the number of vertices of the polygon is equal to:

[CDS 2016(I)]

- (a) 10
- (b) 9
- (c) 8
- (d) 7

116. Consider the following statements:

- I. If $n \geq 3$ and $m \geq 3$ are distinct positive integers, then the sum of the exterior angles of a regular polygon of m sides is

different from the sum of the exterior angles of a regular polygon of n sides.

- II. Let m, n be integers such that $m > n \geq 3$. Then the sum of the interior angles of a regular polygon of m sides is greater than the sum of the interior angles of a regular polygon of n sides, and their sum is $(m+n)\frac{\pi}{2}$.

Which of the above statements is/are correct?

[CDS 2016(I)]

- (a) 1 Only
- (b) 2 Only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

117. Consider the following statements:

- 1. There exists a regular polygon whose exterior angles is 70° .
 - 2. Let $n \geq 5$. Then the exterior angle of any regular polygon of n sides is acute.
- Which of the above statements is/are correct?

[CDS 2016(I)]

- (a) 1 Only
- (b) 2 Only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

118. In a circle of radius 2 unit, a diameter AB intersects a chord of length 2 units perpendicularly at P. If $AP > BP$, then AP is equal to:

[CDS 2016(I)]

- (a) $(2 + \sqrt{5})$ units
- (b) $(2 + \sqrt{3})$ units
- (c) $(2 + \sqrt{2})$ units
- (d) 3 units

119. A truck moves along a circular path and describes 100m when it has traced out 36° at the centre. The radius of the circle is equal to:

[CDS 2016(I)]

- (a) $\frac{100}{\pi} m$
- (b) $\frac{250}{\pi} m$
- (c) $\frac{500}{\pi} m$
- (d) $\frac{600}{\pi} m$

120. A tangent is drawn from an external point O to a circle of radius 3 units at P such that $DP = 4$ units. If O is the centre of the circle, the sine of the angle ODP is:

[CDS 2016(I)]

- (a) $\frac{4}{5}$
- (b) $\frac{3}{4}$
- (c) $\frac{3}{5}$
- (d) $\frac{1}{2}$

121. Consider a circle with centre at O and radius 7cm. Let or be a chord of length 2cm

and let P be the midpoint of QR. Let CD be another chord of the circle passing through P such that $\angle PCQ$ is acute. If M is the midpoint of CD and $MP = \sqrt{24}$ cm, then which of the following statements are correct?

1. If $CP = 135^\circ$
 2. If $CP = m$ cm and $PD = n$ cm, then m and n are the roots of the quadratic equation $x^2 - 10x + 1 = 0$
 3. The ratio of the area of triangle OPR to the area of triangle OMP is $1 : 2\sqrt{2}$.
- Select the correct answer using the code given below:

[CDS 2016(I)]

- (a) 1 and 2 only (b) 2 and 2 only
(c) 1 and 3 only (d) 1, 2 and 3

122. Consider a circle with center at C. Let OP, OQ denote respectively the tangents to the circle drawn from a point O outside the circle. Let R be a point on OP and S be a point on OQ such that $OR \times SQ = OS \times RP$. Which of the following statement is/are correct?

1. If X is the circle with centre at O and radius OR, and Y is the circle with centre at O and radius OS, then $X = Y$.
 2. $\angle POC + \angle QCO = 90^\circ$
- Select the correct answer using the code given below:

[CDS 2016(I)]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

123. A circular path is made from two concentric circular rings in such a way that the smaller ring when allowed to roll over the circumference of the bigger ring, it takes three full revolutions. If the area of the pathway is equal to n times the area of the smaller ring, then n is equal to:

[CDS 2016(I)]

- (a) 4 (b) 6
(c) 8 (d) 10

124. Let ABC be a right angled triangle with $BC = 5$ cm, and $AC = 12$ cm. Let D be a point on the hypotenuse AB such that $\angle BCD = 30^\circ$. What is length of CD?

[CDS 2016(II)]

- (a) $\frac{60}{13}$ cm (b) $\frac{17}{2}$ cm
(c) $\frac{120}{5+12\sqrt{2}}$ cm (d) $\frac{120}{5+12\sqrt{3}}$ cm

125. In an equilateral triangle another equilateral triangle is drawn inside joining the mid-point of the sides of given equilateral triangle and the process is continued up to 7 times. What is the ratio of area of fourth triangle to that of seventh triangle?

[CDS 2016(II)]

- (a) 256:1 (b) 128:1
(c) 64 :1 (d) 16:1

126. The sides of a triangle are given by $\sqrt{a^2+b^2}, \sqrt{c^2+a^2}$ and $(b+c)$ where, a, b, c are positive. What is the area of the triangle equal to?

[CDS 2016(II)]

- (a) $\frac{\sqrt{a^2+b^2+c^2}}{2}$
(b) $\frac{\sqrt{a^2b^2+b^2c^2+c^2a^2}}{2}$
(c) $\frac{a(b+c)}{2}$
(d) $\frac{\sqrt{3(a^2b^2+b^2c^2+c^2a^2)}}{2}$

127. Two circles touch externally and sum of their areas is 130π cm² and distance between their centres is 14 cm. What is the difference in the radii of the circles?

[CDS 2016(II)]

- (a) 5 cm (b) 6 cm
(c) 7 cm (d) 8 cm

128. In a circle of radius 3 units, a diameter AB, intersects a chord of length 2 units perpendicularly at P. If $AP > BP$, then what is the ratio of AP to BP?

[CDS 2016(II)]

- (a) $3 + \sqrt{10} : 3 - \sqrt{10}$ (b) $3 + \sqrt{8} : 3 - \sqrt{8}$
(c) $3 + \sqrt{3} : 3 - \sqrt{3}$ (d) $3 : \sqrt{3}$

129. If two tangents inclined at an angle 60° are drawn to a circle of radius 3 cm, then what is the length of each tangent?

[CDS 2016(II)]

- (a) $3\sqrt{3}$ cm (b) $\sqrt{3}$ cm
(c) 6 cm (d) $2\sqrt{2}$ cm

130. ABCDA is a con-cyclic quadrilateral of a circle ABCD with radius r and centre at O. If AB is the diameter and CD is parallel and half of AB and if the circle complete one rotation about the center O, then the locus

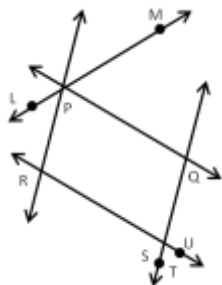
of the middle point of CD is a circle of radius:

[CDS 2016(II)]

- (a) $\frac{3r}{2}$ (b) $\frac{2r}{3}$
(c) $\frac{2\sqrt{3}r}{3}$ (d) $\frac{\sqrt{3}r}{2}$

131. In the figure given below, PQ is parallel to RS and PR is parallel to QS. If $\angle LPR = 35^\circ$ and $\angle UST = 70^\circ$, then what is $\angle MPQ$ equal to?

[CDS 2017(I)]



- (a) 55° (b) 70°
(c) 75° (d) 80°

132. Which one of the following triples does not represent the sides of a triangle?

[CDS 2017(I)]

- (a) (3,4,5) (b) (4,7,10)
(c) (3,6,8) (d) (2,3,6)

133. The angles of a triangle are in the ratio 2 : 4:3. The smallest angle of the triangle is

[CDS 2017(I)]

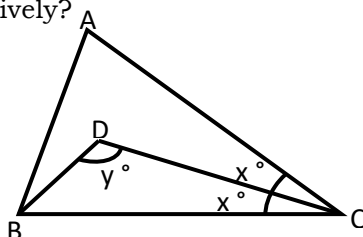
- (a) 20° (b) 40°
(c) 50° (d) 60°

134. ABC is a triangle and D is a point on the side BC. If BC = 12 cm, BD = 9 cm and $\angle ADC = \angle BAC$, then the length of AC is equal to

[CDS 2017(I)]

- (a) 5 cm (b) 6 cm
(c) 8 cm (d) 9 cm

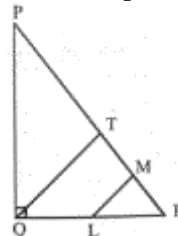
135. In the figure given below, $\angle A = 80^\circ$ and $\angle ABC = 60^\circ$. BD and CD bisect angles B and C respectively. What are the values of x and y respectively?



[CDS 2017(I)]

- (a) 10 and 130 (b) 10 and 125
(c) 20 and 130 (d) 20 and 125

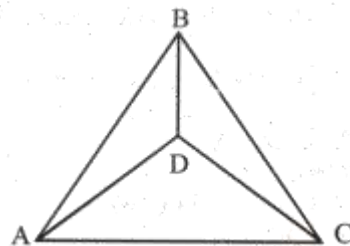
136. In the figure given below, PQR is a non-isosceles right angled triangle, right angled at Q. If LM and QT are parallel and $QT = PT$, then what is $\angle RLM$ equal to?



[CDS 2017(I)]

- (a) $\angle PQT$ (b) $\angle LRM$
(c) $\angle RML$ (d) $\angle QPT$

137. In the figure given below, ABC is a triangle with AB = BC and D is an interior point of the triangle ABC such that $\angle DAC = \angle DCA$.



Consider the following statements:

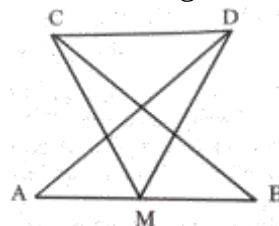
1. Triangle ADC is an isosceles triangle.
2. D is the centroid of the triangle ABC
3. Triangle ABD is congruent to the triangle CBD.

Which of the above statements are correct?

[CDS 2017(I)]

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

138. In the figure given below, M is the mid-point of AB and $\angle DAB = \angle CBA$ and $\angle AMC = \angle BMD$. Then the triangle ADM is congruent to the triangle BCM by:



[CDS 2017(I)]

- (a) SAS rule (b) SSS rule
(c) ASA rule (d) AAA rule

- 139.** ABCD is a square X is the mid-point of AB and Y is the mid-point of BC.

Consider the following statements

1. Triangles ADX and BAY are congruent.
2. $\angle DXA = \angle AYB$.
3. DX is inclined at an angle 60° with AY.
4. DX is not perpendicular to AY.

Which of the above statements are correct?

[CDS 2017(I)]

- (a) 2, 3 and 4 only (b) 1, 2 and 4 only
(c) 1, 3 and 4 only (d) 1 and 2 only

- 140.** Let ABCD be a rectangle. Let P, Q, R, S be the mid-points of sides AB, BC, CD, DA respectively. Then the quadrilateral PQRS is a:

[CDS 2017(I)]

- (a) Square
(b) Rectangle, but need not be a square
(c) Rhombus, but need not be a square
(d) Parallelogram, but need not be a rhombus

- 141.** ABCDEF is a regular polygon. Two poles at C and D are standing vertically and subtend angles of elevation 30° and 60° at A respectively. What is the ratio of the height of the pole at C to that of the pole at D:

[CDS 2017(I)]

- (a) 1 : 1 (b) $1 : 2\sqrt{3}$
(c) $2\sqrt{3} : 1$ (d) $2 : \sqrt{3}$

- 142.** In a trapezium ABCD, AB is parallel to CD and the diagonals intersect each other at O. What is the ratio of OA to OC equal to?

[CDS 2017(I)]

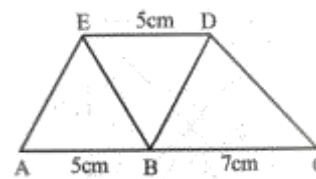
- (a) Ratio of OB to OD
(b) Ratio of BC to CD
(c) Ratio of AD to AB
(d) Ratio of AC to BD

- 143.** ABCD is a rectangle. The diagonals AC and BD intersect at O. If AB = 32 cm and AD = 24 cm, then what is OD equal to?

[CDS 2017(I)]

- (a) 22 cm (b) 20 cm
(c) 18 cm (d) 16 cm

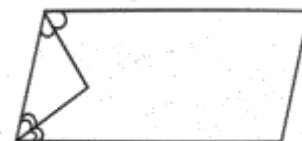
- 144.** In the figure given below, AC is parallel to ED and AB = DE = 5 cm and BC = 7 cm. What is the area ABDE : area BDE : area BCD equal to ?



[CDS 2017(I)]

- (a) 10:5:7 (b) 8:4:7
(c) 2:1:2 (d) 8:4:5

- 145.** In the figure given below, PQRS is a parallelogram. PA bisects angle P and SA bisects angle S. What is angle PAS equal to?



[CDS 2017(I)]

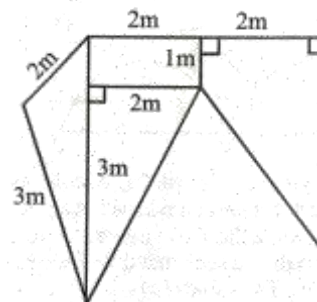
- (a) 60° (b) 75°
(c) 90° (d) 100°

- 146.** Two parallel chords of a circle whose diameter is 13 cm are respectively 5 cm and 12 cm in length. If both the chords are on the same side of the diameter, then the distance between these chords is

[CDS 2017(I)]

- (a) 5.5 cm (b) 5 cm
(c) 3.5 cm (d) 3 cm

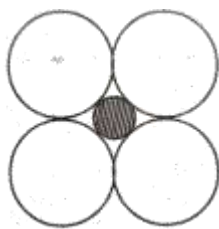
- 147.** A field is divided into four regions as shown in the given figure. What is the area of the field in square metres?



[CDS 2017(I)]

- (a) $6 + \frac{3}{4}\sqrt{5}$ (b) $5 + \frac{3}{2}\sqrt{3}$
(c) $9 + \frac{3}{4}\sqrt{15}$ (d) $7 + 2\sqrt{2}$

- 148.** In the figure given below, D is the diameter of each circle. What is the diameter of the shaded circle?



[CDS 2017(I)]

- (a) $D(\sqrt{2}-1)$ (b) $D(\sqrt{2}+1)$
(c) $D(\sqrt{2}+2)$ (d) $D(2-\sqrt{2})$

- 149.** Let P, Q, R be the mid-points of sides AB, BC, CA respectively of a triangle ABC. If the area of the triangle ABC is 5 square units, then the area of the triangle PQR is:

[CDS 2017(I)]

- (a) $\frac{5}{3}$ square units
(b) $\frac{5}{2\sqrt{2}}$ square units
(c) $\frac{5}{4}$ square units
(d) 1 square unit

- 150.** Consider the following statements in respect of three, straight lines, A, B and C on a plane.

1. If A and C are parallel and B and C are parallel; then A and B are parallel.
 2. If A is perpendicular to C and B is perpendicular to C; then A and B are parallel.
 3. If the acute angle between A and C is equal to the acute angles between B and C; then A and B are parallel.
- Which of the above statements are correct?

[CDS 2017(II)]

- (a) 1, 2 and 3 (b) 1 and 2 only
(c) 1 and 3 only (d) 2 and 3 only

- 151.** In a triangle ABC, AD is perpendicular on BC. If $\angle BAC = 90^\circ$, $AB = c$, $BC = a$, $CA = b$ and $AD = p$, then which one of the following is correct?

[CDS 2017(II)]

- (a) $p = abc$ (b) $p^2 = bc$
(c) $p = bc/a$ (d) $p = ab/c$

- 152.** In an equilateral triangle ABC, BD is drawn perpendicular to AC. What is BD^2 equal to?

[CDS 2017(II)]

- (a) AD^2 (b) $2AD^2$
(c) $3AD^2$ (d) $4AD^2$

- 153.** If PL, QM and RN are the altitudes of triangle PQR whose ortho-centre is O, then Q is the ortho-centre of the triangle

[CDS 2017(II)]

- (a) OPQ (b) OQR
(c) PLR (d) OPR

- 154.** In triangle ABC, $\angle C = 90^\circ$ and CD is the perpendicular from C to AB.

If $(CD)^{-2} = (BC)^{-2} + (CA)^{-2}$ then which one of the following is correct?

[CDS 2017(II)]

- (a) $BC \cdot CD = AB \cdot CA$
(b) $AB \cdot BC = CD \cdot CA$
(c) $CA^2 + CB^2 = 2(AD^2 + CD^2)$
(d) $AB \cdot CD = BC \cdot CA$

- 155.** In a triangle ABC, the medians AD and BE intersect at G. A line DF is drawn parallel to BE such that F is on AC. If $AC = 9$ cm, then what is CF equal to?

[CDS 2017(II)]

- (a) 2.25 cm (b) 3 cm
(c) 4.5 cm (d) 6 cm

- 156.** In a triangle PQR, X is a point on PR and Y is a point on QR such that $PR = 10$ cm, $RX = 4$ cm, $YR = 2$ cm, $QR = 5$ cm.

Which one of the following is correct?

[CDS 2017(II)]

- (a) XY is parallel to PQ
(b) $PQ = 2XY$
(c) $PX = QY$
(d) $PQ = 3XY$

- 157.** One-fifth of the area of a triangle ABC is cut off by a line DE drawn parallel to BC such that D is on AB and E is on AC. If $BC = 10$ cm, then what is DE equal to?

[CDS 2017(II)]

- (a) $\sqrt{5}$ cm (b) $2\sqrt{5}$ cm
(c) $3\sqrt{5}$ cm (d) $4\sqrt{5}$ cm

- 158.** Consider the following statements

1. The point of intersection of the perpendicular bisectors of the side of a triangle may lie outside the triangle.
2. The point of intersection of the perpendiculars drawn from the vertices to the opposite sides of a triangle may lie on two sides. Which of the above statements is/are correct?

[CDS 2017(II)]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

- 159.** If a point O is interior of a rectangle ABCD is joined with each of the vertices A, B, C and D, then $OB^2 + OD^2$ will be equal to?

[CDS 2017(II)]

- (a) $2OC^2 + OA^2$ (b) $2OC^2 - OA^2$
(c) $OC^2 + OA^2$ (d) $OC^2 + 2OA^2$

- 160.** The diagonals of a rhombus are of length 20 cm and 48 cm. What is the length of a side of the rhombus?

[CDS 2017(II)]

- (a) 13 cm (b) 26 cm
(c) 36 cm (d) 39 cm

- 161.** A closed polygon has six sides and one of its angles is 30° greater than each of the other five equal angles. What is the value of one of the equal angles?

[CDS 2017(II)]

- (a) 55° (b) 115°
(c) 150° (d) 175°

- 162.** AB and CD are parallel chords of a circle 3 cm apart. If $AB = 4$ cm, $CD = 10$ cm, then what is the radius of the circle?

[CDS 2017(II)]

- (a) 7 cm (b) $\sqrt{19}$ cm
(c) $\sqrt{29}$ cm (d) 14 cm

- 163.** The diagonals of a cyclic quadrilateral ABCD intersect at P and the area of the triangle APB is 24 square cm. If $AB = 8$ cm and $CD = 5$ cm, then what is the area of the triangle CPD?

[CDS 2017(II)]

- (a) 24 square cm (b) 15 square cm
(c) 12.5 square cm (d) 9.375 square cm

- 164.** The distance between the centres of two circles having radii 9 cm and 4 cm is 13 cm. What is the length of the direct common tangent of these circles?

[CDS 2017(II)]

- (a) 12 cm (b) 11 cm
(c) 10 cm (d) 9.5 cm

- 165.** An arc of a circle subtends an angle 30° at the centre. If the length of the arc is 22 cm, then what is the radius of the circle?

[CDS 2017(II)]

- (a) 5 cm (b) 7 cm
(c) 9 cm (d) 11 cm

- 166.** There are 8 lines in a plane, no two of which are parallel. What is the maximum number of points at which they can intersect?

[CDS 2017(II)]

- (a) 15
(b) 21
(c) 28
(d) None of the above

- 167.** The length of a line segment AB is 2 cm. It is divided into two parts at a point C such that $AC^2 = AB \times CB$. What is the length of CB?

[CDS 2018(I)]

- (a) $3\sqrt{5}$ cm (b) $3 - \sqrt{5}$ cm
(c) $5\sqrt{3}$ cm (d) $\sqrt{5} - 1$ cm

- 168.** The locus of a point equidistant from two intersecting lines is

[CDS 2018(I)]

- (a) A straight line
(b) A circle
(c) A pair of straight lines
(d) None of the above

Consider the following for the next two (02) questions:

In a triangle ABC, a, b and c are the lengths of the sides and p, q and r are the lengths of its medians.

- 169.** Which one of the following is correct?

[CDS 2018(I)]

- (a) $2(p + q + r) = (a + b + c)$
(b) $2(p + q + r) > 3(a + b + c)$
(c) $2(p + q + r) < 3(a + b + c)$
(d) $11(p + q + r) > 10(a + b + c)$

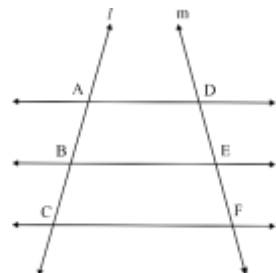
- 170.** Which one of the following is correct?

[CDS 2018(I)]

- (a) $(a + b + c) < (p + q + r)$
(b) $3(p + q + r) > 4(a + b + c)$
(c) $2(p + q + r) < 3(a + b + c)$
(d) $3(a + b + c) > 4(p + q + r)$

- 171.** In the figure given below, p, q, r are parallel lines; l and m are two transversals

[CDS 2018(I)]



Consider the following:

- $AB : AC = DE : DF$
- $AB \times EF = BC \times DE$

Which of the above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

172. ABC is a triangle right angled at C with $BC = a$ and $AC = b$. If p is the length of the perpendicular from C on AB, then which one of the following is correct?

[CDS 2018(I)]

- (a) $a^2 b^2 = p^2 (a^2 + b^2)$
(b) $a^2 b^2 = p^2 (b^2 - a^2)$
(c) $2a^2 b^2 = p^2 (a^2 + b^2)$
(d) $a^2 b^2 = 2p^2 (a^2 + b^2)$

173. Consider the following statements:

1. The orthocenter of a triangle always lies inside the triangle.
2. The centroid of a triangle always lies inside the triangle.
3. The orthocenter of right angled triangle lies on the triangle.
4. The centroid of a right angled triangles lies on the triangle.

Which of the above statements are correct?

[CDS 2018(I)]

- (a) 1 and 2 (b) 1 and 4
(c) 2 and 3 (d) 2 and 4

174. Consider the following statements:

Two triangles are said to be congruent, if

1. Three angles of the triangle are equal to the corresponding three angles of the other triangle.
2. Three sides of one triangle are equal to the corresponding three sides of the other triangle.
3. Two sides and the included angle of one triangle are equal to the corresponding two sides and the included angle of the other triangle.
4. Two angles and the included side of one triangle are equal to the corresponding two angles and the included side of the other triangle.

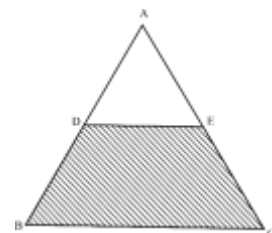
Which of the above statements are correct?

[CDS 2018(I)]

- (a) 1, 2 and 3 (b) 1, 3 and 4
(c) 1, 2 and 4 (d) 2, 3 and 4

175. In the equilateral triangle ABC given below, $AD = DB$ and $AE = EC$. If ℓ is the length of a side of the triangle, then what is the area of the shaded region?

[CDS 2018(I)]



- (a) $\frac{3\sqrt{3}\ell^2}{16}$ (b) $\frac{3\ell^2}{16}$
(c) $\frac{3\sqrt{3}\ell^2}{32}$ (d) $\frac{3\ell^2}{32}$

176. Give that the angles of a polygon are all equal and each angle is a right angle.

Statement-1: The polygon has exactly four sides

Statement-2: The sum of the angles of a polygon having n side is $(3n - 8)$ right angles.

Which one of the following is correct is respect of the above statements?

[CDS 2018(I)]

- (a) Both Statement - 1 and Statements - 2 are true and Statement - 2 is the correct explanation of Statement - 1
(b) Both Statement - 1 and Statement - 3 are true but Statement - 2 is not correct explanation of Statement - 1
(c) Statement - 1 is true but Statement - 2 is false
(d) Statement - 1 is false but Statement - 2 is true

177. The radii of the circles are 4.5 cm and 3.5 cm respectively. The distance between the centers of the circles is 10 cm. What is the length of the transverse common tangent?

[CDS 2018(I)]

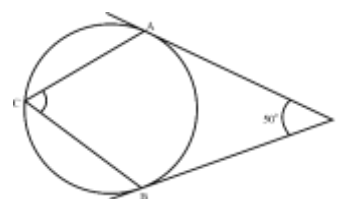
- (a) 4 cm (b) 5 cm
(c) 6 cm (d) 7 cm

178. The locus of the mid-points of the radii of length 16 cm of a circle is

[CDS 2018(I)]

- (a) A concentric circle of radius 8 cm
(b) A concentric circle of radius 16 cm
(c) The diameter of the circle
(d) A straight line passing through the centre of the circle.

179. In the figure given below, XA and XB are

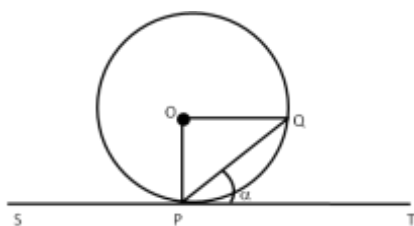


two tangents to a circle. If $\angle AXB = 50^\circ$ and AC is parallel to XB, then what is $\angle ACB$ equal to?

[CDS 2018(I)]

- (a) 70° (b) 65°
(c) 60° (d) 55°

180. In the figure given below, SPT is a tangent to the circle at P and O is the centre of the circle. If $\angle QPT = \alpha$, then what is $\angle POQ$ equal to?

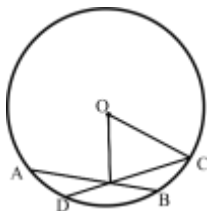


[CDS 2018(I)]

- (a) α (b) 2α
(c) $90^\circ - \alpha$ (d) $180^\circ - 2\alpha$

181. In the figure given below, two equal chords cut at point P. If $AB = CD = 10\text{ cm}$, $OC = 13\text{ cm}$ (O is the centre of the circle and $PB = 3\text{ cm}$, then what is the length of OP?

[CDS 2018(I)]



- (a) 5 cm (b) 6 cm
(c) $2\sqrt{29}\text{ cm}$ (d) $2\sqrt{37}\text{ cm}$

182. ABC is a right angled triangle with base BC and height AB. The hypotenuse AC is four times the length of the perpendicular drawn to it from the opposite vertex. What is $\tan C$ equal to:

[CDS 2018(I)]

- (a) $2 - \sqrt{3}$ (b) $\sqrt{3} - 1$
(c) $2 + \sqrt{3}$ (d) $\sqrt{3} + 1$

183. The number of sides of two regular polygons are in the ratio 5:4. The difference between their interior angles is 9° . Consider the following statements:

- One of them is a pentagon and the other is a rectangle.
- One of them is a decagon and the other is an octagon.

3. The sum of their exterior angles is 720° . Which of the above statements is/are correct?

[CDS 2018(II)]

- (a) 1 only (b) 2 only
(c) 1 and 3 (d) 2 and 3

184. A circle is inscribed in an equilateral triangle of side of length l . The area of any square inscribed in the circle is

[CDS 2018(II)]

- (a) $\frac{l^2}{2}$ (b) $\frac{\sqrt{3}l^2}{4}$
(c) $\frac{l^2}{4}$ (d) $\frac{l^2}{6}$

185. The areas of two similar triangles are $(7 - 4\sqrt{3})\text{ cm}^2$ and $(7 + 4\sqrt{3})\text{ cm}^2$ respectively. The ratio of their corresponding sides is

[CDS 2018(II)]

- (a) $7 - 4\sqrt{3}$ (b) $7 - 3\sqrt{3}$
(c) $5 - \sqrt{3}$ (d) $5 + \sqrt{3}$

186. The chord of a circle is $\sqrt{3}$ times its radius. The angle subtended by this chord at the minor arc is k times the angle subtended at the major arc. What is the value of k ?

[CDS 2018(II)]

- (a) 5 (b) 2
(c) $1/2$ (d) $1/5$

187. In a triangle ABC, the sides AB, AC are produced and the bisectors of exterior angles of $\angle ABC$ and $\angle ACB$ intersect at D. If $\angle BAC = 50^\circ$, then $\angle BDC$ is equal to

[CDS 2018(II)]

- (a) 115° (b) 65°
(c) 55° (d) 40°

188. If two lines AB and CD intersect at O such that $\angle AOC = 5\angle AOD$, then the four angles at O are

[CDS 2018(II)]

- (a) $40^\circ, 40^\circ, 140^\circ, 140^\circ$
(b) $30^\circ, 30^\circ, 150^\circ, 150^\circ$
(c) $30^\circ, 45^\circ, 75^\circ, 210^\circ$
(d) $60^\circ, 60^\circ, 120^\circ, 120^\circ$

189. If a point P moves such that the sum of the squares of its distances from two fixed points A and B is a constant, then the locus of the point P is

[CDS 2018(II)]

- (a) A straight line
(b) A circle

- (c) Perpendicular bisector of AB
(d) An arbitrary curve

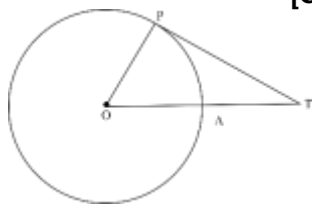
190. If ABC is a right-angled triangle with AC as its hypotenuse, then which one of the following is correct?

[CDS 2018(II)]

- (a) $AC^3 < AB^3 + BC^3$
(b) $AC^3 > AB^3 + BC^3$
(c) $AC^3 \leq AB^3 + BC^3$
(d) $AC^3 \geq AB^3 + BC^3$

191. In the figure give below, the radius of the circle is 6 cm and $AT = 4$ cm. The length of tangent PT is

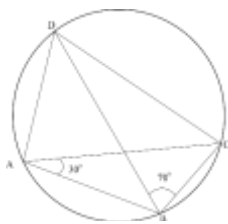
[CDS 2018(II)]



- (a) 6 cm
(b) 8 cm
(c) 9 cm
(d) 10 cm

192. In the figure given below, what is $\angle BCD$ equal to?

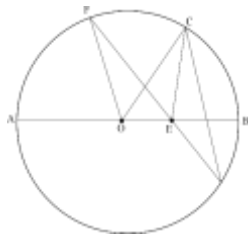
[CDS 2018(II)]



- (a) 70°
(b) 75°
(c) 80°
(d) 90°

193. In the figure give below, AB is the diameter of the circle whose centre is at O. given that $\angle ECD = \angle EDC = 32^\circ$, then $\angle CEF$ and $\angle COF$ respectively are

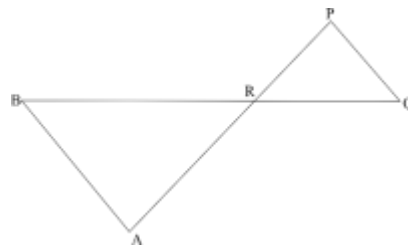
[CDS 2018(II)]



- (a) $32^\circ, 64^\circ$
(b) $64^\circ, 64^\circ$
(c) $32^\circ, 32^\circ$
(d) $64^\circ, 32^\circ$

194. In the figure given below, $\triangle ABR \sim \triangle PQR$. If $PQ = 3$ cm, $AB = 6$ cm, $BR = 8 \cdot 2$ cm and $PR = 5 \cdot 2$ cm, the QR and AR are respectively

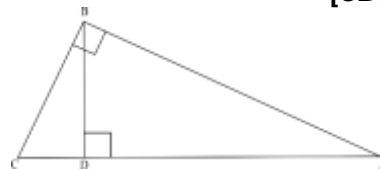
[CDS 2018(II)]



- (a) $8 \cdot 2$ cm, $10 \cdot 4$ cm
(b) $4 \cdot 1$ cm, 6 cm
(c) $2 \cdot 6$ cm, $5 \cdot 2$ cm
(d) $4 \cdot 1$ cm, $10 \cdot 4$ cm

195. In the figure given below, ABC is a triangle with AB perpendicular BC. Further BD is perpendicular to AC. If $AD = 9$ cm and $DC = 4$ cm, then what is the length of BD?

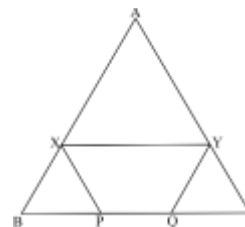
[CDS 2018(II)]



- (a) $13/36$ cm
(b) $36/13$ cm
(c) $13/2$ cm
(d) 6 cm

196. In the figure given below ABC is an equilateral triangle with each side of length 30 cm. XY is parallel to BC, XP is parallel to AC and YQ is parallel to AB. If $XY + XP + YQ$ is 40 cm, then the value of PQ is

[CDS 2018(II)]



- (a) 5 cm
(b) 12 cm
(c) 15 cm
(d) 10 cm

197. If the lengths of two parallel chords in a circle of radius 10 cm are 12 cm and 16 cm, then what is the distance between these two chords?

[CDS 2019(I)]

- (a) 1 cm or 7 cm
(b) 2 cm or 14 cm
(c) 3 cm or 21 cm
(d) 4 cm or 28 cm

198. The corners of a square of side 'a' are cut away so as to form a regular octagon. What is the side of the octagon?

[CDS 2019(I)]

- (a) $a(\sqrt{2} - 1)$
(b) $a(\sqrt{3} - 1)$
(c) $\frac{a}{\sqrt{2}+2}$
(d) $\frac{a}{3}$

- 199.** Three consecutive integers form the lengths of a right-angled triangle. How many sets of such three consecutive integers is/are possible?

[CDS 2019(I)]

- (a) Only one (b) Only two
(c) Only three (d) Infinitely many

- 200.** Two circles are drawn with the same centre. The circumference of the smaller circle is 44 cm and that of the bigger circle is double the smaller one. What is the area between these two circles?

[CDS 2019(I)]

- (a) 154 square cm (b) 308 square cm
(c) 462 square cm (d) 616 square cm

- 201.** The perimeter of a right-angled triangle is k times the shortest side. If the ratio of the other side to hypotenuse is 4 : 5, the what is the value of k ?

[CDS 2019(I)]

- (a) 2 (b) 3
(c) 4 (d) 5

- 202.** The angles of a triangle are in the ratio 1 : 1 : 4. If the perimeter of the triangle is k times its largest side, then what is the value of k ?

[CDS 2019(I)]

- (a) $1 + \frac{2}{\sqrt{3}}$ (b) $1 - \frac{2}{\sqrt{3}}$
(c) $2 + \frac{2}{\sqrt{3}}$ (d) 2

- 203.** In a circle of radius 8 cm, AB and AC are two chords such that $AB = AC = 12$ cm. What is the length of chord BC?

[CDS 2019(I)]

- (a) $2\sqrt{6}$ cm (b) $3\sqrt{6}$ cm
(c) $3\sqrt{7}$ cm (d) $6\sqrt{7}$ cm

- 204.** Consider the following statements:

1. An isosceles trapezium is always cyclic
 2. Any cyclic parallelogram is a rectangle.
- Which of the above statements is/are correct?

[CDS 2019(I)]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

- 205.** A ladder is resting against a vertical wall and its bottom is 2.5 m away from the wall. If it slips 0.8 m down the wall, then its bottom will move away from the wall by 1.4 m. What is the length of the ladder?

[CDS 2019(I)]

- (a) 6.2 m (b) 6.5 m
(c) 6.8 m (d) 7.5 m

- 206.** Two equal circles intersect such that each passes through the centre of the other. If the length of the common chord of the circles is $10\sqrt{3}$ cm, the what is the diameter of the circle?

[CDS 2019(I)]

- (a) 10 cm (b) 15 cm
(c) 20 cm (d) 30 cm

- 207.** Consider the following statements:

1. The number of circles that can be drawn through three non-collinear points is infinity.
 2. Angle formed in minor segment of a circle is acute.
- Which of the above statements is/are correct?

[CDS 2019(I)]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

- 208.** Consider the following inequalities in respect of any triangle ABC:

1. $AC - AB < BC$
 2. $BC - AC < AB$
 3. $AB - BC < AC$
- Which of the above are correct?

[CDS 2019(I)]

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

- 209.** Consider the following statements

1. The perimeter of a triangle is greater than the sum of its three medians
 2. In any triangles ABC, if D is any point on BC, then $AB + BC + CA > 2AD$.
- Which of the above statements is/are correct?

[CDS 2019(I)]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Consider the following for the next two (02) item:

An equilateral triangle ABC is inscribed in a circle of radius $20\sqrt{3}$ cm.

- 210.** What is the length of the side of the triangle?

[CDS 2019(I)]

- (a) 30 cm (b) 40 cm
(c) 50 cm (d) 60 cm

- 211.** The centroid of the triangle ABC is at a distance d from the vertex A. What is d equal to ?

[CDS 2019(I)]

- (a) 15 cm (b) 20 cm
(c) $20\sqrt{3}$ cm (d) $30\sqrt{3}$ cm

- 212.** Three parallel lines x , y and z are cut by two transversals m and n . Transversal m cuts the lines x , y , z at P , Q , R respectively and transversal n cut the line x , y , z at L , M , N respectively, if $PQ = 3\text{cm}$, $QR = 9\text{cm}$ and $MN = 10.5$, then what is the length of LM ?

[CDS 2019(II)]

- (a) 3 cm (b) 3.5 cm
(c) 4 cm (d) 4.5 cm

- 213.** Which one of the following is correct in respect of a right angled triangle?

[CDS 2019(II)]

- (a) Its orthocenter lies inside the triangle.
(b) Its orthocenter lies outside the triangle
(c) Its orthocenter lies on the triangle
(d) It has no orthocenter

- 214.** Let the bisector of the angle BAC of a triangle ABC meet BC in X . Which one of the following is correct?

[CDS 2019(II)]

- (a) $AB < BX$
(b) $AB > BX$
(c) $AX = CX$
(d) None of the above

- 215.** A line segment AB is the diameter of a circle with centre at O having radius 6.5 cm. Point P is in the plane of the circle such that $AP = x$ and $BP = y$. In which one of the following cases the point does not lie on the circle?

[CDS 2019(II)]

- (a) $x = 6.5$ cm and $y = 6.5$ cm
(b) $x = 12$ cm and $y = 5$ cm
(c) $x = 5$ cm and $y = 12$ cm
(d) $x = 0$ and $y = 13$ cm

- 216.** The perimeters of two similar triangles ABC and PQR are 75 cm and 50 cm respectively. If the length of one side of the triangle PQR is 20 cm, then what is the length of corresponding side of the triangle ABC ?

[CDS 2019(II)]

- (a) 25 cm (b) 30 cm
(c) 40 cm (d) 45 cm

- 217.** Let $PQRS$ be a parallelogram whose diagonals PR and QS intersect at O . If triangles QRS is an equilateral triangle

having a side of length 10 cm, then what is the length of the diagonal PR ?

[CDS 2019(II)]

- (a) $5\sqrt{3}$ cm (b) $10\sqrt{3}$ cm
(c) $15\sqrt{3}$ cm (d) $20\sqrt{3}$ cm

- 218.** If ℓ is the length of the median of an equilateral triangle, then what is its area?

[CDS 2019(II)]

- (a) $\frac{\sqrt{3}\ell^2}{3}$ (b) $\frac{\sqrt{3}\ell^2}{2}$
(c) $\sqrt{3}\ell^2$ (d) $2\ell^2$

- 219.** A piece of wire is in the form of a sector of a circle of radius 20 cm, subtending an angle 150° at the centre. If it is bent in the form of a circle, then what will be its radius?

[CDS 2019(II)]

- (a) $\frac{19}{3}$ cm
(b) 7 cm
(c) 8 cm
(d) None of the above

- 220.** Suppose P , Q and R are the mid-points of sides of a triangle of area 128 cm^2 . If a triangle ABC is drawn by joining the mid-points of sides of triangle PQR , then what is the area of triangle ABC ?

[CDS 2019(II)]

- (a) 4 cm^2 (b) 8 cm^2
(c) 16 cm^2 (d) 32 cm^2

- 221.** Let two lines p and q be parallel, Consider two points B and C on the line p and two points D and E on the line q . The line through B and E intersects the line through C and D at A in between the two lines p and q . If $AC : AD = 4 : 9$, then what is the ratio of area of triangle ABC to that of triangle ADE ?

[CDS 2019(II)]

- (a) 2 : 3 (b) 4 : 9
(c) 16 : 81 (d) 1 : 2

- 222.** ABC is a triangles right angled at B . If $AB = 5\text{cm}$ and $BC = 10\text{cm}$, then what is the length of the perpendicular drawn from the vertex B to the hypotenuse?

[CDS 2019(II)]

- (a) 4 cm (b) $2\sqrt{5}$ cm
(c) $\frac{4}{\sqrt{5}}$ cm (d) 8 cm

- 223.** If the length of the hypotenuse of a right angled triangle is 10 cm, then what is the

maximum area of such a right angled triangle?

[CDS 2019(II)]

- (a) 100 cm^2 (b) 50 cm^2
(c) 25 cm^2 (d) 10 cm^2

224. A piece of wire of length 33 cm is bent into an arc of a circle of radius 14 cm. What is the angle subtended by the arc at the centre of the circle?

[CDS 2019(II)]

- (a) 75° (b) 90°
(c) 135° (d) 150°

225. If one side of a right-angled triangle (with all sides' integers) is 15 cm, then what is the maximum perimeter of the triangle?

[CDS 2019(II)]

- (a) 240 cm (b) 225 cm
(c) 113 cm (d) 112 cm

226. Consider a trapezium $ABCD$, in which AB is parallel to CD and AD is perpendicular to CD and AD is perpendicular to AB . If the trapezium has an in-circle which touches AB at E and CD at F , where $EB = 25 \text{ cm}$ and $FC = 16 \text{ cm}$, then what is the diameter of the circle?

[CDS 2019(II)]

- (a) 16 cm (b) 25 cm
(c) 36 cm (d) 40 cm

227. $ABCD$ is a plate in the shape of a parallelogram. EF is the line parallel to DA and passing through the point of intersection O of the diagonals AC and BD . further, E lies on DC and F lies on AB . The triangular portion DOE is cut out from the plate $ABCD$. What is the ratio of area of remaining portion of the plate to the whole?

[CDS 2020(I)]

- (a) $\frac{5}{8}$ (b) $\frac{5}{7}$
(c) $\frac{3}{4}$ (d) $\frac{7}{8}$

228. Two circles of radii 20 cm and 16 cm intersect and the length of common chord is 24 cm. If d is the distance between their centres, then which one of the following is correct?

[CDS 2020(I)]

- (a) $d < 26 \text{ cm}$
(b) $26 \text{ cm} < d < 27 \text{ cm}$
(c) $27 \text{ cm} < d < 28 \text{ cm}$
(d) $d > 28 \text{ cm}$

229. In a circle of radius 5 cm, AB and AC are two chords such that $AB = AC = 8 \text{ cm}$. What is the length of chord BC ?

[CDS 2020(I)]

- (a) 9 cm (b) 9.2 cm
(c) 9.6 cm (d) 9.8 cm

230. $ABCD$ is a parallelogram where AC and BD are the diagonals. If $\angle BAD = 60^\circ$, $\angle ADB = 90^\circ$, then what is BD^2 equal to?

[CDS 2020(I)]

- (a) $\frac{3}{5} AB^2$ (b) $\frac{3}{4} AB^2$
(c) $\frac{1}{2} AB^2$ (d) $\frac{2}{3} AB^2$

231. A line through the vertex A of a parallelogram $ABCD$ meets DC in P and BC produced in Q . If P is the mid-point of DC then which of the following is/are correct?

1. Area of $\triangle PDA$ is equal to that of $\triangle PCQ$
2. Area of $\triangle QAB$ is equal to twice that of $\triangle PCQ$

Select the correct answer using the code given below:

[CDS 2020(I)]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

232. The lengths of sides of a triangle are $3x, 4\sqrt{y}, 5\sqrt{z}$, where $3x < 4\sqrt{y}, 5\sqrt{z}$. If one of the angles is 90° , then what are the minimum integral values of x, y, z respectively?

[CDS 2020(I)]

- (a) 1, 2, 3 (b) 2, 3, 4
(c) 1, 1, 1 (d) 3, 4, 5

233. What is the maximum number of circum-circles that a triangle can have?

[CDS 2020(I)]

- (a) 1 (b) 2
(c) 3 (d) Infinite

234. If an arc of a circle of radius 6 cm subtends a central angle measuring 30° , then which one of the following is an approximate length of the arc?

[CDS 2020(I)]

- (a) 3.14 cm (b) 2.15 cm
(c) 2.14 cm (d) 2 cm

235. A ladder 5 m long is placed in a room so as to reach a point 4.8 m high on a wall and on turning the ladder over to the opposite side of the wall without moving the base it

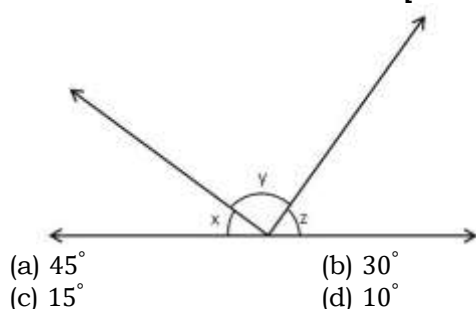
reaches a point 1.4 m high. What is the breadth of the room?

[CDS 2020(I)]

- (a) 5.8 m (b) 6 m
(c) 6.2 m (d) 7.5 m

236. In the given figure, if $\frac{y}{x} = 6$ and $\frac{z}{x} = 5$, then what is the value of x ?

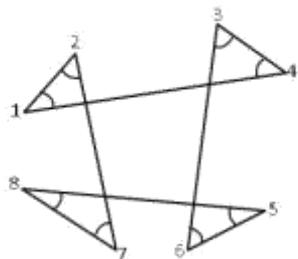
[CDS 2020(I)]



- (a) 45° (b) 30°
(c) 15° (d) 10°

237. Angles are shown in the given figure. What is value of $\angle 1 + \angle 2 + \angle 3 + \angle 4 + \angle 5 + \angle 6 + \angle 7 + \angle 8$?

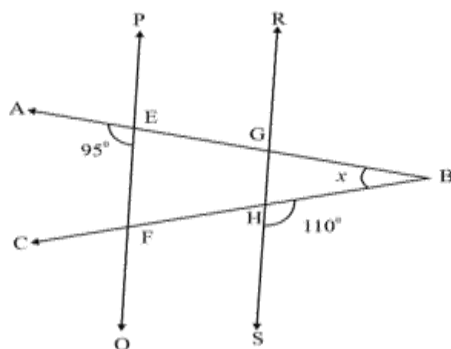
[CDS 2020(I)]



- (a) 240° (b) 360°
(c) 540° (d) 720°

238. In the given figure PQ is parallel to RS, $\angle AEF = 95^\circ$, $\angle BHS = 110^\circ$, and $\angle ABC = x^\circ$. Then what is the value of x ?

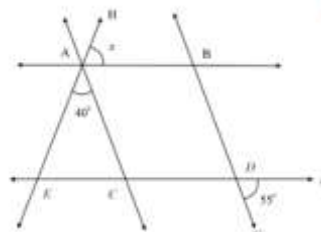
[CDS 2020(I)]



- (a) 15 (b) 25
(c) 30 (d) 35

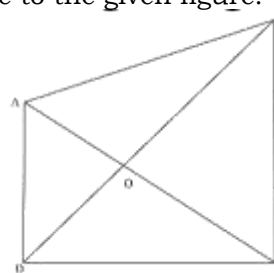
239. In the given figure AB is parallel to CD and AC is parallel to BD. If $\angle EAC = 40^\circ$, $\angle FDG = 55^\circ$, $\angle HAB = x^\circ$, then what is the value of x ?

[CDS 2020(I)]



- (a) 85 (b) 80
(c) 75 (d) 65

240. Consider the following statements with reference to the given figure:



1. The sum of the areas of $\triangle AOD$ and $\triangle BOC$ is equal to the sum of the areas of $\triangle AOB$ and $\triangle DOC$.

2. $\angle AOD = \angle BOC$

3. $AB + BC + CD + DA > AC + BD$

Which of the above statements are correct?

[CDS 2020(I)]

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

241. A road curve is to be laid out on a circle. What radius should be used if the track is to change direction by 42° in distance of 44 m?

(Assume $\pi = \frac{22}{7}$)

[CDS 2020(II)]

- (a) 60 m (b) 66 m
(c) 75 m (d) 80 m

242. In a quadrilateral ABCD, $\angle B = 90^\circ$ and $AB^2 + BC^2 + CD^2 - AD^2 = 0$, then what is $\angle ACD$?

[CDS 2020(II)]

- (a) 30° (b) 60°
(c) 90° (d) 120°

243. In a $\triangle ABC$, $AC = 12\text{cm}$, $AB = 16\text{cm}$ and AD is the bisector of $\angle A$. If $BD = 4\text{cm}$, then what is DC equal to?

[CDS 2020(II)]

- (a) 2 cm (b) 3 cm
(c) 4 cm (d) 5 cm

244. ABCD is a cyclic quadrilateral. The bisectors of the angles A, B, C and D cut

the circle at P, Q, R and S respectively.
What is $\angle PQR + \angle RSP$ equal to?

[CDS 2020(II)]

- (a) 90° (b) 135°
(c) 180° (d) 270°

245. ABC is an equilateral triangle. The side BC is trisected at D such that $BC = 3 BD$. What is the ratio of AD^2 to AB^2 ?

[CDS 2020(II)]

- (a) 7 : 9 (b) 1 : 3
(c) 5 : 7 (d) 1 : 2

246. Consider the following statements:

1. The diagonals of a trapezium divide each other proportionally.
2. Any line drawn parallel to the parallel sides of a trapezium divides the non-parallel sides proportionally.

Which of the above statements are correct.

[CDS 2020(II)]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

247. What is the area of right-angled triangle, if the radius of the circum-circle is 5 cm and altitude drawn to the hypotenuse is 4 cm?

[CDS 2020(II)]

- (a) 20 cm^2 (b) 18 cm^2
(c) 16 cm^2 (d) 10 cm^2

248. In a triangle, values of all the angles are integers (I degree measure). Which one of the following cannot be the proportion of their measures?

[CDS 2020(II)]

- (a) 1 : 2 : 3 (b) 3 : 4 : 5
(c) 5 : 6 : 7 (d) 6 : 7 : 8

249. If the sum of all interior angles of a regular polygon is twice the sum of all its exterior angles, then the polygon is

[CDS 2020(II)]

- (a) Hexagon (b) Octagon
(c) Nonagon (d) Decagon

250. A bicycle wheel makes 5000 revolutions in moving 11 km. What is the radius of the wheel? (Assume $\pi = \frac{22}{7}$)

[CDS 2020(II)]

- (a) 17.5 cm (b) 35 cm
(c) 70 cm (d) 140 cm

251. In a triangles ABC, if $2 \angle A = 3 \angle B = 6 \angle C$, then what is $\angle A + \angle C$ equal to?

[CDS 2020(II)]

- (a) 90° (b) 120°
(c) 135° (d) 150°

252. The lengths of the sides of a right-angled triangle are consecutive even integers (in cm). What is the product of these integers?

[CDS 2020(II)]

- (a) 60 (b) 120
(c) 360 (d) 480

253. A circle is inscribed in a triangle ABC. It touches the sides AB and AC at M and N respectively. If O is the centre of the circle and $\angle A = 70^\circ$, then what is $\angle MON$ equal to?

[CDS 2020(II)]

- (a) 90° (b) 100°
(c) 110° (d) 120°

254. A triangle and parallelogram have equal areas and equal bases. If the altitude of the triangle is k times the altitude of the parallelogram, what is the value of k?

[CDS 2020(II)]

- (a) 4 (b) 2
(c) 1 (d) $\frac{1}{2}$

255. AD is the median of the triangle ABC. If P is any point on AD, then which one of the following is correct?

[CDS 2020(II)]

- (a) Area of triangle PAB is greater than the area of triangle PAC
(b) Area of triangle PAB is equal to area of triangle PAC
(c) Area of triangle PAB is one-fourth of the area of triangle PAC
(d) Area of triangle PAB is half of the area of triangle PAC

256. What is the area of a segment of a circle of radius r subtending and angle θ at the centre?

[CDS 2020(II)]

- (a) $\frac{1}{2} r^2 \theta$
(b) $\frac{1}{2} r^2 \left(\theta - 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} \right)$
(c) $\frac{1}{2} r^2 \left(\theta - \sin \frac{\theta}{2} \cos \frac{\theta}{2} \right)$
(d) $\frac{1}{2} r^2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$

257. ABC is a triangle right-angled at C. Let P be any point on AC and Q be any point BC. Which of the following statements is/are correct?

1. $AQ^2 + BP^2 = AB^2 + PQ^2$

2. $AB = 2PQ$

Select the correct answer using the code given below:

[CDS 2020(II)]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

- 258.** The sum of the squares of sides of a right-angled triangle is 8,450 square units. What is the length of its hypotenuse?

[CDS 2020(II)]

- (a) 50 units (b) 55 units
(c) 60 units (d) 65 units

- 259.** In a trapezium ABCD, AB is parallel to DC. The diagonals AC and BD intersect at P. If $AP : PC = 4$; $(4x-4)$ and $BP : PD = (2x-1) : (2x+4)$, then what is the value of x ?

[CDS 2021(I)]

- (a) 4 (b) 3
(c) $\frac{3}{2}$ (d) 2

- 260.** $\triangle ABC$ is similar to $\triangle DEF$. The perimeters of $\triangle ABC$ and $\triangle DEF$ are 40 cm and 30 cm respectively. What is the ratio of $(BC + CA)$ to $(EF + FD)$ equal to?

[CDS 2021(I)]

- (a) 5 : 4 (b) 4 : 3
(c) 3 : 2 (d) 2 : 1

- 261.** Two isosceles triangles have equal vertical angle and their areas are in the ratio 4.84:5.29. What is the ratio of their corresponding heights?

[CDS 2021(I)]

- (a) 11 : 23 (b) 23 : 25
(c) 22 : 23 (d) 484 : 529

- 262.** ABC is a triangle right angled at A and AD is perpendicular BC. If $BD = 8$ cm and $DC = 12.5$ cm, then what is AD equal to?

[CDS 2021(I)]

- (a) 7.5 cm (b) 8.5 cm
(c) 9 cm (d) 10 cm

- 263.** The sides of a right-angled triangle are in the ratio $x:(x-1):(x-18)$. What is the perimeter of the triangle?

[CDS 2021(I)]

- (a) 28 units (b) 42 units
(c) 56 units (d) 84 units

- 264.** ABC is a triangle right angled at B. Let M and N be two points on AB such that $AM = MN = NB$. Let P and Q be two points on AC such that PM is parallel to QN and QN is

parallel to CB. If $BC = 12$ cm, then what is $(PM + QN)$ equal to?

[CDS 2021(I)]

- (a) 10 cm (b) 11 cm
(c) 12 cm (d) 13 cm

- 265.** AB and CD are the diameters of a circle which intersect at P. Join AC, CB, BD, and DA. If $\angle PAD = 60^\circ$, then what is $\angle BPD$ equal to?

[CDS 2021(I)]

- (a) 30° (b) 60°
(c) 90° (d) 120°

- 266.** An equilateral triangle ABC and a scalene triangle DBC are inscribed in a circle on same side of the arc. What is $\angle BDC$, equal to?

[CDS 2021(I)]

- (a) 30° (b) 45°
(c) 60° (d) 90°

- 267.** The sides of a triangle ABC are 4 cm, 6 cm and 8 cm. With the vertices of the triangles as centres, three circles are drawn each touching the other two externally. What is the sum of the radii of the three circles?

[CDS 2021(I)]

- (a) 6 cm (b) 7 cm
(c) 9 cm (d) 10 cm

- 268.** Let PAB be a secant to a circle intersecting the circle at A and B. Let PT be the tangent segment. If $PA = 9$ cm and $PT = 12$ cm, then what is AB equal to?

[CDS 2021(I)]

- (a) 5 cm (b) 6 cm
(c) 7 cm (d) 9 cm

- 269.** If the perimeter of a right-angled triangle is 30 cm and the hypotenuse is 13 cm then what is the area of the triangle?

[CDS 2021(I)]

- (a) 24 cm^2 (b) 27 cm^2
(c) 30 cm^2 (d) 36 cm^2

- 270.** ABC is a triangle right angled at C. Let p be the length of the perpendicular drawn from C on AB. If $BC = 6$ cm and $CA = 8$ cm, then what is the value of P?

[CDS 2021(I)]

- (a) 5.4 cm (b) 5 cm
(c) 4.8 cm (d) 4.2

- 271.** ABCD is a trapezium in which AB is parallel to DC and $2AB = 3DC$. The diagonals AC and BD intersect at O. What

is the ratio of the area of $\triangle AOB$ to that of $\triangle DOC$?

[CDS 2021(I)]

- (a) 2 : 1 (b) 3 : 2
(c) 4 : 1 (d) 9 : 4

- 272.** A circle touches all the four sides of a quadrilateral ABCD. If AB = 9 cm, BC = 8 cm and CD = 12, then what is DA equal to?

[CDS 2021(I)]

- (a) 14 cm (b) 13 cm
(c) 12 cm (d) 11 cm

Consider the following for the next three (03) items that follows:

ABC is a triangle with sides AB = 6 cm, BC = 10 cm and CA = 8 cm. What vertices A, B and C as centres, three circles are drawn each touching the other two externally.

- 273.** What is the sum of the radii of the circles?

[CDS 2021(II)]

- (a) 10.4cm (b) 11.2cm
(c) 12cm (d) 13cm

- 274.** What is the length of the altitude of the triangle drawn from vertex A on BC?

[CDS 2021(II)]

- (a) 2.4cm (b) 3cm
(c) 4cm (d) 4.8cm

- 275.** If P, Q and R are the areas of sectors at A, B and C within the triangle respectively, then which of the following is/are correct?

1. $P = \pi \text{ cm}^2$

2. $9Q + 4R = 36\pi \text{ cm}^2$

Select the correct answer using the code given below:

[CDS 2021(II)]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Consider the following for the next two (02) items that follow:

ABC is triangle in which AB = AC and D is any point on BC.

- 276.** Which one of the following is correct?

[CDS 2021(II)]

- (a) $AB^2 - AD^2 = AD \times BD$
(b) $AC^2 - AD^2 = BD \times CD$
(c) $AB^2 - AD^2 = 2 AD \times BD$
(d) $AC^2 - AD^2 = 2 BD \times CD$

- 277.** If AD = 5 cm, BD = 4 cm and CD = 6 cm, then what is AB equal to?

[CDS 2021(II)]

- (a) 7cm (b) 6.5cm
(c) 6cm (d) 5.5cm

Consider the following for the next three (03) items that follow:

ABC is a triangle with AB = 1.6cm, BC = 6.3 cm and CA = 6.5 cm. Let P and Q be the mid points of AB and BC respectively.

- 278.** What is $AB^2 + 4BQ^2$ equal to?

[CDS 2021(II)]

- (a) 41.25cm² (b) 42.25cm²
(c) 43.75cm² (d) 44.25cm²

- 279.** What is $AQ^2 + CP^2$ equal to?

[CDS 2021(II)]

- (a) AC^2 (b) $1.2 AC^2$
(c) $1.25 AC^2$ (d) $1.5 AC^2$

- 280.** What is $4(CP^2 - AQ^2)$ equal to?

[CDS 2021(II)]

- (a) 101.39 cm² (b) 111.39 cm²
(c) 121.39 cm² (d) 131.39 cm²

Consider the following for the next two (02) items that follow:

AB is a diameter of a circle with centre O. Radius OP is perpendicular to AB. Let Q be any point on arc PB.

- 281.** What is $\angle BAP$ equal to?

[CDS 2021(II)]

- (a) 30° (b) 40°
(c) 45° (d) 60°

- 282.** What is $\angle AQP$ equal to?

[CDS 2021(II)]

- (a) 30° (b) 40°
(c) 45° (d) 60°

- 283.** In a triangle ABC, AB = 16cm, AC = 12 cm and AD is the bisector of $\angle A$. If BD = 4, then what is CD equal to?

[CDS 2022(I)]

- (a) 2cm (b) 2.5cm
(c) 3cm (d) 3.5cm

- 284.** An equilateral triangle of side x is inscribed in a circle of radius y. Which one of the following is correct?

[CDS 2022(I)]

- (a) $2y = x$ (b) $2y = \sqrt{3}x$
(c) $\sqrt{3}y = 2x$ (d) $\sqrt{3}y = x$

- 285.** ABC is a triangle right angled at B. Let D be the midpoint on AC. If BD = 6.5cm, then what is $AB^2 + BC^2$ equal to?

[CDS 2022(I)]

- (a) 144 square cm (b) 169 square cm
(c) 196 square cm (d) 225 square cm

- 286.** In a right triangle ABC, BD is perpendicular on hypotenuse AC. If AC = 9 cm and AD = 4 cm, then what is AB + BC approximately equal to?

[CDS 2022(I)]

- (a) 12cm (b) 12.2cm
(c) 12.4cm (d) 12.6cm

- 287.** In a triangle ABC, AD is the bisector of $\angle BAC$. If AB = 12 cm, BD = 10 cm and DC = 5cm, then what is the perimeter of the triangle?

[CDS 2022(I)]

- (a) 30cm (b) 31cm
(c) 33cm (d) 35cm

- 288.** What is the radius of the circle inscribed in a triangle whose sides are 4 cm, 7.5cm and 8.5cm?

[CDS 2022(I)]

- (a) 1.5cm (b) 2cm
(c) 2.5cm (d) 3cm

- 289.** An equilateral triangular sheet is formed by joining 9 equilateral triangular sheets each of area $9\sqrt{3}\text{cm}^2$. What is the height of the bigger triangular sheet?

[CDS 2022(I)]

- (a) $9\sqrt{3}$ cm (b) 18cm
(c) $18\sqrt{3}$ cm (d) 27cm

- 290.** ABCD is a trapezium in which AB is parallel to DC. Let E and F be the midpoints on AD and BC respectively. If EF=10 cm and AB-DC =4cm, then what is the value of AB $\times$ DC?

- (a) 84 square cm (b) 96 square cm
(c) 100 square cm (d) 108 square cm

- 291.** ABCD is a parallelogram with AB = 15cm and AD = 8cm. If θ is the acute angle between AB and AD, then what is the area of the parallelogram in square cm?

[CDS 2022(I)]

- (a) $60 \sin \theta$ (b) $120 \sin \theta$
(c) $60 \cos \theta$ (d) $120 \cos \theta$

- 292.** The diagonal of a square is $12\sqrt{2}$ cm and the area of an equilateral triangle is $64\sqrt{3}$ square cm. Which of the following statements is/are correct?

1. The square and the triangle have the same perimeter.
2. Four times the area of the square is equal to $3\sqrt{3}$ times the area of the triangle.

Select the correct answer using the code given below:

[CDS 2022(I)]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

- 293.** Consider the following statements:

1. The sum of any two sides of a triangle is less than twice the median drawn to the third side.
 2. The perimeter of a triangle is greater than the sum of the three medians.
- Which of the above statements is/are correct?

[CDS 2022(I)]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

- 294.** Let D, E and F be the midpoints of the sides BC, CA and AB respectively of a triangle ABC. Triangle DEF is congruent to which of the following triangles?

1. AEF
2. FBD
3. EDC

Select the correct answer using the code given below:

[CDS 2022(I)]

- (a) 1 only (b) 2 and 3 only
(c) 3 only (d) 1, 2 and 3

- 295.** In a triangle ABC, AB = AC and BC is produced to D such that $\angle ACD = x$, then what is $\angle BAC$ equal to?

[CDS 2022(I)]

- (a) $2x - 90^\circ$ (b) $2x - 180^\circ$
(c) $180^\circ - 2x$ (d) $\frac{x}{2}$

- 296.** ABC is a triangle right angled at B with AC = 2BC. If $\angle A = x$, then what is $\angle C$ equal to?

[CDS 2022(I)]

- (a) $\frac{x}{2}$ (b) $2x$
(c) $\sqrt{2}x$ (d) $\sqrt{3}x$

- 297.** Consider the following statements:

1. If two chords AB and AC of a circle are equal, then the centre of the circle lies on the angle bisector of angle CAB.
 2. If two concentric circles are intersected by a line at A, B, C and D respectively, then AC = BD.
- Which of the above statements is/are correct?

[CDS 2022(I)]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

298. A circle of radius 25cm has a chord of length 48 cm. What is the length of the perpendicular drawn from the centre of the circle to the chord?

[CDS 2022(I)]

- (a) 5cm (b) 5.5cm
(c) 6.5cm (d) 7cm

299. The perpendicular dropped from a vertex of a right-angled triangle upon the hypotenuse divides it into two segments of lengths 9 units and 16 units respectively. What is the length of the perpendicular?

[CDS 2022(I)]

- (a) 6 units (b) 8 units
(c) 10 units (d) 12 units

300. What is the ratio of interior angle to exterior angle of a regular polygon of n sides?

[CDS 2022(I)]

- (a) n (b) $\frac{n-1}{2}$
(c) $\frac{n-2}{2}$ (d) $\frac{2(n-2)}{3}$

Consider the following for the next three (03) items that follow:

A triangle ABC with sides $AB = 15\text{cm}$, $BC = 9\text{cm}$, $CA = 12\text{cm}$ is inscribed in a circle.

301. What is $\cos^2 A + \cos^2 B + \cos^2 C$ equal to?

[CDS 2022 (II)]

- (a) $\frac{3}{4}$ (b) 1
(c) $\frac{5}{4}$ (d) 2

302. What is $\sin^2 A + \sin^2 B + \sin^2 C$ equal to?

[CDS 2022 (II)]

- (a) 2 (b) $\frac{5}{4}$
(c) 1 (d) $\frac{3}{4}$

303. What is the radius of the circle?

[CDS 2022 (II)]

- (a) 4.5cm (b) 6cm
(c) 7.5cm (d) 15cm

304. ABCD is a parallelogram. A circle through A, B and C intersects CD (produced) at E. Which of the following is/are correct?

1. $AE = AD$
2. $CD = DE$

Select the correct answer using the code given below:

[CDS 2022 (II)]

- (a) only 1 (b) only 2
(c) both 1 and 2 (d) neither 1 nor 2

305. Consider the following statements:

1. In an equilateral triangle, the centroid and centre of circumcircle coincide
 2. Angle bisectors of a cyclic quadrilateral from another cyclic quadrilateral
 3. Every cyclic parallelogram is a rectangle
- Which of the statements given above are correct.

[CDS 2022 (II)]

- (a) only 1 and 2 (b) only 2 and 3
(c) only 1 and 3 (d) 1, 2 and 3

306. ABCD is a trapezium in which AB is parallel to DC. The vertices A, B, C and D pass through a circle. Which of the following are correct?

1. $AD = BC$
2. $\angle A + \angle C = 180^\circ$
3. $\angle A + \angle D = 180^\circ$

Select the correct answer using the code given below:

[CDS 2022 (II)]

- (a) only 1 and 2 (b) only 2 and 3
(c) only 1 and 3 (d) 1, 2 and 3

307. ABCD is a cyclic quadrilateral. AB and DC when produced, meet in E. Which of the following statements is/are correct?

1. $\triangle EBC$ is similar to $\triangle EAD$
2. $\angle CBE + \angle DAE = 180^\circ$.

Select the correct answer using the code given below:

[CDS 2022 (II)]

- (a) only 1 (b) only 2
(c) both 1 and 2 (d) neither 1 nor 2

308. In a triangle ABC, DE is a line segment which intersects AB at D and AC at E such that DE is parallel to BC. The line segment divides the triangle in two parts of equal

area. What is $\frac{BD}{AB}$ equal to?

[CDS 2022 (II)]

- (a) $\frac{\sqrt{2}-1}{2}$ (b) $\frac{\sqrt{2}-1}{\sqrt{2}}$
(c) $\frac{\sqrt{3}-1}{\sqrt{3}}$ (d) $\sqrt{2}$

- 309.** A right angled triangle ABC is inscribed in a circle of radius 10cm. The altitude drawn to the hypotenuse AC is of length 8cm. If $AB = x$ cm and $BC = y$ cm, then what is the value of xy ?

[CDS 2022 (II)]

- (a) 60 (b) 80
(c) 120 (d) 160

- 310.** A circle is inscribed in a triangle ABC. It touches the sides BC, CA, AB at D, E, F respectively. What is $\angle EDF$ equal to?

[CDS 2022 (II)]

- (a) $90^\circ - A$ (b) $90^\circ - \frac{(B+C)}{2}$
(c) $90^\circ - 2A$ (d) $90^\circ - \left(\frac{A}{2}\right)$

- 311.** Triangle ABC is right-angled at A and AD is perpendicular to BC. If $BD = 7.5$ cm and $DC = 10$ cm, then what is AD equal to?

[CDS 2022 (II)]

- (a) 5cm (b) $5\sqrt{2}$ cm
(c) $5\sqrt{3}$ cm (d) 10cm

- 312.** The perpendicular AD on the base BC of a triangle ABC intersects BC at D so that $DB = 3CD$. Which one of the following is correct?

[CDS 2022 (II)]

- (a) $2(AB+AC)(AB-AC) = BC^2$
(c) $3(AB+AC)(AB-AC) = 2BC^2$
(c) $4(AB+AC)(AB-AC) = 3BC^2$
(d) $5(AB+AC)(AB-AC) = 4BC^2$

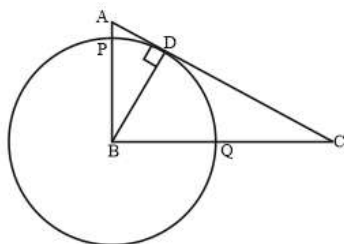
- 313.** Triangle ABC is right-angled at C and $AC = \sqrt{3}BC$. What is $\angle ABC$ equal to?

[CDS 2022 (II)]

- (a) 30° (b) 45°
(c) 60° (d) 75°

Consider the following for the next three (03) items that follow:

In the triangle ABC, $AB = 6$ cm, $BC = 8$ cm and $AC = 10$ cm. The perpendicular dropped from B meets the side AC at D. A circle of radius BD (with centre B) cuts AB and BC at P and Q respectively as shown in the figure.



- 314.** What is the length of QC?

[CDS - 2023 (1)]

- (a) 4.4cm (b) 4.2cm
(c) 3.6cm (d) 3.2cm

- 315.** If $\angle ABD = \theta$, then what is $\sin \theta$ equal to?

[CDS - 2023 (1)]

- (a) 0.4 (b) 0.5
(c) 0.6 (d) 0.8

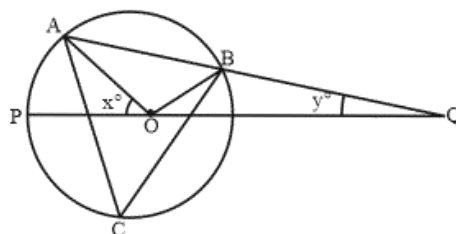
- 316.** What is the radius of the circle?

[CDS - 2023 (1)]

- (a) 5cm (b) 4.8cm
(c) 4.4cm (d) 4cm

Consider the following for the next two (02) items that follows:

In the following figure, a triangle ABC is inscribed in a circle with centre at O. Let $\angle POA = x^\circ$ and $\angle OQB = y^\circ$. Further $OB = BQ$.



- 317.** What is the relation between x and y ?

[CDS - 2023 (1)]

- (a) $x=y$ (b) $2x=3y$
(c) $x=3y$ (d) $3x=4y$

- 318.** If $y = 15$, then what is $\angle ACB$ equal to?

[CDS - 2023 (1)]

- (a) 30° (b) 40°
(c) 45° (d) 60°

- 319.** ABCD is a square field with $AB = x$. A vertical pole OP of height $2x$ stands at the centre O of the square field. If $\angle APO = \theta$, then what is $\cot \theta$ equal to?

[CDS - 2023 (1)]

- (a) $\sqrt{2}$ (b) 2
(c) $2\sqrt{2}$ (d) $3\sqrt{2}$

- 320.** Let ABC be a right-angled triangle with sides 5cm, 12cm and 13cm. If p is the length of the perpendicular drawn from vertex A on the hypotenuse BC, then what is the value of $13p$?

[CDS - 2023 (1)]

- (a) 24 (b) 48
(c) 60 (d) 90

For the next ten (3) items that follows:

Each item contains a question followed by two Statement. Answer each item using the following instructions:

[CDS – 2023 (1)]

(a) choose this option if the question can be answered by one of the statements alone but not by the other.

(b) choose this option if the question can be answered by either statement alone.

(c) Choose this option if the question can be answered by using both the statements together, but cannot be answered by using either statement alone

(d) Choose this option if the question cannot be answered even by using both statements together.

- 321.** Question : Can a circle be drawn through the points A, B and C?

Statement I: $AB = 5\text{cm}$, $BC = 5\text{cm}$, $CA = 6\text{cm}$

Statement II: $AB = 3\text{cm}$, $BC = 4\text{cm}$, $CA = 7\text{cm}$.

- 322.** Question: Is triangle Δ right angled?

Statement I: The length of the line segment joining the mid-points of two sides of Δ is half of the third side of Δ .

Statement II: The angles of Δ are in the ratio 1:2:3.

- 323.** The lengths of two longer sides of the triangle Δ are 25 cm and 24cm.

Question: What is the length of the shortest side?

Statement I: The angles of Δ are in the ratio 1:2:3

Statement II: The length of the perpendicular drawn on the longest side of Δ from its opposite vertex is 6.72cm.

- 324.** The arch of a bridge is in the form of an arc of a circle. If the span of the bridge is 40m and height in the middle is 8m, then what is the radius of curvature of the bridge?

[CDS-2023 (2)]

- (a) 25m (b) 27m
(c) 29m (d) 31m

- 325.** If a, b and c are the sides of a right-angled triangle, where $a > b > c$, then what is the value of the expression.

$(a+b+c) (a+b-c) (a-b+c) (a-b-c)?$

[CDS-2023 (2)]

- (a) $4b^2c^2$ (b) $-4b^2c^2$
(c) $-2a^2b^2$ (d) $-4a^2b^2$

- 326.** Consider the following statements:

1. The angle in a sector greater than a semi-circle is less than a right angle

2. If two sides of a pair of opposite sides of a cyclic quadrilateral are equal, then its diagonals are also equal.

Which of the statements given above is/are correct ?

[CDS-2023 (2)]

- (a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2

- 327.** In a triangle ABC, angle $B = 90^\circ$ and p is the length of the perpendicular from B to AC. If $BC = 10\text{cm}$ and $AC = 12\text{cm}$, then what is the value of p?

[CDS-2023 (2)]

- (a) $\frac{5\sqrt{11}}{3}$ (b) $\frac{10\sqrt{11}}{3}$
(c) $\frac{40}{\sqrt{61}}$ (d) $\frac{12}{25}$

- 328.** A triangle has sides 13cm, 14cm and 15cm long. What is the length of the smallest altitude of the triangle?

[CDS-2023 (2)]

- (a) 11cm (b) 11.2cm
(c) 12cm (d) 12.2cm

- 329.** The hypotenuse AC of a right-angled triangle ABC is $3\sqrt{5}\text{cm}$. If AB is doubled and BC is trip such that ABC remains a right-angle triangle, the hypotenuse becomes 15cm. What is $AB + BC$ equal to?

[CDS-2023 (2)]

- (a) 10cm (b) 9cm
(c) $2\sqrt{5}\text{cm}$ (d) 8cm

Consider the following for the next two (02) items that follow:

Two parallel chords AB and CD of a circle are of lengths 60 cm and 80cm respectively. They are on the same side of the centre O and 10cm apart.

[CDS-2023 (2)]

- 330.** What is the diameter of the circle?

- (a) 120cm (b) 110cm
(c) 100cm (d) 90cm

- 331.** If the chord AB subtends an angle α and chord CD subtends as angle β at the centre O, then what is the value of $\tan\left(\frac{\beta}{2}\right) - \tan\left(\frac{\alpha}{2}\right)?$

[CDS-2023 (2)]

- (a) $\frac{3}{4}$ (b) $\frac{5}{12}$

(c) $1/2$

(d) $7/12$

332. Consider the following for the next question.

Mark option (a) if the question can be answered by using one of the statements alone, but cannot be answered using the other statement alone.

Mark option (b) if the question can be answered by using either statements alone.

Mark option (c) if the question can be answered by using both the statements together, but cannot be answered using either statement alone.

Mark option (d) if the question cannot be answered even by using both the statements together.

[CDS-2023 (2)]

Question: What is the area of the triangle inscribed in a semi circle with the diameter as the base?

Statement I:

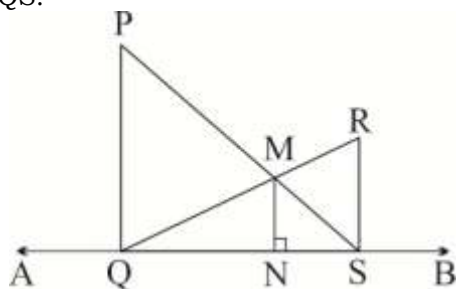
The diameter of semi circle is 20 cm

Statement II:

Two shorter sides of the triangle are 12 cm and 16 cm

Consider the following for the next two (02) items that follow:

Let two parallel line segments $PQ = 5$ cm and $RS = 3$ cm be perpendicular to a horizontal line AB , as shown in the figure given below. The point of intersection of PS and QR is M and MN is perpendicular to QS .



[CDS-2024 (1)]

333. What is the length of MN ?

- | | |
|----------------------|-----------------------|
| (a) $\frac{3}{8}$ cm | (b) $\frac{5}{8}$ cm |
| (c) $\frac{9}{8}$ cm | (d) $\frac{15}{8}$ cm |

334. What is the ratio of the area of the quadrilateral $PQNM$ to the area of the quadrilateral $PSNM$?

(a) $\frac{200}{117}$

(b) $\frac{212}{117}$

(c) $\frac{275}{117}$

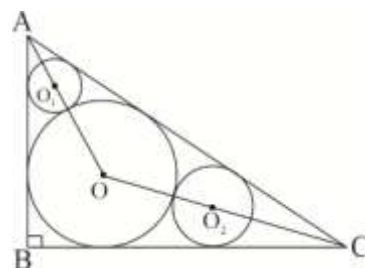
(d) $\frac{250}{117}$

Consider the following for the next three (03) items that follow:

ABC is a right-angled triangle with $\angle ABC = 90^\circ$. The centre of the incircle of the given triangle is at O , whose radius is 2 cm. Two more circles with centres at O_1 and O_2 , touch this circle and the two sides of shown in the figure given below.

Further $MA : MC = 2 : 3$.

[CDS-2024 (1)]



335. What is $AB + BC$ equal to?

- | | |
|-----------|-----------|
| (a) 10 cm | (b) 12 cm |
| (c) 13 cm | (d) 14 cm |

336. What is the radius of the circle with centre at O_1 ?

- | | |
|--------------------|--------------------|
| (a) $4 - \sqrt{5}$ | (b) $1 + \sqrt{5}$ |
| (c) $2 + \sqrt{5}$ | (d) $3 - \sqrt{5}$ |

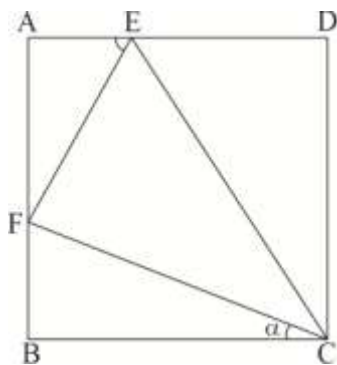
337. What is the radius of the circle with centre at O_2 ?

- | | |
|---------------------------------|---------------------------------|
| (a) $5 - \sqrt{10}$ | (b) $1 + 2\sqrt{5}$ |
| (c) $\frac{22 - 4\sqrt{10}}{9}$ | (d) $\frac{22 - 2\sqrt{10}}{9}$ |

Consider the following for the next two (02) items that follow:

A triangle CEF is drawn inside a square $ABCD$ as shown in the figure given below. Given: $CF = 8$ cm, $EF = 6$ cm and $CE = 10$ cm.

[CDS-2024 (1)]



- 338.** What is the area of the square ?
 (a) $\frac{512}{17}$ square cm (b) $\frac{625}{13}$ square cm
 (c) $\frac{1024}{17}$ square cm (d) $\frac{1296}{13}$ square cm
- 339.** What is $\tan \alpha + \tan \beta$ equal to
 (a) $\frac{13}{16}$ (b) $\frac{15}{16}$
 (c) $\frac{17}{16}$ (d) $\frac{17}{4}$
- 340.** What is the area of triangle CDE?
 (a) $\frac{416}{17}$ square cm (b) $\frac{312}{13}$ square cm
 (c) $\frac{208}{17}$ square cm (d) $\frac{156}{13}$ square cm

Consider the following for the next four items that follow:

Each item contains a Question followed by two Statements. Answer each item using the following instructions:
 Choose option

- (a) If the Question can be answered by one of the Statements alone, but not by the other. **1.**
 (b) If the Question can be answered by either Statement alone **2.**
 (c) If the Question can be answered by using both the Statements together, but cannot be answered by using either Statement alone.
 (d) If the Question can be answered even by using both Statements together

[CDS-2024 (1)]

- 341.** A circle touches all the four sides AB, BC, CD, DA of a quadrilateral ABCD
 Question: What is the perimeter of the quadrilateral?
 Statement-I: $AB + DC = 10$ cm

Statement-II: $AD + BC = 10$ cm

- 342.** Question: what is the ratio of the lengths of diagonals of a rhombus?
 Statement-I: One diagonal of the rhombus is equal to its side
 Statement-II: The longer diagonal of the rhombus is equal to $\sqrt{3}$ times its side
- 343.** The chord of a circle of radius R touches at a point on the circumference of a concentric circle of radius r. The length of the chord is 24 units.
 Question: What are the values of r and R?
 Statement-I: r is an integer
 Statement-II: R is an integer.
- 344.** P, Q, R, S are the mid-points of sides AB, BC, CD, DA respectively of a quadrilateral ABCD.
 Question: What is the difference in the area of the quadrilateral ABCD and the area of the quadrilateral PQRS?
 Statement-I: Area of the quadrilateral ABCD is 100 square unit.
 Statement-II: Area of the quadrilateral PQRS is 50 square unit.
- 345.** In a quadrilateral ABCD, $AB = 6$ m, $BC = 18$ cm, $CD = 6$ cm and $DA = 10$ cm. If the diagonal $BD = x$, then which one of the following is correct?
[CDS-2024 (1)]
 (a) $8 < x < 12$ (b) $12 < x < 16$
 (c) $16 < x < 18$ (d) $18 < x < 20$
- 346.** In a quarter circle of radius R, a circle of radius r is inscribed. What is the ratio of R to r?
[CDS-2024 (1)]
 (a) $(\sqrt{2}+1):1$ (b) $(\sqrt{3}+1):1$
 (c) $3:2$ (d) $5:4$
- 347.** In a quadrilateral ABCD, $AB = BC$ and $CD = DA$; AC and BD are diagonals such that $AC = 6$ cm and $BD = 12$ cm. What is the area of the quadrilateral?
[CDS-2024 (1)]
 (a) 24 square cm (b) 30 square cm
 (c) 36 square cm (d) 40 square cm
- 348.** If the difference between the interior and exterior angles of a regular polygon is 144° , then what is the number of sides of the polygon?
[CDS-2024 (1)]

(a) 12

(b) 16

(c) 18

(d) 20

ANSWER KEY

1.	b	2.	d	3.	d	4.	d	5.	d	6.	d	7.	d	8.	d	9.	c	10.	b
11.	c	12.	c	13.	c	14.	d	15.	c	16.	d	17.	b	18.	b	19.	b	20.	d
21.	b	22.	a	23.	c	24.	c	25.	c	26.	c	27.	b	28.	b	29.	b	30.	a
31.	a	32.	d	33.	a	34.	d	35.	a	36.	a	37.	b	38.	c	39.	d	40.	d
41.	d	42.	d	43.	c	44.	b	45.	b	46.	b	47.	d	48.	d	49.	d	50.	a
51.	c	52.	a	53.	c	54.	a	55.	c	56.	d	57.	d	58.	b	59.	d	60.	a
61.	a	62.	c	63.	a	64.	a	65.	a	66.	d	67.	d	68.	c	69.	d	70.	c
71.	a	72.	a	73.	c	74.	c	75.	d	76.	a	77.	c	78.	b	79.	c	80.	d
81.	c	82.	a	83.	b	84.	c	85.	b	86.	c	87.	a	88.	a	89.	c	90.	d
91.	c	92.	a	93.	c	94.	d	95.	c	96.	d	97.	c	98.	d	99.	a	100.	d
101.	a	102.	b	103.	c	104.	b	105.	d	106.	d	107.	d	108.	d	109.	b	110.	c
111.	c	112.	b	113.	c	114.	b	115.	b	116.	a	117.	b	118.	b	119.	c	120.	c
121.	c	122.	c	123.	c	124.	d	125.	c	126.	c	127.	b	128.	d	129.	a	130.	d
131.	c	132.	d	133.	b	134.	b	135.	c	136.	b	137.	c	138.	c	139.	d	140.	c
141.	b	142.	a	143.	b	144.	a	145.	c	146.	c	147.	c	148.	a	149.	c	150.	b
151.	c	152.	c	153.	d	154.	d	155.	a	156.	a	157.	b	158.	a	159.	c	160.	b
161.	b	162.	c	163.	d	164.	a	165.	b	166.	c	167.	b	168.	c	169.	c	170.	c
171.	c	172.	a	173.	c	174.	d	175.	a	176.	c	177.	c	178.	a	179.	b	180.	b
181.	d	182.	c	183.	d	184.	d	185.	a	186.	b	187.	b	188.	b	189.	b	190.	b
191.	b	192.	c	193.	b	194.	d	195.	d	196.	d	197.	b	198.	a	199.	a	200.	c
201.	c	202.	a	203.	d	204.	c	205.	b	206.	c	207.	d	208.	d	209.	c	210.	d
211.	c	212.	b	213.	c	214.	b	215.	a	216.	b	217.	d	218.	b	219.	d	220.	c
221.	c	222.	a	223.	c	224.	b	225.	b	226.	b	227.	d	228.	b	229.	c	230.	b
231.	a	232.	c	233.	a	234.	a	235.	c	236.	c	237.	b	238.	a	239.	a	240.	b
241.	c	242.	c	243.	b	244.	a	245.	a	246.	c	247.	a	248.	d	249.	a	250.	b
251.	b	252.	d	253.	c	254.	b	255.	b	256.	b	257.	a	258.	d	259.	b	260.	b
261.	c	262.	d	263.	c	264.	c	265.	d	266.	c	267.	c	268.	a	269.	c	270.	c
271.	d	272.	d	273.	c	274.	d	275.	c	276.	b	277.	a	278.	b	279.	c	280.	b
281.	c	282.	c	283.	c	284.	d	285.	b	286.	d	287.	c	288.	a	289.	a	290.	b
291.	b	292.	c	293.	b	294.	d	295.	b	296.	b	297.	c	298.	d	299.	d	300.	c
301.	b	302.	a	303.	c	304.	a	305.	d	306.	d	307.	a	308.	b	309.	b	310.	d
311.	c	312.	a	313.	c	314.	d	315.	c	316.	b	317.	c	318.	d	319.	c	320.	c
321.	b	322.	a	323.	b	324.	c	325.	b	326.	c	327.	a	328.	b	329.	c	330.	c
331.	d	332.	a	333.	d	334.	c	335.	d	336.	d	337.	c	338.	c	339.	d	340.	a
341.	b	342.	b	343.	c	344.	b	345.	b	346.	a	347.	c	348.	d				

PRACTICE SHEET-1

1. The diagonal of a square is $4\sqrt{2}$ cm. The diagonal of another square whose area is double that of the first square is:
 (1) $8\sqrt{2}$ cm (2) 16 cm
 (3) $\sqrt{32}$ cm (4) 8 cm
2. The breadth of a rectangular hall is three-fourth of its length. If the area of the floor is 768 sq. m., then the difference between the length and breadth of the hall is:
 (1) 8 metres (2) 12 metres
 (3) 24 metres (4) 32 metres
3. The length of a plot is five times its breadth. A playground measuring 245 square metres occupies half of the total area of the plot. What is the length of the plot?
 (1) $35\sqrt{2}$ metres (2) $175\sqrt{2}$ metres
 (3) 490 metres (4) $5\sqrt{2}$ metres
4. The difference between the length and breadth of a rectangle is 23 m. If its perimeter is 206 m, then its area is
 (1) 1520 m^2 (2) 2420 m^2
 (3) 2480 m^2 (4) 2520 m^2
5. A path of uniform width runs round the inside of a rectangular field 38 m long and 32 m wide. If the path occupies 600 m^2 , then the width of the path is
 (1) 30 m (2) 5 m
 (3) 18.75 m (4) 10 m
6. The length and breadth of a rectangle are increased by 20% and 25% respectively. The increase in the area of the resulting rectangle will be:
 (1) 60% (2) 50%
 (3) 40% (4) 30%
7. The length of a room floor exceeds its breadth by 20 m. The area of the floor remains unaltered when the length is decreased by 10 m but the breadth is increased by 5 m. The area of the floor (in square metres) is :
 (1) 280 (2) 325
 (3) 300 (4) 420
8. The sides of a triangle are 3 cm, 4 cm and 5 cm. The area (in cm^2) of the triangle formed by joining the mid points of this triangle is:
 (1) 6 (2) 3
 (3) $\frac{3}{2}$ (4) $\frac{3}{4}$
9. From a point in the interior of an equilateral triangle, the length of the perpendiculars to the three sides are 6 cm, 8 cm and 10 cm respectively. The area of the triangle is
 (1) 48 cm^2 (2) $16\sqrt{3} cm^2$
 (3) $192\sqrt{3} cm^2$ (4) $192 cm^2$
10. The area of an isosceles triangle is 4 square unit. If the length of the third side is 2 unit, the length of each equal side is
 (1) 4 units (2) $2\sqrt{3}$ units
 (3) $\sqrt{17}$ units (4) $3\sqrt{2}$ units
11. 360 sq. cm and 250 sq. cm are the area of two similar triangles. If the length of one of the sides of the first triangle be 8 cm, then the length of the corresponding side of the second triangle is
 (1) $6\frac{1}{5}$ cm (2) $6\frac{1}{3}$ cm
 (3) $6\frac{2}{3}$ (4) 6 cm
12. The perimeter of an isosceles triangle is 544 cm and each of the equal sides is $\frac{5}{6}$ times the base. What is the area (in cm^2) of the triangle?
 (1) 38172 (2) 18372
 (3) 31872 (4) 13872
13. The perimeter of a rhombus is 100 cm. If one of its diagonals is 14 cm, then the area of the rhombus is
 (1) 144 cm^2 (2) 225 cm^2
 (3) 336 cm^2 (4) 400 cm^2
14. If the measure of one side and one diagonal of a rhombus are 10 cm and 16 cm respectively, then its area (in cm^2) is :
 (1) 60 (2) 64
 (3) 96 (4) 100

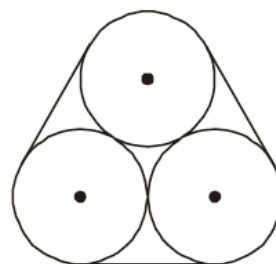
15. The ratio of the length of the parallel sides of a trapezium is 3:2. The shortest distance between them is 15 cm. If the area of the trapezium is 450 cm^2 , the sum of the length of the parallel sides is
 (1) 15 cm (2) 36 cm
 (3) 42 cm (4) 60 cm
16. A parallelogram has sides 15 cm and 7 cm long. The length of one of the diagonals is 20 cm. The area of the parallelogram is
 (1) 42 cm^2 (2) 60 cm^2
 (3) 84 cm^2 (4) 96 cm^2
17. In $\triangle ABC$, D and E are the points of sides AB and BC respectively such that $DE \parallel AC$ and $AD : DB = 3 : 2$. The ratio of area of trapezium ACED to that of $\triangle BED$ is
 (1) 4 : 15 (2) 15 : 4
 (3) 4 : 21 (4) 21 : 4
18. ABCD is a trapezium in which $AB \parallel DC$ and $AB = 2 CD$. The diagonals AC and BD meet at O. The ratio of area of triangles AOB and COD is
 (1) 1 : 1 (2) $1 : \sqrt{2}$
 (3) 4 : 1 (4) 1 : 4
19. The areas of a square and a rectangle are equal. The length of the rectangle is greater than the length of any side of the square by 5 cm and the breadth is less by 3 cm. Find the perimeter of the rectangle.
 (1) 17 cm (2) 26 cm
 (3) 30 cm (4) 34 cm
20. If the side of a square is increased by 25%, then its area is increased by:
 (1) 25% (2) 55%
 (3) 40.5% (4) 56.25%
21. ABC is an isosceles right angled triangle with $\angle B = 90^\circ$. On the sides AC and AB, two equilateral triangles ACD and ABE have been constructed. The ratio of area of $\triangle ABE$ and $\triangle ACD$ is
 (1) 1 : 3 (2) 2 : 3
 (3) 1 : 2 (4) 1 : 2
22. Two triangles ABC and DEF are similar to each other in which $AB = 10 \text{ cm}$, $DE = 8 \text{ cm}$. Then the ratio of the area of triangles ABC and DEF is
 (1) 4 : 5 (2) 25 : 16
 (3) 64 : 125 (4) 4 : 7
23. If $\triangle ABC$ is similar to $\triangle DEF$ such that $BC = 3 \text{ cm}$, $EF = 4 \text{ cm}$ and area of $\triangle ABC = 54 \text{ cm}^2$, then the area of $\triangle DEF$ is :
 (1) 66 cm^2 (2) 78 cm^2
 (3) 96 cm^2 (4) 54 cm^2
24. In $\triangle ABC$, O is the centroid and AD, BE, CF are three medians and the area of $\triangle AOE = 15 \text{ cm}^2$, then area of quadrilateral BDOF is
 (1) 20 cm^2 (2) 30 cm^2
 (3) 40 cm^2 (4) 25 cm^2
25. A straight line parallel to the base BC of the triangle ABC intersects AB and AC at the points D and E respectively. If the area of the $\triangle ABE$ be 36 sq.cm , then the area of the $\triangle ACD$ is
 (1) 18 sq.cm (2) 36 sq.cm
 (3) 18 cm (4) 36 cm
26. ABC is a right angled triangle, B being the right angle. Mid-points of BC and AC are respectively B' and A'. The ratio of the area of the quadrilateral AA'B'B' to the area of the triangle ABC is
 (1) 1 : 2 (2) 2 : 3
 (3) 3 : 4 (4) N. A.
27. Two triangles ABC and PQR are congruent. If the area of $\triangle ABC$ is 60 sq. cm , then area of $\triangle PQR$ will be
 (1) 60 sq.cm (2) 30 sq.cm
 (3) 15 sq.cm (4) 120 sq.cm
28. Two sides of a parallelogram are 20 cm and 25 cm. If the altitude corresponding to the side of length 25 cm is 10 cm, then the altitude corresponding to the other pair of sides is
 (1) 10.5 cm (2) 12 cm
 (3) 12.5 cm (4) 10 cm
29. The diagonal of a quadrilateral shaped field is 24m and the perpendiculars dropped on it from the remaining opposite vertices are 8m and 13m. The area of the field is
 (1) 252 m^2 (2) 156 m^2
 (3) 96 m^2 (4) 1152 m^2
30. Two isosceles triangles have equal vertical angles and their areas are in the ratio 9:16. Then the ratio of their corresponding heights is-
 (1) 4.5 : 8 (2) 4 : 3
 (3) 8 : 4.5 (4) 3 : 4

ANSWER KEY

1.	(4)	2.	(1)	3.	(1)	4.	(4)	5.	(2)	6.	(2)	7.	(3)	8.	(3)	9.	(3)	10.	(3)
11.	(3)	12.	(4)	13.	(3)	14.	(3)	15.	(4)	16.	(3)	17.	(4)	18.	(3)	19.	(4)	20.	(4)
21.	(3)	22.	(2)	23.	(3)	24.	(2)	25.	(2)	26.	(3)	27.	(1)	28.	(3)	29.	(1)	30.	(4)

PRACTICE SHEET-2

- The area of the ring between two concentric circles, whose circumference are 88 cm and 132 cm, is:
 (1) 780 cm² (2) 770 cm²
 (3) 715 cm² (4) 660 cm²
- Four equal circles each of radius 'a' units touch one another. The area enclosed between them ($\pi = 22/7$), in square units, is
 (1) $3a^2$ (2) $\frac{6}{7}a^2$
 (3) $\frac{41}{7}a^2$ (4) $\frac{1}{7}a^2$
- The area of circle whose radius is 6 cm is trisected by two concentric circles. The radius of the smallest circle is
 (1) $2\sqrt{3}$ cm (2) $2\sqrt{6}$ cm
 (3) 2 cm (4) 3 cm
- The radius of circle A is twice that of circle B and the radius of circle B is twice that of circle C. Their area will be in the ratio
 (1) 16 : 4 : 1 (2) 4 : 2 : 1
 (3) 1 : 2 : 4 (4) 1 : 4 : 16
- The four equal circles of radius 4 cm drawn on the four corners of a square touch each other externally. Then the area of the portion between the square and the four sectors is
 (1) $9(\pi - 4)$ sq. cm.
 (2) $16(\pi - 4)$ sq. cm.
 (3) $9(4 - \pi)$ sq. cm.
 (4) $16(4 - \pi)$ sq. cm.
- The area of a circle is proportional to the square of its radius. A small circle of radius 3 cm is drawn within a larger circle of radius 5 cm. Find the ratio of the area of the annular zone to the area of the larger circle. (Area of the annular zone is the difference between the area of the larger circle and that of the smaller circle).
 (1) 9 : 16 (2) 9 : 25
 (3) 16 : 25 (4) 16 : 27
- The diameter of two circles are the side of a square and the diagonal of the square. The ratio of the area of the smaller circle and the larger circle is
 (1) 1 : 2 (2) 1 : 4
 (3) $\sqrt{2} : \sqrt{3}$ (4) 1 : $\sqrt{2}$
- The area of the largest circle, that can be drawn inside a rectangle with sides 18 cm. by 14 cm, is
 (1) 49 cm² (2) 154 cm²
 (3) 378 cm² (4) 1078 cm²
- A circle is inscribed in an equilateral triangle of side 8 cm. The area of the portion between the triangle and the circle is
 (1) 11 cm² (2) 10.95 cm²
 (3) 10 cm² (4) 10.50 cm²
- If the difference between areas of the circum-circle and the in-circle of an equilateral triangle is 44 cm², then the area of the triangle is
 (1) 28 cm² (2) $7\sqrt{3}$ cm²
 (3) $14\sqrt{3}$ cm² (4) 21 cm²
- Three circles of diameter 10 cm each, are bound together by a rubber band, as shown in the figure.



The length of the rubber band, (in cm) if it is stretched as shown, is

- (1) 30 (2) $30 + 10\pi$
(3) 10π (4) $60 + 20\pi$

12. A circular road runs around a circular ground. If the difference between the circumference of the outer circle and the inner circle is 66 metres, the width of the road is:

- (1) 10.5 metres (2) 7 metres
(3) 5.25 metres (4) 21 metres

13. A person observed that he required 30 seconds less time to cross a circular ground along its diameter than to cover it once along the boundary. If his speed was 30 m/minute, then the radius of the circular ground.

- (1) 5.5 m (2) 7.5 m
(3) 10.5 m (4) 3.5 m

14. If the four equal circles of radius 3 cm touch each other externally, then the area of the region bounded by the four circles is

- (1) $4(9 - \pi)$ sq.cm (2) $9(4 - \pi)$ sq.cm
(3) $5(6 - \pi)$ sq.cm (4) $6(5 - \pi)$ sq.cm

15. The area of a circle is increased by 22 cm^2 when its radius is increased by 1 cm. The original radius of the circle is

- (1) 3 cm (2) 5 cm
(3) 7 cm (4) 9 cm

16. The radii of two circles are 5cm and 12cm. The area of a third circle is equal to the sum of the area of the two circles. The radius of the third circle is :

- (1) 13 cm (2) 21 cm
(3) 30 cm (4) 17 cm

17. The perimeter of a semicircular path is 36 m. Find the area of this semicircular path.

- (1) 42sq.m (2) 54 sq. m
(3) 63 sq.m (4) 77 sq. m

18. The ratio between the area of two circles is 4 : 7. What will be the ratio of their radii?

- (1) $2 : \sqrt{7}$ (2) $4 : 7$
(3) $16 : 49$ (4) $4 : \sqrt{7}$

19. Three circles of radius a, b, c touch each other externally. The area of the triangle formed by joining their centre is

- (1) $\sqrt{(a+b+c)abc}$

(2) $(a + b + c) \sqrt{ab + bc + ca}$

(3) $ab + bc + ca$

(4) None of the above

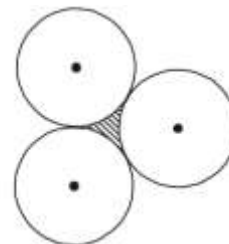
20. The area of a circle is proportional to the square of its radius. A small circle of radius 3 cm is drawn within a larger circle of radius 5 cm. Find the ratio of the area of the annular zone to the area of the larger circle. (Area of the annular zone is the difference between the area of the larger circle and that of the smaller circle).

- (1) 9 : 16 (2) 9 : 25
(3) 16 : 25 (4) 16 : 27

21. The diameter of two circles are the side of a square and the diagonal of the square. The ratio of the area of the smaller circle and the larger circle is

- (1) 1 : 2 (2) 1 : 4
(3) 2 : 3 (4) 1 : 2

22. Three circles of equal radius ' a ' cm touch each other. The area of the shaded region is:



(a) $\left(\frac{\sqrt{3} + \pi}{2}\right)a^2$

(b) $\left(\frac{6\sqrt{3} - \pi}{2}\right)a^2$

(c) $(\sqrt{3} - \pi)a^2$

(d) $\left(\frac{2\sqrt{3} - \pi}{2}\right)a^2$

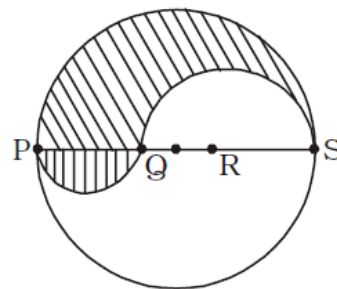
23. The radii of two circles are 10 cm and 24 cm. The radius of a circle whose area is the sum of the area of these two circles is

- (1) 36 cm (2) 17 cm
(3) 34 cm (4) 26 cm

24. The perimeter of a rhombus is 60 cm and one of its diagonal is 24 cm. The area of the rhombus is

- (1) 108 sq. cm. (2) 216 sq. cm.
(3) 432 sq. cm. (4) 206 sq. cm.

25. The ratio of circumference and diameter of a circle is $22 : 7$. If the circumference be $1\frac{4}{7}$ m, then the radius of the circle is :
- (1) $\frac{1}{3}$ m (2) $\frac{1}{2}$ m
(3) $\frac{1}{4}$ m (4) 1 m
26. Four circles of equal radii are described about the four corners of a square so that each touches two of the other circles. If each side of the square is 140 cm then area of the space enclosed between the circumference of the circle is Take
- (1) 4200 cm² (2) 2100 cm²
(3) 7000 cm² (4) 2800 cm²
27. The perimeter of a triangle is 67 cm. The first side is twice the length of the second side. The third side is 11 cm more than the second side. Find the length of the shortest side of the triangle.
- (1) 12 cm. (2) 14 cm.
(3) 17 cm. (4) 25 cm.
28. The radius of a wheel is 25 cm. How many rounds it will take to complete 11 km.
- (1) 5000 (2) 6000
(3) 7000 (4) 4000
29. If the perimeter of circle A is equal to perimeter of semi circle B, what is the ratio of their areas?
- (1) $(\pi + 2)^2 : 2\pi^2$ (2) $2\pi^2 : (\pi + 2)^2$
(3) $(\pi + 2)^2 : 4\pi^2$ (4) $4\pi^2 : (\pi + 2)^2$
30. PS is a diameter of a circle of radius 6 cm. In the diameter PS, Q and R are two points such that PQ, QR and RS are all equal. Semicircles are drawn on PQ and QS as diameter (as shown in the figure). The perimeter of shaded portion is :



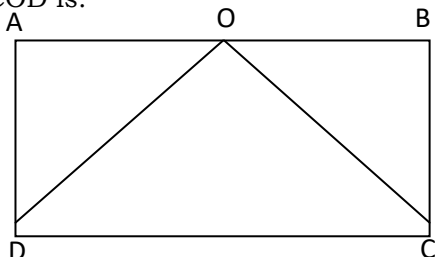
- (1) $15\frac{6}{7}$ cm (2) $75\frac{3}{7}$
(3) $37\frac{5}{7}$ cm (4) $18\frac{6}{7}$ cm

ANSWER KEY

1.	(2)	2.	(2)	3.	(1)	4.	(1)	5.	(4)	6.	(3)	7.	(1)	8.	(2)	9.	(2)	10.	(3)
11.	(2)	12.	(1)	13.	(4)	14.	(2)	15.	(1)	16.	(1)	17.	(4)	18.	(1)	19.	(1)	20.	(3)
21.	(1)	22.	(4)	23.	(4)	24.	(2)	25.	(3)	26.	(1)	27.	(2)	28.	(3)	29.	(3)	30.	(4)

CDS PYQ

1. In the figure given below, the area of rectangle ABCD is 100 sq. cm. O is any point on AB and CD = 20cm. Then, the area of $\triangle COD$ is:



[CDS 2013(I)]

- (a) 40 sq cm (b) 45 sq cm
(c) 50 sq cm (d) 80 sq cm

2. If an isosceles right angled triangle has area 1 sq unit, then what is its perimeter?

[CDS 2013(I)]

- (a) 3 units (b) $2\sqrt{2}+1$ units
(c) $(\sqrt{2}+1)$ units (d) $2(\sqrt{2}+1)$ units

3. A circular water fountain 6.6 m in diameter is surrounded outside by a path of width 1.5m. The area of this path (in sq m) is:

[CDS 2013(I)]

- (a) 13.62π (b) 13.15π
(c) 12.15π (d) None of these

4. The area of a rectangular field is 4500 sq m. If its length and breadth are in the ratio 9:5, then its perimeter is:

[CDS 2013(I)]

- (a) 90m (b) 150m
(c) 280m (d) 360m

5. The area of a square inscribed in a circle of radius 8cm is:

[CDS 2013(I)]

- (a) 32 sq cm (b) 64 sq cm
(c) 128 sq cm (d) 256 sq cm

6. The short and long hands of a clock are 4 cm and 6 cm long, respectively. Then, the ratio of distances travelled by tips of short hand in 2 days and long hand in 3 days is:

[CDS 2013(I)]

- (a) 4:9 (b) 2:9
(c) 2:3 (d) 1:27

7. The arc AB of the circle with centre at O and radius 10cm has length 16cm. What is the area of the sector bounded by the radii OA, OB and the arc AB?

[CDS 2013(I)]

- (a) 40π sq cm (b) 40 sq cm
(c) 80 sq cm (d) 20π sq cm

8. The minute hand of a watch is 2.5 cm long. The distance its extreme end transverses in 40 min is:

[CDS 2013(I)]

- (a) $10\pi/3$ cm (b) $3\pi/10$ cm
(c) $10/3$ cm (d) 10cm

9. If the area of a regular hexagon is $96\sqrt{3}$ sq cm, then its perimeter is:

[CDS 2013(I)]

- (a) 36cm (b) 48cm
(c) 54cm (d) 64cm

10. What is the area of a circle whose area is equal to that of a triangle with sides 7cm, 24cm and 25cm?

[CDS 2013(II)]

- (a) 80cm^2 (b) 84cm^2
(c) 88cm^2 (d) 90cm^2

11. If the area of an equilateral triangle is x and its perimeter is y, then which one of the following is correct?

[CDS 2013(II)]

- (a) $y^4 = 432x^2$ (b) $y^4 = 21x^2$
(c) $y^2 = 432x^2$ (d) None of these

12. The area of an isosceles $\triangle ABC$ with $AB = AC$ and altitude $AD = 3$ cm is 12 sq cm. What is its perimeter?

[CDS 2013(II)]

- (a) 18cm (b) 16cm
(c) 14cm (d) 12cm

13. How many 200mm lengths can be cut from 10m of ribbon?

[CDS 2013(II)]

- (a) 50 (b) 40
(c) 30 (d) 20

14. What is the area between a square of side 10 cm and two inverted semi-circular, cross sections each of radius 5cm inscribed in the square?

[CDS 2013(II)]

- (a) 17.5cm^2 (b) 18.5cm^2
(c) 20.5cm^2 (d) 21.5cm^2

15. The perimeter of a rectangle having area equal to 144cm^2 and sides in the ratio 4 : 9 is:

[CDS 2013(II)]

- (a) 52cm (b) 56cm
(c) 60cm (d) 64cm

16. One side of a parallelogram is 8.06 cm and its perpendicular distance from opposite side is 2.08 cm. What is the approximate area of the parallelogram?

[CDS 2013(II)]

- (a) 12.56cm^2 (b) 14.56cm^2
(c) 16.76cm^2 (d) 22.56cm^2

17. If the diagonals of a rhombus are 4.8 cm and 1.4 cm, then what is the perimeter of the rhombus?

[CDS 2013(II)]

- (a) 5cm (b) 10cm
(c) 12cm (d) 20cm

18. A regular hexagon is inscribed in a circle of radius 5cm. If x is the area inside the circle but outside the regular hexagon, then which one of the following is correct?

[CDS 2013(II)]

- (a) $13\text{cm}^2 < x < 15\text{cm}^2$
(b) $15\text{cm}^2 < x < 17\text{cm}^2$
(c) $17\text{cm}^2 < x < 19\text{cm}^2$
(d) $19\text{cm}^2 < x < 21\text{cm}^2$

19. The area of a rectangle lies between 40cm^2 and 45cm^2 . If one of the sides is 5 cm, then its diagonal lies between:

[CDS 2014(I)]

- (a) 8 cm and 10cm
(b) 9 cm and 11cm
(c) 10 cm and 12 cm
(d) 11 cm and 13 cm

20. How many circular plates of diameter d be taken out of a square plate of side $2d$ with minimum loss of material?

[CDS 2014(I)]

- (a) 8 (b) 6
(c) 4 (d) 2

21. What is the total area of three equilateral triangles inscribed in a semi-circle of radius 2cm ?

[CDS 2014(I)]

- (a) 12cm^2 (b) $\frac{3\sqrt{3}}{4}\text{cm}^2$
(c) $\frac{9\sqrt{3}}{4}\text{cm}^2$ (d) $3\sqrt{3}\text{cm}$

22. The area of sector of a circle of radius 36cm is $72\pi\text{cm}^2$. The length of the corresponding arc of the sector is:

[CDS 2014(I)]

- (a) πcm (b) $2\pi\text{cm}$
(c) $3\pi\text{cm}$ (d) $4\pi\text{cm}$

23. A square is inscribed in a circle of diameter $2a$ and another square is circumscribing circle. The difference between the area of outer and inner squares is:

[CDS 2014(I)]

- (a) a^2 (b) $2a^2$
(c) $3a^2$ (d) $4a^2$

24. ABC is a triangle right angled at A. $AB = 6\text{cm}$ and $AC = 8\text{cm}$. Semi-circles drawn (outside the triangle) on AB, AC and BC as diameters which enclose areas x , y square units, respectively. What is $x + y - z$ equal to?

[CDS 2014(I)]

- (a) 48cm^2 (b) 32cm^2
(c) 0 (d) None of these

25. Consider an equilateral triangle of a side of unit length. A new equilateral triangle is formed by joining the mid points of one,

then a third equilateral triangle is formed by joining the mid points of seconds. The process is continued. The perimeter of all triangles, thus formed is:

[CDS 2014(I)]

- (a) 2 units (b) 3 units
(c) 6 units (d) Infinity

26. What is the area of the larger segment of a circle formed by a chord of length 5cm subtending an angle of 90° at the centre?

[CDS 2014(I)]

- (a) $\frac{25}{4}\left(\frac{\pi}{2}+1\right)\text{cm}^2$ (b) $\frac{25}{4}\left(\frac{\pi}{2}-1\right)$
(c) $\frac{25}{4}\left(\frac{3\pi}{2}+1\right)$ (d) None of these

27. A rectangle of maximum area of drawn inside a circle of diameter 5cm. What is the maximum area of such a rectangle?

[CDS 2014(I)]

- (a) 25cm^2 (b) 12.5cm^2
(c) 12cm^2 (d) None of these

28. If AB and CD are two diameters of a circle of radius r and they are mutually perpendicular, then what is the ratio of the area of the circle to the area of the $\triangle ACD$?

[CDS 2014(I)]

- (a) $\frac{\pi}{2}$ (b) π
(c) $\frac{\pi}{4}$ (d) 2π

29. If every side of an equilateral triangle is doubled, then the area of new triangle becomes k times the area of the old one. What is k equal to?

[CDS 2014(II)]

- (a) $\sqrt{3}$ (b) 2
(c) 4 (d) 8

30. A railroad curve is to be laid on a circle. What radius (approx) should be used. If the track is to change direction by 25° in a distance of 120m?

[CDS 2014(II)]

- (a) 300 m (b) 280 m
(c) 275 m (d) 264 m

31. The sides of a triangular field are 41m, 40m and 9m. The number of rose bend that can be prepared in the field if each rose bed, on an average, needs 900 square cam space, is:

[CDS 2015(I)]

- (a) 2000 (b) 1800
(c) 900 (d) 800

32. The ratio of the outer and inner perimeters of a circular path is 23:22. If the path is 5m wide, the diameter of the inner circle is:

[CDS 2015(I)]

- (a) 55m (b) 110m
(c) 220m (d) 230m

33. For equal-sized maximum circular plate are cut off from a square paper sheet of area 784 square cm. The circumference of each plate is:

[CDS 2015(I)]

- (a) 11cm (b) 22cm
(c) 33cm (d) 44cm

34. A boy is cycling such that the wheels of the cycle are making 140 revolutions per minute. If the radius of the wheel is 30 cm, the speed of the cycle is:

[CDS 2015(II)]

- (a) 15.5km/hour (b) 15.84km/hour
(c) 16km/hour (d) 16.36km/hour

35. There are 437 fruit plants in an orchard planted in rows. The distance between any two adjacent rows in 2m and the distance between any two adjacent plants in 2m. Each rows has the same number of plants. There is 1m clearance on all sides of the orchard. What is the cost of fencing the area at the rate of Rs100 per meter.

[CDS 2015(II)]

- (a) Rs 15,600
(b) Rs 16,800
(c) Rs 18,200
(d) More information is required

36. The circumference of a circle is 100cm. The side of the square inscribed in the circle is:

[CDS 2015(II)]

- (a) $50\sqrt{2}$ cm (b) $\frac{100}{\pi}$ cm
(c) $\frac{50\sqrt{2}}{\pi}$ cm (d) $\frac{100\sqrt{2}}{\pi}$ cm

37. The diameter of a wheel that makes 452 revolutions to move 2km and 26 decameter is equal to:

[CDS 2015(II)]

- (a) $1\frac{9}{22}m$ (b) $1\frac{13}{22}m$
(c) $2\frac{5}{11}m$ (d) $2\frac{7}{11}m$

38. A square is inscribed in a right triangle with legs x and y and has common right angle with the triangle. The perimeter of the square is given by:

[CDS 2015(II)]

- (a) $\frac{2xy}{x+y}$ (b) $\frac{4xy}{x+y}$
(c) $\frac{2xy}{\sqrt{x^2+y^2}}$ (d) $\frac{4xy}{\sqrt{x^2+y^2}}$

39. The area of a trapezium is 336 cm^2 . If its parallel sides are in the ratio 5:7 and the perpendicular distance between them is 14cm, then the smaller of the parallel sides is:

[CDS 2015(II)]

- (a) 20cm (b) 22cm
(c) 24cm (d) 26cm

40. ABCD is a square. If the sides AB and CD are increased by 30%, sides BC and AD are increased by 20%, then the area of the resulting rectangle exceeds the area of the square by:

[CDS 2015(II)]

- (a) 50% (b) 52%
(c) 54% (d) 56%

41. From a rectangular sheet of sides 18 cm and 14cm, a semicircular portion with smaller side as diameter is taken out. Then the area of the remaining sheet will be:

[CDS 2015(II)]

- (a) 98cm^2 (b) 100cm^2
(c) 108cm^2 (d) 175cm^2

42. A square and an equilateral triangle have equal perimeter. If the diagonal of the square is $12\sqrt{2} \text{ cm}$, then the area of the triangle is:

[CDS 2015(II)]

- (a) $24\sqrt{2}$ (b) $24\sqrt{3}$
(c) $48\sqrt{3}$ (d) $64\sqrt{3}$

43. The number of rounds that a wheel of diameter $7/11$ meter will make in traversing 4 km will be:

[CDS 2016(I)]

- (a) 500 (b) 1000
(c) 1700 (d) 2000

44. The base of an isosceles triangle is 300 unit and each of its equal sides is 170 units. Then the area of the triangle is:

[CDS 2016(I)]

- (a) 9600 square units
(b) 1000 square units
(c) 12000 square units
(d) None of the above

45. Four equal discs are placed such that each one touches two others. If the area of empty space enclosed by them is $150/847$ square centimeter, then the radius of each disc is equal to:

[CDS 2016(I)]

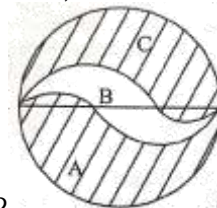
- (a) $7/6 \text{ cm}$ (b) $5/6 \text{ cm}$
(c) $1/2 \text{ cm}$ (d) $5/11 \text{ cm}$

46. A square is inscribed in a right-angled triangle with legs p and q, and has a common right angle with the triangle. The diagonals of the square is given by:

[CDS 2016(I)]

- (a) $\frac{pq}{p+2a}$ (b) $\frac{pq}{2p+q}$
(c) $\frac{\sqrt{2}pq}{p+q}$ (d) $\frac{2pq}{p+q}$

47. A circle of 3m radius is divided into three areas by semicircles of radii 1m and 2m as shown in the figure above. The ratio of the three areas A, B and C will be:

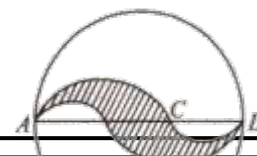


[CDS 2016(I)]

- (a) 2:3:2 (b) 1:1:1
(c) 4:3:4 (d) 1:2:1

48. AD is the diameter of a circle with area 707 m^2 and $AB = BC = CD$ as shown in the figure above. All curves inside the circle are semicircles with their diameters on AD. What is the cost of leveling the shaded region at the rate of 63 per square meter?

[CDS 2016(I)]



- (a) 29,700
(b) 22,400
(c) 14,847
(d) None of the above

49. If a square of side x and an equilateral triangle of side y are inscribed in a circle, then what is the ratio of x to y ?

[CDS 2016(II)]

- (a) $\sqrt{\frac{2}{3}}$ (b) $\sqrt{\frac{3}{2}}$
(c) $\frac{3}{\sqrt{2}}$ (d) $\frac{\sqrt{2}}{3}$

50. A circle and a square have the same perimeter. Which one of the following is correct?

[CDS 2016(II)]

- (a) The areas are equal
(b) The area of the circle is larger
(c) The area of the square is $\frac{\pi}{2}$ times area of circle
(d) The area of the square is π times area of circle

51. What is the area of a triangle with sides of length 12cm, 13cm and 5cm?

[CDS 2016(II)]

- (a) 30cm^2 (b) 35cm^2
(c) 40cm^2 (d) 42cm^2

52. What is area of largest triangle inscribed in a semi circle of radius r units?

[CDS 2016(II)]

- (a) r^2 square units (b) $2r^2$ square units
(c) $3r^2$ square unit (d) $4r^2$ square unit

53. The diameter of the front wheel of an engine is $2x$ cm and that of rear wheel is $2y$ cm. To cover the same distance, what is the number of times the rear wheel revolves when front wheel revolves n times?

[CDS 2016(II)]

- (a) n/xy (b) ny/x
(c) nx/y (d) xy/n

54. What is the number of round that a wheel of diameter $\frac{5}{11}$ m will make in traversing 7km?

[CDS 2016(II)]

- (a) 3300 (b) 3500
(c) 4400 (d) 4900

55. If a quadrilateral has an inscribed circle, then the sum of a pair of opposite sides equal.

[CDS 2016(II)]

- (a) Half the sum of the diagonals
(b) Sum of the other pair of opposite sides
(c) Sum of two adjacent sides
(d) None of the above

56. The wheel of a car are of diameter 80cm each. The car is travelling at a speed of 66km/hr. What is the number of complete revolutions each wheel makes in 10 minutes?

[CDS 2016(II)]

- (a) 4275 (b) 4350
(c) 4375 (d) 4450

57. If the perimeter of a circle is equal to that of a square, then what is the ratio of area of circle to that of square?

[CDS 2016(II)]

- (a) 22:7 (b) 14:11
(c) 7:22 (d) 11:14

58. If the perimeter of a rectangle is 10 cm and the area is 4cm^2 , then its length is:

[CDS 2017(I)]

- (a) 6cm (b) 5cm
(c) 4.5cm (d) 4cm

59. The areas of two circular field are in the ratio 16:49. If the radius of the bigger field is 14m, then what is the radius of the smaller field?

[CDS 2017(I)]

- (a) 4m (b) 8m
(c) 9m (d) 10m

60. The area of a regular hexagon of side 'a' is equal to:

[CDS 2017(I)]

- (a) $\frac{\sqrt{2}}{3} a^2$ square units
(b) $\frac{3\sqrt{3}}{2} a^2$ square units
(c) $\frac{1}{3} a^2$ square units
(d) $\frac{\sqrt{3}}{2} a^2$ square units

61. If each of the dimensions of a rectangle is increased by 200%, the area is increased by:

[CDS 2017(I)]

- (a) 300% (b) 400%

- (c) 600% (d) 800%

- 62.** Three circles each of radius 3.5 cm touch one another. The area subtended between them is:

[CDS 2017(I)]

- (a) $6(\sqrt{3}\pi - 2)$ (b) $6(2\pi - \sqrt{3}\pi)$
(c) $\frac{49}{8}(2\sqrt{3} - \pi)$ (d) $\frac{49}{8}(\sqrt{3} - \pi)$

- 63.** A copper wire when bent in the form of a square encloses an area of 121 cm. If the same wire is bent in the form of a circle, it encloses an area equal to

[CDS 2017(I)]

- (a) 121 cm² (b) 144 cm²
(c) 154 cm² (d) 168 cm²

- 64.** The radius of a circle is increased so that its circumference increases by 15%. The area of the circle will increase by

[CDS 2017(I)]

- (a) 31.25% (b) 32.25%
(c) 33.25% (d) 34.25%

- 65.** An isosceles triangle is drawn outside on one of the sides of a square as base in such a way that the perimeter of the complete figure is $\frac{7}{6}$ times the perimeter of the original square. What is the ratio of area of the triangle to the area of the original square?

[CDS 2017(II)]

- (a) 1:1 (b) 2:2
(c) 1:2 (d) 1:3

- 66.** What is the area of the triangle whose sides are 51 cm, 37cm and 20cm?

[CDS 2017(II)]

- (a) 300 square cm (b) 305 square cm
(c) 306 square cm (d) 307 square cm

- 67.** Two straight lines AB and AC include an angle. A circle is drawn in this angle which touches both these lines. One more circle is drawn which touches both these lines as well as the previous circle. If the area of the bigger circle is 9 times the area of the smaller circle, then what must be the angle A?

[CDS 2017(II)]

- (a) 45° (b) 60°
(c) 75° (d) 90°

- 68.** The diameters of two given circles are in the ratio 12:5 and the sum of their areas is equal to the area of a circle of diameter 65cm. What are their radii?

[CDS 2017(II)]

- (a) 12cm and 5cm
(b) 24cm and 10cm
(c) 60 cm and 25cm
(d) 30cm and 12.5cm

- 69.** Segment QR of length r is a tangent at Q to a circle of radius r with center at P. What is the area of the part of the triangle PQR, which is outside the circular region?

[CDS 2017(II)]

- (a) $\frac{\pi r^2}{16}$ (b) $\frac{r^2}{2} - \frac{\pi r^2}{8}$
(c) $\frac{r^2}{2} - \frac{\pi r^2}{16}$ (d) $\frac{r^2}{4} - \frac{\pi r^2}{8}$

- 70.** If the length of a side of a square is increased by 8cm, its area increases by 120 square cm. What is the length of a side of the square?

[CDS 2018(I)]

- (a) 2.5cm (b) 3.5cm
(c) 4.5cm (d) 5.5cm

- 71.** A rectangular pathway having width 4.5m and length 10m will have to be tiled using square tiles of side 50cm. Each packet of such tiles contains 20 pieces and costs Rs100. What will be the total cost of tiles for the pathway?

[CDS 2018(I)]

- (a) Rs 1,200 (b) Rs 1,100
(c) Rs 1,000 (d) Rs 900

- 72.** The product of the length of the diagonals of a square is 50 square units. What is the length of a side of the square?

[CDS 2018(I)]

- (a) $5\sqrt{2}$ units (b) 5 units
(c) 10 units (d) $2\sqrt{5}$ units

- 73.** Two equal circular regions of greatest possible area are cut off from a given circular sheet of area A. What is the remaining area of the sheet?

[CDS 2018(I)]

- (a) A/2 (b) A/3
(c) 3A/5 (d) 2A/5

- 74.** A square and an equilateral triangle have the same perimeter. If the diagonal of the square is $6\sqrt{2}$ cm, the what is the area of the triangle?

[CDS 2018(I)]

- (a) $12\sqrt{2}$ cm² (b) $12\sqrt{3}$ cm²
(c) $16\sqrt{2}$ cm² (d) $16\sqrt{3}$ cm²

- 75.** What is the area of the region bound externally by a square if side of length 'a' and internally by a square of side of length 'a' and internally by a circle passing through the four corners of the square?

[CDS 2018(I)]

- (a) $(\pi - 1)a^2$ square units
(b) $\frac{(\pi-1)a^2}{2}$ square units
(c) $(\pi - 2)a^2$ square units
(d) $\frac{(\pi-2)a^2}{2}$ square units

- 76.** A region of area A bounded by a circle C is divided into n regions, each of area A/n by drawing circle of radii $r_1, r_2, r_3, \dots, r_{n-1}$ such that $r_1 < r_2 < r_3 < \dots < r_{n-1}$ concentric with the circle C. If $p_m = \frac{r_{m+1}}{r_m}$ where $m = 1, 2, 3, \dots, (n-2)$, then which one of the following is correct?

[CDS-2018-1]

- (a) p increases as m increases
(b) p decreases as m increases
(c) p remains constant as m increases
(d) p increases for some values of m as m increases and then decreases thereafter

- 77.** A wire is in the form of a circle of radius 98cm. A square is formed out of the wire. What is the length of a side of the square? (Use $\pi = 22/7$)

[CDS-2018-1]

- (a) 146cm (b) 152cm
(c) 154cm (d) 156cm

- 78.** What is the area of the largest circular disc cut from a square of side $\frac{2}{\sqrt{\pi}}$ units?

[CDS-2018-1]

- (a) π square units (b) 1 square unit
(c) π^2 square units (d) 2 square units

- 79.** If base and hypotenuse of a right triangle are $(u^2 - v^2)$ and $(u^2 + v^2)$ respectively and the area of the triangle is 2016 square units, then the perimeter of the triangle may be:

[CDS 2018(II)]

- (a) 224 units (b) 288 units
(c) 448 units (d) 576 units

- 80.** Let S be the parallelogram obtained by joining the mid points of the parallelogram T. Consider the following statements:

1. The ratio of area of T to that of S is 2 : 1
2. The perimeter of S is half of the sum of diagonals T

Which of the above statements is/are correct?

[CDS 2018(II)]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

- 81.** The sides of a triangle are 5 cm, 6 cm and 7 cm. The area of the triangle is approximately:

[CDS 2018(II)]

- (a) 14.9cm² (b) 14.7cm²
(c) 14.5 cm² (d) 14.3 cm²

- 82.** There is a path of width 5 m around a circular plot of land whose area is 144π m². The total area of the circular plot including the path surrounding it is:

[CDS 2018(II)]

- (a) 349m² (b) 289m²
(c) 209m² (d) 149π m²

- 83.** An equilateral triangle, a square and a circle have equal perimeter. If T, S and C denote the area of the triangle, area of the square and area of the circle respectively, then which one of the following is correct?

[CDS 2018(II)]

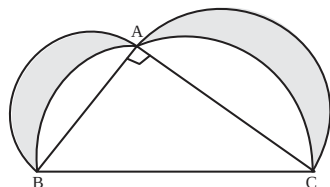
- (a) $T < S < C$ (b) $S < T < C$
(c) $C < S < T$ (d) $T < C < S$

- 84.** The area of the region bounded externally by a square of side 2a cm and internally by the circle touching the four sides of the square is:

[CDS 2018(II)]

- (a) $(4-\pi)a^2$ (b) $(\pi-2)a^2$
(c) $(8-\pi)a^2/2$ (d) $(\pi-2)a^2/2$

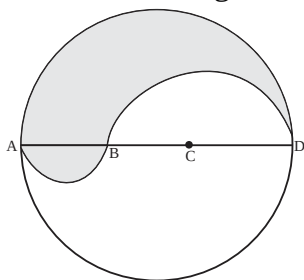
- 85.** In the figure given below, ABC is a right-angled triangle where $\angle A = 90^\circ$, $AB = p$ cm and $AC = q$ cm. On the three sides as diameters semicircles are drawn as shown in the figure. The area of the shaded portion, in square cm, is:



[CDS 2018(II)]

- (a) pq (b) $\pi(p^2 + q^2)/2$
(c) $\pi(p^2 + q^2)$ (d) $pq/2$

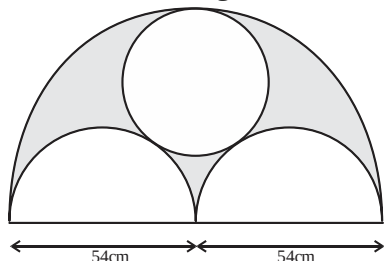
- 86.** In the figure given below, ABCD is the diameter of a circle of radius 9 cm. The lengths AB, BC and CD are equal. Semicircles are drawn on AB and BD as diameters as shown in the figure. What is the area of the shaded region?



[CDS 2018(II)]

- (a) 9π (b) 27π
(c) 36π (d) 81π

- 87.** In the figure given below, the diameter of bigger semicircle is 108cm. What is the area of the shaded region?

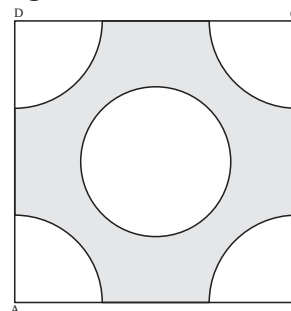


[CDS 2018(II)]

- (a) $201\pi\text{cm}^2$ (b) $186.3\pi\text{cm}^2$
(c) $405\pi\text{cm}^2$ (d) $769.5\pi\text{cm}^2$

- 88.** In the figure given below, ABCD is a square of side 4 cm. Quadrants of a circle of diameter 2 cm are removed from the four

corners and a circle of diameter 2 cm is also removed. What is the area of the shaded region?



[CDS 2018(II)]

- (a) $5\frac{7}{9}\text{cm}^2$ (b) $7\frac{7}{9}\text{cm}^2$
(c) $9\frac{5}{9}\text{cm}^2$ (d) $9\frac{5}{6}\text{cm}^2$

- 89.** In a rectangle, length is three times its breadth. If the length and the breadth of the rectangle are increased by 30% and 10% respectively, then its perimeter increases by :

[CDS 2019(I)]

- (a) $\frac{40}{3}\%$ (b) 20%
(c) 25% (d) 27%

- 90.** What is the percentage decrease in the area of a triangle if its each side is halved?

[CDS 2019(I)]

- (a) 75% (b) 50%
(c) 25% (d) No change

- 91.** Considering two opposite vertices of square of side 'a' as centres, two circular area are drawn within the square joining the other two vertices, thus forming two sectors. What is the common area in these two sectors?

[CDS 2019(I)]

- (a) $a^2 \left(\pi + \frac{1}{2} \right)$ (b) $a^2 \left(\pi - \frac{1}{2} \right)$
(c) $a^2 \left(\frac{\pi}{2} - 1 \right)$ (d) $a^2 \left(\frac{\pi}{2} + 1 \right)$

- 92.** A rectangular red carpet of size 6ft $\times$ 12 ft has a dark red border 6 inches wide. What is the area of the dark red border?

[CDS 2019(I)]

- (a) 9 square feet (b) 15 square feet
(c) 17 square feet (d) 18 square feet

93. A 12 m long wire is cut into two pieces, one of which is bent into a circle and the other into a square enclosing the circle. What is the radius of the circle?

[CDS 2019(I)]

- (a) $\frac{12}{\pi+4}$ (b) $\frac{6}{\pi+4}$
(c) $\frac{3}{\pi+4}$ (d) $\frac{6}{\pi+2\sqrt{2}}$

94. The hypotenuse of a right-angled triangle is 10 cm and its area is 24 cm². If the shorter side is halved and the longer side is doubled, the new hypotenuse becomes:

[CDS 2019(I)]

- (a) $\sqrt{245}$ cm (b) $\sqrt{255}$ cm
(c) $\sqrt{265}$ cm (d) $\sqrt{275}$ cm

Direction (for next three): ABCD is a quadrilateral with AB = 9cm, BC = 40cm, CD = 28cm, DA = 15 cm and angle ABC is a right angle.

95. What is the area of triangle ADC?

[CDS 2019(I)]

- (a) 126cm² (b) 124cm²
(c) 122cm² (d) 120cm²

96. What is the area of quadrilateral ABCD?

[CDS 2019(I)]

- (a) 300cm² (b) 306cm²
(c) 312cm² (d) 316cm²

97. What is the difference between perimeter of triangle ABC and perimeter of triangle ADC?

[CDS 2019(I)]

- (a) 4cm (b) 5cm
(c) 6cm (d) 7cm

98. The perimeter of a triangle is 22cm. Through each vertex of the triangle, a straight line parallel to the opposite side is drawn. What is the perimeter of triangle formed by these lines?

[CDS 2019(II)]

- (a) 33cm (b) 44cm
(c) 66cm (d) 88cm

99. The sides AD, BC of a trapezium ABCD are parallel and the diagonals AC and BD meet O. If the area of triangle AOB is 3 cm² and the area of triangle BDC is 8cm², then what is the area of triangle AOD?

[CDS 2019(II)]

- (a) 8cm² (b) 5cm²
(c) 3.6cm² (d) 1.8cm²

100. An equilateral triangle and a square are constructed using metallic wires of equal length. What is the ratio of area of triangle to that of square?

[CDS 2019(II)]

- (a) 3:4 (b) 2:3
(c) $4\sqrt{3}:9$ (d) $2\sqrt{3}:9$

101. All the four sides of a parallelogram are of equal length. The diagonals are in the ratio 1:2. If the sum of the lengths of the diagonals is 12cm, then what is the area of the parallelogram?

[CDS 2019(II)]

- (a) 9cm² (b) 12cm²
(c) 16cm² (d) 25cm²

102. The length and breadth of a rectangle are increased by 20% and 10% respectively. What is the percentage increase in the area of the rectangle?

[CDS 2019(II)]

- (a) 32% (b) 30%
(c) 25% (d) 15%

103. A square is drawn such that its vertices are lying on a circle of radius 201 mm. What is the ratio of area of circle to that of square?

[CDS 2019(II)]

- (a) 11:7 (b) 7:11
(c) 20:19 (d) 19:20

104. What is the ratio of the area of a square inscribed in a semicircle of radius r to the area of square inscribed in a circle of radius r?

[CDS 2019(II)]

- (a) 1:2 (b) 2:5
(c) 2:3 (d) 3:5

105. The area of a sector of a circle of radius 4cm is 25.6cm^2 . What is the radian measure of the arc of the sector?

[CDS 2019(II)]

- (a) 2.3 (b) 3.2
(c) 3.3 (d) 3.4

106. What is the area of the triangle having side lengths $\frac{y}{z} + \frac{z}{x}, \frac{z}{x} + \frac{x}{y}, \frac{x}{y} + \frac{y}{z}$?

[CDS 2020(I)]

- (a) $\frac{(x+y+z)^2}{xyz}$ (b) $\frac{\sqrt{xyz}}{x+y+z}$
(c) $\sqrt{\frac{x}{y} + \frac{y}{z} + \frac{z}{x}}$ (d) $\sqrt{\frac{xy+yz+zx}{xyz}}$

107. If the angles of a triangle are 30° and 45° and the included side is $(\sqrt{3}+1)\text{cm}$, then what is the area of the triangle?

[CDS 2020(I)]

- (a) $(\sqrt{3}+1)\text{cm}^2$ (b) $(\sqrt{3}+3)\text{cm}^2$
(c) $\frac{1}{2}(\sqrt{3}+1)\text{cm}^2$ (d) $2(\sqrt{3}+1)\text{cm}^2$

108. Two circles touch internally. The sum of their areas is $136\pi\text{ cm}^2$ and distance between their centres is 4cm. What are the radii of the circles?

[CDS 2020(I)]

- (a) 11cm, 7cm (b) 10cm, 6cm
(c) 9cm, 5cm (d) 8cm, 4cm

109. If area of a circle and a square are same, then what is the ratio of their perimeters?

[CDS 2020(I)]

- (a) $2\sqrt{\pi}$ (b) $\sqrt{\pi}$
(c) $\frac{\sqrt{\pi}}{2}$ (d) $\frac{\sqrt{\pi}}{4}$

110. A square and a rectangle have equal areas. If one side of the rectangle is of length numerically equal to the square of the

length of the side of the square then the other side of the rectangle is:

[CDS 2020(I)]

- (a) Square root of the side of the square
(b) Half the side of the square
(c) Of unit length
(d) Double the side of the square

111. The length and breadth of a rectangle are in the ratio 4:3. Then what is the ratio of the area of the triangle formed by the parts of the diagonals with a long side to the area of the triangle formed by the parts of diagonals with a short side?

[CDS 2020(I)]

- (a) 3:4 (b) 4:3
(c) 16:9 (d) 1:1

112. Suppose a region is formed by removing a sector of 20° from a circular region of radius 30 feet. What is the area of this new region?

[CDS 2020(I)]

- (a) 150π square feet (b) 550π square feet
(c) 650π square feet (d) 850π square feet

113. If the diagonals of a rhombus are x and y, then what is its area?

[CDS 2020(I)]

- (a) $\frac{xy}{2}$ (b) $\frac{xy}{4}$
(c) xy (d) x^2-y^2

114. What is the area of the largest square plate cut from a circular disk of radius one unit?

[CDS 2020(I)]

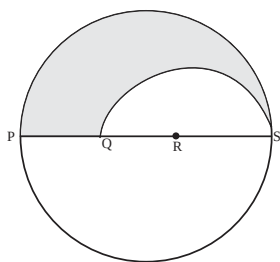
- (a) 4 square units (b) $2\sqrt{2}$ square units
(c) π square units (d) 2 square units

115. ABCD is a quadrilateral such that $AD = DC = CA = 20$ units, $BC = 12$ units and $\angle ABC = 90^\circ$. What is the approximate area of the quadrilateral ABCD?

[CDS 2020(I)]

- (a) 269 sq. units (b) 300 sq. units
(c) 325 sq. units (d) 349 sq. units

116.

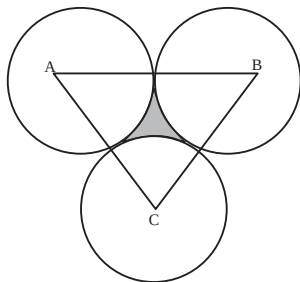


Let PQRS be the diameter of a circle of radius 9 cm. The length PQ, QR and RS are equal. Semi-Circle is drawn with QS as diameter (as shown in the given figure). What is the ratio of the shaded region to that of the unshaded region?

[CDS 2020(I)]

- (a) 25:121 (b) 5:13
(c) 5 : 18 (d) 1:2

117.

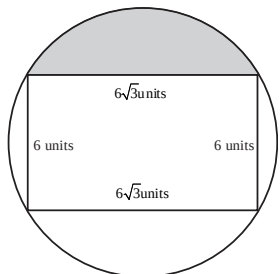


What is the area of the shaded region in the given figure, if the radius of each of the circles is 2cm?

[CDS 2020(I)]

- (a) $4\sqrt{3} - 2\text{cm}^2$ (b) $\sqrt{3} - \pi\text{cm}^2$
(c) $\sqrt{3} - \frac{\pi}{2}\text{cm}^2$ (d) $2\pi - 2\sqrt{3}\text{cm}^2$

118.



In the given figure, what is the area of the shaded region?

[CDS 2020(I)]

- (a) $9(\pi - \sqrt{3})\text{square units}$
(b) $3(4\pi - 3\sqrt{3})\text{square units}$

(c) $3(3\pi - 4\sqrt{3})\text{square units}$

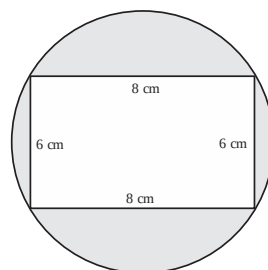
(d) $9(\sqrt{3} - \pi)\text{square units}$

119. ABCD is a trapezium, where AB is parallel to DC. If AB = 4cm, BC = 3cm, CD = 7cm and DA = 2cm, then what is the area of the trapezium?

[CDS 2020(I)]

- (a) $22\sqrt{\frac{2}{3}}\text{cm}^2$ (b) $22\sqrt{\frac{3}{2}}\text{cm}^2$
(c) $22\sqrt{3}\text{cm}^2$ (d) $\frac{22\sqrt{2}}{3}\text{cm}^2$

120.



What is the approximate area of the shaded region in the figure given?

[CDS 2020(I)]

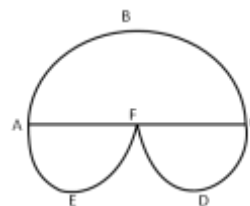
- (a) 15.3cm² (b) 25.5cm²
(c) 28.4cm² (d) 30.5cm²

121. A circle of diameter 8cm is placed in such a manner that it touches two perpendicular lines. Then another smaller circle is placed in the gap such that it touches the lines and the circle. What is the diameter of the smaller circle?

[CDS 2020(I)]

- (a) $4(3 - \sqrt{2})\text{cm}$ (b) $4(3 - 2\sqrt{2})\text{cm}$
(c) $8(3 - \sqrt{2})\text{cm}$ (d) $8(3 - 2\sqrt{2})\text{cm}$

122.



In the given figure, there are three semi circle ABC, AEF and CDF. The distance

between A and C is 28 units and F is the midpoint of AC. What is the total area of the three semi circles?

[CDS 2020(I)]

- (a) 924 square units (b) 824 square units
(c) 624 square units (d) 462 square units

123. The two sides of a triangle are 40cm and 41cm. If the perimeter of the triangle is 90cm, what is its area?

[CDS 2020(II)]

- (a) 90 cm² (b) 135 cm²
(c) 150 cm² (d) 180 cm²

124. The diagonals of a rhombus differ by 2 units and its perimeter exceeds the sum of the diagonals by 6 units. What is the area of the rhombus?

[CDS 2020(II)]

- (a) 48 square units (b) 36 square units
(c) 24 square units (d) 12 square units

125. The length of a rectangle is increased by 10% and breadth is decreased by 10%. Then the area of new rectangle is

[CDS 2020(II)]

- (a) neither increased nor decreased
(b) increased by 1%
(c) decreased by 1%
(d) decreased by 10%

126. If the perimeter of a circle and square are equal, then what is the ratio of the area of the circle to that of the square?

[CDS 2020(II)]

- (a) 1 : π (b) 2 : π
(c) 3 : π (d) 4 : π

127. Areas of two squares are in the ratio $m^2:n^4$, what is the ratio of their perimeter?

[CDS 2020(II)]

- (a) $m : n$ (b) $n : m$
(c) $m : n^2$ (d) $m^2 : n$

128. Four circular coins of equal radius are placed with their centres coinciding four vertices of a square. Each coin touches two other coins. If the uncovered area of the square is 42 cm², then what is the radius of each coin

[CDS 2020(II)]

- (a) 5 cm (b) 7 cm
(c) 10 cm (d) 14 cm

129. A sector is cut from a circle of radius 21 cm. If the length of the arc of the sector is 55cm, then what is the area of the sector?

[CDS 2021(I)]

- (a) 577.5cm² (b) 612.5 cm²
(c) 705.5 cm² (d) 725.5 cm²

130. A wire is in the form of a circle of radius 70 cm. If it is bent in the form of a rhombus, then what is its side length? $\left(Take \pi = \frac{22}{7} \right)$.

[CDS 2021(II)]

- (a) 55cm (b) 75cm
(c) 95cm (d) 110cm

131. If the perimeter of a semicircular park is 360m, then what is its area? $\left(Take \pi = \frac{22}{7} \right)$

[CDS 2021(II)]

- (a) 3850 m² (b) 7700m²
(c) 11550 m² (d) 15400 m²

132. The radius of circum-circle of a right angled triangle is 10cm and the altitude drawn to the hypotenuse is 8cm. What is the area of the triangle?

[CDS 2021(II)]

- (a) 60cm² (b) 80cm²
(c) 100cm² (d) 120cm²

133. Two circles touch externally. The sum of their areas is 41π square cm. If the distance between their centres is 9cm, then what is difference between their diameters?

[CDS 2021(II)]

- (a) 1cm (b) 1.5cm
(c) 2cm (d) 4cm

134. Let p be the area of a square X and q be the area of the square formed on the diagonal of the square X. What is the value of $\frac{p}{q}$?

[CDS 2021(II)]

- (a) $\frac{1}{8}$ (b) $\frac{1}{4}$

(c) $\frac{1}{3}$

(d) $\frac{1}{2}$

- 135.** The area of a rhombus is 336 square cm. If the length of one of its diagonals is 48cm, then what is the perimeter of the rhombus?

[CDS 2021(II)]

- (a) 200cm (b) 120cm
(c) 100cm (d) 90cm

- 136.** The minute hand of a clock is 21cm long. What is the area on the face of the clock described by the minute hand between 10.10 a.m and 10.30 a.m.? (take $\pi = \frac{22}{7}$).

[CDS 2021(II)]

- (a) 231 cm² (b) 331 cm²
(c) 462 cm² (d) 492 cm²

Consider the following for the next two (02) items that follow:

A chord of a circle of radius 2.1 cm subtends an angle of 120° at the centre. (take $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.732$)

- 137.** What is the approximate area of minor segment of the circle?

[CDS 2021(II)]

- (a) 2.71 cm² (b) 2.42 cm²
(c) 1.91 cm² (d) 1.71 cm²

- 138.** What is the approximate area of major segment of the circle?

[CDS 2021(II)]

- (a) 10.05 cm² (b) 10.15 cm²
(c) 11.05 cm² (d) 11.15 cm²

- 139.** A person wishes to fence 375 m² rectangular garden. He has 65m of barbed wire and is able to fence only three sides of the garden. What is the perimeter of the garden?

[CDS 2021(II)]

- (a) 80m (b) 84m
(c) 90m (d) 100m

- 140.** A bicycle wheel of radius 35cm makes n revolutions in moving 11km. What is the value of n? (take $\pi = \frac{22}{7}$)

[CDS 2022(I)]

- (a) 500 (b) 1000
(c) 2500 (d) 5000

- 141.** A farmland is in the shape of a rhombus. The perimeter of the land is 100 m and the

length of one of the diagonals is 40m. The land is divided into four equal parts. What is the area of each part?

[CDS 2022(I)]

- (a) 150 square meter
(b) 225 square meter
(c) 300 square meter
(d) 450 square meter

- 142.** If the perimeter of an isosceles right triangle is $4(2 + \sqrt{2})$ cm, then what is its area in square cm?

[CDS 2022(I)]

- (a) 8 (b) 12
(c) 16 (d) 24

- 143.** A rectangle of length 10 units and breadth 8 unit is split into two squares each of area x square units and two rectangles each of area y square units. Consider the following statements:

1.y is always greater than x

2.y can be 15 square units.

Which of the above statement is/are correct?

[CDS 2022(I)]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

- 144.** A square sheet is formed by joining n identical square sheets of same size. If the length of the diagonal of the bigger square sheet so formed is m, then what is the side length of a smaller square sheet?

[CDS 2022(I)]

- (a) $\frac{m}{\sqrt{n}}$ (b) $\frac{m}{2\sqrt{n}}$
(c) $\frac{m}{\sqrt{2n}}$ (d) $\frac{\sqrt{2}m}{\sqrt{n}}$

- 145.** What is the area of the region enclosed by three identical circles (each of radius 4cm) touching each other?

[CDS 2022(I)]

- (a) $\frac{8\pi}{3}$ square cm
(b) $\left(16\sqrt{3} - \frac{8\pi}{3}\right)$ square cm
(c) $(16\sqrt{3} - 8\pi)$ square cm

(d) $\frac{16\pi}{\sqrt{3}}$ square cm

- 146.** X, Y and Z are three equilateral triangles. The sum of the areas of X and Y is equal to the area of Z. If the side lengths of X and Y are 6cm and 8cm respectively, then what is the side length of Z?

[CDS 2022 (II)]

- (a) 9cm (b) 9.5cm
(c) 10cm (d) 10.5cm

- 147.** What is the time taken by a person to cover one round of a circular park of diameter 210 m if he walks at a speed of 6km/hr? ($\pi = \frac{22}{7}$)

[CDS 2022 (II)]

- (a) 6.6 minutes (b) 5.5 minutes
(c) 4.4 minutes (d) 3.3 minutes

- 148.** In an equilateral triangle of side $2\sqrt{3}$ cm, a circle is inscribed touching the sides. What is the area of the remaining portion of the triangle?

[CDS 2022 (II)]

- (a) $(2\sqrt{3} - \pi)$ square cm
(b) $(3\sqrt{3} - \pi)$ square cm
(c) $(4\sqrt{3} - 2\pi)$ square cm
(d) $(4\sqrt{3} - \pi)$ square cm

- 149.** A triangle has side lengths x cm, x + 13cm and x + 26cm. If its area is 126 square cm, then what is the value of x?

[CDS 2022 (II)]

- (a) 18 (b) 17
(c) 16 (d) 15

- 150.** Two parallel sides of a trapezium are 29cm and 21cm. Non-parallel sides are equal and each is of length 8.5 cm. What is the area of the trapezium?

[CDS 2022 (II)]

- (a) 187.5 square cm
(b) 227.5 square cm
(c) 375 square cm
(d) 455 square cm

- 151.** A square copper plate of side 16cm weighs 128 gm. A circular disc of diameter 14 cm is cut off from the plate. What is the weight of the remaining part? ($\pi = \frac{22}{7}$)

[CDS 2022 (II)]

- (a) 48gm (b) 49gm
(c) 50gm (d) 51gm

- 152.** The area of a rhombus is 96 square cm and one of its diagonals is of length 12cm. What is the perimeter of the rhombus?

[CDS 2022 (II)]

- (a) 36cm (b) 40cm
(c) 44cm (d) 48cm

- 153.** Two circles touch externally. The sum of their areas is 89π square cm and the distance between their centres is 13cm. What is the difference in their radii?

[CDS 2022 (II)]

- (a) 2cm (b) 2.5cm
(c) 3cm (d) 3.5cm

- 154.** A square and a rectangle have same perimeter. They differ in areas by 1 square cm. The length of the rectangle exceeds its breadth by:

[CDS 2022 (II)]

- (a) 1cm (b) 2cm
(c) 3cm (d) 4cm

- 155.** Two rectangles are of same area equal to 480 square cm. They differ in lengths by 6cm and breadths by 4cm. What is the difference in their perimeters?

[CDS 2022 (II)]

- (a) 2cm (b) 4cm
(c) 6cm (d) 10cm

Consider the following for the next two (02) items that follow:

A chord of length l of a circle makes an angle 90° at the centre of the circle.

- 156.** What is the area of the minor segment?

[CDS 2022 (II)]

- (a) $\frac{l^2}{2} \left(\pi - \frac{1}{2} \right)$ (b) $\frac{l^2}{4} \left(\pi - \frac{1}{2} \right)$
(c) $\frac{l^2}{4} \left(\frac{\pi}{2} - 1 \right)$ (d) $\frac{l^2}{2} \left(\frac{\pi}{2} - \frac{1}{2} \right)$

- 157.** What is the area of the major segment?

[CDS 2022 (II)]

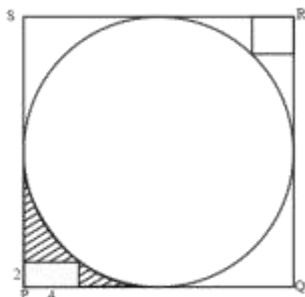
- (a) $\frac{l^2}{4} \left(\frac{3\pi}{2} + 1 \right)$ (b) $\frac{l^2}{4} \left(\frac{3\pi}{2} - 1 \right)$

(c) $\frac{l^2}{2} \left(\frac{3\pi}{2} + 1 \right)$

(d) $\frac{l^2}{2} \left(\frac{3\pi}{2} - 1 \right)$

Consider the following for the next three (03) items that follow:

In the figure given below, a circle is inscribed in a square PQRS. A rectangle at the corner P that measures 4cm × 2cm and a square at the corner R are drawn.



158. What is the area of the circle?

[CDS – 2023 (1)]

- (a) 100π square cm (b) 96π square cm
(c) 50π square cm (d) 48π square cm

159. What is the area of the smaller square?

[CDS – 2023 (1)]

- (a) $50(3 - \sqrt{2})$ square cm
(b) $25(3 - 2\sqrt{2})$ square cm
(c) $25(3 + 2\sqrt{2})$ square cm
(d) $50(3 - 2\sqrt{2})$ square cm

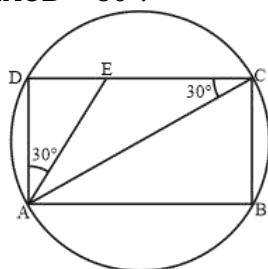
160. What is the area of the shaded region?

[CDS – 2023 (1)]

- (a) $(96 - 25\pi)$ square cm
(b) $(92 - 25\pi)$ square cm
(c) $(96 - 16\pi)$ square cm
(d) $(92 - 16\pi)$ square cm

Consider the following for the next two (02) items that follow:

In the following figure, a rectangle ABCD is inscribed in a circle of radius r. given $\angle DAE = 30^\circ$ and $\angle ACD = 30^\circ$.



161. What is the ratio of the area of the circle to the area of the rectangle?

[CDS – 2023 (1)]

(a) $\frac{\pi}{\sqrt{2}}$

(b) $\frac{\pi}{\sqrt{3}}$

(c) $\frac{2\pi}{\sqrt{3}}$

(d) $\frac{3\pi}{\sqrt{2}}$

162. What is the area of $\triangle AEC$?

[CDS – 2023 (1)]

(a) $\frac{r^2}{\sqrt{3}}$

(b) $\frac{r^2}{2\sqrt{3}}$

(c) $\frac{r^2}{3\sqrt{3}}$

(d) $\frac{3r^2}{\sqrt{3}}$

163. The plinth of a house has an area of 200 square metres. It is rectangular in shape and its length and breadth are in the ratio 2:1. The owner of the house extends the terrace by 1 m on each side. What is the percentage of area that has increased in the terrace relative to the plinth?

[CDS – 2023 (1)]

- (a) 40% (b) 32%
(c) 20% (d) 15.5%

164. A square sheet of side length 44 cm is rolled along one of its sides to form a cylinder by making opposite edges just to touch each other. What is the volume of the cylinder? (Take $\pi = \frac{22}{7}$).

[CDS – 2023 (1)]

- (a) 6776 cubic cm (b) 6248 cubic cm
(c) 5896 cubic cm (d) 5680 cubic cm

165. The perimeter and the area of a right-angled triangle are 36 cm and 54 square cm respectively. What is the length of the hypotenuse?

[CDS – 2023 (1)]

- (a) 12cm (b) 14cm
(c) 15cm (d) 16cm

166. The perimeter of a sector of a circle of radius 5.2cm is 16.4 cm. What is the area of the sector?

[CDS – 2023 (1)]

- (a) 15.6 square cm (b) 15 square
(c) 14.4 square cm (d) 14.1 square cm

167. What is the area of the circle (approximately) inscribed in a triangle with side lengths 12cm, 16cm and 20cm?

[CDS – 2023 (1)]

- (a) 48 square cm (b) 50 square cm
(c) 52 square cm (d) 54 square cm

168. A floor of a big hall has dimensions 30m 60cm and 23m 40cm. Its is to be paved with square tiles of same size. What is the minimum number of tiles required?

[CDS – 2023 (1)]

- (a) 30 (b) 36
(c) 169 (d) 221

169. How long will a man take to walk around the boundary of a square field of area 25 hectares at the rate of 5km/hr?

[CDS – 2023 (1)]

- (a) 36 minutes (b) 30 minutes
(c) 24 minutes (d) 18 minutes

170. Let x be the area of a square inscribed in a circle of radius r and y be the area of an equilateral triangle inscribed in the same circle. Which one of the following is correct?

[CDS – 2023 (1)]

- (a) $9x^2 = 16y^2$ (b) $27x^2 = 64y^2$
(c) $36x^2 = 49y^2$ (d) $16x^2 = 21y^2$

171. If the length of a rectangle is increased by $66\frac{2}{3}\%$, then by what percent should the width of the rectangle be decreased in order to maintain the same area?

[CDS – 2023 (1)]

- (a) 50% (b) 45%
(c) 40% (d) 35%

172. What is the maximum area that can be covered by three non-intersecting circles drawn inside a rectangle of sides 8 cm and 12cm?

[CDS – 2023 (1)]

- (a) 16π square cm (b) 18π square cm
(c) 20π square cm (d) 24π square cm

173. A triangle ABC has been divided into four smaller triangles P, Q, R, S whose perimeters are 16cm, 12cm, 4cm and 12cm respectively. P, R and S contain the vertices A, B and C respectively. What is the perimeter of the triangle ABC?

[CDS – 2023 (1)]

- (a) 18cm (b) 20cm
(c) 22cm (d) 24cm

174. OABC is a rhombus whose three vertices lie on a circle with centre at O. If the area of the rhombus is $32\sqrt{3}$ square cm, then what is the radius of the circle?

[CDS – 2023 (1)]

- (a) 4cm (b) 6cm
(c) 8cm (d) 16cm

175. Question: What is the area of the circle C?

Statement I: an arc of length 7 cm subtends an angle 30° at the centre of C.

Statement II: A chord of length 10cm subtends an angle 90° at the centre of C.

[CDS – 2023 (1)]

(a) choose this option if the question can be answered by one of the statements alone but not by the other.

(b) choose this option if the question can be answered by either statement alone.

(c) Choose this option if the question can be answered by using both the statements together, but cannot be answered by using either statement alone

(d) Choose this option if the question cannot be answered even by using both statements together.

176. A chord PQ of the circle C divides it into two segments such that 3 times the area of the major segment is 4 times the area of the minor segment.

Question: What is the radius of C?

Statement I: Area of the minor segment is 66 square cm.

Statement II: Area of the major segment is 88 square cm.

[CDS – 2023 (1)]

(a) choose this option if the question can be answered by one of the statements alone but not by the other.

(b) choose this option if the question can be answered by either statement alone.

(c) Choose this option if the question can be answered by using both the statements together, but cannot be answered by using either statement alone

(d) Choose this option if the question cannot be answered even by using both statements together.

177. An isosceles triangle has its base length $2a$ and its height is h . On each side of the triangle a square is drawn external to the triangle. What is the area of the figure thus formed?

[CDS-2023 (2)]

- (a) $6a^2 + 2h^2 + 2ah$ (b) $6a^2 + 2h^2 + ah$
(c) $4a^2 + 2h^2 + ah$ (d) $6a^2 + h^2 + ah$

- 178.** A pendulum swings through an angle of 9° and its end describes an arc of length 14.3cm. What is the length of the pendulum? (Take $\pi = \frac{22}{7}$)

[CDS-2023 (2)]

- (a) 88cm (b) 91cm
(c) 95cm (d) 98cm

- 179.** The corners of an equilateral triangular plate were cut in such a manner that it forms a regular hexagonal plate. What is the ratio of the area of the triangular plate to the area of the hexagonal plate?

[CDS-2023 (2)]

- (a) 2:1 (b) 3:2
(c) 4:3 (d) 5:3

- 180.** Two equal arcs of different circles C_1 and C_2 subtend angles of 60° and 75° respectively, at the centres. What is the ratio of the radius of C_1 to the radius of C_2 ?

[CDS-2023 (2)]

- (a) 4:5 (b) 5:4
(c) 1:1 (d) 3:2

- 181.** ABC is a triangle with sides $AB = 41\text{cm}$, $BC = 28\text{cm}$ and $CA = 15\text{cm}$. If D, E and F are the mid points of AB, BC and CA respectively, then what is the area of the triangle DEF?

[CDS-2023 (2)]

- (a) 63 square cm (b) 45 square cm
(c) 31.5 square cm (d) 22.5 square cm

- 182.** The circumference of a circle exceeds the diameter by 16.8 cm. What is the diameter of the circle? (Take $\pi = \frac{22}{7}$)

[CDS-2023 (2)]

- (a) 6.24cm (b) 6.42cm
(c) 7.64cm (d) 7.84cm

- 183.** What is the area of the region between the concentric circles if the chord of the out circle of length 14cm is a tangent of the inn circle? (Take $\pi = \frac{22}{7}$)

[CDS-2023 (2)]

- (a) 125 square cm (b) 132 square cm
(c) 144 square cm (d) 154 square cm

Consider the following for the next two (02) items that follow:

A line segment AB is bisected at C and semi circles S_1 , S_2 and S_3 are drawn respectively on AB, AC and CB as diameters such that they all lie on same side of AB. A circle S is drawn touching internally S_1 and externally S_2 and S_3 .

- 184.** If r is the radius of S and R is the radius of S_2 , then which one of the following is correct?

[CDS-2023 (2)]

- (a) $R=3r$ (b) $R=2r$
(c) $3R=4R$ (d) $2R=3r$

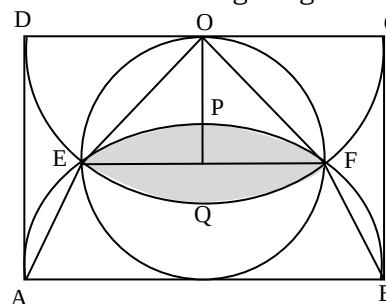
- 185.** If m is the area of the circle S and n is the area of semi-circle S_1 , then which one of the following is correct?

[CDS-2023 (2)]

- (a) $9m=2n$ (b) $9m=4n$
(c) $3m=2n$ (d) $7m=3n$

Consider the following for the next three (03) items that follow:

Consider two identical semicircles and one circle inscribed in a rectangle of length 10 cm as shown in the figure given below.



(Take $\pi = 3.14$ and $\sqrt{2} = 1.4$)

[CDS-2024 (1)]

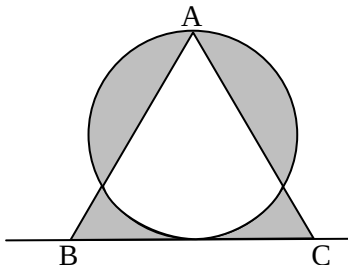
- 186.** What is the area of triangle EOF?
- (a) $12.5\sqrt{3}$ square cm (b) $6.25\sqrt{3}$ square cm
(c) 12.5 square cm (d) 6.25 square cm

- 187.** What is the area of trapezium AEFB?
- (a) 30 square cm (b) 25 square cm
(c) 20 square cm (d) 18.75 square cm

- 188.** What is the area of the shaded region?
- (a) 14.75 square cm (b) 14.25 square cm
(c) 7.225 square cm (d) 7.625 square cm

Consider the following for the next two (02) items that follow:

Consider a circle of area 9π square unit and an equilateral triangle ABC as shown in the figure given below.



[CDS-2024 (1)]

189. What is the length of the side of the triangle ABC?

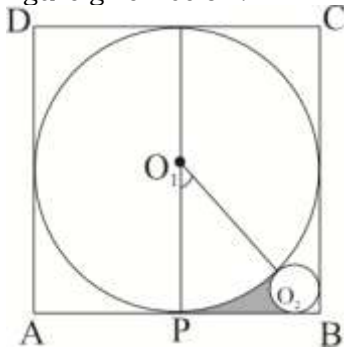
- (a) $2\sqrt{3}$ unit (b) $4\sqrt{3}$ unit
(c) $6\sqrt{3}$ unit (d) $8\sqrt{3}$ unit

190. What is the area of the shaded region?

- (a) $6(\pi + \sqrt{3})$ square unit
(b) $3(\pi + 2\sqrt{3})$ square unit
(c) $1.5(3\pi + 8\sqrt{3})$ square unit
(d) $6(\pi + 2\sqrt{3})$ square unit

Consider the following for the next three (03) items that follow:

Two circles with centres at O_1 and O_2 touching each other are placed inside a rectangle of sides 9 cm and 8 cm as shown in the figure given below:



[CDS-2024 (1)]

191. What is the sum of the areas of the two circles?

- (a) 17π square unit
(b) 16.75π square unit
(c) 16.5π square unit
(d) 16.25π square unit

192. Which one of the following is correct in respect of angle θ ?

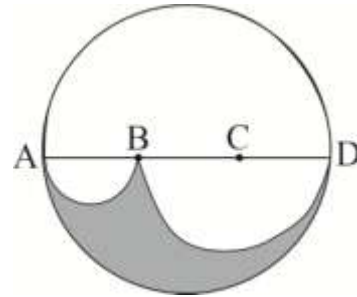
- (a) $0^\circ < \theta < 30^\circ$ (b) $30^\circ < \theta < 45^\circ$
(c) $45^\circ < \theta < 60^\circ$ (d) $60^\circ < \theta < 90^\circ$

193. What is the area of the shaded region?

- (a) $\frac{240 - 10\pi - \pi\theta}{24}$ square unit
(b) $\frac{240 - 6\pi - \pi\theta}{24}$ square unit
(c) $\frac{120 - 12\pi - \pi\theta}{24}$ square unit
(d) $\frac{240 - 12\pi - \pi\theta}{24}$ square unit

Consider the following for the next two (02) items that follow:

Let ABCD be the diameter of a circle of radius 6 cm. The lengths AB, BC and CD are equal. Semi-circles are drawn with AB and BD as diameters as shown in the figure given below.



[CDS-2024 (1)]

194. What is the ratio of the area of the shaded region to that of the non-shaded region?

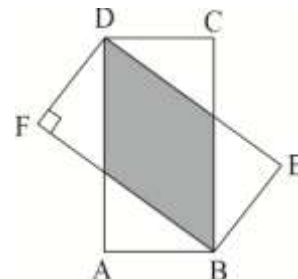
- (a) 2:7 (b) 2:5
(c) 3:5 (d) 5:8

195. What is the perimeter of the shaded region?

- (a) 24π cm (b) 18π cm
(c) 15π cm (d) 12π cm

Consider the following for the next two (02) items that follow:

Consider the two identical rectangles ABCD and BEDF as shown in the figure given below. Let AB = 1 cm and BC = 2 cm.



[CDS-2024 (1)]

196. What is the area of the overlapping region?

- (a) $\frac{8}{5}$ square cm (b) $\frac{5}{4}$ square cm

- (c) $\frac{4}{5}$ square cm (d) $\frac{3}{4}$ square cm

197. What is the area of the non-overlapping region?

- (a) $\frac{3}{4}$ square cm (b) $\frac{11}{4}$ square cm
(c) $\frac{3}{2}$ square cm (d) $\frac{5}{4}$ square cm

Consider the following for the next two (02) items that follow:

In a pie-diagram (with radius 7 cm), the central angles of the sectors are in the ratio 2 : 3 : 7 : 5 : 1.

(Take $\pi = \frac{22}{7}$)

[CDS-2024 (1)]

198. If P is the area of the smallest sector and Q is the area of the largest sector, then what is P + Q equal to?

- (a) $88/3$ square cm (b) $77/3$ square cm
(c) $149/6$ square cm (d) $616/9$ square cm

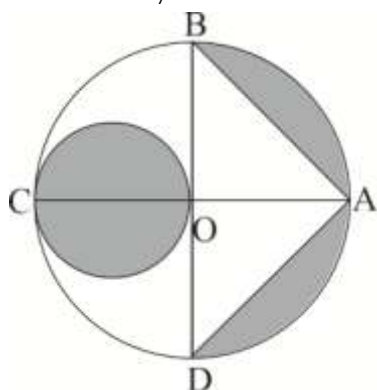
199. If p is the perimeter of the smallest sector, then what is the value of 9p?

- (a) 142 cm (b) 148 cm
(c) 156 cm (d) 221 cm

Consider the following for the next two (02) items that follow:

ABCD is a circle with centre O and taking OC as a diameter, a circle is drawn as shown in the figure given below. Let OB = 7

cm. (use $\pi = \frac{22}{7}$)



200. What is the area of the shaded region?

- (a) 38.5 square cm (b) 48 square cm
(c) 52.5 square cm (d) 66.5 square cm

201. What is the ratio of the area of the shaded region to the area of the non-shaded region?

- (a) $\frac{19}{25}$ (b) $\frac{18}{25}$
(c) $\frac{17}{25}$ (d) $\frac{16}{25}$

202. The chord AB of a circle with centre at O is $2\sqrt{3}$ times the height of the minor segment. If P is the area of the sector OAB and Q is the area of the minor segment of the circle, then what is the approximate value of $\frac{P}{Q}$?

(Take $\sqrt{3} = 1.7$ and $\pi = 3.14$)

[CDS-2024 (1)]

- (a) 1.4 (b) 1.7
(c) 2.2 (d) 2.6

203. In a right-angled triangle ABC, AB = 15 cm, BC = 20 cm and AC = 25 cm. Further, BP is the perpendicular on AC. What is the difference in the area of triangles PAB and PCB?

[CDS-2024 (1)]

- (a) 40 square cm (b) 42 square cm
(c) 45 square cm (d) 48 square cm

204. What is the area of the region between two concentric circles, if the length of a chord of the outer circle touching the inner circle at a particular point of its circumference is 14 cm? (Take $\pi = \frac{22}{7}$)

[CDS-2024 (1)]

- (a) 154 square cm
(b) 144 square cm
(c) 132 square cm
(d) Cannot be determined due to insufficient data

205. ABC is a right-angled triangle, right-angled at B such that AB = 6 cm and BC = 8 cm. What is the perimeter of the square inscribed in the triangle ABC with maximum area?

[CDS-2024 (1)]

- (a) $24/7$ cm (b) $96/7$ cm
(c) 24 cm (d) 32 cm

206. Area of a rectangle with length x and breadth y is P and area of a parallelogram (which is strictly not a rectangle) with adjacent sides of length x and y is Q. Question: Is $P > Q$?

Statement-I: $x : y = 2 : 1$

Statement-II: The angle between the two adjacent sides of the parallelogram is 60°

[CDS-2024 (1)]

(a) If the Question can be answered by one of the Statements alone, but not by the other.

(b) If the Question can be answered by either Statement alone

(c) If the Question can be answered by using both the Statements together, but cannot be answered by using either Statement alone.

(d) If the Question can be answered even by using both Statements together

ANSWER KEY

1.	c	2.	d	3.	c	4.	c	5.	c	6.	d	7.	c	8.	a	9.	b	10.	b
11.	a	12.	a	13.	a	14.	a	15.	a	16.	c	17.	b	18.	a	19.	b	20.	c
21.	d	22.	d	23.	b	24.	c	25.	c	26.	c	27.	c	28.	b	29.	c	30.	c
31.	a	32.	c	33.	d	34.	b	35.	b	36.	c	37.	b	38.	b	39.	a	40.	d
41.	d	42.	d	43.	d	44.	c	45.	b	46.	c	47.	d	48.	c	49.	a	50.	b
51.	a	52.	a	53.	b	54.	d	55.	b	56.	c	57.	b	58.	d	59.	b	60.	b
61.	d	62.	c	63.	c	64.	b	65.	d	66.	c	67.	b	68.	d	69.	b	70.	b
71.	b	72.	a	73.	a	74.	d	75.	d	76.	b	77.	c	78.	b	79.	b	80.	a
81.	b	82.	b	83.	a	84.	a	85.	d	86.	c	87.	c	88.	c	89.	c	90.	a
91.	c	92.	c	93.	b	94.	c	95.	a	96.	b	97.	c	98.	b	99.	d	100.	c
101.	c	102.	a	103.	a	104.	b	105.	b	106.	c	107.	c	108.	b	109.	c	110.	c
111.	d	112.	d	113.	a	114.	d	115.	a	116.	b	117.	a	118.	b	119.	d	120.	d
121.	b	122.	d	123.	d	124.	c	125.	c	126.	d	127.	c	128.	b	129.	a	130.	d
131.	b	132.	b	133.	c	134.	d	135.	c	136.	c	137.	a	138.	d	139.	a	140.	d
141.	a	142.	a	143.	b	144.	c	145.	c	146.	c	147.	a	148.	b	149.	d	150.	a
151.	d	152.	b	153.	c	154.	b	155.	b	156.	c	157.	a	158.	a	159.	d	160.	b
161.	b	162.	a	163.	b	164.	a	165.	c	166.	a	167.	b	168.	d	169.	c	170.	b
171.	c	172.	d	173.	b	174.	c	175.	b	176.	b	177.	b	178.	b	179.	b	180.	b
181.	c	182.	d	183.	d	184.	d	185.	a	186.	c	187.	a	188.	b	189.	b	190.	*
191.	a	192.	c	193.	d	194.	a	195.	d	196.	b	197.	c	198.	d	199.	b	200.	d
201.	a	202.	b	203.	b	204.	a	205.	b	206.	b								

PRACTICE SHEET-1

1. If the height of a given cone be doubled and radius of the base remains the same, the ratio of the volume of the given cone to that of the second cone will be
 (1) 2 : 1 (2) 1 : 8
 (3) 1 : 2 (4) 8 : 1
2. If the radius of the base of a cone be doubled and height is left unchanged, then ratio of the volume of new cone to that of the original cone will be:
 (1) 1 : 4 (2) 2 : 1
 (3) 1 : 2 (4) 4 : 1
3. Each of the measure of the radius of base of a cone and that of a sphere is 8 cm. Also, the volume of these two solids are equal. The slant height of the cone is
 (1) $8\sqrt{17}$ cm (2) $4\sqrt{17}$ cm
 (3) $34\sqrt{2}$ cm (4) 34 cm.
4. A cone of height 15 cm and base diameter 30 cm is carved out of a wooden sphere of radius 15 cm. The percentage of wasted wood is:
 (1) 75% (2) 50%
 (3) 40% (4) 25%
5. In a right circular cone, the radius of its base is 7 cm and its height 24 cm. A cross-section is made through the midpoint of the height parallel to the base. The volume of the upper portion is
 (1) 169 cm^3 (2) 154 cm^3
 (3) 1078 cm^3 (4) 800 cm^3
6. If the area of the base of a cone is 770 cm^2 and the area of the curved surface is 814 cm^2 , then its volume (in cm^3) is:
 (1) $213\sqrt{5}$ (2) $392\sqrt{5}$
 (3) $550\sqrt{5}$ (4) $616\sqrt{5}$
7. Volume of two cones are in the ratio 1 : 4 and their diameters are in the ratio 4 : 5. The ratio of their height is
 (1) 1 : 5 (2) 5 : 4
 (3) 5 : 16 (4) 25 : 64
8. The height of the cone is 30 cm. A small cone is cut off at the top by a plane parallel to its base. If its volume is $\frac{1}{27}$ of the volume of the cone, at what height, above the base, is the section made?
 (1) 6 cm (2) 8 cm
 (3) 10 cm (4) 20 cm
9. The radius of the base and height of a right circular cone are in the ratio 5 : 12. If the volume of the cone is $314\frac{2}{7}\text{ cm}^3$, the slant height (in cm) of the cone will be
 (1) 12 (2) 13
 (3) 15 (4) 17
10. Two solid right cones of equal height and of radii r_1 and r_2 are melted and made to form a solid sphere of radius R. Then the height of the cone is
 (1) $\frac{4R^2}{r_1^2 + r_2^2}$ (2) $\frac{4R}{r_1 + r_2}$
 (3) $\frac{4R^3}{r_1^2 + r_2^2}$ (4) $\frac{R^2}{r_1^2 + r_2^2}$
11. The ratio of radii of two cone is 3 : 4 and the ratio of their height is 4 : 3. Then the ratio of their volume will be
 (1) 3 : 4 (2) 4 : 3
 (3) 9 : 16 (4) 16 : 9
12. If a right circular cone is separated into solids of volumes V_1, V_2, V_3 by two planes parallel to the base, which also trisect the altitude, then $V_1 : V_2 : V_3$ is
 (1) 1 : 2 : 3 (2) 1 : 4 : 6
 (3) 1 : 6 : 9 (4) 1 : 7 : 19
13. If the radii of the circular ends of a truncated conical bucket which is 45cm high be 28 cm and 7 cm, then the capacity of the bucket in cubic centimetre is
 (1) 48510 (2) 45810
 (3) 48150 (4) 48051
14. The ratio of height and the diameter of a right circular cone is 3 : 2 and its volume is 1078 cc , then its height is :
 (1) 7 cm (2) 14 cm
 (3) 21 cm (4) 28 cm

15. The radius of the base of a right circular cone is doubled keeping its height fixed. The volume of the cone will be:
 (1) Three times of the previous volume
 (2) Four times of the previous volume
 (3) 2 times of the previous volume
 (4) Double of the previous volume
16. The heights of two cones are in the ratio 1 : 3 and the diameters of their base are in the ratio 3 : 5. The ratio of their volume is
 (1) 3 : 25 (2) 4 : 25
 (3) 6 : 25 (4) 7 : 25
17. The base of a right circular cone has the same radius a as that of a sphere. Both the sphere and the cone have the same volume. Height of the cone is
 (1) $3a$ (2) $4a$
 (3) $\frac{7}{4}a$ (4) $\frac{7}{3}a$
18. The circumference of the base of a 16 cm height solid cone is 33 cm. What is the volume of the cone in cm^3 ?
 (1) 1028 (2) 616
 (3) 462 (4) 828
19. The perimeter of the base of a right circular cone is 8 cm. If the height of the cone is 21 cm, then its volume is:
 (1) $108\pi \text{ cm}^3$ (2) $112\pi \text{ cm}^3$
 (3) $112\pi \text{ cm}^3$ (4) $108\pi \text{ cm}^3$
20. If the volume of two right circular cones are in the ratio 4 : 1 and their diameter are in the ratio 5 : 4, then the ratio of their height is
 (1) 25 : 16 (2) 25 : 64
 (3) 64 : 25 (4) 16 : 25
21. The volume of a conical tent is 1232 cu. m and the area of its base is 154 sq. m. Find the length of the canvas required to build the tent, if the canvas is 2 m in width.
 (1) 270 m (2) 272 m
 (3) 276 m (4) 275 m
22. If the ratio of the diameters of two right circular cones of equal height be 3 : 4, then the ratio of their volume will be
 (1) 3 : 4 (2) 9 : 16
 (3) 16 : 9 (4) 27 : 64
23. A hollow spherical metallic ball has an external diameter 6 cm and is $\frac{1}{2}$ cm thick. The volume of the ball (in cm^3) is
 (1) $41\frac{2}{3}$ (2) $37\frac{2}{3}$
 (3) $47\frac{2}{3}$ (4) $40\frac{2}{3}$
24. The sum of radii of two spheres is 10 cm and the sum of their volume is 880 cm^3 . What will be the product of their radii?
 (1) 21 (2) $26\frac{1}{3}$
 (3) $33\frac{1}{3}$ (4) 70
25. If the radius of a sphere is doubled, its volume becomes
 (1) double (2) four times
 (3) six times (4) eight times
26. The radii of two spheres are in the ratio 3 : 2. Their volume will be in the ratio :
 (1) 9 : 4 (2) 3 : 2
 (3) 8 : 27 (4) 27 : 8
27. The total surface area of a solid hemisphere is $108\pi \text{ cm}^2$. The volume of the hemisphere is
 (1) $72\pi \text{ cm}^3$ (2) $144\pi \text{ cm}^3$
 (3) $108\sqrt{6} \text{ cm}^3$ (4) $54\sqrt{6} \text{ cm}^3$
28. The largest sphere is carved out of a cube of side 7 cm. The volume of the sphere (in cm^3) will be
 (1) 718.66 (2) 543.72
 (3) 481.34 (4) 179.67
29. The surface areas of two spheres are in the ratio 4 : 9. Their volumes will be in the ratio
 (1) 2 : 3 (2) 4 : 9
 (3) 8 : 27 (4) 64 : 729
30. A sphere and a hemisphere have the same volume. The ratio of their radii is
 (1) 1 : 2 (2) 1 : 8
 (3) $1 : \sqrt{2}$ (4) $1 : \sqrt[3]{2}$

ANSWER KEY

1.	(3)	2.	(4)	3.	(1)	4.	(1)	5.	(2)	6.	(4)	7.	(4)	8.	(4)	9.	(2)	10.	(3)
11.	(1)	12.	(4)	13.	(1)	14.	(3)	15.	(2)	16.	(1)	17.	(2)	18.	(3)	19.	(2)	20.	(3)
21.	(4)	22.	(2)	23.	(3)	24.	(2)	25.	(4)	26.	(4)	27.	(2)	28.	(4)	29.	(3)	30.	(4)

PRACTICE SHEET-2

- A solid sphere of 6 cm diameter is melted and recast into 8 solid spheres of equal volume. The radius (in cm) of each small sphere is
 (1) 1.5 (2) 3
 (3) 2 (4) 2.5
- The total surface area of a sphere is 8π square unit. The volume of the sphere is
 (1) $\frac{8\sqrt{2}}{3}\pi$ cubic unit
 (2) $\frac{8}{3}\pi$ cubic unit
 (3) $8\sqrt{3}\pi$ cubic unit
 (4) $\frac{8\sqrt{3}}{5}\pi$ cubic unit
- Area of the base of a pyramid is 57 sq.cm. and height is 10 cm, then its volume (in cm^3), is
 (1) 570 (2) 390
 (3) 190 (4) 590
- There is a pyramid on a base which is a regular hexagon of side 2a cm. If every slant edge of this pyramid is of length $\frac{5a}{2}$ cm, then the volume of this pyramid is
 (1) $3a^3 \text{ cm}^3$ (2) $3\sqrt{2} a^3 \text{ cm}^3$
 (3) $3\sqrt{3} a^3 \text{ cm}^3$ (4) $6a^3 \text{ cm}^3$
- The base of a right pyramid is a square of side 40 cm long. If the volume of the pyramid is 8000 cm^3 , then its height is:
 (1) 5 cm (2) 10 cm
 (3) 15 cm (4) 20 cm
- The base of a right prism is a trapezium. The length of the parallel sides are 8 cm and 14 cm and the distance between the parallel sides is 8 cm. If the volume of the prism is 1056 cm^3 , then the height of the prism is
 (1) 44 cm (2) 16.5 cm
 (3) 12 cm (4) 10.56 cm
- The height of a right prism with a square base is 15 cm. If the area of the total surface of the prism is 608 sq. cm, its volume is
 (1) 910 cm^3 (2) 920 cm^3
 (3) 960 cm^3 (4) 980 cm^3
- The base of a right prism is an equilateral triangle of side 8 cm and height of the prism is 10 cm. Then the volume of the prism is
 (1) $320\sqrt{3}$ cubic cm (2) $160\sqrt{3}$ cubic cm
 (3) $150\sqrt{3}$ cubic cm (4) $300\sqrt{3}$ cubic cm
- The base of right prism is a triangle whose perimeter is 28 cm and the in radius of the triangle is 4 cm. If the volume of the prism is 366 cc, then its height is
 (1) 6 cm (2) 8 cm
 (3) 4 cm (4) None of these
- If the base of a right pyramid is triangle of sides 5 cm, 12 cm, 13 cm and its volume is 330 cm^3 , then its height (in cm) will be
 (1) 33 (2) 32
 (3) 11 (4) 22
- The diameter of the moon is assumed to be one fourth of the diameter of the earth. Then the ratio of the volume of the earth to that of the moon is
 (1) 64 : 1 (2) 1 : 64
 (3) 60 : 7 (4) 7 : 60
- A conical vessel whose internal radius is 12 cm and height 50 cm is full of liquid. The contents are emptied into a cylindrical vessel with radius (internal) 10 cm. The height to

which the liquid rises in the cylindrical vessel is:

- (1) 25cm (2) 20cm
(3) 24cm (4) 22cm

13. The volume of a right circular cylinder is equal to the volume of that right circular cone whose height is 108 cm and diameter of base is 30 cm. If the height of the cylinder is 9 cm, the diameter of its base is

- (1) 30 cm (2) 60 cm
(3) 50 cm (4) 40 cm

14. The total surface area of a cube and a sphere are equal. What will be the ratio between their volume?

- (1) $\pi : 6$ (2) $\sqrt{\pi} : \sqrt{6}$
(3) $\sqrt{6} : \sqrt{\pi}$ (4) $6 : \pi$

15. A rectangular paper sheet of dimensions 22 cm $\times$ 12 cm is folded in the form of a cylinder along its length. What will be the volume of this cylinder?

- (1) 460 cm³ (2) 462 cm³
(3) 624 cm³ (4) 400 cm³

16. The ratio of the volume of a cube to that of a sphere, which will fit exactly inside the cube, is

- (1) $\pi : 6$ (2) $6 : \pi$
(3) $6 : \pi$ (4) $\pi : 3$

17. The volume of a sphere and a right circular cylinder having the same radius are equal, The ratio of the diameter of the sphere to the height of the cylinder is

- (1) 3 : 2 (2) 2 : 3
(3) 1 : 2 (4) 2 : 1

18. The size of a rectangular piece of paper is 100 cm $\times$ 44 cm. A cylinder is formed by rolling the paper along its length. The volume of the cylinder is

- (1) 4400 cm³ (2) 15400 cm³
(3) 35000 cm³ (4) 144 cm³

19. A cone, a hemisphere and a cylinder stand on equal bases and have the same height. The ratio of their respective volume is

- (1) 1 : 2 : 3 (2) 2 : 1 : 3
(3) 1 : 3 : 2 (4) 3 : 1 : 2

20. The height of a cylinder and that of a cone are in the ratio 2 : 3 and the radii of their bases in the ratio 3 : 4. The ratio of their volume will be

- (1) 1 : 9 (2) 2 : 9
(3) 9 : 8 (4) 3 : 8

21. Find the length of the largest rod that can be placed in a room 16m long, 12m broad and $10\frac{2}{3}$ m. high.

- (1) 23 m. (2) 68 m.
(3) $22\frac{2}{3}$ m. (4) $22\frac{1}{3}$ m.

22. If the volume of two cubes are in the ratio 27 : 64, then the ratio of their total surface area is :

- (1) 27 : 64 (2) 3 : 4
(3) 9 : 16 (4) 3 : 8

23. Find the length of the longest rod that can be placed in a hall of 10 m length, 6 m breadth and 4 m height.

- (1) $2\sqrt{38}$ m (2) $4\sqrt{38}$ m
(3) $2\sqrt{19}$ m (4) 19 m

24. The volume of a cuboid is twice the volume of a cube. If the dimensions of the cuboid are 9 cm, 8 cm and 6 cm, the total surface area of the cube is :

- (1) 72 cm² (2) 216 cm²
(3) 432 cm² (4) 108 cm²

25. The length, breadth and height of a room is 5m, 4m and 3m respectively. Find the length of the largest bamboo that can be kept inside the room.

- (1) 5 m (2) 60 m
(3) 7 m (4) $5\sqrt{2}$ m

26. The length of the longest rod that can be placed in a room which is 12 m long, 9 m broad and 8 m high is

- (1) 27 m (2) 19 m
(3) 17 m (4) 13 m

27. A cube of edge 5 cm is cut into cubes each of edge of 1 cm. The ratio of the total surface area of one of the small cubes to that of the large cube is equal to :

- (1) 1 : 125 (2) 1 : 5
(3) 1 : 625 (4) 1 : 25

28. The perimeter of the floor of a room is 18 m. What is the area of the walls of the room, if the height of the room is 3 m ?

- (1) 21 m² (2) 42 m²
(3) 54 m² (4) 108 m²

29. The length (in metres) of the longest rod that can be put in a room of dimensions $10\text{ m} \times 10\text{ m} \times 5\text{ m}$ is
 (1) $15\sqrt{3}$ (2) 15
 (3) $10\sqrt{2}$ (4) $5\sqrt{3}$
30. The floor of a room is of size $4\text{ m} \times 3\text{ m}$ and its height is 3 m. The walls and ceiling of the room require painting. The area to be painted is
 (1) 66 m^2 (2) 54 m^2
 (3) 43 m^2 (4) 33 m^2

ANSWER KEY

1.	(1)	2.	(1)	3.	(3)	4.	(3)	5.	(3)	6.	(3)	7.	(4)	8.	(3)	9.	(4)	10.	(1)
11.	(1)	12.	(3)	13.	(2)	14.	(2)	15.	(2)	16.	(2)	17.	(1)	18.	(3)	19.	(1)	20.	(3)
21.	(3)	22.	(3)	23.	(1)	24.	(2)	25.	(4)	26.	(3)	27.	(4)	28.	(3)	29.	(2)	30.	(2)

CDS PYQ

1. What is the quantity of cloth required to roll up to form a right circular tent whose base is of radius 12m and height 5m?
[CDS-2013-1]
(a) 40π sq m (b) 60π sq m
(c) 78π sq m (d) 156π
2. The length, breadth and height of a box are respectively 14m, 12m and 13m. The length of the greatest rod that can be put in it is:
[CDS-2013-1]
(a) 22.32m (b) 22.56m
(c) 20m (d) 19.5m
3. The volume of the material of a hemispherical shell with outer and inner radii 9cm and 7 cm, respectively is approximately:
[CDS-2013-1]
(a) 808cu cm (b) 800cu cm
(c) 816 cu cm (d) 824 cu cm
4. The ratio of surface area to diameter of a sphere whose volume is 36π cu cm is:
[CDS-2013-1]
(a) 3π (b) 6π
(c) 6 (d) None of these
5. A cylindrical tube open at both ends is made of metal. The internal diameter of the tube is 6cm and length of the tube is 10cm. If the thickness of the metal used is 1 cm, then the outer curved surface area of the tube is:
[CDS-2013-1]
(a) 140π sq cm (b) 146.5π sq. cm
(c) 70π sq cm (d) None of these
6. The volume of a right circular cone of height 3 cm and slant height 5 cm is:
[CDS-2013-1]
(a) 49.3cu cm (b) 50.3 cu cm
(c) 52 cu cm (d) 53 cu cm
7. A bucket is of a height 25cm. Its top and bottom radii are 20 cm and 10cm, respectively. Its capacity (in L) is:
[CDS-2013-1]
(a) $17.5\pi/3$ (b) 17.5π
(c) 20π (d) 25π
8. The height of a cylinder is 15cm. The lateral surface area is 660 sq. cm. Its volume is:
[CDS-2013-1]
(a) 1155 cu cm (b) 1215cu cm
(c) 1230 cu cm (d) 2310 cu cm
9. From a solid wooden right circular cylinder, a right circular cone whose radius and height are same as the radius and height of the cylinder, respectively is carved out. What is the ratio of the volume of the utilized wood to that of the wasted wood?
[CDS-2013-1]
(a) 1:2 (b) 2:1
(c) 2:3 (d) 1:3
10. A solid spherical ball of iron of radius 4cm is melted to form spheres of radius 2cm each. The number of spheres, so formed is:
[CDS-2013-1]
(a) 8 (b) 9
(c) 10 (d) 16
11. If the heights and the area of the base of a right circular cone and a pyramid with square base are the same, then they have:
[CDS-2013-1]
(a) Same volume and same surface area
(b) Same surface area but different volumes
(c) Same volume but different surface area
(d) Different volumes and different surface area
12. A swimming pool 70 meter long 44 meter wide and 3 m deep is field by water flowing from a pipe at the rate of 30800cm^3 per second the time taken to fill the swimming pool is:
[CDS-2013-1]
(a) $\frac{71}{2}$ hour (b) 80 hours
(c) $\frac{250}{3}$ hours (d) None of these
13. The diameter of the Moon is approximately one-fourth of the diameter of the Earth. What is the ratio (approximate) of their volumes?
[CDS-2013-2]
(a) 1:16 (b) 1:64
(c) 1:4 (d) 1:128
14. If the total surface area of a cube is 6 sq units, then what is the volume of the cube?

[CDS-2013-2]

- (a) 1 cu unit (b) 2 cu units
(c) 4 cu units (d) 6 cu units

15. Let A be a pyramid on a square base and B be a cube. If a, b and c denote the number of edges, number of faces and number of corners, respectively. They, the result $a = b + c$ is true for:

[CDS-2013-2]

- (a) Only A (b) Only B
(c) Both A and B (d) Neither A nor B

16. What is the volume of a cone having a base of radius 10cm and height 21 cm?

[CDS-2013-2]

- (a) 2200cm³ (b) 3000cm³
(c) 5600cm³ (d) 6600cm³

17. A conical cap has base diameter 24cm and height 16cm. What is the cost of painting the surface of the cap at the rate of 70 paise per sq cm?

[CDS-2013-2]

- (a) Rs520 (b) Rs524
(c) Rs528 (d) Rs532

18. What is the whole surface area of a cone of base radius 7 cm and height 24cm?

[CDS-2013-2]

- (a) 654sq cm (b) 704 sq cm
(c) 724 sq cm (d) 964 sq cm

19. A tent is in the form of a right circular cylinder surmounted by a cone. The diameter of the cylinder is 24m. The height of the cylindrical portion is 11m, while the vertex of the cone is 16m above the ground. What is the area of the curved surface for conical portion?

[CDS-2013-2]

- (a) 3434/9 sq m (b) 3431/8 sq cm
(c) 3432/7 sq m (d) 3234/7 sq cm

20. If x is the curved surface area and y is the volume of a right circular cylinder, then which one of the following is correct?

[CDS-2013-2]

- (a) Only the ratio of the height to radius of the cylinder is independent of x
(b) Only the ratio of height to radius of the cylinder is independent of y
(c) Either (a) or (b)
(d) Neither (a) nor (b)

21. A rectangular field is 22m long and 10m wide. Two hemispherical pitholes of radius 2m are dug from two places and the mud is spread over the remaining part of the field. The rise in the level of the field is:

[CDS 2013-2]

- (a) $\frac{8}{93}m$ (b) $\frac{16}{93}m$
(c) $\frac{13}{93}m$ (d) $\frac{23}{93}m$

22. A hospitals room is to accommodate 56 patients. It should be done in such a way that every patient gets 2.2 m² of floor and 8.8m² of space. If the length of the room is 14m, then breadth and the height of the room are respectively:

[CDS 2013-2]

- (a) 8.8m, 4m (b) 8.4m, 4.2m
(c) 8m, 4m (d) 7.8m, 4.2m

23. A cylinder is surmounted by a cone at one end, a hemisphere at the other end. The common radius is 3.5 cm, the height of the cylinder is 6.5 cm and the total height of the structure is 12.8cm. The volume V of the structure lies between:

[CDS-2014-1]

- (a) 370 cm³ and 380 cm³
(b) 380cm³ and 390 cm³
(c) 390 cm³ and 400 cm³
(d) None of these

24. The diameter of the base of a cone is 6 cm and its altitude is 4cm. What is the approximate curved surface area of the cone?

[CDS-2014-1]

- (a) 45cm² (b) 47cm²
(c) 49cm² (d) 51cm²

25. A drainage tile is a cylindrical shell 21 cm long. The inside and outside diameter are 4.5 cm and 5.1 cm, respectively. What is the volume of the clay required for the tile?

[CDS-2014-1]

- (a) $6.96\pi\text{cm}^3$ (b) $6.76\pi\text{cm}^3$
(c) $5.76\pi\text{cm}^3$ (d) None of these

26. For a plot of land of 100m × 80m, the level to be raised by spreading the earth from stack of a rectangular base 10m × 8m with vertical section being a trapezium of height 2m. The top of the stack is 8m × 5m. By how many centimeter can be level raised?

- [CDS-2014-1]**
- (a) 3cm (b) 2.5cm
(c) 2 cm (d) 1.5cm
- 27.** A cube has each edge 2 cm and a cuboid is 1cm long, 2cm wide and 3 cm high. The paint in a certain container is sufficient to paint an area equal to 54cm^2 . Which one of the following is correct?
- [CDS-2014-1]**
- (a) Both cube and cuboid can be painted
(b) Only cube can be painted
(c) Only cuboid can be painted
(d) Neither cube nor cuboid can be painted
- 28.** A cone of radius r cm and height h cm is divided into two parts by drawing a plane through the middle point of its height and parallel to the base. What is the ratio of the volume of the original cone to the volume of the smaller cone?
- [CDS-2014-1]**
- (a) 4:1 (b) 8:1
(c) 2:1 (d) 6:1
- 29.** If 64 identical small spheres are made out of big sphere of diameter 8cm, then what is surface area of each small sphere?
- [CDS-2014-1]**
- (a) πcm^2 (b) $2\pi\text{cm}^2$
(c) $4\pi\text{cm}^2$ (d) $8\pi\text{cm}^2$
- 30.** The dimensions of a field are 15m by 12m. A pit 8m long, 2.5 m wide and 2m deep is dug in one corner of the field and the earth removed is evenly spread over the remaining area of the field. The level of the field is raised by:
- [CDS-2014-1]**
- (a) 15cm (b) 20 cm
(c) 25cm (d) $\frac{200}{9}\text{cm}$
- 31.** What is the diameter of the largest circle lying on the surface of a sphere of surface area 616 sq cm?
- [CDS-2014-1]**
- (a) 14cm (b) 10.5cm
(c) 7cm (d) 3.5cm
- 32.** The volume of a hollow cube is $216x^3$. What surface area of the largest sphere which be enclosed in it?
- [CDS-2014-1]**
- (a) $18\pi x^2$ (b) $27\pi x^2$
(c) $36\pi x^2$ (d) $72\pi x^2$
- 33.** A cylinder circumscribes a sphere. What is the ratio of volume of the sphere to that of the cylinder?
- [CDS-2014-2]**
- (a) 2:3 (b) 1:2
(c) 3:4 (d) 3:2
- For the next three (03) items that follow:**
A right triangle having hypotenuse 25cm and legs in the ratio 3 : 4 is made to revolve about its hypotenuse ($\pi = 3.14$)
- 34.** What is the volume of the double cone so formed?
- [CDS-2014-2]**
- (a) 3124 cm^3 (b) 3424 cm^3
(c) 3768 cm^3 (d) 3924 cm^3
- 35.** What is the surface area of the double cone so formed?
- [CDS-2014-2]**
- (a) 1101.2cm^2 (b) 1111.4cm^2
(c) 1310.4 cm^2 (d) 1318.8 cm^2
- 36.** Consider the following statements:
1.The volume of the cone generated when the triangle is made to revolve about its longer leg is same as the volume of the cone generated when the triangle is made to revolve about its shorter leg.
2.The sum of volume of the cone generated when the triangle is made to revolve about its longer leg and the volume of the cone generated when the triangle is made to revolve about its shorter leg is equal to the volume of the double cone generated when the triangle is made to revolve about its hypotenuse?
Which of the above statements is / are correct?
- [CDS-2014-2]**
- (a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2
- 37.** Consider the following statements in respect of four spheres A, B, C and D having respective radii 6, 8, 10 and 12cm.
1.The surface area of sphere C is equal to the sum of surface area of sphere A and B.
2.The volume of sphere D is equal to the sum of volumes of sphere A, B and C.
Which of the above statements is/are correct?
- [CDS-2014-2]**
- (a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor 2

38. The diameter of a metallic sphere is 6cm. The sphere is melted and drawn into a wire of uniform circular cross section. If the length of the wire is 36m, then what is its radius equal to?

[CDS-2014-2]

- (a) 0.1cm (b) 0.01cm
(c) 0.0001cm (d) 1.0cm

39. The area of the three adjacent faces of a cuboidal box are x , $4x$ and $9x$ sq unit. What is the volume of the box?

[CDS 2014(2)]

- (a) $6x^2$ cu unit (b) $6x^{3/2}$ cu unit
(c) $3x^{3/2}$ cu unit (d) $2x^{3/2}$ cu unit

40. What is the number of pairs of perpendicular planes in a cuboids?

[CDS 2014(2)]

- (a) 4 (b) 8
(c) 12 (d) None of these

41. How many equilateral triangles can be formed by joining any three vertices of a cube?

[CDS 2014(2)]

- (a) 0 (b) 7
(c) 8 (d) None of these

42. What is the maximum distance between two points of a cube of side 2 cm?

[CDS 2014(2)]

- (a) $\sqrt{3}$ (b) $2\sqrt{3}$
(c) $4\sqrt{3}$ (d) $2\sqrt{2}$

For the next two (02) items that follow:

A toy is in the form of a cone mounted on the hemisphere with the same radius. The diameter of the base of the conical portion is 12cm and its height is 8cm.

43. What is the total surface area of the toy?

[CDS 2014(2)]

- (a) 132π cm² (b) 112π cm²
(c) 96π cm² (d) 66π cm²

44. What is the volume of the toy?

[CDS 2014(2)]

- (a) 180π cm³ (b) 240π cm³
(c) 300π cm³ (d) 320π cm³

45. A cylindrical vessel of radius 4cm contains water. A solid sphere of radius 3cm is lowered into the water until it is completely immersed. The water level in the vessel will rise by:

[CDS-2015-1]

- (a) 1.5cm (b) 2cm
(c) 2.25cm (d) 4.5cm

46. If the radius of a sphere is increased by 10%, then the volume will be increased by:

[CDS-2015-1]

- (a) 33.1% (b) 30%
(c) 50% (d) 10%

47. A sphere and a cube have same surface area. The ratio of square of their volumes is:

[CDS-2015-1]

- (a) $6:\pi$ (b) $5:\pi$
(c) 3:5 (d) 1:1

48. The radius of a sphere is equal to the radius of the base of a right circular cone, and the volume of the sphere is double the volume of the cone. The ratio of the height of the cone to the radius of its base is:

[CDS-2015-1]

- (a) 2:1 (b) 1:2
(c) 2:3 (d) 3:2

49. Water flows through a cylindrical pipe of internal diameter 7cm at the rate of 5m/s. The time, in minutes, the pipe would take to fill an empty rectangular tank $4m \times 3m \times 2.31m$ is:

[CDS-2015-1]

- (a) 28 (b) 24
(c) 20 (d) 12

50. The total outer surface area of a right circular cone of height 24cm with a hemisphere of radius 7 cm upon its base is:

[CDS-2015-1]

- (a) 327π square cm (b) 307π square cm
(c) 293π square cm (d) 273π square cm

51. A rectangular block of wood having dimensions $3m \times 2m \times 1.75$ has to be painted on all its faces. The layer of paint must be 0.1 mm thick. Paint comes in cubical boxes having their edges equal to 10cm. The minimum number of boxes of paint to be purchased is:

[CDS-2015-1]

- (a) 5 (b) 4
(c) 3 (d) 2

52. The diagonals of three faces of a cuboid are 13, $\sqrt{281}$ and 20 linear units. Then the total surface area of the cuboid is:

[CDS-2015-1]

- (a) 650 square units (b) 658 square units
(c) 664 square units (d) 672 square units

53. A rectangular paper of 44cm long and 6 cm wide is rolled to form a cylinder of height equal to width of the paper. The radius of the base of the cylinder so rolled is:

[CDS-2015-1]

- (a) 3.5cm (b) 5cm
(c) 7cm (d) 14cm

54. If three metallic sphere of radii 6cm, 8cm and 10 cm are melted to form a single sphere, then the diameter of the new sphere will be:

[CDS-2015-1]

- (a) 12cm (b) 24cm
(c) 30cm (d) 36cm

55. If the height of a right circular cone is increased by 200% and the radius of the base is reduced by 50% then the volume of the cone:

[CDS-2015-1]

- (a) Remains unaltered
(b) Decreases by 25%
(c) Increases by 25%
(d) Increases by 50%

56. A pipe with square cross-section is supplying water to a cistern which was initially empty. The area of cross section is 4cm² and the nozzle velocity of water is 40 m/s. The dimensions of the cistern are 10m×8m×6m. Then the cistern will be full in:

[CDS-2015-2]

- (a) 9.5hours (b) 9hours
(c) 8hours 20 min (d) 8 hours

57. A hollow cylindrical drum has internal diameter of 30cm and a height of 1m. What is the maximum number of cylindrical boxes of diameter 10 cm and height 10cm each that can be packed in the drum?

[CDS-2015-2]

- (a) 60 (b) 70
(c) 80 (d) 90

58. Consider the following statements:

- 1.If the height of a cylinder is doubled, the area of the curved surface is doubled
- 2.If the radius, of a hemispherical solid is doubled, its total surface area becomes fourfold?

Which of the above statements is/are correct?

[CDS-2015-2]

- (a) 1Only (b) 2 Only
(c) Both 1 and 2 (d) Neither 1 nor 2

59. A large water tank has the shape of a cube. If 128 m³ of water is pumped out, the water level goes down by 2m. Then the maximum capacity of the tank is:

[CDS-2015-2]

- (a) 512m³ (b) 480m³
(c) 324m³ (d) 256m³

60. From the solid gold in the form of a cube of side length 1cm, spherical solid balls each having the surface area $\pi^{1/3}$ cm² are to be made. Assuming that there is no loss of the material in the process of making the balls, the maximum number of balls made will be:

[CDS-2015-2]

- (a) 3 (b) 4
(c) 6 (d) 9

61. Water is filled in a container in such a manner that its volume doubles every 5 minutes. If it takes 30 minutes for the container to be full, in how much time will it be one fourth full?

[CDS-2015-2]

- (a) 7.5 minutes (b) 15 minutes
(c) 20 minutes (d) 17.5 minutes

62. 30 metallic cylinders of same size are melted and cast in the form of cones having the same radius and height as those of the cylinders.

Consider the following statements:

Statement I: A maximum of 90 cones will be obtained

Statement II: The curved surface of the cylinder can be flattened in the shape of a rectangle but the curved surface of the cone when flattened has the shape of triangle.

Which one of the following is correct in respect of the above?

[CDS-2015-2]

- (a)Both statement I and Statement II are correct and Statement II is the correct explanation of Statement I
(b)Both statement I and Statement II are correct and Statement II is not the correct explanation of Statement I.
(c)Statement I is correct but Statement II is not correct
(d)Statement I is not correct but Statement II is correct

63. A water tank, open at the top, is hemispherical at the bottom and cylindrical

above it. The radius is 12m and the capacity is $3312\pi\text{m}^3$. The ratio of the surface area of the spherical and cylindrical portions is:

[CDS-2015-2]

- (a) 3:5 (b) 4:5
(c) 1:1 (d) 6:5

64. The areas of three mutually perpendicular faces of a cuboid are x , y , z . If V is the volume, then xyz is equal to:

[CDS-2015-2]

- (a) V (v) V^2
(c) $2V$ (d) $2V^2$

65. Let V be the volume of an inverted cone with vertex at origin and the axis of the cone is along positive y -axis. The cone is filled with water up to half of its height. The volume of water is:

[CDS-2015-2]

- (a) $\frac{V}{8}$ (b) $\frac{V}{6}$
(c) $\frac{V}{3}$ (d) $\frac{V}{2}$

66. If the surface area of a cube is 13254cm^2 , then the length of its diagonal is:

[CDS-2015-2]

- (a) $44\sqrt{2}\text{ cm}$ (b) $44\sqrt{3}\text{ cm}$
(c) $47\sqrt{2}\text{ cm}$ (d) $47\sqrt{3}\text{ cm}$

67. How many spherical bullets each of 4 cm in diameter can be made out of a cube of lead whose edge is 44cm?

[CDS-2015-2]

- (a) 2541 (b) 2551
(c) 2561 (d) 2571

68. A river 2.5 m deep and 45m wide is flowing at the speed of 3.6km/hour. The amount of water that runs into the sea per minutes is:

[CDS-2015-2]

- (a) 6650m^3 (b) 6750m^3
(c) 6850m^3 (d) 6950m^3

69. Three rectangles R_1 , R_2 and R_3 have the same area. Their lengths x_1 , x_2 and x_3 respectively are such that $x_1 < x_2 < x_3$. If V_1 , V_2 and V_3 are the volumes of the cylinders formed from the rectangles R_1 , R_2 and R_3 respectively by joining the parallel sides along the breadth, then which one of the following is correct?

[CDS 2015-2]

- (a) $V_3 < V_2 < V_1$ (b) $V_1 < V_3 < V_2$
(c) $V_1 < V_2 < V_3$ (d) $V_3 < V_1 < V_2$

70. How many right angled triangles can be formed by joining the vertices of a cuboid?

[CDS 2015-2]

- (a) 24
(b) 28
(c) 32
(d) None of the above

71. The area of four walls of a room is 120m^2 . The length of the room is twice its breadth. If the height of the room is 4 m, what is area of the floor?

[CDS-2016-2]

- (a) 40m^2 (b) 50m^2
(c) 60m^2 (d) 80m^2

72. The ratio of the curved surface area to the total surface area of a right circular cylinder is 1:2. If the total surface area is 616cm^2 , what is the volume of the cylinder?

[CDS-2016-2]

- (a) 539cm^3 (b) 616cm^3
(c) 1078cm^3 (d) 1232cm^3

73. A cubic meter of copper weighing 9000kg is rolled into a square bar 9 m long. An exact cube is cut off from the bar. How much does the cube weigh?

[CDS-2016-2]

- (a) 1000kg (b) $\frac{1000}{3}\text{ kg}$
(c) 300kg (d) $\frac{500}{3}\text{ kg}$

74. Into a conical tent of radius 8.4m and vertical height 3.5m, how many full bags of wheat can be emptied, if space required for the wheat in each bag is 1.96m^3 ?

[CDS-2016-2]

- (a) 264 (b) 201
(c) 132 (d) 105

75. A building is the form of a cylinder surmounted by a hemispherical dome on the diameter of the cylinder. The height of the building is three times the radius of the base of the cylinder. The building contains $67\frac{1}{21}\text{m}^3$ of air. What is the height of the building?

[CDS-2016-2]

- (a) 6m (b) 4m
(c) 3m (d) 2m

76. The radius of the base and the height of a solid right circular cylinder are in the ratio 2:3 and its volume is 1617 cm^3 . What is the total surface area of the cylinder?

[CDS-2016-2]

- (a) 462cm^2 (b) 616cm^2
(c) 770cm^2 (d) 786cm^2

77. The cost of painting a spherical vessel of diameter 14cm is Rs.8008. What is the cost of painting per square centimeter?

[CDS-2016-2]

- (a) Rs.8 (b) Rs.9
(c) Rs.13 (d) Rs.14

78. A drinking glass of height 24cm is in the shape of frustum of a cone and its diameters of its bottom and top circular ends are 4cm and 18 cm respectively. If we take capacity of the glass as $\pi x \text{ cm}^3$, then what is the value of x?

[CDS-2016-2]

- (a) 824 (b) 1236
(c) 1628 (d) 2472

79. Rain water from a roof $22\text{m} \times 20\text{m}$ drains into a cylindrical vessel having diameter of base 2m and height 3.5m. If the vessel is just full, what is the rainfall?

[CDS-2016-2]

- (a) 3.5cm (b) 3cm
(c) 2.5cm (d) 2cm

80. The height of a cone is 60cm. A small cone is cut of at the top by a plane parallel to the base and its volume is $\frac{1}{64}$ the volume of original cone. What is the height from the base at which the section is made?

[CDS-2016-2]

- (a) 15cm (b) 20cm
(c) 30cm (d) 45cm

81. What is the volume of a sphere of radius 3cm?

[CDS-2016-2]

- (a) $36\pi \text{ cm}^3$ (b) $18\pi \text{ cm}^3$
(c) $9\pi \text{ cm}^3$ (d) $6\pi \text{ cm}^3$

Directions (for next three): A tent of a circus is made of canvas and is in the form of right circular cylinder and right circular cone above it. The height and diameter of the cylindrical part of the tent are 5 m and 126m respectively. The total height of the tent is 21 m.

82. What is the slant height of the cone?

[CDS-2016-2]

- (a) 60m (b) 65m
(c) 68m (d) 70m

83. What is the curved surface area of the cylinder?

[CDS-2016-2]

- (a) 1980m^2 (b) 2010m^2
(c) 2100m^2 (d) 2240m^2

84. How many square metres of canvas are used?

[CDS-2016-2]

- (a) 14450 (b) 14480
(c) 14580 (d) 14850

85. If the radius of a right circular cone is increased by p% without increasing its height, then what is the percentage increase in the volume of the cone?

[CDS-2017-1]

- (a) p^2 (b) $2p^2$
(c) $\frac{p^2}{100}$ (d) $p\left(2 + \frac{p}{100}\right)$

86. If the surface area of a sphere is reduced to one-ninth of the area, its radius reduces to:

[CDS-2017-1]

- (a) One-fourth (b) One-third
(c) One fifth (d) One ninth

87. Ice-cream, completely filled in a cylinder of diameter 35cm and height 32cm, is to be served by completely filling identical disposable cones of diameter 4cm and height 7 cm. The maximum number of persons that can be served in this way is:

[CDS-2017-1]

- (a) 950 (b) 1000
(c) 1050 (d) 1100

88. A ball of radius 1 cm is put into a cylindrical pipe so that it fits inside the pipe. If the length of the pipe is 14m, what is the surface area of the pipe?

[CDS-2017-1]

- (a) 2200 square cm
(b) 4400 square cm
(c) 8800 square cm
(d) 17600 square cm

89. A cylinder of height 2x is circumscribed by a sphere of radius 2x such that the circular ends of the cylinder are two small circles on the sphere. What is the ratio of the curved

surface area of the cylinder to the surface area of the sphere?

[CDS-2017-2]

- (a) $\sqrt{3}:4$ (b) $\sqrt{3}:3$
(c) $\sqrt{3}:2$ (d) $\sqrt{3}:1$

90. A cylindrical vessel 60cm in diameter is partially filled with water. A sphere 30cm in diameter is gently dropped into the vessel and is completely immersed. To what further height will the water in the cylinder rise?

[CDS-2017-2]

- (a) 20cm (b) 15cm
(c) 10cm (d) 5cm

91. The vertical angle of a right circular cone is $\frac{\pi}{2}$ and the slant height is $\sqrt{2}r$ cm. What is the volume of the cone in cubic cm?

[CDS-2017-2]

- (a) πr^3 (b) $9\pi r^3$
(c) $\frac{\pi r^3}{3}$ (d) $3\pi r^3$

92. The radii of the frustum of a right circular cone are in the ratio 2:1. What is the ratio of the volume of the frustum of the cone to that of the whole cone?

[CDS-2017-2]

- (a) 1:8 (b) 1:4
(c) 3:4 (d) 7:8

93. From a solid cylinder whose height is 8cm and of base radius 6cm, a conical cavity of height 8cm and a base radius 6cm is formed by hollowing out. What is the inner surface area of the cavity?

[CDS-2017-2]

- (a) 6π square cm (b) 8π square cm
(c) 10π square cm (d) 60π square cm

94. A tent has been constructed which is in the form of a right circular cylinder surmounted by a right circular cone whose axis coincides with the axis of the cylinder. If the radius of the base of the cylinder is 50m, the height of the cylinder is 10m and the total height of the tent is 15m, then what is the capacity of the tent in cubic metres?

[CDS-2017-2]

- (a) 37500π (b) $\frac{87500\pi}{3}$

(c) $\frac{26500\pi}{3}$

(d) 25000π

95. Two rectangular sheet of sizes $2\pi \times 4\pi$ and $\pi \times 5\pi$ are available. A hollow right circular cylinder can be formed by joining a pair of parallel sides of any sheet. What is the maximum possible volume of the circular cylinder that can be formed this way?

[CDS-2017-2]

- (a) $4\pi^2$ (b) $8\pi^2$
(c) $1.25\pi^2$ (d) $6.25\pi^2$

96. A hollow cube is formed by joining six identical squares. A rectangular cello tape of length 4 cm and breadth 0.5 cm is used for joining each pair of edges. What is the total area of cello tape used?

[CDS 2017-2]

- (a) 12 square cm (b) 24 square cm
(c) 36 square cm (d) 48 square cm

97. The radius and slant height of a right circular cone are 5 cm and 13 cm respectively. What is the volume of the cone?

[CDS-2018-1]

- (a) $100\pi\text{cm}^3$ (b) $50\pi\text{cm}^3$
(c) $65\pi\text{cm}^3$ (d) $169\pi\text{cm}^3$

98. If the ratio of the radius of the base of a right circular cone to its slant height is 1:3, what is the ratio of the total surface area to the curved surface area?

[CDS-2018-1]

- (a) 5:3 (b) 3:1
(c) 4:1 (d) 4:3

99. A right circular cone is sliced into a smaller cone and a frustum of a cone by a plane perpendicular to its axis. The volume of the smaller cone and the frustum of the cone are in the ratio 64:61. Then their curved surface areas are in the ratio:

[CDS-2018-1]

- (a) 4:1 (b) 16:9
(c) 64:61 (d) 81:64

100. In a room whose floor is a square of side 10m, an equilateral triangular table of side 2 m is placed. Four book shelves of size 4m $\times$ 1m $\times$ 9m are also placed in the room. If half of the rest of the area in the room is to be carpeted at the rate of Rs100 per square metre, what is the cost of carpeting (approximately)?

[CDS-2018-1]

- (a) Rs7600 (b) Rs5635
(c) Rs4113 (d) Rs3200

101. What is the volume of a cone of maximum volume cut out from a cube of edge $2a$ such that their bases are on the same plane?

[CDS-2018-1]

- (a) πa^2 (b) $\frac{\pi a^3}{3}$
(c) $\frac{2\pi a^3}{3}$ (d) $\frac{3\pi a^3}{4}$

102. There are as many square centimeter in the surface area of a sphere as there are cubic centimeter in its volume. What is the radius of the sphere?

[CDS-2018-1]

- (a) 4cm (b) 3cm
(c) 2cm (d) 1cm

103. The curved surface area of right circular cone is 1.76m^2 and its base diameter is 140cm. What is the height of the cone?

[CDS-2018-1]

- (a) 10cm (b) $10\sqrt{2}$ cm
(c) $20\sqrt{2}$ cm (d) $10\sqrt{15}$ cm

104. A cube of maximum volume (each corner touching the surface from inside) is cut from a sphere. What is the ratio of the volume of the cube to that of the sphere?

[CDS-2018-1]

- (a) $3:4\pi$ (b) $\sqrt{3}:2\pi$
(c) $2:\sqrt{3}\pi$ (d) $4:3\pi$

105. If the ratio of the circumference of the base of a right circular cone of radius r to its height is 3:1, then what is the area of the curved surface of the cone?

[CDS-2018-1]

- (a) $3\pi r^2$ (b) $\frac{2\pi r^2 \sqrt{4\pi^2 + 9}}{3}$
(c) $\frac{\pi r^2 \sqrt{\pi^2 + 1}}{3}$ (d) $\frac{\pi r^2 \sqrt{4\pi^2 + 9}}{3}$

106. The surface area of closed cylindrical box is 352 square cm. If its length is 10cm, then what is its diameter? $\left(\text{Use } \pi = \frac{22}{7}\right)$

[CDS-2018-1]

- (a) 4cm (b) 8cm
(c) 9.12cm (d) 19.26cm

107. Walls (excluding roof and floors) of 5 identical rooms having length, breadth and height 6m, 4m, and 2.5m respectively are to be painted. Out of five rooms two rooms have one square window each having a side 2.5m. Paints are available only in cans of 1 litre; and 1 litre of paint can be used for painting 20 square metres. The number of cans required for painting is:

[CDS-2018-1 & 2]

- (a) 10 (b) 12
(c) 13 (d) 14

108. The lateral surface area of a cone is 462cm^2 . Its slant height is 35cm. The radius of the base of the cone is:

[CDS-2018-2]

- (a) 8.4cm (b) 6.5cm
(c) 4.2cm (d) 3.2cm

109. A semi circular plate is rolled up to form a conical surface. The angle between the generator and the axis of the cone is:

[CDS-2018-2]

- (a) 60° (b) 45°
(c) 30° (d) 15°

110. A solid right cylinder is of height π cm. If its lateral surface area is half its total surface area, then the radius of its base is:

[CDS-2018-2]

- (a) $\pi/2$ (b) πcm
(c) $1/\pi\text{cm}$ (d) $2/\pi\text{cm}$

111. A rectangular block of length 20 cm, breadth 15 cm and height 10 cm is cut up into exact number of equal cubes. The least possible number of cubes will be:

[CDS-2018-2]

- (a) 12 (b) 16
(c) 20 (d) 24

112. If the diagonal of a cube is of length l , then the total surface area of the cube is:

[CDS-2018-2]

- (a) $3l^2$ (b) $\sqrt{3}l^2$
(c) $\sqrt{2}l^2$ (d) $2l^2$

113. Two cones have their heights in the ratio 1:3. If the radii of their bases are in the ratio 3:1, then the ratio of their volumes will be:

[CDS-2018-2]

- (a) 1:1 (b) 2:1
(c) 3:1 (d) 9:1

Direction for (next two):

A cube is inscribed in a sphere. A right circular cylinder is within the cube touching all the vertical faces. A right circular cone is inside the cylinder. Their heights are same and the diameter of the cone is equal to that of cylinder.

- 114.** What is the ratio of the volume of the sphere to that of the cone?

[CDS-2019-1]

- (a) $6\sqrt{3}:1$ (b) $7:2$
(c) $3\sqrt{3}:1$ (d) $5\sqrt{3}:1$

- 115.** What is the ratio of the volume of the cube to that of the cylinder?

[CDS-2019-1]

- (a) 4:3 (b) 21:16
(c) 14:11 (d) 45:32

- 116.** Consider the following statements:

1. The surface area of the sphere is $\sqrt{5}$ times the curved surface area of the cone.
 2. The surface area of the cube is equal to the curved surface area of the cylinder.
- Which of the above statements is/are correct?

[CDS-2019-1]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Direction for (next two):

The sum of length, breadth and height of a cuboid is 22 cm and the length of its diagonal is 14cm.

- 117.** What is the surface area of the cuboid?

[CDS-2019-1]

- (a) 288cm^2
(b) 216cm^2
(c) 144cm^2
(d) Cannot be determined due to insufficient data

- 118.** If S is the sum of the cubes of the dimensions of the cuboid and V is its volume, then what is $(S-3V)$ equal to?

[CDS-2019-1]

- (a) 572cm^3
(b) 728cm^3
(c) 1144cm^3
(d) None of the above

- 119.** The volume of a spherical balloon is increase by 700%. What is the percentage increase in its surface area?

[CDS-2019-1]

- (a) 300% (b) 400%
(c) 450% (d) 500%

- 120.** A hollow right circular cylindrical vessel of volume V whose diameter is equal to its height, is completely filled with water. A heavy sphere of maximum possible volume is then completely immersed in the vessel. What volume of water remains in the vessel?

[CDS-2019-2]

- (a) $\frac{V}{2}$ (b) $\frac{V}{3}$
(c) $\frac{2V}{3}$ (d) $\frac{V}{4}$

- 121.** Two cylinder of equal volume have their heights in the ratio 2:3. What is the ratio of their radii?

[CDS-2019-2]

- (a) $\sqrt{3}:1$ (b) $\sqrt{3}:\sqrt{2}$
(c) $2:\sqrt{3}$ (d) $\sqrt{3}:2$

- 122.** A right circular cylinder has a diameter of 20 cm and its curved surface area is 1000cm^2 . What is the volume of the cylinder?

[CDS-2019-2]

- (a) 4000cm^3 (b) 4500cm^3
(c) 5000cm^3 (d) 5200cm^3

- 123.** A hollow sphere of external and internal diameters 6cm and 4cm respectively is melted into a cone of base diameter 8cm. What is the height of the cone?

[CDS-2019-2]

- (a) 4.75cm (b) 5.50cm
(c) 6.25cm (d) 6.75cm

- 124.** A solid metallic cylinder of height 10 cm and radius 6cm is melted to make two cones in the ratio of volume 1:2 and of the same height as 10cm. What is the percentage increase in the flat surface area?

[CDS-2019-2]

- (a) 25% (b) 50%
(c) 75% (d) 100%

- 125.** Three copper spheres of radii 3cm, 4cm and 5 cm are melted to form a large sphere. What is its radius?

[CDS-2019-2]

- (a) 12cm (b) 10cm
(c) 8cm (d) 6cm

- 126.** The volume of a hemisphere is 155232m^3 . What is the radius of the hemisphere?

[CDS-2019-2]

- (a) 40cm (b) 42cm
(c) 38cm (d) 36cm

- 127.** A bucket is in the form of a truncated cone. The diameters of the base and top of the bucket are 6cm and 12cm respectively. If the height of the bucket is 7cm, what is the capacity of the bucket?

[CDS-2019-2]

- (a) 535cm^3 (b) 462cm^3
(c) 234cm^3 (d) 166cm^3

- 128.** A right circular cone has height 8cm. If the radius of its base is 6cm, then what is its total surface area?

[CDS-2019-2]

- (a) $96\pi\text{cm}^2$ (b) $69\pi\text{cm}^2$
(c) $54\pi\text{cm}^2$ (d) $48\pi\text{cm}^2$

- 129.** Six cubes, each with 12cm edge are joined end to end. What is the surface area of resulting cuboid?

[CDS-2019-2]

- (a) 3000cm^2 (b) 3600cm^2
(c) 3744cm^2 (d) 3777cm^2

- 130.** The area of three adjacent faces of a cuboid are x, y and z. If V is the volume of the cuboid, then which one of the following is correct?

[CDS-2019-2]

- (a) $V=xyz$ (b) $V^2=xyz$
(c) $V^3=xyz$ (d) $V=(xyz)^2$

- 131.** A thin rod of length 24 feet is cut into rods of equal size and joined so as to form a skeleton cube. What is the area of one of the faces of the largest cube thus constructed?

[CDS-2019-2]

- (a) 25 square feet (b) 24 square feet
(c) 9 square feet (d) 4 square feet

- 132.** The thickness of a cylinder is 1 foot, the inner radius of the cylinder is 3feet and height is 7feet. To paint the inner surface it

requires one litre of a particular colour. How much quantity of the same colour is required to paint all the surface of the cylinder?

[CDS-2020-1]

- (a) $\frac{7}{3}$ litre (b) $\frac{3}{2}$ litre
(c) $\frac{8}{3}$ litre (d) $\frac{10}{3}$ litre

- 133.** How many cubic metre of earth is to be dug out to dig a well of radius 1.4m and depth 5m?

[CDS-2020-1]

- (a) 30.2cubic metre (b) 30.4 cubic metre
(c) 30.6cubic metre (d) 30.8 cubic metre

- 134.** Out of 4 identical balls of radius r, 3 balls are placed on a plane such that each ball touches the other two balls. The 4th ball is placed on them such that this ball touches all the three balls. What is the distance of center of 4th ball from the plane?

[CDS-2020-1]

- (a) $2\sqrt{\frac{2}{3}}r$ unit (b) $\frac{\sqrt{3}+2\sqrt{2}}{\sqrt{2}}r$ unit
(c) $\frac{r}{3-2\sqrt{2}}r$ unit (d) $\frac{\sqrt{3}+2\sqrt{r}}{\sqrt{3}}r$ unit

- 135.** A right circular cylinder just encloses a sphere. If p is the surface area of the sphere and q is the curved surface area of the cylinder, then which one of the following is correct?

[CDS-2020-1]

- (a) $p=q$ (b) $p=2q$
(c) $2p=q$ (d) $2p=3q$

- 136.** If H, C and V are respectively the height, curved surface area and volume of a cone, then what is $3\pi VH^3 + 9V^2$ equal to?

[CDS-2020-2]

- (a) C^2H^2 (b) $2C^2H^2$
(c) $5C^2H^2$ (d) $7C^2H^2$

- 137.** How many solid lead balls each of diameter 2mm can be made from a solid lead ball of radius 8cm?

[CDS-2020-2]

- (a) 512 (b) 1024
(c) 256000 (d) 512000

- 138.** The surface areas of two sphere are in the ratio 1:4. What is the ratio of their volumes?

[CDS-2020-2]

- (a) 1:16 (b) 1:12
(c) 1:10 (d) 1:8

139. The length, breadth and height of a brick are 20cm, 15 cm and 10cm respectively. The number of bricks required to construct at wall with dimensions 45m length, 0.15m breadth and 3 m height is:

[CDS-2020-2]

- (a) 12450
(b) 11250
(c) 6750
(d) None of the above

140. The volumes of two cones are in the ratio 1:4 and their diameters are in the ratio 4:5. What is the ratio of their heights?

[CDS-2020-2]

- (a) 25:64 (b) 16:25
(c) 9:16 (d) 5:9

141. If p, q, r, s and t represent length, breadth, height, surface area and volume of a cuboid respectively, then what is $\frac{1}{p} + \frac{1}{q} + \frac{1}{r}$ equal to?

[CDS-2020-2]

- (a) $\frac{s}{t}$ (b) $\frac{2t}{s}$
(c) $\frac{s}{2t}$ (d) $\frac{2s}{t}$

142. The radii of the flat circular faces of a bucket are x and 2x. If the height of the bucket is 3x, what is the capacity of the bucket? (assume $\pi = \frac{22}{7}$).

[CDS-2020-2]

- (a) $11x^3$ (b) $22x^3$
(c) $44x^3$ (d) $55x^3$

143. A river 3m deep and 40 m wide is flowing at the rate of 2 km/h and falls into the sea. What is the amount of water in litres that will fall into the sea from this river in a minute?

[CDS-2020-2]

- (a) 40,00,000 litres (b) 4,00,000 litres
(c) 40,000 litres (d) 4,000 litres

144. A cloth of 3m width is used to make a conical tent 12 m in diameter with a slant height of 7m. What is the length of the cloth? $\left(\text{take } \pi = \frac{22}{7} \right)$

[CDS-2021-1]

- (a) 21m (b) 28m
(c) 44m (d) 66m

145. A sphere of diameter 6cm is dropped into a cylindrical vessel partly filled with water. The radius of the vessel is 6cm. If the sphere is completely submerged in water, then by how much will the surface level of water be raised?

[CDS-2021-1]

- (a) 0.5cm (b) 1cm
(c) 1.5cm (d) 2cm

146. The surface area of a cube is equal to that of sphere. If x is the volume of the cube and y is the volume of the sphere, then what is $x^2 : y^2$ equal to?

[CDS-2021-1]

- (a) $\pi:6$ (b) $6:\pi$
(c) $\pi:3$ (d) $3:\pi$

147. A cone and a hemisphere have equal bases and equal volumes. What is the ratio of the height of the cone to the radius of the hemisphere?

[CDS-2021-1]

- (a) 1:1 (b) 2:1
(c) 3:2 (d) 4:3

148. A solid sphere of diameter 60mm is melted to stretch into a wire of length 144cm. What is the diameter of the wire?

[CDS-2021-1]

- (a) 0.5cm (b) 1cm
(c) 1.5cm (d) 2cm

149. The ratio of the radius of base to the height of a cylinder is 2:3. If the volume of the cylinder is 1617 cm^2 , then what is the curved surface area of the cylinder? $\left(\text{take } \pi = \frac{22}{7} \right)$.

[CDS-2021-1]

- (a) 242cm^2 (b) 385cm^2
(c) 462cm^2 (d) 770cm^2

150. A metal solid cube of edge 24cm is melted and made into three small cubes. If the edges of two small cubes are 12 cm and 16cm, then what is the surface area of the third small cube?

[CDS-2021-1]

- (a) 1200cm^2 (b) 1800cm^2
(c) 2400cm^2 (d) 3600cm^2

- 151.** A conical vessel whose internal radius is 5cm and height 24cm is full of water. The water is emptied into a cylindrical vessel with internal radius 10cm. What is the height to which the water rises?

[CDS-2021-1]

- (a) 1cm (b) 2cm
(c) 3cm (d) 4cm

- 152.** A metal solid cube of side 22 cm is melted to make a cone of height 21 cm. What is the 'radius of the base of the cone?

$\left(\text{take } \pi = \frac{22}{7} \right)$

[CDS-2021-1]

- (a) 11cm (b) 16.5cm
(c) 22cm (d) 27.5cm

- 153.** A cone of height 24 cm has a curved surface area 550 cm². What is the ratio of its radius to slant height? $\left(\text{take } \pi = \frac{22}{7} \right)$

[CDS-2021-1]

- (a) $\frac{5}{12}$ (b) $\frac{5}{13}$
(c) $\frac{7}{25}$ (d) $\frac{7}{27}$

- 154.** A rectangular paper is 44cm long and 22cm wide. Let x be the volume of the largest cylinder formed by rolling the paper along its length and y be the volume of the largest cylinder formed by rolling the paper along its width. What is the ratio of x to y?

$\left(\text{take } \pi = \frac{22}{7} \right)$

[CDS-2021-1]

- (a) 1:1 (b) 2:1
(c) 1:2 (d) 3:2

- 155.** A hollow spherical shell is made up of a metal of density 3g/cm³. If the internal and external radii are 5 cm and 6cm respectively, then what is the mass of the shell?

$\left(\text{take } \pi = \frac{22}{7} \right)$

[CDS-2021-1]

- (a) 1144g (b) 1024g
(c) 840g (d) 570g

- 156.** The difference between the outside and the inside surface area of a cylindrical pipe 14cm long is 44 cm². The pipe is made of 99cm³ of metal. If R is the outer radius and

r is the inner radius of the pipe, then what is (R + r) equal to? (Take $\pi = \frac{22}{7}$)

[CDS 2021 (I)]

- (a) 9cm (b) 7.5cm
(c) 6cm (d) 4.5cm

- 157.** A hemispherical bowl of internal radius 18cm contains a liquid. The liquid is filled in small cylindrical bottles of internal radius 3cm and internal height 4cm. What is the number of bottles used to empty the bowl?

[CDS-2021-2]

- (a) 54 (b) 81
(c) 108 (d) 135

- 158.** A hollow spherical shell is made of a metal of density 7 g/cm³. If its internal and external radii are 3 cm and 6cm respectively, then what is the mass of the shell? (take $\pi = \frac{22}{7}$).

[CDS-2021-2]

- (a) 2772g (b) 3322g
(c) 4433g (d) 5544g

- 159.** A cone of height 16cm and diameter 14cm is mounted on a hemisphere of same diameter. What is the volume of the solid thus formed? (take $\pi = \frac{22}{7}$)

[CDS-2021-2]

- (a) 1540 cm³ (b) 1078 cm³
(c) 1048 cm³ (d) 770 cm³

- 160.** 3 cubes each of volume 343 cm³ are joined end to end. What is the total surface area of the resulting cuboid?

[CDS-2021-2]

- (a) 343 cm² (b) 350 cm²
(c) 686 cm² (d) 700 cm²

- 161.** A cubical block of side 14cm is surmounted by a hemisphere of radius 7cm. What is the total surface area of the solid thus formed? (take $\pi = \frac{22}{7}$)

[CDS-2021-2]

- (a) 1330 cm² (b) 1306 cm²
(c) 1296 cm² (d) 1256 cm²

- 162.** How many silver coins, 3.5cm in diameter and of thickness 4 mm, must be melted to form a cuboid of dimension 21cm × 11 cm × 7 cm? (take $\pi = \frac{22}{7}$)

[CDS-2021-2]

- (a) 420 (b) 210
(c) 200 (d) 168

- 163.** A tub is in the shape of a frustum of a cone. The radii of two circular ends of the tub are 105 cm and 42 cm. If the vertical height of the tub is 16 cm, what is its slant height?

[CDS-2021-2]

- (a) 63.5 cm (b) 65 cm
(c) 73.5 cm (d) 75 cm

- 164.** ABC is a triangle right-angled at B with AB = 8 cm and BC = 6 cm. It is made to revolve about its side BC. What is the approximate total surface area of the cone so formed? (take $\pi = \frac{22}{7}$)

[CDS-2021-2]

- (a) 452 cm² (b) 440 cm²
(c) 432 cm² (d) 420 cm²

- 165.** A solid rod consists of a cylinder of height 20 cm and radius 7 cm. It is surmounted by another solid cylinder of height 10 cm and radius 3.5 cm. If 1 cubic metre of rod weighs 10000 kg, what is the mass of the rod?

(take $\pi = \frac{22}{7}$)

[CDS-2021-2]

- (a) 34.65 kg (b) 31.56 kg
(c) 3.465 kg (d) 3.156 kg

Consider the following for the next three (03) items that follow:

A solid consisting of a right circular cone of radius x and height $2x$ standing on a hemisphere of radius x (take $\pi = \frac{22}{7}$)

- 166.** The volume of the solid is equal to that of a:

[CDS-2021-2]

- (a) Sphere of radius x
(b) Sphere of diameter x
(c) Cylinder of radius x
(d) Cylinder of radius $\sqrt{2}x$

- 167.** What is the approximate total surface area of the solid?

[CDS-2021-2]

- (a) $11.2x^2$ (b) $12.5x^2$
(c) $13.3x^2$ (d) $15.1x^2$

- 168.** The solid is placed upright in a right circular cylinder full of water such that it touches the bottom. If the internal radius of cylinder is x and height is $3x$, what is the approximate volume of water left in the cylinder?

[CDS-2021-2]

- (a) $5.04x^3$ (b) $5.09x^3$
(c) $5.14x^3$ (d) $5.24x^3$

- 169.** The length and breadth of a room are 21 m and 16 m respectively. If the length of the longest rod that can be placed in the room is 29 m, then what is the height of the room?

[CDS 2021-2]

- (a) 10 m (b) 11 m
(c) 12 m (d) 13 m

- 170.** Water is trickling out of a completely filled cylindrical tank of height 1 m and diameter 2 m. Every second a spherical droplet 2 cm. Every seconds a spherical droplet of 1 cm radius trickles down from the bottom of the tank. The tank will be emptied in approximately.

[CDS 2022-1]

- (a) 280 hours (b) 260 hours
(c) 230 hours (d) 210 hours

- 171.** The length, breadth and height of a cuboid are in the ratio 27 : 8 : 1. The cuboid is melted and recast into a cube. If p is the surface area of the cuboid and q is the surface area of the cube, then what is p/q equal to?

[CDS 2022-1]

- (a) $\frac{247}{108}$ (b) $\frac{251}{108}$
(c) $\frac{503}{216}$ (d) $\frac{505}{216}$

- 172.** In a shower, 5 cm of rain falls. What is the volume of water that falls on 2 hectare area of land?

[CDS 2022-1]

- (a) 100 cubic meter
(b) 1000 cubic meter
(c) 4000 cubic meter
(d) 10000 cubic meter

- 173.** The radius and height of a right circular cone are in the ratio 3 : 7. If the volume of the cone is 528 cm³, then what is the height of the cone? (Take $\pi = \frac{22}{7}$)

[CDS 2022-1]

- (a) 3.5 cm (b) 7.0 cm
(c) 10.5 cm (d) 14.0 cm

- 174.** A cylindrical pipe has inner diameter of 14 cm. Water flows through it at a rate of

154 litres per minute. What is the speed of water in km/hr? (Take $\pi = \frac{22}{7}$)

[CDS 2022-1]

- (a) 0.5 (b) 0.6
(c) 0.8 (d) 1

175. A spherical ball of lead 6cm in diameter is melted and recast into three spherical balls. The diameters of the balls are 2cm, 4cm and x cm. Which one of the following is correct?

[CDS 2022-1]

- (a) $5\text{cm} < x\text{cm} < 5.2\text{cm}$
(b) $5.2\text{cm} < x\text{cm} < 5.4\text{cm}$
(c) $5.4\text{cm} < x\text{cm} < 5.6\text{cm}$
(d) $5.6\text{cm} < x\text{cm} < 5.8\text{cm}$

176. A rectangular sheet is of length x and breadth y. If p is the volume of the cylinder formed by rolling the sheet along its breadth and q is the volume of the cylinder formed by rolling the sheet along its length, and $q = 2p$, then which one of the following is correct?

[CDS 2022-1]

- (a) $x = y$ (b) $2x = 3y$
(c) $x = 2y$ (d) $3x = 4y$

177. A tall cylindrical container with circular base of radius 18cm contains a good quantity of water. Metal balls each of radius 0.9cm are immersed in it. How many balls are required to raise the water level by 3cm?

[CDS 2022-1]

- (a) 100 (b) 500
(c) 1000 (d) 1500

178. The surface area of a cube of length x is equal to the surface area of a sphere of radius y. Consider the following statements:

1. $2x > 3y$
2. The volume of the cube is greater than the volume of the sphere.

Which of the above statements is/are correct?

[CDS 2022-1]

- (a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

179. A closed box is built of wood of uniform thickness. Its external dimensions are

12cm, 10cm and 8cm. If the inner surface area is 376 square cm, then what is the thickness of the wood?

[CDS 2022 (II)]

- (a) 0.5cm (b) 1cm
(c) 1.5cm (d) 2cm

180. The height of a cone is 30cm. A small cone is cut off at the top by a plane parallel to its base. If its volume is $\frac{1}{27}$ of the volume of the given cone, then what is the height of the frustum of the cone?

[CDS 2022 (II)]

- (a) 10cm (b) 12cm
(c) 18cm (d) 20cm

181. A rectangular metal sheet is of length 24 cm and breadth 18cm. From each of its corners a square of side x cm is cut off and an open box is made of the remaining sheet. If the volume of the box is 640 cubic cm, then what is the value of x?

[CDS 2022 (II)]

- (a) 2 (b) 3
(c) 4 (d) 6

182. Let x, y, z be the length, breadth, height of a cuboid. If its volume is 400 cubic cm and total surface area is 340 square cm, then what is $x^{-1} + y^{-1} + z^{-1}$ equal to?

[CDS 2022 (II)]

- (a) $\frac{17}{20}$ (b) $\frac{17}{40}$
(c) $\frac{7}{10}$ (d) $\frac{9}{10}$

183. The length of a room is $\frac{21}{16}$ times its breadth and breadth is $\frac{4}{7}$ times its height. If H is the height of the room and L is the longest rod that can be placed in the room, then which one of the following is correct?

[CDS 2022 (II)]

- (a) $12L = 29H$ (b) $9L = 25H$
(c) $7L = 23H$ (d) $8L = 13H$

Consider the following for the next three (03) items the follow:

A conical vessel of radius 12cm and height 16cm is filled with water. A sphere is lowered into water and its size is such that

it touches the sides of the vessel and it is just immersed.

184. What is the radius of the sphere?

[CDS 2022 (II)]

- (a) 5cm (b) 6cm
(c) 6.5cm (d) 7cm

185. How much water will remain in the vessel after the overflow?

[CDS 2022 (II)]

- (a) $288\pi\text{mL}$ (b) $360\pi\text{mL}$
(c) $480\pi\text{mL}$ (d) $500\pi\text{mL}$

186. What is the ratio of lateral surface area of the vessel to the surface area of the sphere?

[CDS 2022 (II)]

- (a) $\frac{4}{3}$ (b) $\frac{3}{2}$
(c) $\frac{5}{3}$ (d) 2

187. The diameter of a sphere made of copper is 3cm. The sphere is melted and recast into a wire. If the length of the wire is half metre, then what is the diameter of the wire?

[CDS 2022 (II)]

- (a) 0.3cm (b) .045cm
(c) 0.6cm (d) 0.75cm

188. The volume of a cuboid is 3600 cubic cm. The area of two adjacent faces are 225 square cm and 144 square cm. What is the area of the other adjacent face?

[CDS – 2023 (1)]

- (a) 400 square cm (b) 360 square cm
(c) 320 square cm (d) 300 square cm

189. Two times the total surface area of a solid right circular cylinder is three times its curved surface area. If h is the height and r is the radius of the base of the cylinder, then which one of the following is correct?

[CDS – 2023 (1)]

- (a) $h=r$ (b) $h=2r$
(c) $2h=3r$ (d) $3h=4r$

190. A solid iron ball is melted and 64 smaller solid balls of equal size are made using the entire volume of iron. What is the ratio of the surface area of the larger ball to the sum of the surface area of all the smaller balls?

[CDS – 2023 (1)]

- (a) 0.25 (b) 0.5
(c) 0.75 (d) 1

191. The surface area of a cube is increased by 25%. If p is the percentage increase in its length, then which one of the following is correct?

[CDS – 2023 (1)]

- (a) $16 < p < 18$ (b) $14 < p < 16$
(c) $12 < p < 14$ (d) $10 < p < 12$

192. A solid cube is cut into two cuboids of equal volume. What is the ratio of total surface area of the given cube to that of one the cuboids?

[CDS – 2023 (1)]

- (a) 2:1 (b) 3:2
(c) 4:3 (d) 5:3

193. The length of a diagonal of a cuboid is 11 cm. The surface area is 240 square cm. What is the sum of its length, breadth and height?

[CDS – 2023 (1)]

- (a) 16cm (b) 17cm
(c) 18cm (d) 19cm

194. Three solid led spheres of diameters 6cm, 8cm and 10cm are melted together and recast as a solid sphere. What is the percentage diminution of the surface area as compared to the sum of the surface area of the three sphere?

[CDS-2023 (2)]

- (a) 25% (b) 26%
(c) 27% (d) 28%

195. A solid sphere of radius 3cm is melted to form a hollow cylinder of height 4cm and external diameter 10cm. What is the thickness of the cylinder?

[CDS-2023 (2)]

- (a) 0.42 cm (b) 0.46cm
(c) 0.5cm (d) 1.00cm

196. A sphere of radius 5cm is dropped in a right circular cylindrical vessel partly filled with water. The radius of the cylindrical vessel is 10cm. If the sphere is completely submerged in water, by how much will the level of water, rise in the cylindrical vessel?

[CDS-2023 (2)]

- (a) $\frac{5}{3}\text{cm}$ (b) $\frac{5}{2}\text{cm}$
(c) 1cm (d) $\frac{5}{6}\text{cm}$

197. What is the radius of the sphere passing through the corners of the cuboid with edges 8cm, 12cm and 24cm?

[CDS-2023 (2)]

- (a) 10.5cm (b) 14cm
(c) 21cm (d) 28cm

198. A lamp shade is in the shape of a part of a cone and its top and bottom ends are circles whose circumferences are respectively 30cm and 40cm. The perpendicular distance between the ends is 6 cm. If the cone were to be completed, then how far would its vertex be from the top end?

[CDS-2023 (2)]

- (a) 20cm (b) 18cm
(c) 12cm (d) 9cm

Consider the following for the next two (02) items that follow:

A right conical cap just covers two spheres placed one above the other on a table such that it touches both the spheres. Let r be the radius of the smaller sphere and R be the radius of the bigger sphere. Let 2θ be the vertical angle of the cone.

199. What is the height of the cone?

[CDS-2023 (2)]

- (a) $\frac{2r^2}{R-r}$ (b) $\frac{2R^2}{R-r}$
(c) $\frac{2(r^2+R^2)}{R-r}$ (d) $\frac{r^2+R^2}{R-r}$

200. What is the radius of the base of the cone?

[CDS-2023 (2)]

- (a) $\frac{2r^2 \tan \theta}{R-r}$ (b) $\frac{2R^2 \tan \theta}{R-r}$
(c) $\frac{2(r^2+R^2) \tan \theta}{R-r}$ (d) $\frac{(r^2+R^2) \tan \theta}{R-r}$

201. The length, breadth and height of a cuboid are increased by 10%, 20% and 50% respectively. What is the percentage increase in volume of the cuboid?

[CDS-2024 (1)]

- (a) 100% (b) 99%
(c) 98% (d) 50%

202. A cube whose edge is 14 cm long has on each of its faces a circle of 7 cm radius painted yellow. What is the total area of unpainted surface? (Take $\pi = \frac{22}{7}$)

[CDS-2024 (1)]

- (a) 126 square cm (b) 189 square cm
(c) 252 square cm (d) 315 square cm

203. From a circular metal plate of radius 7 cm and thickness 0.16 mm, a sector is cut off containing an angle 150° . The remaining piece is molded into a spherical bead of radius r . What is the value of r in cm?

[CDS-2024 (1)]

- (a) 0.35 (b) 0.7
(c) 1.05 (d) 1.4

ANSWER KEY

1.	d	2.	b	3.	a	4.	b	5.	d	6.	b	7.	a	8.	d	9.	a	10.	a
11.	c	12.	c	13.	b	14.	a	15.	d	16.	a	17.	c	18.	b	19.	c	20.	d
21.	c	22.	a	23.	a	24.	b	25.	d	26.	d	27.	a	28.	b	29.	c	30.	c
31.	a	32.	c	33.	d	34.	c	35.	d	36.	d	37.	c	38.	a	39.	b	40.	b
41.	d	42.	b	43.	a	44.	b	45.	c	46.	a	47.	a	48.	a	49.	b	50.	d
51.	c	52.	c	53.	c	54.	a	55.	b	56.	c	57.	d	58.	c	59.	a	60.	c
61.	c	62.	c	63.	b	64.	b	65.	a	66.	c	67.	a	68.	b	69.	c	70.	d
71.	b	72.	c	73.	b	74.	c	75.	a	76.	c	77.	c	78.	a	79.	c	80.	d
81.	a	82.	b	83.	a	84.	d	85.	d	86.	b	87.	c	88.	c	89.	a	90.	d
91.	c	92.	d	93.	d	94.	b	95.	b	96.	b	97.	a	98.	d	99.	b	100.	c
101.	c	102.	b	103.	d	104.	c	105.	d	106.	b	107.	b	108.	c	109.	c	110.	b
111.	d	112.	d	113.	c	114.	a	115.	c	116.	d	117.	a	118.	c	119.	a	120.	b
121.	b	122.	c	123.	a	124.	b	125.	d	126.	b	127.	b	128.	a	129.	c	130.	b
131.	d	132.	c	133.	d	134.	d	135.	a	136.	a	137.	d	138.	d	139.	c	140.	a
141.	c	142.	b	143.	d	144.	c	145.	b	146.	a	147.	b	148.	b	149.	c	150.	c
151.	b	152.	c	153.	c	154.	c	155.	a	156.	d	157.	c	158.	d	159.	a	160.	c
161.	a	162.	a	163.	b	164.	a	165.	a	166.	a	167.	c	168.	d	169.	c	170.	d
171.	b	172.	b	173.	d	174.	b	175.	b	176.	c	177.	c	178.	c	179.	b	180.	d
181.	c	182.	b	183.	a	184.	b	185.	c	186.	c	187.	c	188.	a	189.	b	190.	a
191.	b	192.	b	193.	d	194.	d	195.	d	196.	a	197.	b	198.	a	199.	b	200.	b
201.	c	202.	c	203.	b														

CAPF PYQ

1. A solid spherical ball made of iron is melted and two new balls are made whose diameters are in the ratio of 1:2. The ratio of the volume of the smaller new ball to the original ball is:

- (a) 1:3
(c) 2:9

- [CAPF 2022]
(b) 1:5
(d) 1:9

ANSWER KEY

1.	d																		
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PRACTICE SHEET-1

1. In circular measure, the value of the angle $11^\circ 15'$ is
 (1) $\frac{\pi}{16}$ (2) $\frac{\pi}{8}$
 (3) $\frac{\pi}{4}$ (4) $\frac{\pi}{32}$
2. In a triangle ABC, $\angle ABC = 75^\circ$ and $\angle ACB = \frac{\pi}{4}$. The circular measure of $\angle BAC$ is
 (1) $\frac{5\pi}{12}$ radian (2) $\frac{\pi}{3}$ radian
 (3) $\frac{\pi}{6}$ radian (4) $\frac{\pi}{2}$ radian
3. The circular measure of an angle of an isosceles triangle is $\frac{5\pi}{9}$. Circular measure of one of the other angles must be:
 (1) $\frac{5\pi}{18}$ (2) $\frac{5\pi}{9}$
 (3) $\frac{2\pi}{9}$ (4) $\frac{4\pi}{9}$
4. The degree measure of 1 radian (taking $\pi = 22/7$) is
 (1) $57^\circ 61' 22''$ (approx.)
 (2) $57^\circ 16' 22''$ (approx.)
 (3) $57^\circ 22' 16''$ (approx.)
 (4) $57^\circ 32' 16''$ (approx.)
5. Consider the following statements
 I. 1° in radian measure is less than 0.03 radians.
 II. 1 radian in degree measure is greater than 45°
 Which of the above statements is/are correct?
 (a) Only II (b) Only I
 (c) Both I and II (d) Neither 1 nor 2
6. If the sum of two angles is 135° and their difference is $\frac{\pi}{12}$, then the circular measure of the greater angle is
 (1) $\frac{2\pi}{3}$ (2) $\frac{3\pi}{5}$
 (3) $\frac{5\pi}{12}$ (4) $\frac{\pi}{3}$
7. If the sum and difference of two angles are $22/9$ radian and 36° respectively, then the value of smaller angle in degree taking the value of π as $22/7$ is:
 (1) 52° (2) 60°
 (3) 56° (4) 48°
8. The circular measure of the included angle formed by the hour hand and minute hand of a clock at 3 p.m. will be:
 (1) $\frac{\pi}{4}$ (2) $\frac{\pi}{3}$
 (3) $\frac{5\pi}{12}$ (4) $\frac{\pi}{2}$
9. What is the measure of central angle of the arc whose length is 11 cm and radius of the circle is 14 cm?
 (1) 45° (2) 60°
 (3) 75° (4) 90°
10. The radian measure of 1 degree (taking $\pi = 22/7$) is
 (1) 0.016° (2) 0.017°
 (3) 0.018° (4) 0.019°
11. What is the angle between the hands of clock when time is 7:20pm?
 (1) 100° (2) 120°
 (3) 108° (4) 90°
12. What is the angle between the hands of clock when time is 1:40pm?
 (1) 150° (2) 120°
 (3) 120° (4) 170°
13. What is the angle between the hands of clock when time is 3:30am?
 (1) 90° (2) 80°
 (3) 75° (4) 60°
14. At what time between 2 and 3 o'clock will the hands of a clock be together?
 (1) $10\frac{10}{11}$ min, past 2
 (2) 10 min, past 2
 (3) $20\frac{10}{11}$ min, past 2
 (4) 12min, past 2

15. How many times do the hands of a clock coincide in a day?
 (1) 20 (2) 21
 (3) 22 (4) 24
16. In circular measure, the value of the angle $22^{\circ}30'$ is
 (1) $\frac{\pi}{16}$ (2) $\frac{\pi}{8}$
 (3) $\frac{\pi}{4}$ (4) $\frac{\pi}{32}$
17. In circular measure, the value of the angle $9^{\circ}22'30''$ is
 (1) $\frac{7\pi}{96}$ (2) $\frac{5\pi}{96}$
 (3) $\frac{11\pi}{48}$ (4) $\frac{5\pi}{48}$
18. In degree measure, the value of the angle $\frac{\pi}{16}$ is
 (1) $22^{\circ}30'$ (2) $11^{\circ}15'$
 (3) $30^{\circ}30'$ (4) $15^{\circ}30'$
19. In degree measure, the value of the angle $\frac{2\pi}{5}$ is
 (1) 72° (2) 45°
 (3) 36° (4) 20°
20. In degree measure, the value of the angle $\frac{3\pi}{8}$ is
 (1) $72^{\circ}30'$ (2) 45°
 (3) $67^{\circ}30'$ (4) 20°
21. Between 6 and 7 O' clock, both hour hand and minute hand will coincide past 6 O'clock between
 (1) 31-32 minutes (2) 32-33 minutes
 (3) 33-34 minutes (4) 34-35 minutes
22. A clock is started at noon. By 10 minutes past 5, the hour hand has turned through:
 (1) 145° (2) 150°
 (3) 155° (4) 160°
23. How many times are the hands of clock at right angle in a day?
 (1) 22 (2) 24
 (3) 44 (4) 48
24. A clock gains 20 seconds for every 3 hours of time. If a clock is set at a correct time of 2 am on Friday, what would it indicate at 6:30 pm, Saturday?
 (1) 6.32.00pm (2) 6.32.46pm
 (3) 6.34.30pm (4) 6.38.56 pm
25. A person goes to a market between 7pm and 8pm. When he comes back, he finds that the hour hand and the minute hand of the clock have interchanged their positions. For how much time (approximately) was he out of his house?
 (a) 55.38 minutes (b) 55.48 minutes
 (c) 55.57 minutes (d) 55.67 minutes
26. Find the time between 8am and 9am when the hours and minutes hand would form a 90 degree between themselves?
 (1) $25\frac{7}{11}$ minutes past 8 am
 (2) $27\frac{3}{11}$ minutes past 8 am
 (3) $29\frac{8}{11}$ minutes past 8 am
 (4) $24\frac{5}{11}$ minutes past 8 am
27. A clock strikes once 1 O'clock, twice at 2 O'clock and thrice at 3 O'clock and so on. If it takes 8 seconds to strike at 5 O'clock, the time taken by it to strike at 10 O'clock is:
 (a) 14 seconds
 (b) 16 seconds
 (c) 18 seconds
 (d) None of the above
28. The degree measure of 3 radian (taking $\pi = \frac{22}{7}$) is
 (1) $171^{\circ}47'66''$ (approx.)
 (2) $171^{\circ}48'06''$ (approx.)
 (3) $171^{\circ}49'66''$ (approx.)
 (4) $171^{\circ}49'06''$ (approx.)
29. Consider the following:
 1. $57^{\circ} > 1^{\circ}$ 2. $1^{\circ} < 0.02^{\circ}$ 3. $45^{\circ} < 1^{\circ}$
 Which of the above are correct?
 (a) 1 and 2 only (b) 2 and 3 only
 (c) 1 and 3 only (d) 1, 2 and 3
30. In circular measure, the value of the angle $5^{\circ}37'30''$ is
 (1) $\frac{\pi}{16}$ (2) $\frac{\pi}{8}$
 (3) $\frac{\pi}{4}$ (4) $\frac{\pi}{32}$

ANSWER KEY

1.	(1)	2.	(2)	3.	(3)	4.	(2)	5.	(3)	6.	(3)	7.	(1)	8.	(4)	9.	(1)	10.	(2)
11.	(1)	12.	(4)	13.	(3)	14.	(1)	15.	(3)	16.	(2)	17.	(2)	18.	(2)	19.	(1)	20.	(3)
21.	(2)	22.	(3)	23.	(3)	24.	(3)	25.	(a)	26.	(2)	27.	(3)	28.	(4)	29.	(2)	30.	(4)

PRACTICE SHEET-2

- If $3\tan\theta = 4$, then $\sqrt{\frac{1-\sin\theta}{1+\sin\theta}}$ is equal to
 (1) $1/2$ (2) $2/3$
 (3) $1/3$ (4) none of these
- What is the value of $\log(\tan 1^\circ) + \log(\tan 2^\circ) + \log(\tan 3^\circ) + \dots + \log(\tan 89^\circ)$ equal to
 (1) 0 (2) 1
 (3) 2 (4) -1
- If $\sec\theta = \frac{13}{5}$, then $\frac{2\sin\theta - 3\cos\theta}{4\sin\theta - 9\cos\theta}$ is equal to
 (1) $1/3$ (2) -3
 (3) 3 (4) $\sqrt{3}$
- If $\sin\theta = \frac{a^2 - b^2}{a^2 + b^2}$, then $\tan\theta$ is equal to
 (1) $\frac{a^2 + b^2}{2ab}$ (2) $\frac{2ab}{a^2 - b^2}$
 (3) $\frac{a^2 - b^2}{2ab}$ (4) $\frac{ab}{a^2 + b^2}$
- If $\cos\theta = 0.96$, then $\frac{1}{\sin\theta} + \frac{1}{\tan\theta}$ is equal to?
 (1) 0.98 (2) 3
 (3) 4 (4) 7
- If $0 < x < 45^\circ$ and $45^\circ < y < 90^\circ$ then which one of the following is correct?
 (1) $\sin x = \sin y$ (2) $\sin x < \sin y$
 (3) $\sin x > \sin y$ (4) $\sin x \leq \sin y$
- What is the value of $[(1 - \sin^2\theta) \sec^2\theta + \tan^2\theta](\cos^2\theta + 1)$ when $0 < \theta < 90^\circ$?
 (1) 2 (2) > 2
 (3) ≥ 2 (4) < 2
- If $7\cos^2\theta + 3\sin^2\theta = 4$, $0 < \theta < \frac{\pi}{2}$, then what is the value of $\tan\theta$?
 (1) $\sqrt{7}$ (2) $7/3$
 (3) 3 (4) $\sqrt{3}$
- If $0 \leq \theta \leq \frac{\pi}{2}$ and $p = \sec^2\theta$, then which one of the following is correct?
 (1) $p < 1$ (2) $p = 1$
 (3) $p > 1$ (4) $p \geq 1$
- What is the value of $\cos 1^\circ \cos 2^\circ \cos 3^\circ \dots \cos 90^\circ$?
 (1) $1/2$ (2) 0
 (3) 1 (4) 2
- What is $\cot 15^\circ \cot 20^\circ \cot 70^\circ \cot 75^\circ$ equal to?
 (1) -1 (2) 0
 (3) 1 (4) 2
- If $\frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta} = 2$, with $0 < \theta < 90^\circ$, then what is θ equal to?
 (1) 30° (2) 45°
 (3) 60° (4) 75°
- The smallest side of a right angled triangle has length 2 cm. The tangent of one acute angle is $3/4$. What is the hypotenuse of the triangle?
 (1) 5 cm (2) 2.5 cm
 (3) 1.25 cm (4) $10/3$ cm
- Which one of the following statements is true in respect of the expression $\sin 31^\circ + \sin 32^\circ$?
 (1) Its value is 0
 (2) Its value is 1
 (3) Its value is less than 1
 (4) Its value is greater than 1.
- If $\alpha + \beta = \frac{\pi}{2}$ and $\sin\alpha = \frac{1}{2}$, then β is
 (1) 30° (2) 60°
 (3) 45° (4) $22\frac{1}{2}^\circ$

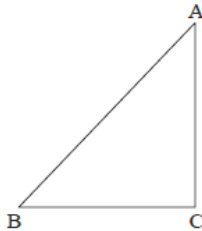
16. Consider the following statements
I. 1° in radian measure is less than 0.03 radians.
II. 1 radian in degree measure is greater than 45°
Which of the above statements is/are correct?
(1) Only II (2) Only I
(3) Both I and II (4) Neither 1 nor 2
- Direction (for next three)**
- Let $\sin(A + B) = 1$ and $\sin(A - B) = \frac{1}{2}$,
where $A, B = \frac{1}{2}$, where $A, B \in \left[0, \frac{\pi}{2}\right]$
17. What is the value of A?
(1) 30° (2) 60°
(3) 45° (4) $22\frac{1}{2}^\circ$
18. What is the value of $\tan(A + 2B) \tan(2A + B)$?
(1) -1 (2) 0
(3) 1 (4) 2
19. What is the value of $\sin^2 A - \sin^2 B$?
(1) 0 (2) $1/2$
(3) 1 (4) 2
20. The value of $\cos 25^\circ - \sin 25^\circ$ is
(1) Positive but less than 1
(2) Positive but greater than 1
(3) Negative
(4) 0
21. If $\tan \theta = 2$, then the value of $\frac{\sin \theta}{\sin^3 \theta + \cos^3 \theta}$ is
(1) $2/9$ (2) $5/9$
(3) $10/9$ (4) $50/9$
22. If $\sin \theta + \cos \theta = 1$, then $\sin \theta \cdot \cos \theta$ is equal to
(1) 0 (2) 1
(3) $1/2$ (4) $-1/2$
23. If $\frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta} = 3$, then the value of $\sin^4 \theta - \cos^4 \theta$ is
(1) $4/3$ (2) $3/4$
(3) $5/3$ (4) $3/5$
24. The value of $\sec^2 17^\circ - \frac{1}{\tan^2 73^\circ} - \sin 17^\circ \sec 73^\circ$ is
(1) 1 (2) 0
(3) -1 (4) 2
25. If $x = a \cos \theta \cos \alpha$, $y = a \cos \theta \sin \alpha$ and $z = a \sin \theta$, then the value of $(x^2 + y^2 + z^2)$ is
(1) $2a^2$ (2) $4a^2$
(3) $9a^2$ (4) a^2
26. Which one of the following is correct?
(a) There is only one θ with $0^\circ < \theta < 90^\circ$ such that $\sin \theta = a$, where a is a real number.
(b) There is more than one θ with $0^\circ < \theta < 90^\circ$ such that $\sin \theta = a$, where a is a real number.
(c) There is no θ with $0^\circ < \theta < 90^\circ$ such that $\sin \theta = a$, where a is a real number.
(d) There are exactly θ 's with $0^\circ < \theta < 90^\circ$ such that $\sin \theta = a$, where a is a real number
27. If $\tan \theta = \tan 30^\circ \cdot \tan 60^\circ$ and θ is an acute angle, then 2θ is equal to
(1) 30° (2) 45°
(3) 90° (4) 0°
28. The value of $(1 + \tan^2 \theta)(1 - \sin^2 \theta)$ is
(1) 2 (2) 1
(3) -1 (4) -2
29. If $r \sin \theta = 1$, $r \cos \theta = \sqrt{3}$ then the value of $(r^2 \tan \theta)$ is
(1) 4 (2) $1/\sqrt{3}$
(3) $4/\sqrt{3}$ (4) $4\sqrt{3}$
30. If $\sin \theta = \sqrt{3}/2$ and $0^\circ < \theta < 90^\circ$, then the value of $\tan(\theta - 15^\circ)$ is
(1) 1 (2) $\sqrt{3}$
(3) $1/\sqrt{3}$ (4) $\sqrt{2}$

ANSWER KEY

1.	(3)	2.	(1)	3.	(3)	4.	(3)	5.	(4)	6.	(2)	7.	(2)	8.	(4)	9.	(4)	10.	(2)
11.	(3)	12.	(2)	13.	(4)	14.	(4)	15.	(2)	16.	(3)	17.	(2)	18.	(3)	19.	(2)	20.	(1)
21.	(3)	22.	(1)	23.	(4)	24.	(2)	25.	(4)	26.	(1)	27.	(3)	28.	(2)	29.	(3)	30.	(1)

PRACTICE SHEET-3

1. What is $\frac{\sin^6 \theta - \cos^6 \theta}{\sin^2 \theta - \cos^2 \theta}$ equal to?
 (1) $\sin^4 \theta - \cos^4 \theta$ (2) $1 - \sin^2 \theta \cos^2 \theta$
 (3) $1 + \sin^2 \theta \cos^2 \theta$ (4) $1 - 3\sin^2 \theta \cos^2 \theta$
 2. If $\sin \theta + \cos \theta = 1$, then what is the value of $\sin \theta \cos \theta$?
 (1) 2 (2) 0
 (3) 1 (4) $1/2$
 3. What is $\sqrt{\frac{1+\sin \theta}{1-\sin \theta}}$ equal to?
 (1) $\sec \theta - \tan \theta$ (2) $\sec \theta + \tan \theta$
 (3) $\operatorname{cosec} \theta + \cot \theta$ (4) $\operatorname{cosec} \theta - \cot \theta$
 4. $\sin^2 \theta - 3 \sin \theta + 2 = 0$ will be true if
 (1) $0 \leq \theta < 90$ (2) $0 < \theta < 90$
 (3) $\theta = 0^\circ$ (4) $\theta = 90^\circ$
 5. In a $\triangle ABC$, $\angle ABC = 90^\circ$, $\angle ACB = 30^\circ$, $AB = 5$ cm. What is the length of AC ?
 (1) 10 cm (2) 5 cm
 (3) $5\sqrt{2}$ cm (4) $5\sqrt{3}$ cm
 6. If $7 \cos^2 \theta + 3 \sin^2 \theta = 4$ and $0 < \theta < 90^\circ$, then what is the value of $\tan \theta$?
 (a) $\sqrt{7}$ (b) $7/3$
 (c) 3 (d) $\sqrt{3}$
 7. For what value of θ is $(\sin \theta + \operatorname{cosec} \theta) = 2.5$, where $0 < \theta < 90^\circ$?
 (1) 30° (2) 45°
 (3) 60° (4) 90°
 8. $\frac{\cos \theta}{1-\sin \theta} - \frac{\cos \theta}{1+\sin \theta} = 2$, is satisfied by which of the following value of θ ?
 (1) $\frac{\pi}{2}$ (2) $\frac{\pi}{3}$
 (3) $\frac{\pi}{4}$ (4) $\frac{\pi}{6}$
 9. Consider the following:
 1. $\frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta + \sin^2 \theta} = \cos^2 \theta (1 + \tan \theta)(1 - \tan \theta)$
 2. $\frac{1 + \sin \theta}{1 - \sin \theta} = (\tan \theta + \sec \theta)^2$
- Which of the above are correct?
 (1) only 1 (2) only 2
 (3) Both 1 and 2 (4) Neither 1 nor 2
10. If x is the angle of first quadrant such that $\operatorname{cosec}^4 x = 17 + \cot^4 x$, then what is the value of $\sin x$?
 (1) $1/3$ (2) $1/4$
 (3) $1/9$ (4) $1/16$
 11. Consider the following equations:
 I. $\operatorname{cosec}^2 x + \sec^2 x = \operatorname{cosec}^2 x \sec^2 x$
 II. $\sec^2 x + \tan^2 x = \sec^2 x \tan^2 x$
 III. $\operatorname{cosec}^2 x + \tan^2 x = \cot^2 x + \sec^2 x$
 Which of the above statements are correct?
 (1) I and II (2) II and III
 (3) I and III (4) All of these
 12. If the given figure, $BC = 15$ cm and $\sin B = 4/5$. What is the value of AB ?



 (1) 25 cm (2) 20 cm
 (3) 5 cm (4) 4 cm
 13. If $\sin^2 60^\circ + \cos^2 (3x - 9^\circ) = 1$, then what is the value of x ?
 (1) 24° (2) 23°
 (3) 22° (4) 21°
 14. What is the expression: $(\sin^4 x - \cos^4 x + 1) \operatorname{cosec}^2 x$ equal to?
 (1) 1 (2) 2
 (3) 0 (4) -1
 15. Assertion (A): $\tan 50^\circ > 1$.
 Reason (R): $\tan \theta > 1$ for $0^\circ < \theta < 90^\circ$.
 (1) A and R are correct and R is correct explanation of A.
 (2) A and R are correct but R is not correct explanation of A.
 (3) A is true but R is false.
 (4) A is true but R is false.

16. Which one of the following is correct?
 (1) $\sin 35^\circ > \cos 55^\circ$ (2) $\cos 61^\circ > 1/2$
 (3) $\sin 32^\circ > 1/2$ (4) $\tan 44^\circ > 1$
17. Which of the following expression for $0^\circ < \theta < 90^\circ$ is/are independent of θ ?
 I. $\cos\theta(1 - \sin\theta)^{-1} + \cos\theta(1 + \sin\theta)^{-1}$
 II. $\cos\theta(1 + \operatorname{cosec}\theta)^{-1} + \cos\theta(\operatorname{cosec}\theta - 1)^{-1}$
 Select the correct answer using the codes given below:
 (1) Only I (2) Only II
 (3) Both I and II (4) Neither I nor II
18. If $p = \sin^{10} x$, then which one of the following is correct for any value x ?
 (1) $p \geq 1$ (2) $0 \leq p \leq 1$
 (3) $1 \leq p \leq 2$ (4) None of these
19. If $\sin(x - y) = 1/2$ and $\cos(x + y) = 1/2$, then what is the value of x ?
 (1) 15° (2) 30°
 (3) 45° (4) 60°
20. If $2x^2 \cos 60^\circ - 4 \cot^2 45^\circ - 2 \tan 60^\circ = 0$, then what is the value of x ?
 (1) 2 (2) 3
 (3) $\sqrt{3} - 1$ (4) $\sqrt{3} + 1$
21. Under which one of the following conditions is the trigonometrically identity $\frac{\sin x}{(1 + \cos x)} = \frac{(1 - \cos x)}{\sin x}$ true?
 (1) x is not a multiple of 360°
 (2) x is not an odd multiple of 180°
 (3) x is not a multiple of 180°
 (4) None of the above
22. If A, B, C and D are the successive angles of a cyclic quadrilateral, then what is $\cos A + \cos B + \cos C + \cos D$ are equal to?
 (1) 4 (2) 2
 (3) 1 (4) 0
23. If $x = \operatorname{cosec} \theta - \sin \theta$ and $y = \sec \theta - \cos \theta$, then the relation between x and y is
 (1) $x^2 + y^2 + 3 = 1$
 (2) $x^2 y^2 (x^2 + y^2 + 3) = 1$
 (3) $x^2 (x^2 + y^2 - 5) = 1$
 (4) $y^2 (x^2 + y^2 - 5) = 1$
24. If $2y \cos \theta = x \sin \theta$ and $2x \sec \theta - y \operatorname{cosec} \theta = 3$ then what is the value of $(x^2 + 4y^2)$?
 (1) 4 (2) 1
 (3) 2 (4) 5
25. If $\cos \theta + \sin \theta = m$ and $\sec \theta + \operatorname{cosec} \theta = n$ then the value of $n(m^2 - 1)$ is equal to:
 (1) $2m$ (2) mn
 (3) $4mn$ (4) $2n$
26. If x is an acute angle and $2\sin x + 15\cos^2 x = 7$ then the value of $\cot x$ is
 (1) $4/3$ (2) $4/5$
 (3) $5/4$ (4) $3/4$
27. If $0^\circ < A < 90^\circ$, then the value of $\tan^2 A + \cot^2 A - \sec^2 A \operatorname{cosec}^2 A$ is
 (1) 0 (2) 1
 (3) 2 (4) -2
28. If $\sec \theta + \tan \theta = 2$, then the value of $\sin \theta$ is (assume that $0 < \theta < 90^\circ$)
 (1) 0.4 (2) 0.5
 (3) 0.6 (4) 0.8
29. The value of $(1 + \cot \theta - \operatorname{cosec} \theta)(1 + \tan \theta + \sec \theta)$ is equal to
 (1) 1 (2) 2
 (3) 0 (4) -1
30. $\frac{\tan \theta}{1 - \cot \theta} + \frac{\cot \theta}{1 + \tan \theta} =$
 (1) $1 - \tan \theta - \cot \theta$ (2) $1 + \tan \theta - \cot \theta$
 (3) $1 - \tan \theta + \cot \theta$ (4) $1 + \tan \theta + \cot \theta$

ANSWER KEY

1.	(2)	2.	(2)	3.	(2)	4.	(4)	5.	(1)	6.	(4)	7.	(1)	8.	(3)	9.	(3)	10.	(1)
11.	(3)	12.	(1)	13.	(2)	14.	(2)	15.	(3)	16.	(3)	17.	(4)	18.	(2)	19.	(2)	20.	(4)
21.	(3)	22.	(2)	23.	(2)	24.	(1)	25.	(1)	26.	(4)	27.	(4)	28.	(3)	29.	(2)	30.	(2)

PRACTICE SHEET-4

1. The value of $\sin^2 1^\circ + \sin^2 5^\circ + \sin^2 9^\circ + \dots + \sin^2 89^\circ$ is
 (1) $11\frac{1}{2}$ (2) $11\sqrt{2}$
 (3) 11 (4) $11/\sqrt{2}$
2. The numerical value of $\cot 18^\circ \left(\cot 72^\circ \cos^2 22^\circ + \frac{1}{\tan 72^\circ \sec^2 68^\circ} \right)$ is
 (1) 1 (2) $\sqrt{2}$
 (3) 3 (4) $1/\sqrt{3}$
3. If $\angle A$ and $\angle B$ are complementary to each other, then the value of $\sec^2 A + \sec^2 B - \sec^2 A \cdot \sec^2 B$ is
 (1) 1 (2) -1
 (3) 2 (4) 0
4. $\sin^2 5^\circ + \sin^2 6^\circ + \dots + \sin^2 84^\circ + \sin^2 85^\circ = ?$
 (1) $39\frac{1}{2}$ (2) $40\frac{1}{2}$
 (3) 40 (4) $39\frac{1}{\sqrt{2}}$
5. The value of $\frac{\sin 39^\circ}{\cos 51^\circ} + 2 \tan 11^\circ \tan 31^\circ \tan 45^\circ + \tan 59^\circ \tan 79^\circ - 3(\sin^2 21^\circ + \sin^2 69^\circ)$ is:
 (1) 2 (2) -1
 (3) 1 (4) 0
6. If $\frac{\cos^2 \theta}{\cot^2 \theta - \cos^2 \theta} = 3$ and $0^\circ < \theta < 90^\circ$, then the value of θ is
 (1) 30° (2) 45°
 (3) 60° (4) None of these
7. If $A = \tan 11^\circ \tan 29^\circ$, $B = 2 \cot 61^\circ \cot 79^\circ$, then:
 (1) $A = 2B$ (2) $A = -2B$
 (3) $2A = B$ (4) $2A = -B$
8. If $0^\circ < \theta < 90^\circ$, the value of $\sin \theta + \cos \theta$ is
 (1) equal to 1 (2) greater than 1
 (3) less than 1 (4) equal to 2
9. The value of $\cot \theta \cdot \tan(90^\circ - \theta) - \sec(90^\circ - \theta) \operatorname{cosec} \theta + (\sin^2 25^\circ + \sin^2 65^\circ) + \sqrt{3}(\tan 5^\circ \tan 15^\circ \cdot \tan 30^\circ \cdot \tan 75^\circ \cdot \tan 85^\circ)$ is :
 (1) 1 (2) -1
 (3) 2 (4) 0
10. If $\tan\left(\frac{\pi}{2} - \frac{\theta}{2}\right) = \sqrt{3}$, then the value of $\cos \theta$ is
 (1) 0 (2) $1/\sqrt{2}$
 (3) $1/2$ (4) 1
11. If $\sec \theta + \tan \theta = \sqrt{3}$, then $\tan 3\theta$ is
 (1) not defined (2) $1/\sqrt{3}$
 (3) $1/\sqrt{2}$ (4) $\sqrt{3}$
12. If $\sin(60^\circ - \theta) = \cos(\psi - 30^\circ)$, then the value of $\tan(\psi - \theta)$ is (assume that θ and ψ are both positive acute angles with $\psi < 60^\circ$ and $\psi > 30^\circ$).
 (1) $1/\sqrt{3}$ (2) 0
 (3) $\sqrt{3}$ (4) 1
13. If $a \sin \theta + b \cos \theta = c$ then the value of $a \cos \theta - b \sin \theta$ is :
 (1) $\pm \sqrt{-a^2 + b^2 + c^2}$ (2) $\pm \sqrt{a^2 + b^2 - c^2}$
 (3) $\pm \sqrt{a^2 - b^2 - c^2}$ (4) $\pm \sqrt{a^2 - b^2 + c^2}$
14. If $\sin(A - B) = \frac{1}{2}$ and $\cos(A + B) = \frac{1}{2}$ where $A > B > 0$ and $A + B$ is an acute angle, then the value B is
 (1) $\frac{\pi}{6}$ (2) $\frac{\pi}{12}$
 (3) $\frac{\pi}{4}$ (4) $\frac{\pi}{2}$
15. $0 \leq \theta \leq 90^\circ$ and $\cos \theta + \sqrt{3} \sin \theta = 2$, then what is the value of θ ?
 (1) $\frac{\pi}{3}$ (2) $\frac{\pi}{4}$
 (3) $\frac{\pi}{6}$ (4) $\frac{\pi}{2}$
16. If $\tan \alpha = n \tan \beta$ and $\sin \alpha = m \sin \beta$, then $\cos^2 \alpha$ is
 (1) $\frac{m^2}{n^2 + 1}$ (2) $\frac{m^2}{n^2}$

- (3) $\frac{m^2-1}{n^2-1}$ (4) $\frac{m^2+1}{n^2+1}$
17. The equation $\cos^2 \theta = \frac{(x+y)^2}{4xy}$ is only possible when
 (1) $x = -y$ (2) $x > y$
 (3) $x = y$ (4) $x < y$
18. If $\sin \alpha + \cos \beta = 2$ ($0^\circ \leq \beta \leq \alpha \leq 90^\circ$), then $\sin\left(\frac{2\alpha+\beta}{3}\right) =$
 (1) $\sin \frac{\alpha}{2}$ (2) $\cos \frac{\alpha}{3}$
 (3) $\sin \frac{\alpha}{3}$ (4) $\cos \frac{2\alpha}{3}$
19. If $\sin \alpha \sec(30^\circ + \alpha) = 1$ ($0 < \alpha < 60^\circ$), then the value of $\sin \alpha + \cos 2\alpha$ is
 (1) 1 (2) $\frac{2+\sqrt{3}}{2\sqrt{3}}$
 (3) 0 (4) $\sqrt{2}$
20. If $\tan \theta = 1$, then the value of $\frac{8\sin \theta + 5\cos \theta}{\sin^3 \theta - 2\cos^3 \theta + 7\cos \theta}$ is
 (1) 2 (2) $2\frac{1}{2}$
 (3) 3 (4) 0.8
21. What is the value of $\sin 390^\circ$
 (1) $1/2$ (2) $-\sqrt{3}/2$
 (3) $-1/2$ (4) $\sqrt{3}/2$
22. What is the value of $\cot 585^\circ$
- (1) 1 (2) -1
 (3) $\sqrt{3}$ (4) $-\sqrt{3}$
23. What is the value of $\cos 1110^\circ$
 (1) $1/2$ (2) $-\sqrt{3}/2$
 (3) $-1/2$ (4) $\sqrt{3}/2$
24. What is the value of $\sin 930^\circ$
 (1) $1/2$ (2) $-\sqrt{3}/2$
 (3) $-1/2$ (4) $\sqrt{3}/2$
25. What is the value of $\sec 330^\circ$
 (1) 2 (2) -2
 (3) $2/\sqrt{3}$ (4) $-2/\sqrt{3}$
26. What is the value of $\cos 660^\circ$
 (1) $1/2$ (2) $\sqrt{3}/2$
 (3) $-1/2$ (4) $\sqrt{3}/2$
27. What is the value of $\sin(-60^\circ)$
 (1) $1/2$ (2) $-\sqrt{3}/2$
 (3) $-1/2$ (4) $\sqrt{3}/2$
28. What is the value of $\cos(-390^\circ)$
 (1) $1/2$ (2) $-\sqrt{3}/2$
 (3) $-1/2$ (4) $\sqrt{3}/2$
29. What is the value of $\tan(-450^\circ)$
 (1) 1 (2) -1
 (3) 0 (4) not defined
30. What is the value of $\sec(-225^\circ)$
 (1) 2 (2) -2
 (3) $\sqrt{2}$ (4) $-\sqrt{2}$

ANSWER KEY

1.	(1)	2.	(1)	3.	(4)	4.	(2)	5.	(4)	6.	(3)	7.	(3)	8.	(2)	9.	(1)	10.	(3)
11.	(1)	12.	(3)	13.	(2)	14.	(2)	15.	(1)	16.	(3)	17.	(3)	18.	(2)	19.	(1)	20.	(1)
21.	(1)	22.	(1)	23.	(4)	24.	(3)	25.	(3)	26.	(1)	27.	(2)	28.	(4)	29.	(4)	30.	(4)

PRACTICE SHEET-5

1. If $\sin A = \frac{1}{\sqrt{5}}$, $\cos B = \frac{3}{\sqrt{10}}$; where A, B being positive acute angles, then what is (A+B) equal to:
 - (1) $\frac{\pi}{6}$
 - (2) $\frac{\pi}{4}$
 - (3) $\frac{\pi}{3}$
 - (4) $\frac{\pi}{2}$
2. What is the value of $\frac{\cos 10^\circ - \sin 10^\circ}{\cos 10^\circ + \sin 10^\circ}$?
 - (1) $\tan 35^\circ$
 - (2) $\tan 10^\circ$
 - (3) $\frac{1}{\sqrt{2}}$
 - (d) 1
3. What is the value of $\sin\left(\frac{5\pi}{12}\right)$?
 - (1) $\frac{\sqrt{3}+1}{2\sqrt{2}}$
 - (2) $\frac{\sqrt{6}+\sqrt{2}}{4}$
 - (3) $\frac{\sqrt{3}+\sqrt{2}}{4}$
 - (4) $\frac{\sqrt{6}+1}{2}$
4. If $\cot \alpha$ and $\cot \beta$ are the roots of equation $x^2 - 3x + 2 = 0$, then what is $\cot(\alpha + \beta)$ equal to?
 - (1) $1/2$
 - (2) $1/3$
 - (3) 2
 - (4) 3
5. What is $\sin(\alpha + \beta) - 2\sin\alpha\cos\beta + \sin(\alpha - \beta) = ?$
 - (1) 0
 - (2) $2\sin\alpha$
 - (3) $2\sin\beta$
 - (4) $\sin\alpha + \sin\beta$
6. If $2\tan A = 3\tan B = 1$, then what is $\tan(A-B)$ equal to?
 - (1) $1/5$
 - (2) $1/6$
 - (3) $1/7$
 - (4) $1/9$
7. What is the value of $\frac{\sin 34^\circ \cos 236^\circ - \sin 56^\circ \sin 124^\circ}{\cos 28^\circ \cos 88^\circ + \cos 178^\circ \sin 208^\circ}$?
 - (1) -2
 - (2) -1
 - (3) 2
 - (4) 1
8. $\tan 54^\circ$ can be expressed as
 - (1) $\frac{\sin 9^\circ + \cos 9^\circ}{\sin 9^\circ - \cos 9^\circ}$
 - (2) $\frac{\sin 9^\circ - \cos 9^\circ}{\sin 9^\circ + \cos 9^\circ}$
 - (3) $\frac{\cos 9^\circ + \sin 9^\circ}{\cos 9^\circ - \sin 9^\circ}$
 - (4) $\frac{\sin 36^\circ}{\cos 36^\circ}$
9. What is the value of $\tan 75^\circ + \cot 75^\circ$?
 - (1) 2
 - (2) 4
 - (3) $2\sqrt{3}$
 - (4) $4\sqrt{3}$
10. What is $\frac{2\tan\theta}{1+\tan^2\theta}$ equal to?
 - (1) $\cos 2\theta$
 - (2) $\tan 2\theta$
 - (3) $\sin 2\theta$
 - (4) $\operatorname{cosec} 2\theta$
11. If $\frac{\sin(x+y)}{\sin(x-y)} = \frac{a+b}{a-b}$, then what is $\frac{\tan x}{\tan y}$ equal to?
 - (1) $\frac{a}{b}$
 - (2) $\frac{b}{a}$
 - (3) $\frac{a+b}{a-b}$
 - (4) $\frac{a-b}{a+b}$
12. What is $\sin 105^\circ + \cos 105^\circ$ equal to?
 - (1) $\sin 50^\circ$
 - (2) $\cos 50^\circ$
 - (3) $1/\sqrt{2}$
 - (4) 0
13. What is $\frac{\sin 5x - \sin 3x}{\cos 5x + \cos 3x}$ equal to?
 - (1) $\sin x$
 - (2) $\cos x$
 - (3) $\tan x$
 - (4) $\cot x$
14. If $\sin x = \frac{1}{\sqrt{5}}$, $\sin y = \frac{1}{\sqrt{10}}$, where $0 < x < \frac{\pi}{2}$, $0 < y < \frac{\pi}{2}$, then what is $(x+y)$ equal to?
 - (1) π
 - (2) $\pi/2$
 - (3) $\pi/4$
 - (4) 0
15. If $\tan(\alpha + \beta) = 2$ and $\tan(\alpha - \beta) = 1$, then $\tan(2\alpha)$ is equal to:
 - (1) -3
 - (2) -2
 - (3) $-\frac{1}{3}$
 - (d) 1
16. What is $\frac{1 - \tan 2^\circ \cot 62^\circ}{\tan 152^\circ - \cot 88^\circ}$ equal to?
 - (1) $\sqrt{3}$
 - (2) $-\sqrt{3}$
 - (3) $\sqrt{2}-1$
 - (4) $1 - \sqrt{2}$

17. Consider the following statements:
 1.If ABC is an equilateral triangle, then $3\tan(A+B)\tan C = 1$
 2.If ABC is a triangle in which $A = 78^\circ$, $B = 66^\circ$, then $\tan\left(\frac{A+C}{2}\right) < \tan A$
 3.If ABC is any triangle, then $\tan\left(\frac{A+B}{2}\right)\sin\left(\frac{C}{2}\right) < \cos\left(\frac{C}{2}\right)$
 which of the above statements is/are correct?
 (1) Only 1 (2) Only 2
 (3) 1 and 2 (4) 2 and 3
18. What is $\frac{\cos 7x - \cos 3x}{\sin 7x - 2\sin 5x + \sin 3x}$ equal to?
 (1) $\tan x$ (2) $\cot x$
 (3) $\tan 2x$ (4) $\cot 2x$
19. If $A + B + C = \pi$, then what is $\cos(A+B) + \cos C$ equal to?
 (1) 0 (2) $2\cos C$
 (3) $\cos C - \sin C$ (4) $2\sin C$
20. What is $\cos 20^\circ + \cos 100^\circ + \cos 140^\circ$ equal to?
 (1) 2 (2) 1
 (3) $\frac{1}{2}$ (4) 0
21. What is $\sin^2(3\pi) + \cos^2(4\pi) + \tan^2(5\pi)$ equal to?
 (1) 0 (2) 1
 (3) 2 (4) 3
22. Consider the following statements:
 I. Value of $\sin\theta$ oscillates between -1 and 1
 II. Value of $\cos\theta$ oscillates between 0 to 1
 Which of the above statements is/are correct?
 (1) Only I (2) Only II
 (3) Both I and II (4) Neither I nor II
23. What is $(1 - \sin^2 \theta)(1 + \tan^2 \theta)$ equal to?
 (1) $\sin^2 \theta$ (2) $\cos^2 \theta$
 (3) $\tan^2 \theta$ (4) 1
24. What is $\sin^2 20^\circ + \sin^2 70^\circ$ equal to?
 (1) 1 (2) 0
 (3) -1 (4) $\frac{1}{2}$
25. If $\cos x = 1/3$. Then, what is $\sin x \cdot \cot x \cdot \operatorname{cosec} x \cdot \tan x$ equal to?
 (1) $2/3$ (2) $3/2$
 (c) 2 (4) 1
26. What is $\frac{\cot 54^\circ}{\tan 36^\circ} + \frac{\tan 20^\circ}{\cot 70^\circ}$ equal to?
 (1) 0 (2) 1
 (3) 2 (4) 3
27. What is $\tan 15^\circ$ equal to?
 (1) $2 - \sqrt{3}$ (2) $2 + \sqrt{3}$
 (3) $1 - \sqrt{3}$ (4) $1 + \sqrt{3}$
28. What is the value of $\tan 105^\circ$?
 (1) $\frac{\sqrt{3}+1}{\sqrt{3}-1}$ (2) $\frac{\sqrt{3}+1}{1-\sqrt{3}}$
 (3) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$ (4) $\frac{\sqrt{3}+2}{\sqrt{3}-1}$
29. If $\sin\theta + 2\cos\theta = 1$, then what is $2\sin\theta - \cos\theta$ equal to?
 (1) 0 (2) 1
 (3) 2 (4) 4
30. What is $\frac{1 - \tan^2 \frac{x}{2}}{1 + \tan^2 \frac{x}{2}}$ equal to?
 (1) $\sin x \cos x$ (2) $\tan x$
 (3) $\sin x$ (4) $\cos x$

ANSWER KEY

1.	(2)	2.	(1)	3.	(1)	4.	(2)	5.	(1)	6.	(3)	7.	(1)	8.	(3)	9.	(2)	10.	(3)
11.	(1)	12.	(3)	13.	(3)	14.	(3)	15.	(1)	16.	(2)	17.	(2)	18.	(2)	19.	(1)	20.	(4)
21.	(2)	22.	(1)	23.	(4)	24.	(1)	25.	(4)	26.	(3)	27.	(1)	28.	(2)	29.	(3)	30.	(4)

PRACTICE SHEET-6

1. The minimum value of $2 \sin^2 \theta + 3 \cos^2 \theta$ is
 (1) 0 (2) 3
 (3) 2 (4) 1
2. If $\cos x + \cos y = 2$, the value of $\sin x + \sin y$ is
 (1) 0 (2) 1
 (3) 2 (4) -1
3. If $0^\circ < \theta < 90^\circ$, the value of $\sin \theta + \cos \theta$ is
 (1) equal to 1 (2) greater than 1
 (3) less than 1 (4) equal to 2
4. Maximum value of $(2 \sin \theta + 3 \cos \theta)$ is
 (1) 2 (2) $\sqrt{13}$
 (3) $\sqrt{15}$ (4) 1
5. The minimum value of $4 \tan^2 \theta + 9 \cot^2 \theta$ is equal to
 (1) 0 (2) 5
 (3) 12 (4) 13
6. If θ be acute and $\tan \theta + \cot \theta = 2$, then the value of $\tan^5 \theta + \cot^5 \theta$ is
 (1) 1 (2) 2
 (3) 3 (4) 4
7. If $\sin \theta + \operatorname{cosec} \theta = 2$, then value of $\sin^{100} \theta + \operatorname{cosec}^{100} \theta$ is equal to :
 (1) 1 (2) 2
 (3) 3 (4) 100
8. What is the minimum value of $8 \sin^2 \theta + 18 \operatorname{cosec}^2 \theta$
 (1) 18 (2) 24
 (3) 3 (4) 100
9. If $A = \sin^2 \theta + \cos^4 \theta$, for any value of θ , then the value of A is
 (1) $1 \leq A \leq 2$ (2) $3/4 \leq A \leq 1$
 (3) $13/16 \leq A \leq 1$ (4) $3/4 \leq A \leq 13/16$
10. The greatest value of $\sin^4 \theta + \cos^4 \theta$ is
 (1) $\sqrt{2}$ (2) 3
 (3) $1/2$ (4) 1
11. If $\sec \theta = x + \frac{1}{4x}$ ($0^\circ < \theta < 90^\circ$) then $\sec \theta + \tan \theta$ is equal to
 (1) $\frac{x}{2}$ (2) $2x$
- (3) x (4) $\frac{1}{2x}$
12. $\sin^2 \theta - 3 \sin \theta + 2 = 0$ will be true if
 (1) $0 \leq \theta \leq 90$ (2) $0 < \theta < 90$
 (3) $\theta = 0^\circ$ (4) $\theta = 90^\circ$
13. If $\frac{\pi}{4} < \theta < \frac{\pi}{2}$, then what is $\sqrt{1 - 2 \sin \theta \cos \theta}$ equal to?
 (a) $\cos \theta - \sin \theta$ (b) $\sin \theta - \cos \theta$
 (c) $\pm(\cos \theta - \sin \theta)$ (d) $\cos \theta \sin \theta$
14. If $0 < x < \pi/2$, then $(\cos x + \sec x)$ is:
 (a) > 2 (b) < 2
 (c) ≥ 2 (d) ≤ 2
15. If $\tan^2 \alpha = 1 + 2 \tan^2 \beta$, then $\sqrt{2} \cos \alpha - \cos \beta$ is equal to
 (1) 0 (2) $\sqrt{2}$
 (3) 1 (4) -1
16. If $0 < x < \frac{\pi}{2}$ and $\sec x = \operatorname{cosec} y$, then the value of $\sin(x + y)$ is
 (1) 0 (2) 1
 (3) $1/2$ (4) $1/\sqrt{3}$
17. If $\cos^4 \theta - \sin^4 \theta = \frac{2}{3}$, then the value of $2 \cos^2 \theta - 1$ is
 (1) 0 (2) 1
 (3) $2/3$ (4) $3/2$
18. If $\frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta} = 3$, then the value of $\sin^4 \theta - \cos^4 \theta$ is
 (a) $1/5$ (2) $2/5$
 (3) $3/5$ (4) $4/5$
19. If $2 \sin\left(\frac{\pi x}{2}\right) = x^2 + \frac{1}{x^2}$, then the value of $\left(x - \frac{1}{x}\right)$ is
 (1) -1 (2) 2
 (3) 1 (4) 0

20. Which one of the following is true for $0^\circ < \theta < 90^\circ$?
 (1) $\cos\theta \leq \cos^2\theta$ (2) $\cos\theta > \cos^2\theta$
 (3) $\cos\theta < \cos^2\theta$ (4) $\cos\theta \geq \cos^2\theta$
21. If $(\sin\alpha + \operatorname{cosec}\alpha)^2 + (\cos\alpha + \sec\alpha)^2 = k + \tan^2\alpha + \cot^2\alpha$, then the value of k is
 (1) 1 (2) 7
 (3) 3 (4) 5
22. If $0^\circ < \theta < 90^\circ$ and $\sec\theta = 3 \operatorname{cosec}^2\theta$, then θ is
 (1) $\frac{\pi}{6}$ (2) $\frac{\pi}{4}$
 (3) $\frac{\pi}{3}$ (4) $\frac{\pi}{5}$
23. $\sqrt{\frac{1+\sin\theta}{1-\sin\theta}} + \sqrt{\frac{1-\sin\theta}{1+\sin\theta}}$ is equal to
 (1) $2 \cos\theta$ (2) $2 \sin\theta$
 (3) $2 \cot\theta$ (4) $2 \sec\theta$
24. If $0^\circ < \theta < 90^\circ$ then the value of $\tan^2\theta + \cot^2\theta - \sec^2\theta \operatorname{cosec}^2\theta$ is
 (1) 0 (2) 1
 (3) 2 (4) -2
25. If α and β are positive acute angles, $\sin(4\alpha - \beta) = 1$ and $\cos(2\alpha + \beta) = 1/2$, then the value of $\sin(\alpha + 2\beta)$ is
 (1) 0 (2) 1
 (3) $\sqrt{3}/2$ (4) $1/\sqrt{2}$
26. The minimum value of $\cos^2 x + \cos^2 y - \cos^2 z$ is:
 (a) -1 (b) 0
 (c) 1 (d) 2
27. If $r \sin\theta = \frac{7}{2}$ and $r \cos\theta = \frac{7\sqrt{3}}{2}$, then the value of r is
 (1) 4 (2) 3
 (3) 5 (4) 7
28. If $0 \leq \theta \leq 90^\circ$ and $4 \cos^2\theta - 4\sqrt{3}\cos\theta + 3 = 0$ then the value of θ is
 (1) 30° (2) 45°
 (3) 90° (4) 60°
29. If $\sec\theta - \cos\theta = 3/2$ where θ is a positive acute angle, then the value of $\sec\theta$ is
 (1) $-1/2$ (2) 1
 (3) 2 (4) 0
30. If $\tan(5x-10^\circ) = \cot(5y+20^\circ)$, then the value of $(x+y)$ is
 (1) 15° (2) 16°
 (3) 24° (4) 20°

ANSWER KEY

1.	(3)	2.	(1)	3.	(2)	4.	(2)	5.	(3)	6.	(2)	7.	(2)	8.	(2)	9.	(2)	10.	(4)
11.	(2)	12.	(4)	13.	(2)	14.	(3)	15.	(1)	16.	(2)	17.	(3)	18.	(3)	19.	(4)	20.	(2)
21.	(2)	22.	(3)	23.	(4)	24.	(4)	25.	(4)	26.	(1)	27.	(4)	28.	(1)	29.	(3)	30.	(2)

CDS PYQ

1. If $\cos A + \cos^2 A = 1$, then what is the value of $2(\sin^2 A + \sin^4 A)$?

[CDS-2013-1]

- (a) 4 (b) 2
(c) 1 (d) $\frac{1}{2}$

2. $(1-\tan A)^2 + (1+\tan A)^2 + (1-\cot A)^2 + (1+\cot A)^2$ is equal to:

[CDS-2013-1]

- (a) $\sin^2 A \cos^2 A$ (b) $\sec^2 A \operatorname{cosec}^2 A$
(c) $2 \sec^2 A \operatorname{cosec}^2 A$ (d) None of these

Directions (for next four): Read the following information carefully to answer the questions that follow.

The angles A, B, C and D of a quadrilateral ABCD are in the ratio 1:2:4:5.

A=30°, B=60°, C=120°, D=150°

3. What is the value of $\cos(A + B)$?

[CDS-2013-1]

- (a) 0 (b) $\frac{1}{2}$
(c) 1 (d) None of these

4. What is the value of $\operatorname{cosec}(C - D + B)$?

[CDS-2013-1]

- (a) 1 (b) 2
(c) 3 (d) 4

5. Consider the following statements:

I. ABCD is a cyclic quadrilateral

II. $\sin(B - A) = \cos(D - C)$

Which of the above statement is/are correct?

[CDS-2013-1]

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

6. What is the value of $\sec^2 D - \tan^2 D$?

[CDS-2013-1]

- (a) $\frac{1}{2}$ (b) $\frac{2}{3}$
(c) 1 (d) None of these

7. What is the value of $\frac{\tan A - \sin A}{\sin^3 A}$?

[CDS-2013-1]

- (a) $\frac{\sec A}{1 - \cos A}$ (b) $\frac{\sec A}{1 + \cos^2 A}$
(c) $\frac{\sec A}{1 + \cos A}$ (d) None of these

8. Consider the following $\sin \theta \leq \theta \leq 90^\circ$.

I. The value of $\sin \theta + \cos \theta$ is always greater than 1

II. The value of $\tan \theta + \cot \theta$ is always greater than 1

Which of the above statement is/are correct?

[CDS-2013-1]

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

9. If $\sin A = \frac{3}{5}$ and A is an acute angle, then $\tan A + \sec A$ is equal to:

[CDS-2013-1]

- (a) 0 (b) 1
(c) 2 (d) -1

10. If $\sin \theta = \frac{x^2 - y^2}{x^2 + y^2}$, then which one of the following is:

[CDS-2013-1]

- (a) $\cos \theta = \frac{2xy}{x^2 - y^2}$ (b) $\cos \theta = \frac{2xy}{x^2 + y^2}$
(c) $\cos \theta = \frac{x - y}{x^2 + y^2}$ (d) $\cos \theta = \frac{xy(x - y)}{x^2 + y^2}$

11. If $a^2 = \frac{1 + 2\sin \theta \cos \theta}{1 - 2\sin \theta \cos \theta}$, then what is the value of

$$\frac{a+1}{a-1}?$$

[CDS-2013-1]

- (a) $\sec \theta$ (b) 1
(c) 0 (d) $\tan \theta$

12. If $5 \sin \theta + 12 \cos \theta = 13$, then what is $5 \cos \theta - 12 \sin \theta$ equal to?

[CDS-2013-2]

- (a) -2 (b) -1
(c) 0 (d) 1

13. If $4 \tan \theta = 3$, then what is $\frac{4 \sin \theta - \cos \theta}{4 \sin \theta + 9 \cos \theta}$ equal to?

[CDS-2013-2]

- (a) $\frac{1}{2}$ (b) $\frac{1}{3}$
(c) $\frac{1}{4}$ (d) $\frac{1}{6}$

14. If $\sin \theta - \cos \theta = 0$, then what is $\sin^4 \theta + \cos^4 \theta$ equal to?

[CDS-2013-2]

- (a) 1 (b) $\frac{3}{4}$
(c) $\frac{1}{2}$ (d) $\frac{1}{4}$

15. Consider the following statements:

I. $\tan \theta$ increases faster than $\sin \theta$ as θ increases

II. The value of $\sin \theta + \cos \theta$ is always greater than I

Which of the above statement is/are correct?

[CDS-2013-2]

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

16. What is $\frac{(\sin \theta + \cos \theta)(\tan \theta + \cot \theta)}{\sec \theta + \operatorname{cosec} \theta}$ equal to?

[CDS-2013-2]

- (a) 1 (b) 2
(c) $\sin \theta$ (d) $\cos \theta$

17. What is $\frac{\cos^2(45^\circ + \theta) + \cos^2(45^\circ - \theta)}{\tan(60^\circ + \theta) \tan(30^\circ - \theta)}$ equal to?

[CDS-2013-2]

- (a) -1 (b) 0
(c) 1 (d) 2

18. What is $\sin^6 \theta + \cos^6 \theta + 3 \sin^2 \theta \cos^2 \theta$ equal to?

[CDS-2013-2]

- (a) 0 (b) 1
(c) 2 (d) 4

19. What is $\frac{(1 + \sec \theta - \tan \theta) \cos \theta}{(1 + \sec \theta + \tan \theta)(1 - \sin \theta)}$ equal to?

[CDS-2013-2]

- (a) 1 (b) 2
(c) $\tan \theta$ (d) $\cot \theta$

20. If $\sin \theta + \cos \theta = \sqrt{3}$, then what is $\tan \theta + \cot \theta$ equal to?

[CDS-2013-2]

- (a) 1 (b) $\sqrt{2}$
(c) 2 (d) $\sqrt{3}$

21. If $\tan \theta + \sec \theta = m$, then what is $\sec \theta$ equal to?

[CDS-2013-2]

- (a) $\frac{m^2 - 1}{2m}$ (b) $\frac{m^2 + 1}{2m}$
(c) $\frac{m + 1}{m}$ (d) $\frac{m^2 + 1}{m}$

22. What is $\operatorname{cosec}(75^\circ + \theta) - \sec(15^\circ - \theta)$ equal to?

[CDS-2013-2]

- (a) 0 (b) 1
(c) $2 \sin \theta$ (d) $2 \cos \theta$

23. If $\triangle ABC$ is right angled at C, then what is $\cos(A + B) + \sin(A + B)$ equal to?

[CDS-2013-2]

- (a) 0 (b) $\frac{1}{2}$
(c) 1 (d) 2

24. If α , β and γ are acute angles such that $\sin \alpha = \frac{\sqrt{3}}{2}$, $\cos \beta = \frac{\sqrt{3}}{2}$ and $\tan \gamma = 1$, then what is $\alpha + \beta + \gamma$ equal to?

[CDS-2013-2]

- (a) 105° (b) 120°
(c) 135° (d) 150°

25. The value of $\cos 25^\circ - \sin 25^\circ$ is:

[CDS-2014-1]

- (a) Positive but less than I
(b) Positive but greater than I

- (c) Negative
(d) 0

26. In a right angled $\triangle ABC$, right angle at B, if $\cos A = \frac{4}{5}$, then what is the value of $\sin C$?

[CDS-2014-1]

- (a) $\frac{3}{5}$ (b) $\frac{4}{5}$
(c) $\frac{3}{5}$ (d) $\frac{2}{5}$

27. If α and β are complementary angle, then what is $\sqrt{\cos \alpha \operatorname{cosec} \beta - \cos \alpha \sin \beta}$ equal to?

[CDS-2014-1]

- (a) $\sec \beta$ (b) $\cos \alpha$
(c) $\sin \alpha$ (d) $-\tan \beta$

28. If $2\cot \theta = 3$, then what is $\frac{2\cos \theta - \sin \theta}{2\cos \theta + \sin \theta}$ equal to?

[CDS-2014-1]

- (a) $\frac{2}{3}$ (b) $\frac{1}{3}$
(c) $\frac{1}{2}$ (d) $\frac{3}{4}$

29. If $\sin \theta \cos \theta = \frac{1}{2}$, then what is $\sin^6 \theta + \cos^6 \theta$ equal to?

[CDS-2014-1]

- (a) 1 (b) 2
(c) 3 (d) $\frac{1}{4}$

30. If $\sec \theta + \tan \theta = 2$, then what is the value of $\sec \theta$?

[CDS-2014-1]

- (a) $\frac{3}{2}$ (b) $\sqrt{2}$
(c) $\frac{5}{\sqrt{2}}$ (d) $\frac{5}{4}$

31. What is $\operatorname{cosec} (75^\circ + \theta) - \sec (15^\circ - \theta) - \tan (55^\circ + \theta) + \cot (35^\circ - \theta)$ equal to?

[CDS-2014-1]

- (a) -1 (b) 0

- (c) 1 (d) $\frac{3}{2}$

32. If $\sin \theta + 2\cos \theta = 1$, where $0 < \theta < \frac{\pi}{2}$, what is $2\sin \theta - \cos \theta$ equal to?

[CDS-2014-1]

- (a) -1 (b) $\frac{1}{2}$
(c) 2 (d) 1

33. If $\cos x + \sec x = 2$, then what $\cos^n x + \sec^n x$ equal to, where n is a positive integer?

[CDS-2014-1]

- (a) 2 (b) 2^{n-2}
(c) 2^{n-1} (d) 2^n

34. What is $\sin 25^\circ \sin 35^\circ \sec 65^\circ \sec 55^\circ$ equal to?

[CDS-2014-1]

- (a) -1 (b) 0
(c) $\frac{1}{2}$ (d) 1

35. If $\tan 8\theta = \cot 2\theta$, where $\theta < 8\theta < \frac{\pi}{2}$, then what is the value of $\tan 5\theta$?

[CDS-2014-1]

- (a) $\frac{1}{\sqrt{3}}$ (b) 1
(c) $\sqrt{3}$ (d) 0

36. If $\sin (A + B) = 1$ where, $0 < B < 45^\circ$, then what is $\cos (A - B)$ equal to?

[CDS-2014-1]

- (a) $\sin 2B$ (b) $\sin B$
(c) $\cos 2B$ (d) $\cos B$

37. At what point of time after 3 O'clock, hour hand and the minute hand of a clock occur at right angles for the first time?

[CDS-2014-1]

- (a) 9 O'clock (b) $4\text{h } 37\frac{1}{6}\text{ min}$
(c) $3\text{h } 30\frac{8}{11}\text{ min}$ (d) $3\text{h } 32\frac{8}{11}\text{ min}$

38. If $0 < \theta < \frac{\pi}{4}$, then what is $\sqrt{1-2\sin\theta\cos\theta}$ equal to?

[CDS-2014-2]

- (a) $\cos\theta - \sin\theta$ (b) $\sin\theta - \cos\theta$
(c) $\pm(\cos\theta - \sin\theta)$ (d) $\cos\theta \sin\theta$

39. If $\tan\theta + \cot\theta = 2$, then what is $\sin\theta + \cos\theta$ equal to:

[CDS-2014-2]

- (a) $\frac{1}{2}$ (b) $\frac{1}{\sqrt{3}}$
(c) $\sqrt{2}$ (d) 1

40. What is $\frac{\sec x}{\cot x + \tan x}$ equal to?

[CDS-2014-2]

- (a) $\sin x$ (b) $\cos x$
(c) $\tan x$ (d) $\cot x$

41. What is $\frac{\sin x - \cos x + 1}{\sin x + \cos x - 1}$ equal to?

[CDS-2014-2]

- (a) $\frac{\sin x - 1}{\cos x}$ (b) $\frac{\sin x + 1}{\cos x}$
(c) $\frac{\sin x - 1}{\cos x + 1}$ (d) $\frac{\sin x + 1}{\cos x + 1}$

42. What is $(\sin^2 x - \cos^2 x)(1 - \sin^2 x \cos^2 x)$ equal to?

[CDS-2014-2]

- (a) $\sin^4 x - \cos^4 x$ (b) $\sin^6 x - \cos^6 x$
(c) $\cos^8 x - \sin^8 x$ (d) $\sin^8 x - \cos^8 x$

43. What is $(\sin x \cos y + \cos x \sin y)(\sin x \cos y - \cos x \sin y)$ equal to?

[CDS-2014-2]

- (a) $\cos^2 x - \cos^2 y$ (b) $\cos^2 x - \sin^2 y$
(c) $\sin^2 x - \cos^2 y$ (d) $\sin^2 x - \sin^2 y$

44. What is $(1 + \cot x - \operatorname{cosec} x)(1 + \tan x + \sec x)$ equal to?

[CDS-2014-2]

- (a) 1 (b) 2
(c) $\sin x$ (d) $\cos x$

45. What is $(\operatorname{cosec} x - \sin x)(\sec x - \cos x)(\tan x + \cot x)$ equal to?

[CDS-2014-2]

- (a) $\sin x + \cos x$ (b) $\sin x - \cos x$
(c) 2 (d) 1

46. Consider the following statements:

I. $\sin 1^\circ > \sin 1$

II. $\cos 1^\circ < \cos 1$

Which of the above statements is /are correct?

[CDS-2014-2]

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

47. If $\sin x + \operatorname{cosec} x = 2$, then what is $\sin^9 x + \operatorname{cosec}^9 x$ equal to?

[CDS-2014-2]

- (a) 2 (b) 18
(c) 512 (d) 1024

48. If $\sin x + \cos x = p$ and $\sin^3 x + \cos^3 x = q$, then what is $p^3 - 3p$ equal to?

[CDS-2014-2]

- (a) 0 (b) $-2q$
(c) $2q$ (d) $4q$

49. If $\tan A + \cot A = 4$, then $\tan^4 A + \cot^4 A$ equal to:

[CDS-2014-2]

- (a) 110 (b) 191
(c) 80 (d) 194

50. If $\tan(A + B) = \sqrt{3}$ and $\tan A = 1$, then $\tan(A - B)$ is equal to:

[CDS-2015-1]

- (a) 0 (b) 1
(c) $\frac{1}{\sqrt{3}}$ (d) $\sqrt{2}$

51. If $\cos A = \tan B$, $\cos B = \tan C$ and $\cos C = \tan A$ then $\sin A$ is equal to:

[CDS-2015-1]

- (a) $\frac{\sqrt{5}-1}{4}$ (b) $\frac{\sqrt{5}-1}{2}$
(c) $\frac{\sqrt{3}-1}{4}$ (d) none

52. If $\frac{3 - \tan^2 A}{1 - 3\tan^2 A} = K$ where K is a real number, then $\operatorname{cosec} A(3 \sin A - 4 \sin^3 A)$ is equal to:

[CDS-2015-1]

- (a) $\frac{2K}{K-1}$
(b) $\frac{2K}{K-1}$, where $\frac{1}{3} \leq K \leq 3$
(c) $\frac{2K}{K-1}$, where $K < \frac{1}{3}$ or $K > 3$

(d) $\frac{2K}{K+1}$

53. If $P = \sqrt{\frac{1-\sin x}{1+\sin x}}$, $q = \frac{1-\sin x}{\cos x}$, $r = \frac{\cos x}{1+\sin x}$, then

which of the following is/are correct?

1. $p = q = r$

2. $p^2 = qr$

Select the correct answer using the code given below.

[CDS-2015-1]

(a) Only 1

(b) Only 2

(c) Both 1 and 2

(d) Neither 1 nor 2

54. Consider the following:

1. $\frac{\cos A}{1-\tan A} + \frac{\sin A}{1-\cot A} = \cos A + \sin A$.

2. $(1-\sin A - \cos A)^2 = 2(1-\sin A)(1+\cos A)$
Which of the above is/are identify / identities ?

[CDS-2015-1]

(a) 1 Only

(b) 2 Only

(c) Both 1 and 2

(d) Neither 1 nor 2

55. ABC is a triangle right angled at B and AB : BC = 3 : 4. What is $\sin A + \sin B + \sin C$ equal to?

[CDS-2015-1]

(a) 2

(b) $\frac{11}{5}$

(c) $\frac{12}{5}$

(d) 3

56. The value of $\operatorname{cosec}^2 67^\circ + \sec^2 57^\circ - \cot^2 33^\circ - \tan^2 23^\circ$ is:

[CDS-2015-1]

(a) $2\sqrt{2}$

(b) 2

(c) $\sqrt{2}$

(d) 0

57. Consider the following statements:

1. There exists at least one value of x between 0 and $\pi/2$ which satisfies the equation $\sin^4 x - 2\sin^2 x - 1 = 0$.

2. $\sin 1.5$ is greater than $\cos 1.5$. Which of the above statement is/are correct?

[CDS-2015-1]

(a) 1 Only

(b) 2 Only

(c) Both 1 and 2

(d) Neither 1 nor 2

58. If $\sin x + \cos x = c$ then $\sin^6 x + \cos^6 x$ is equal to:

[CDS-2015-1]

(a) $\frac{1+6c^2-3c^4}{16}$

(b) $\frac{1+6c^2-3c^4}{4}$

(c) $\frac{1+6c^2+3c^4}{16}$

(d) $\frac{1+6c^2+3c^4}{4}$

59. Consider the following statements:

1. There exists no value of x such that

$$\frac{1}{1-\sin x} = 4 + 2\sqrt{3}, 0 < x < \frac{\pi}{2}$$

2. $\sin x = 3^{\sin^2 x}$ does not hold good for any real x . Which of the above statement is/are correct?

[CDS-2015-1]

(a) 1 Only

(b) 2 Only

(c) Both 1 and 2

(d) Neither 1 nor 2

60. The complement angle of 80° is:

[CDS-2015-1]

(a) $\frac{\pi}{9}$

(b) $\frac{5\pi}{9}$

(c) $\frac{\pi}{18}$

(d) $\frac{9}{5\pi}$

61. A triangle ABC is inscribed in a circle. If sum of the squares of sides of the triangle is equal to twice the square of the diameter, then $\sin^2 A + \sin^2 B + \sin^2 C$ is equal to?

[CDS-2015-1]

(a) 2

(b) 3

(c) 4

(d) None of these

62. At 8:30, the hour hand and the minute hand of a clock form an angle of:

[CDS-2015-1]

(a) 80°

(b) 75°

(c) 70°

(d) 60°

63. A person goes to a market between 4 pm and 5 pm. When he comes back, he finds that the hour hand and minute hand have interchanged their position. For how much time (approximately) was he out of his house?

[CDS 2015 (1)]

(a) 55.25 minutes

(b) 55.30 minutes

(c) 55.34 minutes

(d) 55.38 minutes

64. Consider the following statements:

1. $\frac{1+\tan^2 \theta}{1+\cot^2 \theta} = \left(\frac{1-\tan \theta}{1-\cot \theta}\right)^2$ is true for all $0 < \theta < \frac{\pi}{2}$, $\theta \neq \frac{\pi}{4}$

$\frac{\pi}{2}$, $\theta \neq \frac{\pi}{4}$

2. $\cot \theta - \frac{1}{\tan \theta}$ is true for $\theta = 45^\circ$ only

Which of the above statements is/are correct?

[CDS-2015-2]

- (a) 1 Only (b) 1
(c) $\tan^2 \theta$ (d) $\sin^2 \theta$

65. If $x = a \cos \theta$ and $y = b \cot \theta$, then $(ax^{-1} - by^{-1})(ax^{-1} + by^{-1})$ is equal to:

[CDS-2015-2]

- (a) 0 (b) 1
(c) $\tan^2 \theta$ (d) $\sin^2 \theta$

66. $\frac{\cos \theta}{1 - \sin \theta}$ is equal $\left(\text{where } \theta \neq \frac{\pi}{2} \right)$:

[CDS-2015-2]

- (a) $\frac{\tan \theta - 1}{\tan \theta + 1}$ (b) $\frac{1 + \sin \theta}{\cos \theta}$
(c) $\frac{\tan \theta + 1}{\tan \theta - 1}$ (d) $\frac{1 + \cos \theta}{\sin \theta}$

67. If $\tan(x + 40)^\circ \tan(x + 20)^\circ \tan(3x)^\circ \tan(70 - x)^\circ \tan(50 - x)^\circ = 1$, then the value of x is equal to:

[CDS-2015-2]

- (a) 30 (b) 20
(c) 15 (d) 10

68. If θ is an acute angle and $\sin \theta \cos \theta = 2 \cos^3 \theta - 1.5 \cos \theta$, then what is $\sin \theta$ equal to?

[CDS-2015-2]

- (a) $\frac{\sqrt{5} - 1}{4}$ (b) $\frac{1 - \sqrt{5}}{4}$
(c) $\frac{\sqrt{5} + 1}{4}$ (d) $-\frac{\sqrt{5} + 1}{4}$

69. Consider the following statements:

1. $\sin 66^\circ$ is less than $\cos 66^\circ$
2. $\sin 26^\circ$ is less than $\cos 26^\circ$

Which of the above statement is/are correct?

[CDS-2015-2]

- (a) 1 Only (b) 2 Only
(c) Both 1 and 2 (d) Neither 1 nor 2

70. If a and b are positive, then the relation $\sin \theta = \frac{2a + 3b}{3b}$ is:

[CDS-2015-2]

- (a) Not possible
(b) Possible only if $a = b$
(c) Possible if $a > b$
(d) Possible if $a < b$

71. If $\tan \theta + \sec \theta = 2$, then $\tan \theta$ is equal to:

[CDS-2015-2]

- (a) $\frac{3}{4}$ (b) $\frac{5}{4}$

- (c) $\frac{3}{2}$ (d) $\frac{5}{2}$

72. A clock is started at noon. By 10 minutes past 5, through what angle, the hour hand moves?

[CDS-2015-2]

- (a) 160° (b) 145°
(c) 150° (d) 150°

73. The minimum value of $\cos^2 x + \cos^2 y - \cos^2 z$ is:

[CDS-2015-2]

- (a) -1 (b) 0
(c) 1 (d) 2

74. The value of:

$32 \cot^2 \left(\frac{\pi}{4} \right) - 8 \sec^2 \left(\frac{\pi}{3} \right) + 8 \cos^3 \left(\frac{\pi}{6} \right)$ is equal to:

[CDS-2015-2]

- (a) $\sqrt{3}$ (b) $2\sqrt{3}$
(c) 3 (d) $3\sqrt{3}$

75. $\left(\frac{\sin 35^\circ}{\cos 55^\circ} \right)^2 - \left(\frac{\cos 55^\circ}{\sin 35^\circ} \right)^2 + 2 \sin 30^\circ$ is equal to:

[CDS-2016-1]

- (a) -1 (b) 0
(c) 1 (d) 2

76. If $\frac{x}{a} - \frac{y}{b} \tan \theta = 1$ and $\frac{x}{a} \tan \theta + \frac{y}{b} = 1$, then the value of $\frac{x^2}{a^2} + \frac{y^2}{b^2}$ is:

[CDS-2016-1]

- (a) $2 \sec^2 \theta$ (b) $\sec^2 \theta$
(c) $2 \cos^2 \theta$ (d) $2 \cos^2 \theta$

77. Consider the following:

1. $\sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}} = \operatorname{cosec} \theta - \cot \theta$

2. $\sqrt{\frac{1 + \cos \theta}{1 - \cos \theta}} = \operatorname{cosec} \theta + \cot \theta$

Which of the above is/are identity identities?

[CDS-2016-1]

- (a) 1 Only (b) 2 Only
(c) Both 1 and 2 (d) Neither 1 nor 2

78. If $p = \cot \theta + \tan \theta$ and $q = \sec \theta - \cos \theta$, then $(p^2 q)^{2/3} (q^2 p)^{2/3}$ is equal to:

[CDS-2016-1]

- (a) 0 (b) 1
(c) 2 (d) 3

79. If $\frac{\cos^2 \theta - 3\cos \theta + 2}{\sin^2 \theta} = 1$, where $0 < \theta < \frac{\pi}{2}$, then

which of the following statements is/are correct?

1. There are two values of θ satisfying the above equation

2. $\theta = 60^\circ$ is satisfied by the above equation

Select the correct answer using the code given below:

[CDS-2016-1]

- (a) 1 Only (b) 2 Only
(c) Both 1 and 2 (d) Neither 1 nor 2

80. Which of the following is correct in respect of the equation $3 - \tan^2 \theta = \alpha (1 - 3\tan^2 \theta)$? (given that α is a real number)

[CDS-2016-1]

(a) $\alpha \in \left[\frac{1}{3}, 3\right]$

(b) $\alpha \in \left[-\infty, \frac{1}{3}\right] \cup [3, \infty]$

(c) $\alpha \in \left(-\infty, \frac{1}{3}\right] \cup [3, \infty)$

(d) None of the above

81. If $\tan \theta + \cot \theta = \frac{4}{\sqrt{3}}$, where $0 < \theta < \frac{\pi}{2}$, then $\sin \theta + \cos \theta$ is equal to:

[CDS-2016-1]

(a) 1 (b) $\frac{\sqrt{3}-1}{2}$

(c) $\frac{\sqrt{3}+1}{2}$ (d) $\sqrt{2}$

82. A person goes to a market between 4pm and 5pm. When he comes back, he finds that the hour hand and the minute hand of the clock have interchanged their positions. For how much time (approximately) was he out of his house?

[CDS 2016 (1)]

- (a) 55.38 minutes (b) 55.48 minutes
(c) 55.57 minutes (d) 55.67 minutes

83. If $\sin \theta + \cos \theta = \frac{\sqrt{7}}{2}$, then what is $\sin \theta - \cos \theta$ equal to:

[CDS-2016-2]

- (a) 0 (b) $\frac{1}{2}$
(c) 1 (d) $\sqrt{2}$

84. If $\sin x + \sin^2 x = 1$, then what is the value of $\cos^8 x + 2\cos^6 x + \cos^4 x$?

[CDS-2016-2]

- (a) 0 (b) 1
(c) 1 (d) 4

85. What is the value of: $\operatorname{cosec}^2 68^\circ + \sec^2 56^\circ - \cot^2 34^\circ - \tan^2 22^\circ$?

[CDS-2016-2]

- (a) 0 (b) $1/2$
(c) 1 (d) 2

86. If $2y \cos \theta = x \sin \theta$ and $2x \sec \theta - y \operatorname{cosec} \theta = 3$, then what is $x^2 + 4y^2$ equal to:

[CDS-2016-2]

- (a) 1 (b) 2
(c) 4 (d) 8

87. If $\sin \theta + \cos \theta = \frac{1+\sqrt{3}}{2}$ where $0 < \theta < \frac{\pi}{2}$, then what is $\tan \theta + \cot \theta$ equal to?

[CDS-2016-2]

- (a) $\frac{\sqrt{3}}{4}$ (b) $\frac{1}{\sqrt{3}}$
(c) $\sqrt{3}$ (d) $\frac{4}{\sqrt{3}}$

88. If $A = \sin^2 \theta + \cos^4 \theta$ where $0 < \theta < \pi/2$, then which one of the following is correct?

[CDS-2016-2]

- (a) $1 \leq A \leq 2$ (b) $\frac{3}{4} \leq A \leq 1$
(c) $\frac{13}{16} \leq A \leq 2$ (d) $\frac{3}{4} \leq A \leq \frac{13}{16}$

89. What is $\frac{\cot A + \operatorname{cosec} A - 1}{\cot A - \operatorname{cosec} A + 1}$ equal to?

[CDS-2016-2]

- (a) $\frac{1+\cos A}{\sin A}$ (b) $\frac{1-\cos A}{\sin A}$
(c) $\frac{1+\sin A}{\cos A}$ (d) $\frac{1-\sin A}{\cos A}$

90. Consider the following:

1. $\sin 1^\circ > \sin 1^c$
2. $\cos 1^\circ < \cos 1^c$
3. $\tan 1^\circ > \tan 1^c$

Which of the above are not correct?

[CDS-2016-2]

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

91. If $\tan^2 x + \frac{1}{\tan^2 x} = 2$ and $0^\circ < x < 90^\circ$, then what is the value of x ?

[CDS-2016-2]

- (a) 15° (b) 30°
(c) 45° (d) 60°

92. Consider the following:

1. $\frac{\cos 75^\circ}{\sin 15^\circ} + \frac{\sin 12^\circ}{\cos 78^\circ} - \frac{\cos 18^\circ}{\sin 72^\circ} = 1$
2. $\frac{\cos 35^\circ}{\sin 55^\circ} - \frac{\sin 11^\circ}{\cos 79^\circ} + \cos 28^\circ \operatorname{cosec} 62^\circ = 1$
3. $\frac{\sin 80^\circ}{\cos 10^\circ} - \sin 59^\circ \sec 31^\circ = 1$

Which of the above are correct?

[CDS-2016-2]

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

93. What is the value of $\tan 1^\circ \tan 2^\circ \tan 3^\circ \tan 4^\circ \dots \tan 89^\circ$?

[CDS-2016-2]

- (a) 0 (b) 1
(c) 2 (d) $\sqrt{3}$

94. What is the minimum value of: $9\tan^2 \theta + 4\cot^2 \theta$?

[CDS-2017-1]

- (a) 6 (b) 9
(c) 12 (d) 13

95. If $x \sin \theta = y \cos \theta = \frac{2z \tan \theta}{1 - \tan^2 \theta}$, then what is $4z^2 (x^2 + y^2)$ equal to:

[CDS-2017-1]

- (a) $(x^2 + y^2)^3$ (b) $(x^2 - y^2)^2$
(c) $(x^2 - y^2)^2$ (d) $(x^2 + y^2)^2$

96. If $\cos \theta_1 + \cos \theta_2 + \cos \theta_3 = 3$, then what is $\sin \theta_1 + \sin \theta_2 + \sin \theta_3$ equal to?

[CDS-2017-1]

- (a) 0 (b) 1
(c) 2 (d) 3

97. What is the value of θ which satisfies the equation $\cos \theta + \tan \theta = 1$?

[CDS-2017-1]

- (a) 0° (b) 1
(c) 45° (d) 60°

98. What is the value of:

$$\sin x \sqrt{\frac{1}{1 + \cos x} + \frac{1}{1 - \cos x}} ?$$

[CDS-2017-1]

- (a) $\sqrt{2}$ (b) $2\sqrt{2}$
(c) $\sqrt{2} \tan x$ (d) 0

99. What is $\frac{\cos^4 A - \sin^4 A}{\cos^2 A - \sin^2 A}$ equal to?

[CDS-2017-1]

- (a) $\cos^2 A - \sin^2 A$ (b) $\cos A - \sin A$
(c) 1 (d) 2

100. If $7\sin^2 x + 3\cos^2 x = 4$, $0 < x < 90^\circ$, then what is the value of $\tan x$?

[CDS-2017-1]

- (a) $\sqrt{2}$ (b) 1
(c) $\frac{\sqrt{3}}{2}$ (d) $\frac{1}{\sqrt{3}}$

101. If D is the number of degrees and R is the number of radians in an angle θ , then which one of the following is correct?

[CDS-2017-1]

- (a) $\pi D = 180R$ (b) $\pi D = 90R$
(c) $\pi R = 180D$ (d) $\pi R = 90D$

102. If $x = a \cos \theta + b \sin \theta$ and $y = a \sin \theta - b \cos \theta$, then what is $x^2 + y^2$ equal to?

[CDS-2017-1]

- (a) $2ab$ (b) $a + b$
(c) $a^2 + b^2$ (d) $a^2 - b^2$

103. If a triangle has sides 5, 13 and 12 units and θ is the acute angle of the triangle, then what is the value of $(\sin \theta + \cos \theta)$?

[CDS-2017-2]

- (a) $\frac{5}{13}$ (b) $\frac{7}{13}$
(c) $\frac{12}{13}$ (d) $\frac{17}{13}$

104. If $0 < x < \pi/2$, then $(\sin x + \operatorname{cosec} x)$ is:

[CDS-2017-2]

- (a) > 2 (b) < 2
(c) ≥ 2 (d) ≤ 2

105. If $\sin \theta = \frac{m^2 - n^2}{m^2 + n^2}$ and $0 < \theta < \frac{\pi}{2}$, then what is value of $\cos \theta$?

[CDS-2017-2]

- (a) $\frac{2mn}{m^2 + n^2}$ (b) $\frac{2mn}{m^2 - n^2}$
(c) $\frac{m^2 + n^2}{2mn}$ (d) $\frac{m^2 - n^2}{2mn}$

106. If $A = \frac{\sin 45^\circ - \sin 30^\circ}{\cos 45^\circ + \cos 60^\circ}$ and $B = \frac{\sec 45^\circ - \tan 45^\circ}{\operatorname{cosec} 45^\circ + \cot 45^\circ}$ then what one of the following is correct?

[CDS-2017-2]

- (a) $A = B$ (b) $A > B > 0$
(c) $A < B$ (d) $B < A < 0$

107. Consider the following statements:

1. If $45^\circ < \theta < 60^\circ$, then $\sec^2 \theta + \operatorname{cosec}^2 \theta = \alpha^2$ for some real number $\alpha > 1$.

2. If $0^\circ < \theta < 45^\circ$, then $\frac{1 + \cos \theta}{1 - \cos \theta} = x^2$ for some real number $x > 2$.

3. If $0^\circ < \theta < 45^\circ$, then $\frac{\cos \theta}{1 - \tan \theta} + \frac{\sin \theta}{1 - \cos \theta} \geq 2$

What is the number of true statement?

[CDS-2017-2]

- (a) Zero (b) One
(c) Two (d) Three

108. The value of $\frac{\sin 1^\circ}{\sin 1^\circ}$ where 1° represent 1 radian is:

[CDS-2017-2]

- (a) Equal to 1
(b) Less than 1
(c) Greater than 1 but less than 2
(d) Greater than 2

109. If θ measured in radians is the angle between the hour hand and the minute hand of a clock when the time is 4:36pm then which one of the following is correct?

[CDS-2017-2]

- (a) $\frac{3\pi}{5} < \theta < \frac{4\pi}{5}$ (b) $\frac{2\pi}{5} < \theta < \frac{3\pi}{5}$
(c) $\frac{\pi}{5} \leq \theta \leq \frac{2\pi}{5}$ (d) $\frac{7\pi}{15} \leq \theta \leq \frac{8\pi}{15}$

110. What is $\sin^4 \theta - \cos^4 \theta$ equal to for any real number θ ?

[CDS-2018-1]

- (a) 1 (b) $1 - 2 \sin^2 \theta$
(c) $2 \cos^2 \theta + 1$ (d) $1 - 2 \cos^2 \theta$

111. What is $\cot 1^\circ \cot 23^\circ \cot 45^\circ \cot 67^\circ \cot 89^\circ$ equal to?

[CDS-2018-1]

- (a) 0 (b) 1
(c) $\frac{1}{2}$ (d) $\frac{1}{3}$

112. Consider the following statements:

1. $(\sec^2 \theta - 1)(1 - \operatorname{cosec}^2 \theta) = 1$

2. $\sin \theta (1 + \cos \theta)^{-1} + (1 + \cos \theta) (\sin \theta)^{-1} = 2 \operatorname{cosec} \theta$

Which of the above is/are correct?

[CDS-2018-1]

- (a) 1 Only (b) 2 Only
(c) Both 1 and 2 (d) Neither 1 nor 2

113. If $\sec x \operatorname{cosec} x = 2$, then what is $\tan^n x + \cot^n x$ equal to?

[CDS-2018-1]

- (a) 2 (b) 2^{n+1}
(c) 2^n (d) 2^{n-1}

114. If $\cos x + \cos^2 x = 1$, then what is $\sin^2 x + \sin^4 x$ equal to?

[CDS-2018-1]

- (a) 1 (b) 1.5
(c) 2 (d) 3

115. If $\sin A + \cos A = p$ and $\sin^3 A + \cos^3 A = q$, then which one of the following is correct?

[CDS-2018-1]

- (a) $p^3 - 3p + q = 0$ (b) $q^3 - 3q + 2p = 0$
(c) $p^3 - 3p + 2q = 0$ (d) $p^3 + 3p + 2q = 0$

116. If $x = \frac{\sec^2 \theta - \tan \theta}{\sec^2 \theta + \tan \theta}$, then which one of the following is correct?

[CDS-2018-1]

- (a) $\frac{1}{3} < x < 3$ (b) $x \in \left[\frac{1}{3}, 3\right]$
(c) $-3 < x < -\frac{1}{3}$ (d) $\frac{1}{3} \leq x \leq 3$

117. ABC is a right angled triangle with base BC and height AB. The hypotenuse AC is four times the length of the perpendicular drawn to it from the opposite vertex. What is $\tan C$ equal to?

[CDS-2018-1]

- (a) $2 - \sqrt{3}$ (b) $\sqrt{3} - 1$
(c) $2 + \sqrt{3}$ (d) $\sqrt{3} + 1$

118. What angle does the hour hand of a clock describe in 10 minutes of time?

[CDS-2018-1]

- (a) 1° (b) 5°
(c) 6° (d) 10°

119. If $\cos \theta = \frac{1}{\sqrt{5}}$, where $0 < \theta < \frac{\pi}{2}$, then

$\frac{2 \tan \theta}{1 - \tan^2 \theta}$ is equal to:

[CDS-2018-2]

- (a) $\frac{4}{3}$ (b) $-\frac{4}{3}$
(c) $\frac{1}{3}$ (d) $-\frac{2}{3}$

120. If $0 < \theta < 90^\circ$, $0 < \phi < 90^\circ$ and $\cos \theta < \cos \phi$, then which one of the following is correct?

[CDS-2018-2]

- (a) $\theta < \phi$
- (b) $\theta > \phi$
- (c) $\theta + \phi = 90^\circ$
- (d) No conclusion can be drawn

121. Let $\sin(A + B) = \frac{\sqrt{3}}{2}$ and $\cos B = \frac{\sqrt{3}}{2}$, where A, B are acute angles. What is $\tan(2A - B)$ equal to?

[CDS-2018-2]

- (a) $1/2$
- (b) $\sqrt{3}$
- (c) $\frac{1}{\sqrt{3}}$
- (d) 1

122. Consider the following statements:

1. If $\frac{\cos \theta}{1 - \sin \theta} + \frac{\cos \theta}{1 + \sin \theta} = 4$, where $0 < \theta < 90^\circ$, then $\theta = 60^\circ$

2. If $3 \tan \theta + \cot \theta = 5 \operatorname{cosec} \theta$, where $0 < \theta < 90^\circ$, then $\theta = 60^\circ$

Which of the statements given above is/are correct?

[CDS-2018-2]

- (a) 1 Only
- (b) 2 Only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

123. Consider the following statements:

1. $\cos^2 \theta = 1 - \frac{p^2 + q^2}{2pq}$, where p, q are non-zero real numbers, is possible only when $p = q$.

2. $\tan^2 \theta = \frac{4pq}{(p+q)^2} - 1$, where p, q are non-zero real numbers, is above is/are correct?

[CDS-2018-2]

- (a) 1 Only
- (b) 2 Only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

124. Consider the following statements:

1. $\cos \theta + \sec \theta$ can never be equal to 1.5

2. $\sec^2 \theta + \operatorname{cosec}^2 \theta$ can never be less than 4

Which of the statements given above is/are correct?

[CDS-2018-2]

- (a) 1 Only
- (b) 2 Only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

125. If $\sin^2 x + \sin x = 1$, when what is the value of $\cos^{12} x + 3 \cos^{10} x + 3 \cos^8 x + \cos^6 x$?

[CDS-2018-2]

- (a) -1
- (b) 0
- (c) 1
- (d) 8

126. If $3 \sin \theta + 5 \cos \theta = 4$, then what is the value of $(3 \cos \theta - 5 \sin \theta)^2$?

[CDS-2018-2]

- (a) 9
- (b) 12
- (c) 16
- (d) 18

127. If $\cot \theta (1 + \sin \theta) = 4m$ and $\cot \theta (1 - \sin \theta) = 4n$, then which one of the following is correct?

[CDS-2018-2]

- (a) $(m^2 + n^2) = mn$
- (b) $(m^2 - n^2)^2 = mn$
- (c) $(m^2 - n^2)^2 = m^2 n^2$
- (d) $(m^2 + n^2)^2 = m^2 n^2$

128. The minute hand of a clock overtakes the hour hand after every 72 minutes of correct time. How much time does the clock lose or gain in a day of normal time?

[CDS-2018-2]

- (a) lose $121 \frac{9}{11}$ minutes
- (b) lose $157 \frac{1}{11}$ minutes
- (c) gain $121 \frac{9}{11}$ minutes
- (d) gain $157 \frac{1}{11}$ minutes

129. If $\tan x = 1$, $0 < x < 90^\circ$, then what is the value of $2 \sin x \cos x$?

[CDS-2019-1]

- (a) $1/2$
- (b) 1
- (c) $\frac{\sqrt{3}}{2}$
- (d) $\sqrt{3}$

130. What is the value of $\sin 46^\circ \cos 44^\circ + \cos 46^\circ \sin 44^\circ$?

[CDS-2019-1]

- (a) $\sin 2^\circ$
- (b) 0
- (c) 1
- (d) 2

131. Suppose $0 < \theta < 90^\circ$, then for every θ , $4 \sin^2 \theta + 1$ is greater than or equal to:

[CDS-2019-1]

- (a) 2
- (b) $4 \sin \theta$
- (c) $4 \cos \theta$
- (d) $4 \tan \theta$

132. What is the value of $\tan 1^\circ \tan 2^\circ \tan 3^\circ \dots \tan 89^\circ$?

[CDS-2019-1]

- (a) 0
- (b) 1
- (c) 2
- (d) ∞

- 133.** There are two parallel streets each directed North to South. A person in the first street travelling from South to North wishes to take the second street which is on his right side. At some place, he makes at 150° turn to the right and he travels for 15 minutes at the speed of 20 km/hr. After that he takes a left turn of 60° and travels for 20 minutes at the speed of 30 km/hr in order to meet the second street. What is the distance between the two streets?
[CDS-2019-1]
(a) 7.5km (b) 10.5km
(c) 12.5km (d) 15km
- 134.** If $3 \tan \theta = \cot \theta$ where $0 \leq \theta \leq \pi/2$, then what is the value of θ ?
[CDS-2019-1]
(a) $\pi/6$ (b) $\pi/4$
(c) $\pi/3$ (d) $\pi/2$
- 135.** What is the value of $\sin^2 25^\circ + \sin^2 65^\circ$?
[CDS-2019-1]
(a) 0 (b) 1
(c) 2 (d) 4
- 136.** What is the value of $\sin^6 \theta + \cos^6 \theta + 3 \sin^2 \theta \cos^2 \theta - 1$?
[CDS-2019-1]
(a) 0 (b) 1
(c) 2 (d) 4
- 137.** Consider the following for real numbers α , β , γ and δ :
[CDS-2019-1]
1. $\sec \alpha = \frac{1}{4}$
2. $\tan \beta = 20$
3. $\operatorname{cosec} \gamma = \frac{1}{2}$
4. $\cos \delta = 2$
How many of the above statements are not possible
(a) One (b) Two
(c) Three (d) Four
- 138.** If $\cos^2 x + \cos x = 1$, then what is the value of $\sin^{12} x + 3 \sin^{10} x + 3 \sin^8 x + \sin^6 x$?
[CDS-2019-2]
(a) 1 (b) 2
(c) 4 (d) 8
- 139.** If $0 < \theta < 90^\circ$, $\sin \theta = \frac{3}{5}$ and $x = \cot \theta$, then what is the value of:
 $1 + 3x + 9x^2 + 27x^3 + 81x^4 + 243x^5$?
[CDS-2019-2]
(a) 941 (b) 1000
- (c) 1220 (d) 1365
- 140.** What is The value of $\frac{\sin 19^\circ}{\cos 71^\circ} + \frac{\cos 73^\circ}{\sin 17^\circ}$?
[CDS-2019-2]
(a) 0 (b) 1
(c) 2 (d) 4
- 141.** $\log_{10} (\cos \theta) + \log_{10} (\sin \theta) + \log_{10} (\tan \theta) + \log_{10} (\cot \theta) + \log_{10} (\sec \theta) + \log_{10} (\operatorname{cosec} \theta)$?
[CDS-2019-2]
(a) -1 (b) 0
(c) 0.5 (d) 1
- 142.** A wheel makes 360 revolutions in one minute. What is the number of radians it turns in one second?
[CDS-2020-1]
(a) 4π (b) 6π
(c) 12π (d) 16π
- 143.** What is the least value of $(25 \operatorname{cosec}^2 x + \sec^2 x)$?
[CDS-2020-1]
(a) 40 (b) 36
(c) 26 (d) 24
- 144.** Let $0 < \theta < 90^\circ$ and $100\theta = 90^\circ$. If $\alpha = \prod_{n=1}^{99} \cot n\theta$, then which one of the following is correct?
[CDS-2020-1]
(a) $\alpha = 1$ (b) $\alpha = 0$
(c) $\alpha > 1$ (d) $0 < \alpha < 1$
- 145.** If $\tan 6\theta = \cot 2\theta$, where $0 < 6\theta < \pi/2$, then what is the value of $\sec 4\theta$?
[CDS-2020-1]
(a) $\sqrt{2}$ (b) 2
(c) $\frac{2}{\sqrt{3}}$ (d) $\frac{4}{3}$
- 146.** What is the value of $\sin^2 6^\circ + \sin^2 12^\circ + \sin^2 18^\circ + \dots + \sin^2 84^\circ + \sin^2 90^\circ$?
[CDS-2020-1]
(a) 1 (b) 2
(c) 4 (d) 8
- 147.** What is $\frac{\cos \theta}{1 + \sin \theta} + \frac{1}{\cot \theta}$ equal to?
[CDS-2020-1]
(a) $\operatorname{cosec} \theta$ (b) $\sec \theta$
(c) $\sec \theta + \operatorname{cosec} \theta$ (d) $\operatorname{cosec} \theta - \cot \theta$

148. What is $\frac{\sin \theta - \cos \theta + 1}{\sin \theta + \cos \theta - 1} - \frac{\sin \theta + 1}{\cos \theta}$ equal to?

[CDS-2020-1]

- (a) 0 (b) 1
(c) $2 \sin \theta$ (d) $2 \cos \theta$

149. What is $(\tan x + \tan y)(1 - \cot x \cot y) + (\cot x + \cot y)(1 - \tan x \tan y)$ equal to?

[CDS-2020-1]

- (a) 0 (b) 1
(c) 2 (d) 4

150. What is $\sqrt{\frac{\sec x - \tan x}{\sec x + \tan x}}$ equal to?

[CDS-2020-1]

- (a) $\frac{1}{\sin x + \cos x}$ (b) $\frac{1}{\tan x + \cot x}$
(c) $\frac{1}{\sec x + \tan x}$ (d) $\frac{1}{\operatorname{cosec} x + \cot x}$

151. If θ lies in the first quadrant and $\cot \theta = \frac{63}{16}$, then what is the value of $(\sin \theta + \cos \theta)$?

[CDS-2020-1]

- (a) 1 (b) $\frac{69}{65}$
(c) $\frac{79}{65}$ (d) 2

152. What is the value of $\frac{1 - 2 \sin^2 \theta \cos^2 \theta}{\sin^4 \theta + \cos^4 \theta} + 4$ equal to?

[CDS-2020-1]

- (a) 0 (b) 1
(c) 2 (d) 5

153. A rectangle is 48cm long and 14 cm wide. If the diagonal makes an angle θ with the longer side, then what is $(\sec \theta + \operatorname{cosec} \theta)$ equal to?

[CDS-2020-1]

- (a) $\frac{775}{168}$ (b) $\frac{725}{168}$
(c) $\frac{375}{84}$ (d) $\frac{325}{84}$

154. If $\operatorname{cosec} \theta - \sin \theta = m$ and $\sec \theta - \cos \theta = n$, then what is $m^{\frac{4}{3}} n^{\frac{2}{3}} + m^{\frac{2}{3}} n^{\frac{4}{3}}$ equal to?

[CDS-2020-2]

- (a) 0 (b) 1
(c) mn (d) $m^2 n^2$

155. If $\cos \theta + \sec \theta = k$, then what is the value of $\sin^2 \theta - \tan^2 \theta$?

[CDS-2020-2]

- (a) $4-k$ (b) $4-k^2$
(c) k^2-4 (d) k^2+2

156. ABC is a triangle inscribed in a semicircle of diameter AB. What is $\cos(A+B) + \sin(A+B)$ equal to?

[CDS-2020-2]

- (a) 0 (b) $1/4$
(c) $1/2$ (d) 1

157. Consider the following statements:

1. $\sin \theta = x + 1/x$ is possible for some real value of x .

2. $\cos \theta = x + 1/x$ is possible for some real value of x

Which of the above statements is/are correct?

[CDS-2020-2]

- (a) 1 Only (b) 2 Only
(c) Both 1 and 2 (d) Neither 1 nor 2

158. What is the magnitude (in radian) of the interior angle of a regular pentagon?

[CDS-2020-2]

- (a) $\frac{\pi}{5}$ (b) $\frac{2\pi}{5}$
(c) $\frac{3\pi}{5}$ (d) $\frac{4\pi}{5}$

159. The difference between two angles is 15° and the sum of the angles in radian is $\frac{5\pi}{12}$.

The bigger angle is k times the smaller angle. What is k equal to?

[CDS-2020-2]

- (a) $\frac{4}{3}$ (b) $\frac{3}{2}$
(c) $\frac{6}{5}$ (d) $\frac{7}{6}$

160. Consider the following statements:

1. The equation $2 \sin^2 \theta - \cos \theta + 4 = 0$ is possible for all θ .

2. $\tan \theta + \cot \theta$ cannot be less than 2, where $0 < \theta < \pi/2$.

Which of the above statements is/are correct?

[CDS-2020-2]

- (a) 1 Only (b) 2 Only
(c) Both 1 and 2 (d) 2

161. What is the maximum value of $3 \sin \theta - 4$?

[CDS-2020-2]

- (a) -4 (b) -1
(c) 0 (d) 1

162. If $\sin \theta + \cos \theta = \sqrt{2}$, then what is $\sin^6 \theta + \cos^6 + 6\sin^2 \theta \cos^2 \theta$ equal to?

[CDS-2020-2]

- (a) $\frac{1}{4}$ (b) $\frac{3}{4}$
(c) 1 (d) $\frac{7}{4}$

163. What is the least value of $9 \sin^2 \theta + 16 \cos^2 \theta$?

[CDS-2020-2]

- (a) 0 (b) 9
(c) 16 (d) 26

164. If $\cos 47^\circ + \sin 47^\circ = k$, then what is the value of $\cos^2 47^\circ - \sin^2 47^\circ$?

[CDS-2020-2]

- (a) $k\sqrt{2-k^2}$ (b) $-k\sqrt{2-k^2}$
(c) $k\sqrt{1-k^2}$ (d) $-k\sqrt{1-k^2}$

165. If $\operatorname{cosec} \theta - \sin \theta = p^3$ and $\sec \theta - \cos \theta = q^3$, then what is the value of $\tan \theta$?

[CDS-2020-2]

- (a) $\frac{p}{q}$ (b) $\frac{q}{p}$
(c) pq (d) p^2q^2

166. If $0 \leq \alpha \leq \beta \leq 90^\circ$ such that $\cos(\alpha - \beta) = 1$, then what is $\sin \alpha - \sin \beta + \cos \alpha - \cos \beta$ equal to?

[CDS-2020-2]

- (a) -1 (b) 0
(c) 1 (d) 2

167. Consider the following statements:

1. The value of $\cos 61^\circ + \sin 29^\circ$ cannot exceed 1.
 2. The value of $\tan 23^\circ - \cot 67^\circ$ is less than 0.
- Which of the above statements is/are correct?

[CDS-2020-2]

- (a) 1 Only (b) 2 Only
(c) Both 1 and 2 (d) Neither 1 nor 2

168. Consider the following inequalities:

1. $\sin 1^\circ < \cos 57^\circ$
 2. $\cos 60^\circ > \sin 57^\circ$
- Which of the above is/are correct?

[CDS-2021-1]

- (a) 1 Only (b) 2 Only

- (c) Both 1 and 2 (d) Neither 1 nor 2

169. If $p = \sec \theta - \tan \theta$ and $q = \operatorname{cosec} \theta + \cot \theta$, then what is $p + q$ ($p-1$) equal to?

[CDS-2021-1]

- (a) -1 (b) 0
(c) 1 (d) 2

170. If $\operatorname{cosec} \theta - \cot \theta = m$, then what is $\operatorname{cosec} \theta$ equal to?

[CDS-2021-1]

- (a) $m + \frac{1}{m}$ (b) $m - \frac{1}{m}$
(c) $\frac{m}{2} + \frac{2}{m}$ (d) $\frac{m}{2} + \frac{1}{2m}$

171. Let ABC be a triangle right angled at C, then what is $\tan A + \tan B$ equal to?

[CDS-2021-1]

- (a) $\frac{a}{bc}$ (b) $\frac{a^2}{bc}$
(c) $\frac{b^2}{ca}$ (d) $\frac{c^2}{ab}$

172. Let $\cos \alpha + \cos \beta = 2$ and $\sin \alpha + \sin \beta = 0$, where $0 \leq \alpha \leq 90^\circ$, $0 \leq \beta \leq 90^\circ$. What is the value of $\cos 2\alpha - \cos 2\beta$?

[CDS-2021-1]

- (a) 0
(b) 1
(c) 2
(d) Cannot be determined due to insufficient data

173. If $\sec \theta + \cos \theta = \frac{5}{2}$, where $0 \leq \theta \leq 90^\circ$, then what is the value of $\sin^2 \theta$?

[CDS-2021-1]

- (a) $\frac{1}{4}$ (b) $\frac{1}{2}$
(c) $\frac{3}{4}$ (d) 1

174. What is $(1 + \cot \theta - \operatorname{cosec} \theta)(1 + \tan \theta + \sec \theta)$ equal to?

[CDS-2021-1]

- (a) 4 (b) 3
(c) 2 (d) 1

175. If $6 + 8 \tan \theta = \sec \theta$ and $8 - 6 \tan \theta = k \sec \theta$, then what is the value of k^2 ?

[CDS-2021-1]

- (a) 11 (b) 22
(c) 77 (d) 99

- 176.** What is the least value of $3\sin^2 \theta + 4 \cos^2 \theta$?
[CDS-2021-1]
(a) 5 (b) 4
(c) 3 (d) 2
- 177.** If $\sin \theta \cos \theta = k$, where $0 \leq \theta \leq \pi/2$, then which one of the following is correct?
[CDS-2021-1]
(a) $0 \leq k \leq 1$ (b) $0 \leq k \leq 0.5$ only
(c) $0.5 \leq k \leq 1$ only (d) $0 < k < 1$
- 178.** If $p = \sin^2 \theta + \cos^4 \theta$ for $0 \leq \theta \leq \pi/2$, then consider the following statements:
1. p can be less than $\frac{3}{4}$
2. p can be more than 1
Which of the above statements is/are correct?
[CDS-2021-1]
(a) 1 Only (b) 2 Only
(c) Both 1 and 2 (d) Neither 1 nor 2
- 179.** What is the ratio of the greatest to the smallest value of $2 - 2 \sin x - \sin^2 x$, $0 \leq x \leq \pi/2$?
[CDS-2021-1]
(a) -3 (b) -1
(c) 1 (d) 3
- 180.** If the equation $x^2 + y^2 - 2xy \sin^2 \theta = 0$ contains real solution for x and y , then:
[CDS-2021-1]
(a) $x = y$ (b) $x = -y$
(c) $x = 2y$ (d) $2x = y$
- 181.** Between 3 and 4 O' clock, both hour hand and minute hand will coincide past 3 O'clock between
[CDS-2021-2]
(a) 15-16 minutes (b) 16-17 minutes
(c) 17-18 minutes (d) 18-19 minutes
- 182.** What is the minimum value of $\cos^3 \theta + \sec^3 \theta$, where $0^\circ \leq \theta < 90^\circ$?
[CDS-2021-2]
(a) 0 (b) 1
(c) 2 (d) None of these
- 183.** If $14\sin^2 \theta + 10\cos^2 \theta = 11$, where $0^\circ < \theta < 90^\circ$, then what is the value of $\tan \theta + \cot \theta$?
[CDS-2021-2]
(a) $\frac{4}{\sqrt{3}}$ (b) $\frac{2}{\sqrt{3}}$
(c) $\sqrt{3}$ (d) $2\sqrt{3}$
- 184.** What is $\frac{\sin^3 \theta + \cos^3 \theta}{\sin \theta + \cos \theta} + \frac{\sin^3 \theta - \cos^3 \theta}{\sin \theta - \cos \theta}$ equal to?
[CDS-2021-2]
(a) 0 (b) 1
(c) 2 (d) 4
- 185.** What is the maximum value of $1 + 2\sin^2 \theta \cos^2 \theta - \sin^4 \theta - \cos^4 \theta$, where $0^\circ < \theta < 90^\circ$?
[CDS-2021-2]
(a) 1 (b) 2
(c) 3 (d) 4
- 186.** If $\frac{\cos^2 \theta - 3\cos \theta + 2}{\sin^2 \theta} = 1$, where $0^\circ < \theta < 90^\circ$ then what is $\sin^2 \theta + \cos \theta$ equal to?
[CDS-2021-2]
(a) $5/4$ (b) $3/2$
(c) $7/4$ (d) 2
- 187.** Consider the following:
1. $\sin^4 \theta - \sin^2 \theta = \cos^4 \theta - \cos^2 \theta$
2. $\sin^4 \theta + \cos^4 \theta = 1 + 2\sin^2 \theta \cos^2 \theta$
3. $\tan^4 \theta + \tan^2 \theta = \sec^4 \theta - \sec^2 \theta$
Which of the above are identities?
[CDS-2021-2]
(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3
- 188.** What is the value of $\sin 24^\circ \sin 66^\circ - \cos 24^\circ \cos 66^\circ + \tan 24^\circ \tan 66^\circ - \cot 24^\circ \cot 66^\circ$?
[CDS-2021-2]
(a) 0 (b) 1
(c) 2 (d) 3
- 189.** If $x = p \sin A \cos B$, $y = p \sin A \sin B$ and $z = p \cos A$, then what is the value of $x^2 + y^2 + z^2$?
[CDS-2021-2]
(a) $-p^2$ (b) 0
(c) p^2 (d) $2p^2$
- 190.** If $x = m \sec A + n \tan A$ and $y = m \tan A + n \sec A$, then what is $x^2 - y^2$ equal to?
[CDS-2021-2]
(a) $m^2 - n^2$ (b) $m^2 + n^2$
(c) $m^2 + n^2 - mn$ (d) $m^2 - n^2 + mn$
- 191.** If for some θ lying between 0° and 90° , $\tan \theta = 1$, then what is the value of $\sin^2 \theta - 2 \sin \theta \cos \theta$?
[CDS-2021-2]
(a) -1 (b) 0
(c) $1/2$ (d) $-1/2$

192. What is $\frac{2\sin^3\theta - \sin\theta}{\cos\theta - 2\cos^3\theta}$, ($0^\circ < \theta < 90^\circ$) equal to ?

[CDS-2021-2]

- (a) $\sin\theta$ (b) $\cos\theta$
(c) $\tan\theta$ (d) $\cot\theta$

193. If A, B and C are interior angles of a triangle ABC, then what is $\tan\left(\frac{B+C}{2}\right) + \sin\left(\frac{B+C}{2}\right) - \cot\left(\frac{A}{2}\right) - \cos\left(\frac{A}{2}\right)$ equal to?

[CDS-2021-2]

- (a) 0 (b) $1/2$
(c) $\sin\left(\frac{A+B+C}{2}\right)$ (d) $\tan\left(\frac{A+B+C}{2}\right)$

194. In a triangle ABC, right-angled at B, AB + BC = $10(1+\sqrt{3})$ cm and length of hypotenuse is 20cm. What is the value of $\tan A + \cot A$?

[CDS-2021-2]

- (a) $\frac{4}{\sqrt{3}}$ (b) $\frac{2}{\sqrt{3}}$
(c) $\sqrt{3}$ (d) $2\sqrt{3}$

195. If A, B, C are acute angles and $\sin(B+C-A) = \cos(C+A-B) = \tan(A+B-C) = 1$ then what is (A + B + C) equal to?

[CDS-2022-1]

- (a) 90° (b) 120°
(c) 135° (d) 150°

196. If A, B, C, D are the angles of a cyclic quadrilateral, then what is the value of the following?

$$\sin\left(\frac{A+C}{2}\right) + \sin\left(\frac{B+D}{2}\right)$$

[CDS-2022-1]

- (a) 2 (b) 1
(c) 0 (d) -1

197. What is $(\sec^2\alpha + \tan\alpha \cdot \tan\beta - \tan^2\alpha)^2 + (\tan\alpha - \tan\beta)^2 - \sec^2\alpha \cdot \sec^2\beta$ equal to?

[CDS-2022-1]

- (a) -1 (b) 0
(c) 1 (d) 2

198. If $\tan\theta + \sec\theta = 3$, then what is the value of $3\tan\theta + 9\sec\theta$?

[CDS-2022-1]

- (a) 15 (b) 17
(c) 19 (d) 21

199. Consider the following:

$$1. \sqrt{\sec^2\theta + \operatorname{cosec}^2\theta} = \tan\theta + \cot\theta, \text{ where } 0 < \theta < 90^\circ$$

$$2. \sqrt{\tan^2\theta + \cot^2\theta + 4} = \sec\theta + \operatorname{cosec}\theta, \text{ where } 0 < \theta < 90^\circ.$$

Which of the above is/are an identity/identities?

[CDS-2022-1]

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

200. What is the length of the chord of a unit circle which subtends an angle 2θ at the centre, where $\theta < 45^\circ$?

[CDS-2022-1]

- (a) $\sin 2\theta$ (b) $\cos 2\theta$
(c) $2\sin\theta$ (d) $2\cos\theta$

201. If $\tan^2\theta + 3\sec\theta - 9 = 0$ where $0 < \theta < 90^\circ$, then what is the value of $12\cot^2\theta + 3\operatorname{cosec}\theta$?

[CDS-2022-1]

- (a) $(\sqrt{3} + 1)^2$ (b) $(\sqrt{3} + 2)^2$
(c) $(2\sqrt{3} + 1)^2$ (d) $(3\sqrt{3} + 1)^2$

202. If $\frac{\cos\theta}{\operatorname{cosec}\theta + 1} + \frac{\cos\theta}{\operatorname{cosec}\theta - 1} = 2$ where $0 < \theta < 90^\circ$, then what is the value of $\sin^4\theta + \cos^4\theta$?

[CDS-2022-1]

- (a) 2 (b) 1
(c) $\frac{1}{2}$ (d) $\frac{1}{4}$

203. If $x = \frac{1 - \cos\theta + \sin\theta}{1 + \sin\theta}$ then what is $\frac{\sin\theta + \cos\theta - 1}{\cos\theta}$ equal to?

[CDS-2022-1]

- (a) $\frac{1}{x}$ (b) x
(c) $1 + x$ (d) $x - 1$

204. If $\cos(x + y) = 0$ and $\sin(x - y) = \frac{1}{2}$, where $x, y \in \left[0, \frac{\pi}{2}\right]$, then what is the value of $\cot(2x - y)$?

[CDS-2022-1]

- (a) 0 (b) $\frac{1}{2}$
(c) 1 (d) 2

205. What is the minimum value of $\sin^4\theta + \cos^4\theta - 2\sin^2\theta \cos^2\theta$?

[CDS-2022-1]

- (a) 0
(b) 1

- (c) 1
(d) Minimum value does not exist

206. If $\cos\theta + \sec\theta - 2 = 0$, where $0 \leq \theta < \frac{\pi}{2}$, then what is the value of $\cos^4\theta + \sec^4\theta - 2$?

[CDS-2022-1]

- (a) -2 (b) -1
(c) 0 (d) 2

207. If $y = \cos^2 x + \sec^2 x$, where $0 \leq x < \frac{\pi}{2}$, then which one of the following is correct?

[CDS-2022-1]

- (a) $0 < y < 0.5$ (b) $0.5 \leq y < 1$
(c) $1 \leq y < 2$ (d) $y \geq 2$

208. What is the value of the following?

$$\frac{2 \sin 68^\circ}{\cos 22^\circ} - \frac{2 \cot 15^\circ}{5 \tan 75^\circ} - \frac{3 \tan 20^\circ \tan 40^\circ \tan 45^\circ \tan 50^\circ \tan 70^\circ}{5}$$

[CDS-2022-1]

- (a) -1 (b) 0
(c) 1 (d) 5

209. What is the angle between the hour hand and the minute hand of a clock when the clock shows 4 hours 40 minutes?

[CDS 2022 (II)]

- (a) 80° (b) 100°
(c) 120° (d) 220°

210. In a triangle ABC, $\angle A = 2\theta$, $\angle B = C = 4\theta$ and θ satisfies $4 \sin^2 \theta + 2 \sin \theta - 1 = 0$. What is the ratio of BC to AB?

[CDS 2022 (II)]

- (a) $\sqrt{5} - 1$ (b) $\frac{(\sqrt{5} - 1)}{2}$
(c) $\frac{(\sqrt{5} - 1)}{4}$ (d) $2(\sqrt{5} - 1)$

211. If $\tan \theta = \frac{\sqrt{q^2 - p^2}}{p}$; $0 < \theta < 90^\circ$, then what is $\sec \theta + \cos \theta + 2$ equal to?

[CDS 2022 (II)]

- (a) $\frac{p^2 + q^2}{pq}$ (b) $\frac{(p+q)^2}{pq}$
(c) $\frac{(p+q)^2}{2pq}$ (d) $\frac{(p-q)^2}{pq}$

212. If $11 \sin \theta + 60 \cos \theta = 61$; $0 < \theta < 90^\circ$, then what is the value of $\sqrt{660(\tan \theta + \cot \theta)}$?

[CDS 2022 (II)]

- (a) 61 (b) $61\sqrt{2}$
(c) 122 (d) $122\sqrt{2}$

213. What is the diameter of a circle inscribed in a regular polygon of 15 sides with side length unity?

[CDS 2022 (II)]

- (a) $0.5 \cot 12^\circ$ (b) $\cot 12^\circ$
(c) $0.5 \tan 12^\circ$ (d) $\tan 12^\circ$

214. If $x = \frac{1 + \sin \theta}{\cos \theta}$, then what is $\frac{\tan \theta + \sec \theta - 1}{\tan \theta - \sec \theta + 1}$ equal to?

[CDS 2022 (II)]

- (a) -x (b) x
(c) 2x (d) $\frac{x}{2}$

215. Consider the following:

- $\frac{\tan \theta + \sin \theta}{\tan \theta - \sin \theta} = \frac{\sec \theta + 1}{\sec \theta - 1}$, where $0 < \theta < \frac{\pi}{2}$
- $\frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta + \sin^2 \theta} = \frac{2 \tan \theta}{\tan^2 \theta + 1}$, where $0 < \theta < \frac{\pi}{2}$

Which of the above is/are identities?

[CDS 2022 (II)]

- (a) only 1 (b) only 2
(c) both 1 and 2 (d) neither 1 nor 2

216. If $\frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta} = 5$, where $0 < \theta < \frac{\pi}{2}$, $\theta \neq \frac{\pi}{4}$, then what is the value of $\frac{2 \sin \theta + 3 \cos \theta}{3 \sin \theta - 2 \cos \theta}$?

[CDS 2022 (II)]

- (a) $\frac{8}{5}$ (b) 2
(c) $\frac{12}{5}$ (d) 3

217. If $\frac{1}{\operatorname{cosec} \theta - \cot \theta} - \frac{1}{\sin \theta} = x$, then what is $\frac{1}{\operatorname{cosec} \theta + \cot \theta} - \frac{1}{\sin \theta}$ equal to, where $0 < \theta < \frac{\pi}{2}$?

[CDS 2022 (II)]

- (a) -x (b) x
(c) $\frac{1}{x}$ (d) $-\frac{1}{x}$

218. What is $2 \sin^6 \theta + 2 \cos^6 \theta - 3 \sin^4 \theta - 3 \cos^4 \theta$ equal to?

[CDS 2022 (II)]

- (a) -1 (b) 0
(c) 1 (d) 2

219. What is the minimum value of $3600 \sec^2 \theta + 121 \operatorname{cosec}^2 \theta$, where $0 < \theta < \frac{\pi}{2}$?

[CDS 2022 (II)]

- (a) 1320 (b) 2401
(c) 3721 (d) 5041

220. What is $\sqrt{\left(\frac{1+\sin \theta}{\cos \theta}\right)^2 + \left(\frac{\cos \theta}{1+\sin \theta}\right)^2} - 2$ equal to, where $0 < \theta < \frac{\pi}{2}$?

[CDS 2022 (II)]

- (a) $\tan \theta$ (b) $\cot \theta$
(c) $2 \tan \theta$ (d) $2 \cot \theta$

221. In a triangle ABC, $\angle A = 60^\circ$. What is $AB^2 + AC^2 - BC^2$ equal to?

[CDS 2022 (II)]

- (a) AB.AC (b) AB.BC
(c) AC.BC (d) 2AB.AC

222. What is the value of $\frac{\cos^2 32^\circ + \cos^2 58^\circ}{\sec^2 50^\circ - \cot^2 40^\circ} + 4 \tan 13^\circ \tan 37^\circ \tan 53^\circ \tan 77^\circ$?

[CDS - 2023 (1)]

- (a) 2 (b) 3
(c) 4 (d) 5

223. What is the value of $(1 + \cot^2 \theta)(1 + \cos \theta)(1 - \cos \theta) - (1 + \tan^2 \theta)(1 + \sin \theta)(1 - \sin \theta)$?

[CDS - 2023 (1)]

- (a) -1 (b) 0
(c) 1 (d) 2

224. If $2 \cos^2 \theta + \sin \theta - 2 = 0$, $0 < \theta \leq \frac{\pi}{2}$, then what is the value of θ ?

[CDS - 2023 (1)]

- (a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$
(c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

225. Consider the following:

$$1. \cos^4 \theta - \sin^4 \theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}, 0 < \theta < \frac{\pi}{2}$$

$$2. \operatorname{cosec} \theta + \cot \theta = \frac{1}{\operatorname{cosec} \theta - \cot \theta}, 0 < \theta < \frac{\pi}{2}$$

$$3. \cos^2 \theta - \sin^2 \theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}, 0 < \theta < \frac{\pi}{2}$$

Which of the above equations are identities.

[CDS - 2023 (1)]

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

226. If $\sin \theta = \frac{12}{13}$, then what is the value of $(\tan \theta + \sec \theta)^2 (\operatorname{cosec} \theta - \cot \theta)^{-2}$, $0 < \theta < \frac{\pi}{2}$?

[CDS - 2023 (1)]

- (a) $\frac{121}{4}$ (b) $\frac{169}{9}$
(c) $\frac{196}{9}$ (d) $\frac{225}{4}$

227. If $\tan^8 \theta + \cot^8 \theta = m$, then what is the value of $\tan \theta + \cot \theta$?

[CDS - 2023 (1)]

- (a) $\sqrt{\sqrt{m+2}+2}$ (b) $\sqrt{\sqrt{m+4}+2}$
(c) $\sqrt{\sqrt{\sqrt{m+2}+2}+2}$ (d) $\sqrt{\sqrt{\sqrt{m+4}+2}+2}$

228. What is the minimum value of $6 - 4 \sin \theta$, $0 \leq \theta \leq \frac{\pi}{2}$?

[CDS - 2023 (1)]

- (a) 1 (b) 2
(c) 4 (d) 6

229. What is the value of x that satisfies $4 \cos^2 30^\circ + 2x \sin 30^\circ - \cot^2 30^\circ - 6 \tan 15^\circ \tan 75^\circ = 0$?

[CDS - 2023 (1)]

- (a) 1 (b) 2
(c) 3 (d) 6

230. Which of the following is/are identity/identities?

$$1. \frac{\sin^3 \theta + \cos^3 \theta}{\sin \theta + \cos \theta} + \sin \theta \cos \theta = 1; 0 < \theta < \pi/2$$

$$2. 1 - \sin^6 \theta = \cos^2 \theta (\cos^4 \theta + 3 \sin^2 \theta)$$

Select the correct answer using the code given below:

[CDS-2023 (2)]

- (a) 1 only (b) 2 only
(c) both 1 and 2 (d) neither 1 nor 2

231. If $7 \sin^4 \theta + 9 \cos^4 \theta + 42 \sin^2 \theta = 16$, $0 < \theta < \pi/2$, then what is $\tan \theta$ equal to?

[CDS-2023 (2)]

- (a) 1 (b) $\sqrt{2}$

(c) $\sqrt{3}$

(d) $\frac{1}{\sqrt{3}}$

232. If $\tan A + \cot A = 2$, where $0 < A < 90^\circ$, then what is the value of $\tan^2 A + \tan^3 A + \tan^4 A + \dots + \tan^n A$?

[CDS-2023 (2)]

(a) 1

(b) $n-2$

(c) $n-1$

(d) n

233. If $3 \sin \theta + 5 \cos \theta = 5$, then what is the value of $5 \sin \theta - 3 \cos \theta$ equal to?

[CDS-2023 (2)]

(a) 5

(b) -3

(c) -2

(d) 0

234. What is $3(\sin x - \cos)^4 + 6(\sin x + \cos)^2 + 4(\sin x)^6 + 4(\cos x)^6$ equal to?

[CDS-2023 (2)]

(a) 9

(b) 11

(c) 13

(d) 15

235. What is the value of $\sin \theta + \cos \theta$, if θ satisfies the equation $\cot^2 \theta - (\sqrt{3}+1) \cot \theta + \sqrt{3} = 0$; $0 < \theta < \pi/4$?

[CDS-2023 (2)]

(a) $\sqrt{2}$

(b) 2

(c) $\frac{\sqrt{3}+1}{2}$

(d) $\frac{\sqrt{3}-1}{2}$

236. Which one of the following is a value of θ , if θ satisfies the equation $\tan 2\theta \tan 4\theta - 1 = 0$; $0 < \theta < \pi/2$?

[CDS-2023 (2)]

(a) $\pi/12$

(b) $\pi/15$

(c) $\pi/6$

(d) $\pi/5$

237. If $\tan x = \frac{\sin \theta + \cos \theta}{\sin \theta - \cos \theta}$, $\frac{\pi}{2} < \theta < \frac{3\pi}{2}$, then what is $\sqrt{2} \sin x$ equal to?

[CDS-2023 (2)]

(a) $\sin \theta + \cos \theta$

(b) $\sin \theta - \cos \theta$

(c) $\frac{\sin \theta + \cos \theta}{2}$

(d) $\frac{\sin \theta - \cos \theta}{2}$

238. How many values of θ will satisfy the equation $(\sin^2 \theta - 4 \sin \theta + 3)(4 - \cos^2 \theta + 4 \sin \theta) = 0$, where $0 < \theta < \pi/2$?

[CDS-2023 (2)]

(a) none

(b) only one

(c) only two

(d) only three

239. If $x \sin^3 \theta + y \cos^3 \theta = \sin \theta \cos \theta$ and $x \sin \theta - y \cos \theta = 0$, for every $\theta \in \left(0, \frac{\pi}{2}\right)$, then what is $x^2 + y^2$ equal to?

[CDS-2023 (2)]

(a) 0

(b) 1

(c) 2

(d) 3

240. For what relation between a and b is the equation $\sin \theta = \frac{a+b}{2\sqrt{ab}}$ possible?

[CDS-2023 (2)]

(a) $a = b$

(b) $a \leq b$

(c) $a \geq b$

(d) $a > b$

241. If $\tan (3A) = \cot (A - 22^\circ)$ where $3A$ is an acute length, then what is the value of A ?

[CDS-2024 (1)]

(a) 25°

(b) 27°

(c) 28°

(d) 30°

242.

If $\frac{\sin \theta - \cos \theta + 1}{\sin \theta + \cos \theta - 1} = p \sec \theta + q \tan \theta$,

where $0 < \theta < \frac{\pi}{2}$, then what is $p + q$ equal to?

[CDS-2024 (1)]

(a) 0

(b) 1

(c) 2

(d) 4

243. Consider the following statements

1. In a triangle ABC, if

$\sin A + \sin B + \sin C = \frac{3\sqrt{3}}{2}$, then the

triangle can be equilateral

2. In a triangle ABC, if

$\cos A + \cos B + \cos C = \frac{3}{2}$, then the

triangle can be equilateral.

Which of the statements given below is/are correct?

[CDS-2024 (1)]

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

244. If $3 \sin \theta + 5 \cos \theta = 5$, then what is the value of $5 \sin \theta - 3 \cos \theta$?

[CDS-2024 (1)]

(a) -3

(b) -2

(c) 5

(d) 8

245. What is the minimum value of $\frac{\sin^2 A + 5 \sin A + 1}{\sin A}$ for $0 < A < \frac{\pi}{2}$?

[CDS-2024 (1)]

(a) 3

(b) 5

(c) 7

(d) 9

246. What is the difference between the greatest value and the least value of $\cos^2 \theta + 3 \sin^2 \theta + 2$?

[CDS-2024 (1)]

- (a) 4
(b) 3
(c) 2
(d) 1

247. If $t = \cos 79^\circ$, then what is $\operatorname{cosec} 79^\circ (1 - \cos 79^\circ)$ equal to?

[CDS-2024 (1)]

- (a) $\sqrt{\frac{1+t}{1-t}}$
(b) $\frac{t}{\sqrt{1-t^2}}$
(c) $\sqrt{\frac{1-t^2}{t}}$
(d) $\sqrt{\frac{1-t}{1+t}}$

ANSWER KEY

1.	b	2.	c	3.	a	4.	b	5.	d	6.	c	7.	c	8.	b	9.	c	10.	b
11.	d	12.	c	13.	d	14.	c	15.	a	16.	a	17.	c	18.	b	19.	a	20.	a
21.	b	22.	a	23.	c	24.	c	25.	a	26.	b	27.	c	28.	c	29.	d	30.	d
31.	b	32.	c	33.	a	34.	d	35.	b	36.	a	37.	d	38.	a	39.	c	40.	a
41.	b	42.	d	43.	d	44.	b	45.	d	46.	d	47.	a	48.	b	49.	d	50.	c
51.	d	52.	c	53.	c	54.	c	55.	c	56.	b	57.	b	58.	b	59.	b	60.	c
61.	a	62.	b	63.	d	64.	a	65.	b	66.	b	67.	c	68.	a	69.	b	70.	a
71.	a	72.	b	73.	a	74.	d	75.	c	76.	d	77.	c	78.	b	79.	b	80.	d
81.	c	82.	a	83.	b	84.	c	85.	d	86.	c	87.	d	88.	b	89.	a	90.	d
91.	c	92.	d	93.	b	94.	c	95.	b	96.	a	97.	a	98.	a	99.	c	100.	d
101.	a	102.	c	103.	d	104.	a	105.	a	106.	a	107.	c	108.	a	109.	b	110.	d
111.	b	112.	b	113.	a	114.	a	115.	c	116.	d	117.	a	118.	b	119.	b	120.	b
121.	c	122.	c	123.	c	124.	c	125.	c	126.	d	127.	b	128.	a	129.	b	130.	c
131.	b	132.	b	133.	c	134.	a	135.	b	136.	a	137.	c	138.	a	139.	d	140.	c
141.	b	142.	c	143.	b	144.	a	145.	a	146.	d	147.	b	148.	b	149.	a	150.	c
151.	c	152.	d	153.	a	154.	b	155.	b	156.	d	157.	d	158.	c	159.	b	160.	b
161.	b	162.	d	163.	b	164.	b	165.	b	166.	b	167.	a	168.	a	169.	a	170.	d
171.	d	172.	a	173.	c	174.	c	175.	d	176.	c	177.	b	178.	d	179.	a	180.	a
181.	b	182.	c	183.	a	184.	c	185.	a	186.	a	187.	c	188.	a	189.	c	190.	a
191.	d	192.	c	193.	a	194.	a	195.	c	196.	a	197.	b	198.	c	199.	a	200.	c
201.	a	202.	c	203.	b	204.	a	205.	a	206.	c	207.	d	208.	c	209.	b	210.	b
211.	a	212.	a	213.	b	214.	a	215.	c	216.	a	217.	a	218.	c	219.	c	220.	a
221.	a	222.	d	223.	b	224.	a	225.	b	226.	d	227.	c	228.	b	229.	d	230.	c
231.	d	232.	c	233.	b	234.	c	235.	c	236.	a	237.	a	238.	c	239.	b	240.	a
241.	c	242.	c	243.	c	244.	a	245.	c	246.	c	247.	d						

PRACTICE SHEET

1. The angles of elevation of the top of a tower from the top and foot of a pole are respectively 30° and 45° is h_T is the height of the tower and h_p is the height of the pole, then which of the following are correct?
 1. $\frac{2h_p h_T}{3 + \sqrt{3}} = h_p^2$
 2. $\frac{h_T - h_p}{\sqrt{3} + 1} = \frac{h_p}{2}$
 3. $\frac{2(h_p + h_T)}{h_p} = 4 + \sqrt{3}$

Select the correct answer using the code given below:

(a) 1 and 3 only (b) 2 and 3 only
(c) 1 and 2 only (d) 1, 2 and 3
2. From the top of a lighthouse, 100 m high, the angle of depression of a boat is $\tan^{-1}\left(\frac{5}{12}\right)$. What is the distance between the boat and the lighthouse?

(a) 120 m (b) 180 m
(c) 240 m (d) 360 m
3. The top of a hill when observed from the top and bottom of a building of height h is at angles of elevation p and q respectively. What is the height of the hill?

(a) $\frac{h \cot q}{\cot q - \cot p}$ (b) $\frac{h \cot p}{\cot p - \cot q}$
(c) $\frac{2h \tan p}{\tan p - \tan q}$ (d) $\frac{2h \tan q}{\tan q - \tan p}$
4. A moving boat is observed from the top of a cliff of 150 m height. The angle of depression of the boat changes from 60° to 45° in 2 minutes. What is the speed of the boat in metres per hour?

(a) $\frac{4500}{\sqrt{3}}$ (b) $\frac{4500(\sqrt{3}-1)}{\sqrt{3}}$
(c) $4500\sqrt{3}$ (d) $\frac{4500(\sqrt{3}+1)}{\sqrt{3}}$
5. Two poles are 10 m and 20 m high. The line joining their tops makes an angle of 15° with the horizontal. The distance between the poles is approximately equal to

(a) 36.3 m (b) 37.3 m
(c) 38.3 m (d) 39.3 m
6. A vertical tower standing on a leveled field is mounted with a vertical flag staff of length 3 m. From a point on the field, the angles of elevation of the bottom and tip of the flag staff are 30° and 45° , respectively. Which one of the following gives the best approximation to the height of the tower?

(a) 3.90 m (b) 4.00 m
(c) 4.10 m (d) 4.25 m
7. The angle of elevation of the top of a tower from a point 20 m away from its base is 45° . What is the height of the tower?

(a) 10 m (b) 20 m
(c) 30 m (d) 40 m
8. The angles of elevation of the top of a tower standing on a horizontal plane from two points on a line passing through the foot of the tower at distances 49 m and 36 m are 43° and 47° respectively. What is the height of the tower?

(a) 40 m (b) 42 m
(c) 45 m (d) 47 m
9. A lamp post stands on a horizontal plane. From a point situated at a distance 150 m from its foot, the angle of elevation of the top is 30° . What is the height of the lamp post?

(a) 50 m (b) $50\sqrt{3}$ m
(c) $\frac{50}{\sqrt{3}}$ m (d) 100 m
10. From an aeroplane above a straight road the angles of depression of two positions at a distance 20 m apart on the road are observed to be 30° and 45° . The height of the aeroplane above the ground is

(a) $10\sqrt{3}$ m (b) $10(\sqrt{3}-1)$ m
(c) $10(\sqrt{3}+1)$ m (d) 20 m
11. If the angles of elevation of the top of a tower from two places situated at distances 21 m and x m from the base of the tower are 45° and 60° respectively, then what is the value of x ?

- (a) $7\sqrt{3}m$ (b) $7 - \sqrt{3}m$
(c) $7 + \sqrt{3}m$ (d) 14
- 12.** A person standing on the bank of a river observes that the angle subtended by a tree on the opposite of bank is 60° . When he returns 40 m from the bank, he finds the angle to be 30° . What is the breadth of the river?
(a) 60 m (b) 40 m
(c) 30 m (d) 20 m
- 13.** The angle of elevation of the top of a tower of height H from the foot of another tower in the same plane is 60° and the angle of elevation of the top of the second tower from the foot of the first tower is 30° . If is the height of the other tower, then which one of the following is correct?
(a) $H=2h$ (b) $H = \sqrt{3} h$
(c) $H = 3h$ (d) None of these
- 14.** A man walks 10 m towards a lamp post and notices that the angle of elevation of the top of the post increases from 30° to 45° . the height of the lamp post is
(a) 10 m (b) $(5\sqrt{3}+5)m$
(c) $(5\sqrt{3}-5)m$ (d) $(10\sqrt{3}+10)m$
- 15.** The shadow of a tower standing on a level plane is found to be 50 m longer when the Sun's elevation is 30° than when it is 60° . The height of the tower is
(a) 25 m (b) $25\sqrt{3} m$
(c) 50 m (d) none of these
- 16.** The top of a hill observed from the top and bottom of a building of height h is at angles of elevation α and β respectively. The height of the hill is
(a) $\frac{h \cot \beta}{\cot \beta - \cot \alpha}$ (b) $\frac{h \cot \alpha}{\cot \alpha - \cot \beta}$
(c) $\frac{h \cot \alpha}{\tan \alpha - \tan \beta}$ (d) None of these
- 17.** From the top of a lighthouse 70 high with its base at sea level, the angle of depression of a boat is 15° . The distance of the boat from the foot of the lighthouse is
(a) $70(2 - \sqrt{3})m$ (b) $70(2 + \sqrt{3})m$
(c) $70(3 - \sqrt{3})m$ (d) $70(3 + \sqrt{3})m$
- 18.** Two poles are 10 m and 20 m high. The line joining their tips makes an angle of 15° with the horizontal. What is the distance between the poles?
(a) $10(\sqrt{3}-1)m$ (b) $5(4+2\sqrt{3})m$
(c) $20(\sqrt{3}+1)$ (d) $10(\sqrt{3}+1)m$
- 19.** The angle of elevation of a tower at a level ground is 30° . The angle of elevation becomes θ when moved 10 m towards the tower. If the height of tower is $5\sqrt{3} m$, then what is the value of θ ?
(a) 45° (b) 60°
(c) 75° (d) None
- 20.** From the top of a building of height h m, the angle of depression of an object on the ground is θ . What is the distance (in m) of the object from the foot of the building?
(a) $h \cot \theta$ (b) $h \tan \theta$
(c) $h \cos \theta$ (d) $h \sin \theta$
- 21.** The angle of elevation of the tip of a flag staff from a point 10 m due South of its base is 60° . What is the height of the flag staff correct to nearest meter?
(a) 15 m (b) 16 m
(c) 17 m (d) 18 m
- 22.** What is the angle subtended by 1 m pole at distance 1 km on the ground in sexagesimal measure?
(a) $\frac{9}{50\pi}$ degree (b) $\frac{9}{5\pi}$ degree
(c) 3.4 min (d) 3.5 min
- 23.** A tower of height 15 m stands vertically on the ground. From a point on the ground the angle of elevation of the top of the tower is found to be 30° . What is the distance of the point from the foot of the tower?
(a) $15\sqrt{3} m$ (b) $10\sqrt{3}$
(c) $5\sqrt{3} m$ (d) 30
- 24.** At a point 15 m away from the base of a 15 m high house, the angle of elevation of the top is
(a) 90° (b) 60°
(c) 45° (d) 30°
- 25.** A vertical tower stands on a horizontal plane and is surmounted by a vertical flag staff of elevation of the bottom of the flag

staff is β and that of the top is α . What is the height of the tower?

- (a) $\frac{h \tan \beta}{\tan \alpha}$ (b) $\frac{h \tan \beta}{\tan \alpha + \tan \beta}$
 (c) $\frac{h \cos \beta}{\tan \alpha - \tan \beta}$ (d) $\frac{h}{\cos(\alpha - \beta)}$

- 26.** An aeroplane flying at a height of 300 m above the ground passes vertically above another plane at an instant when the angles of elevation of two planes from the same point on the ground are 60° and 45° , respectively. What is the height of the lower plane from the ground?

- (a) 50 m (b) $\frac{100}{\sqrt{3}}$ m
 (c) $100\sqrt{3}$ m (d) $150(\sqrt{3} + 1)$ m

- 27.** A man standing on the bank of a river observes that the angle of elevation of the top of a tree just on the opposite bank is 60° . The angle of elevation is 30° from a point at a distance y m from the bank. What is the height of the tree?

- (a) y m (b) $2y$ m
 (c) $\frac{\sqrt{3}y}{2}$ m (d) $\frac{y}{2}$ m

- 28.** Two poles are 10 m and 20 m high. The line joining their tops makes an angle of 15° with the horizontal. What is the approximate distance between the poles?

- (a) 35.3 m (b) 37.3 m
 (c) 41 m (d) 44 m

- 29.** From the top of a lighthouse 120 m above the sea, the angle of depression of a boat is 15° . What is the distance of the boat from the lighthouse?

- (a) 400 m (b) 421 m
 (c) 444 m (d) 460 m

- 30.** The angle of elevation of the top of a flag post from a point 5 m away from its base is 75° . What is the approximate height of the flag post?

- (a) 15 m (b) 17 m
 (c) 19 m (d) 21 m

ANSWER KEY

1.	(c)	2.	(c)	3.	(b)	4.	(b)	5.	(b)	6.	(c)	7.	(b)	8.	(b)	9.	(b)	10.	(c)
11.	(c)	12.	(d)	13.	(c)	14.	(b)	15.	(b)	16.	(b)	17.	(b)	18.	(b)	19.	(b)	20.	(a)
21.	(c)	22.	(a)	23.	(a)	24.	(c)	25.	(a)	26.	(c)	27.	(c)	28.	(b)	29.	(c)	30.	(c)

CDS PYQ

DIRECTIONS (for next four): Read the following information carefully to answer the questions that follow As seen from the top and bottom of a building of height h m, the angles of elevation of the top of a tower of height $\frac{3+\sqrt{3}}{2}h$ are α and β , respectively.

[CDS 2013 (I)]

1. If $\beta = 30^\circ$, then what is the value of $\tan \alpha$?
(a) $1/2$ (b) $1/3$
(c) $1/4$ (d) none of these
2. If $\alpha = 30^\circ$, then what is the value of $\tan \beta$?
(a) 1 (b) $1/2$
(c) $1/3$ (d) none of these
3. If $\alpha = 30^\circ$ and $h = 30$ m, then what is the distance between the base of the building and the base of the tower?
(a) $15+15\sqrt{3}$ m (b) $30+15\sqrt{3}$ m
(c) $45+15\sqrt{3}$ m (d) None of these
4. If $\beta = 30^\circ$ and if θ is the angle of depression of the foot of the tower as seen from the top of the building, then what is the value of $\tan \theta$?
(a) $\frac{(3-\sqrt{3})}{3\sqrt{3}}$ (b) $\frac{(3+\sqrt{3})}{3\sqrt{3}}$
(c) $\frac{(2-\sqrt{3})}{3\sqrt{3}}$ (d) none of these
5. A spherical balloon of radius r subtends angle 60° at the eye of an observer. If the angle of elevation of its centre is 60° and h is the height of the centre of the balloon, then which one of the following is correct?

[CDS 2013 (II)]

- | | |
|------------|-------------------|
| (a) $h=r$ | (b) $h=\sqrt{2}r$ |
| (c) $h=3r$ | (d) $h=2r$ |

6. What is the angle of elevation of the Sun, when the shadow of a pole of height x m is $\frac{x}{\sqrt{3}}$ m?

[CDS 2013 (II)]

- | | |
|----------------|----------------|
| (a) 30° | (b) 45° |
| (c) 60° | (d) 75° |

7. The shadow of a tower standing on a level plane is found to be 50 m longer when the Sun's elevation is 30° , then when it is 60° . What is the height of the tower?

[CDS 2014 (I)]

- | | |
|---------------------|--------------------|
| (a) 25m | (b) $25\sqrt{3}$ m |
| (c) $25/\sqrt{3}$ m | (d) 30 m |

8. The angle of elevation of the top of a tower 30 m high from the foot of another tower in the same plane is 60° and the angle of elevation of the top of the second tower from the foot of the first tower is 30° . The distance between the two towers in m times the height of the shorter tower. What is m equal to?

[CDS 2014 (I)]

- | | |
|------------------|------------------|
| (a) $\sqrt{2}$ m | (b) $\sqrt{3}$ m |
| (c) $1/2$ | (d) $1/3$ |

9. From a certain point on a straight road, a person observe a tower in the West direction at a distance of 200 m. He walks some distance along the road and finds that the same tower is 300 m South of him. What is the shortest distance of the tower from the road?

[CDS 2014 (II)]

- | | |
|-------------------------------|-------------------------------|
| (a) $\frac{300}{\sqrt{13}}$ m | (b) $\frac{500}{\sqrt{13}}$ m |
| (c) $\frac{600}{\sqrt{13}}$ m | (d) $\frac{900}{\sqrt{13}}$ m |

10. If from the top of a post a string twice the length of the post is stretched tight to a point on the ground, then what angle will the string make with the post?

[CDS 2014 (II)]

- | | |
|---------------|-------------|
| (a) $\pi/6$ | (b) $\pi/4$ |
| (c) $5\pi/12$ | (d) $\pi/3$ |

11. The angles of elevation of the top of a tower from two points P and Q at distances m^2 and n^2 respectively, from the base and in the same straight line with it are complementary. The height of the tower is

[CDS 2015 (I)]

- | | |
|------------------|----------------|
| (a) $(mn)^{1/2}$ | (b) $mn^{1/2}$ |
| (c) $m^{1/2}n$ | (d) mn |

- 12.** The angle of elevation of a cloud from a point 200 m above a lake is 30° and the angle of depression of its reflection in the lake is 60° . The height of the cloud is
[CDS 2015 (I)]
(a) 200 m (b) 300 m
(c) 400 m (d) 600 m
- 13.** From the top of a tower, the angles of depression of two objects P and Q (situated on the ground on the same side of the tower) separated at a distance of $100(3-\sqrt{3})$ m are 45° and 60° respectively. The height of the tower is
[CDS 2015 (I)]
(a) 200 m (b) 250 m
(c) 300 m (d) none of these
- 14.** An aeroplane flying at a height of 3000 m passes vertically above another aeroplane at an instant when the angles of elevation of the two planes from some point on the ground are 60° and 45° respectively. Then the vertical distance between the two planes is
[CDS 2015 (II)]
(a) $1000(\sqrt{3}-1)$ m (b) $1000\sqrt{3}$ m
(c) $(1000(3-\sqrt{3}))$ m (d) $3000\sqrt{3}$ m
- 15.** A pole is standing erect on the ground which is horizontal. The tip of the pole is tied tight with a rope of length $\sqrt{12}$ m to a point on the ground. If the rope is making 30° with the horizontal, then the height of the pole is
[CDS 2015 (II)]
(a) $2\sqrt{3}$ m (b) $3\sqrt{2}$ m
(c) 3 m (d) $\sqrt{3}$ m
- 16.** Two observers are stationed due north of a tower (of height x metre) at a distance y metre from each other. The angles of elevation of the tower observed by them are 30° and 45° respectively. Then x/y is equal to
[CDS 2016 (I)]
(a) $\frac{\sqrt{2}-1}{2}$ (b) $\frac{\sqrt{2}-1}{2}$
(c) $\frac{\sqrt{3}+1}{2}$ (d) 1
- 17.** A man from the top of a 100 m high tower sees a car moving towards the tower at an angle of depression 30° . After some time, the angle of depression becomes 60° . What is the distance travelled by the car during this time?
[CDS 2016 (II)]
(a) $100\sqrt{3}$ m (b) $\frac{200\sqrt{3}}{3}$ m
(c) $\frac{100\sqrt{3}}{3}$ m (d) $200\sqrt{3}$ m
- 18.** Two men on either side of a tower 75 m high observe the angle of elevation of the top of the tower to be 30° and 60° . What is the distance between the two men?
[CDS 2016 (II)]
(a) $100\sqrt{3}$ m (b) $75\sqrt{3}$ m
(c) $\frac{100\sqrt{3}}{3}$ m (d) $60\sqrt{3}$ m
- 19.** If the length of the shadow of a tower is equal to its height, then what is the Sun's altitude at that time?
[CDS 2016 (II)]
(a) 15° (b) 30°
(c) 45° (d) 60°
- 20.** A pole stands vertically inside a triangular park ABC. If the angle of elevation of the top of the pole from each corner of the park is same, then in the triangle ABC, the foot of the pole is at the
[CDS 2016 (II)]
(a) Centroid (b) Circumference
(c) Incentre (d) Orthocentre
- 21.** An aeroplane flying at a height of 300 m above the ground passes vertically above another plane at an instant when the angles of elevation of the two planes from the same point on the ground are 60° and 45° respectively. What is the height of the lower plane from the ground?
[CDS 2017 (I)]
(a) $100\sqrt{3}$ m (b) $\frac{100}{\sqrt{3}}$ m
(c) $50\sqrt{3}$ m (d) $150(\sqrt{3}+1)$ m
- 22.** From the top of a building 90 m high, the angles of depression of the top and the bottom of a tree are 30° and 45° respectively. What is the height of the tree?
[CDS 2017 (I)]
(a) $30\sqrt{3}$ m (b) $90-30\sqrt{3}$ m
(c) $90+30\sqrt{3}$ m (d) $60+30\sqrt{3}$ m
- 23.** From an aeroplane vertically over a straight horizontal road, the angles of depression of

two consecutive kilometre- stones on the opposite sides of the aeroplane are observed to be A and B. The height of the aeroplane above the road is

[CDS 2017 (I)]

- | | |
|---|---|
| (a) $\frac{\tan \alpha + \tan \beta}{\tan \alpha \tan \beta}$ | (b) $\frac{\tan \alpha \tan \beta}{\tan \alpha + \tan \beta}$ |
| (c) $\frac{\cot \alpha \cot \beta}{\cot \alpha + \cot \beta}$ | (d) $\frac{\cot \alpha + \cot \beta}{\cot \alpha \cot \beta}$ |

24. Let AB represent a building of height h metre with A being its top, B being its bottom. Let A'B' represent a tower of height (h + x) metre (x > 0) with A' being its top and B' being its bottom. Let BB' = d metre. Let the angle of elevation of A' as seen from A be 45°.

Consider the following statements:

Statement I: $h + x > d$

Statement II: The angle of depression of B as seen from A' is less than 45° Which one of the following is correct in respect of the above statements?

[CDS 2017 (II)]

- (a) Both Statement I and Statement II are true and Statement II is the correct explanation of Statement I
 (b) Both Statement I and Statement II are true but Statement II is not the correct explanation of Statement I
 (c) Statement I is true but Statement II is false
 (d) Statement I is false but Statement II is true 58.

25. A man, standing at a point X on the bank XY of a river that cannot be crossed, observes a tower to be N α° E on the opposite parallel bank. He then walks 200 m along the bank to the point Y towards East, and finds the tower to be N β° W. From these observations, the breadth of the river will be (Given that $\tan \alpha^\circ = 2$ and $\tan \beta^\circ = 0.5$)

[CDS 2017 (II)]

- | | |
|----------|----------|
| (a) 60 m | (b) 70 m |
| (c) 80 m | (d) 90 m |

26. Each side of a square subtends an angle of 60° at the tip of a tower of height h metres standing at the centre of the square. If l is the length of each side of the square, then what is h^2 equal to?

[CDS 2018 (I)]

- | | |
|----------------------|----------------------|
| (a) $2l^2$ | (b) $\frac{l^2}{2}$ |
| (c) $\frac{3l^2}{2}$ | (d) $\frac{2l^2}{3}$ |

27. From a height of h units, a man observes the angle of elevation as α and angle of depression as β of the top and the bottom respectively of a tower of height H (>4h). To what further height should he climb so that the values of angle of elevation and angle of depression get interchanged for the top and bottom of the tower?

[CDS 2018 (I)]

- | | |
|----------------|----------------|
| (a) H-h units | (b) H-2h units |
| (c) H-3h units | (d) H-4h units |

28. On the top of a hemispherical dome of radius r, there stands a flag of height h. From a point on the ground, the elevation of the top of the flag is 30°. After moving a distance d towards the dome, when the flag is just visible, the elevation is 45°. The ratio of h to r is equal to:

[CDS 2018 (II)]

- | | |
|-------------------------------------|---|
| (a) $\sqrt{2} - 1$ | (b) $\frac{\sqrt{3}+1}{2\sqrt{2}}$ |
| (c) $\frac{\sqrt{3}+1}{2\sqrt{2}}d$ | (d) $\frac{(\sqrt{3}+1)(\sqrt{2}-1)}{2\sqrt{2}}d$ |

29. Consider a regular hexagon ABCDEF. Two towers are situated at B and C. The angle of elevation from A to the top of the tower at B is 30°, and the angle of elevation to the top of the tower at C is 45°. What is the height of towers at B and C?

[CDS-2019-1]

- | | |
|-------------------|--------------------|
| (a) 1: $\sqrt{3}$ | (b) 1:3 |
| (c) 1:2 | (d) 1:2 $\sqrt{3}$ |

30. The angles of elevation of the tops of two pillars of heights h and 2h from a point P on the line joining the feet of the two pillars are complementary. If the distances of the foot of the pillars from the point P are x and y respectively, then which one of the following is correct?

[CDS-2019-2]

- | | |
|--------------------|----------------------|
| (a) $2h^2 = x^2 y$ | (b) $2h^2 = xy^2$ |
| (c) $2h^2 = xy$ | (d) $2h^2 = x^2 y^2$ |

31. A tree of height 15 m is broken by wind in such a way that its top touches the ground and makes an angle 30° with the ground.

What is the height from the ground to the point where tree is broken?

[CDS-2020-1]

- (a) 10m (b) 7m
(c) 5m (d) 3m

32. On a plane area there are two vertical towers separated by 100 feet apart. The shorter tower is 40 feet tall. A pole of length 6 feet stands on the line joining the base of two towers so that the tip of the towers and tip of the pole are also on the same line. If the distance of the pole from the shorter tower is 75feet, then what is the height of the taller tower (approximately)?

[CDS-2020-1]

- (a) 85feet (b) 110feet
(c) 125feet (d) 140fet

33. The angles of elevation of the top of a tower from two points at distances p and q from the base and on the same straight line are 27° and 63° respectively. What is the height of the tower?

[CDS-2020-1]

- (a) pq (b) $\sqrt{pq}$
(c) $\frac{pq}{2}$ (d) $\frac{\sqrt{pq}}{2}$

34. A pole on the ground leans at 60° with the vertical. At a point x metre away from the base of the pole on the ground, two halves of the pole subtended the same angle. If the pole and point are in the same vertical plane, then what is the length of the pole?

[CDS-2021-1]

- (a) $\sqrt{2}x$ metre (b) $\sqrt{3}x$ metre
(c) 2x metre (d) $2\sqrt{2}x$ metre

35. A vertical tower standing at the corner of a rectangular field subtends angles of 60° and 45° at the two nearer corners. If θ is the angle that the tower subtends at the farthest corner, then what is $\cot\theta$ equal to?

[CDS-2021-1]

- (a) $\frac{1}{2}$ (b) 2
(c) $\frac{2}{\sqrt{3}}$ (d) $\frac{4}{\sqrt{3}}$

36. A ladder 10m long reaches a point 10m below the top of a vertical flagstaff. From foot of the ladder, the elevation of top of the

flagstaff is 60° . What is the height of flagstaff?

[CDS-2021-2]

- (a) 12m (b) 15m
(c) 16m (d) 20m

37. From an aero-plane flying above a river at an altitude of 1200m, it is observed that the angles of depression of opposite points on the two banks of a river are 30° and θ . If the width of the river is 3000m, then which one of the following is correct?

[CDS-2021-2]

- (a) $\theta < 30^\circ$ (b) $30^\circ < \theta < 45^\circ$
(c) $45^\circ < \theta < 60^\circ$ (d) $60^\circ < \theta < 90^\circ$

38. An aeroplane is observed to the approaching the airport. It is at a distance of 10km from the point of observation on the ground and makes an angle of elevation θ . If the aeroplane is at a height of 8km above the ground, then which one of the following is correct?

[CDS-2022-1]

- (a) $0 < \theta < 30^\circ$ (b) $30^\circ < \theta < 45^\circ$
(c) $45^\circ < \theta < 60^\circ$ (d) $60^\circ < \theta < 90^\circ$

39. The angle of elevation of the top of a tower of height x meter from a point on the ground is found to be 60° . By going y meter away from that point, it becomes 30° . Which one of the following relations is correct?

[CDS-2022-1]

- (a) $x = y$ (b) $2x = 3y$
(c) $2x = \sqrt{3}y$ (d) $2y = \sqrt{3}x$

Consider the following for the next two (02) items that follow:

The angle of elevation of a cloud at C from a point (P), H metres above the surface of a lake is 30° . The height of the cloud from the surface of the lake is 2H metres. Let θ be the angle of depression of the reflection of the cloud in the lake from the point P.

40. What is the value of θ ?

[CDS 2022 (II)]

- (a) 30°
(b) 45°
(c) 60°
(d) cannot be determined due to insufficient data

41. What is PC equal to?

[CDS 2022 (II)]

- (a) H metres (b) $\sqrt{2}H$ metres
(c) $\sqrt{3}H$ metres (d) $2H$ metres

42. A person on the top of a vertical tower observes a car moving at a uniform speed coming directly towards it. If it takes 6 minutes for the angle of depression to change from 30° to 45° , and further t minutes to reach the tower, which one of the following is correct?

[CDS - 2023 (1)]

- (a) $7.7 < t < 8$ (b) $8 < t < 8.3$
(c) $8.3 < t < 8.6$ (d) $8.6 < t < 8.9$

43. A woman is standing on the deck of a ship, which is h (in metres) above water level. She observes the angle of elevation of the top of a tower as 60° and the angle of depression of the base of the tower as 30° . What is the height of the tower?

[CDS - 2023 (1)]

- (a) $2h$ (b) $3h$
(c) $4h$ (d) $5h$

Consider the following for the next two (02) items that follow:

A flagstaff stands on the top of a vertical tower. The angle of elevation of the top of the flagstaff from a certain place on the same horizontal level with the base of the tower is found to be α . Advancing a distance d towards the tower in the same horizontal plane, the angle of elevation of the top of the flagstaff is observed to be β and that of the top of the tower is observed to be γ . Let H be the height of the top of the flagstaff from the base of the tower and h be the height of the tower.

44. Which one of the following is correct?

[CDS-2023 (2)]

- (a) $H \tan \gamma - h \tan \beta = 0$ (b) $h \tan \gamma - H \tan \beta = 0$

- (c) $H \tan \gamma - h \tan \alpha = 0$ (d) $h \tan \gamma - H \tan \alpha = 0$

45. Which one of the following is correct?

[CDS-2023 (2)]

- (a) $d = \frac{H(\cot \alpha - \cot \beta)}{2}$ (b) $d = \frac{H(\tan \alpha - \tan \beta)}{2}$
(c) $d = H(\cot \alpha - \cot \beta)$ (d) $d = H(\tan \alpha - \tan \beta)$

Consider the following for the next two (02) items that follow:

AB is a straight road leading to the foot P of a tower of height h . Q is at distance x from P and R is at a distance y from Q. (R is farther from P and Q. R, Q are on the same side). The angle of elevation of the top of the tower at Q is twice of that at R. (use the formula $\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$)

46. Which one of the following is correct?

[CDS-2023 (2)]

- (a) $x=y$
(b) $x < y$
(c) $x > y$
(d) cannot be concluded due to insufficient data

47. Which one of the following is correct?

[CDS-2023 (2)]

- (a) $h^2 = x^2 - y^2$ (b) $h^2 = x^2 + y^2$
(c) $h^2 = 2(y^2 - x^2)$ (d) $h^2 = y^2 - x^2$

48. The angles of elevation of the top of a tower from two points A and B at a distance of x m and $(x + 5)$ m from the base of the tower of height 6 m and in the same straight line with it are complementary. What is the value of x ?

[CDS-2024 (1)]

- (a) 4 m (b) 5 m
(c) 6 m (d) 9 m

ANSWER KEY

1.	b	2.	a	3.	c	4.	a	5.	c	6.	c	7.	b	8.	b	9.	c	10.	d
11.	d	12.	c	13.	c	14.	c	15.	d	16.	c	17.	b	18.	a	19.	c	20.	a
21.	a	22.	b	23.	b	24.	c	25.	c	26.	b	27.	c	28.	a	29.	b	30.	c
31.	c	32.	a	33.	b	34.	b	35.	c	36.	b	37.	c	38.	b	39.	c	40.	c
41.	d	42.	b	43.	c	44.	a	45.	c	46.	b	47.	d	48.	a				

PRACTICE SHEET-1

1. What type of classification is needed to enumerate the female population of India?
(1) Geographical (2) Chronological
(3) Qualitative (4) Quantitative
2. For less than ogive, the cumulative frequencies are plotted against which of the following?
(a) Upper limits of class intervals
(b) Lower limits of the class intervals
(c) Mid-points of the class intervals
(d) Either (b) or (c)
3. The cumulative frequency curve of a frequency distribution with 6 classes and total frequency 60 is a straight line. Consider the following statements:
I. The first and the last classes have a frequency of 10 each.
II. Both the middle classes have a total frequency of 30.
III. The frequency distribution does not have a mode.
Which of the above statements are incorrect?
(a) I and II (b) I and III
(c) II and III (d) I, II and III
4. In Statistics, a suitable graph for representing the partitioning of total into subparts is
(a) An ogive (b) A pictograph
(c) A histogram (d) A pie chart
5. Which one of the following statements is not correct with reference to a histogram?
(a) Frequency curve is obtained by joining the mid-points of the top of the adjacent rectangles with smooth curves
(b) Histogram is drawn for continuous data
(c) The height of the bar is proportional to the frequency of that class
(d) Mode of the distribution can be obtained from the histogram
6. Assume that population densities of 5 major states of India are given. Which one of the following diagrams is suitable to represent the data?
(a) Single bar diagram
(b) Percentage bar diagram
(c) Pie diagram
(d) Since population density is a ratio, it cannot be represented by any diagram
7. Which one of the following represents statistical data?
(a) The names of all owners of shops located in a shopping complex
(b) A list giving the names of all states of India
(c) A list of all European countries and their respective capital cities
(d) The volume of a rainfall in certain geographical area, recorded every month for 24 consecutive months
8. Data on percentage distribution of area of land in acres owned by households in two districts of a particular state are as follows:

Land Holding	District A	District B
0.01 - 0.99	5.62	13.53
1.0 - 2.49	18.35	21.84
2.5 - 7.49	47.12	39.32
7.5 - 12.49	19.34	12.15
12.5 - 19.99	7.21	7.43
20.0 - 29.99	2.36	5.73

What is the appropriate diagram to represent the above data?
(a) Pie diagram
(b) Histogram
(c) Bar chart
(d) None of the above
9. Examples of data are given below:
I. Information on households collected by an investigator by door to door visits.
II. Data on the percentage of literates, sexwise, for the different districts of a state collected from records of the census of India.
III. General information about families, collected by telephonic interviews.
Which one of the following in respect of the above is correct?
(a) I and II are primary data
(b) I and III are primary data
(c) II and III are primary data
(d) I, II and III are primary data
10. Consider the following statements in respect of histogram.
I. Histogram is an equivalent graphical representation of the frequency distribution.

II. Histogram is suitable for continuous random variables, where the total frequency of an interval is evenly distributed over the interval.

Which of the statements given above is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

11. Consider the following statements in respect of a histogram:

I. The histogram consists of vertical rectangular bars with a common base such that there is no gap between consecutive bars.

II. The height of the rectangle is determined by the frequency of the class it represents.

Which of the statements given above is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

12. Consider the following statements in respect of class intervals of grouped frequency distribution.

1. Class intervals need not be mutually exclusive.

2. Class intervals should be exhaustive

3. Class intervals need not be of equal width

Which of the above statements are correct?

- (a) Both 1 and 2 (b) Both 2 and 3
(c) Both 1 and 3 (d) 1, 2 and 3

13. Consider the following statements in respect of a histogram.

1. The total area of the rectangles in a histogram is equal to the total area bounded by the corresponding frequency polygon and the axis.

2. When class intervals are unequal in a frequency distribution, the area of the rectangle is proportional to the frequency.

Which of the above statements is/are correct?

- (a) Only 1 (b) Only 2
(c) Both 1 and 2 (d) Neither 1 nor

14. Consider the following statements in respect of histogram.

I. The histogram is a suitable representation of a frequency distribution of a continuous variable.

II. The area included under the whole histogram is the total frequency.

Which of the above statement (s) is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither 1 nor 2

15. The cumulative frequency of the largest observed value must always be:

- (a) Less than the total number of observations
(b) Greater than the total number of observations
(c) Equal to total number of observations
(d) Equal to mid-point of the last class interval

16. Consider the following statements:

I. A continuous random variable can take all values in an interval.

II. A random variable which takes a finite number of values is necessary discrete.

III. Construction of a frequency distribution is based on data which are discrete.

Which of the above statements are correct?

- (a) Both I and II (b) Both II and III
(c) Both I and III (d) All of these

17. Consider the following statements in respect of histogram.

I. Pie diagram is suitable for categorical data.

II. The arc length of a sector of a pie diagram is proportional to the value of the component represented by the sector.

Which of the above statement (s) is/are correct?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither 1 nor 2

18. A Histogram is a graphical presentation of a frequency distribution of a

- (1) individual series (2) discrete series
(3) continuous series (4) none of these

19. Which of the following diagram is drawn by joining mid-points of the tops of all rectangles in a histogram:

- (1) frequency distribution
(2) frequency polygon
(3) frequency curve
(4) none of these

20. What is the shape of 'less than ogive'

- (1) Rising upward (2) falling downward
(3) Parallel to X-axis (4) Parallel to Y axis

21. Which of the following is a shape of frequency distribution curve?

- (1) A-shaped (2) B-shaped
(3) U shaped (4) All the above

22. Normal curves are also known as:

- (1) J - shaped

(2) L - shaped

(3) U - shaped

(4) bell-shaped curves

23. In a pie diagram, there are four slices with angles 150° , 90° , 60° and 60° . A new pie diagram is formed by deleting one of the slices having angle 60° in the given pie diagram. In the new pie diagram

(a) The largest slice has angle 150°

(b) The smallest slice has angle 70°

(c) The largest slice has angle 180°

(d) The smallest slice has angle 90°

24. Which one of the following is not correct?

The proportion of various items in a pie diagram is proportional to the

(a) Areas of slices

(b) Angles of slices

(c) Lengths of the curved arcs of the slices

(d) Perimeters of the slices

25. Consider the following statements in respect of histogram.

I. Primary data involve more time and more expense.

II. Investigator is a person who plans and conducts an empirical investigation independently.

Which of the above statement (s) is/are correct?

(a) Only I

(b) Only II

(c) Both I and II

(d) Neither 1 nor 2

26. Consider the following statements in respect of histogram.

I. Positively skewed curves have their tail more spread towards right.

II. Histogram is drawn only for equal class intervals.

Which of the above statement (s) is/are correct?

(a) Only I

(b) Only II

(c) Both I and II

(d) Neither 1 nor 2

27. Data can be represented in which of the following forms?

1. Textual form

2. Tabular form

3. Graphical form

Select the correct answer using the codes given below:

(a) 1 and 2

(b) 2 and 3

(c) 1 and 3

(d) 1, 2 and 3

28. Which one of the following can be obtained from an ogive?

(a) Mean

(b) Median

(c) Geometric

(d) Mode

29. A set of annual numerical data, comparable over the years, is given for the last 12yr.

Consider the following statements.

I. The data is best represented by a broken line graph, each corner (turning point) representing the data of one year.

II. Such a graph depicts the chronological change and also enables one to make a short-term forecast.

Select the correct answer using the codes given below:

(a) Only I

(b) Only II

(c) Both I and II

(d) Neither I nor II

30. Which term is not related with statistics?

(a) Data

(b) Histogram

(c) Domain

(d) Range

ANSWER KEY

1.	(4)	2.	(1)	3.	(2)	4.	(4)	5.	(4)	6.	(3)	7.	(4)	8.	(3)	9.	(2)	10.	(3)
11.	(3)	12.	(1)	13.	(3)	14.	(1)	15.	(3)	16.	(1)	17.	(3)	18.	(3)	19.	(2)	20.	(1)
21.	(3)	22.	(4)	23.	(3)	24.	(4)	25.	(3)	26.	(1)	27.	(4)	28.	(2)	29.	(3)	30.	(3)

PRACTICE SHEET-2

Direction for next three:

Study the following table and answer the questions.

School	No. of students scoring marks less than 50%	Percentage of students scoring marks more than 50%	No. of students appeared
A	240	55	600
B	220	40	400
C	300	20	375
D	280	10	350
E	210	25	300

- The ratio of the total number of students scoring marks less than 50% to that of scoring marks exactly 50% is
(1) 50 : 3 (2) 25 : 2
(3) 25 : 4 (4) 35 : 2
- Which school has the highest number of students scoring exactly 50% marks?
(1) D (2) E
(3) B (4) A
- The total number of students scoring 50% or more marks is
(1) 1250 (2) 875
(3) 775 (4) 675

Direction for next three:

Study the following table and answer the questions.

Source of Income	Schools				
	A	B	C	D	E
Tuition Fee.	120	60	210	90	120
Term Fee	24	12	45	24	30
Donation	54	21	60	51	60
Grants	60	54	120	42	55
Miscellaneous	12	3	15	3	15
Total	270	150	450	210	280

- For school E, what per cent of the income from miscellaneous is the income from donation?
(1) 25% (2) 40%
(3) $\frac{1}{4}$ % (4) 400%
- Which school has the highest percentage of income from tuition fee out of its total income?

- (1) A (2) B
(3) C (4) D

6. In case of how many schools, is the income by way of tuition fee, is less than four times of term fee

- (1) 0 (2) 1
(3) 2 (4) 3

Direction for next three:

Study the following table and answer the questions.

Food Grains Production in a country in 1999 (in lakh tons)					
State	Rice	Wheat	Jowar	Pulses	Others
P	45	103	—	27	29
Q	48	86	73	19	15
R	59	32	67	14	31
S	41	37	59	21	15
T	37	22	41	13	11
U	68	15	12	—	18
V	57	8	7	12	10
W	38	28	31	22	45

- Which State had the highest grain production
(1) P (2) Q
(3) R (4) S
- What was the proportion of rice production to wheat production in the country ?
(1) 1 : 1 (2) 0.8 : 1
(3) 1.2 : 1 (4) 2 : 1
- Jowar was the most important food grain in the State/States :
(1) Q, R, S (2) Q
(3) R, S (4) R, S, T

Direction for next three:

The following table shows the production of food grains (in million tonnes) in a State for the period 1988 to 1992. Read the table and answer the following questions.

Year	Production in million tonnes				Total
	Wheat	Rice	Maize	Other Cereals	
1988	580	170	150	350	1250
1989	600	220	234	400	1454
1990	560	240	228	420	1448
1991	680	300	380	460	1820
1992	860	260	340	500	1960
Total	3280	1190	1332	2130	7932

10. During 1990, the percentage of decrease in production of maize as against the previous year was:

- (1) 2.63% (2) 2.56%
(3) 2.71% (4) 2.47%

11. In 1991, the increase in production over the previous year was maximum for :

- (1) Wheat (2) Rice
(3) Maize (4) Other Cereals

12. The increase in the production of other cereals (over the previous year) was minimum during the year :

- (1) 1989 (2) 1990
(3) 1991 (4) 1992

Direction for next three:

Study the following table and answer the questions.

Years	Men	or Women	Children	Total	Increase (+) decrease (-) over preceding year
1988	65104	60387	—	146947	—
1989	70391	62516	—	—	+ (11630)
1990	—	63143	20314	153922	—
1991	68395	—	21560	—	— (5337)
1992	71274	659935	23789	16098	—

13. The number of children in 1988 is :

- (1) 31236 (2) 125491
(3) 14546 (4) 21456

14. The total population in 1989 is :

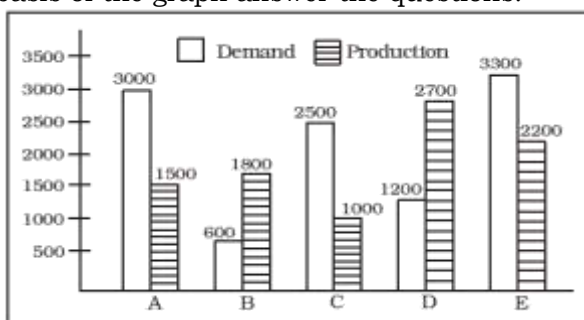
- (1) 144537 (2) 158577
(3) 146947 (4) 149637

15. Number of children in 1989 is :

- (1) 25670 (2) 14040
(3) 13970 (4) 15702

Direction for next three:

The following chart represents Demand and Production for 5 companies ABCDE. On the basis of the graph answer the questions.



16. If company A desires to meet the demand by purchasing surplus production of company, then the most suitable company is :

- (1) C (2) D
(3) E (4) B

17. If $x\%$ of demand for company C equals demand for company B, then x equals

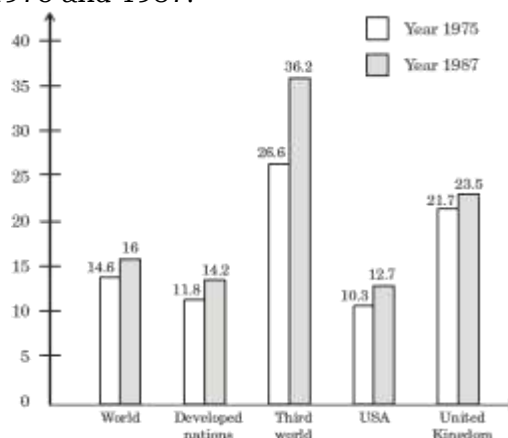
- (1) 24 (2) 20
(3) 60 (4) 4

18. If the production of company D is h times of the production of company A. Then h equals :

- (1) 1.5 (2) 2.5
(3) 1.2 (4) 1.8

Direction for next three:

Study the following Bar graph and answer the questions. The Bar Graph gives the annual rates of inflation in percentages for 1975 and 1987.



19. From 1975 to 1987, inflation rate increased in the third world countries approximately by

- (1) 10% (2) 20%
(3) 30% (4) 36%

20. The change in rate of inflation was least in which of the following?

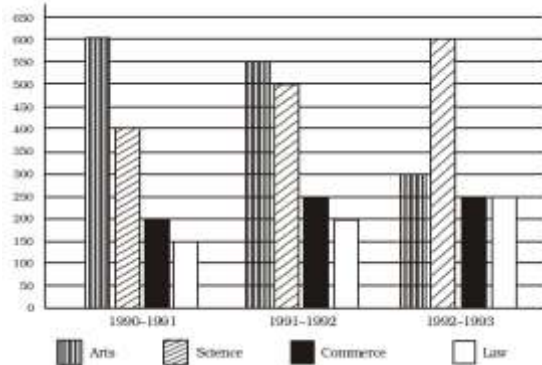
- (1) Developed Nations (2) United Kingdom
(3) World (4) Third world

21. Comparing the figures for USA vis-a-vis the developed nations, it can be concluded that.

- (1) USA had better control on inflation
(2) Developed nations had better control on inflation
(3) The inflation rate continues to be the same for USA and developed nations
(4) No conclusions can be drawn

Direction for next three:

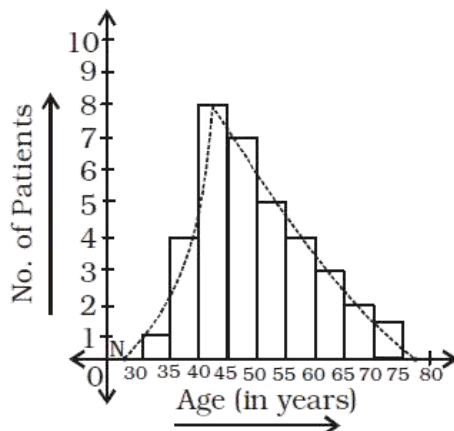
Shown below is the multiple bar diagram depicting the changes in the student's strength of a college in four faculties from 1990-91 to 1992-93. (Scale 1 cm = 100)



22. How much per cent was the increase in science students in 1992-93 over 1990-91?
 (1) 50% (2) 150%
 (3) 66.66% (4) 75%
23. The respective ratio of the number of commerce students to the number of law students from 1990-91 to 1992-93 is :
 (1) 6 : 7 (2) 7 : 6
 (3) 14 : 1 (4) 2 : 8
24. During which year the strength of arts faculty was minimum?
 (1) 1990-91 (2) 1991-92
 (3) 1992-93 (4) None of these

Direction for next Four:

The diagram shows the age-distribution of the patients admitted to a hospital on a particular day. Study the diagram and answer the questions.

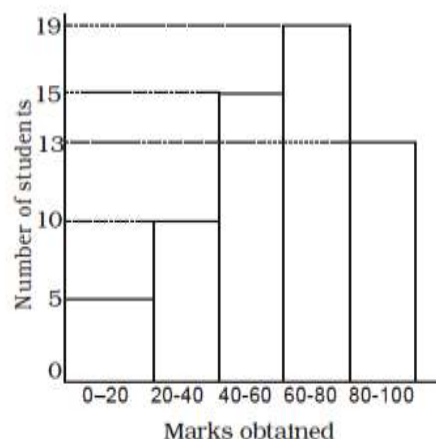


25. Total number of patients of age more than 55 years, who got admitted to the hospital is
 (1) 4 (2) 7
 (3) 9 (4) 10

26. Number of patients of age more than 40 years and less than 55 years, who got admitted to the hospital on that day is
 (1) 20 (2) 30
 (3) 15 (4) 12
27. Percentage of patients of age less than 45 years, who got admitted to the hospital on that day is approximately equal to
 (1) 14% (2) 20%
 (3) 37% (4) 62%
28. About 11% of the patients who got admitted to the hospital on that particular day were of age
 (1) either between 35 years and 40 years or between 55 years and 60 years
 (2) between 60 years and 65 years
 (3) between 35 years and 40 years
 (4) between 35 years and 40 years and between 55 years and 60 years.

Direction for next two:

In each of the following questions, the following histogram shows the relationship between the marks obtained by the students and the number of students in an examination. Study the histogram and answer the questions.



29. The percentage of students who obtained 40 marks or less is
 (1) 25
 (2) less than 25, but not 15
 (3) more than 24
 (4) 15
30. The ratio of the number of students who obtained 60 or more marks to that of students who obtained 60 or less marks is

- (1) 15 : 16 (2) 15 : 19
(3) 14 : 17 (4) 16 : 15

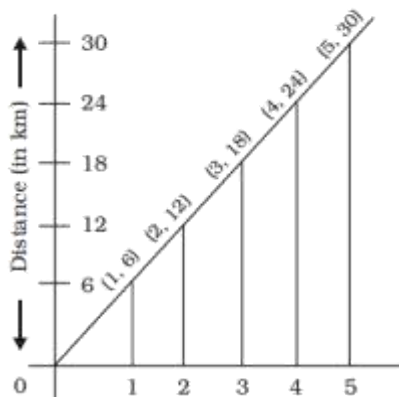
ANSWER KEY

1.	(2)	2.	(1)	3.	(2)	4.	(1)	5.	(3)	6.	(2)	7.	(2)	8.	(3)	9.	(4)	10.	(2)
11.	(3)	12.	(1)	13.	(4)	14.	(2)	15.	(1)	16.	(2)	17.	(1)	18.	(4)	19.	(4)	20.	(2)
21.	(1)	22.	(1)	23.	(2)	24.	(3)	25.	(4)	26.	(1)	27.	(3)	28.	(4)	29.	(2)	30.	(1)

PRACTICE SHEET-3

Direction for next three:

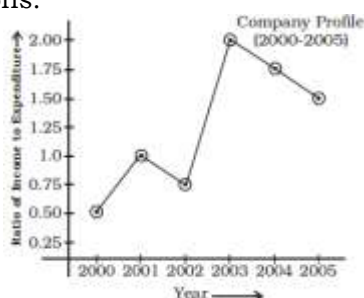
The graph given here shows a car following the linear path with uniform speed. Study the graph and answer the questions.



- The speed of the car is
(1) 12 km/hr (2) 6 km/hr
(3) 18 km/hr (4) 24 km/hr
- The speed of the car (in metres per minute) is
(1) 60 (2) 100
(3) 600 (4) 1000
- The distance travelled by the car in 4.5 hours is
(1) 27 km (2) 30 km
(3) 36 km (4) 40 km

Direction for next two:

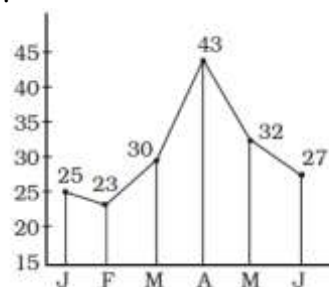
Study the following graph and answer the questions.



- Find the percentage decrease in income from 2001 to 2002.
(1) 50% (2) 33%
(3) 37.5% (4) Data inadequate

- If the income shows positive growth every year throughout the period (2000 - 2005), then in how many years the expenditure shows a positive growth?
(1) 5 (2) 3
(3) 4 (4) 2

- Given is a line graph showing the number of accidents in a city during the first 6 months of 1999.

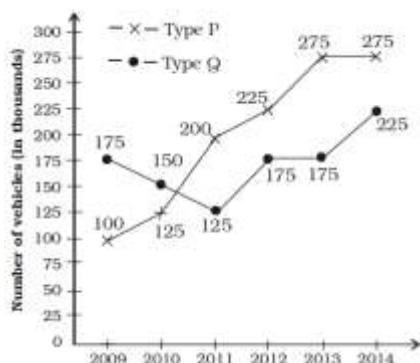


The decrease % of accidents from May to June is

- (1) 15.375% (2) 15.125%
(3) 15.625% (4) 15.875%

Direction for next three:

The following graph shows production (in thousands) of two types (P and Q) of vehicles by a factory over the years 2009 to 2014. Study the graph and answer the given questions.



7. In how many of the given years, was the production of Type P vehicles of the company more than the average production of this type vehicles in the given years?

(1) 3 (2) 4
(3) 2 (4) 5

8. Approximate percentage decrease in production of Type Q vehicles from 2010 to 2011 is

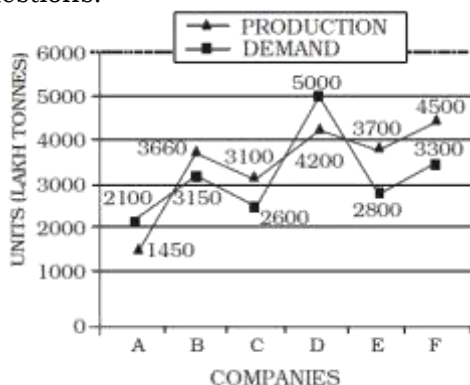
(1) 10.1 (2) 16.7
(3) 14.3 (4) 12.5

9. The total production of Type P vehicles in the years 2009 and 2011 is what percent of total production of Type Q vehicles in 2010 and 2014?

(1) 75 (2) 69.25
(3) 80 (4) 81.25

Direction for next three:

In the following questions, the Graph shows the demand and production of different companies. Study the graph and answer the questions.



10. The demand of company B is what percentage of the production of company F?

(1) 60% (2) 70%
(3) 80% (4) 50%

11. What is the difference between the average demand and the average production of the companies (in lakh tonnes)? [Approximately]

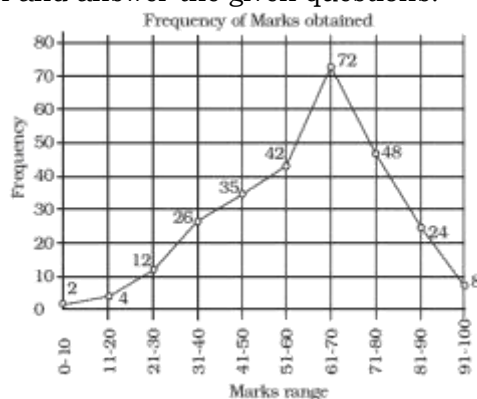
(1) 200 (2) 325
(3) 275 (4) 250

12. The production of company A is approximately what percent of the demand of company C?

(1) 50% (2) 65%
(3) 60% (4) 55%

Direction for next three:

The marks obtained by 273 examinees are shown by the frequency polygon. Given that mean marks is 59.5. Study the frequency polygon and answer the given questions.



13. The number of examinees getting more than average marks is

(1) 72 (2) 105
(3) 152 (4) 164

14. Percentage of the students who get above 80% marks is

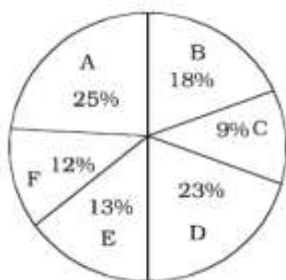
(1) 9.81 (2) 10.53
(3) 11.28 (4) 11.72

15. Percentage of the students who got marks above 60% and below 80% is

(1) 43.95 (2) 48.39
(3) 51.06 (4) 56.84

Direction for next three:

The Pie chart given here represents the domestic expenditure of a family in per cent. Study the chart and answer the following questions if the total monthly income of the family is 33,650.

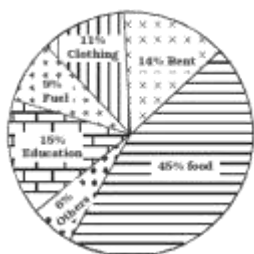


A:Expenditure on food
B:Expenditure on house-rent
C:Expenditure on entertainment
D:Expenditure on education and maintenance of children
E:Medical and miscellaneous expenditure
F:Deductions towards provident fund

16. The house rent per month is :
(1) 6000 (2) 6152
(3) 6057 (4) 6048
17. The annual savings in the form of provident fund would be
(1) 48,456 (2) 48,540
(3) 44,856 (4) 45,480
18. After provident fund deductions and payment of house rent, the total monthly income of the family remains
(1) 23,545 (2) 24,435
(3) 23,555 (4) 25, 355

Direction for next three:

The pie chart given below shows the spending of a family on various heads during a month. Study the graph and answer the following questions.



19. If the total income of the family is 25,000, then the amount spent on Rent and Food together is
(1) 17,250 (2) 14, 750
(3) 11, 250 (4) 8, 500
20. What is the ratio of the expenses on Education to the expenses on Food?

- (1) 1 : 3 (2) 3 : 1
(3) 3 : 5 (4) 5 : 3

21. Expenditure on Rent is what percent of expenditure on Fuel?

- (1) 135% (2) 156%
(3) 167% (4) 172%

Direction for next three:

Study the pie chart and answer questions.



22. The ratio of the total amount spent for food and medicine is

- (1) 1 : 2 (2) 3 : 1
(3) 12 : 5 (4) 11 : 5

23. If the total amount spent on the family during the year was 50,000, how much they spent for buying clothes?

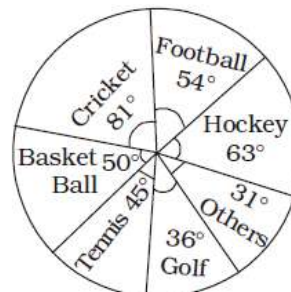
- (1) 6,250 (2) 6,500
(3) 7,250 (4) 7,500

24. If the total amount spent on the family during the year was 35,000, the amount spent for study of children and food together was

- (1) 19,250 (2) 19,500
(3) 19,750 (4) 19,850

Direction for next three:

The Pie Chart shows the expenditure of a country on various sports during a particular year. Study the graph and answer the questions



25. If the total amount spent on cricket and hockey together is 80,000, the total amount spent on sports is

- (1) 1,00,000 (2) 2,00,000
(3) 2,50,000 (4) 3,00,000

26. How much per cent more is spent on Hockey than that on Golf?

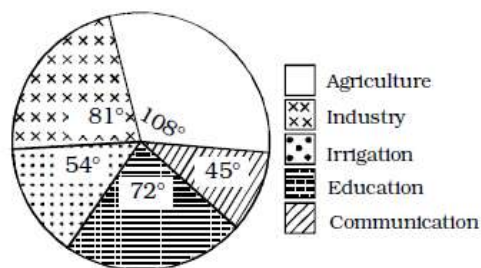
- (1) 27% (2) 35%
(3) 37.5% (4) 75%

27. How much per cent less is spent on football than that on cricket?

- (1) 22.22% (2) 27%
(3) 33.33% (4) 37.5%

Direction for next three:

The pie-chart shows the proposed outlay for different sectors during a Five-Year plan of Government of India. Total outlay is Rs. 40,000 crores. By reading the pie-chart answer the following three questions.



28. What is the proposed outlay for Education?

- (1) Rs. 6000 crores (2) Rs. 8000 crores
(3) Rs. 9000 crores (4) Rs. 7000 crores

29. If the proposed outlay of Irrigation is $x\%$ of the proposed outlay of Agriculture, then x is equal to

- (1) 50% (2) 15%
(3) 25% (4) 75%

30. What is the ratio between the proposed outlay of Irrigation and Communication?

- (1) 9 : 8 (2) 3 : 2
(3) 9 : 5 (4) 6 : 5

ANSWER KEY

1.	(2)	2.	(2)	3.	(1)	4.	(4)	5.	(4)	6.	(3)	7.	(1)	8.	(2)	9.	(4)	10.	(2)
11.	(3)	12.	(4)	13.	(3)	14.	(4)	15.	(1)	16.	(3)	17.	(1)	18.	(3)	19.	(2)	20.	(1)
21.	(2)	22.	(3)	23.	(1)	24.	(1)	25.	(2)	26.	(4)	27.	(3)	28.	(2)	29.	(1)	30.	(4)

PRACTICE SHEET-4

1. The median of the observations 22, 24, 33, 37, $x + 1$, $x + 3$, 46, 47, 57, 58 in ascending order is 42. What are the values of 5th and 6th observations respectively?

- (a) 42, 45 (b) 41, 43
(c) 43, 46 (d) 40, 40

2. The class marks in a frequency table are given to be 5, 10, 15, 20, 25, 30, 35, 40, 45, 50. The class limits of the first five classes are

- (a) 3-7, 7-13, 13-17, 17-23, 23-27
(b) 2.5-7.5, 7.5-12.5, 12.5-17.5, 17.5-22.5, 22.5-27.5
(c) 1.5-8.5, 8.5-11.5, 11.5-18.5, 18.5-21.5, 21.5-28.5
(d) 2-8, 8-12, 12-18, 18-22, 22-28

3. Consider the following discrete frequency distribution:

x	1	2	3	4	5	6	7	8
f	3	15	45	57	50	36	25	9

What is the value of median of the distribution?

- (a) 4 (b) 5
(c) 6 (d) 7

4. The following observation has been arranged in ascending order:

29, 32, 48, 52, x , $x + 3$, 71, 75, 80, 92

If the median of data is 61.5, then what is the value of x ?

- (a) 54 (b) 60
(c) 62 (d) 56

5. What is the geometric means of the observations?

125, 729, 1331?

- (a) 495 (b) 1485
(c) 2221 (d) None of these
6. What is the median of a numbers 4.6, 0, 9.3, -4.8, 7.6, 2.3, 12.7, 3.5, 8.2, 6.1, 3.9, 5.2?
(a) 3.8 (b) 4.9
(c) 5.7 (d) 6.0
7. Consider the following statements:
I.The sum of deviations from mean is always zero.
II.The sum of absolute deviations is minimum when taken around median.
Which of the above statement is/are correct?
(a) I only (b) 2 only
(c) Both I and II (d) Neither I nor II
8. Consider the following statements:
I.Mean is independent of change in scale and changes in origin
II.Variance is independent of change in scale but not in origin.
Which of the above statements is/are correct?
(a) I only (b) II only
(c) Both I and II (d) Neither I nor II
9. Let $\bar{x}$ be the mean of $x_1, x_2, x_3, \dots, x_n$. If $x_i = a + cy_i$ for some constants a and c , then what will be the mean of $y_1, y_2, y_3, \dots$,
(a) $a + c\bar{x}$ (b) $a - \frac{1}{c}\bar{x}$
(c) $\frac{1}{c}\bar{x} - a$ (d) $\frac{\bar{x} - a}{c}$
10. If x_1 and x_2 are positive quantities, then the condition for the difference between the arithmetic mean and the geometric mean to be greater than 1 is:
(a) $x_1 + x_2 > 2\sqrt{x_1x_2}$
(b) $\sqrt{x_1} + \sqrt{x_2} > \sqrt{2}$
(c) $|\sqrt{x_1} - \sqrt{x_2}| > \sqrt{2}$
(d) $x_1 + x_2 < 2(\sqrt{x_1x_2} + 1)$

DIRECTIONS (for next two): Read the following information carefully to answer the questions that follow.

The average age of 6 persons living in a house is 23.5 years. Three of them are majors and their average age is 42 years. The difference in ages of the three minor children is same.

11. What is the mean of the ages of minor children?
(a) 3 years (b) 4 years
(c) 5 years (d) 6 years
12. What is the median of the ages of minor children?
(a) 3 years
(b) 5 years
(c) 7 years
(d) Cannot be determined
13. Consider the following statements:
1.The mean and median are equal in symmetric distribution.
2.The range is the difference between the maximum value and the minimum value in the data.
3.The sum of the area of the rectangles in the histogram is equal to the total area bounded by the frequency polygon and that horizontal axis.
Which of the above statements are correct?
(a) Both 1 and 2 (b) Both 2 and 3
(c) Both 1 and 3 (d) All 1, 2 and 3

Direction (for next two): The following table gives the frequency distribution of life length in hours of 100 electric bulbs having median life 20 h.

Life of bulbs (in hours)	Number of bulbs
8-13	7
13-18	x
18-23	40
23-28	y
28-33	10
33-38	2

14. What is the missing frequency 'x'?
(a) 31 (b) 27
(c) 24 (d) 14
15. What is the missing frequency 'y'?
(a) 27 (b) 24
(c) 14 (d) 11
16. A random sample of 20 people is classified in the following table according to their ages:

Age	Frequency
15-25	2
25-35	4
35-45	6
45-50	5
55-65	3

What is the mean age of this group of people?

- (a) 41.0 (b) 41.5
(c) 42.0 (d) 42.5

17. The scores of 15 students in an examination were recorded as 10, 5, 8, 16, 18, 20, 8, 10, 16, 20, 18, 11, 16, 14 and 12. After calculating the mean, median and mode, an error is found. One of the values is wrongly written as 16 instead of 18. Which of the following measures of central tendency will change?

- (a) Mean and median
(b) Median and mode
(c) Only mode
(d) Mean and mode

18. What is the mean deviation from the mean of the numbers 10, 9, 21, 16, 24?

- (a) 5.2 (b) 0.5
(c) 0.6 (d) 0.7

19. For the data 3, 5, 1, 6, 5, 9, 5, 2, 8, 6 the mean, median and mode are x, y and z respectively.

Which one of the following is correct?

- (a) $x = y \neq z$ (b) $x \neq y = z$
(c) $x \neq y \neq z$ (d) $x = y = z$

20. Sets A, B and C contain 5 numbers each. The medians of the numbers in these sets are 3, 8 and 11, respectively. What is the median of the combined 15 numbers of the three sets?

- (a) 8
(b) 7
(c) $22/3$
(d) Cannot be determined

21. The mean of five numbers is 30. If one number is excluded, their mean becomes 28. The excluded number is:

- (a) 28 (b) 30
(c) 35 (d) 38

22. Consider the following types of data:

- I. Marks of students who appeared for a test of 100 marks.
II. Collar sizes of 200 shirts sold in a week.
III. Monthly incomes of 250 employees of a factory. For which of the above data, mode is a suitable measure of central tendency?

- (a) Both I and II (b) Only II
(c) Both I and III (d) I, II and III

Directions (for next three): The number of telephone calls received in 245 successive

one minute intervals at an exchange is given below in the following frequency distribution.

Number	0	1	2	3	4	5	6	7
Frequency	14	21	25	43	51	40	39	12

23. What is the mean of the distribution?

- (a) 3.76 (b) 3.84
(c) 3.96 (d) 4.05

24. What is the median of the distribution?

- (a) 3.5 (b) 4
(c) 4.5 (d) 5

25. What is the mode of the distribution?

- (a) 3 (b) 4
(c) 5 (d) 6

26. The cumulative frequency of the largest observed value must always be:

- (a) Less than the total number of observations
(b) Greater than the total number of observations
(c) Equal to total number of observations
(d) Equal to mid-point of the last class interval

27. Some measure central tendency for n discrete observations are given below:

- I. Arithmetic mean
II. Geometric mean
III. Harmonic mean
IV. Median

A desirable property of a measure of central tendency is, if every observation is multiplied by c, then the measure of central tendency is also multiplied by c, where $c < 0$.

Which of the above measures satisfy the property.

- (a) I, II and III (b) I, II and IV
(c) III and IV (d) I, II, III and IV

Directions (for next two): The following table gives the continuous frequency distribution of a continuous variable X.

Class Interval	Frequency
0-10	5
10-20	10
20-30	20
30-40	5
40-50	10

28. What is the median of the above frequency distributions?

- (a) 23 (b) 24
(c) 25 (d) 26

29. What is the mean of the above frequency distribution?

- (a) 25 (b) 26
(c) 27 (d) 28

30.

Class Interval	1-5	6-10	11-15	16-20
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Frequency	3	7	6	5
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Consider the following statements in respect of the above frequency distribution.

- I. The median is contained in the modal class.
II. The distribution is bell-shaped

Which of the above

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

ANSWER KEY

1.	(b)	2.	(c)	3.	(b)	4.	(b)	5.	(a)	6.	(b)	7.	(c)	8.	(d)	9.	(d)	10.	(c)
11.	(c)	12.	(b)	13.	(d)	14.	(b)	15.	(c)	16.	(b)	17.	(d)	18.	(a)	19.	(d)	20.	(d)
21.	(d)	22.	(b)	23.	(a)	24.	(b)	25.	(b)	26.	(c)	27.	(d)	28.	(c)	29.	(b)	30.	(b)

CDS PYQ

1. Which of the following are the examples of discrete variables?

- I. Number of errors per page in a book
II. Height of individuals measured in centimeter
III. Waiting time to failure of electric bulbs
IV. Number of levels on branches of a tree

Select the correct answer using the codes given below

[CDS 2013 (1)]

- (a) Only I (b) I and IV
(c) III and IV (d) II and IV

Directions (for next three): Read the following information carefully to answer the questions that follow:

In a frequency distribution having class intervals 0-10, 10-20, 20-30 and 30-40 the respective frequency are x , $x+8$, $x-2$, and $x-4$ and the arithmetic mean of the distribution is 17.8.

2. The value of x is:

[CDS 2013 (1)]

- (a) 3 (b) 6
(c) 8 (d) 12

3. The median lies in which one of the following class intervals?

[CDS 2013 (1)]

- (a) 0-10 (b) 10-20

- (c) 20-30 (d) 30-40

4. The modal class is:

[CDS 2013 (1)]

- (a) 0-10 (b) 10-20
(c) 20-30 (d) 30-40

5. Consider the following statements:

- I. A frequency distribution condenses the data and reveals its important features
II. A frequency distribution is an equivalent representation of original data

Which of the above statement is/are correct?

[CDS 2013 (1)]

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

6. Which one of the following statements is correct?

[CDS 2013 (1)]

- (a) A frequency polygon is obtained by connecting the corner points of the rectangles in a histogram
(b) A frequency polygon is obtained by connecting the mid points of the tops of the rectangles in a histogram
(c) A frequency polygon is obtained by connecting the corner points of the class intervals in a histogram
(d) None of the above

7. The average weight of a class of 15 boys and 10 girls is 38.4kg if the average weight of boys is 40 kg. Then what is the average weight of girls?

[CDS 2013 (1)]

- (a) 36.5kg (b) 35kg
(c) 36kg (d) 35.6kg

8. The average of u, v, w, x, y and z is 10. What is the average of $u + 10, v + 20, w + 30, x + 40, y + 50$ and $z + 60$?

[CDS 2013 (2)]

- (a) 30 (b) 35
(c) 40 (d) 45

9. If m is the mean of p, q, r, s, t and v , then what is $(p-m) + (q-m) + (r-m) + (s-m) + (t-m) + (v-m)$ equal to?

[CDS 2013 (2)]

- (a) 0 (b) s
(c) $\frac{(p+v)}{2}$ (d) None of these

Directions (next two): Read the following information carefully and answer the questions given below:

The median of the following distribution is 14.4 and the total frequency is 20.

Class Interval	0-6	6-12	12-18	18-24	
	1	1	2	3	
	2	8	4	0	
Frequency	4	x	5	y	1

10. What is x equal to?

[CDS 2013 (2)]

- (a) 4 (b) 5
(c) 6 (d) 7

11. What is the relation between x and y ?

[CDS 2013 (2)]

- (a) $2x = 3y$ (b) $3x = 2y$
(c) $x = y$ (d) $2x = y$

12. Methods of presentation of data are:

[CDS 2013 (2)]

- (a) tables only (b) graphs only
(c) diagrams only (d) all of the above

13. Consider the following:

I. The arithmetic mean of two unequal positive numbers is always greater than their geometric mean

II. The geometric mean of two unequal positive numbers is always greater than their harmonic mean.

Which of the above statements is/are correct?

[CDS 2014 (1)]

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

14. Consider the following statements in respect of a discrete set of numbers:

I. The arithmetic mean uses all the data is always uniquely defined.

II. The median uses only one or two numbers from the data and may not be unique.

Which of the above statements is/are correct?

[CDS 2014 (1)]

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

15. The geometric mean of $(x_1, x_2, x_3, \dots, x_n)$ is x and the geometric mean of $(y_1, y_2, y_3, \dots, y_n)$ is y . Which of the following is/are correct?

I. The geometric mean of $(x_1, y_1, y_2, x_3, y_3, \dots, x_n, y_n)$ is XY .

II. The geometric mean of $\left(\frac{x_1}{y_1}, \frac{x_2}{y_2}, \frac{x_3}{y_3}, \dots, \frac{x_n}{y_n}\right)$ is $\frac{X}{Y}$.

Select the correct answer using the code given below:

[CDS 2014 (1)]

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

16. The following table gives 'less than' type frequency distribution of income per day.

Income (in ₹) less than	Number of person
1500	100
1250	80
1000	70
750	55
500	32
250	12

What is the model class?

[CDS 2014 (1)]

- (a) 250-500 (b) 500-750
(c) 75-1000 (d) None of these

17. Which of the following items of information is a good example of statistical data?

[CDS 2014 (1)]

- (a) A table of logarithms of numbers
(b) A list of names of 120 students of a class
(c) A list of annual incomes of the members of a club
(d) Holiday list of the offices of Government of India in the year 2013.

18. Consider the following in respect of variate which takes value 2, 2, 2, 2, 7, 7, 7 and 7.

I. The median is equal to mean

II. The mode is both 2 and 7

Which of the above statements is/are correct?

[CDS 2014 (1)]

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

19. Consider the following statements pertaining to a frequency polygon of a frequency distribution of a continuous variable having seven class intervals of equal width:

I. The original frequency distribution can be reconstructed from the frequency polygon

II. The frequency polygon touches the x-axis in its extreme right and extreme left.

Which of the above statements is/are correct?

[CDS 2014 (1)]

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

20. The mean of the following distribution is 18.

Class Interval	Frequency
11-13	3
13-15	6
15-17	9
17-19	13
19-21	f
21-23	5
23-25	4

What is the value of f?

[CDS 2014 (1)]

- (a) 8 (b) 9
(c) 10 (d) 11

21. The class which has maximum frequency is known as:

[CDS 2014 (2)]

- (a) Median class (b) Mean class
(c) Modal class (d) None of these

22. Consider the following statements related to cumulative frequency polygon of a frequency distribution, the frequencies being cumulated from the lower end of the range.

1. The cumulative frequency polygon gives an equivalent representation of frequency distribution table

2. The cumulative frequency polygon is a closed polygon with one horizontal and one vertical side

Which of the above statements is/are correct?

[CDS 2014 (2)]

- (a) Only 1 (b) Only 2
(c) Both I and 2 (d) Neither 1 nor 2

23. Consider the following data:

1. Number of complaints lodged due to road accidents in a state within a year for 5 consecutive years.

2. Budgetary allocation of the total available funds to the various items of expenditure

Which of the above data is/are suitable for representation of a pie diagram?

[CDS 2014 (2)]

- (a) Only 1 (b) Only 2
(c) Both I and 2 (d) Neither 1 nor 2

24. When we take class intervals on the X-axis and corresponding frequencies on the Y-axis and draw rectangles with the area proportional to the frequencies of the respective class intervals, the graph so obtained is called:

[CDS 2014 (2)]

- (a) Bar diagram
(b) Frequency curve
(c) Ogive
(d) None of the above

25. If x_i 's are the mid points of the class intervals of grouped data, f_i 's are the corresponding frequencies at $\bar{x}$ is the mean, then what is $\sum f_i (x_i - \bar{x})$ equal to?

[CDS 2014 (2)]

- (a) 0 (b) -1
(c) 1 (d) 2

26. Ten observations 6, 14, 15, 17, $x + 1$, $2x - 13$, 30, 32 and 43 are written in ascending order. The median of the data is 24. What is the value of x ?

[CDS 2014 (2)]

- (a) 15 (b) 18
(c) 20 (d) 24

27. If A, G and H are the arithmetic, geometric and harmonic means between a and b respectively, then which one of the following relationship is correct?

[CDS 2015 (1)]

- (a) G is the geometric mean between A and H
(b) A is the arithmetic mean between G and H
(c) H is the harmonic mean between A and G
(d) None of the above

28. The geometric mean of three positive numbers a, b, c is 3 and the geometric mean of another three positive numbers d, e, f is 4. Also, at least three elements in the set (a, b, c, d, e, f) are distinct. Which one of the following inequalities gives the best information about M , the arithmetic mean of the six numbers?

[CDS 2015 (1)]

- (a) $M > 2\sqrt{3}$
(b) $M > 3.5$
(c) $M \geq 3.5$
(d) It is not possible to set any precise lower limit of M

29. There are five parties A, B, C, D and E in an election. Out of total 100000 votes cast, 36000 were cast to party A, 24000 to party B, 18000 to party C, 7000 to party D and rest to party E. What angle will be allowed for party E in the pie chart?

[CDS 2015 (1)]

- (a) 15° (b) 54°
(c) 60° (d) 72°

For the next (4) items that follow:

Consider the following frequency distribution:

Class	Frequency
0-10	4
10-20	5
20-30	7
30-40	10
40-50	12
50-60	8
60-70	4

30. What is the mean of the distribution?

[CDS 2015 (1)]

- (a) 37.2 (b) 38.1
(c) 39.2 (d) 40.1

31. What is the median class?

[CDS 2015 (1)]

- (a) 20-30 (b) 30-40
(c) 40-50 (d) 50-60

32. What is the median of the distribution?

[CDS 2015 (1)]

- (a) 37 (b) 38
(c) 39 (d) 40

33. What is the mode of the distribution?

[CDS 2015 (1)]

- (a) 38.33 (b) 40.66
(c) 42.66 (d) 43.33

34. If a variable takes discrete values $a + 4, a - 3.5, a - 2.5, a - 3, a - 2, a + 0.5, a + 5$ and $a - 0.5$ where $a > 0$, then the median of the data set is:

[CDS 2015 (2)]

- (a) $a - 2.5$ (b) $a - 1.25$
(c) $a - 1.5$ (d) $a - 0.75$

35. If each of a n numbers $x_i = i$ ($i=1, 2, 3, \dots, n$) is replaced by $(i + 1) x_i$, then the new mean is:

[CDS 2015 (2)]

- (a) $\frac{n+3}{2}$ (b) $\frac{n(n+3)}{2}$

(c) $\frac{(n+1)(n+2)}{3n}$

(d) $\frac{(n+1)(n+2)}{3}$

36. The weighted arithmetic mean of first 10 natural numbers whose weights are equal to the corresponding numbers is equal to:

[CDS 2015 (2)]

- (a) 7 (b) 14
(c) 35 (d) 38.5

37. The election result in which six parties contested was depicted by a pie chart. Party A had an angle 135° on this pie chart. If it secured 21960 votes, how many valid votes in total were cast?

[CDS 2016 (1)]

- (a) 51240 (b) 58560
(c) 78320 (d) 87840

38. The mean and median of 5 observations are 9 and 8 respectively. If 1 is subtracted from each observation, then the new mean and the new median will respectively be:

[CDS 2016 (1)]

- (a) 8 and 7
(b) 9 and 7
(c) 8 and 9
(d) cannot be determined due to insufficient data

39. The age distribution of 40 children is as follows:

Age (in years)	5-	6-	7-	8-	9-	10-
					10	11
No. of Children	4	7	9	12	6	2

Consider the following statements in respect of the above frequency distribution.

- The median of the age distribution is 7 years
 - 70% of the children are in the age group 6-9 years
 - The modal age of the children is 8 years
- Which of the above statements are correct?

[CDS 2016 (1)]

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

40. Suppose $x_i = \lambda^i$ for $0 \leq i \leq 10$, where $\lambda > 1$. Which one of the following is correct?

[CDS 2016 (1)]

- (a) $AM < \text{Median}$ (b) $GM < \text{Median}$
(c) $GM = \text{Median}$ (d) $AM = \text{Median}$

41. Suppose $x_i = \frac{1}{i}$ for $i = 1, 2, 3, \dots, 11$.

Which one of the following is not correct?

[CDS 2016 (1)]

- (a) $AM > 1/6$ (b) $GM > 1/6$
(c) $HM > 1/6$ (d) $\text{Median} = HM$

42. For $x > 0$, if a variable takes discrete values $x + 4, x - 3.5, x - 2.5, x - 3, x - 2, x + 0.5, x - 0.5, x + 5$, then what is the median?

[CDS 2016 (2)]

- (a) $x - 1.25$ (b) -0.5
(c) $x + 0.5$ (d) $x + 1.25$

43. The median of a set of 9 distinct observations is 20.5. If each of the largest 4 observations of the set is increased by 2, then the median of the new set:

[CDS 2016 (2)]

- (a) Is increased by 2
(b) Is decreased by 2
(c) Is two times the original median
(d) Remains the same as that of original set

44. Number of credit cards held by an individual can be treated as:

[CDS 2016 (2)]

- (a) Qualitative data
(b) Discrete data
(c) Categorical data
(d) None of the above

45. A pie chart depicts the classification of total funds of an organization according to different sources of funds. A particular sector of pie chart for corporate tax has 108° angle at the centre. What is the percentage of income from corporate tax to total funds?

[CDS 2016 (2)]

- (a) 20% (b) 25%
(c) 30% (d) 35%

46. Consider the following statements:

1. The classes of type 15-19, 20-24, 25-29 etc. are exclusive classes.

2. The classes of type 15-20, 20-25, 25-30 etc. are inclusive classes.

Which of the above statements is/are correct?

[CDS 2016 (2)]

- (a) 1 Only (b) 2 Only
(c) Both 1 and 2 (d) Neither 1 nor 2

47. Suppose the class interval 10-15 has frequency 30, then what is the frequency density of this class interval?

[CDS 2016 (2)]

- (a) 2 (b) 3
(c) 5 (d) 6

48. Consider the following distribution:

Class	Frequency
0-20	17
20-40	28
40-60	32
60-80	f
80-100	19

If the mean of the above distribution is 50, what is the value of f?

[CDS 2017 (1)]

- (a) 24 (b) 34
(c) 56 (d) 96

49. In a pie diagram, there are four slices with angles 150° , 90° , 60° and 60° . A new pie diagram is formed by deleting one of the slices having angle 60° in the given pie diagram. In the new pie diagram:

[CDS 2017 (1)]

- (a) The largest slice has angle 150°
(b) The smallest slice has angle 70°
(c) The largest slice has angle 180°
(d) The smallest slice has angle 90°

50. In an asymmetrical distribution, if the mean and median of the distribution are 270 and 220 respectively, then the mode of the data is:

[CDS 2017 (1)]

- (a) 120 (b) 220
(c) 280 (d) 370

51. An individual purchases three qualities of pencils. The relevant data is given below:

Quality Pencil	Price per pencil	Money spent (in Rs.)
A	1.00	50
B	1.50	X
C	2.00	20

It is known that the average price per pencil is Rs 1.25. What is the value of x?

[CDS 2017 (1)]

- (a) Rs 10 (b) Rs 30
(c) Rs 40 (d) Rs 60

52. Consider the following frequency distribution:

X	Frequency	Cumulative Frequency
1	8	8
2	10	18
3	f_1	29
4	f_2	45

What are the values of f_1 and f_2 respectively:

[CDS 2017 (2)]

- (a) 10 and 17 (b) 17 and 10
(c) 11 and 16 (d) 16 and 11

For the next five (05) items that follow:

In a University there are 1200 students studying four different subjects Mathematics, Statistics, Physics and Chemistry. 20% of the total number of students are studying Mathematics, one-fourth of the total number of students are studying Physics, 320 students are studying Statistics and remaining students are studying Chemistry. Three-fifth of the total number of students studying Chemistry are girls. 150 boys are studying Mathematics. 60% of student studying Physics are boys. 250 girls are studying Statistics.

53. What is the total number of boys studying Statistics and Physics?

[CDS 2017 (2)]

- (a) 180 (b) 240
(c) 250 (d) 310

54. The number of girls studying Statistics is what percentage (approximate) of the total number of students studying Chemistry?

[CDS 2017 (2)]

- (a) 58.8 (b) 73.5
(c) 78.7 (d) 80.6

55. In which subjects is the difference between the number of boys and girls equal to?

[CDS 2017 (2)]

- (a) Mathematics and Chemistry
(b) Statistics and Chemistry
(c) Mathematics and Physics
(d) Mathematics and Statistics

56. What is the difference between the number of boys studying Mathematics and the number of girls studying Physics?

[CDS 2017 (2)]

- (a) 20 (b) 30
(c) 60 (d) 80

57. What is the ratio of the total number of boys to the total number of girls?

[CDS 2017 (2)]

- (a) 67:83 (b) 17:26
(c) 27:19 (d) 189:179

58. Frequency density of a class is computed by the ratio:

[CDS 2017 (2)]

- (a) Class frequency to the class width
(b) Class frequency to total frequency
(c) Class frequency to of total number of classes
(d) Cumulative frequency up to that class to total frequency

59. The pie diagrams on the monthly expenditure of two families A and B are drawn with radii of two circles taken in the ratio 16:9 to compare their expenditure.

Which one of the following is the appropriate data used for the above mentioned pie diagram?

[CDS 2017 (2)]

- (a) Rs 16,000 and Rs 9,000
(b) Rs 8,000 and Rs 4,500
(c) Rs 25,600 and Rs 8,100
(d) Rs 4,000 and Rs 3,000

60. Consider the following statements:

Statement I: The value of a random variable having the highest frequency is mode

Statement II: Mode is unique

Which one of the following is correct in respect of the above statements?

[CDS 2017 (2)]

- (a) Both Statements I and II are true and statement II is the correct explanation of statement I.

(b) Both Statement I and II are true and statement II is not the correct explanation of statement I.

(c) Statement I is true but statement II is false

(d) Statement I is false but statement II is true

61. Which one of the following is not correct?

The proportion of various items in a pie diagram is proportional to the

[CDS 2017 (2)]

- (a) Areas of slices
(b) Angles of slices
(c) Lengths of the curved arcs of the slices
(d) Perimeter of the slices

62. The geometric mean of x and y is 6 and the geometric mean of x , y and z is also 6. Then the value of z is:

[CDS 2017 (2)]

- (a) 12 (b) $\sqrt{6}$
(c) 6 (d) $3\sqrt{6}$

63. The total number of live births in a specific locality during different months of a specific year was obtained from the office the Birth Registrar. This set of data may be called:

[CDS 2017 (2)]

- (a) Primary data (b) Secondary data
(c) Recorded data (d) Countable data

64. The heights (in cm) of 5 students are 150, 161, 165, 144 and 155. What are the values of mean and median (in cm) respectively?

[CDS 2017 (2)]

- (a) 165 and 161 (b) 155 and 155
(c) 160 and 155 (d) 155 and 161

65. In the following table of inverse variation, what are the values of A , B and C respectively?

M	15	-6	2	C
N	-4	A	B	60

[CDS 2018 (1)]

- (a) 10, -30, -1 (b) 10, -1, 30
(c) -30, 10, -1 (d) -1, -30, -10

66. A Pie Chart is drawn for the following data:

Sector	Percentage
Agriculture and Rural Development	12.9
Irrigation	12.5
Energy	27.2
Industry and Minerals	15.4
Transport and Communication	15.9

Social Services	16.1
-----------------	------

What is the angle (approximate) subtended by the Social Services Sector at the centre of the circle?

[CDS 2018 (1)]

- (a) 45° (b) 46°
(c) 58° (d) 98°

67. The arithmetic mean of two numbers is 10 and their geometric mean is 8. What are the two numbers?

[CDS 2018 (1)]

- (a) 15, 5 (b) 12, 8
(c) 16, 4 (d) 18, 2

68. If H is the harmonic mean of P and Q, then the value of $\frac{H}{P} + \frac{H}{Q}$ is:

[CDS 2018 (2)]

- (a) 1 (b) 2
(c) $\frac{P+Q}{PQ}$ (d) $\frac{PQ}{P+Q}$

69. Data on ratings of hotels in a city is measured on:

[CDS 2018 (2)]

- (a) Nominal scale (b) Ordinal scale
(c) Interval scale (d) Ratio scale

70. The median of 19 observations in 30. Two more observations are made and the values of these are 8 and 32. What is the median of the 21 observation?

[CDS 2018 (2)]

- (a) 32
(b) 30
(c) 20
(d) Cannot be determined due to insufficient data

71. As the number of observations and classes increases, the shape of a frequency polygon.

[CDS 2018 (2)]

- (a) Tends to become jagged
(b) Tends to become increasingly smooth
(c) Stays the same
(d) Varies only if data become more reliable

72. Consider the following statements:

Statement I: Median can be computed even when the end intervals of a frequency distribution are open.

Statement II: Median is a positional average. Which one of the following is correct in respect of the above statements?

[CDS 2018 (2)]

(a) Both Statements I and Statement II are true and Statement II is the correct explanation of statement I

(a) Both Statements I and Statement II are true and Statement II is the correct explanation of statement I

(b) Both Statements I and Statement II are true and Statement II is not the correct explanation of statement I.

(c) Statement I is true but statement II is false
(d) Statements I is false but statement II is true

Consider the following for the next three (03) items:

In a certain town of population size 1,00,000 three types of newspapers (I, II and III) are available. The percentages of the people in the town who read these papers are as follows:

Newspapers	Proportion of Readers
I	10%
II	30%
III	5%
Both I and II	8%
Both II and III	4%
Both I and III	2%
All the three (I, II, III)	1%

73. What is the number of people who read only one newspaper?

[CDS 2019 (1)]

- (a) 20,000 (b) 25,000
(c) 30,000 (d) 35,000

74. What is the number of people who read at least two newspapers?

[CDS 2019 (1)]

- (a) 12,000 (b) 13,000
(c) 14,000 (d) 15,000

75. What is the number of people who do not read any of these newspapers?

[CDS 2019 (1)]

- (a) 62,000 (b) 64,000
(c) 66,000 (d) 68,000

76. Consider the following grouped frequency distribution:

x	f
0-10	8
10-20	12
20-30	10
30-40	p
40-50	9

If the mean of the above data is 25.2, then what is the value of p?

[CDS 2019 (1)]

- (a) 9 (b) 10
(c) 11 (d) 12

77. Consider the following frequency distribution:

x	f
8	6
5	4
6	5
10	8
9	9
4	6
7	4

What is the median for the distribution?

[CDS 2019 (1)]

- (a) 6 (b) 7
(c) 8 (d) 9

78. Diagrammatic representation of data includes which of the following:

1. Bar diagram
2. Pie diagram
3. Pictogram

Select the correct answer using the code given below:

[CDS 2019 (1)]

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

79. The data collected from which one of the following methods is not a primary data?

[CDS 2019 (1)]

- (a) By direct personal interviews
(b) By indirect personal interviews
(c) By schedules sent through enumerators
(d) From published thesis

80. The monthly expenditure of a person is Rs 6,000. The distribution of expenditure on various items is as follows:

	Items of expenditure	Amount (in Rs)
1.	Food	2,000
2.	Clothing	660
3.	Fuel and rent	1200
4.	Education	480
5.	Miscellaneous	1660

If the above data is represented by a percentage bar diagram of height 15cm, then what are the lengths of the two segments of the bar diagram corresponding to education and miscellaneous respectively?

[CDS 2019 (1)]

- (a) 1.25cm and 5cm
(b) 1.2cm and 4.15cm
(c) 1.2cm and 3.5cm
(d) 4.15 cm and 6cm

81. Which one of the following pairs is correctly matched?

[CDS 2019 (1)]

- (a) Median Graphical location
(b) Mean Graphical location
(c) Geometric mean Ogive
(d) Mode Ogive

82. The following pairs relate to frequency distribution of a discrete variable and its frequency polygon. Which one of the following pairs is not correctly matched?

[CDS 2019 (1)]

(a) Base line of the polygon	X-axis
(b) Ordinates of the vertices	Class frequency of the polygon
(c) Abscissa of the vertices	Class marks of The polygon frequency
(d) Area of the polygon	total frequency of the

Directions for the following two (02) items

Read the following frequency distribution for two series of observations and answer the two items that follow:

Class interval	Frequency	
	Series-I	Series-II
10-20	20	4
20-30	15	8
30-40	10	4
40-50	x	2x
50-60	y	y
total	100	100

83. What is the mean of frequency distribution of Series-I?

[CDS 2019 (2)]

- (a) 33-6 (b) 35-6
(c) 37-6 (d) 39-6

84. What is the mode of the frequency distribution of Series-II?

[CDS 2019 (2)]

- (a) 26 (b) 36
(c) 46 (d) 56

Directions for the following two (04) items

Read the following information and answer the four items that follow:

Let the distribution of number of scooters of companies X and Y sold 5 showrooms (A, B, C, D and E) in a certain year be denoted by S1 and the distribution of number of scooters of only company X and by the five showrooms in the same year be denoted by S2.

Show Room	A	B	C	D	E	Total number of scooters sold
S1 (in %)	1	2	1	5	1	6400
S2 (in %)	2	1	2	3	8	3000

85. Number of scooters of company Y sold by showroom E is what per cent of the number of scooters of both companies sold by showroom C?

[CDS 2019 (2)]

- (a) 52 (b) 54
(c) 55 (d) 56

86. Number of scooters of both the companies sold by showroom B is what per cent more than the number of scooters of company X and sold by showroom A?

[CDS 2019 (2)]

- (a) $78\frac{2}{3}$ (b) $83\frac{1}{3}$
(c) $86\frac{2}{3}$ (d) $88\frac{1}{3}$

87. What is the average number of scooters of company Y sold by the showroom A, C and E?

[CDS 2019 (2)]

- (a) $461\frac{1}{3}$ (b) $431\frac{1}{3}$
(c) $426\frac{1}{3}$ (d) $416\frac{1}{3}$

88. What is the difference between the number of scooters of both companies sold by showroom A and total number of scooters of company X sold by showroom B and E together?

[CDS 2019 (2)]

- (a) 416 (b) 426
(c) 432 (d) 436

Directions for the following four (04) items:

Read the following information and answer the four items that follow:

The data shows that Indian roads are turning deadlier over the year.

Year	2014	2015	2016	2017
Number of bikers killed	40957	46070	52750	48746
Number of pedestrian s killed	12330	13894	15746	20457
Number of cyclists killed	4037	3125	2585	3559

89. What was the average number of pedestrian killed per day in the year 2017?

[CDS 2019 (2)]

- (a) 51 (b) 53
(c) 54 (d) 56

90. What is the approximate percentage change in the pedestrians' fatalities during the period 2014-17?

[CDS 2019 (2)]

- (a) 66% (b) 68%
(c) 71% (d) 76%

91. What is the average number of bikers killed daily in road accidents in the year 2017?

[CDS 2019 (2)]

- (a) 163 (b) 152
(c) 147 (d) 134

92. What is the average number of cyclists killed daily in road accidents in 2017?

[CDS 2019 (2)]

- (a) 10 (b) 12
(c) 19 (d) 21

93. What is the median of the data 3, 5, 9, 4, 6, 11, 18?

[CDS 2020 (1)]

- (a) 6 (b) 6.5
(c) 7 (d) 7.5

94. In a pi-diagram there are three sectors. If the ratio of the angles of the sectors is 1:2:3, then what is the angle of the largest sector?

[CDS 2020 (1)]

- (a) 200° (b) 180°
(c) 150° (d) 120°

Directions for the following three (03) items:

Read the following information and answer the three items that follow:

The following data presents count of released convicts who have served prison terms (X), those who have received some educational or technical training during their term (Y) and those who were offered Company placement (Z) respectively, from six different jails A, B, C, D, E and F, in the year 2010.

	X	Y	Z
A	86	45	25
B	1305	903	461
C	2019	940	474
D	1166	869	416
E	954	544	254
F	1198	464	174

95. Jails with highest and smallest percentage of trained convicts are respectively:

[CDS 2020 (1)]

- (a) F and D (b) D and F
(c) C and A (d) D and A

96. Jail with highest placement rate of trained convicts is:

[CDS 2020 (1)]

- (a) F (b) D
(c) B (d) A

97. Jails from which more than half of the trained convicts are offered jobs, are:

[CDS 2020 (1)]

- (a) A, B and C (b) A, B and D
(c) A, D and E (d) A, E and F

98. For two observations, the sum of S and product is P. What is the harmonic mean of these two observations?

[CDS 2020 (1)]

- (a) $\frac{2S}{P}$ (b) $\frac{S}{(2P)}$
(c) $\frac{2P}{S}$ (d) $\frac{P}{(2S)}$

99. The maximum marks in a test are converted from 250 to 50 for the purpose of an Internal Assessment. The highest marks scored were 170 and lowest marks were 70. What is the difference between the maximum and minimum marks scored in the Internal Assessment.

[CDS 2020 (2)]

- (a) 15 (b) 17
(c) 20 (d) 24

100. Fifteen candidates appeared in an examination. The marks of the candidates who passed in the examination are 9, 6, 7, 8, 8, 9, 6, 5, 4 and 7. What is the median of marks of all the fifteen candidates?

[CDS 2020 (2)]

- (a) 6 (b) 6.5
(c) 7 (d) 7.5

101. If the yield (in gm) of barley from 7 plots of size one square yard each, were found to be 180, 191, 175, 111, 154, 141 and 176, then what is the median yield?

[CDS 2020 (2)]

- (a) 111gm (b) 154gm
(c) 175gm (d) 176gm

102. Which one of the following measures of central tendency will be used to determine the average size of the shoe sold in the shop?

[CDS 2020 (2)]

- (a) Arithmetic mean (b) Geometric
(c) Median (d) Mode

103. When the class intervals have equal width, the height of a rectangle in a histogram represents:

[CDS 2020 (2)]

- (a) Width of the class
(b) Lower class limit
(c) Upper class limit
(d) Frequency of the class

104. What is the algebraic sum of the deviations from the mean of a set of values 25, 65, 73, 75, 83, 76, 17, 15, 7, 14?

[CDS 2020 (2)]

- (a) -1 (b) 0
(c) 1 (d) 2

105. What is the median of 2, 4, 6, ..., 100?

[CDS 2020 (2)]

- (a) 48 (b) 49
(c) 50 (d) 51

106. The harmonic mean and the geometric mean of two numbers are 10 and 12 respectively. What is their arithmetic mean?

[CDS 2020 (2)]

- (a) $\frac{25}{3}$ (b) $\sqrt{120}$
(c) 11 (d) 14.4

107. Consider the following data with regard to production of cars (in lakhs):

	Year 2015	Year 2016
Country A	35	38
Country B	45	47
Country C	88	93
Country D	75	79
Country E	58	60.9

In which of the countries, the production of cars has increased by more than or equal to 5% in 2016 over 2015?

[CDS 2021 (1)]

- (a) B and E (b) A, C and D only
(c) A, C, D and E (d) A, D and E only

108. The following table shows the marks of 90 students in a test of 80 marks:

Marks	Number of Students
1-10	5
11-20	8
21-30	10
31-40	13
41-50	18
51-60	17
61-70	12
71-80	7

The percentage of students who have obtained less than or equal to 50% marks is:

[CDS 2021 (1)]

- (a) 30% (b) 40%
(c) 45% (d) 60%

109. What is the median of the following data?
2, 3, -1, 2, 6, 8, 9

[CDS 2021 (1)]

- (a) 2 (b) 3
(c) 4 (d) 5

Directions:

For the next four (04) items, consider the following data with regard to different types (I, II, III, IV, V) of multivitamin tables produced in a company (in lakhs)

Year	I	II	III	IV	V
2000	160	80	70	90	75

2001	200	150	85	160	100
2002	135	35	44	95	85
2003	240	95	120	80	120
2004	180	110	85	95	115
2005	210	150	100	92	110

110. Which product is produced least over the years 2000-2005?

[CDS 2021 (1)]

- (a) Type II (b) Type III
(c) Type IV (d) Type V

111. In which one of the following pairs of years, the difference in total number of tablets produced between them is minimum?

[CDS 2021 (1)]

- (a) (2003, 2005) (b) (2001, 2005)
(c) (2003, 2004) (d) (2000, 2002)

112. The ratio of percentage drop in total production in 2004 compared to 2001 to that 2000 compared to 2001, is:

[CDS 2021 (1)]

- (a) $\frac{1}{3}$ (b) $\frac{1}{4}$
(c) $\frac{1}{2}$ (d) $\frac{1}{5}$

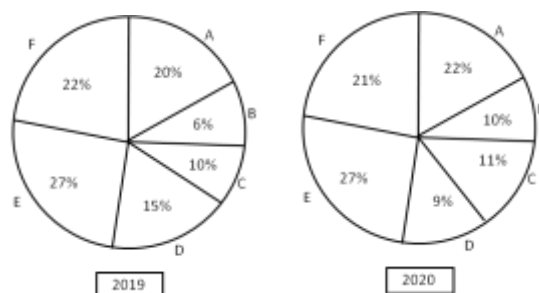
113. In which year, the production of Type I is more than the sum of the production of Type II and Type IV?

[CDS 2021 (1)]

- (a) 2001 (b) 2002
(c) 2003 (d) 2004

Consider the following for the next four (04) items that follow:

The following Pie-Charts show the percentage of different categories (A, B, C, D, E and F) of employees in company in the year 2019 and 2020. The total number of employees was 4000 in 2019 and 5000 in 2020.



- 114.** There are two categories of employees in 2019 whose total strength remained the same in 2020. What are these two categories?

[CDS 2021 (2)]

- (a) C and B (b) E and A
(c) E and F (d) C and D

- 115.** In which one of the following categories of employees, the percentage change in number of employees in 2020 was maximum as compared to 2019?

[CDS 2021 (2)]

- (a) A (b) B
(c) C (d) D

- 116.** What was the percentage increase of category -E employees in 2020 as compared to that in 2019?

[CDS 2021 (2)]

- (a) 20% (b) 25%
(c) 30% (d) 40%

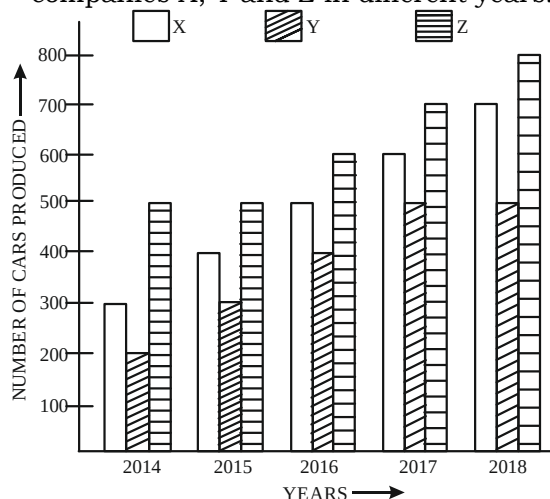
- 117.** There are two categories of employees whose strength increased by same percentage in 2020. What are the two categories?

[CDS 2021 (2)]

- (a) E and F (b) C and E
(c) A and E (d) A and C

Consider the following for the next four (04) items that follows:

The following Bar-Chart gives the production of cars by three different companies X, Y and Z in different years:



- 118.** The percentage increase in the total production of cars from 2014 to 2018 was:

[CDS 2021 (2)]

- (a) 50% (b) 100%
(c) 150% (d) 200%

- 119.** Consider the production of each company separately. In how many instances was the percentage increase in production of cars over the previous year's production greater than 20%?

[CDS 2021 (2)]

- (a) 5 (b) 6
(c) 7 (d) 8

- 120.** In which year was the percentage increase in production of cars minimum as compared to its previous year?

[CDS 2021 (2)]

- (a) 2015 (b) 2016
(c) 2017 (d) 2018

- 121.** In one of the years the percentage increase in production of cars was minimum as compared to its previous year. What was the minimum percentage?

[CDS 2021 (2)]

- (a) $11\frac{1}{9}\%$ (b) $11\frac{2}{9}\%$
(c) $13\frac{1}{3}\%$ (d) 20%

Direction: Consider the following data for the two (02) items that follows:

The table below gives the age-wise population percentage of a city:

Age group	Percentage
Below 30 years	14.00
30-34.99	29.75
35-39.99	26.25
40-44.99	0
45-49.99	18.50
50 years and above	11.50

The number of persons below the age of 40 years is 10.5 lakhs

- 122.** What is the total population of the city (in lakhs)?

[CDS 2022 (1)]

- (a) 21 (b) 18
(c) 15 (d) 12

- 123.** If the ratio of taxpayers to other persons in the same age group below 30 years is 1 : 2, then what is the number of taxpayers (in lakhs) in that age group?

[CDS 2022 (1)]

- (a) 0.4 (b) 0.7
(c) 0.85 (d) 1.05

Direction: Consider the following data for the two (02) items that follow:

The expenditure (in lakhs of rupees) of a company for the years 2011 to 2017 is as under:

Year	Expenditure
2011	13.8
2012	15.4
2013	10.4
2014	13.1
2015	15.8
2016	17.2
2017	19.4

- 124.** How many times the increase in expenditure in a year exceeded by more than 15% as compared to previous year?

[CDS 2022 (1)]

- (a) 2 (b) 3
(c) 4 (d) 5

- 125.** In which year, the percentage increase in expenditure is maximum as compared to its previous year?

[CDS 2022 (1)]

- (a) 2012 (b) 2014
(c) 2015 (d) 2017

Direction: Consider the following for the two (02) items that follow:

The budget allocations represented in a pie diagram under five different heads A, B, C, D and E are respectively 40%, 18%, 9%, 25% and 8%. The total budget allocation is ₹300.4 lakhs.

- 126.** How much less amount is allocated to A and C together as compared to B, D and E together?

[CDS 2022 (1)]

- (a) ₹3.004 lakhs (b) ₹4.005 lakhs
(c) ₹6.008 lakhs (d) ₹8.010 lakhs

- 127.** How much amount will be increased on A if the total budget is increased by three times?

[CDS 2022 (1)]

- (a) ₹360.48 lakhs (b) ₹300.36 lakhs
(c) ₹240.32 lakhs (d) ₹180.40 lakhs

- 128.** The arithmetic mean and the geometric mean of two positive numbers p and q ($p > q$) are A and G respectively. Which one of the following is correct?

[CDS 2022 (1)]

- (a) $A > G$ (b) $G > A$
(c) $A = G$ (d) $A^2 = G$

Consider the following for the next three (03) items that follow:

x	0	1	2	3	4	5
f	46	p	q	25	10	5

Total frequency is 200 and mean of the distribution is 1.46.

- 129.** What is the value of p?

[CDS 2022 (II)]

- (a) 70 (b) 72
(c) 76 (d) 78

- 130.** What is the value of q?

[CDS 2022 (II)]

- (a) 32 (b) 34
(c) 36 (d) 38

- 131.** What is the median of the distribution?

[CDS 2022 (II)]

- (a) 1 (b) 2
(c) 3 (d) 4

- 132.** If x is the harmonic mean between y and z, then which one of the following is correct?

[CDS 2022 (II)]

- (a) $xy + xz - yz = 0$
(b) $xy + xz - 2yz = 0$
(c) $xy + xz + yz = 0$
(d) $xy + xz - 4yz = 0$

- 133.** What is the median of all possible factors of 120?

[CDS 2022 (II)]

- (a) 10 (b) 11
(c) 12 (d) 13.6

- 134.** The sum of deviations of a set of n values measured from 50 is -10 and the sum of deviations of the values measured from 46 is 70. What is the mean of the values?

[CDS 2022 (II)]

- (a) 48.5 (b) 49.0
(c) 49.5 (d) 50.0

- 135.** Consider the following table in respect of students of 4 schools who appeared in a test:

School	Name of students	Average marks in the test
I	60	60
II	50	80
III	50	40
IV	x	50

If the average marks of the students of all four schools are 58, then how many students appeared from School-IV?

[CDS 2022 (II)]

- (a) 38 (b) 40
(c) 42 (d) 44

- 136.** The data of different natural numbers 4, 7, 10, 14, $2x+3$, $2x+5$, 22, 23, 30, 50 are in ascending order. How many possible values are there for the median of the data for various values of x ?

[CDS 2022 (II)]

- (a) only one value
(b) only two values
(c) only three values
(d) five values

Consider the following data for the next two (02) items that follows:

Class	0-30	30-60	60-90	90-120
Frequency	4	5	7	4

- 137.** What is the mode of the distribution?
[CDS – 2023 (1)]
(a) 60 (b) 72
(c) 75 (d) 80

- 138.** If the median (P) and mode (Q) satisfy the relation $7(Q-P) = 9R$, then what is the value of R?

[CDS – 2023 (1)]

- (a) 6 (b) 5
(c) 3 (d) 1

Consider the following data for the next two (02) items that follow:

Class	40-50	50-60	60-70	70-80
Frequency	4	3	1	2

- 139.** What is the mean of the distribution?
[CDS – 2023 (1)]
(a) 51 (b) 52
(c) 54 (d) 56

- 140.** If M is the median, then what is the value $3M$?

[CDS – 2023 (1)]

- (a) $53\frac{1}{3}$ (b) 60
(c) 160 (d) 180

- 141.** The sum of deviations of n numbers from 10 and 20 are a , b respectively. If $\frac{b}{a} = -4$, then what is the mean of these n numbers?

[CDS-2023 (2)]

- (a) 12 (b) 14
(c) 16 (d) 18

- 142.** If the median of observations 12, 1, 8, 54, 61, 28, 45, 35, 21, 17 is M, then what is the value of $2M + 5$?

[CDS-2023 (2)]

- (a) 12 (b) 28
(c) 52 (d) 54

Consider the following for the next two (02) items that follow:

A grouped frequency distribution is given below:

Weekly wages in Rupees (₹)	Number of workers
2050-2550	5
2550-3050	10
3050-3550	k
3550-4050	8
4050-4550	2
4550-5050	10

- 143.** If average weekly wages earned by a worker is Rs.3520, then what is the value of k ?

[CDS-2023 (2)]

- (a) 10 (b) 12
(c) 15 (d) 20

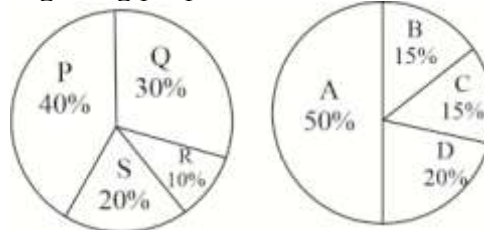
- 144.** What is the median (approximate value) of the distribution?

[CDS-2023 (2)]

- (a) Rs.3263 (b) Rs.3383
(c) Rs.3413 (d) Rs.3483

Consider the following for the next three (03) items that follow:

The following Pie-Chart-I shows the people migrating to Delhi from different Indian States (P, Q and R are three different States and S is the combined group of other States) and Pie-Chart-II indicates the different age groups A, B, C and D of these migrating people for each State.



- 145.** If the people coming from a particular State belong to S are 15% and 24,000 in number, then what is the total number of migrating people belonging to the age group B?

[CDS-2024 (1)]

- (a) 1.2 lac (b) 1.25 lac
(c) 1.30 lac (d) 1.50 lac .

146. What is the maximum of differences between the number of people coming from different groups P, Q, R and S?

[CDS-2024 (1)]

- (a) 1.6 lac (b) 1.8 lac
(c) 2.4 lac (d) 2.6 lac

147. What is the difference between number of people coming from R having age group A and those coming from Q having age group D?

[CDS-2024 (1)]

- (a) 6,000 (b) 8,000
(c) 12,000 (d) 18,000

148. Let the positive numbers $a_1, a_2, a_3, \dots, a_{3n}$ be in GP. If P is the GM of $a_1, a_2, a_3, \dots, a_n$ and Q is the GM of $a_{n+1}, a_{n+2}, a_{n+3}, \dots, a_{3n}$, then what is the GM of $3n$ numbers?

[CDS-2024 (1)]

- (a) P^2Q (b) PQ^2
(c) $\sqrt{PQ}$ (d) $P^{1/3}Q^{2/3}$

149. The sum of two positive numbers is 40. If the GM of these two numbers is lower than their AM by 20%, then what is the difference between the two numbers?

[CDS-2024 (1)]

- (a) 12 (b) 18
(c) 24 (d) 28

150. Consider the following data:

110, 41, 43, 95, 127, 99, 61, 92, 71, 93, 110, 36

If 93 is replaced by 94, then consider the following statements:

1. The difference between new median and old median is 1
2. The difference between new mean and old mean is less than 0.1
3. The difference between new mode and old mode is zero

Which of the statements given above are correct?

[CDS-2024 (1)]

- (a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

ANSWER KEY

1.	b	2.	d	3.	b	4.	b	5.	c	6.	b	7.	c	8.	d	9.	a	10.	a
11.	b	12.	d	13.	c	14.	a	15.	c	16.	b	17.	d	18.	c	19.	c	20.	a
21.	c	22.	a	23.	c	24.	d	25.	a	26.	c	27.	a	28.	b	29.	b	30.	a
31.	b	32.	c	33.	d	34.	b	35.	d	36.	a	37.	b	38.	a	39.	c	40.	c
41.	c	42.	a	43.	d	44.	c	45.	c	46.	c	47.	d	48.	a	49.	c	50.	a
51.	b	52.	c	53.	c	54.	b	55.	c	56.	b	57.	a	58.	a	59.	a	60.	c
61.	d	62.	c	63.	b	64.	b	65.	a	66.	c	67.	c	68.	b	69.	b	70.	b
71.	b	72.	a	73.	a	74.	a	75.	d	76.	c	77.	c	78.	d	79.	d	80.	b
81.	a	82.	d	83.	c	84.	c	85.	c	86.	c	87.	a	88.	d	89.	d	90.	a
91.	d	92.	a	93.	a	94.	b	95.	b	96.	d	97.	a	98.	b	99.	c	100.	a
101.	c	102.	d	103.	d	104.	b	105.	d	106.	d	107.	c	108.	b	109.	b	110.	b
111.	a	112.	c	113.	c	114.	d	115.	b	116.	b	117.	d	118.	b	119.	a	120.	d
121.	a	122.	c	123.	b	124.	a	125.	b	126.	c	127.	c	128.	a	129.	c	130.	d
131.	b	132.	b	133.	b	134.	c	135.	b	136.	d	137.	b	138.	a	139.	d	140.	C
141.	a	142.	d	143.	c	144.	b	145.	a	146.	c	147.	b	148.	d	149.	c	150.	b

CAPF PYQ

1. There is a group of 5 people among which there is one couple. In how many ways can these 5 people be seated in a row having 5 chairs if the couple is to be seated next to each other ?
[CAPF 2022]
(a) 24 (b) 48
(c) 60 (d) 120
2. A coin is tossed 3 times. The probability of getting exactly 2 heads is ?
[CAPF 2022]
(a) $\frac{1}{3}$ (b) $\frac{3}{8}$
(c) $\frac{1}{2}$ (d) $\frac{5}{8}$

ANSWER KEY

1.	b	2.	b																
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